Marketing Research

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Contents

Preface

No prerequisites required to read this book

Chapter 1

Introduction

This is not an actual book or guide on marketing research, but more like a placeholder for major constructs in marketing as well as interesting articles in marketing.

Feel free to comment or suggest changes to the content of this book.

Differences between goods and services:

- Intangibility
- Complexity
- Heterogeneity

Dimensions of Qualitative Research:

1. Ontology: "is a philosophical belief system about the nature of social reality- what can be known and how".

2. Epistemology: "is a philosophical belief system about who can be a knower".

3. Methodology (theoretical perspective): "an account of social reality or some component of it that extends further than what has been empirically investigated":

    - Post-positivist: causal relationship can be tested (i.e., disprove)
    - Interpretive: assumes the world is constantly being constructed through group interactions;hence social reality can be understood via the perspectives of social actors enmeshed in meaning-making activities".
    - Critical: view social reality as an ongoing construction, and suggest that discourses created in shifting fields of social power shape social reality and its study.

4. Methods: "technique for gathering evidence".

Procedures recommended for literature review papers:

- (Katsikeas et al., 2016)
- (Lipsey and Wilson, 2001)

This part is based on professor Ajay Kohli's workshop on theory

Purpose of theory is to

1. Explain
2. Predict
3. Control

We need to have theory because it's economical and can help us figure out many other manifestations.

Definition: Theory is hypothesized/expected description of causal relationships among construct

Parts of theory:

1. Constructs (labels, definitions)
2. Statement stating relationship among constructs
3. Arguments to why X causes Y (including assumptions)
4. Boundary conditions

Developing theoretical arguments

1. Main effect: X -> A -> Y
    1. Both arrows are assumptions that can be agreed upon provided prior theory or anecdotal evidence
    2. Assumptions are taken to be true
    3. Propositions are yet to be true
    4. Difference between a proposition and assumption is the Goldilocks distance between X and Y

One proposition can become a theory

Conceptual models = linkages without arguments

Framework = linkages between categories of variables, theory at abstract level

Framework is important because you can come up with more holes (generative power)

How to argue for U and inverted-U effect (Haans et al., 2016)

1.1 Approaches to Research

1.1.1 Theory First Approaches

1.1.2 Empiric First-Apporaches

(Golder et al., 2023)

- Theory-First Approach is when researchers "borrowed refined, or developed and tested empirically" (p. 319)
- Stages of the Empiric First Research
    1. Identify Opportunity (grounded in a real-world phenomenon, observation):
        1. Explore the real word for inspiration, novelty and relevance
        2. Evaluate Aptness fo EF Approach
    2. Explore Terrain (obtaining and analyze data)
        1. Generative/Compile/Acquire/Data
        2. Understand the Data
        3. Define the Scope
    3. Advance Understanding (get valid insights without developing or testing theory)
        1. Uncover Empirical Regularities
        2. Formulate Conceptual and Theoretical Insights
        3. Advise Stakeholders

1.1.3 A Shift to Observable-to-construct links

1.1.4 Conceptual Contributions

(Kindermann et al., 2023)

1.1.5 Disruption-Driven Anomalies Apporach

(Swaminathan et al., 2023)

Types of anomalies:

1. Gap between theory and practice
2. Disconnects between theory and empirical evidence
3. Challenges to foundational assumptions underlying an existing paradigms
4. Emergence of entirely new phenomena.

Part I

CONSTRUCTS

Chapter 2

Construct vs. Variable

Table 2.1: measured or perceived vs. reality

		Reality	
		True	False
Measured	True	Correct	Type 1 error (False positive)
	False	Type 2 error (False Negative)	Correct

picture from (Irvine, 2019)

Constructs are things we make up, define clearly

Phenomena are observable variables

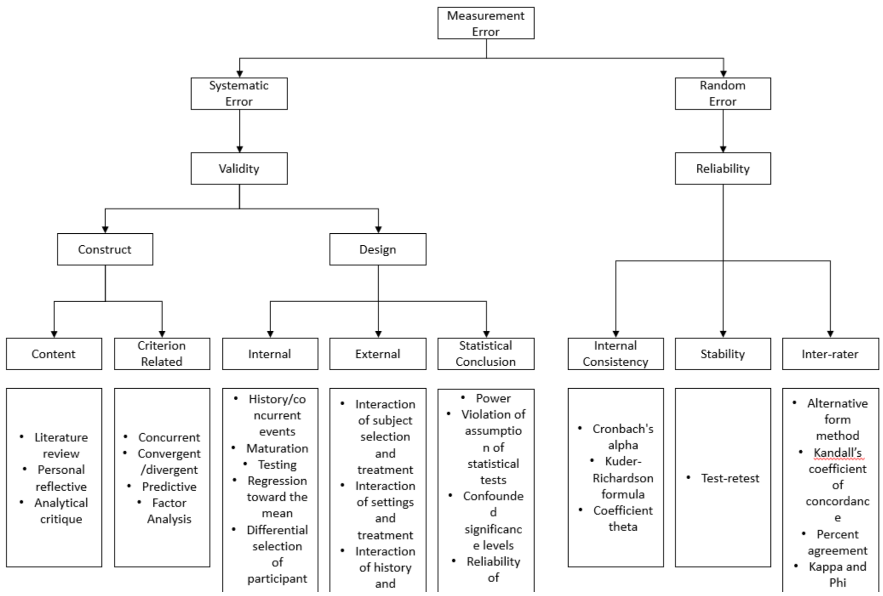


Figure 2.1: Measurement Error

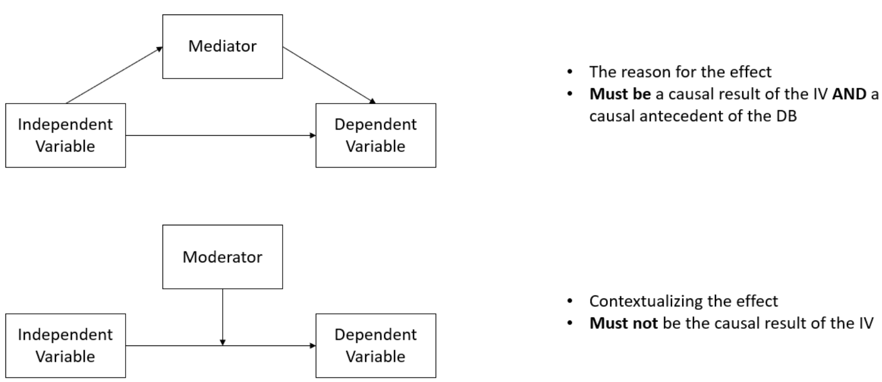


Figure 2.2: Moderator vs. Mediator

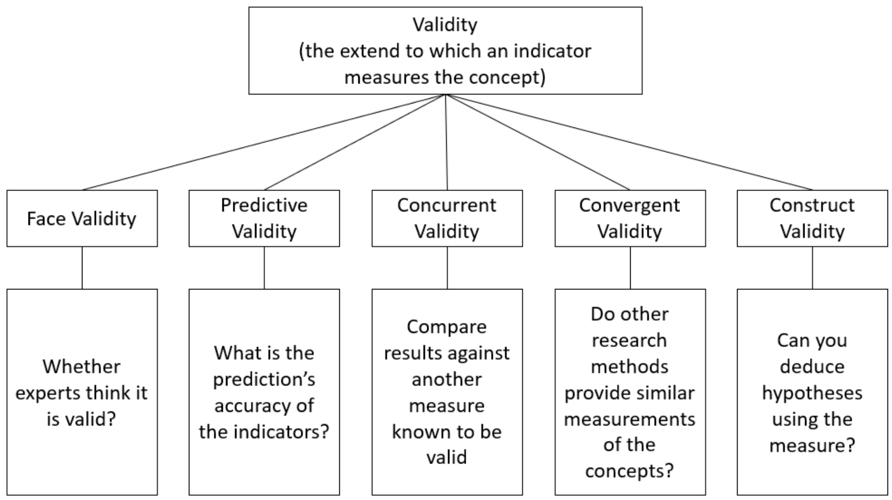


Figure 2.3: Validity

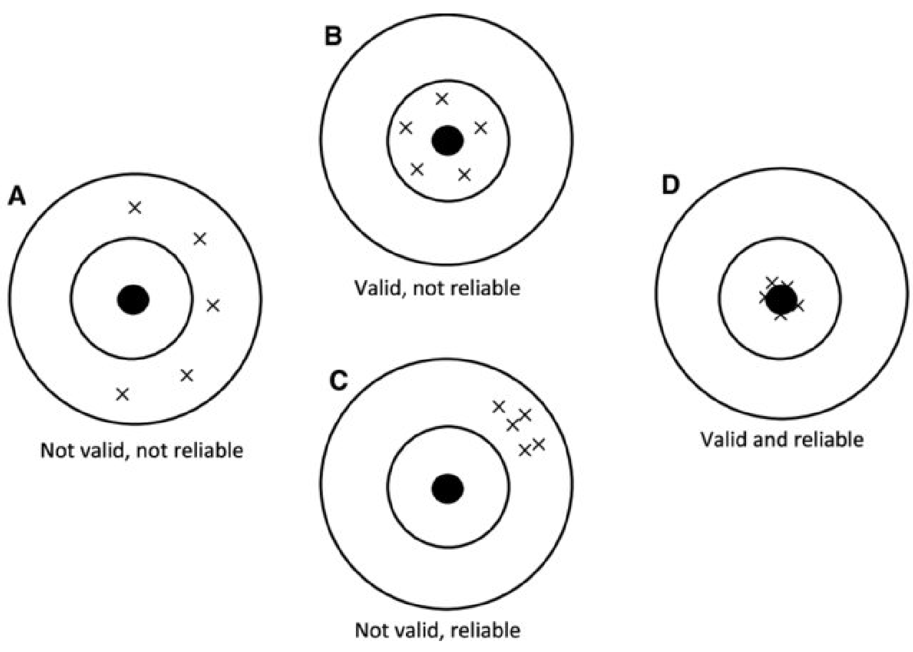


Figure 2.4: Reliability vs. Validity

Chapter 3

Satisfaction

Customer satisfaction was measured using sentiment analysis (Kumar and Zymbler, 2019; Wei et al., 2020).

Another implementation can be found here

Your assessment of satisfaction can come from:

1. Emotional part (Affect)
2. Logical part (Cognitive)

Customer sanctification has an impact on the financial market

- Stock returns
  - Overall market (Fornell et al., 2006)
  - Only in the Internet and computer sector (Jacobson and Mizik, 2009a) (this is the current academic consensus (Jacobson and Mizik, 2009b)).
  - Prospect theory validation: A one-unit increase in customer satisfaction increases abnormal returns by.56 percentage points, while dissatisfaction decreases them by 1.34 percentage points.(Malshe et al., 2020). Short interest (a measure of short seller activity) mediates the effect of customer (dis)satisfaction on abnormal stock returns.
- Risk of stock returns: positive changes in customer satisfaction lead to negative changes in overall and downside systematic (i.e., market movements)) and idiosyncratic risk (i.e., stock returns volatility). (Tuli and Bharadwaj, 2009)

(Oliver and Burke, 1999) Expectation processes in satisfaction formation: A field study

- **Objective**: To examine the role and persistence of expectations and related effects within the expectancy disconfirmation and performance model in the context of a novel restaurant experience.
- **Methodology**:
    1. **Expectation Manipulation**: Preconsumption expectations were set using realistic reviews.
    2. **Service Experience Recording**: Actual dining experiences were documented using protocol methods.
    3. **Postconsumption Measurement**: Perceptions post dining were captured immediately after the experience.
- **Key Findings**:
    1. **Immediate Decline of Expectation's Effect**: The manipulated expectations had an immediate impact that faded over the course of the dining experience.
    2. **Forward Assimilation**: Expectations influenced perceptions of performance.
    3. **Backward Assimilation**: Retrospective expectations were, to an extent, shaped by actual performance.
    4. **Performance Comparisons & Satisfaction**: Differences between expectations and actual experience (disconfirmation) played a pivotal role in influencing satisfaction.
    5. **Model Dimension-Specificity**: The expectancy disconfirmation model works in a dimension-specific manner regarding its components.

(Slotegraaf and Inman, 2004) see Branding

(Gruca and Rego, 2005) Customer satisfaction, cash flow, and shareholder value

Objective:

- Establish a link between customer satisfaction and characteristics of future cash flows (growth and stability) that affect shareholder value.

Data and Methodology:

- Utilized longitudinal data from the American Customer Satisfaction Index (ACSI) and COMPUSTAT.
- Employed hierarchical Bayesian estimation to analyze the relationship.

Key Findings:

1. **Direct Impact:** Customer satisfaction plays a vital role in driving future cash flow growth and reducing its variability.

2. **Shareholder Value:** By influencing cash flow growth and stability, customer satisfaction contributes directly to increasing shareholder value.

Robustness Tests:

1. **Consistency Across Variables:** The relationship between satisfaction and future cash flows remains stable across different firm and industry characteristics.
2. **Multimeasure and Multimethod Estimation:** Further analysis using various measures and methods confirmed the initial findings, emphasizing the robustness of the results.

(Morgan and Rego, 2006) Customer Satisfaction and Loyalty Metrics in Predicting Business Performance

Objective:

- To determine which customer feedback metrics are most valuable in predicting future business performance.

Data and Methodology:

- The study utilizes data from the American Customer Satisfaction Index (ACSI).
- The research period spans from 1994-2000.
- Linkages between six satisfaction and loyalty metrics are assessed against measures from COMPUSTAT and CRSP databases related to various aspects of firms' business performance.

Key Metrics Examined:

1. Top 2 Box customer satisfaction scores.
2. Intention-to-repurchase loyalty scores.
3. Average customer satisfaction scores.
4. Net promoters.
5. Repurchase likelihood.
6. Proportion of customers complaining.

Key Findings:

1. **Most Predictive Metric:** Average satisfaction scores were found to be the most valuable in forecasting future business performance.
2. **Strong Predictive Value:** Top 2 Box satisfaction scores also exhibited a robust predictive capacity.
3. **Moderate Predictive Value:** Metrics like repurchase likelihood and the proportion of customers complaining showed some predictive value,

but this varied based on the specific dimension of business performance considered.

4. **Limited Predictive Value:** Metrics rooted in recommendation intentions (like net promoters) and recommendation behavior (average number of recommendations) demonstrated little to no value in predicting future business performance.

(Rego et al., 2013) Market share - customer satisfaction relationship

- **Objective:** Examine the relationship between market share and customer satisfaction, metrics conventionally utilized for assessing marketing performance.
- **Prevailing Assumption:** A positive correlation exists between market share and customer satisfaction.
- **Existing Literature:** Indicates often a negative or nonsignificant correlation between the two metrics.
- **Methodology:** Utilized an extended temporal analysis in a representative sample from U.S. consumer markets.
- **Key Findings:**
  - A consistently significant negative relationship was observed between market share and customer satisfaction.
  - Customer satisfaction was not a robust predictor of prospective market share.
  - Market share, conversely, was a strong negative predictor of subsequent customer satisfaction.
- **Conditions for Variation:**
  - The predictive power of a firm's customer satisfaction for future market share is enhanced when:
    * Benchmarked against the firm's nearest competitor.
    * The costs associated with customer brand switching are minimal.
- **Underlying Mechanism:** The generally negative relationship between market share and ensuing customer satisfaction is largely mediated by preference heterogeneity.
- **Strategic Implications:**
  - Marketing a broader range of brands attenuates the adverse impact of preference heterogeneity on impending customer satisfaction.
  - A diversified brand portfolio emerges as a tactical resolution for the inherent trade-off between market share and satisfaction.

(Malshe and Agarwal, 2015) found that firm financial leverage can affect marketing outcomes and firm value

- IT decreases customer satisfaction (advertising intensity mediates this effect)
- moderates the relationship between satisfaction and firm value.

Service firms and firms in competitive industries suffer more from financial leverage

Customer Satisfaction can also impact utility profit (i.e., within the utility industry) (i.e., in monopoly industry). (Bhattacharya et al., 2020)

(Bhattacharya et al., 2021) Customer satisfaction and firm profits in monopolies: A study of utilities

- **Objective:** Investigate the correlation between customer satisfaction and the financial performance of utility firms, addressing a gap in literature that predominantly focuses on competitive markets, with monopolistic markets like utilities being largely overlooked.
- **Context:** The utilities sector offers an intriguing environment due to its stringent regulatory requirements that yield precise operating and accounting data.
- **Data & Methodology:**
  - Sourced from U.S. public utility firms.
  - Aims to ascertain the causal dynamics linking customer satisfaction with profitability in the utilities domain.
- **Key Findings:**
  - Contrary to the evident lack of influence of customer satisfaction on future revenues, it harbors a positive correlation with future profitability.
  - The crux of this correlation is embedded in the substantial reduction of operational costs affiliated with higher customer satisfaction. Specifically, costs related to utility firm distribution, customer service, and general administration diminish with increased satisfaction.
  - Post hoc evaluations align with two pivotal assertions:
    1. Customer satisfaction mitigates the costs related to direct customer interactions and fosters enhanced employee engagement by curbing the challenges tied to managing dissatisfied customers.
    2. A satisfied customer base is synonymous with heightened levels of trust and collaboration.

Chapter 4

Innovation

(Chandrasekaran et al., 2020) introduce a model for successive technologies adoption that accounts for:

- Different rate of technology disengagement
- Different rate of technology adoption

Diffusion

Bass Model (Bass, 1969)

$$h(t) = p + qF(t)$$

where

- $h(t)$ = rate of adoption, given not adopted so far
- $F(t)$ = cumulative fraction of adopters
- p = coefficient of innovation, tendency to adopt independent social contagion
- q = coefficient of imitation tendency to adopt due to social contagion
- m = market size

After going public, firms increase their innovation levels (i.e., have more innovation), but those innovation are less risky (i.e., fewer breakthrough). (Wies and Moorman, 2015)

There is an inverted U-shape relationship between product market competition and innovation (Aghion et al., 2005)

A way to track technology adoption life cycle (Meade and Rabelo, 2004)

(Amabile et al., 1990) found evaluation expectation lowers creativity

(Srinivasan et al., 2009)

- An empirical study that provides evidence for the link between product innovation and marketing investments on stock returns
- Investors react favorably to pioneering innovation as compared to minor updates (7 times greater), and their associated advertising support is 9 times more effective
- Perceived quality of the new product also affect stock returns
- Price promotions have a negative impact on stock returns, which be interpreted that it signals weak demand.

(Slotegraaf and Pauwels, 2008) found that Permanent and cumulative sales effects are greater for brands with higher equity and more product releases, while brands with less equity benefit more from product introductions.

(You et al., 2020) found that CEO and CMO characteristics (e.g., personalities, demographics, experiences, values) can affect innovation and stock returns

(Hitt et al., 1991)

- Acquisition activity has a negative impact in R&D inputs and outputs (e.g., number of patents).

(Sorescu et al., 2003)

- Dominant firms introduce fewer radical innovations than non-dominant firms.
- Financial rewards for radical innovations vary greatly across firms and are tied closely to a firm's resource base.
- Firms that provide higher levels of marketing and technology support and have greater depth and breadth in their product portfolio gain more from their radical innovations.

(Cao et al., 2022)

- Firms can use their innovation potential as a credible signal of their quality at the time of an initial public offering (IPO).
- Innovation potential is positively associated with the initial value of the IPO and with first-day IPO returns, and negatively associated with the extent to which insiders sell their shares at the time of the IPO.
- Patents have a stronger impact on insider sales than preannouncements and generic references to future innovation, while preannouncements have the strongest impact on first-day IPO returns.

(Mishra and Slotegraaf, 2013) Acquisitions and Their Impact on a Firm's Innovation Base

- **Research Context:**

  - Innovation is key to sustainable competitive advantage.
  - Firms aim to establish an innovation base: a collection of inventions, ideas, and discoveries to propel future innovations.
  - Acquisitions as a strategy to build this innovation base.
- **Literature Gap:**
  - Mixed findings on the influence of acquisitions on post-acquisition inventions.
- **Research Focus:**
  - The impact of the type of acquisition, acquisition behavior over time, and characteristics of inventions on post-acquisition innovation outcomes.
- **Methodology:**
  - Analysis of 352 firms spanning five industries over a 17-year period.
- **Key Findings:**
  1. Firms that engage in acquisitions tend to create a more robust innovation base compared to firms that don't.
  2. Type of acquisition matters:
     - **Vertical Acquisitions:** Integrating firms from different stages of the same industry supply chain.
     - **Horizontal Acquisitions:** Integrating firms from the same stage of similar industry supply chains.
  3. The breadth of knowledge within the acquiring firm is pivotal in determining the success of either acquisition type in enhancing the innovation base.

(Mallapragada and Srinivasan, 2017) Franchising Impact on Innovation

- **Context:** Increasing trend to improve innovation by extending firms' boundaries through franchising.
- **Objective:** Explore the effects of:
  - Franchising on organizational innovativeness.
  - Interactions with firm characteristics: franchising experience, firm size, financial leverage, and slack resources.
- **Data Source:**
  - Panel data from 38 U.S. restaurant chains (1992-2005).
- **Methodology:**

  - Nonlinear seemingly unrelated regression model.

- **Core Findings:**
  - Positive relationship between franchising emphasis and product innovativeness.
  - This effect intensifies for firms with high financial leverage but weakens for those with high slack resources.
  - For process innovativeness, the effect strengthens for firms with high financial leverage. It weakens for large firms, firms with high franchising experience, and high slack resources.

- **Significance:**
  - Franchising has a conditional impact on product and process innovation outcomes.
  - It acts as an alternative method to alliances and joint ventures in influencing organizational innovativeness.

4.1 Innovation Measure

(Chang and Wu, 2021)

1. Obtain information on patenting activities from the National Bureau of Economic Research (NBER) patent database, which provides detailed information for patents granted by the U.S. Patent and Trademark Office (USPTO) from 1976 to 2006.
2. Use the patent data provided by Kogan et al. (2017) to update patent information through 2010.
3. Use patent-based metrics to proxy for firms' innovative activity, as they measure innovation output that incorporates the effectiveness of firms' use of innovation input following Chemmanur et al. (2014).
4. Measure a firm's patenting activity using two measures:
   1. Number of patents applied for and ultimately granted by the USPTO in a given year, using the application year rather than the grant year to measure a firm's innovation in a given year.
   2. Number of subsequent citations received by all patents filed in a given year that captures the quality and influence of innovation.
5. Correct for two types of truncation problem as follows:
   1. End the sample period in 2008 to mitigate the potential truncation issue on the lag between patent application and grant date.
   2. Correct for truncation bias in patent citations using the technology class-year fixed effect approach in Hall et al. (2001, 2005).
6. Use the natural logarithm form of each measure to mitigate bias introduced by outliers.

7. In extended analysis, obtain results based on alternative measures of innovation activities, such as R&D expenses, innovation efficiency originality, and generality to provide a complete picture.
8. In untabulated tests, use raw citations, citations excluding self-citations, adjusted citations based on a weighting index, and the economic value of patents following Kogan et al. (2017).
9. Reach the same conclusion using these alternative measures of patent quality.

4.2 New Product Development

(Moorman and Slotegraaf, 1999) The Contingency Value of Complementary Capabilities in Product Development

- **Objective:** Revisit and advance the interdisciplinary discourse on the direct relationship between organizational capabilities and firm performance by introducing a contingency-based approach centered on the role of external environmental information.
- **Conceptual Premise:** The paper posits that the external environment's informational content modulates how firms utilize their inherent technological and marketing capabilities, particularly concerning their product development efforts – both in magnitude and pace.
- **Methodology:**
  - Conducted a longitudinal quasi-experiment to insulate and discern the interplay between external information cues, organizational capabilities, and product development outcomes.
- **Key Findings:**
  - Data analysis corroborates the proposed framework. Specifically, external environmental information acts as a catalyst, prompting firms to deploy their capabilities in ways that influence their product development trajectories.
  - The nuanced relationship uncovered underscores the flexibility of organizational capabilities, suggesting their primary value might not be in their sheer existence, but in their adaptive deployment in alignment with external cues.

(Sethi et al., 2001) Cross-functional product development teams, creativity, and the innovativeness of new consumer products

- Using survey from cross-functional teams in new product development.
- **Objective:** The study examines the impact of cross-functional team characteristics and contextual influences on product innovativeness, given its crucial role in preventing new product failure.

- **Main Findings:**
  1. **Primary Cause of Product Failure:** A lack of innovativeness is the main reason for new product failure. Innovativeness is defined as providing uniquely meaningful benefits.
  2. **Factors Positively Affecting Innovativeness:**
     - Strong superordinate identity within the team.
     - Encouragement to take risks.
     - Influence from customers.
     - Active project monitoring by senior management.
  3. **Social Cohesion:** While beneficial up to a point, high levels of social cohesion among team members decrease innovativeness.
  4. **Interactions Between Factors:**
     - The positive effect of superordinate identity on innovativeness is amplified when teams are encouraged to take risks.
     - This positive effect is reduced with increased social cohesion.
  5. **Functional Diversity:** This doesn't significantly influence innovativeness.
- **Implications:** The authors highlight the importance of understanding these dynamics for both management practice and future research on product development teams.

(Slotegraaf and Atuahene-Gima, 2011) Value of Stability in New Product Development (NPD) Teams

- **Background:**
  - The significant role of cross-functional teams in innovation.
  - Despite firms adopting cross-functional setups for NPD teams, they encounter challenges.
- **Research Gap:**
  - Understanding the role of stability in NPD teams, especially its effect on decision-making processes.
- **Objective:**
  - To decipher how stability within NPD teams impacts decision-making processes, and consequently, the success of new products.
- **Methodology:**
  - A process-based model to investigate the influence of team stability.

  - Sample: Cross-functional NPD project teams from 208 high-tech firms.

- **Key Findings:**
  1. **Stability and its Impact**: The relationship between NPD team stability and team-level debate & decision-making comprehensiveness is curvilinear (i.e., having a turning point).
  2. **Decision-Making Dynamics**: Team-level debate positively influences decision comprehensiveness.
  3. **Product Success**: Decision comprehensiveness positively impacts new product advantage, but this is particularly pronounced at high levels of comprehensiveness.

(Grewal et al., 2013) Environments, Unobserved Heterogeneity, Effect of Market Orientation on Outcomes for High-TechFirms

- **Context:**
  - Studying the interaction between market orientation and environment facets is a key focus in market orientation literature.
  - Previous empirical studies have shown mixed findings regarding this interaction.
- **Main Proposition:**
  - The inconsistency in findings might be due to unaccounted unobserved heterogeneity that potentially hides the true relationships.
  - Unobserved heterogeneity might arise from higher order interactions involving factors like firm size and innovativeness.
- **Methodology:**
  - Two studies presented focusing on firm performance or new product performance as the outcome.
  - Studies consider:
    1. Market orientation, technological turbulence, market dynamism, and their interactions as explanatory variables.
    2. Use finite mixture regression models to estimate relationships while factoring in unobserved heterogeneity in the form of latent regimes (segments).
    3. Incorporate firm size and innovativeness as profiling variables in the model.
- **Core Findings:**

    - Disaggregate models (multi-regime solutions) provide the best fit in both studies.
    - The effects vary across latent regimes, suggesting that potential aggregation bias might be present in previous empirical studies.
    - There's a need for disaggregated analyses to study marketing phenomena accurately.

- **Theoretical Implications:**
    - The findings hint towards the possibility of forming theories that take into account unobserved heterogeneity and potentially higher order interaction effects.

(Wu et al., 2015) Stock Market Reactions to Horizontal Collaborations in New Product Development (NPD)

- **Background:**
    - Rising trend of firms partnering with competitors for NPD.
    - Limited research on how the stock market perceives these collaborations.
- **Objective:**
    - Analyze stock market reactions to horizontal collaborations across various NPD phases: initiation, development, and commercialization.
- **Methodology:**
    - Analysis of a unique dataset containing 831 NPD announcements related to horizontal collaborations over a span of 12 years.
- **Key Findings:**
    1. **Initiation Phase**: Stock market typically reacts positively to horizontal collaborations in the NPD initiation phase.
    2. **Development and Commercialization Phases**: The reaction is generally negative during the development and commercialization phases of NPD.
    3. **Asymmetrical Moderation**: The innovativeness of the new product and the relative market and technological strengths of the collaborating competitor moderate these effects.

(Kim and Slotegraaf, 2016) Enhancing New Product Development Through Consumer Co-Creation and Brand-Embedded Interaction

- **Research Context:**
    - Emphasizes the importance of accessing market knowledge promptly for effective new product development (NPD).

  - Previous research promotes consumer co-creation during the NPD process, typically viewing consumer input as static.
- **Study Proposition:**
  - Challenging the notion of consumer-generated content as fixed, the study explores how firms can elevate the quality of consumer suggestions by promoting relevant information exchange during co-creation.
- **Concept Introduction:**
  - **Brand-Embedded Interaction:** A process encouraging consumers to generate NPD ideas that satisfy both consumer needs and are in line with the brand's aspirations and abilities.
- **Methodology:**
  - Two quasi-field experiments on Twitter.
- **Key Findings:**
  1. Dynamic interaction and personalized engagement during co-creation are instrumental.
  2. Such interactions enable consumers to produce more "constructive" NPD ideas – ideas beneficial for both the consumer and the firm.

(Srinivasan et al., 2018) Corporate board interlocks and new product introductions

- **Context:**
  - Firms' boards of directors influence various strategic outcomes.
  - However, the impact of boards on new products, a pivotal aspect of organizational adaptation, hasn't been deeply studied.
- **Core Propositions:**
  - The study focuses on "board interlock centrality", which denotes how connected a firm's board members are to boards of other companies.
  - The central argument is that this centrality provides firms with enhanced market intelligence, thus facilitating the introduction of incremental new products.
  - The motivation-opportunity-ability theory is used to suggest that two facets of board leadership can moderate this effect:
    * **Internal vs. External Leadership:** How the leadership of the board is oriented.
    * **Marketing Leadership:** The presence and influence of marketing expertise in leadership roles.
- **Methodology:**

  - The hypotheses are tested using data from publicly-listed US consumer packaged goods firms. This sector is chosen because it frequently sees incremental product innovations.

- **Core Findings:**
  - Board interlock centrality indeed boosts new product introductions.
  - This effect is magnified when:
    * There's strong internal leadership.
    * The board has internal marketing leadership.
    * The firm's CEO has a marketing background.
  - Conversely, the effect diminishes when there's a pronounced intra-industry external leadership presence.
  - These results underscore the importance of board interlocks on innovation outcomes, enriching the discourse on marketing leadership, board interlocks, and social networks.

(Molner et al., 2018) Market Scoping

- Managers' market-scoping mindset (manifest as market ambiguity avoidance or acceptance) can shape market space decisions and outcomes
- Avoidance leads to managers' downstream orientation toward end users, leading to technology commercialization failure
- Acceptance leads to managers' upstream orientation, and helps uncover viable market spaces.

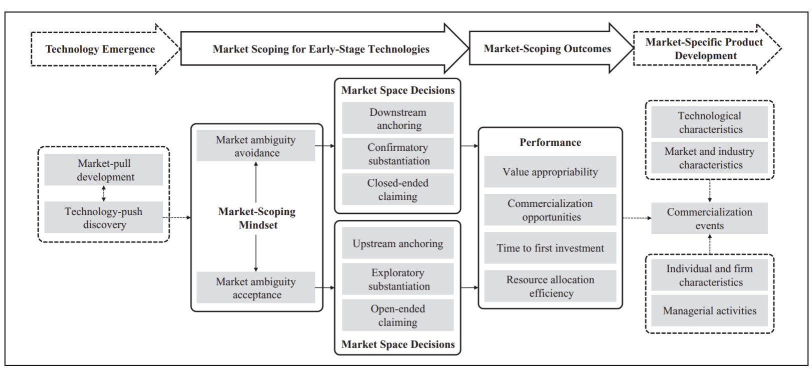


Figure 1. A conceptual model of market scoping for early-stage technologies.
Notes: The dashed boxes and arrows present constructs and relationships that are not explicitly theorized in this article (they are outside the scope of the proposed conceptualization). We depict them for nomological comprehensiveness and to illustrate links with extant theoretical work in this area.

Figure 4.1: p. 39

(Ho-Dac et al., 2020) Marketing Continuous Improvement Products (CIPs): The Role of Development Information

- **Introduction:**
  - **Continuous Improvement Products (CIPs)**: Products designed for post-purchase improvements without the need for full replacement.
  - **Challenge**: Securing customer adoption of CIPs.
- **Research Design:**
  - Employed multimethod research across two different types of studies to examine factors influencing CIP adoption.
- **Key Findings:**
  1. **Development Progress Information**: Sharing updates or information about the progress of a product's development increases the adoption of the current version of the CIP.
  2. **Perceived Commitment**: The relationship between sharing development information and increased adoption is mediated by customers perceiving a stronger commitment from developers toward continuous improvement of the product.
  3. **Role of Product Familiarity**: The influence of development information on adoption, through perceived developer commitment, is stronger for individuals familiar with the product. In other words, the more familiar customers are with a product, the more they value and trust the developer's commitment based on the shared development progress.

(Harz et al., 2021) Virtual Reality in NPD: durable goods

- Prelaunch sales forecasting approach
- Virtual reality support behavioral consistency between participant's info search, preferences, and buying behaviors
- Research questions
  - Does virtual reality improve pre-launch sales forecasting?
  - Why do VR simulations (vs. traditional studio tests) lead to advantage in forecasting?
- Data:
  - GfK sales data
- Virtual Reality:
  - Visualization capability: "the ability to simulate new products, customer touchpoints, and environments in a comprehensive, realistic, and engaging manner." (p. 158)

        * Simulation scope
        * Similarity to reality
        * Immersion
    - Automated-tracking capability: Tracking of behavior
        * Interactivity
- Virtual reality types
    - Lab virtual reality
    - Online virtual reality
- Extended Macro-Flow Model
    - models forecast the sales of new products over time before market launch (Urban et al., 1990)
    - Need 3 states
        * Behavioral states: attitudes and actions through which consumers flow in their purchase journal
        * the flows between these states
        * determines of these flows

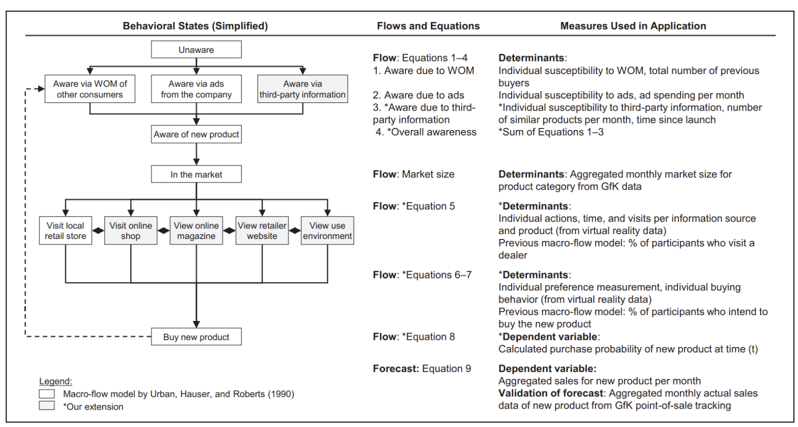


Figure 2. Behavioral states, flows, and determinants in the extended macro-flow model.
Notes: The behavioral states shown on the left are simplified. For a detailed depiction, see Figure W1.1 in Web Appendix.

Figure 4.2: Figure 2 (p. 162)

- Studies 1a and 1b: 2 durables' products, use GfK for data collection. lab virtual reality, online virtual reality. (good prediction after adjusting for advertising).

  - Benchmark macro-flow model: info acceleration.

  - Conjoint analysis, survey on purchase intentions.

  - Better forecasting accuracy comes from virtual purchases with preferences in the purchase decision stage, adding awareness via third-party information

  - At the aggregate and individual levels, this model has better prediction performance

- Study 2: a lab vs. an online virtual reality purchase journey simulation vs. a traditional studio test with real products.

  - VR mechanisms: presence ("the phenomenon of the participant feeling as if they are actually in the simulation" p. 169) and vividness ("the extent to which consumers feel that a simulation is lively and detailed as well as easy to imagine and remember" p. 169).

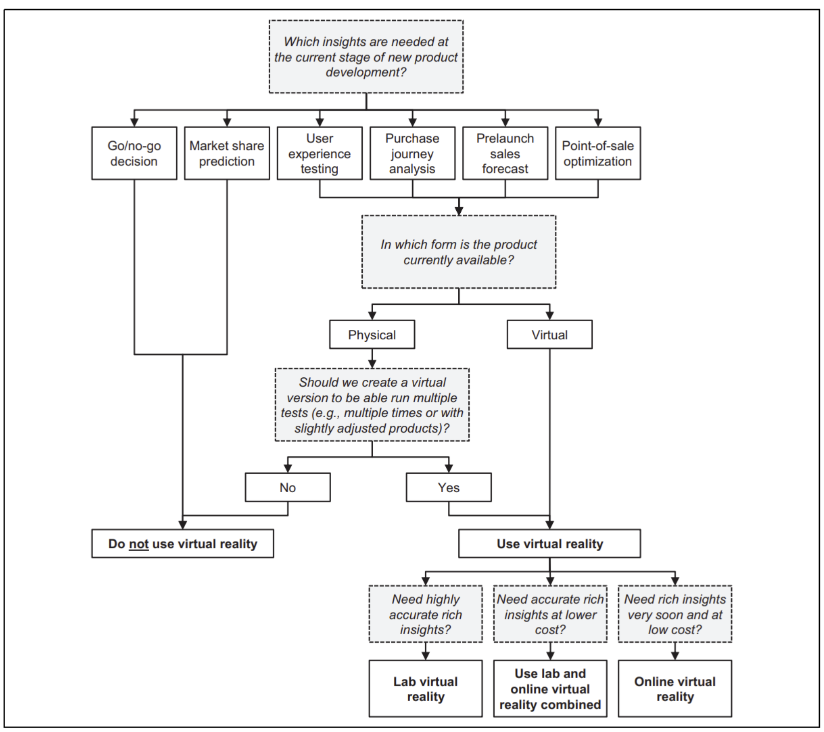


Figure 5. Guidance for managers on when to utilize virtual reality in NPD.

Figure 4.3: p. 176

4.3 Research and Development

(King et al., 2008) Exploring Resource Interactions in Acquisitions and Firm Performance

- **Objective**:
  - Investigate the role of resource interactions between acquiring and target firms in determining post-acquisition performance.
- **Key Findings**:
  1. **Average Acquisition Impact**: Acquisitions, in general, do not automatically lead to higher firm performance.
  2. **Complementary Resources**: Complementary resource profiles between target and acquiring firms can lead to positive abnormal returns.
     - **Marketing-Technology Complementarity**: If the acquiring firm has strong marketing resources and the target firm has robust technology resources, these tend to complement each other, leading to enhanced performance.
     - **Technology-Technology Substitution**: If both the acquiring and target firms have strong technology resources, they can substitute for each other, which may not yield the expected synergistic benefits and might even harm performance.

Chapter 5

Market Entry

(Maesen and Lamey, 2022) Organic Specialist Store Entry effect on Incumbent stores

- Negative Impact of organic specialist store entry on category performance at incumbent generalist stores (in term of sales)
- It also exaggerates the impact of price on sales
- To mitigate this negative effect, the generalist stores can reduce the relative distinctiveness of the entrant along:
  - variety: increase variety in organic product and organic feature
  - price-quality
  - authenticity
- Premium organic products are harmed less than frequently promoted organic products
- Offering an organic specialist brand can mitigate this effect as well

(Bei and Gielens, 2022) first vs. third party operations

- First party operation (wholesale to platform) decreases share, while third-party (selling on platform) operation increases shares.
- For 1P operations, share drops are bigger for brands that can't build trust with customers, especially when there are a lot of other brands and a lot of rogue sellers in the product category.
- the 3P share gains are bigger for non-leading brands with premium priced that have had D2C experience before.

Chapter 6

WOM / Virality

To understand virality, we first need to comprehend its root, which is WOM. Acknowledging that there are subtle differences among WOM, virality, and sharing, we use the three terms interchangeably in this paper. Previous authors provide strong supports for the similarity between virality and WOM (Camarero and San José, 2011; Phelps et al., 2004)

Formally, (Westbrook, 1987, p. 261) defined WOM as "informational communication directed at other consumers about the ownership, usage, or characteristics of particular goods and services or their sellers." In the age of the Internet, WOM evolves to "word of mouse" (or e-WOM). Both virality and WOM take advantages of the network effect to reach wide audiences (Vilpponen et al., 2006), but virality does it better due to its competitive cost, rapid diffusion, and better targeting (Bampo et al., 2008; Feroz Khan and Vong, 2014). Hence, virality is more synonymous with e-WOM.

Virality comes up as a by-product of the Internet, social media platforms, and network theory developments. In network theory, researchers from the domain of sociology or computer science study the structural characteristics/properties of a node (an individual) or a network, such as a node's position in a network or network's cluster. While epidemiologists, management, and marketing researcher study the characteristics of the entity being transmitted, such as virus or message itself. Thus, virality can be driven by content or context of the message (Berger & Milkman, 2012), nature of the spreader, and audience (Hennig-Thurau, Gwinner, Walsh, & Gremler, 2004; Iyengar, van den Bulte, & Valente, 2011), seeding strategies (Kalish, Mahajan, & Muller, 1995; Libai, Muller, & Peres, 2005), and social network structure (Bampo et al., 2008). In this conceptual development, we focus on the content and context drivers of virality.

Virality is defined as the probability of an entity being passed along (Hansen, Arvidsson, Nielsen, Colleoni, & Etter, 2011). The entity can be a message, content, video, knowledge, or disease. Alternatively, (Tucker, 2015) defines virality as “a process whereby an ad is successively shared by viewers.” These two definitions focus on virality as a process; instead, we focus on viral as a construct. Hence, we are in line with (Chandler & Munday, 2016) defining virality (social media virality, social shareability, social media shareability) as “the potential for spreadability of any given content; or particular qualities which are considered to have led to content ‘going viral’.”

For WOM to work, people must share brand-related content, and attention needs to be translated into sales. With these positive associations towards a brand, marketers can increase a customer’s positive brand attitude and later purchase behavior (Faircloth, Capella, & Alford, 2001).

81 percent of a telecom firm indicated that they are likely to refer other to the company; however, only 33 percent actually did. And of those who are referred, only 8 percent became profitable customer. (Kumar et al., 2007)

Even though suffering from the same fate as brand equity, WOM has made it come back under a different alias - virality. The attention that virality has garnered in recent years can be attributed to several reasons:

(1) the dawn of the Internet helps propel marketing into a new paradigm - digital marketing - where earned media is the new king
(2) the introduction of social network platforms such as Facebook, Twitter that facilitate the transmission of information and messages seamlessly and provide an unlimited source of data
(3) The advancement in network theory both theoretically, and empirically.

One platform that can be representative of this digital revolution is Twitter. According to a recent report, Twitter has 145 million monetizable daily active users. Twelve percent of Americans get their news from Twitter. And Twitter has 2 billion video impressions per day (25 Twitter Stats All Marketers Need to Know in 2020. There are more than 500 million Tweets are sent per day. (The 2014 #YearOnTwitter, 2014). One cannot deny the power of online WOM as well as its potential reach; hence, viral marketing has been born with these platforms.

The age group from 18-24 can be more challenging to target since they reported fewer positive responses compared to other age groups (Libert, 2014). Since their threshold is higher due to the overwhelming digital content they encounter, their emotions are harder to activate.

Wide gaps between two opposing opinions can increase future discussion (reviews); however, fluctuation around one opinion does not. Higher informational content moderate positively the relationship between opposing opinions and future reviews. (Nagle and Riedl, 2014)

WOM should be taken under the context of the evolution of conversation. (Berger, 2014)

Contagion/ Virality used to be thought of in term of S-shaped curve, but there are problems with this notion:

1. S-shaped could result from different processes that are not contagion
    1. Population heterogeneity (Chatterjee and Eliashberg, 1990)
    2. Marketing efforts (den Bulte and Lilien, 2001)
2. Selection bias (or survival bias)

Questions:

- When diffusion is "viral"?
- What does viral diffusion look like?
    - (Goel et al., 2012):
        - 90% events have no adoption
        - less than 99% events has less than 2 viral adoptions
- Can we predict it?

Viral is different from Popular (Goel et al., 2015)

- Popular = Broadcast
- Viral = Cascades
    - needs diversity (i.e., differences/ distinction)
- Total number of retweet over time is not a good approximate of virality

"Big data" help understand rare events.

Valuable virality (Akpinar and Berger, 2017)

Establishes causality of appeals and brand integralness to virality.

- Compared to informative appeal ads (focus on product features), emotional appeal ads (drama, mood, music, emotion-eliciting strategies) are more likely to be shared.
- Informative appeals boost brand evaluations and purchase because the brand is an integral part of the ad content.
- Hence, to combine the best of both worlds, emotional brand-integral ads boost sharing while also bolstering brand-related outcomes (embedded the brand into the stories).
- The mechanism that ads affect brand evaluation through two simultaneous mechanisms: persuasive attempt and brand knowledge.

"keeping viewers involved depends in large part on two emotions: joy and surprise"(Teixeira, 2012) - "Research: The Emotions that Make Marketing Campaigns Go Viral"

"ads that produce stable emotional states generally aren't effective at engaging viewers for very long." And "shock may get people to watch an ad privately, it often works against their desire to share the spot. Like Clothing Drive that try to mimic"Swear Jar" by Bud Light" (Teixeira, 2012)

(Westbrook, 1987, p. 261) defined WOM as "informational communication directed at other consumers about the ownership, usage, or characteristics of particular goods and services or their sellers." WOM evolves to "word of mouse" (or e-WOM).

(Berger, 2014):

WOM is goal-driven and serves five key functions:

- Impression Management
- Emotion regulation
- Information acquisition
- Social bonding
- Persuasion

The motivation behind this process is self -serving and even subconscious level. Hence, we can predict the types of news and information people are most likely to discuss.

- Over 70% of everyday speech is about the self, such as personal experiences or relationships (Dunbar et al., 1997). Consistent with previous research, online behaviors exhibit the same pattern: over 70% of social media posts are about self (Naaman et al., 2010).
- At any given point in time, multiple drivers can be present.

6.1 Structural Virality

(Goel et al., 2015)

$S = f(Q) + E$

where

- Q = skill
- E = luck
- S = successful viral process

Measure fraction of variance remaining after conditioning on Q

$$F = \frac{E(var(S|Q))}{var(S)} = 1 - R^2$$

Hence, if

- $R^2 \to 1$, you can succeed in the world we created based on "pure skill"
- $R^2 \to 0$ you can't succeed in the world we created (because it's based on luck)

The more variations in the initial seed, the more likely there is a larger variation in the prediction, which means lower R-squared

They show there is a theoretical limit to the predictive power of predicting cascade event due to its intrinsic dynamic properties.

6.1.1 Network Structure and position

(Ansari et al., 2018)

- Network management on social network
- Networking activities and ego network structure on an actor's online success
- Related literature:
  - Seeding viral marketing campaigns:
  - Contagion and product diffusion: network position and network structure
  - How do network structures impact online success?
- Method: stochastic actor-based models
- Ego Network measures
  - Ego network density
  - Degree centrality
  - Betweenness centrality
  - Closeness centrality (not applicable to ego network)
- Endogeneity
  - Time fixed effects
  - Control function: observed time-varying and firm-specific instrumental variables

  - Instrumental Variables are those associated with the networking activities: independent US social platform on which those focal European firms also main their profile pages. (activities in the US is correlated with activities in Europe on social media, but not directly to the success of European market).

6.1.2 Seeding Strategies

Hinz et al. (2011)

- Despite analytical models and simulations that support seeding strategies don't work, this study finds evidence in small-scale field experiments and real campaign that seeding to well-connected people can lead to 8 times more successful because they are more active in using their greater reach (even though they don't have more influence on their peers than less connected people)

(Chae et al., 2017)

- seeding strategy can help increase online WOM, and decrease conversation about competing products (in the same category) but also focal brand products in other categories.
- data: cosmetics and beauty products in Korea.

6.2 Mechanisms/ Processes

6.2.1 Impression Management

Consumers buy to signal their derived identities to achieve desired impressions (J. Berger & Heath, 2007)

Interpersonal communication (WOM) influences impression management in 3 ways: (1) self-enhancement, (2) identity -signaling, and (3) filling conversational space.

- **Self-enhancement**: human has a tendency to self-enhance (Markus, 1987). Hence, people like to share things to help them look good (Chung & Darke, 2006; Hennig-Thurau et al., 2004). Since bragging too much about oneself might have an opposite effect. People sometimes engage in" humblebragging": brag but self-deprecating at the same time (Sezer et al., 2018). Wojnicki and Godes (2017) found people share content for self-enhancement.
- **Identity-signalling**: they talk about themselves to signal they have certain traits or expertise (G. Packard & Wooten, 2013), such as opinion leaders talk to signal their identity (share to show their knowledge).
- **Filling conversational space**: (small talk) people engage in small talk to avoid pauses and silence in between conversations so that the other

party would not make lousy inferences about the person. (Berger, 2014)

Hence impression management induces people to share things that are

- **Entertaining** (i.e., interesting, surprising, funny or extreme) so that they shared look more interesting, and in-the-know. Interesting products get more word of mouth (Berger and Iyengar, 2013; Berger and Schwartz, 2011; Berger and Milkman, 2012). Moderate controversy is a catalyst for word of mouth to spread (Chen & Berger, 2013). And sometimes, people even exaggerate stories to make things more interesting (C. Heath, 1996); around 60% of the stories are distorted (Marsh and Tversky, 2004). Things that can be considered interesting are those that are either novel, exciting, or violate previous expectations (i.e., surprise) (Berlyne, 2006; Silvia, 2008). Nevertheless, comprehensibility can be a moderate of whether a novel thing is interesting: novel and comprehensible equals interesting, while novel and incomprehensible equal confusing (Silvia, 2008). Novelty refers to things that are new, surprising, exciting, or complex (Berlyne, 1960). Comprehensibility refers to the fact that the novelty must be understandable.
  - (Vosoughi et al., 2018) found that false news are spread faster, farther, deeper and more broadly because they are more novel where true or false based on 6 independent fact-checking organizations. This effect is greater about terrorism, natural disasters, science, urban legends, or financial information. False stories inspire fear, disgust and surprise replies, where as true stories inspire anticipation, sadness, joy and trust. Even after controlling for bots account, their results still hold, which means that most of false stories are driven by humans, not bots.
- **Useful**: because it makes the sharer smart and helpful. Empirical evidence show that useful stories(J. Berger & Milkman, 2012) and higher quality brands (Lovett et al., 2013) are more likely to be shared.
- **Self-concept relevant**: different people see different domain as the optimal signal amplifier such as politics, economics, etc. hence symbolic products are more likely to be shared than utilitarian ones (Chung & Darke, 2006) (even though useful information is helpful, but the signal to be smart and helpful are not appreciated as being interesting and entertaining).
- **Status related**: premium brands are talked about more (Lovett et al., 2013). Knowledge is can also be considered as a status symbol, and people share to signal that they re in the know (Ritson & Elliott, 1999)
- **Unique**: unique products are more likely to be shared, but people with a high need for uniqueness will be less likely to share positive WOM because they do not want people to adopt the product and reduces their uniqueness (Cheema & Kaikati, 2010). Sometimes they even scare others by citing product complexity (Moldovan et al., 2015)

- **Common ground**: sharing common things induces listener to convey that there is interpersonal similarity and facilitates conversation.

- **Emotional valence**: people prefer to share positive things than negative news (Berger and Milkman, 2012). However, in some special cases, negative WOM can be perceived as a desired component such as reviewers are seen as more intelligent (Amabile, 1983). Which is moderated by whether people are refereeing to themselves or others will affect WOM valence (positive vs. negative) (Kamins et al., 1997). Positive WOM when talking oneself conveys a positive image, while negative WOM about others convey that they are relatively better than the other party.

- **Incidental arousal**: unrelated arousal (e.g., running in place) can increase sharing in general Berger and Schwartz (2011) because of the increase in arousal levels.

- **Accessible things**: products with more cued or triggered will be discussed more (J. Berger & Schwartz, 2011). Hence, more advertised products generate more WOM (Onishi and Manchanda, 2012). "top of mind" and "tip of the tongue." Availability bias means that the easier we can recall something, the more likely we are going to talk bout it. Publicly visible products also have more WOM (J. Berger & Schwartz, 2011)

6.2.2 Emotion regulation

WOM help consumers regulate their emotions. Emotion regulation refers to "a person's ability to effectively manage and respond to an emotional experience" (Rolston & Lloyd-Richardson, 2017). and sharing with other (WOM) can facilitate emotion regulation by:

- **Generating social support**: sharing can give you comfort and consolation (Rimé, 2009). For example, after a negative emotional experience, sharing can help increase well-being through perceived social support (Buechel and Berger, 2017)
- **Venting**: WOM (sharing) helps people achieve catharsis when experiencing negative experiences(Alicke et al., 1992): consumers express when they are angry (Wetzer et al., 2007) or dissatisfied (Anderson, 1998).
- **Sensemaking**: "talking can help people understand what they feel and why" (J. Berger, 2014)
- **Reducing dissonance**: even after making a decision, we want to make sure we make the right choice by talking to others tot reduce feelings of doubt (Rosnow, 1980)
- **Taking vengeance:** to punish the company (different from venting, which makes you feel better), consumers share negative consumption experience (Ward & Ostrom, 2006)
- **Rehearsal**: people talk to relive positive experiences (Rimé, 2009)

Emotion regulation drives people to share (1) emotional content (2) influence the valence of the content shared, (3) lead people to share more emotionally arousing content:

- **Emotionality**: more emotional intensity, more sharing (J. Berger & Milkman, 2012), people share more emotional social anecdotes (Peters et al., 2009). But not all emotions increase sharing: for example, shame and guilt decreases sharing (Finkenauer & Rimé, 1998) since sharing those things makes people look bad.
- **Valence**: "Emotion regulation tends to focus on the management of negative emotions" (J. Berger, 2014), but a counter vailing effect is that under impression management, people avoid sharing negative stories since it might reflect on the shares. Sharing negative things can decrease willingness to share (Chen & Berger, 2013)
- **Emotional arousal**:
  - **Negative** side: low arousal emotion (sadness), high arousal (anger or anxiety) should increase the need to vent.
  - **Positive** side: low arousal (contentment), high arousal (awe, excitement or amusement) increase desires for rehearsal.

6.2.3 Information acquisition

WOM helps acquire information.

- **Seeking advice**: (Hennig-Thurau et al., 2004; Rimé, 2009). Rather than trial and error, or direct observation, gossip serves as another form of learning (Baumeister et al., 2004)
- **Resolving problems**: online forums or other online opinion platforms help customers find a solution quickly from others (Hennig-Thurau et al., 2004)

Hence the process of information acquisition would induce to talk more about

- **Risky, important, complex, or uncertainty-ridden decision**: talking to others can reduce risk, simplify complexity, and increase consumer's confidence (Hennig-Thurau et al., 2004)
- **Lack of (trustworthy) information**:

6.2.4 Social bonding

"people have a fundamental desire for social relationships (Baumeister & Leary, 1995), and interpersonal communication fills that need (Hennig-Thurau et al., 2004)". Empirically, people in brand communities because want to connect with others like them (Muniz & O'Guinn, 2001). Sharing facilitates social bonding through

- **Reinforce shared views**: group memberships, and one's place in the social hierarchy. WOM (sharing) allows people to fit in as a group member (J. Berger & Heath, 2007; Ritson & Elliott, 1999)
- **Reducing loneliness and social exclusion**: sharing decrease interpersonal distance and decreases loneliness and exclusion (Maner et al., 2007)

Social bonding drives what people share: people share things are :

- **Common ground**: social bonding drives people to talk about things they have in common because it makes them feel socially connected (Clark & Kashima, 2007)
- **Emotionality**: "sharing an emotional story or narrative increases the chance that others will feel similarly" (J. Berger, 2014). Emotional similarity increase group cohesiveness (Barsade & Gibson, 2007). And emotion sharing and social bonding can be a simultaneous process since previous works found that emotion sharing bonds people (Peters & Kashima, 2007) and high arousal emotion increase social bonding needs (Chen & Berger, 2013)

6.2.5 Persuading others

people use WOM to persuade others to affect their satisfaction or choices (J. Berger, 2014). Persuasive drive people to share things that are

- **Emotionally polarized** (polarized valence): since the goal is to convince something is good or bad (i.e., people share extremely rather than moderately positive(negative) information
- **Arousing content**: "people who want to persuade others may find shared arousing content to incite others to take desired actions" (J. Berger, 2014)

6.3 Moderators

- Two key moderators when different WOM functions have a greater impact:
  - **Communication audience**:
    - Tie Strength
    - Audience size
    - Tie status
  - **Communication Channel**
    - Written vs. oral
    - Identifiability: whether communicators are identifiable. (anonymously)
    - Audience salience: whether the audience is salient during communication (online communication, the audience is less salient).

Social motivation: why we share :(why_some_videos_go_viral_2015)

- Impression management:
  - authority (e.g., "I want to demonstrate my knowledge")
  - Social utility (e.g., this could be useful to my friends)
  - coolhunter (e.g., I want to be the first o tell my friends)
  - Zeitgeist (e.g., It's about a current trend or event)
  - Conversation starting (e.g., I want to start an online conversation)
  - Self-expression (e.g., it says something about me)
  - Social good (e.g., It's for a good cause, and I want to help").
- Information acquisition:
  - Opinion seeking (e.g., "I want to see what my friends think)
- Social bonding:
  - Shared passion (e.g, "it lets me connect with my friends bout a shared interest)
  - Social in real life (e.g., it will help me socialize with my friends offline).

More frequent exposure to perceptually or conceptually related cues increases product accessibility and makes the product easier to process. In turn, this increased accessibility influences product evaluation and choice, which are found to vary directly with the frequency of exposure to conceptually related cues. These results support the hypothesis that conceptual priming effects can have a strong impact on real-world consumer judgment" (Berger and Fitzsimons, 2008)

6.3.1 Tie Strength

(Peng et al., 2018)

- the authors establish that a receiver is more likely to share content from a sender with whom they share more common followees, common followers, or common mutual followers even after accounting for other measures.
- Assess the impact of network overlap across dyads on the level of content sharing in social platforms.
- Network overlap is defined as:
  - Common followees
  - Common followers
  - Common mutual followers
- Network overlap (similarly embeddedness or social cohesion) can influence the sharing propensity :
  - A high number of common followers means that senders and receivers have similar interests and may have a similar propensity to share a common content

  - More common followers and common mutual followers meant that their followers share a similar interest. Hence a receiver thinks content is more suitable for her audience and has a higher propensity to share
  - A user may be less likely to share popular content because many others have already shared it

- Used data on Twitter and Digg
- Use dyadic hazard model

Lee and Kronrod (2020)

- Consensus language: Words and expressions that imply widespread agreement among a group of people about a viewpoint, a product, or a conduct
- The strength tie between the communicator and the receiver of WOM affects the interpretation and persuasiveness of consensus language.
- When employing consensus language, weak links (e.g., distant friends, acquaintances) have more influence than strong ties, because weak relationships inspire views of a broader and more diversified group in agreement, signaling higher validity for the topic at hand.
- For tie strength operationalization in the literature, see table 1 (p. 355)
- Descriptive norms provide social proof 6.4.1 (i.e., social validation)
- For the first experiment, the language of the message was too formal (p. 360). Hence, it's more likely that the strong ties will become suspicious of the news, while weak ties might think it's normal and click on it.

6.4 Drivers of Virality

6.4.1 Social Currency

(Toubia and Stephen, 2013) found that (self) image-related utility is a dominant driver (vs. intrinsic utility) of posting activity

Different people see different domain as the optimal signal amplifier, such as politics, economics

Even though useful information is helpful, the signal of being smart and helpful is not appreciated as being interesting and entertaining (Berger, 2014)

People talk more about status-related products because they enhance their impression.

In extreme cases, they even scare others by citing product complexity (Moldovan, Steinhart, & Ofen, 2015).

Interesting things are discussed more in online settings (J. Berger & Milkman, 2012), but this effect fails to replicate in the case of face-to-face settings (Berger & Schwartz, 2011). Hence, research has found that written communication (e.g., texting, messaging, posting online) leads people to talk more about interesting products or brands than oral communication because they have more time to think and polish what to say, leading to more prevalent self-enhancement motives (Berger & Iyengar, 2013). In other words, not because people do not have self-enhancement motives, but because synchronous conversations do not allow for much time to think and polish ideas, people might talk about mundane things to fill the gaps between conversational turns (Berger, 2014).

6.4.2 Accessibility

Accessibility drives ongoing WOM (i.e., publicly visible products are cued more in the environment – top of mind), regardless of how mundane it might be.

From the theory of semantic process (Collins & Loftus, 1975), priming can help activate related perceptual or conceptual constructs in memory spontaneously (Berger & Fitzsimons, 2008). For example, prior exposure affects the brand choice of both target brand and competing brands (Nedungadi, 1990). In other words, the activation of one brand can lead to others more accessible, which increases the chance of remembered brands to be included in the consideration set. (Lee & Labroo, 2004) argues that ease of processing drives our positive evaluation, which is also consistent with the discrepancy -attribution hypothesis. The Discrepancy-attribution hypothesis states that if people cannot easily attribute the ease of processing to a source, they will likely attribute it to the positive quality of the target. Additionally, (Nedungadi, 1990) found that recently exposed brands are more likely to be included and chosen from consumers' consideration set. Repeated exposure to an object increases favorable ratings (e.g., an object, or person) (Zajonc, 1968), which also apply to attractive people (Moreland & Beach, 1992), and brand (Baker, 1999). Behavioral can be influenced by incidental exposure to stimuli because the primes can activate relevant constructs associated in memory (Christian Wheeler & Petty, 2001; Dijksterhuis & Bargh, 2001; Wheeler & Berger, 2007)

Familiar content are more frequently selected(Kim, 2015)

6.4.3 Emotions

6.4.3.1 Valence

Milkman and Berger (2014) found that contents that are (1) emotional aroused, (2) useful, or (3) interesting have a higher chance of being shared. Consistently, Milkman and Berger (2014) also found that women produce articles that are more likely to be shared because female authors write more comprehensible, useful, and interesting articles.

People share their extreme emotional experience (i.e., either highly satisfied or highly dissatisfied) (Anderson, 1998)

Positive emotion prompted more frequent sharing (Kim, 2015)

6.4.3.2 Arousal

Lastly, people use WOM to persuade others to affect their satisfaction or choice (Berger, 2014). Thus, emotionally polarized (polarized valance) are more likely to be shared due to persuasion reasons. Since the goal is to convince something is good or bad, people share extremely rather moderately positive (negative) information.

Virality is partially driven by physiological arousal (Berger and Schwartz, 2011, Berger and Milkman (2012)). These results hold even when the authors control how surprising, interesting, or practically useful content is (all of which are positively linked to virality) and external drivers of attention (e.g., how prominently content was featured) (Berger and Milkman, 2012). Different emotion evokes different levels of physiological arousal (i.e., activation) (Smith and Ellsworth, 1985). Arousal is an excitatory state that increases action-related behavior, such as helping others (Gaertner and Dovidio, 1977). Hence, we see contents that are physiological arousal get shared more. Specifically, Negative high arousal emotions (e.g., anger or anxiety) are shared more than negative low arousal (e.g., sadness). Moreover, sharing negative experiences can help people vent to regulate their emotions Berger (2014). On the other hand, positive emotion high arousal emotions (e.g., awe, excitement, or amusement) are shared more than positive low arousal (e.g., contentment). Furthermore, sharing for the purpose of rehearsal (i.e., people talk to relive positive experiences) helps people regulate their emotions (Rimé et al., 1991).

Akpinar and Berger (2017) posits that valuable virality is virality contents that are beneficial to the brand.

- Informative appeals are benefit to brands, but less likely to be shared
- Emotional are more likely o be shared, but less beneficial to the brand
- Combining both into emotional-brand-integral ads boost sharing and brand-desirable outcomes.
- Research is based on Unruly data and lab experiments.

Operationalization via language: Yin et al. (2017)

6.4.4 Usefulness

Wojnicki and Godes (2017) found that sharing useful content can help sharers appear more knowledgeable

Fehr et al. (1998) found that people share useful content to generate reciprocity

Homans (1958) posits that sharing useful content can have social exchange value.

Informational utility prompt readers to share. Informational usefulness and novelty exhibited more positive connections with e-mail-specific virality, but emotional evocativeness, content familiarity, and exemplification had a more significant impact in activating social media-based retransmissions. (Kim, 2015)

Practical utility increases newspaper articles' virality (Berger and Milkman, 2012)

6.4.5 Narratives

(Singh and Sonnenburg, 2012)

- Storytelling is important to branding.
- In the social media space, brand cocreate brand performances with costumers.
- The process of improvisation is more critical than the output
- Managing brands on social media is about keeping the brand performance alive

6.5 Other variables

6.5.1 Controversality

- Increase sharing (Kim, 2015)
- The presence of dissonance initiates interaction (Gunawardena et al., 1997) and people are motivated to reduce this dissonance (Festinger, 1957)

6.5.2 Popularity

- Difference between views (popularity) and shares (virality)
  - Popularity usually contaminated by promotion (advertising)

6.5.3 Contractuality

(Lisjak et al., 2021) defined contractuality as " the extend to which a perk is perceived to be conditional on specific behaviors and contingencies dictated by a company."

- Consumers like to have perks with less salient contractuality.
- The existing relationship with the company (e.g., dislike, distrusted) may giving consumers the wrong intention, or let them interpret the park as a manipulative act with ulterior motives.

- Low contractuality perks can foster "real" authentic WOM, but you have to trade off with effectiveness and efficiency (e.g., as in the case of filling out satisfaction survey).

6.5.4 Locus of Control

(Lam and Mizerski, 2005)

- An individual personality construct is locus of control
- Locus of control is defined as "individuals' general and daily expectancies about the causes of their reward and punishment (Lam and Mizerski, 2005, pp. 216), consisting of internal (i.e., one can control his or her lives) and external (i.e., external factors can control their lives such as luck, or fate).
  - Internals tended to be more educated (Lachman and Leff, 1989), higher household incomes (Hoffman et al., 2003), men and those in senior positions (Smith et al., 1997). Internals are more action-oriented, risk-taking.
  - Externals perform avoidance behavior, greater needs for affiliation
- Individuals with high internal locus of control are more likely to engage in WOM communication with their out-groups.
- Individuals with high external locus of control are more likely to engage in WOM with their in-group.

6.5.5 Horizontal/Vertical Individualism

(Choi and Kim, 2019)

- Antecedents of WOM: homophily, tie strength, trust.
- Culture affects individuals' communication attitudes and styles.
- Vertical (i.e., emphasizing hierarchy) and horizontal (i.e., emphasizing equality) dimensions of individualism and collectivism (HVIC) (Singelis et al., 1995)
  - opinion leadership: "the tendency that an individual attempts to influence the decisions of others by giving his or her opinion about products, services, or firms" (Choi and Kim, 2019, pp. 294)
    - socially appropriate (horizontal orientation)
    - show off knowledge (vertical orientation)
  - opinion seeking "the tendency that an individual seeks information or opinions from more knowledgeable others to find out about and/or evaluate products or services" (Choi and Kim, 2019, pp. 294)
    - well-informed decisions (individualism)

    * find values and beliefs of the reference group (collectivism)

  - Horizontal individualism is "a cultural pattern where an autonomous self is postulated but the individual is more or less equal in status with others" (Singelis et al., 1995, pp. 245)
  - Vertical individualism is "a cultural pattern in which an autonomous self is postulated, but individuals see each other as different, and inequality is expected" (Singelis et al., 1995, pp. 245).
  - Horizontal collectivism is "a cultural pattern in which the individual sees the self as an aspect of an in-group" (Singelis et al., 1995, pp. 244).
  - Vertical collectivism is "a cultural pattern in which the individual sees the self as an aspect of an in-group, but the members of the in-group are different from each other, some having more status than others" (Singelis et al., 1995, pp. 244).

- Collectivist culture will have greater levels of opinion seeking, whereas individualist culture has greater information-giving.(Fong and Burton, 2008)
- Results:
  - "horizontal individualism to opinion leadership, vertical individualism to opinion leadership and opinion seeking"
  - "horizontal collectivism to opinion leadership, and vertical collectivism to opinion leadership and opinion seeking"

6.5.6 Linguistic style

- Moore (2012) found how different explaining language can influence the storytellers. Linguistic content in nontraditional WOM (e.g., online reviews, or other online channels) can influence the storyteller.
- language abstraction (Schellekens et al., 2010)
- explaining language (Moore, 2012)
- figurative language (Kronrod and Danziger, 2010)
- markers of modesty or boastful (Packard et al., 2016)
- dispreferred markers (Hamilton et al., 2014)
- personal pronouns (Packard and Wooten, 2013)
- linguistic mimicry (Moore and McFerran, 2017)

6.6 Negative Virality

Herhausen et al. (2019) offer a framework for drivers of negative eWOM with strategies that usually employ by firms to militate negative eWOM. Level of arousal, structural tie strength, and linguistic style match can all affect the firestorm. Optimal strategy: the response must match the intensity of arousal in the negative eWOM

6.7 Articles

- Empirical evidence
  - contagion in consumer packaged goods (Du and Kamakura, 2011)
- Interesting stat:
  - "Articles by business academics, psychologists, and economists, for example, are more likely to be shared than articles by physicists, geneticists, and biochemists" (Milkman and Berger, 2014)
  - "Specifically, men see the same scientific summaries as more comprehensible ($P < 0.001$), interesting ($P < 0.001$), and useful" hence they share more than women (Milkman and Berger, 2014)
  - "though intentionally outrageous videos command attention (Tellis 2004), an ad design of this type ultimately detracts from the ad's persuasiveness" (Tellis, 2004)
  - (Godes and Mayzlin, 2009) the impact of dispersion declines over time; hence should measure WOM early in a product's life. Main finding: higher WOM dispersion leads to higher future ratings. WOM is both precursors and consequences of consumer behavior.
  - Promotional giveaways increase WOM (Berger and Schwartz, 2011)

Berger and Milkman (2012)

- Positive emotional valence content is more likely to be shared
- Activation = physiological arousal induces action.
  - Low arousal = deactivation = relaxation (Feldman Barrett and Russell, 1998)
  - HIgh arousal = activation = activity (Heilman, 1997)

1. Examine 7000 articles from thew New York Times
   1. Examine emotionality, prominent features, interest evoked can affect likelihood to make the most email list. (controlling for practically content usefulness, interestingness, surprise, release timing and author fame (using hits for first author's full name from the number of Google hits), writing complexity, author gender, article length and day dummies).
   2. Robustness: control for article's general topic.
2. Lab experiments

1. Amusement case (fictitious): high arousal
2. Anger case (real): high arousal
3. Sadness (real vs. fictitious): low arousal

All hypotheses are confirmed

Potential confounders: structural virality

How likely they would share a story? (no social risk involved - risk to other weak ties, hence the effect might be inflated, the same thing with the New York Times study)

All experiments have low participant numbers

Tellis et al. (2019)

Two field studies

Information-focused content is less likely to be shared (exception risky contexts)

Positive emotions (e.g., amusement, excitement, inspiration, warmth) are more likely to be shared

Drama elements (e.g., surprise, plot, characters, babies, animals, celebrities) increase arousal, which in turn increases sharing.

Prominent placement of brand name (brand prominence)

Emotional ads are shared more on general platforms (Facebook, Twitter) as compared to professional one (e.g., LinkedIn), while informational ads are more likely to be shared on professional ones.

Optimal length is 1.2 to 1.7 min ads.

Third study: identifies predictors of sharing

(Trusov et al., 2010)

- How to determine Influential users in online social networks

Chapter 7

Sarcasm

- AI-based Learning Techniques for Sarcasm Detection of Social Media Tweets
- Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media

Part II

SUBSTANCE

Chapter 8

Branding

Reviews:

(Keller and Lehmann, 2006)

(Swaminathan et al., 2020) Hyperconnected World

- Hyperconnectivity is "the proliferation of networks of people, devices, and other entities, as well as the continuous access to other people, machines, and organizations, regardless of time or location." (p. 24). 3 aspects
  - Information availability and speed of info dissemination: reassess models of attention and brands as quality signals
  - networks of people and devices and the growth of platforms: loss control of brand meaning.
  - Device-to-device connectivity: branded experiences can be different.
- Two major changes in this world
  - Blurring of branding boundaries: Increased access to information and people is enabling more stakeholders to co-create brand experiences and brand meanings alongside conventional brand owners, and as a result, companies are moving away from single to shared ownership.
    * cocreation of brand meaning
  - Broadening of branding boundaries: Existing brands can increase their geographic reach and societal functions, while new sorts of branded organizations are expanding the branding landscape even more.
    * More and more entities are branded
    * Commercial brands have missions

- Three theoretical perspectives
  - Firm
    * Strategic approach: Actions can be taken by firms strategically
    * Financial approach: measure the effect of brand equity and branding actions on the stock market value.
  - Consumer:
    * Economic approach: brands as market signals
    * Psychological approach: consumer brand equity
  - Society
    * Sociological approach: brands as container of meaning (can be changed)
    * Cultural approach: "branded goods (as cultural meaning producers) enhance consumers' lives" (p. 26)
- Rethinking the roles and the functions of brands
  - Brands as weak quality signals (alternative channels such as online reviews)
  - Brands as mental cues in info-rich environments (system 1 and 2 need to be reassessed because all senses can be activated)
  - Brands as instruments of identity expression (consumers can have multiple personae)
  - Brands as Containers of socially constructed meaning (social brand engagement, stay relevant with social issues)
  - Brands as architects of value in networks (important to have seamless access within complex network - brand network user experience)
  - Brands as catalysts of communities (how to serve as gatekeepers)
  - Brands as Arbiter of Controversy (active brands may benefits, while passive brands will suffer)
  - Brands as stewards of data privacy
- Rethinking brand value cocreation
  - Cocreating brand experiences (beyond product cocreation).
  - Cocreating brand meaning (e.g., outsourcing designs of ads to consumers). To create new brands in a hyper-connected world, marketers need to care about
    * Networked communications: how devices communicate

        - Network value creation: brands are not owned by a sole entity anymore. (question: "What mechanism will replace brand equity to induce loyalty and trust in consumers?" is related to my research on brand equity meta analysis).
        - Data collection: privacy and customization tradeoff.

- Rethinking Brand Management
    - Blurred Control of Brand Positioning and Brand Communication. Changes made by hyperconnectivity
        - Broader set of competitions
        - Broader space and ways to communicate brand messages
        - Outsiders can affect brand positioning
        - More consumer, and more heterogeneity
        - Ad placements by algorithms (brand safety)
    - Blurred Control of Brand Crises
        - Different stakeholders can have different impact on brand crises and on how consumers attributes blame.
    - Blurred Control of brands as Identifiers of Intelligent, Interactive and Networked Devices
        - Branded products in the IoT
        - Virtual and artificial reality and AI make it hard to create brand experience
    - Measuring brand value
        - How to value platform brands? E.g.,
            - Financial approach: brand value = current profit/(interest rate - profit growth rate). not applicable for young networks, or when profit growth rate > interest rate
            - Customer lifetime value for subscription-based brands (based on customer equity theory), but not robust to cancellation
            - Social capital from each user will create platform brand value
        - Which structural of the network have the best valuation? (e.g., close-knit or sparse)
- Rethinking the boundaries of branding
    - Idea brands: " ideologies, initiatives, or other abstract, noncommercial notions that are identified by their stakeholders and the public

at large using the same specific name." (p. 39). How, when and why ideas brands come to life

- People brands (e.g., person brand (Fournier and Eckhardt, 2019), human brand (Thomson, 2006), celebrity brand(Kerrigan et al., 2011)): person and brands are intertwined that is hard to distinguish especially under reputational crises.
- Place Brands: "a network of associations in the consumers' mind based on the visual, verbal, and behavioral expression of a place, which is embodied through the aims, communication, values, and the general culture of the place's stakeholders and the overall place design (Zenker and Braun, 2010, p. 5). Multiple stakeholders involve in the development of place brand.
- Organization as brands: organizations other than corporate ones (e.g., nefarious groups).
- Metrics for newer branded entities:

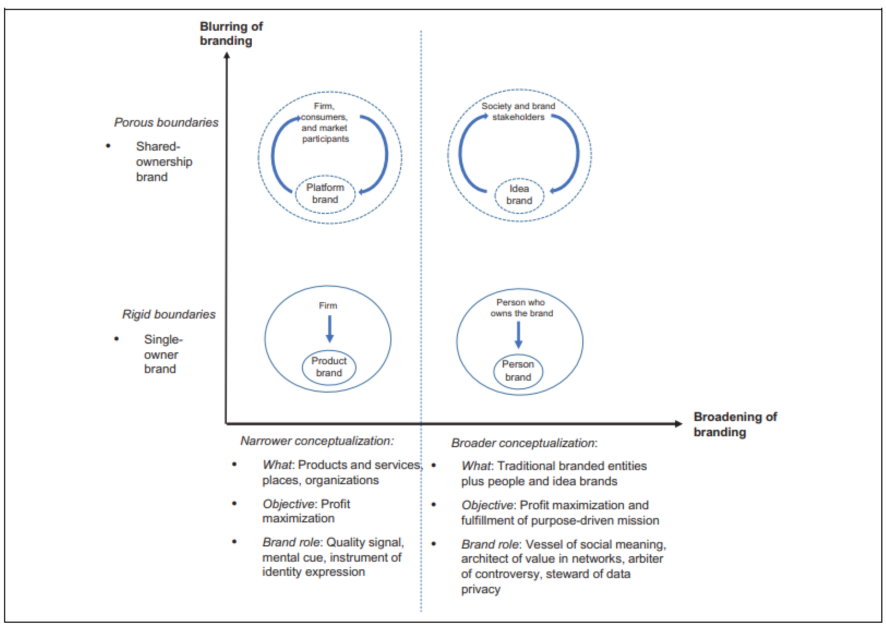


Figure 1. Ownership of branded entities and changes in the branding landscape.
Notes: The figure depicts interactions between various types of brands and their owners and primary stakeholders. Inner circles represent a specific type of brand; outer circles encompass the brand owners or main stakeholders. Dotted circles indicate more porous boundaries of brand ownership. We have labeled the brand owner and used arrows to illustrate whether the relationship between the brand and its owner/stakeholders is unidirectional or dynamic.

Figure 8.1: Figure 1 (p. 28)

(Kotler and Pfoertsch, 2007) encourage B2B firms to adopt a long-term branding strategy since it is positively correlated with stock performance

Belk (1988) posits the construct of extended-self, where possessions contribute and reflect a person's identity (more related to consumer behavior than buyer behavior). We consume products to show our identity such as clothes or music.

Brands (e.g., Nike, Adidas) are getting their own apps instead of using Google Shopping or Amazon (Wichmann et al., 2021)

(Muntinga et al., 2011) classify three types of consumers' online brand-related activities (COBRAs): (1) Consuming (passive consumption), (2) Contributing (engagement via comment, rating), and (3) Creating (upload, publish, write)

8.1 Brand Elements

8.1.1 Brand Name

Pogacar et al. (2021) found that linguistically feminine brand names increase perceived warmth, in turn, increases attitudes and choice share-both hypothetically and consequentially- improves brand outcomes/ performance. The positive effect of feminine brand name on brand performance is lower when subjects are male and when products are utilitarian.

(Pavia and Costa, 1993) For technological items, alphanumeric brand names are more acceptable than for nontechnical products.

(Tsai et al., 2015) The majority of the gain in sales caused by rebranding is attributable to the brand identities before and after rebranding.

8.1.2 Brand Logos

(Henderson and Cote, 1998) High-recognition logos are ones that are natural, harmonic, and intricate, whereas high-image logos are ones that are just somewhat intricate and natural. Complex and elaborate logos are more effective at retaining audience interest and favor.

(Park et al., 2013) Brands with symbols as logos are more effective at providing self-identity/expression advantages than logos consisting solely of brand names.

8.1.3 Brand Slogans

(Dahlén and Rosengren, 2005) Slogans for strong brands are more well-liked and recognizable than slogans for poor brands, independent of respondents' ability to accurately connect them to a brand.

8.2 Brand Marketing

8.2.1 Brand Advertising

(Sethuraman et al., 2011)The average short-term and long-term brand advertising elasticity values are 0.12 and 0.24, respectively. The elasticity of brand advertising has decreased with time, and advertising elasticity is greater for durable products than non-durable goods in the early life cycle stage than in the mature life cycle stage.

(Danaher et al., 2008) The advertising elasticity of the focal brand decreases when one or more rival companies run ads during the same week. If all brand advertising were reduced, each brand's advertising would receive more reaction.

8.2.2 Brand Promotion

(DelVecchio et al., 2006) On average, sales campaigns have little effect on brand choice post-promotion.

(Guyt and Gijsbrechts, 2014) Timing brand promotion out-of-phase with competing chains does not necessarily increase merchants' net revenue gains, especially in areas where brand promotions impact the place of purchase.

(Ailawadi et al., 2001) Heavy consumers of out-of-store brand promotions plan their purchases, are prepared to transfer stores but not brands, have ample storage space, and have a low cognitive demand. Retail brand promotion users feel more financially limited, are impulsive, and are not motivated by conformity motive or cognitive need.

8.3 Brand Equity

Brand equity was operationalized as .... of Perceived Quality, Brand Associations, Brand Awareness, Brand Loyalty:

- a reflective construct (Yoo and Donthu, 2001)
- a formative construct of (Henseler, 2017)
- a consequence/outcome (preferable, and easier to argue with reviewers, but still not the consensus of the field)
- Quantitative measure of brand equity:
  - Kamakura and Russell (1993)
  - Rust et al. (2021)
  - (Mitra and Golder, 2006): (need to read this paper)
- Qualitative measure of brand equity:
  - Yoo and Donthu (2001)
  - Lassar et al. (1995)
  - Keller (1993)

brand and psychological distance: Congruence effect: "information is easier to process, which leads consumers to respond more favorably." (Connors et al., 2020) "consumer spending higher when distance between consumer and brand matched with construal level of marketing communications."

Value-added is the difference between a product's price to consumers and the cost of producing it.

Whereas, brand equity is the difference between a branded product's price to a generic product's price.

This generic product can also have value-added in it in other forms (e.g., convenience)

Brand identity offers a value proposition to customers, where "a brand's value proposition is a statement of the functional, emotional, and self-expressive benefits delivered by the brand that provide value to the customers" (Aaker, 1996, pp.95).

Functional benefits are product attributes that deliver functional utility to customers, while emotional benefits are positive feelings customers feel when using the brand. Ads that provide emotional benefits are higher in effectiveness score compared to functional benefits ads (Aaker, 1996, pp.98).

For example, Evian - "Another day, another chance to feel healthy" ad - which provides emotional benefits of feeling satisfied after a workout. Moreover, self-expressive benefits are when brands provide a means for consumers to express their self-image. A person/ consumer can have multiple roles which associate with multiple self-concepts. Such as a woman, a mother, an author. The difference between emotional benefits and self-expressive benefits can be subtle. Emotional benefits focus on feelings, private setting (e.g., feeling of watching a TV show), past-oriented (memories), transitory, after-use emotional; In contrast, self-expressive benefits focus on self, public settings (e.g., using cars or clothes to signal), aspiration or future-oriented, permanent (i.e., the self links to person's personality), and during-use (wearing fancy clothes to signify a successful person). (cf. (Aaker, 1996, pp.99))

With the advancement in mass production of high -quality products, quality is no longer a dimension of status signal (Holt, 1998)

In the virality framework, the first and foremost driver is social currency. Social currency refers to things worth sharing. People share so that they can let others know that they are in the know. Sharing a new product to assimilate his or her identity with the product and its brand identity, customers are acquiring social currency. Moreover, brand identity is a driver of brand equity, which refers to the value portion that customers are willing to pay above the product's utility value (practical value). Brand identity offers a value proposition to customers, where "a brand's value proposition is a statement of the functional, emotional, and self-expressive benefits delivered by the brand that provide value to the customers" (Aaker, 1996, p. 95). Functional benefits are product attributes that deliver functional utility to customers, while emotional benefits are positive feelings customers feel when using the brand. Ads that provide emotional benefits are higher in effectiveness score compared to functional benefits ads (Aaker, 1996, p. 99). For example, Evian - "Another day, another chance to feel healthy" ad - which provides emotional benefits of feeling satisfied after a workout. Moreover, self-expressive benefits are when brands provide a means for consumers to express their self-image. A person/ consumer can have multiple roles which associate with multiple self-concepts. Such as a woman, a mother, an author. The difference between emotional benefits and self-expressive benefits can be subtle. Emotional benefits focus on feelings, private setting (e.g., feeling of watching a TV show), past-oriented (memories), transitory, after-use emotional; In contrast, self-expressive benefits focus on self, public settings (e.g., using cars or clothes to signal), aspiration or future-oriented, permanent (i.e., the self links to person's personality), and during-use (wearing fancy clothes to signify a successful person). (cf. (Aaker, 1996, p. 99)) Hence, one can see that the brand identity portion that provides self-expressive benefits is social currency while exhaustively different from the brand identity portion that provides functional benefits (i.e., the practical value/ usefulness/ utility) .

We acknowledge that some might view usefulness/ utility as a dimension of brand equity (perceived quality): the perceived quality dimension can signal the true practical value. However, the practical value that drives virality is usually understood, experimented, experienced; while perceived quality is subjective comprehension that is hard to demonstrate. Moreover, quality used to be the status signal. However, with the advancement in mass production of high-quality products, quality is no longer a status signal dimension (Holt, 1998). Thus, we believe that the distinction between the two understandings should be appreciated.

(Bharadwaj et al., 2011; Larkin, 2013; Madden, 2006; Mizik and Jacobson, 2008; Rego et al., 2009) Customer-based brand equity is positively related to a company's stock returns, market value, credit ratings, and leverage, and adversely related to idiosyncratic risk and earnings volatility.

(Tavassoli et al., 2014)

- Employee-based brand equity significantly affects executive pay.
- Executives value being associated with strong brands and are willing to accept lower pay at firms that own strong brands.
- This effect is stronger for chief executive officers and younger executives than for other executives.

8.3.1 Brand Loyalty

- (Berkowitz et al., 1978; Berkowitz, 1978) reviews on Brand Loyalty: Measurement and Management by Jacoby and Chestnut 1978, which pioneered the idea of **attitudinal** and **behavioral loyalty**.

- The Loyalty Economy
- Loyal customers trump powerful brands

8.3.2 Brand Awareness

also known as brand familiarity

(Homburg et al., 2010) found that there exists a correlation between brand awareness and market performance. However, this link is mitigated by market features (product homogeneity and technological volatility) and organizational buyer characteristics (buying center heterogeneity and time pressure in the buying process)

8.3.2.1 Brand Prominence

Butcher et al. (2016) found that consumer views of the quality of luxury items are influenced by increased brand prominence, with quality and respondents' status consciousness influencing the emotional value they obtain from luxury goods. Along with the direct influence of brand prominence, this emotional value has a significant impact on purchase intentions.

(Lee, 2021) reduced emotionality correlates with high-status communication norm, evoking high-status reference groups.

(Han et al., 2010)

- Brand prominence: "reflects the conspicuousness of a brand's mark or logo on a product." (p. 15)

- Brand prominence is "the extent to which a product has visible markings that help ensure observers recognize the brand." (p.17)

- Consumers prefer quiet or versus loud branding because they want to associate vs. disassociate themselves with /from different groups of consumers.

- Luxury goods are those that, in addition to any utilitarian value, confer status on the owner simply by being used or shown.

- Need for status is the "tendency to purchase goods and services for the status or social prestige value that they confer on their owners." (Eastman et al., 1999, p. 41)

- What confers status is the evidence of wealth (i.e., wasteful exhibition = conspicuous consumption)

- Upper-class members spend conspicuous things to distance themselves from the lower class ("invidious comparison"), whereas lower-class members consume conspicuously to align themselves with the higher class ("pecuniary emulation").

- Studies

1. Market data (Bag - LV and Gucci, Car - Mercedes, shoes - LV): inconspicuously branded luxury goods cost more than that of conspicuous branding.
2. Market data (knockoffbag.com and confiscated counterfeit goods from Thailand) counterfeiters copy the lower-priced, louder luxury goods to appeal to poseurs that want to emulate parvenus.
3. Survey (assume people to belong to certain groups based on zipcodes) Patricians pay a premium for signals that only other patricians can decipher
4. Survey (self-categorization to a persona) Social motives explain why groups choose loud or quiet luxury products. Poseurs are more inclined than parvenus to buy fake loud bags when given the chance.

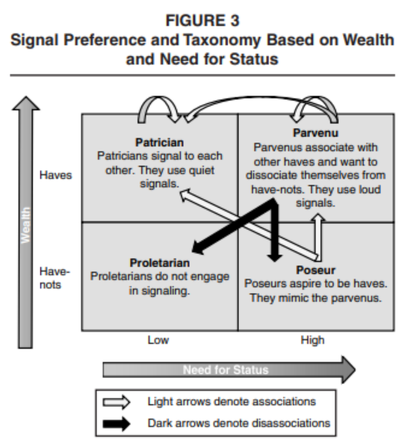


Figure 8.2: Figure 3. p. 17

8.3.3 Brand Associations

"different groups of people can have different or even contradictory associations with the same stimulus" (Wheeler & Berger, 2007)

8.3.3.1 Brand Meaning

Brand meaning is a subset of brand associations (brand associations can also have associations regarding attributes - perceived quality)

Batra (2019)

- Brand meaning refers to "the complete network of brand associations in consumers' minds, produced by the consumer's interactions with the brand

and communications about it." (p. 535)

- Endorsers are conduits of cultural meaning transfer.
- Previous researchers conflated brand meaning with the narrower construct - brand personality.
- Propose possible brand meaning items
- sources of brand meaning (independent variables):
  - visual cues (advertising, packaging)
  - sensory cues (e.g., music)
  - human cues (e.g., endorsers)

8.3.3.2 Brand Image

Successful brand transferability typically depends on attribute similarity and personality similarity (image similarity). Since customers expect a firm's technological competencies to transfer between products with similar makeups, an antecedent of a successful brand's transferability is that products share similar attributes. On the other hand, image similarity is the "relationship of categories in terms of the images of the brands in them" (Batra et al., 1993). For instance, a brand in the entertainment industry with the image of being fun and exciting can potentially extend to soda production. Aaker and Keller (1990) have shown that transferability is higher for complimentary products and hard-to-make products, and no impact for substitutes products, while Batra et al. (1993) showed that brand transferability is higher for a product with a high match on the image (personality dimension) and on product attributes.

Batra and Homer (2004)

- Nonverbalized personality association of celebrity endorsers on fun and sophistication (classiness) can increase consumer beliefs about a brand's fun and classiness. (only under social consumption context, and match between brand image beliefs and product category).
- And ad-created brand image beliefs only influence brand purchase intention, not brand attitudes.

Chu et al. (2021)

- Prestigious brands whose brand image is associated with status and luxury, consumers' attitude toward the product becomes more favorable and their willingness to pay a premium for the product grows as the distance between the visual representations of the product and the consumer increases.
- Popular brands whose brand image is associated with broad appeal and

social connectedness, the closer the distance, the more favorable is consumers' attitude and the higher their willingness to pay a premium.

Chernev and Blair (2015)

- CSR increases not only public relations and customer goodwill, but also customer product evaluation.
- Consumers perceive that products from companies performed CSR to be better performing (even though they can experience and observe the products or when CSRs are unrelated to the company's core business).
- This effect is lowered when consumers believe that company's behavior is driven by self-interest as compared to benevolence.
- Hence, acting good (and people don't know your true intent) can lead to better company's performance.

Liu et al. (2020) built BrandImageNet (based on multi-label deep convolutional neural network model) to predict the presence of perceptual brand attributes in the consumer-created images (which is also consistent with consumer brand perception collected from survey). Hence,this model can monitor brand portrayal in real time to understand consumer brand perceptions and attitudes toward their and competitor brands.

8.3.3.3 Brand Personality

Aaker (1997)

- 5 dimensions of brand personality:
    1. Sincerity
    2. Excitement
    3. Competence
    4. Sophistication
    5. Ruggedness
- Also offers a scale to measure brand personality
- Brand personality is defined as "the set of human characteristics associated with a brand." (p. 347)

Batra et al. (2010)

- Previous research posits that brand with greater brand associations and imagery "fit" will more likely to have successful brand extensions
- If the brand is "atypical" - associations and imagery are broad and abstract compared to the brand's original product category, the brand will be more likely to succeed

- This research offers a Bayesian factor model that can measure that brand-level and category-level random effects, which later helps measure a brand's fit and atypicality.
- Category personality can be masculine or feminine, etc. (Levy, 1959)
- Brand extension considers
  - Fit
    - Concrete product attributes
    - Abstract imagery or personality attributes
  - Atypicality

Grohmann (2009)

- advancement on Aaker (1997)
- scale measuring masculine and feminine brand personality
- Spokespeople shape masculine and feminine brand personality perceptions
- brand personality-self concept congruence affects affective, attitudinal and behavioral consumer responses.
- masculine and feminine brand personality contributes to the brand extension perception

More reference on Actual and the Ideal Self: Malär et al. (2011)

8.3.3.4 Brand Coolness

Warren and Campbell (2014)

- Autonomy refers to "a willingness to pursue one's own course irrespective of the norms, beliefs, and expectations of others" (p. 544)
- Autonomy means diverging from the norm.
- Properties of coolness:
  - Coolness is socially constructed (i.e., "a perception or an attribution bestowed by an audience") (p. 544), which is similar to popularity or status
  - Coolness is subjective and dynamic: can use consensual assessment technique to measure (Amabile, 1983), which is similar to creativity.
  - Coolness is a good thing (positive quality)
  - Coolness is more than positive perception or desirable
- Cool is different from good (being liked) is inferred autonomy

- Autonomy only increase perceptions of coolness under **appropriate context**
- Coolness is " a subjective and dynamic, socially constructed positive trait attributed to cultural objects (people, brands, products, trends, etc.) inferred to be appropriately autonomous." (p. 544)
- Autonomy increases coolness when it's appropriate will depend on
    1. whether a brand diverges from a descriptive or injunctive norm
    2. the perceived legitimacy of the injunctive norm
    3. th extent to which a brand diverges from injunctive and descriptive norms
    4. the extent to which the observer or audience values autonomy
- **Descriptive norm** is" what most people typically do in a particular context" (p. 545).
    - People follow descriptive norm because of uncertain or unclear outcome, and the "normal" course of action is already effective. Hence, divergent could sometimes be more effective.
- **Injunctive norm** is "a cultural ideal or a rule that people are expected to follow" (Cialdini et al., 1990)
    - People follow injunctive norms because it can help build or maintain relationships or social esteem (p. 545)
- Norm legitimacy moderates the appropriateness of divergence from an injunctive norm

Warren et al. (2019)

- Conceptualizes "brand coolness" and a set of characteristics associated with cool brands
- Cool brands are perceived to be extraordinary, aesthetically appealing, energetic, high status, rebellious, original, authentic, subcultural, iconic, and popular
- Develops scale items to measure component of brand coolness and its effect on **consumers' attitudes toward brands, satisfaction, intention to talk about, and willing to pay**.
- Cool brands are dynamic and change over time.
- Cool brands first to a small niche (subcultural, rebellious, authentic, and original)
- Later, when adopted by the masses, cool brands acquire iconic and popular attributes.

8.3.4 Perceived Quality

(Aaker and Jacobson, 1994) Brand quality influences shareholder wealth positively.

(Slotegraaf and Inman, 2004) Customer Equity, Satisfaction, and Quality: Role of Attribute Types Over Time

- **Research Context:**
  - The importance of understanding factors affecting satisfaction and quality, as highlighted by customer equity research.
- **Existing Literature Gap:**
  - Prior studies recognized factors affecting satisfaction and quality.
  - Missed the potential for asymmetric effects over time based on attribute types.
- **Study Focus:**
  - Investigate how consumers' perceptions of satisfaction with product attributes change as they near the end of the warranty period.
- **Attribute Classification:**
  1. **Resolvable Attributes:** Characteristics of a product that can be fixed or adjusted.
  2. **Irresolvable Attributes:** Inherent characteristics of a product that can't be changed or remedied.
- **Key Findings:**
  1. As the warranty end approaches:
  - Satisfaction with resolvable attributes declines faster.
  - However, the influence of these attributes on overall product quality perception intensifies.
  2. On the contrary:
  - Satisfaction with irresolvable attributes decreases at a slower pace.
  - Their impact on overall product quality perception weakens over time.

(Kalra and Li, 2008) Signaling Quality Through Specialization

- **Objective:**
  - To explore how firms, particularly in effort-intensive categories, can utilize specialization as a strategy to signal quality to consumers.
- **Premise:**

  - Companies often brand themselves as specialists, implying they've intentionally forgone alternative opportunities to focus on a specific area.

- **Methodology:**
  - The study models a firm's decision to either provide a single service or offer two services.
  - By opting for a single category, the firm sacrifices potential profit from the secondary category but might realize reduced costs.
  - This cost difference is termed the 'signaling cost' - the extra amount a high-quality firm might spend to differentiate itself from a lower-quality firm.

- **Key Findings:**
  1. **Homogenous Markets:** In markets with similar consumers, a firm of high quality signals its superiority by specializing in just one category.
  2. **Heterogeneous Markets:** In varied markets, a firm can signal its high quality merely using its pricing strategy in both primary and secondary categories. But specialization still remains an effective, secondary quality signal due to its reduced signaling costs.
  3. **Role of Competition:** The likelihood of using specialization as a signaling tool increases in competitive markets.

- **Implications:**
  - Firms operating in effort-intensive markets can harness the strategy of specialization to convey their high-quality status to consumers.
  - While pricing can be a direct quality signal, specialization provides a subtle yet effective cue, especially in diverse markets or those with significant competition.
  - Businesses need to weigh the potential profits from diversification against the benefits of being perceived as a specialized, high-quality provider.

8.4 Brand Authenticity

This construct is to be determined, we still have not consensus whether this is a reflective or formative construct. The first paper (Morhart et al., 2015) is a reflective model, but could not find a latent construct of brand authenticity and they have some troubles with the first order dimensions. On the other hand, (Nunes et al., 2021) recently introduced the formative measurement scale of authenticity. But their arguments on this paper is still questionable.

Morhart et al. (2015)

- Perceived brand authenticity measurement scale
- "PBA arises from the interplay of objective facts (indexical authenticity), subjective mental associations (iconic authenticity), and existential motives connected to a brand (existential authenticity)." (p. 202)
- Definition: PBA is "the extent to which consumers perceive a brand to be faithful and true toward itself and its consumers, and to support consumers being true to themselves." (p. 202)
- Dependent variables: emotional brand attachment, positive WOM
- PBA is the interplay of
  - Objective facts (indexical authenticity) - objectivist perceptive
  - Subjective mental association (iconic authenticity) - constructivist perspective
  - Existential motives connected to a brand (existential authenticity) - existentialist perspective
- Four reflective measures:
  - Continuity
  - Integrity
  - Credibility
  - Symbolism

Nunes et al. (2021)

- Authenticity has 6 formative constructs: accuracy, connectedness, integrity, legitimacy, originality, and proficiency
- Authenticity is defined as "a holistic consumer assessment determined by six component judgments (accuracy, connectedness, integrity, legitimacy originality, and proficiency) whereby the role of each component can change according to the consumption context." (p. 2)
- Definitions of the six components (p. 3)
- **Authenticity** is conceptually different from **consumer attitude**
- Using grounded theory to come up with potential dimension, and verify the measurement model of authenticity based on PLS
- Dependent variables: attitudes, behavioral intentions.

8.5 Brand Relationship

Fournier (1998)

- Brands can serve as relationship partners (p. 344)
  - Interdependence must be present in a relationship (i..e, partners can affect, define and redefine the relationship). (Hinde, 1979)
  - Three ways brands are animated (animism):
    * brand is possessed by the spirit of a past or present other (e.g., spokesperson, significant others who use it, or givers)
    * Anthropomorphization of the brand object with human qualities such as emotionality, thought and volition (p. 345)
    * Perform as active relationship partner
- "Consumer-brand relationships are valid at the level of lived experience" (p. 344)
- Brand can be defined as "a collection of perceptions held in the mind of the consumer." (p. 345)
- Sources of relationship meaning: psychological, socio-cultural and relational
  - 5 socio-cultural contexts: age/cohort, life cycle, gender, family/social network, and culture
- Relationships
  - as Multiplex phenomena
  - in Dynamic Perspective
- **Brand Relationship Quality** BRQ (6 facets):
  - Love/passion
  - Self-connection
  - Commitment
  - Interdependence
  - Intimacy
  - Brand Partner Quality: brand performance in its partnership role.
- Related to Brand Loyalty and Brand Personality

Aggarwal (2004)

- When people form a relationship with brands, they use international relationship norms to guide this relationship.

- There are two relationship types
  - Exchange relationship: (reciprocal favors)
  - Communal relationship: benefits are given to show concern for others' needs.
- Norm violation can influence overall brand evaluations.
- Initial judgments of social stimuli (e.g., people) depend on inferred, abstract information while initial judgment of nonsocial stimuli (e.g., products) depend on concrete attributes because people use self as a frame of reference when comparing to other people (Fong and Markus, 1982)
- Norms of exchange relationship (i.e., quid pro quo): expected return, and prompt repayment
- Norms of communal relationship (to demonstrate a concern for partners and to attend their needs): no expected return, or prompt repayment
- "When consumers form relationships with brands, brands are evaluated as if they are members of a culture and need to conform to its norm" (p. 89)
- The relationship between brands and consumers are more in line with celebrity and fan (p. 89)

Aaker et al. (2010)

- People use warmth and competence (social judgments of people Fiske et al. (2007)) to form perceptions of firms.
- For-profit = competent but less warmth, nonprofits = warm but less competent
- If we can increase competence (through subtle cues that increase credibility) then we can recover willingness to buy products for nonprofits.
- "Warmth judgments include perceptions of generosity, kindness, honesty, sincerity, helpfulness, trustworthiness, and thoughtfulness, whereas competence judgments include confidence, effectiveness, intelligence, capability, skillfulness, and competitiveness" (p. 225) (Aaker, 1997; Yzerbyt et al., 2005)
- Warmth = other-focused, competence = self-focused (Cuddy et al., 2008)
- People trust non-profits more than for-profits (Hansmann, 1981; Arrow, 2020)
- Stereotype is defined as "a shorthand, blanket judgment containing evaluative components." (p. 225)

Puzakova et al. (2013)

- The negative side of brand humanization (i.e., anthropomorphization of a brand): it can decrease consumers' brand evaluations when brand faces negative publicity as compared to non-humanized brands.
- Because brands are living entities (after the humanization process) and attributions are due to stable traits instead of unstable contextual influences in human minds (Gawronski, 2004), it is seen as having intention and responsibility for its actions
- The extent of this negative effect depends on consumer-based factor (e.g., implicit theory of personality):
  - Those who believe in stable human traits (i.e., entity theorists) are more likely to devalue humanized brands because they attribute the wrongdoing to the underlying trait - indicative of future transgression. (p. 82)
  - "Those who believe personality traits as more malleable (i.e., incremental theorist) don't form impression based on a single transgression and do not deem a single misbehavior a predictor of a future pattern of action" (p. 82)
- Leveraging perceptual fluency when products are under human schemas, the product can enjoy greater liking (Delbaere et al., 2011)
- Compensation (vs. denial or apology) is the only effective response among entity theorists.

MacInnis and Folkes (2017)

- A summary of "humanizing brands" literature
- Drivers of Humnaizing brands Epley et al. (2007)
- Human-Focused Perspective (Anthropomorphism). Brands can be perceived as .. with consumers:
  - like
  - part of
  - in a relationship
- Self-Focused Perspective. brands can also be perceived as
  - congruent or
  - connected to the self
- Relationship-Focused Perspective: brand relationships are analogous to human relationships

(Ordabayeva et al., 2022)

- Negative internet reviews from socially distant (but not socially close) individuals may not be as harmful to identity-relevant brands. Because a negative review of an identity-relevant brand can threaten a client's identity, the consumer will seek to strengthen their relationship with the brand.
- They show that this effect does not appear when the review is positive or when the brand is irrelevant.

(Swaminathan et al., 2007)

- Self-concept is "the amount that the brand contributes to one's identity, values, and goals." (p. 248)
- Under independent self-construal, self-concept connection (with its focus on the individual) is more important
- Under interdependent self-construal, brand country-of-origin connections (with its focus on the group) is more important.

(Swaminathan et al., 2009)

- Attachment theory posits two dimensions of attachment style: anxiety and avoidance.
- Level of avoidance predicts anxiety-related brand personality.
  - Under high avoidance and anxiety, individuals favor exciting brands; under low avoidance and anxiety, they prefer sincere brands.

(Aggarwal, 2004) With certain businesses, customers can form enduring relationships that are "humanlike." Strength type (fling vs. partner) and relationship norm (communal vs. exchange) can differ among brand partnerships.

8.5.1 Brand Love

It exists on the same level as Brand Equity and it subsumes Brand Affect

Batra et al. (2012)

- Brands are defined as "the totality of perceptions and feelings that consumers have about any item identified by a brand name, including its identity (e.g., its packaging and logos), quality and performance, familiarity, trust, perception about the emotions and values the brand symbolizes, and user imagery." (p. 1)
- Love emotion is a single, specific feeling, short term and episodic, while love relationship is long-lasting and involves numerous affective, cognitive, and behavioral experiences.
- Brand love is measured based on reflective measurement (reflective indicators of hierarchical organized factors)

- Brand love is a component of brand relationships.

Bagozzi et al. (2016)

- Developed a parsimonious brand love scale
- Reflective Higher-order factor
  - Self-brand integration
  - Positive emotional connection
  - Passion driven behavior
- Used Multitrait-Multimethod Matrix (MTMM) of method bias

8.6 Reputation

Reputation is "a global evaluation of an organization accumulated over a period of time." (quote by (Aaker et al., 2010, p. 225)), original by (Fombrun and Shanley, 1990).

Reputation can have both competence dimension (devine and Halpern, 2001) and warmth (Aaker et al., 2004)

What is the difference between brand equity and brand reputation?

- Brand equity: belongs to the marketing world, where it means the positive (good) part of the firm.
- Brand reputation: belongs to the management world, where it means both the positive and negative parts of the firm.

(Proserpio and Zervas, 2017) Effect of management responses on consumer reviews

- Hotels respond after a negative shock to their ratings
- hotels respond about the same rate to positive, negative, and neutral reviews
- Responding hotels receive 0.12 stars higher in their ratings
- When hotels begin to respond, they receive fewer but longer negative reviews because dissatisfied customers are less likely to post short, unjustifiable comments when they anticipate being scrutinized. Therefore, managers must consider a trade-off: fewer bad evaluations at the expense of lengthier and more thorough negative feedback.
- Data: TripAdvisor hotel ratings.
- Identification strategy (good paper to follow for endogenous treatment)s:

  - Since hotels can choose to respond to reviews and how to respond (non-random treatment).

8.7 Brand Evaluation

- Naylor et al. (2012) defined "mere virtual presence" as whether presence of virtual supporters for a brand (e.g., demographic) is revealed. The mere virtual presence can affect a target consumer's brand evaluation and purchase intention. This effect is moderated by the composition of existing supporters and targeted new supporters and (2) and salience of competitor brands when evaluating the focal brand.

8.8 Brand Favorability

(Zhang and Moe, 2021)

- Account for the measure biases shown by social media posters.
- Use probabilistic graphical model-based
- The measure is correlated with traditional survey-based measures

8.9 Branding Portfolio Management

8.9.1 Brand Architecture/Portfolio

(Aribarg and Arora, 2008)

- Firms may increase portfolio profitability by eliminating versions of lower-tier brands, but the proportion of upper-tier brands to lower-tier brands moderates benefits.

(Rao et al., 2004) Corporate branding strategy association with intangible value (Tobin's q)

- House of brands
- Branded house: greater efficiency and lower customization and cannibalization, more potential risk(Rao et al., 2004)
- Corporate branding strategy is associated with higher Tobin's q, mixed branding strategy is associated with lower values of Tobin's q

(Morgan and Rego, 2009a)

- Brand portfolio characteristics
  - Number of brand owned
  - Number of segments

- Degree to which the brands compete with one another
- Consumer perceptions of the quality and price of the brands

(Strebinger, 2014)

- Corporate branding is favored by both service and consumer durables companies than consumer nondurables ones.

(Hsu et al., 2015)

- An extension of (Rao et al., 2004)
- House of brands
- Branded house
- Sub-branding (e.g., Intel Pentium): has both the brand and corporate brand name (high risk high return option)
- Endorsed branding: the second brand is more prominent graphically than the parent brand (reduced risk).
- Hybrid branding: Combine all four above options.
- The most valuable increase in firm value comes from subbranding, but it also has the most risk.
- Examine idiosyncratic risk
  - Brand reputation risk
  - Brand dilution risk
  - Brand cannibalization risk
  - Brand stretch risk

8.9.1.1 Brand alliances and co-branding

- (Samu et al., 1999): impact of advertising alliances on new brand introductions
  - Degree of complementary, type of differentiation, processing strategies employed by consumers of ad alliances affect ad effectiveness.
- (Cao and Sorescu, 2013): Stock Market Reactions to the Introduction of Cobranded Products
  - Consumer react: signal of firm innovativeness, improved quality, trust
  - Investor react: positively to consistency between co-brand partners and exclusive partnerships.

- (Robinson et al., 2015): Structure of licensing agreements on shareholder value
- (Cao and Yan, 2017): brand value on brand alliance outcome
  - Moderated by the value differential between partner brands and the level of past exploitation fo the target brand
- (Rao et al., 1999; Simonin and Ruth, 1998): A brand alliance's reputation affects its alliance partners.
- (Washburn et al., 2004): When judging the quality of a product with an important quality that can't be seen, consumers' views of the product's quality are improved when it is paired with a second brand that is seen as vulnerable to consumer sanctions.
- (Singh et al., 2020) Preventable, highly controllable, and purposeful crises are bad to the reputation of the culpable ally. Deny reaction is effective in repairing company image despite being inferior to decrease and acknowledge/rebuild responses. In addition, we show that the non-culpable partner suffers from crises only indirectly, as a result of negative post-crisis sentiments against the partnership

(Rindfleisch and Moorman, 2001)

- Although embeddedness increases both the acquisition and utilization of information in alliances, redundancy decreases the acquisition of information but increases its utilization.

(Swaminathan and Moorman, 2009): Marketing Alliances

(Malhotra and Bhattacharyya, 2022) Leveraging Co-Followership Patterns on Social Media to Identify Brand Alliance Opportunities

- Use Twitter followership data, authors identify brand extension or co-branding opportunities based on common followership patterns.
- Introduce brand **transcendence** construct: "measures the extension which a brand's followers overlap with those of other brands in a new category."

8.9.1.2 Brand acquisitions and disposals

Positive abnormal stock market returns when acquirers have stronger marketing capabilities, low product diversification, higher positioning, high cost synergies and low sales synergies (Swaminathan et al., 2022, p. 649)

- (Bahadir et al., 2008): The value of the target company's brands is favorably influenced by its marketing capabilities and its brand portfolio diversification.

- (Jit Singh Mann and Kohli, 2012): Brand Acquisitions Create Wealth for Acquiring Company Shareholders
- (Newmeyer et al., 2016): brand acquisitions

(Wiles et al., 2012): The Effect of Brand Acquisition and Disposal on Stock Returns

- The results of brand acquisition and disposal (not symmetric) depend on
  - marketing capabilities (stronger)
  - channel relationships (Sellers with worse channel partnerships and those selling several brands, brands with lower price/quality positioning, and unrelated products have higher abnormal returns)
  - brand portfolios (buying brands with higher price/quality positioning than their existing portfolio)
- Investors reward purchasers who find cost savings when integrating new brands, but penalize those who find revenue synergies.
- This paper is different from (Bahadir et al., 2008)
  - View from the perspective of shareholder (not manager)
  - Examine brand asset transactions from both the buyer and seller shareholders' perspectives
  - Brand assets acquisition and disposal is different from sale of entire firms
  - Examine net shareholder wealth created.
- Hypotheses
  - A firm's marketing capability is "its ability to define, develop, and deliver value to customers by combining and deploying its available resources." (p. 40)
  - A firm's channel relationships is "the extent and nature of its connections with channel partners." (p. 41)
  - Brand portfolio was looked at from the perspective of quality/price positioning, relative positioning of the brand involved in the transaction and (3) cost vs. revenue synergies
- Method
  - See (Srinivasan and Bharadwaj, 2004)
  - Sample: (n = 322) in 31 B2C industries from 1994 to 2008, exclude non-G7 countries and noncusumser food service channel (not convincing).

- Event search: from SDC platinum database, firms' annual reports, investor relations material, press releases on firms' websites, Factiva search.
- Event description:
    * Disposal event: "An announcement of a sale or pending sale of a brand, identified through a Factiva search of company news releases and press reports" (p. 43)
    * Acquisition event: "An announcement that an agreement has been reached to acquire a brand."
- Confounding event: earnings announcements, stock splits, key executive changes, unexpected stock buybacks, changes in dividends within the two-trading day window surrounding the event. (McWilliams and Siegel, 1997)

- Measures
    - Marketing capability: used an input-output approach
        * Marketing capability is "its ability to use available resources to create market-based, intangible asset value." (p. 43) following (Srivastava et al., 1998a)
        * Use a stochastic frontier estimation (CFE) marketing capability operationalization (Dutta et al., 1999), where the (1) resource inputs are each firm's sales, general and admin, ad expense (from compustat) and number of trademarks owned (from the USPTO database) and (2) resource output variable used the intangible asset value of the firm (Tobin's q) following (Simon and Sullivan, 1993) (might consider using Total Q here) adjusted for the variance accounted for by the firm's technology (R&D spending and the number of patents) and management quality ("maagnement quality" score from Fortune's Most Admired American Companies database where it was available, and top management team compensation relative to the industry average as an alternative management quality indicators for the remaining observations).
        * Regress technology and management quality indicators on the firm's Tobin's q and used the residual from this regression as the market-based value created by the firm output indicator in the SF.
        * Need to verify the face validity of the measure by looking at the correlation with firm's future cash flow performance.
    - Distribution resources and portfolio characteristics and cost vs. revenue synergies measures were weak (based on coders).
    - Control for diversification in business.

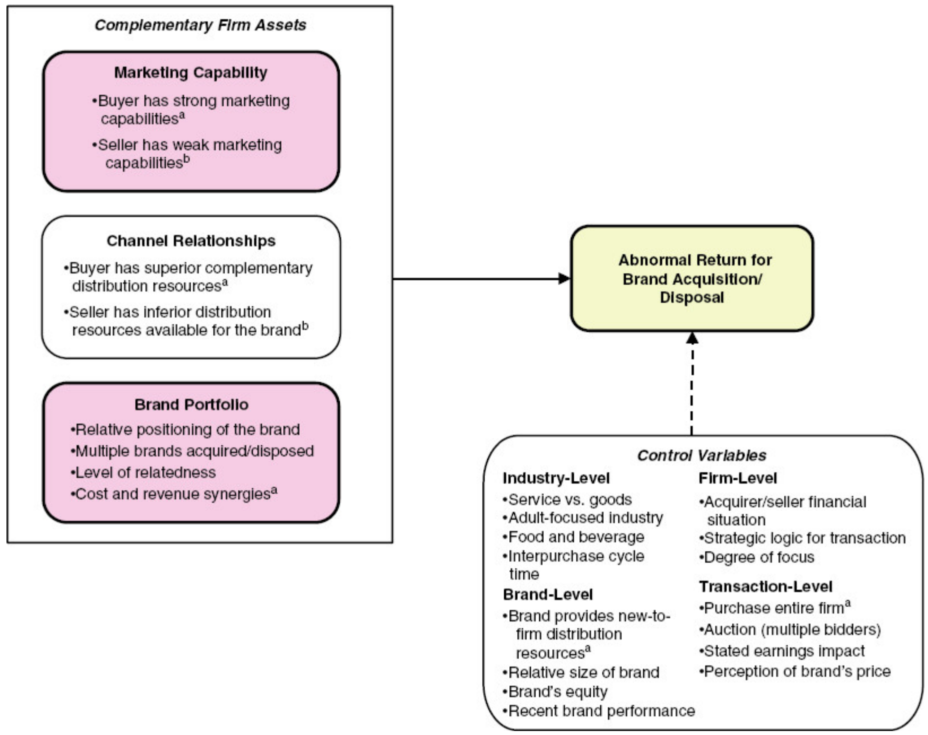


Figure 8.3: Figure 1 (p. 39)

(Biraglia et al., 2022)

- Ten studies demonstrate negative brand reactions can be explained by perceived loss of brand's unique values.
- Negative effect of acquisitions depends on:
  - Acquired brand's values
  - Brand age
  - Leadership continuity
  - Alignment between acquiring and acquired brands
- Research covers a range of methods, designs, product categories, and brands.
- Explanation for negative brand reactions is based on the "values authenticity account."

(Mukherji et al., 2011)

8.9.1.3 Ingredient Branding

(Swaminathan et al., 2011)

- Significant behavioral spillover impact of cobranded product trial on host and ingredient brands.
  - This impact is bigger among non-loyal past consumers of the host and ingredient brands and when perceived fit is higher.
- Data: AC Nielsen scanner panel data

8.9.2 Brand Extensions

(Aaker and Keller, 1990) Consumer Evaluations of Brand Extensions

- The perceived compatibility between the parent brand and an extension product is the single most critical factor in determining the level of success achieved by a brand extension.

(Smith and Park, 1992) Brand Extensions on Market Share and Advertising Efficiency

- Surveys: Product Managers and Consumers
- **Objective:** The study investigates how brand strategy (brand extensions vs. individual brands) impacts new product market share and advertising efficiency.
- **Main Findings:**
  1. **Brand Extensions vs. Individual Brands:** Brand extensions have a higher market share and better advertising efficiency than individual brands.
  2. **Parent Brand Strength:** A strong parent brand positively impacts the market share of its extensions. However, it doesn't influence advertising efficiency.
  3. **Number of Products:** The number of products linked to the parent brand doesn't affect the market share or advertising efficiency of its extensions.
  4. **Similarity of Products:**
     - Market share isn't influenced by the similarity between the extension and other brand-related products.
     - Advertising efficiency is higher when product similarity is based on intrinsic attributes.
  5. **Product Attributes and Market Knowledge:**

   - Both market share and advertising efficiency increase when the brand extension mainly has experience attributes and is in a market where consumers lack knowledge.

6. **Competitive Intensity:**
   - Advertising efficiency is unaffected by competition.
   - Market share increases when there are fewer competitors in the market.
7. **Extension Establishment:** As a brand extension becomes more established, both its market share and advertising efficiency decrease.

(Rangaswamy et al., 1993) Brand equity and the extendibility of brand names

- **Opportunity for Firms**: Brand extensions leverage the equity of existing brand names to boost marketing productivity.
- **Key Management Consideration**: Before an extension, determine the extendibility potential of the brand name.
- **Consumer Utility Components**: We propose that consumer utility for a brand encompasses:
  1. Brand name.
  2. Physical product attributes.
  3. Interaction between the brand name and product attributes.
- **Extendibility Constraint**: The brand's extendibility is affected by its interaction with product attributes in the primary category.
- **Comparison of Two Brands**:
  - If both brands are equally preferred, but one has utility due to brand-attribute interaction, it may be less extendable than a brand without this interaction.
- **Consumer Utility Model**: A model has been crafted for experimental validation using real brands and hypothetical extensions.
- **Findings**: Our experimental outcomes align with our model.
- **Strategy for Max Extendibility**: Brands should focus on enhancing non-product-specific values like quality, style, durability, and reputation linked with their name.

(Dacin and Smith, 1994) Brand Portfolio Characteristics on Consumer Evaluations of Brand Extensions

- **Objective:** The study investigates the influence of brand portfolio characteristics on brand strength, particularly focusing on consumers' confidence in and favorability towards brand extensions.

- **Background:** While more brands are diversifying their product categories, concerns exist about potential weakening of brand strength. Current research in this area is limited.
- **Main Findings:**
  1. **Number of Products and Brand Evaluation:** Laboratory experiments show a positive link between the number of products linked with a brand and the consumers' confidence in, as well as favorability of, evaluations of extension quality. This result was not mirrored in the survey.
  2. **Portfolio Quality Variance:** A consistent observation from both the experiments and the survey was that when there's a decrease in the variance of portfolio quality, a positive association emerges between the number of affiliated products and consumers' confidence in evaluating brand extensions.
- **Implications:** The findings provide insights for both theoretical understanding and practical application in brand management, suggesting that brand portfolio characteristics play a role in shaping consumers' perceptions and evaluations of brand extensions.

(Broniarczyk and Alba, 1994) The Importance of the Brand in Brand Extension

- The impact of perceived fit can be overridden by the influence of key brand associations, which can form the basis of fit between a parent brand and an extension brand.

(Lane and Jacobson, 1995) Stock Market Reactions to Brand Extension Announcements

- Negative impact: when familiarity is disproportionately higher than esteem or vice versa.
- How the stock market reacts to using brands in new ways depends on how well known and respected the brands are.

(Erdem, 1998) Umbrella Branding

(Morrin, 1999) Brand Extensions on Parent Brand Memory Structures and Retrieval Processes

- Transferring brand equity is made easier when there is a high degree of resemblance between the parent brand and the expansion category.

(Klink and Smith, 2001) Threats to the External Validity of Brand Extension Research (Brand Extendibility: Beyond Fit)

- **Objective:** Explore the mismatch between research suggesting brand extendibility relies on perceived fit with the extension category and real-world examples of successful extensions into varied domains.
- **Key Insights:**

1. Past research had limitations: scant extension information, didn't consider consumer adoption tendencies, and only one exposure to proposed extensions.
2. Perceived fit's influence vanishes when more attribute information is provided about the extension.
3. Fit effects mainly concern later product adopters.
4. Exposure to an extension multiple times enhances perceived fit.

- **Implications:** Research design factors previously deemed unimportant can actually reshape our understanding of brand extension dynamics.

(Swaminathan et al., 2001) Brand Extension Introduction on Choice

- brand extension is a strategy that a brand attaches its brand name to a new product in a different product category.
- Contribution:
  - Empirically, there is a positive reciprocal effects of extension trial on parent brand (more intense for prior non-users of the parent brand). Also evidence for potential negative of extension failure on the parent brand (and again for more loyal customers).
  - Experience with parent brand influences extension trial, but not extension repeat.
  - Moderating role of category similarity in the effect of brand extension on parent brand has been seen in an attitudinal context (effect might be overstated in lab setting (Dacin and Smith, 1994)), but non in an actual purchase context.
- Study 1: main effect (brand extension effect on parent brand choice): use binary logit instead of traditional multinomial logit in brand choice modeling because the latter cannot capture incremental effect of the loyalty coefficient.
- Limitation: did not consider sequential introduction in the second study, which is later addressed by (Swaminathan, 2003) (likely similar dataset and the takeaway is that prior experience with the parent brand and intervening extension influences purchase behavior of later brand extension for those with low loyalty towards the parent brand).

(DelVecchio and Smith, 2005) Brand-Extension Price Premiums

- **Objective:** Explore the influence of brand strength on potential price premiums when extending into new product categories.
- **Key Insights:**
  1. **Brand Benefits:** Strong brands in new categories can command higher prices than lower equity brands.

2. **Perceived Risk Reduction:** Recognizable brands diminish the perceived purchase risk for customers.

3. **Price Premiums & Risk:** Price premiums can fluctuate based on the associated purchase risk.

4. **Perceived Fit & Price:** Price premiums are tied to the fit between the brand and the extension category.

5. **Risk Variables:** The relationship between perceived fit and price premiums changes based on financial and social risks linked with the extension product.

(Völckner and Sattler, 2006) Drivers of Brand Extension Success

(Heath et al., 2011) Asymmetric Effects of Extending Brands to Lower and HigherQuality

- The reach of high-quality brands can be significantly greater than that of low-quality brands.

(Mukherji et al., 2011) How Incumbents Take on Acquisitive Entrants

- Corporate acquisitions are a common means by which large firms enter new markets.

- Large acquirers' entry into new markets can affect the strategies and performance of incumbent firms in those markets.

- Incumbent firms are more likely to align their product mix strategy with that of the acquisitive entrant if the incumbent is large, the acquirer's past performance has been strong, and the market served by the incumbent is small. However, large incumbents that deviate from acquirers' product mix strategy perform better than other incumbents do.

(Mathur et al., 2022) The Context (In)Dependence of Low Fit Brand Extensions

- Identify conditions in which low fit brand extension that can be beneficial

- For context dependent individuals, benefit-based on can help increase the evaluation of low fit extensions, but providing attribute-based info can decrease the favorable evaluation of low fit extension via reliance on extension fit

- For context independent individuals, they base their judgment on extension fit regardless of info provided

- Not surprisingly, the high fit extension is unaffected by context dependence and type of information.

8.10 Strategic Branding Decisions

8.10.1 Brand Crisis/ Recovery

Keywords: brand crisis, product harm, crises, firestorm, negative word of mouth

(Ahluwalia et al., 2000) firms responses to negative events can mitigate brand perceptions

(An et al., 2011) found firms communication can affect news.

(Golmohammadi et al., 2021) Complaint Publicization in Social Media

(Borah and Tellis, 2016): nameplate recall effects on another car brand. Brand crises can negatively spill over into other segments and brands within the same brand, and the consequences are more pronounced inside a particular nation.

Other studies:

- (Cleeren et al., 2013)
- (Van Heerde et al., 2007)
- (Dawar and Pillutla, 2000): Customers' past expectations of the company influence how they understand the evidence of the firm's reaction to a brand crisis.
- (Dutta and Pullig, 2011): Effectiveness of corporate responses to brand crises: The role of crisis type and response strategies
- (Pfeffer et al., 2013)
- (Hewett et al., 2016): negative news spiral
- (Dahlén and Lange, 2006): Crises can affect other brands in the same product category, and this is especially true if the brands are comparable.
- (Ertekin et al., 2018): Investors respond poorly to trademark infringement cases, but businesses that prevail in them do well over the long run.
- (Dinner et al., 2018): If a brand crisis occurs in a foreign market, the repercussions relate inverted U-shaped patterns to the mental distance between the home country and the host nation. If a company has built strong marketing capabilities, the impact of psychological distance is greatly lessened.

(Herhausen et al., 2022)

- Active listening and empathy in the firm's response evoke gratitude in high-arousal customers, even if the actual failure is not (yet) recovered.
- Pre-study interviews: 16 social media managers
- Data: Facebook brand community listed on the S&P 500 (Oct 1, 2011 and Jan 31th, 2016): 472,995 negative customer posts across 89 brand communities
  - LIWC text mining dictionary to have intensity of pos and neg words

- Measurement:
  - Arousal: text analysis
  - Strength of structural ties: frequency of communication between users
  - Degree of linguistic style match (LSM): intensity of each of word categories in negative postings
  - Intensity of empathy/ explanation: proportion of affect/explanatory words
  - Control var:
    * Firm level: industry membership, brand familiarity, brand reputation,
    * Community level: community size, member attentiveness and expressiveness, firm engagemetn frequency, average structural tie strength among members, variance in linguistic style,
    * Post level: # of competing inputs at the time of post, sentiment of previous post, post length, post complexity, frequency of complain
    * firm response: respond or not, response time
    * time: year and month
  - Virality: # of likes and comments
  - Dummy var: apology, compensation, suggested communication channel change

8.10.2 Global Brand Strategy

(Yang et al., 2019) When consumers think of themselves as local instead of global, they are more likely to link price to how good they think something is.

(Steenkamp, 2019) A company's implementation of a global brand strategy is greatly affected by its organizational structure and management methods. The more global a brand is seen to be, the higher its perceived quality and prestige, as well as its association with global citizenship, trust, affect, purchase likelihood, and loyalty.

(Torelli and Ahluwalia, 2012) The perceived quality and prestige of a brand, as well as its associations with global citizenship, trust, affect, purchase propensity, and loyalty, increase the more globally recognized it is.

(Gürhan-Canli and Maheswaran, 2000) Depending on the vertical dimension of individualism and collectivism, the influence of nation of origin on product choices varies.

(Strizhakova et al., 2011) Consumers who believe more strongly in achieving global citizenship through global brands are more inclined to utilize brands as identity markers and symbolic signals.

(Aaker and Maheswaran, 1997) Processing differs systematically across cultures, and these variances are caused by changes in cue diagnosticity.

8.10.2.1 Perceived Brand Globalness

E M Steenkamp et al. (2002)

- Perceived brand globalness affects brand purchase (via brand quality, and prestige)
- The effect is moderated by consumer ethnocentrism (CET)
  - CET is "the beliefs held by consumers about the appropriateness, indeed morality, of purchasing foreign-made products' (Shimp and Sharma, 1987, p. 280)
- Alternative route to brand purchase is to become an icon of the local culture

Davvetas et al. (2015)

- Validate results from E M Steenkamp et al. (2002)

Xie et al. (2015)

- Formally introduce perceived brand localness
- With the construct "brand identity expressiveness", the path of brand quality and brand prestige from previous model become null.
- Brand identity expressiveness is "the capability of a particular brand to construct and signal a person's self-identity to himself as well as his social identity to important others." (p. 53)
- Three key needs in defining self-identity (Edson Escalas and Bettman, 2012):
  - self-continuity
  - self-distinctiveness
  - self-enhancement

Steenkamp and de Jong (2010)

- Introduce two concepts:
  - Attitude toward global products (AGP)
  - Attitude toward local products (ALP)

BATRA et al. (2000)

- In the context of developing countries, brands from a nonlocal country of origin are preferred to those that are local because of **social status**.
  - This effect is greater for those who have a greater admiration for developed countries lifestyles.

  - This effect is greater for consumers who are high in susceptibility to normative influence and product categories that carry social signaling values
  - This effect is greater when products are less familiar

(Bronnenberg et al., 2009, 2012)

- Found pattern of consumer preferences for **local brands**
- People carry their local preferences to their new location (i.e., preference persistence)

8.10.3 Competitors

(Zhou et al., 2021) found compliments posts (on Twitter) generated over 10x more likes and retweets than their typical content. Hence, complimenting competitors can have a positive effect on sales and reputation, because the complimenters can be seen as warmer, more friendly and trustworthy. This phenomenon is coined as "brand-to-brand praise." In the case of skeptical consumers and for-profit brands (i.e., those were seen not as warm) and authentic and ingenuous compliments, this effect is largest.

(Wedel and Zhang, 2004) The influence of national brands on store brands across subcategories is stronger than the impact of store brands on national brands.

(Thomadsen, 2012) When a moderate number of consumers were underserved before to the launch of the new product and the new product is positioned so that both rivals' goods appeal to similar sets of customers, firms gain from the arrival of a rival the most.

(Paharia et al., 2014) Compared to when they are competing with brands that are comparable to them or when consumers perceive them outside of a competitive context, support for small companies rises when they are exposed to a competitive threat from large brands.

(Dubé and Manchanda, 2005) Advertising's function is to create categories rather than to steal market share (competitive), however the complementary function of advertising is considerably more pronounced in larger markets than in smaller ones.

8.11 Brand-Consumer Interaction

8.11.1 Brand Experience

(Brakus et al., 2009) The brand experience is comprised of sensory, emotional, cognitive, and behavioral impressions and associated behaviors. Brand experience influences brand purchase and loyalty positively.

(Morgan-Thomas and Veloutsou, 2013) Positive online experiences are determined by credibility and utility, resulting in gratification and the establishment of an online relationship.

(Zarantonello et al., 2013) Consumers place a higher value on brands' functional communications in developing countries, but in developed markets, they place a higher value on brands' experiential communications.

8.11.2 Brand Cocreation

(Fuchs et al., 2013) Cocreation does not improve brand attitudes or brand like in some categories (luxury categories).

(Fuchs et al., 2010) Consumer empowerment and brand identity are both boosted through cocreation.

(Paharia and Swaminathan, 2019) When compared to other customer categories, cocreation does not effectively increase brand liking for some consumer types (high power distance and conservative political orientation) (low power distance, liberal political orientation)

8.11.3 Brand Community

(Muniz and O'Guinn, 2001) Oppositional brand loyalty asserts that support for one brand may occasionally be overshadowed by disdain or disapproval of another.

(Muñiz Jr. and Schau, 2005) As customers readily appropriate and incorporate a well-known religious mythology into a brand they really adore, maybe even worship, religiosity can be related to brand community.

Chapter 9

Virtual Environment

API and webscraping can be used to (Boegershausen et al., 2022) (for technical tutorials, see table W2)

- Study new phenomena in marketing
- Boosting ecological value
- Help advance method to deal with this new type of data
- Improve measurement

Researchers should not create fake accounts to scrape

Threat to internal validity:

- Algorithmic interference (Xu et al., 2019) (e.g., sorting or recommendation system)
- Inter-temporal stability: changes to data sources

Documenting data

- Logbook: note important events
- Write documentation for the data collection process: following (Gebru et al., 2021)
- Consider long-term data archival space: Zenodo, dataverse, Harvard Dataverse, DataCite, re3data
- Let others use your data: following Creative Commons License

Review papers:

(Kannan and Li, 2017) Digital Marketing Framework

(Sridhar and Fang, 2019): Digital, data–rich, and developing market (D3)

(Verhoef and Bijmolt, 2019): Digital business models

(Lamberton and Stephen, 2016)

- look at at changes in academic on digital, social media, and mobile (DSMM) marketing themes from 2000 2015
- DSMM as

	Theme 1: facilitator of individual expression (self-expression and communication)	Theme 2: decision support tools	Theme 3: market intelli-gence source
Era 1 (2000-2005)	with online communities	recommendation agents and comparison matrices • (Häubl and Trifts, 2000) found that with decision aids, consumers enjoy higher-quality alternatives, lower search cots, better choices. • (Brynjolfsson et al., 2003a)online retailers have to respond to a stronger price competition than offline retailers. • Internet boom does not mean price war or choice overload for consumers	help firms predict con-sumer behav-iors.
Era 2: 2005-2010 50% pen-e-tra-tion	Online WOM WOM referred customers have greater long-term value than traditionally acquired customers (Villanueva et al., 2008) (Trusov et al., 2009a)	Theme 2 and 3 merged together. Who drives diffusion: audience + influencers (Trusov et al., 2010) found a way to identify influential users in online social networks	(Katona and Sarvary, 2008) (Katona and Sarvary, 2010)

	Theme 1: facilitator of individual expression (self-expression and communication)	Theme 2: decision support tools	Theme 3: market intelligence source
Era 3: 2011-2015 80% penetration	People are more expressive	UGC as a marketing tool Consumption and production are tradeoffs (people consume more content will produce less content, and vice versa) at individual level (Ghose and Han, 2011)	

- Papers usually include methodological keywords which suggests that the new paradigm new methods to play with new data types (click stream, online chat, etc.)
- During this period DSMM in both worlds (academia and practice) correspond well.

Tonietto and Barasch (2020) show that generating content during an experience increases feelings of immersion and accelerate perceived time, which in turn enhances enjoyment of the experience. Moreover, consumers who are incentivized or motivated by social norms to generate content have the same experiential benefits as those who create content organically.

Lacka et al. (2021) define price impact as" the impact on the variance of stock price." They estimate the permanent and temporary price impacts of the firm-generated Twitter content of S&P 500 IT firms: firm-generated tweets induce both permanent and temporary price impacts, depending on valence and subject matter. Tweets reflecting **only valence or subject matter** concerning consumer or competitor orientation onlly have **temporary** price impacts, while those with both attributes generate permanent price impact. Moreover, negative valence tweets about competitors generate the largest permanent price impacts.

Golmohammadi et al. (2021) show that firm responses to complaints on social media increase the potential public exposure of complaints. This negative effect can outweigh any positive customer care–signaling impact from firm responses. More specifically, a response strategy that has a high level of complaint publicization (e.g., detailed responses through multiple communication exchanges) could decrease **perceived quality** and **firm value**, diminish the positive impact of a firm's own posts, and increase the volume of future complaints. These adverse impacts are stronger for firms that are targeted by retail investors.

Miao et al. (2021) define avatars as "digital entities with anthropomorphic appearance, controlled by a human or software, that are able to interact." (p. 5). The authors also offer a 2x2 typology of avatars (behavioral realism and form realism) from low to high.

Bharadwaj et al. (2021) extracted emotions from salespeople using their livestream videos. In one-to-many screen-mediated communications, salespeople should not display emotions.

Jia et al. (2020) found that there is an inverse size-speed association (i.e., speed-based scaling effect induces people to estimate the size of an object to be smaller when it is animated to move faster) when they see digital ads (because they learned from other animate agents such as animals). To mitigate this effect, ad creator can (1) reduce the similarity between the movement of object in the ads with objects in other domains (e.g., human, animals), (2) give consumers more knowledge about the domain, (3) highlight product size information. Interestingly, the effect can be reserved when a positive size-speed association in the base domain is made accessible.

KOUKOVA et al. (2008) found evidence from experiments that consumers' content consumption preferences (e-book, books, etc) vary across platforms

(Yadav and Pavlou, 2014) reviews marketing in computer-mediated environments

(Grewal et al., 2021) Multimedia Medium (Twitter, Facebook, Instagram) and Modality (auditory, tone, visual, olfactory, gustatory, haptic systems, visual)

1. Data Gathering

- Consumer privacy
- Legal and ethical concerns regarding web scraping

2. Data Analysis

- Feature extraction
- Dimensionality reduction

3. Interpretation

Substantive Challenges

- Sentiment-based targeting
- Location-sensitive targeting
- Computers as Consumers
- Privacy Risk

9.1 Email

(Munz et al., 2020)

- Having donation receivrs with similar names can encourage donors to open and donate more.
- Potential donors with similar surname first letter with donation receivers can also increase generosity

(Sahni et al., 2018)

- personalizing the emails by adding consumer-specific information (e.g., recipient's name) benefit advertisers.
- Personalization increases opening, sales, and reduces unsubscribing probability.

9.2 Social Media

To find relevant hashtags to an issue, consult with

- http://best-hashtags.com/
- https://ritetag.com/

Review

- B2B: (Salo, 2017)

Social Media Metrics and Dashboards (Peters et al., 2013)

(Pelletier et al., 2020)

- Consumer usage motivation for Facebook, Twitter, and Instagram and understanding which platforms consumers prefer to use to co-create with brands
- Types of customer response:
  - Social purpose: interact with others and share content with others
  - Information purpose: seeking out info, news, events, professional business purposes
  - Entertainment purposes: have fun or follow their favorite celeb and pop culture events
  - Convenience purposes just to browse and observe when they are bored
- Uses and gratification theory: benefits extracted from a media source vary based on the different purposes for which a person chooses to consume media. (Severin et al., 1997)
  - Social motivation: people use platforms with more broad ways of communications for social motivation (e.g., Facebook)

  - Informational motivation: users with informational purpose usage will come to the platform with an emphasis on real-time information (Twitter)
  - Entertainment motivation: Hedonic purpose is to escape and enjoy experiences (Instagram). Users with entertainment purposes will go to a platform with an emphasis on hedonic experiences (Instagram)
  - convenience motivation: a source of convenience or distribution (to pass the time). Easy browsing to scroll content (e.g., Instagram). Users will go to the platform with visual content and minimal text (Instagram). People find all platforms (Instagram, Facebook, and Twitter) convenient.
- Users who desire to engage in co-creation will go to a platform with a broader array of communications mechanisms.
- With social media, consumers co-create brand stories and share personal experiences, which can enhance or destroy brand equity
- Findings:
  - For informational purpose, consumer use twitter
  - For social purposes, people like to use Twitter and Instagram
  - Instagram is a primary platform for entertainment motivation and co-creating with brands
  - Facebook has the lowest usage intentions and co-creation (despite being the largest platform and network most widely used by marketers) Even though practitioners have a strong bias toward Facebook as a social media promotional platform, but then one size does not fit all.

(Schweidel and Moe, 2014) Listening in on Social Media:

- Joint modeling of sentiment in social media posts and venue format.
- The content of the post and sentiment towards the brand affects both processes.
- Common monitoring approaches can lead to misleading brand sentiment metrics.
- Model-based measure of brand sentiment serves as a leading indicator of external performance measures

(Appel et al., 2020)

Social media:

- “a collection of software-based digital technologies—usually presented as apps and websites—that provide users with digital environments in which they can send and receive digital content or information over some type of online social network” (p. 80)
- “social media to be a technology-centric—but not entirely technological—ecosystem in which a diverse and complex set of behaviors, interactions,

and exchanges involving various kinds of interconnected actors (individuals and firms, organizations, and institutions) can occur" (p. 80)

Social media at present

1. Platform: major and minor, established and emerging the at providing the underlying tech and business model

2. Use cases: How various kinds of people and orgs are using this tech and for what purpose

This paper categorizes social media as

1. Communication between known others (e.g., family friends)
2. Communication between unknown but share common interests
3. Access and contribute to digital content (e.g., news, gossip, and product reviews)

Future of social media in marketing

1. Immediate future:
    1. Omni-social presence: sharing is embedded on every social platform, embedded in every digital tool. Platforms broaden their platform to encompass the Omni-social world. Affect consumer decision-making process (beginning to end). Social media is shaping the cultural world.
    2. The rise of influencers: increase accessibility and appeal of social media. Micro influencer (more authentic and credible), virtual influencer (i.e., non-human, CGI).
    3. Privacy concerns on social media: no hard privacy definition., more and more people delete social media.
2. Near future
    1. Combating loneliness and isolation: on the rise
    2. Integrated customer care: social-media-based customer care., reduced customer service using humans.
    3. Social media as a political tool:
3. Far future
    1. Increased sensory richness: AR, VR
    2. Online, offline integration and complete convergence: omni channel approach (e.g., mobile app to do AR) online and offline self (self presentation)
    3. Social media by non-humans: AI, social bots, hard to do attribution model, buy likes and shares, erode consumer trust,

(Kietzmann et al., 2011)

Seven functional building blocks:

1. Identity: self-disclosure
2. Conversation
3. Sharing

4. Presence
5. Relationships
6. Reputation: not only people but also content
7. Groups

Differences Matter: the 4 Cs the 4 Cs: cognize, congruity, curate, and chase - relate how firms should develop strategies for monitoring, understanding, and responding to different social media activities

- Cognize: firms should recognize and understand their social media landscape
- Congruity: develop strategies that are congruent with different social media functionality
- Curate: firms should be the curator of social media interactions and content.
- Chase: firms should understand the conversation and info flow of social media content.

(Srinivasan et al., 2015)consumer journey (know-feel-do pathway)

- Studied low-involvement product (fast moving consumer good)
- Earned media drive sales
- higher consumer activity on earned and owned media lead to consumer disengagement (e.g., unlikes)
- On top of traditional marketing sales drivers (distribution, price), earned and owned social media can also explain part of the path to purchase.

(Zhang et al., 2017) Correlation between online shopping and social media

- In the long run, social media usage is positively correlated with shopping activity
- In the short run, immediately after social media usage, online shopping activity is lower.

(Buratti et al., 2018)

- SMM is an accessible and low-cost way to create a competitive advantage even in conservative sectors - B2B services (e.g., tanker shipping companies and ocean carriers).

(Kärkkäinen et al., 2010)

- B2B businesses utilized social media less than B2C businesses.
- The front-end phase of the new product development (NPD) process and the launch/commercialization phase provide the highest social media innovation potential for B2B businesses.

(Herhausen et al., 2022)

- Active listening and empathy in the firm's response evokes gratitude in high-arousal customers, even if the actual failure is not (yet) recovered.

(Colicev et al., 2018) Owned Social Media (OSM) vs. Earned Social Media (ESM)

- The volume of ESM engagement influences brand awareness and purchase intent, but not customer satisfaction.
- Positive and negative valence of the ESM has the greatest influence on customer satisfaction.
- OSM boosts brand recognition and customer happiness, but not purchase intent, demonstrating its nonlinear influence.
- OSM enhances purchase intent for high participation utilitarian brands and for brands with higher repute, suggesting that socially responsible business practices provide OSM greater legitimacy.
- Purchase intent and customer satisfaction impact shareholder value positively.

Definitions:

- Brand-controlled social media: Owned social media (OSM)
- Social media exposure via voluntary, user-generated brand mentions, recommendation, and so on that are not directly generate or control by a company is earned social media (ESM)

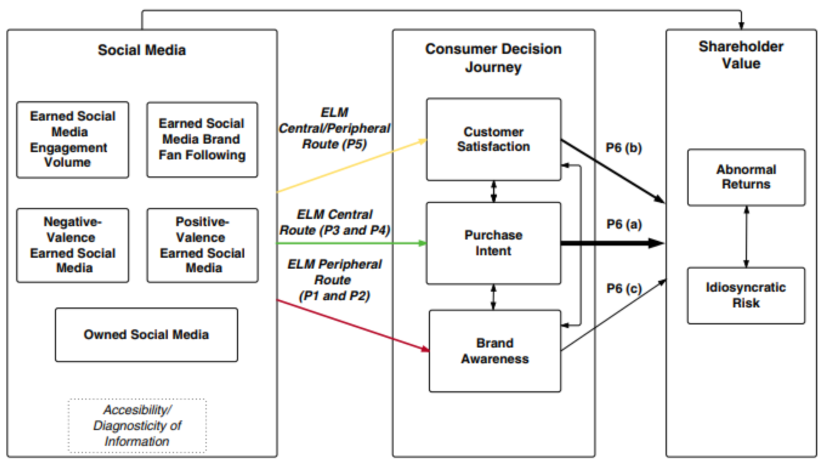
Background

- Research (see table 1 - p. 38) has shown positive impact of social media on consumer mindset metrics (e.g., brand awareness, purchase intent, customer satisfaction) which lead to higher firm performance, but no explicit mediation path.

Data

- Sampling steps
  - obtain detailed OSM and ESM data from a third-party data provider
  - obtain data on consumer mindset metrics
  - Sample brands that follow a coporate branding strategy
  - brand must be listed on the US stock exchange.
- Social Media Measures
  - OSM: From Facebook, Twitter, but not YouTube.
  - Earned Social Media:

FIGURE 1
Conceptual Framework



Notes: The different colors of lines represent the route to persuasion: red = peripheral, green = central, and yellow = mix of peripheral/central. (In the print edition, black = peripheral, gray = central, and dashed = mix.) Thicker (thinner) lines represent higher (lower) effects of CDJ stages on shareholder value.

Figure 9.1: p. 40

TABLE 2
ELM and the Effects of Owned and Earned Social Media on the Stages of CDJ

	Brand Awareness	Purchase Intent	Customer Satisfaction
Consumer motivation to process information	Low	High	Medium
Route to persuasion	Peripheral	Central	Between peripheral and central
Factors that impact consumer persuasion	Accessibility of information	Accessibility and diagnosticity of information	Accessibility and diagnosticity of information
Proposed underlying mechanism of social media impact	Consumers are more affected by the amount and virality of brand-related information than by the actual content of such information.	Consumers are more affected by actual content of the information, which is more diagnostic and requires higher motivation.	Consumers are affected by both the amount of information to reduce cognitive dissonance and by the actual content of similar experiences by other consumers.
Supporting literature	Keller (1993); Nedungadi and Hutchinson (1985)	Cacioppo and Petty (1981); Herr, Kardes, and Kim (1991)	Adams (1961); Festinger (1957); Ma, Sun, and Kekre (2015)
Proposed main drivers within OSM and ESM	ENG volume, BFF, and OSM	Positive- and negative-valence ESM	Both OSM and ESM
Propositions supported	• P_{1a} ✓ • P_{1b} ✓ • P_{1c} ✓ • P_2 ✓	• P_{3a} ✗ • P_{3b} ✗ • P_{4a} ✓ • P_{4b} ✓ • P_{4c} ✓	• P_5 ✓

Figure 9.2: p. 41

    * Brand fan following: Cumulative daily numbers of FB likes, Twitter followers, YouTube subscribers.
    * ENG volume: Daily cumulative number of people talking about this (PTAT) on Facebook, Twitter user re-tweets, YouTube video views
    * positive and negative-valance: based on a composite volume - valence metric that captures the number and popularity of the user posts based on naive Bayes algorithms.
  - Apply factor analysis with Varimax rotation on all metrics within each construct to obtain a one-factor solution for each.
  - Consumer mindset metrics from YouGov.
  - Missing product and brand level heterogeneity (in the Web appendix mentioned that brand's rating on hedonic-utilitarian scale and product purchase involvement), but likely used human coders following (Bart et al., 2014)

Methodology

- Persistence-modeling framework (Dekimpe and Hanssens, 1995)

(Trusov et al., 2009b)

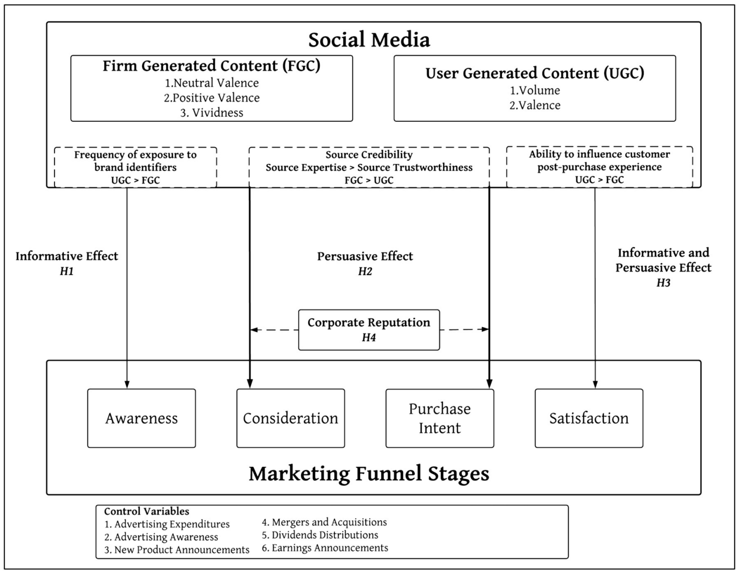
- WOM referrals have stronger and longer carryover effects than traditional marketing
- WOM referrals result in higher response elasticities
- Monetary value of WOM referrals can be calculated through revenue from advertising impressions
- This calculation provides an upper-bound estimate for financial incentives a company may offer to encourage WOM.

9.3 Who-Generated Content

Observational and experimental work on Negative WOM (Sen and Lerman, 2007)

Differences between Firm and User-generated content can be found in Colicev et al. (2019)

Colicev et al. (2019) propose a framework of how FGC and UGC (based on information processing and source credibility) match on marketing funnel stages and related studies. UGC has a strong correlation with **awareness** and **satisfaction** while FGC (specifically vividness) are more effective for **consideration** and purchase intent. While UGC valence dominates UGC volume for these stages. Brands with higher corporate reputation have stronger relationships between dimensions of FGC and the marketing funnel stages.



(Hewett et al., 2016)

- Brand Buzz in the Echoverse
- Echoverse: "the entire communication environment in which a brand/firm operates, with actors contributing and being influenced by each other's actions." (p. 1)
  - Actors
    - Firms: advertising, press releases
    - Consumers: online WOM, attitudes, and behaviors
  - Sources of actions: firms, news media, online WOM, consumer sentiment, and business outcomes
- Data: longitudinal US financial services
- Use: Vector Autoregressive Model with Granger causality
- Can view table 1 for review on echoverse components
- Symmetric echo between traditional media and online WOM.
- Firm communications influence the traditional media, consumer sentiment, business outcomes (e.g., deposits)
- Negative news spiral: Bad news can spread fast
  - Solution: Twitter and press releases can save the day

- Advertising do not directly affect traditional media, online WOM, or consumer sentiment, but directly affect firm outcome.
- Firms care more about online WOM as compared to consumer sentiment (measured by survey)
- Online WOM can affect consumer sentiment, but not the other way around.
- Online WOM affects business outcome., but not the other way around.
- Twitter gone from positive spiral to negative spiral overtime.
- Even though firms counteract negative WOM with positive tweets, the volume of their tweets stay limited.
- At first, Twitter has no impact on business outcome, but later they do have more significant impact on firm performance (p. 15)
- Wells Fargo extremely positive tone backfire in negative firestorm, while consistent (i.e., large volume) of moderate tweets are better.

9.4 Images

- related to 8.3.3.2

Potential Variables:

- Structural Complexity: complexity due to visual features (Pieters et al., 2010). Measured by the Canny Edge Detection technique (Forsythe et al., 2003), because edge information is highly correlated with observed complexity
- Color Complexity: complexity due to variety of colors within an image (Reinecke et al., 2013). Measured by the color variety algorithm (Bing Zhou et al., 2015)
  - $C = \sum_{i=0}^{m} n_i \log(\frac{n_i}{N})$ where
    * m is the number of distinct colors
    * n_i the number of pixels of the i-th color
    * N is the total number of pixels
- Image Permanence: how adept images are at remaining in users' memory (after users are no longer looking at them) (Khosla et al., 2012, 2015). Measured based on AMNet memorability index (Fajtl et al., 2018), which has been verified by (Leyva and Sanchez, 2021) (achieved near human consistency in predicting image memorability).
- Number of faces: based on Google Vision API

- Aspect ratio: height divided by width
- Average HSV value: Average HSV values across all pixels within an image
- Themes: LDA based on keywords (labels) from Google Vision API

Pieters et al. (2007) found that optimal feature design of ads (e.g., brand, text, pictorial, price, and promotion) can be achieved without increased costs. Moreover, the authors also propose two entropy-based measures of clutter effects, where they characterize the salience of feature ads based on Attention Engagement Theory.

Wedel and Kannan (2016) is a review on marketing analytics for data-rich environments. And competition for attention in these environments are more fierce.

Li and Xie (2019) found a significant and robust positive mere presence effect of image content on user engagement in both product categories on Twitter, and high-quality and professionally shot pictures consistently lead to higher engagement on both platforms (e.g., Twitter and Instagram) for both product categories. However, the effect of colorfulness varies by product category, while the presence of human face and image–text fit can induce higher user engagement on Twitter but not on Instagram. And the fit between the image and the accompanying message matters

Jalali and Papatla (2016) study visual UGC. Where color compositions were operationalized as combinations level of hue, chroma, and brightness. Consumer engagement (i.e., click rate) is higher for photos that have higher proportions of green and lower proportions of red and cyan, as well as higher chroma of red and blue.

9.5 Mobile and Smartphone

(Rangone and Renga, 2006) offers a framework in the domain mobile advertising

(Boyd et al., 2019) Influence of Branded Mobile Apps on Firm Value

- **Context:**
  - Increasing trend of firms launching branded mobile apps.
  - Limited knowledge on how these apps influence firm value.
- **Research Method:**
  - Used stock market returns to measure firm value.
  - Assessed the impact of announcements related to branded mobile app launches on firm value.
- **Research Focus:**

  - Investigate the influence of mobile app design on firm value, considering the various touchpoints apps introduce in the customer journey.
  - Examined app features emphasizing:
    1. Peer-to-peer interactions regarding the brand.
    2. Personal-oriented interactions between the customer and the brand.
    3. The purchase phase.

- **Key Findings:**
  1. Announcing the launch of a mobile app positively impacts firm value.
  2. The features highlighted in the app design significantly influence this value creation.

(Cao et al., 2023) Monetizing Free Mobile Apps

Objective:

- Address the challenges faced by non-advertising-based mobile apps when attempting to monetize free services.
- Investigate the effectiveness of different pricing strategies and aspects of product design.

Strategies Evaluated:

1. **Pricing Strategies:**
   - **Hard Landing:** A “pay or churn” approach where users face a paywall.
   - **Soft Landing:** Users continue to access limited free services even after monetization.
2. **Product Design:**
   - Decision to provide exclusive secondary offerings to those users who subscribe.

Methodology:

- A large-scale randomized field experiment with a mobile app firm.
- Assessments of user behavior in response to implemented strategies.
- Customer survey and a separate experiment on the Prolific platform to validate the mechanisms at play.
- A follow-up field experiment to test generalizability.

Key Findings:

1. **Soft Landing:** This approach reduced the willingness of existing users to subscribe.

2. **Exclusive Secondary Offerings:** Offering exclusive benefits to paying users also decreased the willingness to subscribe.

3. **Positive Interaction:** There's a beneficial interaction between the soft landing approach and exclusive secondary offerings, which positively influences subscriptions.

4. **Generalizability:** The results from the second field experiment were consistent with the primary experiment.

Theoretical Mechanisms:

- The findings suggest that users might perceive soft landing strategies as a less aggressive monetization method, which might reduce the immediate perceived need to subscribe.

- Exclusive offerings, while appealing, might create a sense of divide among users. However, when paired with the soft landing strategy, users might view it as a fair trade-off, leading to increased subscriptions.

Managerial Implications:

- While both soft landing and exclusive secondary offerings individually deter subscriptions, their combined application may be beneficial.

- Firms should consider the interplay between pricing strategies and product design features when planning monetization.

Melumad and Pham (2020) found that consumers derive not only function benefits, but also **emotional benefits** from their smartphones. Under stress, consumers want their smartphones, because smartphones give users greater stress relief. Moreover, this pacifying effect is greater for one's own smartphone than other's smartphones.

9.6 Review

Finkelstein and Fishbach (2012) found that as people gain more expertise, they are more likely to respond to negative reviews (i.e., feedback), which is robust for many domains (e.g., language acquisition, environment, marketing). novices are more likely to respond to positive feedback while experts seek and respond to negative feedback

(Büschken and Allenby, 2016) Sentence-based text analysis for customer reviews

- Challenge in analyzing unstructured consumer reviews: making sense of topics expressed

- Proposed a new model for text analysis using sentence structure in the reviews
- Model leads to improved inference and prediction of consumer ratings
- Sentence-based topics found to be more distinguished and coherent than word-based analysis
- Data used from www.expedia.com and www.we8there.com to support findings.

Folse et al. (2016) defines negatively valenced emotional expressions as having intense language, all caps, exclamation points, emoticons in online reviews. Reviews with negatively valenced emotions are viewed as more helpful and damage attitude toward the product when used by experts, but when sued by novices, a negative self-reflection is observed and attitude towards the product is unchanged. Language complexity moderates the adverse effect of expertise on trustworthiness.

(Yazdani et al., 2018) Effect of Reviews by Rank on Product Sales

- **Objective**:
  - Examine how reviews by top-ranked and bottom-ranked reviewers influence product sales.
- **Data Source**:
  - Sales data from 182 new music albums released over approximately three months.
  - User review data sourced from Amazon.com.
- **Methodology**:
  - Use of instrumental variables to account for potential confounding factors in the measurement of online word-of-mouth impacts.
- **Key Findings**:
  1. **Greater Impact by Bottom-Ranked Reviewers**: Reviews from bottom-ranked reviewers have a more pronounced effect on sales than those from top-ranked reviewers.
  2. **Influence of Top-Ranked Reviewers**: While they can act as opinion leaders, their effect on sales is predominantly seen in specific cases such as:
     - New product releases.
     - Products with high variability in existing reviews.
  3. **Driving Factors for Differences in Influence**: The disparity in the influence of top- and bottom-ranked reviewers can be attributed to:

    - **Content**: The nature and quality of the review written.
    - **Identity**: The credibility and reputation of the reviewer.

  4. **Robustness of Results**: The findings remain consistent across:
    - Different product categories like music albums and cameras.
    - Various metrics like sales and sales rank.

- **Implications**:
  - **For Businesses**: It's crucial for businesses to understand that not all reviews have equal impact. While top-ranked reviewers can have authority, it's the bottom-ranked reviewers that can potentially sway a larger audience.
  - **For Consumers**: When making purchasing decisions, consumers should consider the content of reviews and not just the rank of the reviewer.
  - **For Platforms**: Online platforms might consider re-evaluating their reviewer ranking systems or highlighting reviews that can provide the most accurate and influential information to potential buyers.

Dai et al. (2019)

- Based on Amazon reviews and experiments, Consumer reviews are less likely to be trusted for experiential purchases than they are for material ones.
- This impact stems from the idea that ratings of experiences are less representative of the purchase's objective quality than reviews of actual objects. These findings reveal not only how **word of mouth** influences different types of purchases, but also the psychological mechanisms that underpin customers' reliance on consumer reviews. Furthermore, these findings imply that people are less responsive to being instructed what to do than what to have, as one of the first examinations into how people choose among numerous experience and material buying possibilities. (Wang et al., 2021)
- extract and monitor products and attributes info from consumer reviews using machine learning and NLP (to create embedded representation)
  - In customer evaluations, embedded representation characterizes (represents) textual data like particular product features by employing the words that surround such textual data (i.e., the contextual information). Neural networks are used to measure how similar different product attributes are based on what people say about them (i.e., contextual information), which shows similarities and differences in how people use the attributes. From this embedded representation,

the model then picks out multi-level clusters of product attributes that show how abstract the product benefits are at different levels.

- Close the gap between engineered attributes (i.e., concrete attributes) and meta-attributes (abstract attributes)
- Survey can be inconsistent and time consuming.

(Sunder et al., 2019) Drivers of Herding Behavior in Online Ratings

- **Objective:** Investigate how herding effects, driven by reference groups (crowd and friends), impact online ratings.
- **Background:** While post-purchase evaluations are known to influence sales, the nuances of herding in online ratings remain underexplored.
- **Key Insights:**
  1. **Herding Significance:** Herding effects in online ratings are substantial, calling for a detailed understanding of its dynamics.
  2. **Rater Experience:** As raters become more experienced, the impact of the crowd diminishes, while friends' influences grow.
  3. **Divergent Opinions:** Differences in opinions among reference groups lead to varied herding effects based on the reference group and the rater's experience.
  4. **Firm Product Portfolio's Role:** A diverse product range not only boosts perceived quality but also lessens the sway of social influence on ratings.

Nguyen et al. (2020) found that more expertise leads to less extreme evaluations. Reviewing experts have less impact on a brand valence metric (which affects page rank and consumer evaluation). And experts do benefit and harm service providers with their ratings. Hence, excellent experiences may lead to lower ratings from experts (than from novices)

Observational data show that experts are more likely to post negative reviews.

(Schoenmueller et al., 2020) found Polarity self-selection of online reviews and it reduces the informativeness of online reviews

(Banerjee et al., 2021) Q&A section (answered by other consumers and sellers) improve fit and match between products and consumers and reduce negative reviews regarding mismatch.

(Nishijima et al., 2021) interestingly found that Rotten Tomatoes ratings has no affect on box office performance using the categorization of "fresh" or "rotten." They obtain box office performance data from boxofficemojo

(Park et al., 2021) The Fateful First Consumer Review

- valence and volume are not independent

- Positive first review can create long-term advantage in future WOM valance and volume, while negative first reveiw suffers long-term disadvantage in future WOM valance and volume.
- Because of information-availability bias, consumers are entrenched with either positive or negative first reviews which renders difficulty for firms to correct their first negative review.

(Hoskins et al., 2021) Online Review Ratings: Differences Between Niche and Mainstream Brands

- **Objective**:
  - Explore the differences in the drivers of online review ratings between niche and mainstream brands.
- **Data Source**:
  - A unique dataset on the U.S. beer product category.
- **Key Factors Examined**:
  1. **Customer Review Valence**: The overall positive or negative sentiment of customer reviews.
  2. **Professional Critics Review Valence**: Sentiment analysis of reviews by professional critics.
  3. **Community Characteristics**: The nature and characteristics of the online community reviewing the product.
  4. **Location Similarity**: How similar or close in location a reviewer is to a brand or other reviewers.
  5. **Reviewer Characteristics**: Traits, behaviors, and preferences of individual reviewers.
- **Major Findings**:
  1. **Niche Brand Influence**: Niche brands are generally more affected by Online Word of Mouth (OWOM) because consumers typically have less established brand awareness and pre-formed brand imagery.
  2. **Local Preference for Niche**: Reviewers tend to rate a local niche brand more favorably compared to non-local niche brands.
  3. **Professional Critics vs. Online Community**: For the average reviewer, the online community's sentiment has a more profound influence than professional critics.
  4. **Influence of Prior Reviews**: A review from the online community gains more traction when:
     - The reviewer's expertise is high.

    - The prior reviewer shares geographic traits with the subsequent reviewer.

5. **Alignment of Reviewer Sentiments**:
    - Reviewers engaging more with products/brands tend to have sentiments that align with professional critics.
    - Reviewers engaging more with the online community tend to resonate with that community's sentiment.

(Li et al., 2022) Online Reviews on Intergenerational Product Sales

- **Objective**:
    - Investigate how online customer reviews of one product generation influence the sales of another generation within the same product series.
- **Data Source**:
    - Data from intergenerational pairs of point-and-shoot cameras sold on Amazon.com.
- **Methodology**:
    - Joint estimation of the current and previous generation models, with errors clustered at both daily and product levels.
    - Use of instrumental variables to address potential endogeneity concerns related to online word-of-mouth measures.
- **Key Findings**:
    1. **Positive Influence of Previous Generation Reviews**: The valence (or tone) of reviews for the previous product generation positively impacts the sales of the current generation.
    2. **Negative Impact on Previous Generation Sales**: Interestingly, the valence of current generation reviews negatively affects the sales of the previous generation.
    3. **Factors Amplifying the Impact of Previous Generation Valence**: The positive effect of previous generation valence on current generation sales intensifies when:
        - Uncertainty (as measured by the standard deviation) in reviews for the current generation is high.
        - The current generation product receives favorable reviews (high valence).
    4. **Factors Mitigating the Impact of Previous Generation Valence**: The positive effect weakens when:

    - There's higher uncertainty in reviews for the previous generation.
    - The current generation product has been available in the market for a more extended period.

- **Implications**:
    - **For Marketers**: The legacy of a product series plays a crucial role in shaping consumer perceptions and sales of newer versions. Ensuring consistent quality and addressing issues in earlier generations can bolster the success of future releases.
    - **For Online Retailers**: Highlighting positive reviews of previous generations, especially when there's uncertainty around a newer product, can help boost sales.
    - **For Consumers**: Evaluating reviews of earlier versions of a product can provide valuable insights into the expected performance and reliability of the current generation.

(Ordabayeva et al., 2022)

- Negative internet reviews from socially distant (but not socially close) individuals may not be as harmful to identity-relevant brands. Because a negative review of an identity-relevant brand can threaten a client's identity, the consumer will seek to strengthen their relationship with the brand.
- They show that this effect does not appear when the review is positive or when the brand is irrelevant.

(Chen et al., 2022) Order between rating and tipping matters

- If customers rate a service professional before tipping, they will tip less
- If customers tip before rating, the tipping amount is unchanged.
- This negative effect is because customers think that they reward service providers already by reviewing, so they only need to tip a smaller amount
- This negative effect is more pronounced when customers
    - tip from their own pocket
    - have higher categorization flexibility (i.e., considering rating as reward like tipping)
    - think service professionals benefit from rating
- To mitigate this effect, service professionals can highlight the consistency motivation between the rating and tipping sequence.

(He et al., 2022)

- There is a big online market where fake online reviews are sold and bought.

- Fake internet reviews do help product vendors get better ratings and make more sales.
- Most big companies don't buy fake reviews.
- Online marketplaces try to stop fake reviews, but they can't always do it right away.
- Fake reviews (bought on Facebook private groups) is correlated with short-term increase in average rating and number of reviews
- When firms stop purchasing fake views, their average ratings decrease (due to share of one-star reviews), especially for young and low-quality products.

9.7 Slacktivism

Supporting a political or social cause through social media or online petitions, with minimal work or commitment.

9.8 e-Marketplace

(Dholakia and Rego, 1998) Analyzing Marketing Information and Hit-Rates on Commercial Home-Pages

Objectives:

1. To comprehensively and rigorously document the types and nature of marketing information presented on commercial home-pages, aiming to uncover the primary objectives of prevalent commercial websites.
2. To empirically identify key factors of commercial home-pages that contribute to higher visits or hit-rates.

Framework & Methodology:

- The study draws inspiration from Resnik and Stern's "information content" paradigm to assess the informativeness of commercial home pages.
- This approach aims to evaluate how much relevant and valuable information is being provided to visitors of these home pages.

Potential Findings and Implications:

1. **Nature of Information:** The research may unveil specific types of marketing information that are predominant on commercial websites, which can shed light on the marketing priorities and strategies of companies on the web.

2. **Informativeness Evaluation:** By using the "information content" paradigm, it's possible to determine how beneficial and valuable commercial home-pages are from a user's perspective.

3. **Key Determinants of Hit-Rates:** By understanding which factors lead to increased visits, companies can optimize their websites to attract more visitors. This could include insights on website design, content prioritization, user experience, interactive elements, and more.

4. **Value to Stakeholders:** The findings are particularly beneficial for:

    - **Web Page Designers:** Insights can inform best practices in design and content placement to drive user engagement and hit-rates.

    - **Businesses:** Especially for those looking to enhance their online presence, understanding what resonates with online audiences can be pivotal in shaping their online strategies.

(Burke, 2002) Consumer Value of Technology in Shopping

Objective:

- To comprehend the value consumers attribute to innovations in customer interface technology during the shopping process.

Methodology:

- A large-scale national survey involving 2,120 online consumers.

- Exploration of consumer preferences in both online and brick-and-mortar shopping contexts.

- Examination of 128 varied facets of the shopping journey, ranging from traditional elements to novel technological innovations.

Key Findings:

1. **General Satisfaction**:

    - **High Satisfaction**: Consumers generally find current retail offerings satisfactory in terms of convenience, product quality, selection, and perceived value for money.

    - **Areas of Dissatisfaction**:

        - **Service Levels**: Consumers are not entirely pleased with the quality of service they receive.

        - **Product Information**: There's a felt need for better, more detailed product information.

        - **Shopping Speed**: Consumers desire a more swift and seamless shopping experience.

2. **Potential of New Technologies**:

- **Enhancing Experience**: Emerging technologies have the capability to augment the overall shopping experience.
- **Customization is Crucial**: However, the deployment of these technologies should be strategically tailored to:
  - Cater to distinct consumer segments.
  - Address the specific needs and dynamics of various product categories.

3. **Synergy of Media**: The study hints at the combined strength of both interactive and conventional media in steering consumers through their buying journey.

(Rohm and Swaminathan, 2004)

- based on shopping motives (e.g., convenience, physical store orientation, information use in planning and shopping, variety seeking), there are 4 types of shoppers
  - convenience shoppers
  - variety seekers
  - balanced buyers
  - store-orientated shoppers

(Montgomery et al., 2004) Modeling Online Browsing and Path Analysis Using Clickstream Data

- **Background:**
  - Clickstream data offers insights into users' navigation patterns on websites by capturing the sequence or path of pages they view.
- **Research Objective:**
  - This study aims to classify and model path information using a dynamic multinomial probit model of Web browsing.
  - The goal is to ascertain if and how path data can enhance predictions about users' future moves on a website.
- **Methodology:**
  - The research utilized data from a prominent online bookseller.
  - The dynamic multinomial probit model was compared against traditional multinomial probit models and first-order Markov models to gauge its predictive accuracy.
- **Key Findings:**

  - The memory component in the dynamic model is vital for accurately predicting users' paths.
  - Traditional models (multinomial probit and first-order Markov) fall short in predicting paths effectively.
  - The results hint that users' paths may be indicative of their goals on the website, which could aid in forecasting their subsequent movements.
  - When applied to anticipate purchase conversions, the dynamic model demonstrates that purchasers can be predicted with over 40% accuracy after just six page viewings. This is notably superior to the baseline 7% purchase conversion prediction rate achieved without leveraging path information.

- **Relevance:**
  - The findings underscore the potential of clickstream data in personalizing web designs and product suggestions based on users' navigation paths.
  - Businesses can benefit from this approach by offering a more tailored online experience, potentially increasing conversion rates.

(Li et al., 2005) Cross-Selling Sequentially Ordered Products: in Consumer Banking Services

- **Premise:**
  - Customers' buying behaviors follow predictable life cycles. Recognizing these life cycles, firms observe that some products or services are typically bought before others.
- **Research Objective:**
  - This study aims to understand how customer demand for multiple products changes over time and the resulting patterns of sequential product acquisitions when these products have a natural order.
  - The objective is to utilize these insights to enhance cross-selling strategies for firms.
- **Methodology:**
  - A structural multivariate probit model was employed to study customer purchasing behaviors.
  - The research used data from a major bank in the Midwest, focusing on customer purchase patterns of its marketed products.
- **Key Findings:**

    - There exists a discernible pattern in which customers buy certain bank products before moving on to others.
    - Gender and age play roles in influencing purchase decisions:
        * Women and older customers are more influenced by their overall satisfaction with the bank when deciding on acquiring additional services than younger customers and men.
        * Households led by more educated individuals or males tend to progress more rapidly along the financial maturity continuum compared to those led by less educated individuals or females.
- **Implications:**
    - By understanding these buying patterns, firms can tailor their cross-selling strategies more effectively.
    - Recognizing the significance of satisfaction for certain demographic groups (like women and older customers) can help in customizing service delivery to retain and upsell to these segments.
    - Addressing the diverse financial maturity progression rates across demographics allows for better timing and targeting of product offerings.
- **Overall Importance:**
    - Recognizing and acting upon the sequential purchasing behaviors of customers, based on their life cycles, can enhance a firm's ability to successfully cross-sell, ultimately boosting revenue and customer loyalty.

(Li et al., 2009) Internet Auction Features as Quality Signals

- **Background:**
    - Due to the inherent risks and uncertainty of online auctions, companies have introduced tools to help sellers showcase their credibility and the quality of their products to tackle the "lemons" problem (where buyers risk purchasing low-quality or misrepresented goods).
- **Research Objective:**
    - The authors aim to:
        1. Develop a categorization for Internet auction quality and credibility markers based on signaling and auction theories.
        2. Modify and employ Park and Bradlow's 2005 model to study the effectiveness of these indicators on eBay.
        3. Analyze how these indicators influence consumer participation and bidding decisions.

- **Empirical Findings:**
  - The research investigates how certain auction features affect user engagement and choices.
  - It delves into factors that can influence the perceived reliability of these quality markers and also studies the varied responses of different buyer segments to these indicators.
- **Key Contributions:**
  - The research confirms the importance of signaling in influencing consumers' bidding behavior in online auctions.
  - It is the first empirical study of its kind to assess the role of extensive Internet auction institutional tools in tackling the adverse selection issue, enhancing the understanding of the economic rationale behind novel online auction designs.
  - The findings underline the significance of credible signaling in online auctions. By understanding and leveraging these indicators, auction platforms and sellers can instill greater trust, encourage participation, and optimize bidding outcomes.

(Li et al., 2011) Cross-Selling the right Product to the right Customer at the right time

- **Challenge:**
  - Companies seek to enhance the success rate of their cross-selling campaigns.
- **Research Objective:**
  - The authors aim to formulate a customer-response model addressing:
    1. The evolving customer demand for diverse products.
    2. Multiple potential roles of cross-selling solicitations, such as for promotion, advertising, and education.
    3. Varied customer preferences concerning communication channels.
- **Methodological Approach:**
  - The authors employ a stochastic dynamic programming framework.
  - This approach is designed to maximize long-term profits from existing customers by considering the progressive nature of customer demand and the varied functions of cross-selling promotion.
  - The end goal of the model is to determine the best strategies for presenting the right product to the apt customer at the optimal time through the preferred communication channel.

- **Key Insights from a National Bank Data:**
  - Households display different preferences and responsiveness towards cross-selling solicitations.
  - Beyond immediate sales, cross-selling solicitations accelerate households' progression along the financial maturity curve (indicating an educational role) and also foster goodwill (indicating an advertising role).
  - A breakdown analysis reveals that the educational role of solicitations (83%) far outweighs its advertising (15%) and instant promotional (2%) roles.
- **Model's Efficacy and Benefits:**
  - Applying the formulated model leads to tailor-made and dynamic cross-selling campaigns.
  - The outcomes include:
    1. A 56% boost in immediate response rate.
    2. A 149% increase in long-term response rate.
    3. A significant 177% improvement in long-term profitability.
- **Overall Implication:**
  - Adopting the proposed framework enables firms to better understand and cater to their customers, significantly enhancing the efficiency and impact of their cross-selling campaigns. The educational aspect, in particular, plays a dominant role in influencing customer behavior and driving long-term value.

(Zhang et al., 2014) Social Influence on Shopper Behavior Using Video Tracking Data

- **Objective:**
  - The study delves into understanding the role social factors play during a retail shopping experience and how these factors influence a shopper's likelihood to interact with products and make a purchase.
- **Methodology:**
  - A bivariate model is utilized to examine both the behavioral drivers leading to product interaction and purchase simultaneously. This model, implemented within a hierarchical Bayes framework, takes into account both individual shopper characteristics and the broader shopping context.
  - An innovative video tracking system is used to monitor each shopper's journey and actions throughout the store.

- **Key Findings:**
  1. Direct social interactions, such as those with salespeople or conversations with other shoppers, lead to:
     - Longer shopping durations.
     - Enhanced product interaction.
     - An increased probability of making a purchase.
  2. When shoppers are in larger groups, their buying behaviors are more influenced by discussions with their group members than interactions with strangers.
  3. A store that has a moderate number of customers encourages more product interaction. However, beyond a certain threshold of crowd density, the shopping experience becomes less enjoyable and efficient.
  4. The impact of social influences is also determined by factors like:
     - The similarity in demographics between a salesperson and the shopper.
     - The specific type of product category being considered.
  5. There are particular behavioral signs that indicate when a shopper might be more inclined to make a purchase.
- **Implication:**
  - Understanding these social dynamics can help retailers strategize their in-store experiences. By optimizing salesperson interactions, understanding group dynamics, and managing store traffic, retailers can enhance the shopping experience and potentially boost sales.

(Ding et al., 2015) Learning User Real-Time Intent for Optimal Dynamic Web Page Transformation

- **Objective:**
  - The study aims to explore how understanding and adapting to online users' real-time intentions and cart choices can help e-commerce websites increase their conversion rate of visitors into buyers.
- **Methodology:**
  - A dynamic model is introduced which simultaneously learns a user's real-time, unobserved intent from their cart actions and then instantaneously adapts the website page accordingly to improve the chances of a sale.
  - The model tailors its strategies based on:
    1. Individual user's browsing behavior.

    2. Effectiveness of different marketing and web prompts.
    3. Users' comparison shopping activities on other sites.

  - The approach involves not only learning from the user's behavior but also implementing optimal web page changes in real time.
  - Data was sourced from an online retailer and a controlled lab experiment to validate the model's efficacy.

- **Key Findings:**
  1. Real-time learning and adapting to a user's unobserved purchasing intent can significantly lower shopping cart abandonment rates and boost purchase conversions.
  2. By starting optimal page adaptations based on intent after the user's first page view, the shopping cart abandonment rate drops by 32.4%, and the purchase conversion rate rises by 6.9%.
  3. The most efficient time to intervene and make web adaptations is after a user has viewed three pages. This balance ensures that the site has learned enough about the user's intent while also intervening early enough to influence the purchase decision.

- **Implications:**
  - E-commerce platforms can benefit from implementing dynamic models that not only understand but also react in real time to users' purchase intentions. By doing so, they can significantly improve their chances of converting visitors into buyers, optimizing their user experience, and boosting sales.

(Mallapragada et al., 2016) Impact of Product and Website Context on Online Shopping Basket Value

- **Background:**
  - Managing consumer relationships and understanding the determinants of online shopping behavior pose challenges given the numerous influencing factors.
  - The factors of interest include the nature of the product being shopped for (the "what") and the specific context or attributes of the website where shopping takes place (the "where").

- **Research Objective:**
  - This study delves into the effects of these characteristics on the value of an online transaction's basket.
  - Additionally, the study considers the influence of other facets of online browsing, such as the number of page views and the duration of the visit.

- **Methodology:**
  - A multivariate mixed-effects Type II Tobit model is used, featuring a system of equations to elucidate the variance in shopping basket value.
  - The dataset used consists of 773,262 browsing sessions that led to 9,664 transactions. These transactions span 43 product categories and come from 385 distinct websites.
- **Key Findings:**
  - Contextual factors play a crucial role in online browsing.
  - A website's product variety breadth is:
    - Positively related to both the length of visits and the basket values.
    - Negatively linked with the number of page views.
  - The ability of a website to communicate or its communication functionality elevates the basket value, particularly when shopping for hedonic or pleasure-driven products.
- **Relevance:**
  - The research offers actionable insights for online retailers, emphasizing the importance of tailoring product offerings and website strategies based on the observed influences on basket value.

(Zhang et al., 2018) Group Interaction on Shopper Behavior

- **Objective:**
  - To understand how social interactions and group discussions influence shopper behavior in retail contexts.
- **Methodology:**
  - Creation of a dynamic linear model in three stages, analyzing the effects of group discussions on shopper choices and purchases.
  - Empirical study using a unique video tracking and transaction dataset obtained from a specialty apparel store.
- **Key Findings:**
  1. **Influence on Shopper Decisions:** Group conversations profoundly impact where shoppers decide to browse (department or "zone" choice), their likelihood to make a purchase, and the amount they spend over time.

2. **Size Matters:** The size of the shopping group amplifies the influence, especially when deciding where to browse and the conversion from browsing to buying.

3. **Group Dynamics:** Group composition and how cohesive the group is (whether they stick together or split up) moderate the effects of group discussions on shopping behavior.

   – *Mixed-Age vs. Adult Groups:* Mixed-age group discussions tend to influence shopping behavior more strongly across all stages, while adult groups show a slight lingering effect on the final purchase decision.

   – *Repeated Discussions:* Shoppers who engage in multiple conversations in a specific department tend to revisit and buy from that department.

   – *Cumulative Discussions:* The more discussions held store-wide, the higher the overall spending.

4. **Comparing Group vs. Solo Shoppers:** Group shoppers tend to visit more departments than those shopping alone. Mixed-age groups and solo shoppers show a higher buying propensity than groups of just adults or teens.

- **Implications for Retailers:**

  – **Engagement Strategy:** Retailers can create zones or areas designed to promote group interactions, given the observed influence on shopping behavior.

  – **Store Layout:** Recognizing that group shoppers tend to explore more, store layouts can be optimized to encourage group wanderlust and engagement.

  – **Targeting Strategy:** Since mixed-age groups and solo shoppers are more inclined to make a purchase, promotional efforts or assistance can be tailored for these segments.

(Ding and Li, 2019) Rational Herding in Digital Goods Consumption and Purchase

- **Objective:**

  – Understand whether users display herding behavior in the consumption and purchase of serialized digital goods and identify the factors influencing this behavior.

- **Background:**

  – With the rise of digital goods, average individuals increasingly produce serialized content.

    - There's uncertainty regarding the presence and degree of herding behavior in the consumption and purchase of these digital goods.

- **Methodology:**
    - A simultaneous equations model, rooted in herding theory, was designed.
    - The model assesses two competing effects:
        1. **Private Signal Effect:** Impact of individual private signals on quality inference.
        2. **Sequential Actions Effect:** Influence of observed sequential actions of others on herding.
    - The model utilizes a hierarchical Bayes framework.
    - Data from China's leading literature website was used for empirical analysis.
- **Key Findings:**
    1. **Presence of Rational Herding:** Users display rational herding in both the consumption and purchase of digital books on the platform.
    2. **Herding in Purchase:** The tendency to herd is notably stronger when purchasing than when merely consuming.
    3. **Factors Affecting Herding:**
        - **Product Features:** These reduce the herding bias.
        - **Reputation of the Producer:** Amplifies the herding effect.
        - **Competition:** Intensifies the herding behavior.

(Costello and Reczek, 2020)

- P2P platforms (e.g., Uber, Lyft, Airbnb): mediate good and service flow between providers and consumers.
- When P2P platforms focus on providers, then consumers will increase purchase-related outcomes because of increased "empathy lens" (tendency to think about how one's purchase affects the provider), in contrast to "exchange lens" (the reciprocal exchange of money with a firm in return for a good or service).
- 4 pilot studies confirm the prediction that P2P firms are perceived to have greater provider-firm independence.
- Study 1: Field study with a P2P firm (Borrow'd)
- Study 2:

  - a. mediation step via increased perceptions that a purchase helps an individual (P2P - Reliable ride vs. traditional - Reliable Cab).
  - b. the mediation step and outcome do not occur in traditional business model (whether a service provide was an employee or independent contractor)

- Study 3: how study 1 continues to affect willingness to pay for a gift card for a P2P ride service.
- Study 4: compare the prototypical for-profit matchmaker P2P model to (a) P2P forums and (b) cooperatives.

9.9 Social Listening Platforms

According to Forrester Wave's report by Jessica Liu and Sara Dawson (2020), the top 10 social listening platforms (SLPs) are

1. Brandwatch
2. Digimind
3. Linkfluence
4. ListenFirst
5. Meltwater
6. NetBase Quid
7. Sprinklr
8. Synthesio
9. Talkwalker
10. Zignal

9.9.1 Fake Check Platform

- https://mightyscout.com/influencer-lookup
- https://socialauditor.io/
- https://analisa.io/
- https://hypeauditor.com/free-tools/instagram-audit/
- https://www.inbeat.co/fake-follower-checker/
- https://grin.co/influencer-marketing-tools/fake-influencer-tool/
- https://www.fakecheck.co/
- https://www.modash.io/fake-follower-check/

9.10 Live Streaming

(Lin et al., 2021): found that happy streamers can make audience happier and tip more. This effect is bigger for broadcaster with more experience, receive more tips, and popular.

9.11 Augmented Reality

Yim et al. (2017)

- When compared to web-based product presentations, AR provides effective communication benefits by providing better novelty, immersion, enjoyment, and utility, leading to better attitudes regarding medium and purchase intention.
- In the AR condition, immersion mediates the relationship between interactivity/vividness and two performance outcomes: usefulness and enjoyment, whereas on the web, no significant routes between interactivity and immersion, nor between previous media exposure and media novelty are found.
- Two schools define **Interactivity** (p. 91)
  - Technological outcome: focus on speed, mapping, range (i.e., the extent to which one can manipulate content).
  - Users' subjective perceptions: very much related to **motivation** (then what is the contribution of this school of thought?)
- Vividness (i.e., realness, realism, or richness) is "the ability of a technology to produce a sensorially rich mediated environment"

  $$@steuer1992, p.80$$

  (which is very similar to the mapping aspect of Interactivity, then wouldn't it be hard to operationalize these two constructs).
  - Technological perspective: vividness increases depth and breadth of the usage experience
- Model (p.94): it makes sense that media usefulness and enjoyment directly affect attitude toward medium. However, immersion could potentially directly affect purchase intention via the path of attitude toward the product.

9.12 Artificial Intelligence

(Garvey et al., 2021)

- When engaging with an AI agent, consumers respond better in terms of increased purchase likelihood and satisfaction when a product or service offer is poorer than expected.

- Consumers are more likely to respond positively to a human representative if the offer is better than expected.
- When administering offers, AI agents are regarded to have weaker intentions than human agents, which explains for this result
- Marketers may anthropomorphize AI agents to reinforce their perceived intentions, giving them a way to get credit from customers when they make a better offer and avoid blame when they make a bad one.

9.13 Search Engine

(Guan and Cutrell, 2007) found the Google golden triangle where people pay attention the the top 3 organic results and spill over to the fourth one (paid).

(Johnson et al., 2004) On the Depth and Dynamics of Online Search Behavior

- Examination of search behavior across competing e-commerce sites
- Study based on panel data from over 10,000 households and 3 products (books, CDs, air travel)
- On average, households visit 1.2 book sites, 1.3 CD sites, and 1.8 travel sites per active month
- Characterization of search behavior in terms of depth, dynamics, and activity
- Search modeled as a logarithmic process, with limited search across few sites
- Model allows for time-varying dynamics, with only mild evidence of decreasing search over time in one product category
- Results suggest more-active online shoppers tend to search across more sites, driving the dynamics of search.

(Kulkarni et al., 2012) Using online search data to forecast new product sales

- Data on consumer search terms provides valuable measures and indicators of consumer interest
- Can be useful to managers in gauging product interest in a new product launch or consumption interest post-release
- Model of pre-launch search activity developed and linked to early sales
- Model applied to motion picture industry and found to provide significant forecasting power for release week sales
- Advertising data included in model increases explanatory and forecasting power

- Managerial insights offered on how search volume data and the model can be used for new product release planning.

9.14 Customer Engagement

(Harmeling et al., 2016) visualize the mechanism of how customer engagement affects firm performance

(Kumar et al., 2010) categorize 4 components of a customer's engagement value:

- Customer lifetime value (the customer's purchase behavior)
- Customer referral value (incentivized referral of new customers)
- Customer influencer value (customer's behavior to influence other customers)
- Customer knowledge value (value added to the firm by feedback from the customer)

(Bucklin and Sismeiro, 2003)

- browsing behavior modeled form clickstream data
- visitors' propensity to browse is a function of the depth of a site's visit and the number of repeat visits.
- aggregate site metrics cannot capture the individual browsing behavior.

(Reimer et al., 2014)

- Different customer segments respond differently to marketing activities (e.g., coupon promotions, TV, radio, print, Internet ads)
- Different from consumer packaged goods, heavy users of digital music products are less sensitive to price and more sensitive to TV ads, while light users are price sensitive and more likely to opt out of targeted communication.

9.15 Wisdom of the Crowds

(Grewal et al., 2006) Network Embeddedness and Open Source Project Success

- **Background:**
  - Open source environments, where software development relies on a community-driven model, are emerging as credible alternatives to conventional, firm-based models of software development.
  - Within open source environments, the relationships (or network connections) between projects and developers play a crucial role.

- **Research Objective:**
  - This study aims to explore the concept of network embeddedness, which refers to the nature of interconnections among projects and developers.
  - The goal is to determine how various degrees and types of network embeddedness impact the success of open source projects.
- **Methodology:**
  - The research highlights the existence of significant variability in how open source projects and their managers are embedded in networks.
  - The study employs both visual depictions of the affiliation networks and rigorous statistical analysis to:
    * Demonstrate this variability.
    * Investigate the differing network structures across various projects and their managers.
  - A latent class regression analysis is applied to uncover distinct regimes or patterns within the data.
- **Key Findings:**
  - Network embeddedness exerts a profound influence on both the technical and commercial outcomes of projects. However, these influences are intricate.
  - Multiple patterns or regimes of effects are detected. In some of these regimes, network embeddedness promotes project success, while in others, it may be detrimental.
  - Project age and the number of page views offer additional insights into how network embeddedness might steer project outcomes.

(Mallapragada et al., 2012) User-generated open source products: Founder's social capital and time to product release

- **Context:**
  - Open source products are developed using collaborative Internet technologies, often by volunteer users.
  - The timeframe to product release is a key metric for the success of such projects.
  - Open source communities are often bifurcated:
    * **Developer Users:** Contribute to product development.
    * **End Users:** Collaborative testers providing feedback.
- **Core Propositions:**

- The study explores the influence of:
  * The position of project founders within the developer users' social network.
  * The dynamics between developer users and end users.
  * Certain project and product attributes.
- The central aim is to understand how these factors impact the time taken for product release.

- **Methodology:**
  - Hypotheses are developed, informed by the two-community structure of the open source space.
  - Data from 817 development projects on SourceForge, a notable open source platform, is used.
  - A split hazard model is employed to assess the hypotheses.
- **Core Findings:**
  - Results affirm the dual-community concept.
  - A founder's significant position in the developer user community can expedite product release by up to 31%.
  - Projects where end users are highly engaged tend to witness an 11% reduction in release time compared to less engaged projects.

(Mahr et al., 2015) Crowdsourcing and Problem-Solving Styles

- **Context:**
  - Surge in firms leveraging crowdsourcing platforms for innovation-related challenges.
  - Mixed results from crowdsourcing due to limited successful contributions from external experts.
- **Research Objective:**
  - Understand why certain external solvers are more successful than others in crowdsourcing contexts.
- **Theoretical Framework:**
  - **Dual-Processing Theory:** Analyzes two distinct cognitive processes in decision-making.
- **Methodology:**
  - Combination of survey and archival data.
- **Problem-Solving Styles Investigated:**

   1. **Creative Style:** Spontaneous and intuitive.
   2. **Deliberate Style:** Systematic and analytical.

- **Key Findings:**
   1. Both creative and deliberate styles can lead to successful problem-solving, but their effectiveness varies based on two key conditions:
      - **Contextual Familiarity:** Understanding of the problem's context.
      - **Time Investment:** The duration spent on crafting a solution.
   2. **Creative Style:** More effective under high contextual familiarity and shorter time investments.
   3. **Deliberate Style:** More effective under low contextual familiarity and longer time investments.
   4. Using both styles simultaneously results in decreased problem-solving success.

(Herd et al., 2022) Backer Affiliations Effects on Crowdfunding Success?

- **Context:** The rise of crowdfunding as a tool to raise funds for entrepreneurial ventures.
- **Focus:** Examining how affiliations (backers funding the same idea) on crowdfunding platforms affect funding behavior.
- **Core Findings:**
   - Increased total number of backers has a positive effect on funding.
   - Affiliation among backers negatively impacts:
      * Funding amounts.
      * Overall funding success.
   - When affiliated others fund an idea, potential backers may feel less inclined to fund due to "vicarious moral licensing."
- **Data Source:**
   - Data from 2,021 ideas on a major crowdfunding platform.
- **Moderators:**
   - Creator engagement (description & updates) and backer engagement (Facebook shares) reduce the negative effect of affiliation.
- **Robustness:**
   - Effect remains consistent across various instrumental variables, model types, measures of affiliation, and crowdfunding outcomes.

- **Additional Evidence:**
  - Three experiments, a survey, and interviews validate the negative effect of affiliation and its explanation through vicarious moral licensing.
- **Practical Implications:**
  - Creators can counteract negative effects of affiliation through specific language in descriptions and updates.

Chapter 10

Advertising

People don't like intrusive ads, especially when they are vulnerable (Tybout and calkins, 2005, p.143)

Meta-analysis has inclusive evidence for the impact of advertising on firm performance (Sethuraman et al., 2011)

Advertising can increase firm value via consumer mindsets and behavior (Wang et al., 2008)

Advertising does not directly affect traditional media, online WOM, or consumer sentiment, but directly affects firm outcomes. (Hewett et al., 2016, p. 12)

10.1 Behavioral Approach

Visual Persuasion

- (Messaris, 1997) and (Scott and Batra, 2003) are two books on this subject
- Prior distribution for TV advertising elasticity for consumer packaged goods can be found in (Shapiro et al., 2018). A substantial portion of products has statistically insignificant or negative estimates for advertising elasticity. TV ads might not be the best vehicle to reach the customer now.
- (Gordon et al., 2019a) reframe our idea of advertising effect, which should be thought of in the sense of incremental effect above the baseline of an ad on consumer behavior.
- (Gordon et al., 2017) and (Lewis and Rao, 2015) provided evidence that observational data can hardly detect adverting effects from noise (i.e., we could either over or under-estimate).
- The goal standard to measure advertising is a random experiment, but it is not possible. An improvement is to have panel data from different

regions of your market to estimate the baseline for each market. Then, you can uncover the advertising effect.

(McGuire, 1978):

- Information processing model of social influence (e.g., advertising effectiveness)
- Behavioral chain of information profession model. (p. 158) (Markov process) - persuasion
- P(presentation of message) x P(attention to message) x P(Comprehension of Conclusion) x P(Yielding to the Conclusion) x P (retention of the new belief) x P (Behaving on the basis of belief)
  - Presentation of message = # of reach (advertising investment)
  - attention to message = # of true reach or Recognition test
  - comprehension = recall test or semantic differential profiles or checklist tests
  - yielding = attitude change
  - retention = attitude change after time interval
  - behavior = actual observation
- Independent variables (components of persuasive communication): source, message, channel, receiver, and destination (Table 1, p. 167)
- Matrix of persuasion
- Advertising appeals:
  - Positive (gain): reduce anxiety
  - Negative (loss): increase anxiety
- Development in the field: this model assume 1 hierarchy, but later developments change the idea of hierarchy model.

Court et al. (2009)

Traditional funnel:

1. Awareness
2. Familiarity
3. Consideration
4. Purchase
5. Loyalty
   1. Active loyalist

2. Passive loyalist

Consumer decision journey McKinsey (2009):

1. Initial consideration set
2. Active evaluation
3. Decision at moment of purchase
4. Postpurchase experience

- Customer Decision Journey is from the perspective of consumers
- Customer Funnel is still important but it's from the company's perspective.
- Analogy here is that you look from 3D (Customer Decision Journey) to 2D (Customer Funnel).

(Maclnnis and Jaworski, 1989)

1. Hierarchy of effects model (1960s)
2. Multiattribute attitude model and cognitive response models (1970s)

Proposed model:

- Antecedents:
  - Types of needs
    - Utilitarian needs
    - Expressive needs
      - Socially expressive needs: to express, or reflect self-image
      - Experiential needs: to satisfy one's cognitive or sensory
  - Motivation (in CB **involvement**): "the desire to process brand information in the ad." (p. 4)
    - situational
    - enduring
- Processing
  - Attention:
    - higher processing motivation leads to higher attention
    - higher utilitarian needs, more focused on brand attributes
    - higher expressive needs, more focused on symbolic/ experiential value
  - Processing capacity:

    * greater processing motivation leads to greater processing capacity for analyzing the ad
    * greater the processing capacity results in less processing capacity for other task.
  - Level of brand processing matches processing operations
    1. feature analysis: salient properties/features
    2. basic categorization
    3. meaning analysis: basic understanding
    4. information integration: Is this brand association?
    5. role-taking
    6. constructive processes
  - Ability and opportunity are moderators of processing
- Consequences
  - Cognitive responses (thoughts)
  - Emotional responses (feelings)
  - Another classifications of responses:
    * message-related (brand-relevant)
    * execution-related (brand-irreverent)
    * viewing-context-related (environment-related)

Advantages of the proposed model: Incorporate

- Elaboration Likelihood model: (Petty and Cacioppo, 1986)
- Mitchell's brand processing model
- Greenwald and Leavitt's model
- Lutz's typology

Note:

- A_{Ad} mediates A_B based on levels of processing (specifically, meaning analysis, information integration, and role taking - high levels).
- Tradeoff between brand/message-related elements and ad-execution elements
- Cohen and chestnut 1990, behavioral loyalty and attitudinal loyalty stem did not stem from the gap between attitude and actual behavioral.

(Vakratsas and Ambler, 1999)

- Advertising input -> Filters -> Consumer (Cognition, Affect, Experience) -> Behavior
- Market Response Models:
  - Aggregate level (Bass' model): market data
    - 3 exposures per purchase cycle is optimal (p. 29)
  - Individual level: individual brand choice
- Cognitive Information Models (C)
  - There is a differential effect between price (increase) and non-price (decrease) advertising on price sensitivity
- Pure Affect Models (A)
  - Mere exposure theories
    - Response competition and Optimal arousal theories: "wear-in" effect which means it takes time to getting familiar to advertising messages to reach optimal effectiveness.
    - Two-factor theory: wear-out effect: after optimal exposures, the effect of advertising starts to decrease
    - Hence, advertising response has an inverted-U shape
  - Affective responses to advertising:
    - Attitude towards the brand
    - Attitude towards the ad
- Persuasive Hierarchy Models (CA)
  - Cognition -> Affect -> behavior. (CA)
  - Elaboration Likelihood Model (ELM): (Petty and Cacioppo, 1986)
  - Another model is (Maclnnis and Jaworski, 1989)
  - Fishbein-Ajzen (1975) involvement model
  - Batra and Ray (1985): Utilitarian and hedonic effect on attitudes towards the brand
  - Involvement moderates the effect of ad evaluation (persuasion is found in low-involvement consumers with attitude towards the ad).
  - Little support for CA model, but the authors still believe in it.
- Low-involvement (motivation) Hierarchy Models (CEA)
  - Cognition -> Experience -> Affect
- Integrative Models (CAE)

  - Information Integration Response Model (IIRM)
  - Deighton's (1984, 1986) two-stage model:
    * first experience = expectations
    * second stage = product trail/experience
- Hierarchy-Free Models (NH)
- Generalizations:
  1. Experience, affect, cognition are mediators of advertising effects.
  2. "Short-term ad elasticities are small and decrease during the product life cycle (p. 35).

(Wind and Sharp, 2009)

- 23 Empirical Generalizations
- Gaps:
  - Boundary conditions
  - Advertising properties
  - Measurement issues

(Batra and Keller, 2016)

- Considerations for a well-integrated marketing communications program
  - Consistency
  - Complementarity
  - Cross-effects
- Dynamic Expanded Customer Decision Journey
  - traditional media
  - newer media
    * Search ads
    * display ads
    * websites
    * email
    * social media
    * Mobile
- Interaction and Cross-effects
  - Traditional media synergies

- sales force and personal selling interactions
- Online and offline synergies

- Drawbacks
  - Limited outcome variables
  - Limited longitudinal studies
  - Did not account for consumer decision stage
- Media type
  - Paid (TV, print, direct)
  - owned (websites, blogs, apps, social media)
  - earned (WOM, press coverage)
- Factors that affect consumer communication processing (model) (p. 130)
- Communication Matching Model: "matches the expected main and interactive effects of different media options with the communications objectives for a brand"
- Communications Optimization Model
  - Coverage
  - Cost
  - Contribution
  - Commonality: different communications share the same meaning
  - Complementarity
  - Cross-Effects
  - Conformability
- IMC Conceptual Framework

10.1.1 Cognitive and Affective

Cognitive and Affective mediators of Advertising Effects

Evaluative responses = attitude

Cognitive approach -> Affective approach (not only comes from cognitive) -> Behavioral approach (Fishbein & Ajzen, 1975 - theory of planned behavior)

System 2: is kinda of independent of Cognitive, but Affective is system 1.

- Expanded model added subjective norm
- (Zajonc, 1980)

- Evaluation from cognitive approach is multi-attribute model
- Anthony Greenwald: Cognitive Response Theory: what important is what is in the consumer mind when they see the ad.

Conditioning:

- Classical (Pavlovian) conditioning: physiological automatic reaction occurred after being exposed to an unconditioned stimulus.
- Evaluative conditioning: direct transfer of affect from one stimulus to another via a conditioning paradigm.

Affective route:

Moods = diffuse, hard to pin down the source, more long-lasting

Emotion = specific, discrete

(Wright, 1973)

- Three modes of spontaneous cognitive responses to advertising stimulus:
  - Counterargument
  - Source Derogation
  - Support Argument

$$Accecptance = w_{SA} \sum_i SA_i - w_{CA} \sum_j CA_j - w_{SD} \sum_k SD_k$$

Situational Factors

- Content-processing involvement: "stemming from receiver's perception of the relevancy"
- Message Modality: audio, print

(Batra and Ray, 1986b)

- Advertising repetition increases brand attitude and purchase intention when support and counter argument production are low; while under high level of such production, brand attitude and purchase intention level off.
- What happen to make the downturn of advertising repetition earlier or later?
- Appropriate interval: purchase cycle (number of exposure per purchase cycle). 3 exposures per purchase cycle is the optimal number
- Wear-in: how many times it takes for the ad to take effect?
- Wear-out: how many times it takes for the ad to bore you?
  - If you change the ad execution, the wear-out is pushed back.

- Traditional thoughts advertising repetition would always wear out (inverted-U curve between repetition and impact on customer's attitude) because of wearout and mere exposure
- **Ability, motivation and opportunity** are antecedents of cognitive processing

(Batra and Ray, 1986a)

- Antecedents of attitude towards the ad:
- Attitude toward the ad leads changes in brand attitudes (MacKenzie et al., 1986; Mitchell and Olson, 1981)
- In low involvement context, execution cues and source likeability (message-oriented and communicator-oriented) have greater impact on persuasion
- Affect typologies (p. 237)

(Holbrook and Batra, 1987)

- emotional reactions mediate the effect of advertising on attitudes toward ad or brand.
- Why divided two articles? the second study claimed that the last paper's list of positive affective mediators was limited, the second one expands to range of emotions.
- Is there a difference between affect and emotions?

(Burke and Srull, 1988) Competitive interference and consumer memory for advertising

- **Experiment 1: Retroactive Interference**:
  - Objective: Analyze the impact of subsequent ads on memory for an initial ad.
  - Finding: Memory for a brand's ad was hampered by:
    1. Later ads for other products within the same manufacturer's line.
    2. Ads from competing brands in the same product class.
- **Experiment 2: Proactive Interference**:
  - Objective: Examine how prior ads impact memory for subsequent ads.
  - Finding: Analogous interference effects observed, meaning prior ads can disrupt recall of later ads.
- **Experiment 3: Ad Repetition & Competition's Influence**:
  - Objective: Study the link between ad repetition and consumer memory in the face of competition.

- Finding:
    1. Ad repetition positively affected recall when there was minimal or no advertising for analogous products.
    2. Presence of competitive ads altered the positive memory effect of repeated advertising.

(Batra and Ahtola, 1991; Voss et al., 2003) offer scale to measure the hedonic and utilitarian dimensions of consumer attitude

(Gibson, 2008): Affective Responses Mediating Acceptance of Advertising

- Using Implicit Association Test **(???)**
- Evaluative conditioning only influences explicit attitudes when there is no previous strong preference or priori

(Pham et al., 2013)

- ad-evoked feelings positively influence brand attitudes both directly and indirectly (via changes in attitude toward the ad), regardless of involvement with the product category, products types (e.g., durables, nondurables, services, search or experience goods).
- This effect is greater among hedonic products than utilitarian ones.

(Kupor and Tormala, 2015)

- Momentary interruptions can promote persuasion
    - higher for low need for cognitive individuals (motivation to engage in thoughtful processing) than high ones
- In other words, interruptions can increase consumers' processing of a message.
- Interruption amplified arousal (need for completion/ goal pursuit and curiosity)

(Dall'Olio and Vakratsas, 2022) Effect of Advertising Creative strategy on Advertising Elasticity

- offer composite metrics that measure aspects of creative strategy
- Content affect advertising elasticity in the following descending order
    - Experiential content
    - Cognitive content
    - Affective content

10.1.2 Involvement

Two overarching frameworks in this stream of research are:

1. Elaboration LIkelihood Model (Petty and Cacioppo, 1979)
2. Heuristic-Systematic Model H(Chaiken, 1980)

Involvement as a key moderator in advertising effectiveness

From the ELM by (Petty and Cacioppo, 1986) by use the word "motivation" in place of involvement. And if you use the term "motivation", reviewers are less likely to fight with you since involvement is so fragmented

For review and operationalization, check (Muehling et al., 1993) (preferable) or (Andrews et al., 1990) and famous scale is (Zaichkowsky, 1985)

Possible manipulation of involvement:

- ego-involvement: how a product is relevant to you (e.g., pick a free product for you, or for others)

Involvement roughly means "How deeply you are as a consumer wants to think about a product,"

Motivation (2) **Opportunity** (kinda under ability in the original ELM, we as marketers separate this factor) (3) Ability are necessary for elaboration likelihood model (Petty et al., 1983)

Defensive processing is not fully captured under the ELM model: Motivation: not the desire to think, but also the desire to find out the truth, assuming that consumers want to find out the truth.

Involvement vs. Engagement:

- According to (Greenwald and Leavitt, 1984) (p. 583), define audience involvement and actor involvement (should be called engagement).

(Greenwald and Leavitt, 1984)

- derived from Sherif & Hovland (1961) Social Judgment (ego-involvement)
- Enduring involvement vs. situational involvement (Houston & Rothschild, 1977) (A Paradigm for Research on Consumer Involvement)
- Four levels of involvement:
    1. Preattention: little capacity
    2. Focal attention: modest capacity to decipher the message
    3. Comprehension: more capacity to analyze the message
    4. Elaboration: most capacity to integrate the message into the audience's knowledge.
- Antecedents: Situational involvement

- Consequences:
  - Under high involvement: communication can modify beliefs
  - Under low involvement: communication affect perceptions, and can gradually be persuasive after repeated exposure.
  - Under ego-involvement: high involvement is more resistance to persuasion.
- Processes of involvement:
  - High involvement creates link between new info to previous experience or attitude
  - differentiate high vs. involvement by central vs. peripheral routs to persuasion (Petty and Cacioppo, 1986)
  - Mitchell (1979) equates high involvement to arousal/drive
- Involvement stems from
  - Actor (participant) or audience (observer)
  - Distinction: Attentional capacity and attentional arousal
    * Arousal = "a state of wakefulness, general preparation, or excitement that facilitates the performance of well-leaned response." (p. 583)
    * Capacity (also known as effort by Kahneman (1973)) = " a limited resource that must be used to focus on a specific task and that is needed in increasing amounts as the cognitive complexity of a task increases." (p. 583)
  - Levels of processing: influences long-term memories
  - Principles for the control of involvement:
    * Bottom-up (data-driven) processing
    * Top-down (concept-driven) processing
    * Competence (data) limitation
    * Capacity (resource) limitation
  - Effects of involvement
    * Immediate Effects: "analyze codes produced by prior processing"
    * Enduring Effects
      · Preattention: no definitive conclusion

        · Focal attention: (1) Familiar stimuli could be identified as separated objects and (2) Unfamiliar stimuli primes sensory memory traces
        · Comprehension: create traces at the propositional level of representation
        · Elaboration: "substantial freedom of memory and attitude from the specific details of th original message or its setting."
      * Principle of higher-level dominance: the effect of the highest level of involvement is dominant in cases where the effects of different levels oppose one another.
        · Both routes can happen at the same time
        · deeper thinking, play judgment will dominate the net results (weights on whatever route is higher)

(Petty et al., 1983)

- provides evidence for the two routes to persuasion
  – Central route: long-lasting and predictive of behavior
  – Peripheral route: associated with positive or negative cues , can be temporary and unpredictive of behavior.
- Argument quality influences attitudes more under high than low involvement
- Product endorsers (celebrities vs. joe) influences attitudes more under low than high involvement
- Can use this as an example of (1) message content, and (2) executional cues (e.g., endorsers) can influence persuasiveness.

(Batra and Stayman, 1990)

- Mood affects cognitive elaboration, bias the argument quality, peripherally affect brand attitudes.
  – Positive moods reduces elaboration

(Macinnis et al., 2002)

- Under ELM, for the endorsers to have an effect, customers have to have some motivation (require some levels of cognition), while affective processing does not require any motivation. Hence, for consumers have higher ability (know about products because it's mature).
- For mature brands, affectively based executional cues can induce sales
- Advertisement with positive feelings induces sales

(Schivinski et al., 2016)

- Propose consumers' engagement scales (in the context of social media)
- Three dimensions of consumer's engagement based on previous research Muntinga et al. (2011)
  - Consumption (e.g., using)
  - Contribution: (e.g., liking or sharing, participating)
  - Creation: (e.g., posting, producing contents)

McQuarrie (1998): Meta analysis

- Lab experiments (in advertising context) are different from real-world phenomenon because:
  - Forcing exposure
  - Failing to measure choice
  - does not consider competitive ads, decay, repeated exposures or mature/familiar brands.

(Muehling et al., 1993)

- A review on involvement in advertising research
- See figure 1 (p. 43) for involvement conceptualization

10.1.3 Visual Cues

Chu et al. (2021)

- Prestigious brands whose brand image is associated with status and luxury, consumers' attitude toward the product becomes more favorable and their willingness to pay a premium for the product grows as the distance between the visual representations of the product and the consumer increases.
- Popular brands whose brand image is associated with broad appeal and social connectedness, the closer the distance, the more favorable is consumers' attitude and the higher their willingness to pay a premium.

10.2 Econometric Approach

10.2.1 Product Placements

(Fossen and Schweidel, 2019a) Measuring the Impact of Product Placement with Brand-Related Social Media Conversations and Website Traffic

- Investigation of relationship between product placement in TV programs and social media activity/website traffic for featured brand

- Study uses data on nearly 3,000 product placements for 99 brands from fall 2015 television season
- Results show that prominent product placements, especially verbal placements, lead to increases in online conversations and web traffic
- Decreasing returns observed at high levels of prominence
- Most placement modalities not enhanced by TV advertising in close proximity to placement activities.

10.2.2 Deceptive Advertising

Regulatory reports of deceptive advertising have negative impact on stock market performance, but brand reputation can mitigate this negative effect (Wiles et al., 2010)

(Rao, 2022)

- Fake news advertising are those that "mimics the format of its surrounding news content in both digital and print media, promotional segments aired on news programs without disclousre to the public." (p. 534)
- Research question: What happens when the fake news ad path is shutdown?
- Data:
  - Browsing data: comScore
  - Purchase data: Ripoff Report, Complaints Board
  - Ad spend: Neilsen AdSpend data
  - keep only first visits.
- Total site visits = organic visits + fake news ad referrals + regular ad referrals.

$$
\begin{aligned}
Q_{merchant} &= Q_{org} + Q_{fn} + Q_{reg} \\
Q_{org} &= \alpha_{org} + \gamma_1^{fn} Ad_{fn} + \gamma_1^{reg} Ad_{reg} \\
Q_{fn} &= \alpha_{fn} Ad_{fn} + \gamma_2^{reg} Ad_{reg} I(Ad_{fn} > 0) \\
Q_{reg} &= \alpha_{reg} Ad_{reg} + \gamma_3^{fn} Ad_{fn} I(Ad_{reg} > 0)
\end{aligned}
$$

where

- fn = fake news
- reg = regular advertising
- Q = demand

- α = direct effect
- γ = spillover effect
- Two effects of interest of fake news
  - Treatment Effect: directly to Q_{fn}, spillover to organic searchers γ_1^{fn} or regular ad referrals γ_3^{fn}
  - Selection Effect: In the absence of fake news, use other channels (substitution effect), 2 segments
    * Those who will use other channels (e.g., regular)
    * Those who will stop
    * In the presence of fake news,
- Results
  - Fake news ads and regular ads are substitutes (i.e., increase in regular ad referrals after FTC shutdown)
  - Probability of a merchant receiving a complaint declines by 8% after the FTC shutdown (somewhat mechanistically)
  - The selection effect is small (i.e., substitution to the regard ad channel is dominated by the decline in organic demand + fake news ad referrals)
  - Supply side: no change in spend, duration and impressions from the customers, or new fake news ad campaigns from merchants.
- Robustness
  - Negative publicity (from the press) does not explain visits reduction.
  - Google might change algorithm because of the FTC shutdown. Hard to know. Hence, claim that the measured decline is the upper bound of the true impact of the FTC regulations.
  - Might consider using logistic regression in rare events data here (King and Zeng, 2001) (but this is a borderline case since they have 95% 0, whereas rare events are usually 99% 0).

10.2.3 Advertising Effects

Mitra and Lynch, Jr. (1995)

- Advertising affect price elasticity via:
  - the size of consideration set
  - the relative strength of preference

Anand and Shachar (2011)

- Horizontal differentiation: "advertisement (informational role) decreases the consumer's probability of not choosing her best alternative by approximately 10%."

(Gopinath et al., 2013) Blogs and Advertising on Movie Performance

- **Objective**:
  - Examine how pre- and postrelease blog activity and advertising affect the box office performance of 75 movies across 208 U.S. geographic markets.
- **Factors in Consideration**:
  - **Blog Volume**: Quantity of blog discussions.
  - **Blog Valence**: Sentiment orientation of blogs.
  - **Advertising**: Promotional efforts by studios.
- **Demographics**:
  - Market differences are attributed to variations in demographic characteristics which influence access to, exposure to, and response to blogs.
- **Time Phases**:
  1. **Prerelease Factors**: Influence on opening day box office performance.
  2. **Postrelease Factors**: Impact on box office performance one month after release.
- **Methodology**:
  - Use of instrumental variables to account for potential confounders in measurement.
- **Key Findings**:
  1. **Heterogeneity in Effects**: Differences are found across consumer and firm-generated media and across various markets. Variations are influenced by demographics like gender, income, race, and age.
  2. **Release Day Impact**: Prerelease blog volume and advertising have the most pronounced effect on opening day performance.
  3. **Postrelease Impact**: Performance a month post-release is influenced by postrelease blog sentiment and advertising.
  4. **Variation in Elasticities**: Greater variance exists in advertising and postrelease blog valence than in prerelease blog volume across markets.

5. **Most Responsive Markets**: 20 markets are identified as having the highest sensitivities to advertising, prerelease blog volume, and postrelease blog valence.
6. **Market Classification**: Markets are classified based on their responsiveness to these factors, aiding studios in targeting for limited release strategies.
7. **Overlap Analysis**: Studios, at first release, only cater to 53% of the most responsive advertising markets and 44% of the most responsive prerelease blog volume markets, suggesting room for optimization in their limited release strategies.

- **Implications**:
  - Movie studios can greatly benefit from understanding market sensitivities to blogs and advertising.
  - Given the clear impact of both pre- and postrelease blog activity, studios should consider integrating them into their promotional strategies.
  - A more nuanced approach to market targeting can help studios maximize their box office returns, especially in limited release scenarios.

Lewis and Reiley (2014)

(Gopinath et al., 2014) Impact of Online Word of Mouth vs. Advertising on Firm Performance

- **Objective**:
  - Analyze the comparative significance of consumer-generated online word of mouth (OWOM) and advertising on firm performance over time post-product launch.
- **OWOM Distinctions**:
  - **Volume**: Total quantity of conversations.
  - **Valence**: Specific sentiment orientation.
    * **Attribute-oriented**: Focus on product features.
    * **Emotion-oriented**: Relates to feelings or experiences.
    * **Recommendation-oriented**: Direct suggestions or endorsements.
- **Advertising Content Classifications**:
  - **Attribute Advertising**: Rational, feature-based messages.
  - **Emotion Advertising**: Affective, sentiment-focused messages.
- **Analytical Approach**:

  - Utilized a Dynamic Hierarchical Linear Model (DHLM).
  - Benchmarked DHLM against dynamic linear model, vector autoregressive model, and a generalized Bass model.
  - Addressed potential endogeneity concerns in key metrics.

- **Key Insights**:
  1. **OWOM's Direct Impact**: Only the valence of recommendation-based OWOM directly affects sales, emphasizing that not all OWOM types are equally influential.
  2. **Time Factor**: The influence of recommendation OWOM increases with time, while the impact of both attribute and emotion advertising wanes.
  3. **Ad Message Wear-out**: Rational messages (attribute-oriented) exhibit faster wear-out than emotion-oriented advertisements.
  4. **Volume vs. Content**: The overall volume of OWOM doesn't significantly impact sales. Instead, the sentiment or "what people say" is more crucial than sheer quantity or "how much people say".
  5. **OWOM Valence Dynamics**: Initially, recommendation OWOM is influenced by attribute valence. As the product ages, emotion valence becomes dominant.

- **Brands Classification**:
  - Based on the research outcomes, brands can be categorized as:
    * **Consumer Driven**: Where OWOM is the dominant influencer.
    * **Firm Driven**: Where advertising efforts are more impactful.

- **Implication**:
  - Brands need to be aware of the evolving nature of OWOM and adjust their advertising strategies accordingly, emphasizing the kind of message being communicated over mere volume.

Draganska et al. (2013)

Sahni et al. (2014)

Hoban and Bucklin (2015)

Stephens-Davidowitz et al. (2017)

Hartmann and Klapper (2018)

(Fossen and Schweidel, 2017) Advertising Effect on WOM

- Research focuses on the link between TV advertising and online word-of-mouth (WOM).

- Concept introduced: “social TV” (joint TV program consumption and social media activity).
- Goals:
  - Determine how TV ads affect online WOM about brands and programs.
  - Identify what boosts or hinders viewer engagement in social TV.
- Data used:
  - TV advertising instances.
  - Minute-by-minute social media mentions.
- Key findings:
  - TV ads influence online WOM for the brand and the program.
  - Most talked-about programs online aren’t always best for brand engagement.
  - Specific brand, ad, and program traits can promote or deter social TV engagement.

(Fossen and Schweidel, 2019b) Social TV,Advertising, and Sales

- Television viewers often engage in media-multitasking, especially using another screen.
- “Social TV” is defined as simultaneous TV viewing and social media discussions about the program.
- While social TV can increase audience engagement, it might also divert attention from ads.
- Central research question: Are “social shows” (programs with high social TV activity) beneficial for advertisers?
- Study examines:
  - Connection between TV advertising, social TV, online traffic, and online sales.
  - Impact of program-related online discussions on online shopping for advertisers during the show.
- Key findings:
  - Ads in programs with higher social TV activity lead to greater online shopping engagement.
  - Ad responsiveness is influenced by the ad’s mood.
  - Affective ads, especially funny and emotional ones, drive the most online shopping activity.

(Lu et al., 2022) Frenemies: Corporate Advertising Under Common Ownership

- Ownership structure impacts corporate advertising expenditures.
- Mutual fund mergers serve as an exogenous shock to ownership structure.
- Competing firms with common institutional blockholders experience reduced advertising spending.
- Reduction is more likely in industries with higher competition and advertising intensity.
- Greater common ownership and concentrated institutional ownership intensify the effect.
- Firms located in the same state show a stronger impact on advertising strategy.

(Fossen et al., 2022) Political advertising effectiveness

- **Objective**: Understand the connection between TV political ad content and ad effectiveness, focusing on message slant and alignment with primary campaign messaging.
- **Metrics Evaluated**:
  1. Online word-of-mouth (WOM)
  2. Voter preference (via daily polls)
- **Dataset**: 2016 presidential election.
- **Key Insights**:
  1. **Centrist Messages**: Ads with a more moderate tone correlate with increased online WOM and voter preference for the respective candidate.
  2. **Consistency**: Ads mirroring the candidate's primary-election messages associate with gains in online WOM and voter preference.
  3. **Temporal Significance of Consistency**: Ad messaging consistency is crucial in the early campaign phase (before October).
  4. **Post-Primary Messaging Strategy**: While moderation post-primary might benefit candidates, abandoning primary messaging early in the general election can be detrimental.
- **Implication on Extreme Messaging**:
  - Despite its growing popularity, extremist political advertising may harm candidate WOM and preference.

(Jindal and Slotegraaf, 2023) Impact of Marketing Investments on Bankruptcy Spillovers:

- **Objective**:
  - Examine how a firm's marketing investments, specifically in advertising and R&D, influence the spillover effects when a competitor faces bankruptcy.
- **Context**:
  - While the relationship between marketing investments and a firm's own bankruptcy is well-studied, little is known about how these investments affect spillover outcomes from a rival's bankruptcy.
- **Key Findings**:
  1. **Contrasting Effects**: In the realm of bankruptcy spillovers, the typical positive impacts of advertising and R&D can manifest differently—both positively and negatively.
  2. **Advertising**:
     - **Industry Growth as a Moderator**:
       * In **low-growth industries**, advertising tends to decrease a firm's stock return when a competitor files for bankruptcy.
       * Conversely, in **high-growth industries**, advertising can increase a firm's stock return under similar circumstances.
  3. **R&D**:
     - **Industry Concentration as a Moderator**:
       * In **low-concentration industries**, R&D investments decrease a firm's stock return when a rival goes bankrupt.
       * However, in **high-concentration industries**, R&D can enhance the firm's stock return.
  4. **Advertising in High-Concentration Industries**: Advertising exerts a more potent impact on stock returns in industries that have higher concentration.

10.2.4 Estimation of Advertising Effects

Shapiro (2016)

Blake and Coey (2014)

Lewis and Rao (2015)

Lewis et al. (2011)

Guitart and Stremersch (2021)

10.2.5 Spillovers

Kitts et al. (2014)

Joo et al. (2014)

Sahni (2016)

Yang and Ghose (2010)

Ghose et al. (2016)

Rutz and Bucklin (2011)

(Fossen et al., 2021) Spillover Effects of Political Ads on TV

- **Context:** Concerns about the influence of political TV ads on viewers' responses to subsequent advertisements.
- **Objective:** Examine how political TV ads impact:
  - Subsequent ad viewership.
  - Online conversations about subsequent ads.
- **Data Source:**
  - 849 national political TV ads from the 2016 election.
- **Methodology:**
  - Quasi-experimental design to assess the impact of a political ad on the following ad.
- **Core Findings:**
  - Political ads lead to positive spillover effects.
  - Ads after a political ad see an 89% reduction in audience decline, reaching a larger audience.
  - A 3% increase in positive online chatter is observed for ads airing after political ads.
- **Contribution:**
  - Fills the gap in advertising research regarding ad-to-ad spillover effects.
  - Offers insights into the influence of political messages on consumers.

(Ghosh Dastidar et al., 2023) Societal Spillovers of TV Advertising

- **Research Question:** Can TV advertising influence societal outcomes, especially beyond typical marketing results like sales and brand awareness?
- **Context:** Analyzing the effect of TV advertising during the COVID-19 pandemic.

- **Data Source:**
  - Daily advertising and mobility data.
  - 2,194 counties across 204 designated market areas in the U.S.
- **Methodology:**
  - Used a border identification strategy to leverage discontinuities across TV markets.
- **Key Findings:**
  - TV ads with COVID-19 narratives positively affect social distancing behaviors.
  - Effects are 11 times larger in counties without government interventions (e.g., mask mandates, shelter-in-place) compared to those with interventions.
  - Overall effect of government ads on social distancing is nonsignificant, but:
    * Effect is negative with policy interventions.
    * Effect is positive without policy interventions.
- **Robustness:** Findings are consistent across different model specifications, variable definitions, and data considerations.
- **Implications:**
  - TV ads, particularly from brands, can have spillover effects during significant public crises.
  - Brand-sponsored TV ads can compensate for the absence of local government policies.
  - Advertising can potentially enhance public health outcomes and promote societal well-being.

10.2.6 Attribution of Advertising Effects

Simonson et al. (2001)

Lambrecht and Tucker (2013)

Li and Kannan (2014)

Blake et al. (2015)

Zantedeschi et al. (2017)

(Fossen and Bleier, 2021) Online Program Engagement and Audience Size During Television Ads

- Research topic: Connection between television viewers' online program engagement (OPE) and the audience size during ads of those programs.
- Data source:
  - 8417 ad instances.
  - OPE activity metrics (Twitter mentions about the program).
  - Audience size during the advertisements.
- Core findings:
  - Increase in OPE volume and positive deviations from an episode's average OPE before an ad are associated with a larger ad audience size.
  - OPE is indicative of viewers' program involvement, which reduces their likelihood to change channels during ads.
  - Positive OPE deviations result in larger ad audiences, especially for the initial ads in a commercial break.
- Implication: TV networks and advertisers can enhance ad audience size by strategically placing ads during "social episodes" (high OPE volume) and "social moments" (positive OPE deviations).

10.2.7 Advertising Content

Xu et al. (2014)

Teixeira et al. (2014)

Tucker (2015)

Liaukonyte et al. (2015)

Rao and Wang (2015)

Sudhir et al. (2016)

10.2.8 Consumer Demand for Ads

Goldstein et al. (2014)

Wilbur (2016)

Tuchman et al. (2017)

Chapter 11

Communication

11.1 Information Theory

11.1.1 Entropy

Most of the examples are taken from `philentropy` package in R, which can be accessed via this link

11.1.1.1 Discrete Entropy

If $\mathbf{p}_X = (p_{x1}, p_{x2}, ..., p_{xn})$ is the probability distribution of the discrete random variable X with $P(X = x) = p_{xi}$. Then, the **entropy** of the distribution is (Shannon, 1948)

$$H(\mathbf{p}_X) = -\sum_{x_i} p_{xi} \log_2 p_{xi}$$

where

- entropy unit is **bits**, and $H(p) \in [0, 1]$ and max at $p = 0.5$
- or the unit is **nats** (log_e)
- or the unit is **dits** (log_{10})

In general, if you have different bases, you can convert between units. Given bases a and b

$$\log_a k = \log_a b \times log_b k$$

where $\log_a b$ is a constant.

Generalize

$$H_b(X) = log_b a \times H_a(X)$$

Equivalently,

$$H(X) = E(\frac{1}{\log p(x)})$$

Entropy is the measure of how much information gained from learning the value of X (Zwillinger, 1995, pp.262). Entropy depends on the distribution of X.

Example:

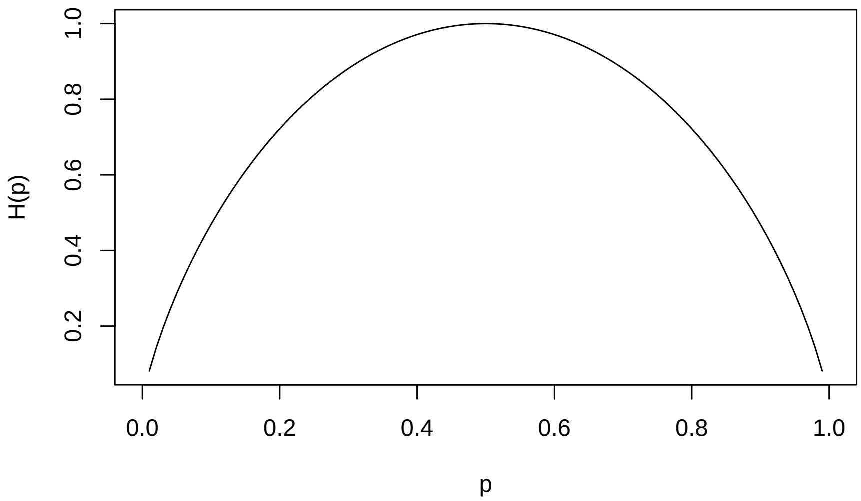
If X has 2 values, then $\mathbf{p} = (p, 1-p)$ and

$$H(\mathbf{p}_X) = H(p, 1-p) = -p\log_2 p - (1-p)\log_2(1-p)$$

which is known as binary entropy function

Plot of p vs. H(p) where max of $H(\mathbf{p}_X) = \log_2 n$ when $X \sim U$

```
hb <- function(gamma) {
    -gamma * log2(gamma) - (1 - gamma) * log2(1 - gamma) # output in bits
}

xs <- seq(0, 1, len = 100)
plot(xs,
     hb(xs),
     type = 'l',
     xlab = "p",
     ylab = "H(p)")
```




```
library("philentropy")
Prob <- 1:10/sum(1:10) # probabilities P(X)
H(Prob) # Shannon's Entropy
```

```
## [1] 3.103643
```

11.1.1.2 Mutual Information

If X and Y are two discrete random variables and $\mathbf{p}_{X \times Y}$ is the their joint distribution, the **mutual information** (i.e., learning of a value of X gives information about the value of Y) of X and Y is

$$I(X,Y) = I(Y,X) = H(\mathbf{p}_X) + H(\mathbf{p}_Y) - H(\mathbf{p}_{X \times Y})$$

where $I(X,Y) \geq 0$ and $I(X,Y) = 0$ iff X and Y are independent (information about X gives no information about Y).

Equivalently,

$$MI(X,Y) = \sum_{i=1}^{n} \sum_{j=1}^{m} P(x_i, y_j) \times \log_b\left(\frac{P(x_i, y_j)}{P(x_i) \times P(y_j)}\right)$$

Properties:

- $I(X;X) = H(X)$ known as self-information

- $I(X;Y) \geq 0$ with equality iff $X \perp Y$

```
P_x <- 1:10/sum(1:10) # distribution P(X)
P_y <- 20:29/sum(20:29) # distribution P(Y)
P_xy <- 1:10/sum(1:10) # joint distribution P(X,Y)
MI(P_x, P_y, P_xy) # Mutual Information
```

```
## [1] 3.311973
```

11.1.1.3 Relative Entropy

The measure of the distance between distribution p and q is called **relative entropy**, denoted by $D(p||q)$

Equivalently, it measures the inefficiency of assuming distribution q when the true distribution is p.

The relative entropy between p(x) and q(x) is

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(X)}$$

where

- $D(p||q) \geq 0$ and equality iff $p = q$

11.1.1.4 Joint-Entropy

$$H(X,Y) = -\sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \times \log_b(P(x_i, y_j))$$

```
P_xy <- 1:10/sum(1:10) # joint distribution P(X,Y)
JE(P_xy) # Joint-Entropy
```

```
## [1] 3.103643
```

11.1.1.5 Conditional Entropy

$$H(Y|X) = -\sum_{i=1}^{n}\sum_{j=1}^{m} P(x_i, y_j) \times \log_b(\frac{P(x_i)}{P(x_i, y_j)})$$

```
P_x <- 1:10/sum(1:10) # distribution P(X)
P_y <- 1:10/sum(1:10) # distribution P(Y)
CE(P_x, P_y) # Joint-Entropy
```

```
## [1] 0
```

Relation among entropy measures

$$H(X,Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$$

Properties:

- Chain rule: $H(X_1, ..., X_n) = \sum_{i=1}^{n} H(X_i|X_{i-1}, ..., X_1)$
- $0 \leq H(Y|X) \leq H(Y)$ Information on X does not increase the entropy measure of Y
- If $X \perp Y$, then $H(Y|X) = H(Y); H(X,Y) = H(X) + H(Y)$
- $H(X) \leq log|n_X|$ where n_X = number of elements in X
- $H(X) = \log(n_X)$ iff $X \sim U$
- $H(X_1, ..., X_n) \leq \sum_i^n H(X_i)$
- $H(X_1, ..., X_n) = \sum_i^n H(X_i)$ iff X_i's are mutually independent.

11.1.1.6 Continuous Entropy

If X is a continuous variable, its entropy is

$$h(\mathbf{X}) = -\int_{R^d} p(\mathbf{x}) \log p(\mathbf{x}) dx$$

where d is the dimension of X.

If X and Y are continuous random variables with density functions $p(\mathbf{x})$ and $q(\mathbf{y})$, the relative entropy is

$$H(X,Y) = \int_{R^d} p(x) \log \frac{p(x)}{q(x)} dx$$

More advance analysis can be access here

11.1.2 Divergence

Three metrics for divergence can be found here:

- Kullback-Leibler
- Jensen-Shannon
- Generalized Jensen-Shannon

11.1.3 Channel Capacity

The information channel capacity of a discrete memoryless channel is

$$C = \max_{p(x)} I(X;Y)$$

which is the highest rate use at which information can be sent without much error.
More information can be found here

Chapter 12

Sales

(Habel et al., 2021) found that consistent with traditional idea, having variable factor in salesperson's compensation plan will increase performance, but also introduce health hazard (more stress which results in emotional exhaustion and more sick days). In cases where salespeople have higher personal ability and more social resources, this harmful effect is mitigated.

12.1 Sales from Rank

(Chevalier and Goolsbee, 2003)

- use an experiment to infer demand
- From low-selling books (where demand was either known or very small) and purchased large quantities (relative to their low demand)
- Data from Amazon
- As the ranks for the books changed, the relationship between the sales rank and demand can be inferred.
- However, this is practical only to very low-selling books.

(Brynjolfsson et al., 2003b)

- With additional demand data from a book publisher, they can derive the relationship between the sales and rank.

(Garg and Telang, 2013)

- Without demand data, they can infer the demand data

On Apple's App Store, there are three rank list:

1. Top-free apps

2. Top-paid apps
3. Top-grossing apps: based on revenue generation

Assuming a Pareto distribution

$$d_{r_p} = b_p \times r_p^{-a_p} | 1 \leq 200$$

where

- d_{r_p} = downloads at rank r_p in the **top-paid**
- b_p = scale factor that is dependent on the total market size for iPad or iPhone apps
- a_p = shape of the Pareto curve

For apps in the top-grossing list, we assume a Pareto distribution for each app (to estimate use a simple truncated OLS regression)

$$\begin{aligned} pd_{r_g} &= p \times d_{r_p} = b_g \times r_g^{-a_g} \\ \log(r_g) &= \frac{1}{a_g} \times \log\left(\frac{b_g}{b_p}\right) + \frac{a_p}{a_g} \times \log(r_p) - \frac{1}{a_g} \times \log(p) \\ \log(r_g) &= \beta_0 + \beta_1 \times \log(r_p) + \beta_2 \log(p) \end{aligned}$$

where

- p = product price
- b_g = scale factor that is dependent on the total market size for iPad or iPhone apps
- d_{r_p} = downloads of the same app in the top-paid list
- Assumption: top-grossing apps generate revenue from upfront pricing only
  - Free and paid apps may have additional purchasable features inside the app, but these purchases will not be included in the paid apps (which is reasonable since paid apps make most of their money from upfront prices) (p. 1256).
  - It's reasonable to assume that in-app features are most common in free apps (p. 1256)
  - Apps' rank and price are assumed to be independent across sectional data (even when an app appears multiple times).
- $a_g = -1/\beta_2$
- $a_p = -\beta_1/\beta_2$
- $\frac{b_g}{b_p} = \exp(-\beta_1/\beta_2)$

- To recover individual values of the scale parameters, we aggregate downloads across apps in a day (since actual downloads for an app is not available) $D_t = \sum d_{r_p} = b_p \sum_{r=1}^{N} r_p^{-a_p}$
- Using the total number of downloads of top ranked apps, the two parameters are
  - $b_p = (\sum_{r_p=1}^{N} d_{r_p})/(\sum_{r_p=1}^{N} r_p^{-a_p})$
  - $b_g = \exp(-\beta_0/\beta_1) \times (\sum_{r_p=1}^{N} d_{r_p})/(\sum_{r_p=1}^{N} r_p^{-a_p})$

Data:

- Apple, Appshopper (shutdown in 2021), AppAnnie (now is data.ai), Mobilewalla
- Period: April 2011 - May 2011
- Data: 200 paid apps, 200 grossing apps and their price data.

(He and Hollenbeck, 2020) On Amazon, which is a generalized results from (Chevalier and Goolsbee, 2003)

12.2 Salespeople

(Sunder et al., 2017) own and peer effects on salesperson turnover behavior

- **Objective:** Delve into the factors influencing salesperson turnover, emphasizing personal performance and peer effects.
- **Background:** Prior studies mostly focused on the repercussions of voluntary turnover. The direct causes, especially the role of personal achievements and peer influences, have been less explored.
- **Key Insights:**
  1. **Framework Proposal:** Introduces a model assessing the impact of individual factors (via identity theory) and peer factors (via social identity theory) on turnover.
  2. **Data Analysis:** Used data from 6,727 salespeople over two years.
  3. **Findings:** Alongside personal performance metrics, peer behaviors, especially peer turnover, significantly affect a salesperson's likelihood to leave.
  4. **Peer vs. Own Effects:** Peer influences have a more pronounced impact than individual factors on turnover.

(Kumar et al., 2014) Measure saleperson's future value

- **Objective:** Introduce a forward-looking metric for sales force evaluation, emphasizing the impact of training and incentive types on future salesperson value.
- **Background:** Traditional sales evaluations focus on retrospective metrics like sales volume. With businesses shifting towards customer-centric views, there's a need for sales strategy adaptation.
- **Key Insights:**
  1. **New Metric Proposal:** A profit-oriented, forward-looking metric is presented to assess salesperson value.
  2. **Method:** Uses a latent class modeling approach to identify different sales force segments.
  3. **Findings:** Sales force segments respond differently to training and incentives, suggesting a universal approach might not be effective.
  4. **Time Horizon Analysis:** The impact of training and incentives can vary based on whether short-term or long-term effects are considered.

Chapter 13

Customer Lifetime Value (CLV)

also known as Lifetime Value (LTV)

Bain & Comapny customer value model

"A year's projected business gives a number that is normally about half of a customer's full lifetime value, although that can vary by industry." (Kumar et al., 2007)

(Kumar et al., 2008) CLV model

$$CLV_i = \sum_{j=T+1}^{T+N} \frac{p(Buy_{ij} = 1) \times \hat{CM}_{ij}}{(1+r)^{j-T}} - \frac{\hat{MC}_{ij}}{(1-r)^{j-T}}$$

where

- CLV_i = lifetime value for customer i,
- $p(Buy_{ij})$ = predicted probability that customer i will purchase in period j,
- CM_{ij} = predicted contribution margin provided by customer i in period j,
- MC_{ij} = predicted marketing costs directed toward customer i in period j,
- j = index for time periods (semiannual or annual)
- T = the end of the calibration or observation time frame (usually 1 year projection),
- N = total number of prediction periods,
- r = semiannual discount factor (e.g., 0.07238 (Kumar et al., 2008) = 15% annual rate)

13.1 Example

This example is subscription customers by Sergey Bryl' in which he adapted model Fader and Hardie (2007)

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.2     v readr     2.1.4
## v forcats   1.0.0     v stringr   1.5.0
## v ggplot2   3.4.3     v tibble    3.2.1
## v lubridate 1.9.2     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
```

```
library(reshape2)
```

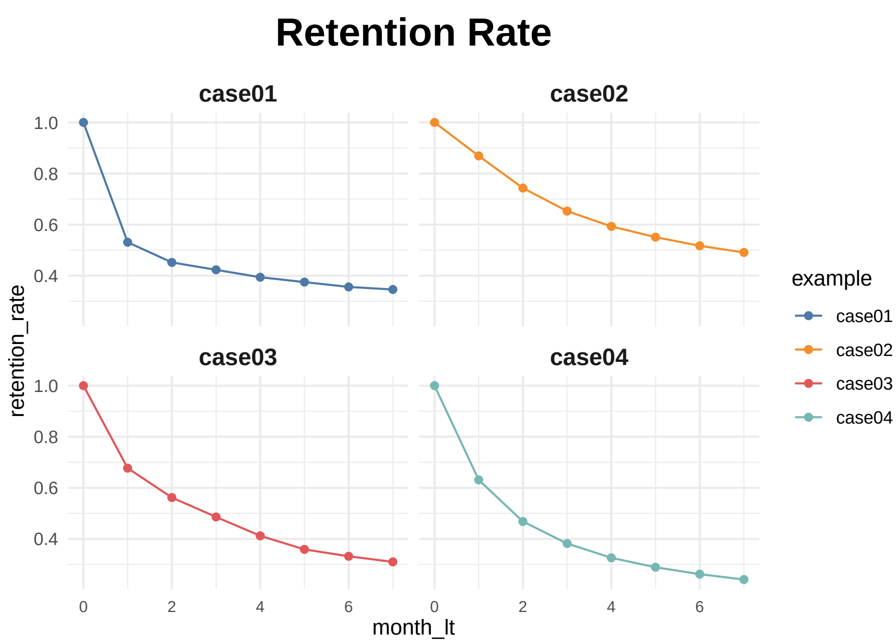
```
##
## Attaching package: 'reshape2'
##
## The following object is masked from 'package:tidyr':
##
##     smiths
```

```
library(MLmetrics)
```

```
##
## Attaching package: 'MLmetrics'
##
## The following object is masked from 'package:base':
##
##     Recall
```

```
# retention rate data
df_ret <- data.frame(month_lt = c(0:7),
                     case01 = c(1, .531, .452, .423, .394, .375, .356, .346),
                     case02 = c(1, .869, .743, .653, .593, .551, .517, .491),
                     case03 = c(1, .677, .562, .486, .412, .359, .332, .310),
                     case04 = c(1, .631, .468, .382, .326, .289, .262, .241)
                     ) %>%
        melt(., id.vars = c('month_lt'), variable.name = 'example', value.name = 'reten

ggplot(df_ret, aes(x = month_lt, y = retention_rate, group = example, color = example))
        theme_minimal() +
        facet_wrap(~ example) +
```

```
        scale_color_manual(values = c('#4e79a7', '#f28e2b', '#e15759', '#76b7b2')) +
        geom_line() +
        geom_point() +
        theme(plot.title = element_text(size = 20, face = 'bold', vjust = 2, hjust = 0.5),
              axis.text.x = element_text(size = 8, hjust = 0.5, vjust = .5, face = 'plain'),
              strip.text = element_text(face = 'bold', size = 12)) +
        ggtitle('Retention Rate')
```



Prediciton when we only have values on certain months

```
# functions for sBG distribution
churnBG <- Vectorize(function(alpha, beta, period) {
        t1 = alpha / (alpha + beta)
        result = t1
        if (period > 1) {
                result = churnBG(alpha, beta, period - 1) * (beta + period - 2) / (alpha + beta +
        }
        return(result)
}, vectorize.args = c("period"))

survivalBG <- Vectorize(function(alpha, beta, period) {
        t1 = 1 - churnBG(alpha, beta, 1)
        result = t1
        if(period > 1){
                result = survivalBG(alpha, beta, period - 1) - churnBG(alpha, beta, period)
```

```
        }
        return(result)
}, vectorize.args = c("period"))

MLL <- function(alphabeta) {
        if(length(activeCust) != length(lostCust)) {
                stop("Variables activeCust and lostCust have different lengths: ",
                     length(activeCust), " and ", length(lostCust), ".")
        }
        t = length(activeCust) # number of periods
        alpha = alphabeta[1]
        beta = alphabeta[2]
        return(-as.numeric(
                sum(lostCust * log(churnBG(alpha, beta, 1:t))) +
                        activeCust[t]*log(survivalBG(alpha, beta, t))
        ))
}


df_ret <- df_ret %>%
        group_by(example) %>%
        mutate(activeCust = 1000 * retention_rate,
               lostCust = lag(activeCust) - activeCust,
               lostCust = ifelse(is.na(lostCust), 0, lostCust)) %>%
        ungroup()


ret_preds01 <- vector('list', 7)
for (i in c(1:7)) {

        df_ret_filt <- df_ret %>%
                filter(between(month_lt, 1, i) == TRUE & example == 'case01')

        activeCust <- c(df_ret_filt$activeCust)
        lostCust <- c(df_ret_filt$lostCust)

        opt <- optim(c(1, 1), MLL)
        retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2],

        df_pred <- data.frame(month_lt = c(0:7),
                              example = 'case01',
                              fact_months = i,
                              retention_pred = retention_pred)
        ret_preds01[[i]] <- df_pred
}
```

```
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(survivalBG(alpha, beta, t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(survivalBG(alpha, beta, t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(survivalBG(alpha, beta, t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(survivalBG(alpha, beta, t)): NaNs produced
```

```
ret_preds01 <- as.data.frame(do.call('rbind', ret_preds01))


ret_preds02 <- vector('list', 7)
for (i in c(1:7)) {

        df_ret_filt <- df_ret %>%
                filter(between(month_lt, 1, i) == TRUE & example == 'case02')

        activeCust <- c(df_ret_filt$activeCust)
        lostCust <- c(df_ret_filt$lostCust)

        opt <- optim(c(1, 1), MLL)
        retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2], c(1:7))),

        df_pred <- data.frame(month_lt = c(0:7),
                              example = 'case02',
                              fact_months = i,
                              retention_pred = retention_pred)
        ret_preds02[[i]] <- df_pred
}
```

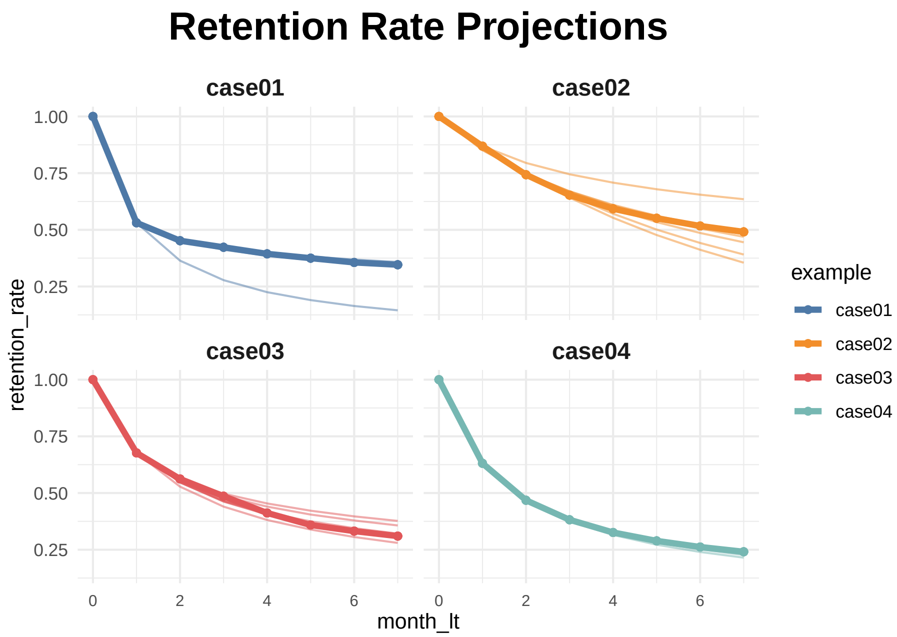
```
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
```

```
## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced

## Warning in log(churnBG(alpha, beta, 1:t)): NaNs produced
```

```
ret_preds02 <- as.data.frame(do.call('rbind', ret_preds02))


ret_preds03 <- vector('list', 7)
for (i in c(1:7)) {

        df_ret_filt <- df_ret %>%
                filter(between(month_lt, 1, i) == TRUE & example == 'case03')

        activeCust <- c(df_ret_filt$activeCust)
        lostCust <- c(df_ret_filt$lostCust)

        opt <- optim(c(1, 1), MLL)
        retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2],

        df_pred <- data.frame(month_lt = c(0:7),
                              example = 'case03',
                              fact_months = i,
                              retention_pred = retention_pred)
        ret_preds03[[i]] <- df_pred
}

ret_preds03 <- as.data.frame(do.call('rbind', ret_preds03))


ret_preds04 <- vector('list', 7)
for (i in c(1:7)) {

        df_ret_filt <- df_ret %>%
                filter(between(month_lt, 1, i) == TRUE & example == 'case04')

        activeCust <- c(df_ret_filt$activeCust)
        lostCust <- c(df_ret_filt$lostCust)

        opt <- optim(c(1, 1), MLL)
        retention_pred <- round(c(1, survivalBG(alpha = opt$par[1], beta = opt$par[2],

        df_pred <- data.frame(month_lt = c(0:7),
                              example = 'case04',
                              fact_months = i,
                              retention_pred = retention_pred)
        ret_preds04[[i]] <- df_pred
```

```
}

ret_preds04 <- as.data.frame(do.call('rbind', ret_preds04))


ret_preds <- bind_rows(ret_preds01, ret_preds02, ret_preds03, ret_preds04)




df_ret_all <- df_ret %>%
        select(month_lt, example, retention_rate) %>%
        left_join(., ret_preds, by = c('month_lt', 'example'))

ggplot(df_ret_all, aes(x = month_lt, y = retention_rate, group = example, color = example)) +
        theme_minimal() +
        facet_wrap(~ example) +
        scale_color_manual(values = c('#4e79a7', '#f28e2b', '#e15759', '#76b7b2')) +
        geom_line(size = 1.5) +
        geom_point(size = 1.5) +
        geom_line(aes(y = retention_pred, group = fact_months), alpha = 0.5) +
        theme(plot.title = element_text(size = 20, face = 'bold', vjust = 2, hjust = 0.5),
              axis.text.x = element_text(size = 8, hjust = 0.5, vjust = .5, face = 'plain'),
              strip.text = element_text(face = 'bold', size = 12)) +
        ggtitle('Retention Rate Projections')
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



calculate the average LTV for case03 based on two historical months with a forecast horizon of 24 months and a subscription price of $1

```
### LTV prediction ###
df_ltv_03 <- df_ret %>%
        filter(between(month_lt, 1, 2) == TRUE & example == 'case03')

activeCust <- c(df_ltv_03$activeCust)
lostCust <- c(df_ltv_03$lostCust)

opt <- optim(c(1, 1), MLL)
retention_pred <- round(c(survivalBG(alpha = opt$par[1], beta = opt$par[2], c(3:24))),

df_pred <- data.frame(month_lt = c(3:24),
                      retention_pred = retention_pred)

df_ltv_03 <- df_ret %>%
        filter(between(month_lt, 0, 2) == TRUE & example == 'case03') %>%
        select(month_lt, retention_rate) %>%
        bind_rows(., df_pred) %>%
        mutate(retention_rate_calc = ifelse(is.na(retention_rate), retention_pred, rete
               ltv_monthly = retention_rate_calc * 1,
               ltv_cum = round(cumsum(ltv_monthly), 2))
# average LTV of $9.33. actual data for the observed periods (from 0 to 2nd months) an
```

13.2 Referral value (CRV)

"Referrals made by customers after a referral-incentive marketing campaign can be attributed to that campaign for about a year." (Kumar et al., 2007). Hence, we usually count those referrals made after one year of a campaign.

2 types of referral:

- Type-one referral: Would not join without referral.

- Type-two referral: still join without referral.

(Kumar et al., 2007) estimated that each customer made about half type-one and half type-two referrals.

Referral value = PV(Type-one referrals) + PV(Type-two referrals)

where

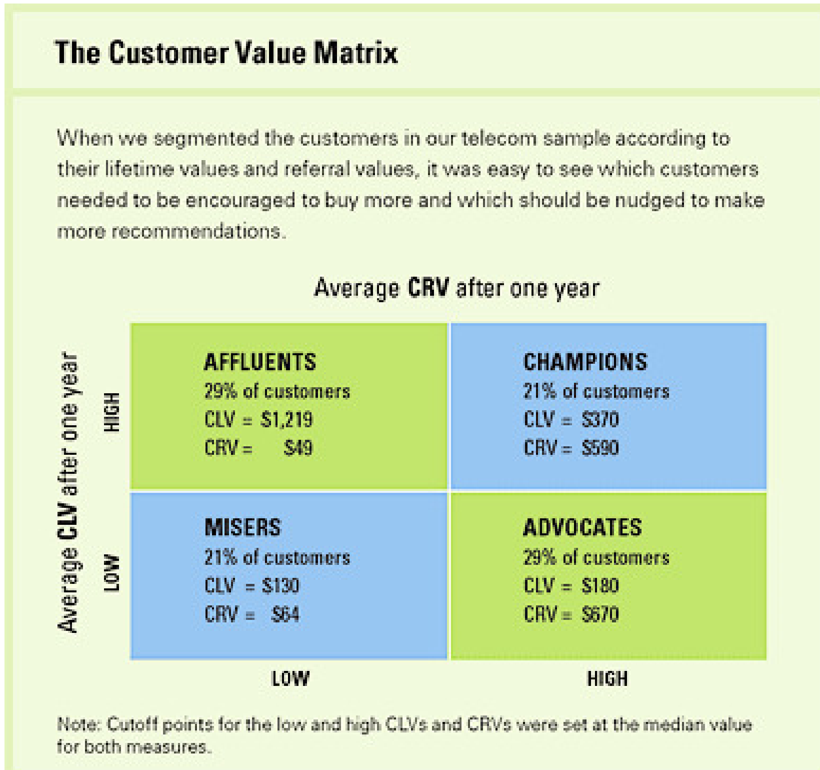
- PV(type-two referrals) = the prevent value of the savings in acquisition costs.

$$
CRV_i = \sum_{t=1}^{T} \sum_{y=1}^{n1} \frac{A_{ty} - a_{ty} - M_{ty} + ACQ1_{ty}}{(1+r)^t} + \sum_{t=1}^{T} \sum_{y=n1+1}^{n2} \frac{ACQ2_{ty}}{(1+r)^t}
$$

where

- T = the number of periods that will be predicted into the future (e.g., typically 1 year)
- A_{ty} = the gross margin contributed by customer y who other otherwise would not have bought the product,
- a_{ty} = the cost of the referral for the customer y (this is what you set up)
- $n1$ = the number of customers who would not join without the referral,
- $n2 - n1$ = the number of customers who would have joined anyway,
- M_{ty} = the marketing costs needed to retain the referred customers,
- r = semiannual discount factor ((Kumar et al., 2008) used .07238 = 15% annual rate)
- $ACQ1_{ty}$ = the savings in acquisition costs from customers who would not join without the referral,
- $ACQ2_{ty}$ = the savings in acquisition cost from customers who would have joined anyway.

Customers with high CLV does not guarantee high CRV (Kumar et al., 2007). Hence, (Kumar et al., 2007) proposed the customer value matrix



(Sunder et al., 2016) CLV in the consumer packaged goods industry

- **Objective:** Propose a framework to assess Customer Lifetime Value (CLV) in the Consumer Packaged Goods (CPG) industry.
- **Challenges Addressed:**
  - **Multiple-Discreteness:** Varied buying patterns.
  - **Brand-Switching:** Customers shifting between brands.
  - **Budget Constrained Consumption:** Customers' purchasing limited by budget.
- **Key Features:**
  - **Bayesian Estimation:** Enables inference of a consumer's latent budget constraint using transaction data alone.
  - **Brand-Level & Category-Level CLV:** Shift from traditional firm-centric CLV models.
- **Implementation:**
  - Applied to carbonated beverages category.

  - Demonstrated superiority over simpler heuristics and traditional CLV models.

- **Policy Simulations:**
  - Explored budget constraint's impact on CLV.
  - Analyzed asymmetric pricing effects and derived managerial insights.

Chapter 14

Celebrity Endorsement

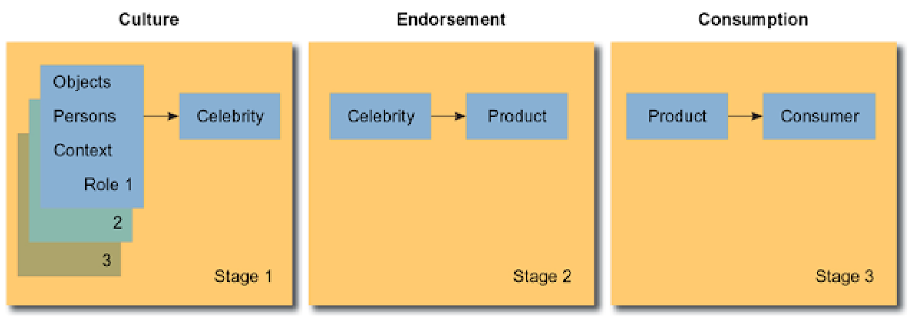
Based on Power Distance Theory, power distance beliefs moderates the positive effect of celebrity endorsers on evaluations of advertising (Winterich et al., 2018)

McCracken (1989)

- Source credibility and source attractiveness model are criticized
- **Meaning transfer** model is then proposed to explain the effectiveness of celebrities as endorsers can be traced to cultural meaning (meaning from celebrity to product to consumer)
- celebrity endorsers are "individuals who enjoy public recognition and use this recognition on behalf of a consumer good by appearing with it in an advertisement." (p.310)
- Mode of endorsement:
  - explicit mode
  - implicit mode
  - imperative mode
  - copresent mode
- Original model contends expertness and trustworthiness affect message effectiveness
  - Expertness = "perceived ability of the source to make valid assertions"
  - Trustworthiness = "perceived willingness of the source to make valid assertions."
- Source attractiveness model contends familiarity, likability, similarity of the source affect message effectiveness.

  - Familiarity = "knowledge of the source through exposure"
  - Likability = "affection for the source as a result of the source's physical appearance and behavior."
  - Similarity = "a supposed resemblance between the source and receiver of the message."

- Cultural meaning and celebrity endorser:
  - Variety of the meanings: status, class, gender, age, personality, and lifestyles
  - Meaning transfer:
    * Under culturally constituted world, meaning from consumer goods to consumers. Advertising is the vehicle to facilitate this transfer.



(p. 315)

McCracken (1986)

- Cultural meaning moves from culturally constituted world to goods to consumers
- Instruments that facilitates this movement are:
  - Advertising
  - The fashion system
  - Four consumption rituals
    * Possession ritual
    * Exchange ritual
    * Grooming ritual
    * Divestment ritual

Chapter 15

Nudges

(Garbinsky et al., 2020) uncovers that people under save because they have positive illusion of financial responsibility (i.e., they thought they are financially more responsible than the average).

(Mrkva et al., 2021) found that choice architecture influences inequities across people. More specifically, people with the following dimensions enjoy the most benefit from nudges:

- Low socioeconomic status (SES)
- Low numerical ability
- Low knowledge

Hence, we should use nudges to reduce SES disparities.

Chapter 16

Marketing-Finance Interface

Review papers

- (Srinivasan and Hanssens, 2009; Kimbrough et al., 2009)
- (Edeling and Fischer, 2016): meta-analysis

Market efficiency

- Weak efficiency: historical information only
- Semi-strong efficiency: historical info + public info in real time
- Strong Efficiency: historical info + public and private info

In marketing, the general consensus is that market efficiency holds in between its weak and semi-strong form

stock market reward firms with higher stock prices as good news about marketing becomes available and vice versa.

Stock market valuation is in sync with product-market valuation (i.e., actions that drive value in product markets should drive firm value)

Methods

- Single equation approach based on ECM: (e.g., (Srinivasan et al., 2009)) recognizing random walk
- Stock return response modeling
- Persistence Modeling (VAR modeling)

Market-based assets are defined as those "arise from the commingling of the firm with entities in its external environment" (Srivastava et al., 1998b, p. 2)

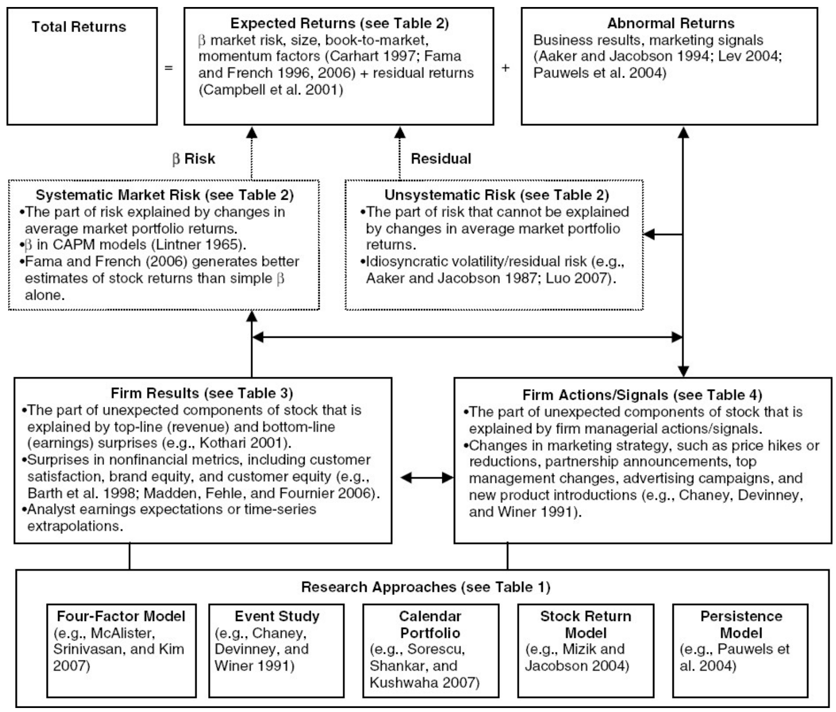


Figure 16.1: Figure 1 (Srinivasan and Hanssens, 2009)

Assets are "any physical, organizational, or human attribute that enables the firm to generate and implement strategies that improve its efficiency and effectiveness in the marketplace" (Srivastava et al., 1998b, p. 3)

Sustained value creation test

1. Convertible (firms can exploit assets)
2. Rare
3. Imperfectly imitable
4. No perfect substitutes

Market-based assets are: relational and intellectual.

- Relational market-based assets are "outcomes of the relationship between a firm and key external stakeholders, including distributors, retailers, end customers, other strategic partners". (Srivastava et al., 1998b)

  - For example, brand equity = relational assets between firms and their customers, while channel equity - relational assets between firms and their channel partners. Brand equity can be created from advertising and superior product functionality.

- Intellectual market-based assets are "the types of knowledge a firm possesses about the environment, such as the emerging and the potential state of market conditions and the entities in it, including competitors, customers, channels, suppliers, and social and political interest group." (Srivastava et al., 1998b)

Intangible assets can be thought of as (Srivastava et al., 1998b, p. 5)

- Stock: "a specific amount or extent of brand equity or knowledge of customers' purchasing criteria possessed by a firm

- Flow: "the extent to which a stock of a particular asset is augmenting or decaying."

Examples of market-based assets

- Customer relationships

- Channel relationships

- Partner relationships

Market-based assets "increase shareholder value by accelerating and enhancing cash flows, lowering the volaitliity and vulnerability of cash flows, and increasing the residual value of cash flows" (Srivastava et al., 1998b, p. 2)

TABLE 1
Assumptions About the Marketing–Finance Interface

	Traditional Assumptions	Emerging Assumptions
Purpose of marketing	Create value for customers; win in the product marketplace	Create and manage market-based assets to deliver shareholder value
Relationship between marketing and finance	Positive product-market results translate into positive financial results	Marketing–finance interface must be managed systematically
Perspective on customers and channels	The object of marketing's actions	A relational asset that must be cultivated and leveraged
Input to marketing analysis	Understanding of the marketplace and organization	Financial consequences of marketing decisions
Conception of assets	Primarily specific to the organization	Result from the commingling of the organization and the environment
Marketing decision-making participants: internal	Principally marketing professionals; others if deemed necessary	All relevant managers irrespective of function or position
Marketing stakeholders: external	Customers, competitors, channels, regulators	Shareholders, potential investors
What is measured	Product-market results; assessments of customers, channels, and competitors	Financial results; configuration of market-based assets
Operational measures	Sales volume, market share, customer satisfaction, return on sales, assets, and equity	Net present value of cash flow; shareholder value

(Srivastava et al., 1998b, p. 2, table 1)

Market-based assets: 3 propositions

1. The more market-based assets external entities can develop, the more satisfied and eager they are to work with the firm, and the more valuable they are to the firm.
2. The more market-based assets pass asset testing, the more external entities benefit.
3. When a company taps or leverages market-based assets to boost cash flows, shareholder value is created.

Generating Customer Value

Different from tangible balance-sheet assets, intangible assets can create long-term sustained customer value by

- satisfying the four resource-based tests (knowledge and relationships pass all the tests)
- adding the value-generating capability of physical assets (Marketing capabilities inherent in organizational processes, such as new product development, order fulfillment, and speed to market, cannot be generated or utilized without **knowledge** of and **relationships** with external entities, such as customers, distributors, suppliers, and other strategic partners. The successful execution of these competencies extends and enhances knowledge and relationships.)

Table 2 Attributes of Balance-Sheet and Off-Balance Sheet Assets

Property	Balance-Sheet Assets	Off-Balance-Sheet Assets
Type of asset	Largely tangible	Largely intangible
Examples	Plant and equipment	Market-based assets such as customer/brand and channel relationships
Can they be bought and sold?	Yes. Tangible property has salvage value.	Yes. For example, AT&T's acquisition of McCaw Cellular.
Can they be leveraged to lower costs?	Yes, by enhancing productivity.	Yes. They can result in lower sales and service costs due to superior knowledge of customers and channels.
Can they be leveraged to command higher prices or share?	Yes. Superior product quality or functionality can be used to justify higher prices.	Yes. Brand and channel equity lead to higher perceived value that may be tapped through price or share premiums.
Can they generate entry barriers?	Yes. Others must make similar investments to be competitive.	Yes. Customer switching costs and loyalty reduce competitive vulnerability.
Can they provide a competitive edge?	Yes. They can make other assets, such as employees, more productive.	Yes, by making other resources more productive (e.g., satisfied buyers are more responsive to marketing efforts).
Can they create options for managers?	Yes, if plant and equipment can be shared across products.	Yes. Satisfied customers are more likely to try brand and category extensions.
Are asset acquisition costs capitalized?	Yes. Plant and equipment can be paid for over several years.	No. Marketing costs are "expensed" and must be justified in the short run.

Figure 16.2: (Srivastava et al., 1998b, p. 9)

- exploiting the benefits of organizational networks (network is "a coordinated set of knowledge sources and cooperative relationships" where the value can be greater than the sum of its individual components (Srivastava et al., 1998b, p. 7))

Asset Valuation Methods and Drivers of Shareholder Value

- Book value (bad because of historical value)
- Replacement value (bad because of measurement issue)
- PE (bad because of accrual accounting measure of firm performance)
- shareholder value approach (current best): discounted future cash flow.
  - Present value of cash flows during the value growth period and long-term residual value of the business at the end of the value growth period.
  - Value of any strategy is determined by
    * **An acceleration of cash flow**s (today's money is preferred than tomorrow's - less risk)
    * **An increase in the level of cash flows** (e.g., higher sales, or lower costs, working capital, fixed investments)

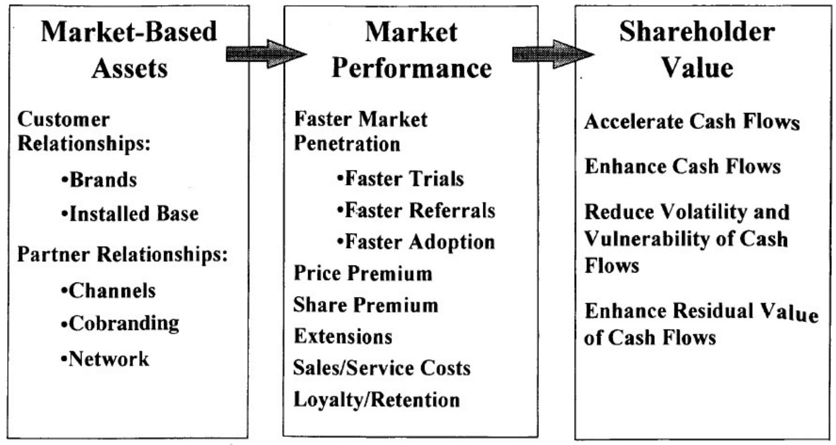


Figure 16.3: (Srivastava et al., 1998b, p. 8 Figure 1)

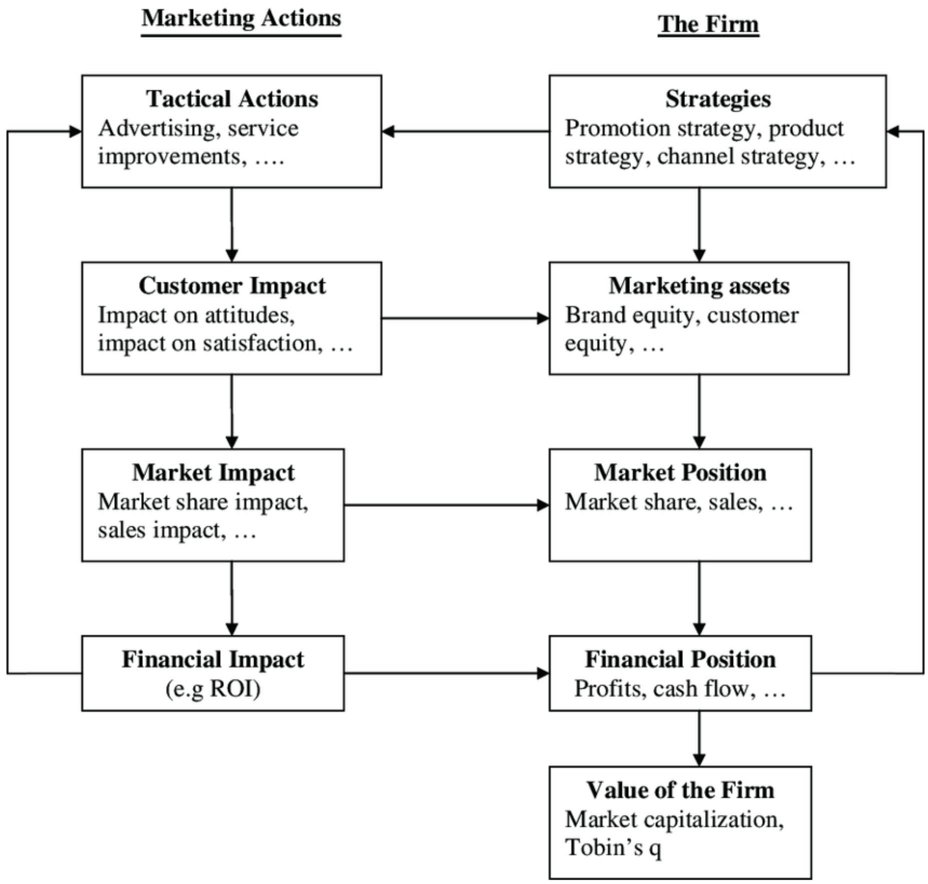
* **A reduction in risk associated with cash flows** (either via volatility or vulnerability of future cash flows), and firm's cost of capital
* **The residual value of the business** (long-term value is improved by larger customer base).

Market-Based Assets and Shareholder value

- Accelerating cash flow: increasing the responsiveness of the marketplace to marketing activity
  - Quicker response to ads from greater brand equity (brand awareness), faster product trial (Keller, 1993)
  - Speed up product life cycle (time-to-market acceptance) from installed base and more partnership to speed to up the product adoption process(Robertson, 1993)
- Enhancing cash flows:
  - generating higher revenues: e.g., product extension
  - Lowering costs: extensions can increase market share and advertising efficiency and lower costs.
  - Lowering working capital requirements: relationship marketing can increase efficiency
  - Lowering fixed capital requirement: relationship marketing can increase efficiency. Networked market-based assets (from cooperative ventures - co-branding, co-marketing alliances) allow partners to leverage each other's networks including customer base.

- Vulnerability and Volatility of Cash Flows
  - The vulnerability of cash flows decreases with increased customer satisfaction loyalty and retention
  - Customers and channel partners relationships enhance stability in operations (from greater sharing of information, automatic ordering and replenishment, lower inventories).
  - Net present value of a cash flow is enhanced with increased switching costs, and lowering of capital requirements from operation integration.
- Residual Value of Cash Flows (those beyond forecast periods)
  - increase with size, loyalty and quality (purchase more) of customer base
  - customer base creates barriers fro competition.
- Cash flow is encouraged to be firm performance measure because of less accrual accounting problems (i.e., historical performance, not adjusted for risk, and easily manipulated)

Marketing actions (e.g, adverting, product innovation) can build long-term assets (e.g., brand equity, customer equity), in turn increase short-term profitability (e.g., increase marketing effectiveness for dollars spent on advertising) (Rust et al., 2004)



The Chain of Marketing Productivity (source (Rust et al., 2004))

Relationship between marketing and finance (Srivastava et al., 1998b)

Stage 1: Market-based Assets

- Customer relationships: Brands, installed base.
- Partner Relationship: channels, co-branding network.

Stage 2:

- Faster Market penetration
- Price premium
- share premium
- extensions
- reduce sales/service costs
- Loyalty/ Retention

Stage 3: Shareholder value:

- Accelerate Cash Flows

- Enhance Cash Flows
- Reduce Volatility and Vulnerability of Cash flows
- Enhance Residual Value of Cash Flows.

(Grewal et al., 2010; McAlister et al., 2016; Morgan and Rego, 2009b) used Tobin's Q in marketing research. since it is "forward-looking, risk-adjusted, and less easily manipulated by managers" (Wies et al., 2019). When probing factors affecting Tobin's Q, it was suggested to control for **financial leverage**, and **cash flows**

Anderson et al. (2018) found that there is a significant improvements from increasing both business skills (marketing and finance). However, the pathway to improve profits from marketing skills is via growth focus (e.g., higher sales, more investments in products, and employees). The pathway to improve profits from the finance skills is via efficiency or cost focus. Hence, the recommendation is that marketing/sales skills have better fit for startups, while finance/accounting skills are more needed in mature companies.

Cheong et al. (2021) found that advertising investments is not only beneficial in terms of customers, but also in terms of investors since it reduces stock price synchronicity (i.e., a firm stock price follows closely to its industry moving)

Lacka et al. (2021) define price impact as" the impact on the variance of stock price." They estimate the permanent and temporary price impacts of the firm-generated Twitter content of S&P 500 IT firms: firm-generated tweets induce both permanent and temporary price impacts, depending on valence and subject matter. Tweets reflecting **only valence or subject matter** concerning consumer or competitor orientation only have **temporary** price impacts, while those with both attributes generate permanent price impact. Moreover, negative valence tweets about competitors generate the largest permanent price impacts.

(Katsikeas et al., 2016) Performance outcomes in marketing

(King and Slotegraaf, 2011) Impact of Firm Investment Decisions on Industry Growth and Volatility

- **Research Context:**
  - Managers' challenge in deciding firm investments aimed at both creating and capturing value.
  - Proposition: Such investment decisions can shape the growth and volatility of an entire industry.
- **Objective:**
  - To determine the impact of aggregate firm investments on the growth and volatility of industry profit and sales.
- **Methodology:**

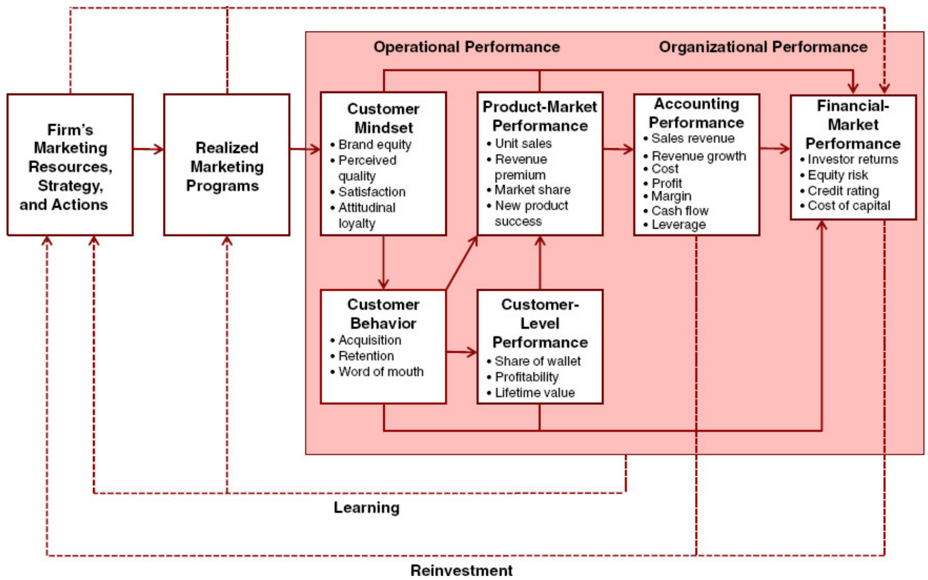


Figure 16.4: Figure 1 (p. 3)

- Analysis of data from 377 industries spanning a 16-year period.

- **Key Findings:**

  1. Firm investments have a noticeable effect on both the growth rate and volatility of their respective industries.

  2. Investment in value creation and value appropriation has intricate relationships with different facets of the industry environment.

16.1 Marketing Value

(Hanssens and Pauwels, 2016) argue for marketing value in a firm and how to appropriate assess its effectiveness and efficiency.

16.2 Business Valuation

(McCarthy et al., 2017) Valuing subscription-based business

- Use DISH network and Sirius XM Holdings data, the authors estimate the overall value of the firm (customer-based corporate valuation)

16.3 M&A

(Singh and Montgomery, 1987)

- Variable of interest:
  - DV: cumulative portfolio abnormal returns
  - IV: relatedness between acquirer and target
- Similarity is the presence of technological or product-market relationships between acquirer and target
- Data: 106 mergers (1975 - 1980)
- Related firms create more value when merging

(Shelton, 1988)

- Variable of interest:
  - DV: merger dollar gains
  - IV: product-market fit (between acquirer and target)
- Similarity is the presence of similar target markets, similar products, or both.
- Data: 218 mergers (1962 - 1983)
- Related firms create more value when merging

(Harrison et al., 1991)

- Variable of interest:
  - DV: ROA
  - IV: Similarity in R&D, capital, and administrative intensities
- Complementary is differences between acquirers and targets in R&D, capital, debt, and administrative intensity
- R&D complementarity boosted unrelated acquisitions.

(Datta et al., 1992)

- Variable of interest:
  - DV: wealth effects
  - IV: # of bids, type of financing, and acquisition
- Similarity is overlap in served markets
- Data: 41 primary studies
- Similar firms create more value when merging

(Ramaswamy, 1997)

- Variable of interest:
  - DV: ROA
  - IV: similarity in market coverage, operational efficiency, marketing activity, client mix, and risk propensity
- Difference between acquirers and targets was based on a distance metric
- Sample: 46 horizontal mergers (banking)
- Differences impede horizontal banking merger success.

(Healy et al., 1997)

- Returns to buyer's shareholders are connected with managers' equity stakes.
- Leveraged buyouts (LBOs) that align management and shareholder interests provide value for purchasers.

(Agrawal et al., 1992; Loughran and Vijh, 1997)

- Cash deals are more popular with investors than stock-financed deals. The bad signaling effect of equity financing suggests managers may offer overvalued stock. This unfavorable stock market reaction to stock financing is permanent.
- Negative returns for acquirers who paid with equity five years post-takeover.

(Hitt et al., 1998)

- Variable of interest:
  - DV: ROA
  - IV: product-market relatedness and resource complementarities
- Complementarity is the presence of related assets
- Data: 24 mergers (case study)
- Resource complementarities contribute to merging success

(Harford, 1999)

- Excess cash acquisitions fail. Announcement period returns are negatively associated with an acquirer's cash.
- A cash reserve eliminates external borrowing. When managers are shielded from the external market, they make irrational investment decisions.

(Larsson and Finkelstein, 1999)

- Variable of interest

  - DV: Synergy realization
  - IV: combination potential, degree of integration achieved, lack of employee resistance

- Data: 112 mergers (case study)
- Complementary operations increase synergy, especially with organizational integration.

(Agrawal and Jaffe, 2000)

- there is evidence for underperformance after M&A performance

(Mitchell and Stafford, 2000)

- Small acquirers see favorable returns three years post-acquisition.

(Sudarsanam and Mahate, 2003)

- Undervalued acquirers with high book-to-market ratios (value' acquirers) have favorable long-run returns, while overvalued acquirers with low book-to-market ratios have negative returns.
- The stock market extrapolates management's ability to make solid purchases from the company's finances. "Glamour" acquirers' managers are generally overconfident in their ability to handle an acquisition, and their actions aren't rigorously checked. Value' acquirers, rigorously scrutinized by other stakeholders, are more cautious in appraising possible targets, leading to wiser decisions.

(Megginson et al., 2004)

- Business similarity between merging firms is linked with positive stock performance.
- Divestitures or spin-offs that dispose of non-core assets or operating divisions boost stockholder value.

(Moeller et al., 2004)

- Small acquirers have larger announcement period returns, regardless of target ownership (public/private).

(Fuller et al., 2002; Conn et al., 2005)

- Buyer shareholders enjoyed big gains when the target was a privately owned corporation or a parent-controlled subsidiary.
- The bidder benefits from the illiquidity discount and enhanced information sharing from concentrated ownership.
- Bidders using stock transfers and other non-cash means did not underperform and saw positive stock market reactions. Private target managers, who are expected to become major stakeholders in the new business, have

a greater motive to supervise the bidder's management, which provides value for the bidder. Deferred tax obligations in a stock offer may lead target ownership to accept a lower price. This leads to increased shareholder returns.

- A lack of awareness around private bids makes it easier for bidders to abandon discussions, reducing hubris-driven judgments. The stock market favors glamour' businesses bidding for private targets.

(Moeller et al., 2005)

- When inorganic growth is no longer sustainable, serial acquirers with high valuations lose money on their purchases.

(Cosh et al., 2006)

- CEO ownership increased long-term returns.

(Swaminathan et al., 2008) found that when merging organizations have poor strategic emphasis alignment, diversity improves value. In contrast, when merging organizations have strong strategic emphasis alignment, value is enhanced when the merger goal is consolidation.

- For a summary of M&A research in marketing up until 2008 (see table 1, p. 34)
- Contributions:
  - Strategic emphasis alignment is an important variable in merger value creation
  - Both resource similarity and resource complementarity create value (but under different merger motives)
  - marketing resources (e..g, advertising relative to R&D) affect M&A value creation.
- Merger motives (p. 40):
  - Consolidation: to gain power by combining two firms with similar products and serving similar markets
  - related diversification: two firms that operate in completely non-overlapping businesses.
  - unrelated diversification: two firms in closely related industries
    - "related diversification in which firms gain access to and acquire a related technology"
    - "related diversification in which firms acquire a related product for product line filling"
- Data: 206 M&A (electronics, chemicals, foods).

- Method: event studies
- Strategic emphasis was operationalized by substracting the firm's R&D expenditures from its advertising expenditures and dividing it by the total assets of the firm in the year preceding the merger (following (Mizik and Jacobson, 2003))
- Strategic emphasis alignment: absolute value fo the difference between the acquirer strategic emphasis and that of the target.

(King et al., 2008) see Research and Development

(Chari et al., 2009)

- Acquirers of domestic targets experience bigger short- and long-term wealth benefits than offshore acquirers. Deals that diversify across borders perform poorly. Returns are higher when the target's domestic product and financial markets move independently of the acquirer's own nation. Stock markets favor developed market acquirers in emerging markets.
- As developing market spreads grow, acquirer returns rise. When majority ownership is transferred, the acquirer, target, and acquirer's shareholders all enjoy stronger profits, especially in R&D- and brand-intensive businesses where intellectual assets are significant.

(Bouwman et al., 2009)

- In terms of two- and three-year buy-and-hold returns, bull market acquisitions underperform bear market acquisitions.
- Managers suffer from hubris during market upswings and overestimate synergies and profits. When the market is pessimistic, managers make more cautious decisions, thus purchases are motivated by realistic prospects of beneficial synergies.

(Swaminathan and Moorman, 2009)

- Contribution:
  - alliance announcements create value for firm
  - When network efficiency and density are moderate, they have the most favorable influence.
  - Network reputation and network centrality have no effect
  - marketing alliance capability (firm's ability to management a network of previous marketing alliances), positively influence value creation.
- Network features:
  - Network centrality: whether the company has partnerships (# of firms with which a firm is directly connected)

- Network efficiency: The company's network offers new capabilities (the degree to which the firm's network of alliances involves firms that possess non-redundant knowledge, skills, and capabilities)
- Network density: the firm's network involves interconnections among firms (the degree of interconnectedness among various actors in a network)
- Network reputation: the firm has a strong reputation (aggregate-level quality ascribed to firms in a firm's network).
- Marketing alliance capability: The company may handle marketing relationships

* To construct the network variables, they were used from the period of 5 years before, but other research use 7 years (Gulati and Gargiulo, 1999; Schilling and Phelps, 2007)
* How firm networks influence firm abnormal returns?
  - Networks multiply alliance benefits
  - Networks facilitate alliance compliance
  - Networks signal firms and alliance quality
* Control variables:
  - Following (Dutta et al., 1999), installed base of customers (firm sales), firm resources devoted to building customer relationships (firm receivables), marketing expenditures (firm selling, general, and administrative expenses), firm advertising expenditures. Then, add technical know-how (R&D expenditure). All of these variables use Koyck lag function, then combine under principal components analysis to avoid multicollinearity.
  - firm alliance experience
  - size of alliance partners (ratio of market cap of the firm to the partner)
  - Intra industry alliances vs. inter industry alliances (following (Rindfleisch and Moorman, 2001))
  - Repeat partnering
  - partner network characteristics (same network variables)
* Selection model and value creation model because partnership may be based on referrals (following (Verbeek and Nijman, 1992)

Umashankar et al. (2021) found that M&A decreases customer satisfaction that cannibalize firm value despite potential efficiencies (due to a shift by executives from customers attention to financial issues). The presence of marketing experts in upper echelons can mitigate this negative effect. Customer dissatisfaction with M&As might offset any synergy and efficiency advantages.

16.4 Stock Return Response Modeling

Valuation model

$$MarketCap_{it} = \sum_{T=t}^{\infty} (\frac{1}{1 + r_{it}})^{T-t} E(CF_T)$$

Equivalently,

$$MarketCap_{it} = (1 + Eret_{it}) MarketCap_{it-1} + \sum_{T=t}^{\infty} (\frac{1}{1 + r_{it}})^{T-t} \Delta E(CF_{iT})$$

where0

- $Eret$ = expected rate of return for an asset
- $\Delta E(CF_{iT})$ = change in the expected cash flows

Hence, the stock return can be written as

$$StockReturn_{it} =$$

16.5 Tobin's Q

Traditional Tobin's q (Rao et al., 2004)

$$q = \frac{MVE + PS + DEBT}{TA}$$

where

- MVE = share price x number of common stock outstanding
- PS = liquidating value of the firm's preferred stock
- DEBT = (short-term liabilities - short-term assets) + book value of long-term debt
- TA = book value of total assets

Tobin's q is scale independent. Hence, it's a good measure of relative market performance

(Peters and Taylor, 2017) Peters and Taylor Total Q

- The neoclassical theory of investment can still be applied to intangible assets.
- Propose a new Tobin's q proxy that accounts for intangible capital, which is called **total q**
  - = the firm's market value divided by the sum of its physical and intangible capital stocks
  - Firm's intangible capital = sum of its knowledge capital and organizational capital
    * R&D spending = an investment in knowledge capital (using the perpetual -inventory method to a firm's past R&D to measure the replacement cost of its knowledge capital)
    * Fraction of past selling, general, and administrative (SG&A) spending is organization capital, which includes human capital, brand, customer relationship,s and distribution systems.
    * A firm's total capital as the sum of its physical and intangible capital (measured at replacement cost
- Intangible capital is costlier than physical capital to adjust
- Download Data from WRDS
- Sample excludes utilities (SIC 4900-4999), financial firms (6000-6999), others (9000+), missing or non-positive book value of assets or sales, and firms with less than $5mil in physical capital. Winsorize all regression variables at the 1% level to remove outliers.

Proposed total q

$$
\begin{aligned}
q_{it}^{tot} &= \frac{V_{it}}{K_{it}^{phy} + K_{it}^{int}} \\
&= \frac{prcc_f \times csho + dltt + dlc - act}{ppegt + K_{it}^{int}}
\end{aligned}
$$

where

- V_{it} = firm's market value = outshining equity (prcc_f times csho)+ book value of debt (dltt + dlc) - firm's current assets (act)
- K_{it}^{phy} = replacement cost of physical capital = ppegt
- K_{it}^{int} = replacement cost of intangible capital

Previous Tobin's q measure did not include the intangible part (Fazzari et al., 1987) (Erickson and Whited, 2011)

Intangible capital can be

- Created internally
  - knowledge knowledge (e.g., patents, software): using perpetual inventory method $G_{it} = (1 - \delta_{RD})G_{i,t-1} + RD_{it}$
    * G_{it} = the end-of-period stock of knowledge capital (for G_{i0} can assume to be 0)
    * δ_{RD} =depreciation rate (use BEA's industry-specific R&D depreciation rates (Li and Hall, 2018) and 15% for missing data)
    * RD_{it} = real expenditures on R&D during the year (if missing treat as 0, (Lev and Radhakrishnan, 2005)
  - brand capital (advertising under selling expense within SG&A)
  - human capital (employee training under general or admin expense within SG&A)
  - Others: Other Intangible Assets (intano) in Compustat
- Purchased externally
  - Goodwill (not separately identifiable e.g., human capital is booked under goodwill).
  - Other Intangible Asset: (e..g, patent, software if separately identifiable)

16.6 Corporate Agility

- Corporate agility defined as "a firm's ability to adapt to environmental changes"
- (Lehn, 2021) hypothesizes that firm agility increases firms' survival rates.
- Since firms have to report accurately to the SEC, this measure is reliable.
- Agility measured as the sensitivity of its responsive strategies to rivals' threats.
  - "A firm's corporate agility is estimated as the sensitivity of its product similarity or dissimilarity (to its rival's products) to the rivals' product similarity (to the firm's products." (p. 9)
- Agility is different from firm flexibility

16.7 Firm Complexity

(Hoitash and Hoitash, 2017) Accounting Reporting Complexity

- Accounting complexity is defined as "the difficulty to understand, prepare, audit, and analyze the financial reports." (p. 262)
- Accounting reporting complexity (ARC) is based on the count of accounting items disclosed in eXtensible Business Reporting Language (XBRL) 10-K filings.
- Operating complexity: measured by the number of business and geographic segments and the existence of foreign operations as measure of complexity
- Linguistic complexity of financial reports (i.e., readability)
  - Fog Index (Gunning et al., 1952)
  - Length of 10-K filings
- Data
  - XBRL from Calbench, criteria:
    * 2011-2014 that filed within 150 days of the fiscal year-end
    * merge with compustat
    * positive sales, and non-missing common shares outstanding
    * assets > 10 mil
    * Exclude missing data on material weaknesses, audit fess, restatements from Audit Analytics
- ARC is inversely associated with financial reporting quality
- ARC is positively associated with audit delay and audit fees

(Hoitash and Hoitash, 2022) Firm complexity

- ARC was used to measure firm complexity
  - It can be dis aggregated to measure the complexity of specific economic activities, (e.g., derivatives or leases)
- Even though (Loughran and McDonald, 2020a,b) measure firm complexity using dictionary-based approach, the texts can sometimes refer to other companies' activities, whereas accounting line items belong to the focal firm
- Accounting disclosures comprise the majority of economic and corporate activity. Consequently, these disclosures can reflect company complexity holistically.
- Other measures of complexity: such as number of operating segments (e.g., https://gist.github.com/joosti/213050de42d6e78f1634) using BUSSEG (business segments) and OPSEG (operating segment) in compustat.

- (Hoitash et al., 2021) find that when complexity increases, analysts struggle to create accurate forecasts.
    - (Garcia et al., 2020) find that CSR are positively correlated with greater ARC.

16.8 Initial Coin Offerings

Data sources

- (Czaja and Röder, 2021)
- (Lyandres et al., 2022)
- (Hsieh and Oppermann, 2021)
- https://research.tokendata.io/
- (Momtaz, 2020)
- (Belitski and Boreiko, 2021)
- (Campino et al., 2022)
- https://zenodo.org/record/4034258#.YzdQbXbMLq4

16.9 Initial Public Offerings

New era for IPO

- Businesses focus on more growth (e.g., number of users) than traditional financial measures and take longer to go public (double the age before going public as compared to 1980s).
- More retail investors due to lo-cost trading platforms.
- Longer but still uninformative data. Proposed triggered disclosure

(Cao et al., 2022)

- Firms can use their innovation potential as a credible signal of their quality at the time of an initial public offering (IPO).
- Innovation potential is positively associated with the initial value of the IPO and with first-day IPO returns, and negatively associated with the extent to which insiders sell their shares at the time of the IPO.
- Patents have a stronger impact on insider sales than preannouncements and generic references to future innovation, while preannouncements have the strongest impact on first-day IPO returns.

Chapter 17

Privacy

Reviews:

- Retailing: (Martin and Palmatier, 2020; Martin et al., 2020)

(Martin and Murphy, 2016) Data privacy in marketing

Definitions of privacy

- Privacy as "a limited access to a consumer's information" (Westin, 1967) with 4 substates: anonymity, solitude, reserve, and intimacy
- claim to control personal information flows

Consumer info privacy: "control of dissemination and use of consumer information including, but not limited to demographic, search history and personal profile information" (p. 136). Examples of consumer privacy violations are unwanted marketing communication, highly targeted advertisements, online tracking.

Privacy concerns are proxies for measuring consumer privacy that is often operationalized as consumers' beliefs, attitudes, and perceptions bout their privacy (survey data) (Malhotra et al., 2004).

Privacy paradox: the disconnect between consumers' declared privacy desires and their actual privacy policies (even though people say that they are concerned about privacy, they still hare their sensitive personal information freely).

Privacy failure (e.g., data breach, hacking intrusion, company loss company): (Malhotra and Kubowicz Malhotra, 2011; Martin et al., 2017)

Privacy can be a dimension for firms to differentiate (Casadesus-Masanell and Hervas-Drane, 2015; Rust et al., 2002)

Privacy in society:

- Federal Trade Commission (FTC) regulates consumer info privacy issues using 2 frameworks
  - Notice-and-Choice Model (including Fair Info Practice Principles)
  - Harm-Based Model (whether harms occurs)
- European Union (EU) Data Protection Directive is a set of consumer info privacy protections (more comprehensive than U.S.-based).
  - The default approach to address consumer information privacy concerns is still industry self-regulation (but after the article was published, EU has GDPR but nothing in the US).
- When firms lack privacy regulations, consumer demand more government intervention (Milberg et al., 2000)
- advertising effectiveness is diminished after EU Data protection directive was implemented (Goldfarb and Tucker, 2011)

Theoretical perspectives of privacy:

- Social Contract Theory: A moral contract regulates the fundamental principles and agreements between a society and an individual. Consumers believe that marketers have kept their end of the social contract when the company gives them with additional value through personalized products or monetary compensation.
- Justice Theory: Procedures designed to preserve customer privacy represent the procedural justice dimension, whereas the outcome of a marketing interaction with privacy implications represents the distributive justice dimension. Fair access to and utilization of information are characteristics of procedural fairness. Fair-appearing policies can address privacy concerns.
  - Theoretical outcomes of distributive justice include any benefits consumers obtain for disclosing their personal information, so jeopardizing their privacy. Common outcomes include customized goods, personalized value, faster customer-business interaction, complimentary services, and even monetary compensation.
  - Parallel privacy paradox indicates that customers like the outcomes delivered by these marketers while simultaneously feeling vulnerable when releasing personal information. (Awad and Krishnan, 2006)
  - Complex privacy policies that impede customer comprehension are perceived as less fair and, thus, decrease consumer confidence and trust in the company (Vail et al., 2008). Procedural privacy is about not only firms' privacy policies but also consumers' interpretation of them.
- Power-Responsibility Equilibrium Theory/ Control Theory: The social obligation of the more powerful partner in a partnership is to foster a

sense of equality (trust and confidence). Consumers respond defensively to perceived power inequalities (i.e., threats to their information privacy). As a result, they may withhold facts or supply false information to a company.

- Social Exchange Theory: When perceived benefits exceed perceived costs, people will disclose personal information. Benefits include individualized marketing offerings or even free services (e.g., consumers regard more frequent marketing communication and targeted advertisements as a tradeoff for valuable services supplied by the marketer).
  - When consumers sense greater personalization value from an information exchange, they are more likely to believe marketers have met their end of the social contract governing the transaction (Chellappa and Sin, 2005).
- Reactance Theory: Privacy concerns heighten reactance
  - Some consumers fear that highly targeted and individualized marketing communications violate their privacy (Tucker, 2013)
  - Ultimately, consumer responses to tailored and personalized marketing communications might appear as communication avoidance, information falsification, unfavorable word-of-mouth, and other negative behaviors (White et al., 2008)
- Behavioral Decision Theory: Consumer perceptions and behaviors about privacy are influenced by multidimensional rational calculations.

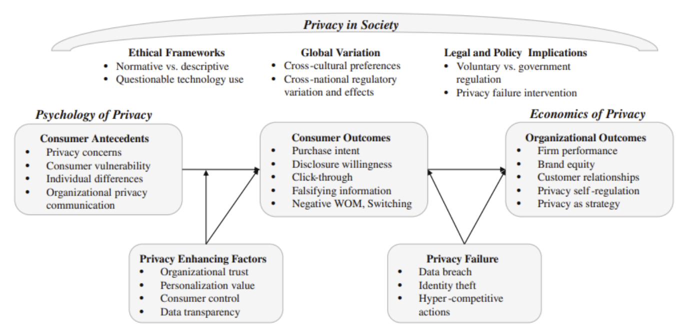


Fig. 1 Data privacy research in marketing: predominant constructs and relationships

Figure 17.1: p. 140

Consumer privacy concerns:

- "a proxy for understanding consumers' feeling about their information privacy" (p. 145)
- Survey scales: see (Smith et al., 1996), more updated version (Malhotra et al., 2004) which includes

  - information collection
  - unauthorized secondary use (internal and external)
  - improper access
  - error protection

- Alternatively, (Sheehan and Hoy, 2000) use a somewhat similar set of dimensions
  - Awareness of information collection
  - Information use
  - Information sensitivity
  - Familiarity with entity
  - Compensation
- Privacy concerns increase over time among both older and younger consumers (but more for older) (Goldfarb and Tucker, 2011)
- privacy concern is a mediators of a website's personalization increasing click-through (Bleier and Eisenbeiss, 2015)
- Firm signals (e.g., privacy seals) increase consumers' willingness to reveal info, and promote positive perceptions about the org (Miyazaki and Krishnamurthy, 2002)
- The broadcast of a privacy policy reduces customer disclosure. In addition, people are inclined to share greater sensitive information when they feel that others have already done so, indicating a decreased risk of disclosure (Acquisti et al., 2012)
- Privacy policy declarations boost the benevolence and integrity elements of trust, but do not increase purchase intent (p. 146)
- Increased purchasing from websites with more stringent privacy practices (i.e., Customers are willing to pay more for privacy) (Tsai et al., 2011)

Trust:

- In contrast to measures to alleviate privacy concerns, which are reactive procedures, firm initiatives to increase trust serve as proactive mechanisms (Wirtz and Lwin, 2009)
- In situations where retailers have personalized or targeted consumer material, trust plays an important role in assuaging privacy worries (Aguirre et al., 2016)
- People do read privacy policies, and those policies influence their trust in that firm (Milne and Culnan, 2004) (84% in the sample of 2500 consumers)

- (Martin et al., 2017) privacy policies can be a good proxy for firms transparency and control, and they are related to firm performance and consumer behaviors.

Personalization:

- Mixed effects on consumers outcome (p. 147)
- Key question: Whether information was provided voluntarily or unwillingly, given marketers' ability to gather data overtly or secretly through a variety of methods.

Control

- Personalized and targeted ad effectiveness is greater when consumers have greater ability (or at least perceived) to control their privacy (Tucker, 2013)
- Consumers' perceived information is the mediator through which self-protection, industry self-regulation, and government mandates reduce privacy concerns. (Xu et al., 2012)
  - But consumers can also disclose too much info, leaving them vulnerable, when they have greater perceived controls (Brandimarte and Acquisti, 2012)

Propositions:

- "Firms that prioritize data privacy in an authentic way will experience positive performance"
- "Firms that involve their customers in the info privacy dialogue will experience positive performance"
- "Firms that implement privacy promoting practices will experiences positive performance under the condition that they align data privacy practices across all aspects of the firm"
- "Firms that focus on what they do right with respect to data privacy will experience positive performance"
- "Firms that commit to data privacy practices over the long-term will experience positive performance"
- "Firms that embody these privacy as strategy tenets will experience higher consumer trust."

17.1 Psychology of Privacy

(Brough et al., 2022) Consumers' response to privacy notices (bulletproof glass effect)

- Study 1: Managers expect a privacy notice will make customers feel more secure
  - Since managers see that privacy notices as formal legal contract (i.e., binding agreements that dictate how a firm can collect, use and store data), they would expect that this formal contract-based approach can enhance consumers' comfort in purchasing a firm's product.
- Studies 2, 3, 5: consistent with bulletproof glass increasing feelings of vulnerability (despite the protection offers), formal privacy notices can decrease consumer trust, and purchase interest (despite objective protection emphasis)
  - On the one hand, privacy is considered a social contract (Consumer perception of privacy invasion versus respect is governed by norm), and firms that respect these expectation can increase consumers' trust (McCole et al., 2010) and increase purchase intention(Eastlick et al., 2006), and those that violate these norms will receive negative WOM (Miyazaki, 2009). Benefits for consumers would not negate this negative reactions (Kim et al., 2019). These social contracts can enhance trust (Kim et al., 2019)
  - On the other hand, formal contracts (instead of social contract) can undermine trust (similar to the bulletproof in unusual settings) (Malhotra and Murnighan, 2002): Evidence from that multi-round trust game that consumers are less likely to trust when a formal contract was created in the first round than without the contract.
- Study 4: when consumers had a priori that they should be distrustful, the unintended consequence of privacy notice did not materialize.
  - When consumers do not have potential harm in mind, when seeing a bulletproof glass (i.e., in unexpected situations), they will react more negatively than when some potential harm is expected
- Study 5 and 6: when benevolence cues were added to privacy notices, the unintended consequence of privacy notice did not materialize
  - To increase trust, one can increase either benevolence or competence. Because benevolence-trust can induce consumers to view privacy notices as social contracts rather than formal contracts (as compared to using competence-trust).
  - Since benevolence-trust is affective where as ability-trust is cognitive, we can use LIWC to score whether a privacy notice regarding its proportion of words that reflect affect process and proportion of words that reflect cognitive process
- Study 7: the presence and conspicuous absence of privacy info are enough too in decrease purchase intent.

  - Use Apple's privacy nutrition labels to examine the presence and conspicuous absence of a privacy notice in terms of triggering decreased purchase interest.
  - Apps were not required to include privacy notices (between 08/2017 and 05/2018), only half of the apps included a privacy policy link (Story et al., 2018)). In 12/2020, Apple mandated privacy labels in the App Store. Hence, (Brough et al., 2022) found that apps with a privacy notice conspicuously absence or present decease app downloads as compared to those without privacy notice.

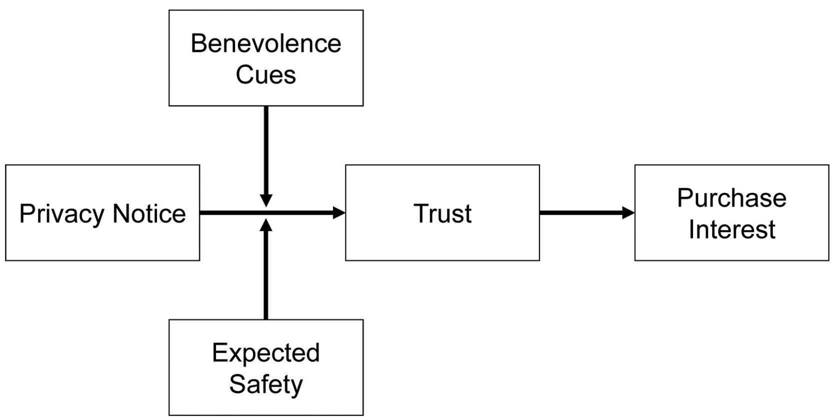


Figure 17.2: p. 742

Even though previous study posits that transparency regarding customer data can **reduce** perceived vulnerability (Martin et al., 2017), (Brough et al., 2022) posit that privacy notices can make costumers feel **more** vulnerable (similar to the bulletproof glass analogy)

The "bulletproof glass effect" is the increased purchase interest resulting from exposure to a privacy notice" (Brough et al., 2022, p. 740)

- When consumers notice that their data have been collected without consent, click-through rates decrease (Kim et al., 2019; Aguirre et al., 2015)
- Consumers prefer to purchase from website with higher levels of privacy protection (Tsai et al., 2011)
- Even though consumers were predicted to be rational in their response to privacy-related info, consumers are more malleable and less intuitive: Consumers are willing to tradeoff privacy in response to
  - architecture and framing (Adjerid et al., 2019; Brandimarte et al., 2013)
  - small inconveniences or incentives (Athey et al., 2017)

  - greater perceived control over personal info (Mourey and Waldman, 2020; Tucker, 2013)

17.2 Organizational Perspective

(Martin et al., 2017): The impacts of a data breach on a company's performance are mitigated by the provision of transparency and control, which compensate for the vulnerabilities highlighted by consumers and mitigate the impact of a data breach on firm performance.

(Martin and Murphy, 2016) The low likelihood of litigation paired with the backdrop of a data breach (a potentially rare event in some industries) suggests that businesses would continue to advocate modest consumer information privacy regulation, keeping the privacy management parts under their control.

17.3 Privacy Paradox

disconnect between consumers' declared privacy desires and their actual privacy policies (even though people say that they are concerned about privacy, they still hare their sensitive personal information freely). (Martin and Murphy, 2016)

(Aguirre et al., 2016)

- Privacy concerns can stem from firm leveraging collected customer data to provide personalized communication.
  - Heighten customer risk perception and decrease trust (Van Slyke et al., 2006)
  - increase skepticism toward and avoidance of ad (Baek and Morimoto, 2012)
  - negative consumer actions (Son and Kim, 2008)
- Privacy can be consider a commodity where consumers can make risk-benefit tradeoff (Sultan et al., 2009)

(Norberg et al., 2007) the privacy paradox

(Johnson et al., 2020)

- Even though consumers express strong privacy concerns in survey, only 0.23% of American ad impressions arise from users who opted out of online behavioral advertising.
- Opt-out users ads get 52% less revenue than that of users with behavioral targeting.
- Without target ads, per American opt-out consumers, publishers and the exchange lose about $8.58
- Opt-out users tend to be more tech savvy.

- Opt-out rates are higher in older and wealthier American cities.

17.3.1 Tradeoff

(Chellappa and Sin, 2005)

- This study examines the usage of online personalization after the trade-off between personalization value and privacy concerns. More specifically, they found that value for personalization will increase the likelihood of using personalized services, but concern for privacy will decrease the likelihood of using personalized services.
  - Trust in vendor positively moderates the use of personalization. Hence, online vendors can leverage trust building activities to acquire and use customer info.
- This study provides support for orthogonality of personalization and privacy, vendors can independently develop personalized products and strategies to alleviate consumers' privacy concerns.
- Personalization is defined as "the ability to proactively tailor products and purchase experience to the tastes of individual consumers based upon their personal and preference information." (p. 181), which depends on
  - Vendors' ability to acquire and process consumer info
  - Consumers' willingness to share info and use personalized services.
- Personalization can increase switching cost (filling out more information if one switches to another competitor) and loyalty (Alba et al., 1997). In the offline it is used to extract implicit price premiums, but in the online context, vendors are expected to do so (p. 183).
  - Online personalization's main benefits are product or service fit and proactive delivery.
- Privacy concerns:
  - "consumer is willing to share her preference information in exchange for apparent benefits, such as convenience, from using personalized products and services" (p. 186). This is a type of social exchange where consumers expect that their expected (personalization) gain is greater than their (privacy) loss
  - Privacy is defined as "an individual's ability to control the terms by which their personal information is acquired and used." (p. 186)

(White, 2004)

- Although individuals with somewhat deep connection views were more likely to reveal "privacy-related" material, they were less likely to reveal humiliating information.

- Although loyal customers found the exchange of privacy-related personal information for customized benefit offerings (relative to noncustomized offerings) attractive, the opposite was true for embarrassing information; these participants appeared to find the exchange of customized offerings for this type of information unattractive.

(Wattal et al., 2012)

- Consumers respond favorably when businesses employ product-based personalisation (where the usage of information is not explicitly mentioned).
- Consumers, on the other side, react unfavorably when businesses are explicit about how they use personally identifiable information (i.e., a personalized greeting).
- consumer familiarity with businesses moderates consumers' unfavorable reactions to personalized greetings.

17.3.2 No tradeoff

(Tucker, 2013) personalized ad and privacy controls in social networks

- found that personalized ad was ineffective before Facebook's introduction of privacy controls, but was twice as effective at attracting users after the shift in Facebook's policy (because they have more control over their personal info)
- when sites can assure consumers that they are in control of their privacy, personalization can generate higher click-through rates.
- DiD from Facebook introduction of more privacy controls for users.

(Walrave et al., 2016) personalization and privacy in social network ad

Contributions:

- Even though this study hypothesize that a moderate level of personalization is optimal in ad effectiveness, they found the highest personalisation condition yielded the most positive response, and privacy concerns did not mitigate its impacts (authors justify that maybe persoanlization is not high enough to trigger the persuasion knowledge). In short, there is no tradeoff between personalization and privacy on brand engagement.

Definitions:

- Personalization in the marketing communication context is defined as "creating persuasive messages that refer to aspects of a person's self" (Maslowska et al., 2011a)
- Evidence of positive impact of personalization on performance
  - on brand responses (Bauer and Lasinger, 2014)

  - on campaign responses (Kim and Sundar, 2012)
  - traditional banner ad context, more positive persuasive effects (Tam and Ho, 2005)
  - email newsletters, more positive persuasive effects (Maslowska et al., 2011b)
  - personalized mobile coupons are evaluated positively and more likely to be redeemed (Xu et al., 2008)
  - Positive relationship with brand engagement (Chu, 2011)

- U-shaped impact of personalization on ad performance:
  - Positive: Personalization can evoke central information processing due to its high self-relevant (Cho and as, 2004). According to information processing theory, consumers are driven to elaborate on relevant messages, which increases attention, elaboration, and strong attitudes (Petty and Cacioppo, 1986). Hence, more personalization, more positive ad response.
  - Negative: personalization increases skepticism toward ad due to the persuasion knowledge model (Friestad and Wright, 1994). Knowing that a brand tries to persuade, one triggers persuasion knowledge which diminish persuasion effects. For example, people resist personalized emails when personal info is highly distintive (White et al., 2008)
    * Consumers privacy concerns can enhance this negative relationship (i.e., reactance).
- Privacy is defined as a "selective control of access to the self or to one's group" (Altman, 1976)
  - People want to strive a balance between openness and closeness (p. 604). When people feel their privacy concerns is threatened by personalization, they have coping strategies to protect their privacy from marketers (Youn, 2009; Grant, 2005)

17.4 Privacy Calculus/ Economics of Privacy

(Bol et al., 2018)

Privacy self-regulation (Martin and Murphy, 2016): more prevalent in the US than EU.

(Bowie and Jamal, 2006)

(Conitzer et al., 2012)

- A monopolist model with a continuum of heterogeneous consumers and the option for them to remain anonymous and avoid being recognized as prior customers, maybe at a cost.
- When customers have the option to keep their anonymity, they all prefer to do so on an individual basis, which generates the most profit for the monopolist.
- Consumers can gain from higher anonymity costs, but only up to a point when the effect turns against them.
- It frequently harms consumers if the monopolist or an unaffiliated third party controls the cost of anonymity.

17.5 GDPR

To code up privacy notice (i.e., detect privacy-related text or not), we can use (Chang et al., 2019) to detect privacy policy under GDPR.

(Janssen et al., 2022)

- Data: 4.1 mil apps on Google Play Store (2016-2019)
- GDPR forces a third of apps exit and app introduction fell by half.
- GPDR reduces
  - consumer surplus
  - app usage
- There is a tradeoff between privacy and innovation.

(Martin et al., 2017)

- Data: mobile app market
- Volatility is defined as the the volatity in top-ranking charts at the category level (a proxy for the competition intensity among apps).
- GDPR increases the volatility of the free app market (pro-competitive effect)
- GDPR decreases the volatility in the paid app market (anti-competitive effect)
- Data privacy regulation
  - Anti-competitive (high compliance costs)
    - Fewer entries
    - more exists
    - Harder for smaller business

  - Pro-competitive (Challenges of data collection of utilization)
    * fines under GDPR is costlier for bigger firms
    * more difficult for established firms to ensure third-party's compliance.

(Kollnig et al., 2021)

- Data: 2 mil Android apps
- GPDR has limited change in the presence of third-party tracking in apps.
- There are still a few big gatekeeper corporations with a disproportionate amount of tracking capability both before and after GDPR

(Betzing et al., 2020)

- when privacy policies are transparent, there is a greater understanding of how data is processed (i.e., comprehension of data processing), but acceptance percentages do not change.

(Kreuter et al., 2020)

- Data: 4,300 invitees and 650 participants
- People are willing to share their data freely even after GDPR.
- No matter how different data requests are made, people are still willing share their data regardless
- Explanations regarding data collection and usage are not read carefully.

(Godinho de Matos and Adjerid, 2022)

- The General Data Protection Regulation (GDPR) is a significant shift in global privacy regulation.
- Focuses on enhanced consumer consent requirements for transparent data allowances.
- Evaluation of the impact of enhanced consent on consumer opt-in behavior and firm behavior and outcomes.
- Utilizing an experiment at a telecommunications provider in Europe, the study finds:
  - Increased opt-in for different data types and uses with GDPR-compliant consent.
  - Consumers did not uniformly increase data allowances and continued to restrict permissions for sensitive information.
  - Increased sales, efficacy of marketing communications, and contractual lock-in after new data allowances provided by consumers.

  - These gains to the firm resulted from the ability to increase targeted marketing efforts for households that were receptive.

- Similar to GPDR, California Consumer Privacy Act (US), The General Data privacy law (Brazil)

Part III

METHODOLOGY

Chapter 18

Metrics

18.1 Finance

18.1.1 Return on Investment (ROI)

$$ROI = \frac{\text{Net Profit}}{\text{Investment}}$$

Similarly, return on marketing investment

$$ROIM = \frac{IRAM - CM}{MS}$$

where

- IRAM = Incremental Revenue Attributable to Marketing
- CM = Cost of the Marketing Investment
- MS = Marketing Spending

If ROMI is positive, then firm is on doing well.

18.1.2 Economic Value Added

- also known as economic profit.
- is a measure based on the residual wealth calculated by deducting its cost of capital from its operating profit, adjusted for taxes on a cash basis.
- measures the value a company generates from its invested funds.
- EVA relies heavily on invested capital. Hence, more suitable for asset-rich companies, whereas companies with intangible assets, such as technology businesses, may not be good candidates.

$$EVA = NOPAT - (InvestedCapital * WACC)$$

where

- NOPAT = Net Operating profit after taxes = Operating Profit x (1-Tax Rate)
- Invested capital = Debt + capital leases + shareholders' equity = Equity + long-term debt at the beginning of the period
- WACC = Weighted average cost of capital (average rate of return a company expects to pay its investors).
  - $WACC = \frac{Ke \times E}{E+D} + \frac{Kd \times (1-t) \times D}{E+D}$
    * Ke = required return on equity
    * Kd(1-t) = after tax return on debt.
- (WACC* capital invested) is also known as a **finance charge**

Invested capital can also be calculated as (Total Assets - Current Liabilities). Hence, the modified version of EVA is:

$$EVA = NOPAT - (\text{total assets} - \text{current liabilities}) * WACC$$

18.1.3 Market Value Added

- is the amount of wealth that a company is able to create for its stakeholders since its foundation.
- (the current market value of the company's stock - the initial invested capital)

$$MVA = \text{market value of shares (or enterprise value)} - \text{book value of shareholders' equity}$$

18.1.4 Unexpected size-adjusted advertising investments

(Chakravarty and Grewal, 2016; Kim and McAlister, 2011; Liu et al., 2017) used

18.1.5 Shareholder Complaints

(Wies et al., 2019) used data from RiskMetrics

18.1.6 Profitability

(Grewal et al., 2008; McAlister et al., 2016)

$$\text{profitability} = \frac{\text{operating income before depreciation}}{\text{total assets}}$$

18.1.7 Firm Size

(Grewal et al., 2010; McAlister et al., 2016; Nezami et al., 2018) used log of total asset in million as firm size

$$\text{firm size} = \log(\text{total assets})$$

18.1.8 Sales Growth

(Grewal et al., 2010; Nezami et al., 2018; Rao et al., 2004)

Percentage change in gross sales = sales growth

18.1.9 Financial Flexibility

18.1.10 Cash flows

(Chakravarty and Grewal, 2011; Malshe and Agarwal, 2015) Cash flows = log(operating cash flow in mil)

18.1.11 Financial Leverage

(Kashmiri and Mahajan, 2017; Chakravarty and Grewal, 2011)

$$\text{financial leverage} = \frac{\text{long-term debt}}{\text{book value of assets}}$$

18.1.12 Stock return

(Markovitch et al., 2005; Chakravarty and Grewal, 2011, 2016) used dummy if stock returns exceed industry averaged stock returns

18.1.13 Financial Flexibility

18.1.14 Book Equity

Following Fama and French's definition:

"BE is the book value of stockholders' equity, plus balance sheet deferred taxes and investment tax credit (if available), minus the book value of preferred stock. Depending on availability, we use the redemption, liquidation, or par value (in that order) to estimate the book value of preferred stock. Stockholders' equity is the value reported by Moody's or Compustat, if it is available. If not, we measure stockholders' equity as the book value of common equity plus the par value of preferred stock, or the book value of assets minus total liabilities (in that order)" (cite (Davis et al., 2000))

```r
library(tidyverse)
seq - coalesce(pstkrv, pstkl, pstk, 0) + coalesce(txdtc, 0)
```

18.1.15 Net Contribution

Net contribution = gross margin x sales - cost of marketing

$$NC = m \times S(a) - ka$$

Table 18.1: All data without CSRP come from Compustat/Fundamentals

Variable	Data Item	Source
Book Value on Equity	PRCC_C x CSHO	PRCC_C: CRSP/Annual Update/CRSP/Compustat Merged/Fundamental Annual/Supplemental Data Items CSHO: CRSP/Annual Update/CRSP/Compustat Merged/Fundamental Annual /Miscellaneous Items
CAPEX	CAPX	Cash Flow Items
Capital Intensity	CAPX / AT	CAPX: Cash Flow Items AT: Balance Sheet Items
Cash	CH	Balance Sheet Items
Cash and short-term Investments	CHE	Balance Sheet Items
Cash Flow	$\frac{IBC+DP}{AT}$	IBC: Cash Flow Items DP: Income Statement Items AT: Balance Sheet Items
Cash Holdings	$\frac{CHE}{AT}$	CHE: Balance Sheet Items AT: Balance Sheet Items
Closing Price (annual, calendar)	PRCC_C	Supplement Data Items
Closing Price (annual, fiscal)	PRCC_F	Supplement Data Items
Cost of Capital	$\frac{XINT}{DLC}$	XINT: Income Statement Items DLC: Balance Sheet Items
Current Assets Total	ACT	Balance Sheet Items

Variable	Data Item	Source
Current Liabilities	LCT	Balance Sheet Items
Debt in Current Liabilities	DLC	Balance Sheet Items
Dividends (common)	DVC	Income Statement Items
Dividends (preferred)	DVP	Income Statement Items
Dividends (total)	DVT	Income Statement Items
Earning per Shares	$\frac{NI}{CSHO}$	NI: Income Statement Items CSHO: Miscellaneous Items
EBIT	EBIT	Income Statement Items
EBITDA	EBITDA	Income Statement Items
Equity Book Value	BKVLPS	Balance Sheet Items
Firm Size	log(AT)	AT: Balance Sheet Items
Income before Extraordinary Items	IB	Income Statement Items
Interest on long-term debt	UXINTD	Income Statement Items
Interest on short-term Debt	UXINST	Income Statement Items
Leverage	$\frac{DLTT+DLC}{SEQ}$	Balance Sheet Items
Long-term Assets	AT - ATC	Balance Sheet Items
Long-term Debts	DLTT	Balance Sheet Items
Market to Book Ratio	$\frac{MKVALT}{BKVLPS}$	MKVALT: Supplement Data Items BKVLPS: Balance Sheet Items
Market Value	MKVALT or CSHO x PRCC_F	MKVALT: Supplement Data Items CSHO: Miscellaneous Items PRCC_F: Supplement Data Items
Net income (Loss)	NI	Income Statement Items
Payout Ratio	$\frac{DVP+DVC+PRSTKC}{IB}$	DVP: Income Statement Items DVC: Income Statement Items PRSTKC: Cash Flow Items IB: Income Statement Items
Property, Plant, and Equipment	PPENT	Balance Sheet Items
Purchase of common and preferred Stocks	PRSTKC	Cash Flow Items
R&D Intensity	$\frac{XRD}{AT}$	XRD: Income Statement Items AT: Balance Sheet Items

Variable	Data Item	Source
ROA	$\frac{NI}{AT}$	NI: Income Statement Items AT: Balance Sheet Items
ROE	NI/(CSHO x PRCC_F)	NI: Income Statement Items CSHO: Miscellaneous Items PRCC_F: Supplement Data Items
ROI	$\frac{NI}{ICAPT}$	NI: Income Statement Items ICAPT: Balance Sheet Items
Sale	SALE	Income Statement Items
Short-term Liabilities	APC	Balance Sheet Items
Stockholders Equity (total)	SEQ	Balance Sheet Items
Tangibility	$\frac{PPENT}{AT}$	PPENT: Balance Sheet Items AT: Balance Sheet Items
Tobin's Q	$AT + (CSHOxPRCC_F) - CEQ/$(AT)	AT: Balance Sheet Items CSHO: Miscellaneous Items PRCC_F Supplement Data Items CEQ: Balance Sheet Items
Total Assets	AT	Balance Sheet Items
Total Equity	PSTKC + CSHO	PSTKC: Balance Sheet Items CSHO: Miscellaneous Items
Total Liabilities	LT	Balance Sheet Items

Information is taken from WRDS Data Items

Variable	Data Item	Data Source
Age		CRSP
Valuation		
Market Cap/ GDP		series GDPA (U.S. Bureau of Economic Analysis)
Tobin's q	(AT (CSHO * PRCC_F) - CEQ)/(AT)	
Market cap (000s)	$prc \times shrout$	CRSP
Small firm	Market cap $<$ \$ 100M	
Revenue Herfindahl (for each 3-digit NAICS in each year)	$\frac{revt_i^2}{\sum_1^n revt}$ where i = the firm, n = firms in the same industry	
Investment		

Variable	Data Item	Data Source
Capital Expenditures / Assets	$\frac{capx}{lag(at)}$	
R&D/Assets	$\frac{xrd}{lag(at)}$	If R&D is missing, set to 0
Fixed Assets/ Assets	$\frac{ppent}{at}$	
Inventory / Assets	$\frac{invt}{at}$	
Cash / Assets	$\frac{che}{at}$	
Profitability		
Operating cashflow / assets	$\frac{oidbp - xint - txt}{lag(at)}$	Operating income before depreciation (oibdp) minus interest (xint) minus taxes (txt), divided by lagged assets
Loss firms	% of firm with net income (ni) < 0	
R&D-adjusted operating cash flow/ assets	$CF/at + RD/at$	
ROA	$\frac{ib}{at}$	
Financing		
Book Leverage	$\frac{dltt+dlc}{at}$	
Market leverage	(dltt + dlc)/(at - ceq + (chso x prcc_f)	
Net Leverage	$\frac{dltt+dlc-che}{aat}$	
Negative net leverage firms	% of firms with Net Leverage < 0	
Interest/ Assets	$\frac{xint}{lag(at)}$	
No debt firms	% with no dltt or dlc	
Net equity issuance	$\frac{sstk-prstkc}{lag(at)}$	
Ownership		
Institutional ownership	% of shares outstanding held by institution	Thomson Financial 13f data

Variable	Data Item	Data Source
Blockholder	% of firms with an institutional owner who holds 10 percent or more of outstanding shares	Thomson Financial 13f data
Payout Policy		
Dividend paying firms	% of firms with dvc > 0	
Dividends / Assets	$\frac{dvc}{lag(at)}$	
Repurchase / Assets	$\frac{prstkc-pstk}{lag(at)}$	
Total payout/ assets	$\frac{dvc+prstkc}{lag(at)}$	
Total payout/ Net income	$\frac{dvc+prstkc}{ni}$	

Info from (Kahle and Stulz, 2017) Appendix

18.1.16 Diversity

Simpson Diversity index measures the market share of each sector

- Equivalent to the Herfindahl index (in economics)

18.2 Marketing

18.2.1 Trust

Measuring trust algorithmically using social media data (Roy et al., 2017)

18.2.2 Sentiment

(Hartmann et al., 2023) Accuracy and Application of Sentiment Analysis

- Sentiment is core to human communication.
- Marketing uses sentiment analysis for:
  - Social media.
  - News articles.
  - Customer feedback.

  - Corporate communication.
- Available sentiment analysis methods:
  - Lexicons: link words/expressions to sentiment scores.
  - Machine learning: complex but potentially more accurate.
- Study introduces an empirical framework to:
  - Evaluate method suitability based on research questions, data, and resources.
- Meta-analysis conducted on:
  - 272 datasets.
  - 12 million sentiment-labeled documents.
- Findings:
  - Transfer learning models top performance.
  - These models may not always meet leaderboard benchmarks.
  - Transfer learning models are, on average, >20% more accurate than lexicons.
- Performance influenced by:
  - Number of sentiment classes.
  - Text length.
- Study offers:
  - SiEBERT - a pre-trained sentiment analysis model.
  - Open-source scripts for easy application.

18.2.3 Purchase Intention

(Hartmann et al., 2021) The Power of Brand Selfies

- RoBERTa-based model: https://huggingface.co/j-hartmann/purchase-intention-english-roberta-large
- Smartphones simplify sharing branded imagery.
- Study categorizes social media brand imagery.
- Identified image types:
  - Packshots (just the product).
  - Consumer selfies (consumer's face with brand).
  - Brand selfies (product held, no visible consumer).

- Convolutional neural networks used to recognize image types.
- Language models analyze social media responses to 250,000+ brand-image posts from 185 brands on Twitter & Instagram.
- Findings:
  - Consumer selfies lead to more likes and comments.
  - Brand selfies induce higher purchase intentions.
- Traditional social media metrics may not fully capture brand engagement.
- Display ad results:
  - Higher click-through rates for brand selfies than consumer selfies.
- Lab experiment indicates self-reference affects image responses.
- Machine learning can decipher marketing insights from multimedia content.
- Image perspective impacts actual brand engagement.

18.2.4 Brand Reputation

- Measuring brand reputation from Twitter data (Rust et al., 2021)

18.2.5 Capabilities

(Dutta et al., 1999) Marketing capability is critical for high-technology markets

Insights

- A company's sales are increased by a solid foundation of innovative technologies that positively affect consumers' perceptions of the benefits of its product's externalities.
- For businesses with a strong technological foundation, marketing capability has the biggest impact on the production of innovation that has been quality-adjusted. The companies that stand to benefit the most from great marketing capabilities are those with a solid R&D foundation.
  - because a company may have tremendous R&D capabilities but be unable to translate them into commercially viable items due to weak marketing skills
- The interaction of marketing and R&D capabilities is the most significant factor in determining a firm's performance.
  - High tech firm need to have both the ability to come up with innovation constantly and the ability to commercialize these innovations.

- A firm's capability is defined as "its ability to deploy the resources (inputs) available to it to achieve the desired objective(s) (output)."
  - Logically, **the higher the functional capability a firm has, the more efficiently it is able to deploy its productive inputs to achieve its functional objectives**.
  - Equivalently, the lower the functional inefficiency, the higher the functional capability of the firm

Operationalization

Marketing Capability

Sales = f(technological base, advertising stock, stock of marketing expenditure, investment in customer relationships, installed base)

Weight for marketing expenditure = 0.5 (Dutta et al., 2005, p. 281)

Advertising stock weight = 0.4 (Peles, 1971) or 0.5 (Wang and Kim, 2017)

R&D Capability

2 dimensions of the quality of technological output

- innovativeness (Trajtenberg, 1990b,a)
- width of applicability (Jaffe et al., 1993)

Quality-adjusted technological output = f(technological base, cumulative R&D expenditure, marketing capability)

R&D expenditure weight = 0.4 (Dutta et al., 2005, p. 281)

Operations Capability

Cost of production = f(output, cost of capital, labor cost, technological base, marketing capability)

Variables

Label	Variable
	Sales Did not use raw patent count because quality matters. Innovative-adjusted technological output The number of times the patents of a firm have been cited (citation-weighted patent count)

Replication

As the time of this writing, the US Patents by WRDS dataset only covers from 2011 to 2019. Hence, I can only replicate the results in this period but not the years in the study (1985 - 1994). Alternatively, you can visit USPTO to download the raw dataset and create your own matching algorithm based on company names.

`wrdsapps_patents_link` is from US Patents by WRDS / Compustat Link

`wrdsapp_patents` is from US Patents by WRDS / Patents

- forward citations counts as of Dec 31st 2019

`wrdsapp_patents_citations` is from US Patents by WRDS / Citations

```
library(tidyverse)
library(lubridate)

totalq <- read.csv(file.path("data/totalq.csv.gz"))

capability <- read.csv(file.path("data/capability.csv.gz")) %>%
    # fix the length of numeric variables
    mutate(gvkey = str_pad(as.character(gvkey),width = 6, pad = "0", side = "left"),
           sic = str_pad(as.character(sic), width = 4, pad = "0", side = "left")) %>%

    # need to have data on advertising, r&d, sga, cogs
    filter(!is.na(xad) & !is.na(xrd) & !is.na(xsga) & !is.na(cogs))


wrdsapps_patents_link <-
    read.csv("data/patents/wrdsapps_patents_link.csv.gz") %>%
    mutate(gvkey = str_pad(
        as.character(gvkey),
        width = 6,
        pad = "0",
        side = "left"
    )) %>%

    # get industry data
    inner_join(capability %>% select(gvkey, sic) %>% distinct(), by = "gvkey")

# view(head(wrdsapps_patents_link))

wrdsapps_patents <- read.csv("data/patents/wrdsapps_patents.gz")
# view(head(wrdsapps_patents))

wrdsapps_patents_citations <- read.csv("data/patents/wrdsapps_patents_citations.csv.gz
    # correct dates
    mutate(grantdate = as.Date(as.character(grantdate), "%Y%m%d"),
           cited_pat_gdate = as.Date(as.character(cited_pat_gdate), "%Y%m%d")) %>%

    # get year
```

```
    # grant year here is the year that the new patent that cite the old one was granted.
    mutate(grantyear = year(grantdate),
           cited_pat_gyear = year(cited_pat_gdate)) %>%

    # get only those patents belong to firms in the Compustat database
    left_join(
        wrdsapps_patents_link %>%
            select(patnum, sic) %>%
            rename(cited_patnum_sic = sic),
        by = c("cited_patnum" = "patnum")
    ) %>%
    left_join(
        wrdsapps_patents_link %>%
            select(patnum, sic) %>%
            rename(patnum_sic = sic),
        by = c("patnum" = "patnum")
    )


# view(head(wrdsapps_patents_citations, 100))

wrdsapps_patents_citations_by_year <- wrdsapps_patents_citations %>%

    select(patnum, grantyear, cited_patnum) %>%

    count(cited_patnum, grantyear) %>%

    rename(citation_count = n)




# check
# wrdsapps_patents_citations_by_year %>%
#     filter(cited_patnum == "01044494") %>%
#     view()


# wrdsapps_patents_citations %>%
#     filter(cited_patnum == "01044494") %>%
#     view()
```

Innovativeness-adjusted technological output:

1. Calculate the average number of citations receive by all patents belonging the firms in the sample within one industry (defined by sic1, sic2, sic3,

sic4)

2. The weight assigned to a firm's patents = # of citations / industry sample average
3. Tech_innv = sum of citation-weighted patents within a firm in a year.

```
# calculate industry average citation based on
# sic1
tech_innv1 <- aggregate(x = tech_innv$citation_count,
                        by = list(tech_innv$sic1),
                        FUN = mean) %>%
    rename(sic1 = 1,
           mean_cite_ind1 = 2)

# sic2
tech_innv2 <- aggregate(x = tech_innv$citation_count,
                        by = list(tech_innv$sic2),
                        FUN = mean) %>%
    rename(sic2 = 1,
           mean_cite_ind2 = 2)

# sic3
tech_innv3 <- aggregate(x = tech_innv$citation_count,
                        by = list(tech_innv$sic3),
                        FUN = mean) %>%
    rename(sic3 = 1,
           mean_cite_ind3 = 2)

# sic4
tech_innv4 <- aggregate(x = tech_innv$citation_count,
                        by = list(tech_innv$sic),
                        FUN = mean) %>%
    rename(sic = 1,
           mean_cite_ind4 = 2)

# merge all four sic types together
tech_innv_sic <- tech_innv4 %>%
    mutate(
        sic3 = substr(sic, 1, 3),
        sic2 = substr(sic, 1, 2),
        sic1 = substr(sic, 1, 1)
    ) %>%
    full_join(tech_innv3, by = "sic3") %>%
    full_join(tech_innv2, by = "sic2") %>%
    full_join(tech_innv1, by = "sic1") %>%

    select(-c(sic1, sic2, sic3))
```

```
# note that a modification from Dutta's paper is that I use the average number of citations rece
# Dutta's paper did not do this because they consider only one industry

# calculate the Tech_innv
tech_innv <- wrdsapps_patents_link %>%
    select(gvkey, patnum, sic) %>%

    inner_join(wrdsapps_patents_citations_by_year, by = c("patnum" = "cited_patnum")) %>%

    # match with the firm and year data
    filter(gvkey %in% capability$gvkey) %>%

    inner_join(tech_innv_sic, by = "sic") %>%

    # create the weight assigned to a firm's patent
    mutate(weight1 = citation_count / mean_cite_ind1,
           weight2 = citation_count / mean_cite_ind2,
           weight3 = citation_count / mean_cite_ind3,
           weight4 = citation_count / mean_cite_ind4) %>%

    group_by(gvkey, grantyear) %>%
    # create the citation-weighted patent count
    summarize(tech_innv1 = sum(weight1),
              tech_innv2 = sum(weight2),
              tech_innv3 = sum(weight3),
              tech_innv4 = sum(weight4)) %>%
    ungroup()

rm(tech_innv_sic,
   tech_innv1,
   tech_innv2,
   tech_innv3,
   tech_innv4)
```

Width-of-applicability-technological output 1. Calculate the proportion of citations received by a patent from firms belonging outside of focal SIC code (sic1, sic2, sic3, sic4) = # of citations received by a patent from firms outside the focal SIC code / the total number of citation the patent received. 2. The weight assigned to a firm's patent = the proportion of outside citations for the patent/ the industry sample average proportion 3. Tech_width = The sum of the "proportion-of-outside citation"-weighted patent in a year for a firm

```
wrdsapps_patents_citations_out_ind <- wrdsapps_patents_citations %>%
    # select patent number with data on industry
    filter(!is.na(cited_patnum_sic) & !is.na(patnum_sic)) %>%
```

```
    # create variables indicating whether the new and old patents belong in the same i
    mutate(sim4 = if_else(patnum_sic == cited_patnum_sic, 1, 0),
       sim3 = if_else(substr(patnum_sic,0,3) == substr(cited_patnum_sic,0,3), 1, 0),
       sim2 = if_else(substr(patnum_sic,0,2) == substr(cited_patnum_sic,0,2), 1, 0),
       sim1 = if_else(substr(patnum_sic,0,1) == substr(cited_patnum_sic,0,1), 1, 0)) %

    select(
        patnum,
        cited_patnum,
        grantyear,
        cited_pat_gyear,
        cited_patnum_sic,
        patnum_sic,
        contains("sim")
    )

total_citation <- wrdsapps_patents_citations_out_ind %>%
    group_by(cited_patnum, cited_pat_gyear) %>%
    count()

# view(head(total_citation))

# get per patent, per year, the number of outside-industry citations
# sic1
outside_citation_sic1 <- wrdsapps_patents_citations_out_ind %>%
    group_by(cited_patnum, cited_pat_gyear, sim1) %>%
    count() %>%
    ungroup() %>%
    # get only citations outside of the patent's industry
    filter(sim1 == 0) %>%
    select(-c(sim1)) %>%
    rename(n_sim1 = n)

# sic2
outside_citation_sic2 <- wrdsapps_patents_citations_out_ind %>%
    group_by(cited_patnum, cited_pat_gyear, sim2) %>%
    count() %>%
    ungroup() %>%
    # get only citations outside of the patent's industry
    filter(sim2 == 0) %>%
    select(-c(sim2)) %>%
    rename(n_sim2 = n)

# sic3
outside_citation_sic3 <- wrdsapps_patents_citations_out_ind %>%
```

```
    group_by(cited_patnum, cited_pat_gyear, sim3) %>%
    count() %>%
    ungroup() %>%
    # get only citations outside of the patent's industry
    filter(sim3 == 0) %>%
    select(-c(sim3)) %>%
    rename(n_sim3 = n)

# sic4
outside_citation_sic4 <- wrdsapps_patents_citations_out_ind %>%
    group_by(cited_patnum, cited_pat_gyear, sim4) %>%
    count() %>%
    ungroup() %>%
    # get only citations outside of the patent's industry
    filter(sim4 == 0) %>%
    select(-c(sim4)) %>%
    rename(n_sim4 = n)


tech_width <- total_citation %>%

    full_join(outside_citation_sic1, by = c("cited_patnum", "cited_pat_gyear")) %>%
    full_join(outside_citation_sic2, by = c("cited_patnum", "cited_pat_gyear")) %>%
    full_join(outside_citation_sic3, by = c("cited_patnum", "cited_pat_gyear")) %>%
    full_join(outside_citation_sic4, by = c("cited_patnum", "cited_pat_gyear")) %>%


    # fill in 0 (those patents without any citations outside of their industry)
    replace(is.na(.), 0) %>%

    mutate(weight1 = n_sim1 / n,
           weight2 = n_sim2 / n,
           weight3 = n_sim3 / n,
           weight4 = n_sim4 / n) %>%

    # get company gvkey for each patent
    inner_join(wrdsapps_patents_link %>%
                   select(gvkey, patnum, sic),
               by = c("cited_patnum" = "patnum")) %>%

    # step 3: get proportion-of-outside citation-weighted patent
    group_by(gvkey, cited_pat_gyear) %>%
    summarize(tech_width1 = sum(weight1),
              tech_width2 = sum(weight2),
              tech_width3 = sum(weight3),
```

```
            tech_width4 = sum(weight4)) %>%
    ungroup()


rm(outside_citation_sic1,
   outside_citation_sic2,
   outside_citation_sic3,
   outside_citation_sic4,
   total_citation)

rm(wrdsapps_patents,
   wrdsapps_patents_citations,
   wrdsapps_patents_citations_by_year,
   wrdsapps_patents_citations_out_ind,
   wrdsapps_patents_link)
```

18.2.5.1 Marketing Capabilities

- Bahadir et al. (2008)
- Marketing Capability: (Xiong and Bharadwaj, 2013)
- Wiles et al. (2012)
- (Mishra and Modi, 2016)
- Dinner et al. (2018)
- Marketing Alliance Capabilities: Swaminathan and Moorman (2009)
- Digitized Selling Capabilities: (Johnson, 2005)
- Big Data Capabilities: (Johnson et al., 2017)
- Social CRM Capabilities: (Trainor et al., 2014)
- Customer relationship capabilities: (Wang and Kim, 2017)

```
library(frontier)

mar_cap <-
    sfa(
        log(sales) ~ sub_market,
        log(adstock) + log(marketingstock) + log(techbase) + log(receivable) + log(ins
        data = capability,
        ineffDecrease = T, # inefficiency decreases the endogenous variable for estima
        timeEffect = T # Error Compoennts Frontier are time-variant
    )

efficiencies(mar_cap)
```

```
# defunct
# install.packages("FEAR")

# install.packages("sfaR")
library(sfaR)
mar_cap <-
    sfacross(
        log(sales) ~ sub_market,
        log(adstock) + log(marketingstock) + log(techbase) + log(receivable) + log(installedbase)
        data = capability
        # ineffDecrease = T, # inefficiency decreases the endogenous variable for estimating a pr
        # timeEffect = T # Error Compoennts Frontier are time-variant
    )

efficiencies(mar_cap)
```

(Morgan et al., 2009) Linking marketing capabilities with profit growth

- **Objective:** Examine the connection between a firm's marketing capabilities and its profit growth, with a particular emphasis on how specific marketing functions impact the composite components of profit growth.
- **Context:** Although profit growth stands as a pivotal determinant of a firm's stock price, there's a limited understanding of how integral marketing capabilities interlink with this growth trajectory.
- **Methodology:**
  - Engaged a cross-industry dataset from 114 firms.
  - Delineated the exploration into three cornerstone marketing capabilities: market sensing, brand management, and customer relationship management (CRM).
  - These capabilities were then juxtaposed with the dual facets of profit growth: revenue growth and margin growth.
- **Key Findings:**
  - The scrutinized marketing capabilities exhibited both direct and synergistic influences on the growth rates of revenue and margin.
  - A pivotal revelation was the counteractive effects of brand management and CRM capabilities on the growth rates of revenue and margin. Specifically, while one capability might promote revenue growth, it might simultaneously inhibit margin growth (and vice versa).
  - Such intricate dynamics imply that a surface-level analysis, overlooking the nuanced contributions to revenue and margin growth, could

obscure the genuine relationships tying marketing capabilities to the overarching profit growth trajectory.

18.2.5.2 Digital Marketing Capabilities

Survey:

- (Wang, 2020)
- (Homburg and Wielgos, 2022)

Review:

(Herhausen et al., 2020)

18.2.5.3 Social Media Strategics Capabilities

(Nguyen et al., 2015)

18.2.5.4 Organization Capabilities

(Grewal and Slotegraaf, 2007) Embeddedness of Organizational Capabilities

- **Core Issue:**
  - Managers need to efficiently use limited resources to build lasting organizational capabilities for a sustainable competitive edge.
- **Challenge:**
  - Neglecting the intricate underlying processes of organizational capabilities can hinder understanding their impact on competitive advantage.
- **Key Insight:**
  - Managerial decisions regarding resource usage directly impact the depth at which capabilities are ingrained in the organization, termed "capability embeddedness."
- **Research Method:**
  - Introduced a hierarchical composed error structure framework.
  - Uses cross-sectional data and can be applied to panel data.
- **Case Study: Retailing**
  - Organizational capability embeddedness directly influences retailer performance.
  - This influence remains significant even when considering both tangible and intangible resources/capabilities.
- **Takeaways:**

1. Acknowledging how resources and capabilities affect performance at various organizational layers helps managers make informed decisions.
2. Essential to recognize whether the objectives of capabilities align (convergent) or differ (divergent).
3. The alignment or misalignment of these objectives can dictate how much embedded capabilities boost firm performance.

Chapter 19

Data

19.1 MongoDB

in `Mongo` shell to check the location of your MongoDB files

```
db.adminCommand("getCmdLineOpts")
```

19.2 WRDS

Retrieve data from WRDS

```r
# to set up connection from R to WRDS (https://wrds-www.wharton.upenn.edu/pages/support/programm
library(RPostgres)
library(dplyr)

# I've set up wrds connection before hand. Please use your username and password here.

# wrds <- dbConnect(Postgres(),
#                   host='wrds-pgdata.wharton.upenn.edu',
#                   port=9737,
#                   dbname='wrds',
#                   sslmode='require',
#                   user='')

# Check variables (column headers) in COMP ANNUAL FUNDAMENTAL
#uses the already-established wrds connection to prepare the SQL query string and save the query
# check avaiable databases: https://wrds-web.wharton.upenn.edu/wrds/tools/variable.cfm?vendor_id=
res <- dbSendQuery(wrds, "select column_name
                   from information_schema.columns
                   where table_schema='compa'
```

```
                   and table_name='funda'
                   order by column_name")
data <- dbFetch(res, n=-1) # fetches the data that results from running the query res 
dbClearResult(res) # closes the connection
head(data)

# select everything
res <- dbSendQuery(wrds, "select * from compa.funda")

# from compa.funda

# only select the following variables
res <- dbSendQuery(wrds, "select gvkey, datadate, fyear, indfmt, consol, popsrc, dataf

# res <- dbSendQuery(wrds, "select gvkey, gp from compa.funda") #check variables from 
data1 <- dbFetch(res, n=-1)
dbClearResult(res)

data = data1 %>%
    distinct(gvkey,datadate,fyear,tic,conm,.keep_all = T)
```

19.3 YouTube

- YouTube Ads
- YouTube Analytics and Reporting APIs: All YouTube Analytics and YouTube Reporting API requests must be authorized by the channel or content owner that owns the requested data

19.3.1 OAuth

Go to link to access your API

```
library(tuber)
app_id = "YOUR APP ID"
app_secret = "YOUR APP SECRET"
yt_oauth(app_id, app_secret)


get_stats(video_id = "N708P-A45D0")
```

Note for Ubuntu user

```
httr::set_config(httr::config( ssl_verifypeer = 0L ) )
```

Get info about a video

```
get_video_details(video_id="N708P-A45D0")
```

Get caption of a video

```
get_captions(video_id="yJXTXN4xrI8")
```

Search Videos

```
res <- yt_search(term = "test")
head(res[, 1:3])
```

Get Comments of a video

```
res <- get_comment_threads(c(video_id="N708P-A45D0"))
head(res)
```

Get All the Comments Including Replies

```
get_all_comments(video_id = "a-UQz7fqR3w")
```

Get statistics of all the videos in a channel

```
a <- list_channel_resources(filter = c(channel_id = "UCT5Cx1l4IS3wHkJXNyuj4TA"), part="contentDet

# Uploaded playlists:
playlist_id <- a$items[[1]]$contentDetails$relatedPlaylists$uploads

# Get videos on the playlist
vids <- get_playlist_items(filter= c(playlist_id=playlist_id))

# Video ids
vid_ids <- as.vector(vids$contentDetails.videoId)

# Function to scrape stats for all vids
get_all_stats <- function(id) {
  get_stats(id)
}

# Get stats and convert results to data frame
res <- lapply(vid_ids, get_all_stats)
res_df <- do.call(rbind, lapply(res, data.frame))

head(res_df)
```

19.3.2 API

```
require(curl)
require(jsonlite)
```

```
library(kableExtra)
library(dplyr)
library(ggplot2)
library(plotly)
library(reshape2)

API_key = "YOUR API KEY"

getstats_video<-function(video_id,API_key){

  url=paste0("https://www.googleapis.com/youtube/v3/videos?part=snippet,statistics&id="
  result <- fromJSON(txt=url)
  salida=list()
  return(data.frame(name=result$items$snippet$channelTitle, result$items$statistics,ti
}

get_playlist_canal<-function(id,API_key,topn=15){

  url=paste0('https://www.googleapis.com/youtube/v3/playlistItems?part=contentDetails&p

  result=fromJSON(txt=url)
  return(data.frame(result$items$contentDetails))
}

getstats_canal<-function(id,API_key){

url=paste0('https://www.googleapis.com/youtube/v3/channels?part=snippet%2CcontentDetai

result <- fromJSON(txt=url)
return(data.frame(name=result$items$snippet$title,result$items$statistics,pl_list_id=r

}

getall_channels<-function(ids,API_key,topn=5){

  videos=lapply(ids,FUN=get_playlist_canal,API_key=API_key,topn=topn) %>% bind_rows()

  stats=lapply(videos[,1],FUN=getstats_video,API_key=API_key)
  stats=bind_rows(stats)
  stats$vid_id=videos[,1]
  return(stats)
}
```

Statistics per Channel

```
can_st=lapply(comp_data$cha_id,FUN = getstats_canal,API_key=API_key)
can_st=bind_rows(can_st)
can_st$viewCount=as.numeric(can_st$viewCount)
can_st[,1:6] %>% kable() %>%kable_styling()
```

```
can_st$viewCount=round(as.numeric(can_st$viewCount)/1000000,2)
p1=can_st %>% ggplot(aes(x=reorder(name,viewCount),y=viewCount,fill=name))+
geom_bar(stat="sum")+guides(size=F)+coord_flip()+scale_fill_manual(values = c("red", "darkblue",
geom_text(inherit.aes = T,aes(label=paste(viewCount,"M")),nudge_y =0,angle = 90)+
labs(x="Total Visualizations(Millions)",y="Visualizations",fill="")+
theme(legend.position = "top")+mytheme3

  ggplotly(p1,tooltip=c("name","viewCount"))  %>%
    layout(legend = list(orientation = "h",x = 0.01, y = -0.1,autosize=F))
```

Information on Individual Videos per Channel

```
var_to_see="dislikeCount" #favoriteCount or commentCount

info=getall_channels(ids = can_st$pl_list_id.uploads,API_key = API_key,topn =20)


 datacond=melt(info[,c(1:6,8)],id.vars = c("name","date"))
 datacond$date=as.Date(datacond$date)
 datacond$value=as.numeric(datacond$value)


ggplot(filter(datacond,variable==var_to_see),aes(x=as.Date(date),y=value,fill=name))+
geom_bar(stat="sum")+labs(x=var_to_see,y="",fill="")+guides(size=FALSE)+scale_fill_manual(values
  scale_x_date(limits =as.Date(c(as.Date(min(datacond$date)),as.Date(Sys.time()))),date_breaks  =
```

19.3.3 Python

Pytube

```
# from pytube import YouTube
# YouTube('https://youtube.com/watch?v=2lAe1cqCOXo').streams.first().download()
```

```
import requests
import json
r = requests.get("http://gdata.youtube.com/feeds/api/standardfeeds/top_rated?v=2&alt=jsonc")
#r.text
data = json.loads(r.text)
data
# for item in data['data']['items']:
#   print "Video Title: %s" % (item['title'])
#   print "Video Category: %s" % (item['category'])
```

```
#   print "Video ID: %s" % (item['id'])
#   print "Video Rating: %f" % (item['rating'])
#   print "Embed URL: %s" % (item['player']['default'])
```

Using BeautifulSoup

```
from requests_html import HTMLSession
from bs4 import BeautifulSoup as bs # importing BeautifulSoup

# sample youtube video url
video_url = "https://www.youtube.com/watch?v=jNQXAC9IVRw"
# init an HTML Session
session = HTMLSession()
# get the html content
response = session.get(video_url)
# execute Java-script
response.html.render(sleep=1)
# create bs object to parse HTML
soup = bs(response.html.html, "html.parser")
# write all HTML code into a file
open("video.html", "w", encoding='utf8').write(response.html.html)
```

Generate random video

```
import json
import urllib.request
import string
import random

count = 50
API_KEY = 'your_key'
random = ''.join(random.choice(string.ascii_uppercase + string.digits) for _ in range(

urlData = "https://www.googleapis.com/youtube/v3/search?key={}&maxResults={}&part=snip
webURL = urllib.request.urlopen(urlData)
data = webURL.read()
encoding = webURL.info().get_content_charset('utf-8')
results = json.loads(data.decode(encoding))

for data in results['items']:
    videoId = (data['id']['videoId'])
    print(videoId)
    #store your ids
```

Random used by YouTube API, test

```
# -*- coding: utf-8 -*-

# Sample Python code for youtube.search.list
# See instructions for running these code samples locally:
# https://developers.google.com/explorer-help/guides/code_samples#python

import os

import google_auth_oauthlib.flow
import googleapiclient.discovery
import googleapiclient.errors

scopes = ["https://www.googleapis.com/auth/youtube.force-ssl"]

def main():
    # Disable OAuthlib's HTTPS verification when running locally.
    # *DO NOT* leave this option enabled in production.
    os.environ["OAUTHLIB_INSECURE_TRANSPORT"] = "1"

    api_service_name = "youtube"
    api_version = "v3"
    client_secrets_file = "YOUR_CLIENT_SECRET_FILE.json"

    # Get credentials and create an API client
    flow = google_auth_oauthlib.flow.InstalledAppFlow.from_client_secrets_file(
        client_secrets_file, scopes)
    credentials = flow.run_console()
    youtube = googleapiclient.discovery.build(
        api_service_name, api_version, credentials=credentials)

    request = youtube.search().list(
        part="snippet",
        maxResults=25,
        q="surfing"
    )
    response = request.execute()

    print(response)

if __name__ == "__main__":
    main()
```

Another way, but this is a dirty crawler, not for production

```
import re, urllib
from random import randint
```

```
def random_str(str_size):
    res = ""
    for i in xrange(str_size):
        x = randint(0,25)
        c = chr(ord('a')+x)
        res += c
    return res

def find_watch(text,pos):
    start = text.find("watch?v=",pos)
    if (start<0):
        return None,None
    end = text.find(" ",start)
    if (end<0):
        return None,None

    if (end-start > 200): #silly heuristics, probably not a must
        return None,None

    return text[start:end-1], start


def find_instance_links():
    base_url = 'https://www.youtube.com/results?search_query='
    url = base_url+random_str(3)
    #print url

    r = urllib.urlopen(url).read()

    links = {}

    pos = 0
    while True:
        link,pos =  find_watch(r,pos)
        if link == None or pos == None:
            break
        pos += 1
        #print link
        if (";" in link):
            continue
        links[link] = 1

    items_list = links.items()

    list_size = len(items_list)
```

```
    selected = randint(list_size/2,list_size-1)
    return items_list[selected][0]


for i in xrange(1000):
    sleep(randint(7,20)) # pause randomly between 7 and 20 seconds
    link = find_instance_links()
    print link
```

19.4 Consumer Expenditure

- A consumer unit: "a single person, or group of person who live together and who share responsibilities for most major expenditures."
- The "Reference Person" is **the first person named** when the respondent is asked, "Who is responsible for owning or renting this home?"
- 2 components to the Consumer Expenditure:
  - Interview survey
    - for "big ticket" and recurring expenditures, and global estimates on others
    - every three months for four quarter
    - rotating-panel survey
    - 6,000 consumer units each quarter
  - Diary survey
    - small-ticket, frequently purchased items
    - independent from the interview survey
    - about 14,000 diaries within families each year
- Table format
  - annual
    - Jan to Dec: since 1984
    - July to June: since 2013
  - two-year (Jan year 1 - Dec year 2)
    - since 1986
- Experimental research
  - all consumer units (available on request)
  - E.g., income tables, generational groups (e.g., millennial, gen X)

19.5 Gender, Age, Nationality

Now, we can make credible prediction about a person's gender, age, and nationality based on his or her name.

The APIs are

- `https://genderize.io/`
- `https://agify.io/`
- `https://nationalize.io/`

19.6 Google Trends

Online search volume and news reporting show a strong correlation, while academic publishing lag behind online public interest due to its delayed review and publication process (Nghiem et al., 2016). This result was interpreted as news serves as a conductor between the research and the public community.

Google Trends only return

Data Granularity	Data Window
Hourly Data	Last 7 days
Daily Data	Less than 9 months
Weekly Data	between 9 months and 5 years
Monthly Data	Longer than 5 years

Since the data is indexed for the chosen time window, for the same keywords with different time windows might result in different index values.

19.6.1 Relative Search

```
library(gtrendsR)

# for proxy
# setHandleParameters(
#     user = "xxxx",
#     password = "*******",
#     domain = "mydomain",
#     proxyhost = "10.111.124.113",
#     proxyport = 8080
# )

keywords = c("7eleven", "3m")
country = c("US")
time = ("2010-01-01 2012-01-30") # earliest is 2004
```

```
channel = "web"

trends = gtrends(keyword = keywords, geo = country, time = time, gprop = channel)

# objects
names(trends)

# this is weekly data only
time_trend <- trends$interest_over_time
```

The value is relative to the maximum volume (not absolute search volume)

channel type ("web" (default), "news", "images", "froogle" , "youtube")

```
library(ggplot2)

plot <-
    ggplot(data = time_trend, aes(
        x = date,
        y = hits,
        group = keyword,
        col = keyword
    )) +
    geom_line() + xlab('Time') + ylab('Relative Interest (weekly)') + theme_bw() +
    theme(
        legend.title = element_blank(),
        legend.position = "bottom",
        legend.text = element_text(size = 12)
    ) + ggtitle("Google Search Volume")
plot
```

Smoothing to remove seasonality

```
plot <-
    ggplot(data = time_trend, aes(
        x = date,
        y = hits,
        group = keyword,
        col = keyword
    )) +
    geom_smooth(span = 0.5, se = FALSE) + xlab('Time') + ylab('Relative Interest') +
    theme_bw() + theme(
        legend.title = element_blank(),
        legend.position = "bottom",
        legend.text = element_text(size = 12)
    ) + ggtitle("Google Search Volume")
```

```
plot
```

Alternatively, we can use the plot function in the `gtrendsR` package readily

```
plot(gtrendsR::gtrends(
    keyword = keywords,
    geo = country,
    time = time
))
```

1. Scaling method (overlapping method) (recommended)

A way to get daily data from `gtrendsR` readily is proposed by Alex Dyachenko

1. Get daily estimates for the window less than 9 months
2. Get monthly estimates for your desired time frame
3. Multiply daily estimates for each month from step 1 by their weights from step 2
4. Concatenation method (normalization/dailydata method)

Daily data are concatenated from 1-month queries and normalized by weekly trends data, which has been done in this post

From this post, we can be sure that the scaling method is better than the normalization method

Rate limit:

- 1,400 requests in 4 hours

```
library(gtrendsR)
library(tidyverse)
library(lubridate)

get_daily_gtrend <-
    function(keyword = c('7eleven', '3M'),
             geo = 'US',
             from = '2013-01-01',
             to = '2013-02-15') {
        if (ymd(to) >= floor_date(Sys.Date(), 'month')) {
            to <- floor_date(ymd(to), 'month') - days(1)

            if (to < from) {
                stop("Specifying \'to\' date in the current month is not allowed")
            }
        }

        aggregated_data <-
```

```
        gtrends(keyword = keyword,
                geo = geo,
                time = paste(from, to))
    if (is.null(aggregated_data$interest_over_time)) {
        print('There is no data in Google Trends!')
        return()
    }

    mult_m <- aggregated_data$interest_over_time %>%
        mutate(hits = as.integer(ifelse(hits == '<1', '0', hits))) %>%
        group_by(month = floor_date(date, 'month'), keyword) %>%
        summarise(hits = sum(hits)) %>%
        ungroup() %>%
        mutate(ym = format(month, '%Y-%m'),
               mult = hits / max(hits)) %>%
        select(month, ym, keyword, mult) %>%
        as_tibble()

    pm <- tibble(
        s = seq(ymd(from), ymd(to), by = 'month'),
        e = seq(ymd(from), ymd(to), by = 'month') + months(1) - days(1)
    )

    raw_trends_m <- tibble()

    for (i in seq(1, nrow(pm), 1)) {
        curr <- gtrends(keyword,
                        geo = geo,
                        time = paste(pm$s[i], pm$e[i]))
        if (is.null(curr$interest_over_time))
            next
        print(paste(
            'for',
            pm$s[i],
            pm$e[i],
            'retrieved',
            count(curr$interest_over_time),
            'days of data (all keywords)'
        ))
        raw_trends_m <- rbind(raw_trends_m,
                              curr$interest_over_time)
    }

    trend_m <- raw_trends_m %>%
        select(date, keyword, hits) %>%
```

```
            mutate(ym = format(date, '%Y-%m'),
                   hits = as.integer(ifelse(hits == '<1', '0', hits))) %>%
            as_tibble()

        trend_res <- trend_m %>%
            left_join(mult_m) %>%
            mutate(est_hits = hits * mult) %>%
            select(date, keyword, est_hits) %>%
            as_tibble() %>%
            mutate(date = as.Date(date))

        return(trend_res)
    }

daily_trend <- get_daily_gtrend(
    keyword = c('7eleven', '3M'),
    geo = 'US',
    from = '2013-01-01',
    to = '2013-02-01'
)
head(daily_trend)
```

This method was used in a research paper (Risteski and Davcev, 2014)

Similarly, without the `gtrendsR` package, you can follow Erik Johansson's method, but the Google no longer has his URL: http://www.google.com/trends/trendsReport?hl=en-US&q= so you might have to figure out the new URL

Other methods include:

- (Liu et al., 2019)

19.6.2 Absolute Search

See Google Trend Dataset available via BigQuery

19.7 Baidu Index

see (Liu et al., 2019) on how to use Java-based spider to get search volume data

Chapter 20

Modeling in Marketing

20.1 Definitions

- Structural in quantitative marketing: An estimation strategy where we assume and impose a structure to the consumer's maximization problem and these parameters are of the consumers' utility functions (Erdem and Keane, 1996).
  - These parameter are policy-invariant.
- Reduced form brand choice model: functions of marketing strategy variables (e..g, marketing mix). Hence, reduced-form models' parameters are variant to policy. So it's hard to study outcome of policy implementation may be reliable.

Uncertainty about product characteristics + learning behavior affect brand choice.

Two models (good in cases where consumer learning is vital to the choice process):

- Dynamic structural model with immediate utility maximization
- Forward-looking dynamic structural model (i.e., consumers don't choose products based on their current utility, but also future utility).

20.2 Quasi-Experimental

(Goldfarb et al., 2022) for a review

- Settings: Weather, border, contract changes, firm policy, life or regulatory changes.

- Nine Stages to go through in a quasi-experimental design
    1. Research Question
    2. Data Question
    3. Identification Strategy
    4. Empirical Analysis
    5. challenges to research design
    6. Robustness
    7. Mechanism
    8. External validity
    9. Unproven or Caveats

20.3 Transformation

20.3.1 Log-transformation

(Manchanda et al., 2004; Wies et al., 2019) used 1 in place of 0 for log-transformation. And also use 0.5 and 0.0001 for sensitivity analysis.

To control for firm size effects, (Wies et al., 2019) scale advertising investments by the firm's total assets in the given year

20.4 Endogeneity

Check Background in strucutral Models section under marketing mix models for more up-to-date academic practice.

Modal paper that address both selection bias and endogenoueity (Frennea et al., 2018)

20.4.1 Control Function

In the context of consumer choice model (Petrin and Train, 2010)

20.5 Variance Info Factors

Sometimes you can decrease the maximum variance inflation factor by removing the squared terms and centering all non-dummy regressors.

Chapter 21

Analytical Models

Marketing models consists of

1. Analytical Model: pure mathematical-based research
2. Empirical Model: data analysis.

"A model is a representation of the most important elements of a perceived real-world system".

Marketing model improves decision-making

- Econometric models
  - Description
  - Prediction
  - Simulation
- Optimization models
  - maximize profit using market response model, cost functions, or any constraints.
- Quasi- and Field experimental analyses
- Conjoint Choice Experiments.

"A decision calculus will be defined as a model-based set of procedures for processing data and judgments to assist a manager in his decision making"(Little, 1976):

- simple
- robust
- easy to control
- adaptive
- as complete as possible
- easy to communicate with

		Type of game	
		Static	Dynamic
Info Content	Complete	Nash	Subgame perfect
	Incomplete	Bayesian Nash	Perfect Bayesian
		(Auctions)	(signaling)

(Moorthy, 1993)

- Mathematical Theoretical Models
- Logical Experimentation
- An environment as a model, specified by assumptions
  - Math assumptions for tractability
  - Substantive assumptions for empirical testing
- Decision support modeling describe how things work, and theoretical modeling present how things should work.
- Compensation package including salaries and commission is a tradeoff between reduced income risk and motivation to work hard.
- Internal and External Validity are questions related to the boundaries conditions of your experiments.
- "Theories are tested by their predictions, not by the realism of their super model assumptions." (Friedman, 1953)

(McAfee and McMillan, 1996)

- Competition is performed under uncertainty
- Competition reveals hidden information
  - Independent-private-values case: selling price = second highest valuation
  - It's always better for sellers to reveal information since it reduces chances of cautious bidding that is resulted from the winner's curse
- Competition is better than bargaining
  - Competition requires less computation and commitment abilities
- Competition creates effort incentives

(Leeflang et al., 2000)

- Types of model:
  - Predictive model
    - Sales model: using time series data

    * Trial rate: using exponential growth.
    * Product growth model: Bass (1969)
  - Descriptive model
    * Purchase incidence and purchase timing : use Poisson process
    * Brand choice: Markov models or learning models.
    * Pricing decisions in an oligopolistic market Howard and Morgenroth (1968)
  - Normative model
    * Profit maximization based on price, adverting and quality (Dorfman and Steiner, 1976), extended by (Roberts et al., 1964; Lambin, 1970)

Later, Little (1970) introduced decision calculus and then multinomial logit model (Guadagni and Little, 1983)

Potential marketing decision automation:

- Promotion or pricing programs
- Media allocation
- Distribution
- Product assortment
- Direct mail solicitation

(Moorthy, 1985)

- Definitions:
  - Rationality = maximizing subjective expected utility
  - Intelligence = recognizing other firms are rational.
  - Rules of the game include
    * # of firms
    * feasible set of actions
    * utilities for each combination of moves
    * sequence of moves
    * the structure of info (who knows what and when?)
  - Incomplete info stems from
    * unknown motivations
    * unknown ability (capabilities)

    * different knowledge of the world.
  - Pure strategy = plan of action
  - A mixed strategy = probability dist of pure strategies.
  - Strategic form representation = sets of possible strategies for every firm and its payoffs.
  - Equilibrium = a list of strategies in which "no firm would like unilaterally to change its strategy."
- Equilibrium is not outcome of a dynamic process.
- Equilibrium Application
  - Oligopolistic Competition
    * Cournot (1838): quantities supplied: Cournot equilibrium. Changing quantities is more costly than changing prices
    * Bertrand (1883): Bertrand equilibrium: pricing.
  - Perfect competition
  - Product Competition: Hotelling (1929): Principle of Minimum Differentiation is invalid.
  - Entry:
    * first mover advantage
    * deterrent strategy
    * optimal for entrants or incumbents
  - Channels
- Perfectness of equilibria
  - Subgame perfectness
  - Sequential rationality
  - Trembling-hand perfectness
  - Application
    * Product and price competition in Oligopolies
    * Strategic Entry Deterrence
- Dynamic games
  - Long-term competition in oligopolies
  - Implicit Collusion in practice : price match from leader firms
- Incomplete Information

- Durable goods pricing by a monopolist
- predatory pricing and limit pricing
- reputation, product quality, and prices
- Competitive bidding and auctions

21.1 Building An Analytical Model

Notes by professor Sajeesh Sajeesh

Step 1: Get "good" idea (either from literature or industry)

Step 2: Assess the feasibility of the idea

- Is it interesting?
- Can you tell a story?
- Who is the target audience?
- Opportunity cost

Step 3: Don't look at the literature too soon

- Even when you have an identical model as in the literature, it's ok (it allows you to think)

Step 4: BUild the model

- Simplest model first: 1 period, 2 product , linear utility function for consumers
- Write down the model formulation
- Everything should be as simple as possible .. but no simpler

Step 5: Generalizing the model

- Adding complexity

Step 6: Searching the literature

- If you find a paper, you can ask yourself why you didn't do what the author has done.

Step 7: Give a talk /seminar

Step 8: Write the paper

21.2 Hotelling Model

(KIM and SERFES, 2006): A location model with preference variety

(Hotelling, 1929)

- Stability in competition
- Duopoly is inherently unstable
- Bertrand disagrees with Cournot, and Edgeworth elaborates on it.
  - because Cournot's assumption of absolutely identical products between firms.

seller try to $p_2 < p_1 c(l-a-b)$

the point of indifference

$$p_1 + cx = p_2 + cy$$

- c = cost per unit of time in each unit of line length
- p = price
- q = quantity
- x, y = length from A and B respectively

$$a + x + y + b = l$$

is the length of the street

Hence, we have

$$x = 0.5(l - a - b + \frac{p_2 - p_1}{c})y = 0.5(l - a - b + \frac{p_1 - p_2}{c})$$

Profits will be

$$\pi_1 = p_1 q_1 = p_1(a+x) = 0.5(l+a-b)p_1 - \frac{p_1^2}{2c} + \frac{p_1 p_2}{2c}\pi_2 = p_2 q_2 = p_2(b+y) = 0.5(l+a-b)p_2 - \frac{p_2^2}{2c} + \frac{p_1 p_2}{2c}$$

To set the price to maximize profit, we have

$$\frac{\partial \pi_1}{\partial p_1} = 0.5(l + a - b) - \frac{p_1}{c} + \frac{p_2}{2c} = 0\frac{\partial \pi_2}{\partial p_2} = 0.5(l - a + b) - \frac{p_2}{c} + \frac{p_1}{2c} = 0$$

which equals

$$p_1 = c(l + \frac{a-b}{3})p_2 = c(l - \frac{a-b}{3})$$

and

$$q_1 = a + x = 0.5(l + \frac{a-b}{3})q_2 = b + y = 0.5(l - \frac{a-b}{3})$$

with the SOC satisfied

In case of deciding locations, socialism works better than capitalism

(d'Aspremont et al., 1979)

- Principle of Minimum Differentiation is invalid

$$\pi_1(p_1, p_2) = \begin{cases} ap_1 + 0.5(l-a-b)p_1 + \frac{1}{2c}p_1p_2 - \frac{1}{2c}p_1^2 & \text{if } |p_1 - p_2| \leq c(l-a-b) \\ lp_1 & \text{if } p_1 < p_2 - c(l-a-b) \\ 0 & \text{if } p_1 > p_2 + c(l-a-b) \end{cases}$$

and

$$\pi_2(p_1, p_2) = \begin{cases} bp_2 + 0.5(l-a-b)p_2 + \frac{1}{2c}p_1p_2 - \frac{1}{2c}p_2^2 & \text{if } |p_1 - p_2| \leq c(l-a-b) \\ lp_2 & \text{if } p_2 < p_1 - c(l-a-b) \\ 0 & \text{if } p_2 > p_1 + c(l-a-b) \end{cases}$$

21.3 Positioning Models

Tabuchi and Thisse (1995)

- Relax Hotelling's model's assumption of uniform distribution of consumers to non-uniform distribution.
- Assumptions:
  - Equal cost
  - Consumers distributed over [0,1]
  - $F(x)$ = cumulative distribution of consumers where $F(1) = 1$ = total population
  - 2 distributions:
    * Traditional uniform density: $f(x) = 1$
    * New: triangular density: $f(x) = 2 - 2|2x - 1|$ which represents consumer concentration
  - Transportation cost = quadratic function of distance.

Hence, marginal consumer is

$$\bar{x} = (p_2 - p_1 + x_2^2 - x_1^2)/2(x_2 - x_1)$$

then when $x_1 < x_2$ the profit function is

$$\Pi_1 = p_1 F(\bar{x})$$

and

$$\Pi_2 = p_2[1 - F(\bar{x})]$$

and vice versa for $x_1 > x_2$, and Bertrand game when $x_1 = x_2$

- If firms pick **simultaneously** their locations, and then **simultaneously** their prices, and consumer density function is log-concave, then there is a unique Nash price equilibrium
  - Under **uniform** distribution, firms choose to locate as far apart as possible (could be true when observing shopping centers are far away from cities), but then consumers have to buy products that are far away from their ideal.
  - Under **triangular** density, no symmetric location can be found, but two asymmetric Nash location equilibrium can still be possible (decrease in equilibrium profits of both firms)
- If firms pick **sequentially** their locations, and pick their prices **simultaneously,**
  - Under both uniform and triangular, first entrant will locate at the market center

Sajeesh and Raju (2010)

- Model satiation (variety-seeking) as a relative reduction in the willingness to pay of the previously purchased brand. also known as negative state dependence
- Previous studies argue that in the presence of variety seeking consumers, firms should enjoy higher prices and profits, but this paper argues that average prices and profits are lower.
  - Firms should charge lower prices in the second period to prevent consumers from switching.

Assumptions:

- Period 0, choose location simultaneously

- Period 1, choose prices simultaneously
- Period 2, firms choose prices simultaneously

Moorthy (1988)

- 2 (identical) firms pick product (quality) first, then price.

Tyagi (2000)

- Extending Hotelling (1929) Tyagi (1999b) Tabuchi and Thisse (1995)
- Two firms enter **sequentially**, and have **different cost structures**.
- Paper shows second mover advantage

KIM and SERFES (2006)

- Consumers can make multiple purchases.
- Some consumers are loyal to one brand, and others consume more than one product.

Shreay et al. (2015)

- Quantity surcharges from different sizes of the same product (i.e., imperfect substitute or differentiated products) can be led by consumer preferences.

21.4 Market Structure and Framework

Basic model utilizing aggregate demand

- Bertrand Equilibrium: Firms compete on price
- Cournot Market structure: Firm compete on quantity
- Stackelberg Market structure: Leader-Follower model

Because we start with the quantity demand function, it is important to know where it's derived from Richard and Martin (1980)

Moorthy (1988)

- studied how two firms compete on product quality and price (both simultaneous and sequential)

21.4.1 Cournot - Simultaneous Games

$TC_i = c_i q_i$ where $i = 1, 2 P(Q) = a - bQ Q = q_1 + q_2 \pi_1 =$ price$\times$quantity$-$cost $= [a - b(q_1 + q_2)]q_1 - c_1 q_1 \pi_2 =$ price$\times$qua

$$\frac{d\pi_1}{dq_1} = a - 2bq_1 - bq_2 - c_1 = 0(1) \tag{21.1}$$

$$\frac{d\pi_2}{dq_2} = a - 2bq_2 - bq_1 - c_2 = 0 \tag{21.2}$$

From (21.1)

$$q_1 = \frac{a - c_1}{2b} - \frac{q_2}{2} = R_1(q_2) \tag{21.3}$$

is called reaction function, for best response function

From (21.2)

$$q_2 = \frac{a - c_2}{2b} - \frac{q_1}{2} = \tag{21.4}$$

$$q_1 = \frac{a - c_1}{2b} - \frac{a - c_2}{4b} + \frac{q_1}{4}$$

Hence,

$$q_1^* = \frac{a - 2c_1 + c_2}{3b} q_2^* = \frac{a - 2c_2 + c_1}{3b}$$

Total quantity is

$$Q = q_1 + q_2 = \frac{2a - c_1 - c_2}{3b}$$

Price

$$a - bQ = \frac{a + c_1 + c_2}{3b}$$

21.4.2 Stackelberg - Sequential games

also known as leader-follower games

- Stage 1: Firm 1 chooses quantity
- Stage 2: Firm 2 chooses quantity

$$c_2 = c_1 = c$$

Stage 2: reaction function of firm 2 given quantity firm 1

$$R_2(q_1) = \frac{a - c}{2b} - \frac{q_1}{2}$$

Stage 1:

$$\pi_1 = [a - b(q_1 + \frac{a-c}{2b} - \frac{q_1}{2})]q_1 - cq_1 = [a - b(\frac{a-c}{2b} + \frac{q_1}{2}]q_1 + cq_1$$

$$\frac{d\pi_1}{dq_1} = 0$$

Hence,

$$\frac{a+c}{2} - bq_1 - c = 0$$

The Stackelberg equilibrium is

$$q_1^* = \frac{a-c}{2b} q_2^* = \frac{a-c}{4b}$$

Under same price (c), Cournot =

$$q_1 = q_2 = \frac{a-c}{3b}$$

Leader produces more whereas the follower produces less compared to Cournot

$$\frac{d\pi_W^*}{d\beta} < 0$$

for the entire quantity range $d < \bar{d}$

As β increases in π_W^* Firm W wants to reduce β.

Low β wants more independent

Firms W want more differentiated product

On the other hand,

$$\frac{d\pi_S^*}{d\beta} < 0$$

for a range of $d < \bar{d}$

Firm S profit increases as β decreases when d is small

Firm S profit increases as β increases when d is large

Firm S profit increases as as product are more substitute when d is large

Firm S profit increases as products are less differentiated when d is large

21.5 More Market Structure

Dixit (1980)

- Based on Bain-Sylos postulate: incumbents can build capacity such that entry is unprofitability
- Investment in capacity is not a credibility threat if incumbents can change their capacity.
- Incumbent cannot deter entry

Tyagi (1999a)

- More retailers means greater competition, which leads to lower prices for customers.
- Effect of $(n+1)$ st retailer entry
  1. Competition effect (lower prices)
  2. Effect on price (i.e., wholesale price), also known as input cost effect
- Manufacturers want to increase wholesale price because now manufacturers have higher bargaining power, which leads other retailers to reduce quantity (bc their choice of quantity is dependent on wholesale price), and increase in prices.

Jerath et al. (2016)

- Organized Retailer enters a market
- Inefficient unorganized retailers exit
- Remaining unorganized retailers increase their prices. Thus, customers will be worse off.

Amaldoss and Jain (2005)

- consider **desire for uniqueness** and **conformism** on pricing conspicuous goods
- Two routes:
  - higher desire for uniqueness leads to higher prices and profits
  - higher desire for conformity leads to lower prices and profits
- Under the analytical model and lab text, consumers' desire for unique is increased from price increases, not the other way around.

Snob:

$$U_A = V - p_A - \theta t_s - \lambda_s(n_A) U_B = V - p_B - (1-\theta)t_s - \lambda_s(n_B)$$

where

- λ_s = sensitivity towards externality.
- θ is the position in the Hotelling's framework.
- t_s is transportation cost.

Conformist

$$U_A = V - p_A - \theta t_s + \lambda_c(n_A) U_B = V - p_B - (1-\theta)t_s + \lambda_c(n_B)$$

Rational Expectations Equilibrium

If your expectations are rational, then your expectation will be realized in equilibrium

Say, Marginal Snob = θ_s and β = number of snob in the market

Snobs

$$U_A^c \equiv U_B^c = \theta_s$$

Conformists

$$U_A^c = U_B^c = \theta_c$$

Then, according to rational expectations equilibrium, we have

$$\beta\theta_s + (1-\beta)\theta_c = n_A \beta(1-\theta_s) + (1-\beta)(1-\theta_c) = n_B$$

where

- $\beta\theta_s$ = Number of snobs who buy from firm A
- $(1-\beta)\theta_c$ = Number of conformists who buy from firm B
- $\beta(1-\theta_s)$ = Number of snobs who buy from firm B
- $(1-\beta)(1-\theta_c)$ = Number of conformists who buy from firm B

which is the rational expectations equilibrium (whatever we expect happens in reality).

In other words, expectation are realized in equilibrium.

The number of people expected to buy the product is endogenous in the model, which will be the actual number of people who will buy it in the market.

We should not think of the expected value here in the same sense as expected value in empirical research ($E(.)$) because the expected value here is without any errors (specifically, measurement error).

- The utility function for snobs is such that overall when price increase for one product, snob will like to buy the product more. When price increases, conformist will reduce the purchase.

Balachander and Stock (2009)

- Adding a Limited edition product has a positive effect on profits (via increased willingness of consumers to pay for such a product), but negative strategic effect (via increasing price competition between brands)

- Under quality differentiation, high-quality brand gain from LE products

- Under horizontal taste differentiation, negative strategic effects lead to lower equilibrium profits for both brands, but they still have to introduce LE products because of prisoners' dilemma

Sajeesh et al. (2020)

- two consumer segments:

  - functionality-oriented

  - exclusivity-oriented

- Firm increase value enhancements when functionality-oriented consumers perceive greater product differentiation

- Firms decrease value enhancements if exclusivity-oriented perceive greater product differentiation

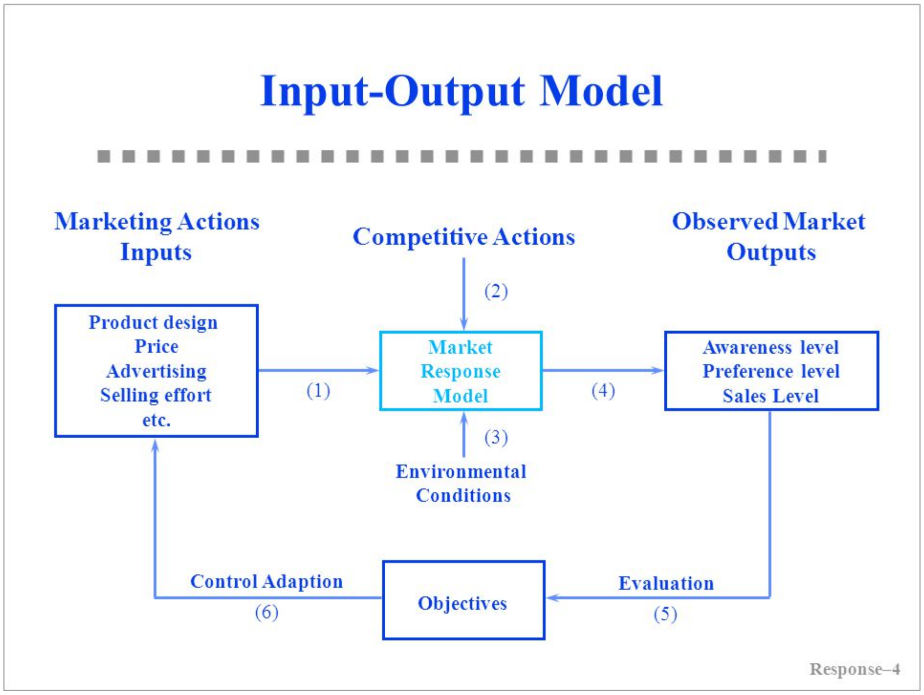
21.6 Market Response Model

Marketing Inputs:

- Selling effort
- advertising spending
- promotional spending

Marketing Outputs:

- sales
- share
- profit
- awareness



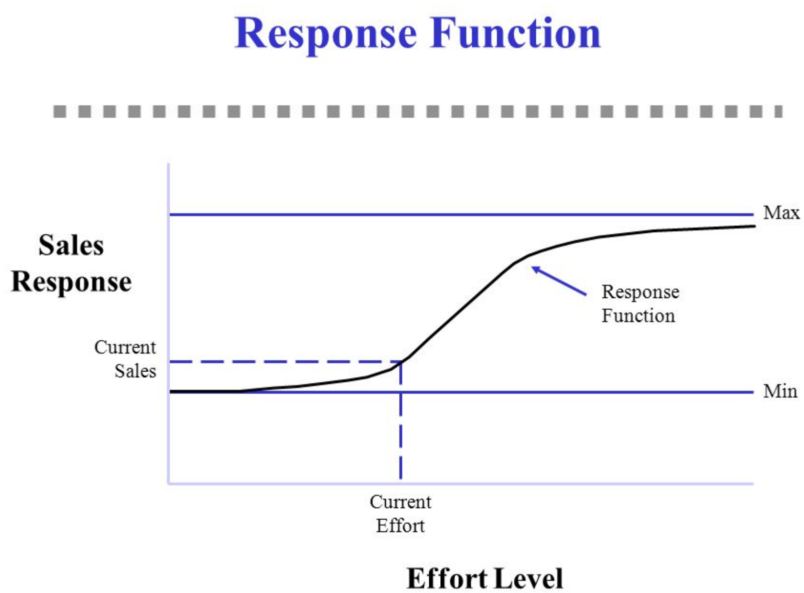
Give phenomena for a good model:

- P1: Dynamic sales response involves a sales **growth rate** and a sales **decay rate** that are different
- P2: Steady-state response can be **concave or S-shaped**. Positive sales at 0 adverting.
- P3: **Competitive effects**
- P4: Advertising effectiveness dynamics due to changes in media, copy, and other factors.
- P5: Sales still increase or fall off even as advertising is held constant.

Saunder (1987) phenomena

- P1: Output = 0 when Input = 0
- P2: The relationship between input and output is linear
- P3: Returns decrease as the scale of input increases (i.e., additional unit of input gives less output)
- P4: Output cannot exceed some level (i.e., saturation)
- P5: Returns increase as scale of input increases (i.e., additional unit of input gives more output)
- P6: Returns first increase and then decrease as input increases (i.e., S-shaped return)
- P7: Input must exceed some level before it produces any output (i.e., threshold)
- P8: Beyond some level of input, output declines (i.e., supersaturation

point)



Aggregate Response Models

- Linear model: $Y = a + bX$
  - Through origin
  - can only handle constant returns to scale (i.e., can't handle concave, convex, and S-shape)
- The Power Series/Polynomial model: $Y = a + bX + cX^2 + dX^3 + ...$
  - can't handle saturation and threshold
- Fraction root model/ Power model: $Y = a + bX^c$ where c is prespecified
  - c = 1/2, called **square root model**
  - c = -1, called **reciprocal model**
  - c can be interpreted as elasticity if a = 0.
  - c = 1, linear
  - c <1, decreasing return
  - c>1, increasing returns
- Semilog model: $Y = a + b \ln X$

  - Good when constant percentage increase in marketing effort (X) result in constant absolute increase in sales (Y)
- Exponential model: $Y = ae^{bX}$ where X >0
  - b > 0, increasing returns and convex
  - b < 0, decreasing returns and saturation
- Modified exponential model: $Y = a(1 - e^{-bX}) + c$
  - Decreasing returns and saturation
  - upper bound = a + c
  - lower bound = c
  - typically used in selling effort
- Logistic model: $Y = \frac{a}{a+e^{-(b+cX)}} + d$
  - increasing return followed by decreasing return to scale, S-shape
  - saturation = a + d
  - good with saturation and s-shape
- Gompertz model
- ADBUDG model (Little, 1970) : $Y = b + (a - b)\frac{X^c}{d+X^c}$
  - c > 1, S-shaped
  - 0 < c < 1
    * Concave
    * saturation effect
    * upper bound at a
    * lower bound at b
  - typically used in advertising and selling effort.
  - can handle, through origin, concave, saturation, S-shape
- Additive model for handling multiple Instruments: $Y = af(X_1) + bg(X_2)$
- Multiplicative model for handling multiple instruments: $Y = aX_1^b X_2^c$ where c and c are elasticities. More generally, $Y = af(X_1) \times bg(X_2)$
- Multiplicative and additive model: $Y = af(X_1) + bg(X_2) + cf(X_1)g(X_2)$
- Dynamic response model: $Y_t = a_0 + a_1X_t + \lambda Y_{t-1}$ where a_1 = current effect, λ = carry-over effect

Dynamic Effects

- Carry-over effect: current marketing expenditure influences future sales

- Advertising adstock/ advertising carry-over is the same thing: lagged effect of advertising on sales

- Delayed-response effect: delays between when marketing investments and their impact
- Customer holdout effects
- Hysteresis effect
- New trier and wear-out effect
- Stocking effect

Simple Decay-effect model:

$$A_t = T_t + \lambda T_{t-1}, t = 1, ...,$$

where

- A_t = Adstock at time t
- T_t = value of advertising spending at time t
- λ = decay/ lag weight parameter

Response Models can be characterized by:

1. The number of marketing variables
2. whether they include competition or not
3. the nature of the relationship between the input variables
    1. Linear vs. S-shape
4. whether the situation is static vs. dynamic
5. whether the models reflect individual or aggregate response
6. the level of demand analyzed
    1. sales vs. market share

Market Share Model and Competitive Effects: $Y = M \times V$ where

- Y = Brand sales models
- V = product class sales models
- M = market-share models

Market share (attraction) models

$$M_i = \frac{A_i}{A_1 + .. + A_n}$$

where A_i attractiveness of brand i

Individual Response Model:

Multinomial logit model representing the probability of individual `i` choosing brand `l` is

$$P_{il} = \frac{e^{A_{il}}}{\sum_j e^{A_{ij}}}$$

where

- A_{ij} = attractiveness of product j for individual i $A_{ij} = \sum_k w_k b_{ijk}$
- b_{ijk} = individual i's evaluation of product j on product attribute k, where the summation is over all the products that individual `i` is considering to purchase
- w_k = importance weight associated with attribute k in forming product preferences.

21.7 Technology and Marketing Structure and Economics of Compatibility and Standards

Amaldoss and Jain (2005)

Balachander and Stock (2009)

Sajeesh et al. (2020)

21.8 Conjoint Analysis and Augmented Conjoint Analysis

More technical on 27.1

Jedidi and Zhang (2002)

- Augmenting Conjoint Analysis to Estimate Consumer Reservation Price
- Using conjoint analysis (coefficients) to derive at consumers' reservation prices for a product in a category.
- Can be applied in the context of
  - product introduction
  - calculating customer switching effect
  - the cannibalization effect
  - the market expansion effect

$$Utility(Rating) = \alpha + \beta_i Attribute_i$$

where α

Netzer and Srinivasan (2011)

- Break conjoint analysis down to a sequence of constant-sum paired comparison questions.
- Can also calculate the standard errors for each attribute importance.

21.9 Distribution Channels

McGuire and Staelin (1983)

Assumptions:

1. Two manufacturing (wholesaling) firms differentiated and competing products: Upstream firms (manufacturers) and downstream channel members (retailers)

3 types of structure:

1. Both manufacturers with privately owned retailers (4 players: 2 manufacturers, 2 retailers)
2. Both vertically integrated (2 manufacturers)
3. Mix: one manufacturer with a private retailer, and one manufacturer with vertically integrated company store (3 players)

Each retail outlet has a downward sloping demand curve:

$$q_i = f_i(p_1, p_2)$$

Under decentralized system (4 players), the Nash equilibrium demand curve is a function of wholesale prices:

$$q_i^* = g_i(w_1, w_2)$$

More rules:

1. Assume 2 retailers respond, but not the competing manufacturer

And unobserved wholesale prices and market is not restrictive, and Nash equilibrium whole prices is still possible.

Under mixed structure, the two retailers compete, and non-integrated firm account for all responses in the market

Under integrated structure, this is a two-person game, where each chooses the retail price

Decision variables are prices (not quantities)

Under what conditions a manufacturer want to have intermediaries

Retail demand functions are assumed to be linear in prices

Demand functions are

$$q'_1 = \mu S[1 - \frac{\beta}{1-\theta}p'_1 + \frac{\beta\theta}{1-\theta}p'_2]$$

$$q'_2 = (1-\mu)S[1 + \frac{\beta\theta}{1-\theta}p'_1 - \frac{\beta}{1-\theta}p'_2]$$

where

- $0 \le \mu, \theta \le 1; \beta, S > 0$
- S is a scale factor, which equals industry demand ($q' \equiv q'_1 + q'_2$) when prices are 0.
- μ = absolute difference in demand
- θ = substutability of products (reflected by the cross elasticities), or the ratio of the rate of change of quantity with respect to the competitor's price to the rate of change of quantity with respect to own price.
  - $\theta = 0$ means independent demands (firms are monopolists)
  - $\theta \to 1$ means maximally substitutable

3 more conditions:

$$P = \{p'_1, p'_2 | p'_i - m' - s' \ge 0, i = 1,2; (1-\theta) - \beta p'_1 \beta\theta p'_2 \ge 0, (1-\theta) + \beta\theta p'_1 - \beta p'_2 \ge 0\}$$

where m', s' are fixed manufacturing and selling costs per unit

To have a set of P, then

$$\beta \le \frac{1}{m' + s'}$$

and to have industry demand no increase with increases in either price then

$$\frac{\theta}{1+\theta} \le \mu \le \frac{1}{1+\theta}$$

After rescaling, the industry demand is

$$q = 2(1-\theta)(p_1 + p_2)$$

Results:

- When each manufacturer is a monopolist ($\theta = 0$), it's twice as profitable for each to sell through its own channel
- When demand is maximally affected by the actions of the competing retailers ($\theta \to 1$), it's 3 times as profitable to have private dealers.
- The breakeven point happens at $\theta = .708$
- **In conclusion, the optimal distribution system depends of the degree of substitubability at the retail level.**

Jeuland and Shugan (2008)

- Quantity discounts is offered because
  1. Cost-based economies of scale
  2. Demand based - large purchases tend to be more price sensitive
  3. Strategic reason- single sourcing
  4. Channel Coordination (this is where this paper contributes to the literature

Moorthy (1987)

- Price discrimination - second degree

Geylani et al. (2007)

Jerath and Zhang (2010)

21.10 Advertising Models

Three types of advertising:

1. Informative Advertising: increase overall demand of your brand
2. Persuasive Advertising: demand shifting to your brand
3. Comparison: demand shifting away from your competitor (include complementary)

n customers distributed uniformly along the Hotelling's line (more likely for mature market where demand doesn't change).

$$U_A = V - p_A - txU_B = V - p_B - t(1-x)$$

For **Persuasive** advertising (highlight the value of the product to the consumer):

$$U_A = A_A V - p_A - tx$$

or increase value (i.e., reservation price).

$$U_A = \sqrt{Ad_A} V - p_A - tx$$

or more and more customers want the product (i.e., more customers think firm A product closer to what they want)

$$U_A = V - p_A - \frac{tx}{\sqrt{Ad_A}}$$

Comparison Advertising:

$$U_A = V - p_A - t\sqrt{Ad_B} x U_B = V - p_B - t\sqrt{Ad_A}(1-x)$$

Find marginal consumers

$$V - p_A - t\sqrt{Ad_B} x = V - p_B - t\sqrt{Ad_A}(1-x)$$

$$x = \frac{1}{t\sqrt{Ad_A} + t\sqrt{Ad_B}}(-p_A + p_B + t\sqrt{Ad_A})$$

then profit functions are (make sure the profit function is concave)

$$\pi_A = p_A x n - \phi Ad_A \pi_B = p_B(1-x)n - \phi Ad_B$$

where

- ϕ = per unit cost of advertising (e.g., TV advertising vs. online advertising in this case, TV advertising per unit cost is likely to be higher than online advertising per unit cost)
- t can also be thought of as return on advertising (traditional Hotelling's model considers t as transportation cost)

Equilibrium prices conditioned on advertising

$$\frac{d\pi_A}{p_A} = -\frac{d}{p_A}()\frac{d\pi_B}{p_B} = \frac{d}{p_B}$$

Then optimal pricing solutions are

$$p_A = \frac{2}{3}t\sqrt{Ad_A} + \frac{1}{3}t\sqrt{Ad_B}p_B = \frac{1}{3}t\sqrt{Ad_A} + \frac{2}{3}t\sqrt{Ad_B}$$

Prices increase with the intensities of advertising (if you invest more in advertising, then you charge higher prices). Each firm price is directly proportional to their advertising, and you will charger higher price when your competitor advertise as well.

Then, optimal advertising (with the optimal prices) is

$$\frac{d\pi_A}{dAd_A}\frac{d\pi_B}{dAd_B}$$

Hence, Competitive equilibrium is

$$Ad_A = \frac{25t^2n^2}{576\phi^2}Ad_B = \frac{25t^2n^2}{576\phi^2}p_A = p_B = \frac{5t^2n}{24\phi}$$

As cost of advertising (ϕ), firms spend less on advertising

Higher level of return on advertising (t), firms benefit more from advertising

With advertising in the market, the equilibrium prices are higher than if there were no advertising.

Since colluding on prices are forbidden, and colluding on advertising is hard to notice, firms could potential collude on advertising (e.g., pulsing).

Assumption:

1. Advertising decision before pricing decision (reasonable because pricing is earlier to change, while advertising investment is determined at the beginning of each period).

Collusive equilibrium (instead of using Ad_A, Ad_B, use Ad - set both advertising investment equal):

$$Ad_A = Ad_B = \frac{t^2n^2}{16\phi^2} > \frac{25t^2n^2}{576\phi^2}$$

Hence, collusion can be make advertising investment equilibrium higher, which makes firms charge higher prices, and customers will be worse off. (more reference Aluf and Shy - check Modeling Seminar Folder - Advertising).

Combine both **Comparison** and **Persuasive** Advertising

$$U_A = V - p_A - tx\frac{\sqrt{Ad_B}}{\sqrt{Ad_A}} U_B = V - p_B - t(1-x)\frac{\sqrt{Ad_A}}{\sqrt{Ad_B}}$$

Informative Advertising

1. Increase number of n customers (more likely for new products where the number of potential customers can change)

How do we think about customers, how much to consume. People consume more when they have more availability, and less when they have less in stock (Ailawadi and Neslin, 1998)

Villas-Boas (1993)

- Under monopoly, firms would be better off to pulse (i.e., alternate advertising between a minimum level and efficient amount of advertising) because of the S-shaped of the advertising response function.

Model assumptions:

1. The curve of the advertising response function is S-shaped
2. Markov strategies: what firms do in this period depends on what might affect profits today or in the future (independent of the history)

Propositions:

1. "If the loss from lowering the consideration level is larger than the efficient advertising expenditures, the unique Markov perfect equilibrium is for firms to advertise, whatever the consideration levels of both firms are."

Nelson (1974)

- Quality of a brand is determined before a purchase of a brand is "search qualities"
- Quality that is not determined before a purchase is "experience qualities"
- Brand risks credibility if it advertises misleading information, and pays the costs of processing nonbuying customers
- There is a reverse association between quality produced and utility adjusted price
- Firms that want to sell more advertise more

- Firms advertise to their appropriate audience, i.e., "those whose tastes are best served by a given brand are those most likely to see an advertisement for that brand" (p. 734).
- Advertising for experience qualities is indirect information while advertising for search qualities is direct information . (p. 734).
- Goods are classified based on quality variation (i.e., whether the quality variation was based on searhc of experience).
- 3 types of goods
  - experience durable
  - experience nondurable
  - search goods
- Experience goods are advertised more than search goods because advertisers increase sales via increasing the reputability of the sellers.
- The marginal revenue of advertisement is greater for search goods than for experience goods (p. 745). Moreover, search goods will concentrate in newspapers and magazines while experience goods are seen on other media.
- For experience goods, WOM is better source of info than advertising (p. 747).
- Frequency of purchase moderates the differential effect of WOM and advertising (e.g., for low frequency purchases, we prefer WOM) (p. 747).
- When laws are moderately enforced, deceptive advertising will happen (too little law, people would not trust, too much enforcement, advertisers aren't incentivized to deceive, but moderate amount can cause consumers to believe, and advertisers to cheat) (p. 749). And usually experience goods have more deceptive advertising (because laws are concentrated here).

Iyer et al. (2005)

- Firms advertise to their targeted market (those who have a strong preference for their products) than competitor loyalists, which endogenously increase differentiation in the market, and increases equilibrium profits
- Targeted advertising is more valuable than target pricing. Target advertising leads to higher profits regardless whether firms have target pricing. Target pricing increased competition for comparison shoppers (no improvement in equilibrium profits). (p. 462 - 463).

Comparison shoppers size:

$$s = 1 - 2h$$

where h is the market size of each firm's consumers (those who prefer to buy product from that firm). Hence, h also represents the differentiation between the two firms

See table 1 (p. 469).

A is the cost for advertising the entire market

r is the reservation price

Chen et al. (2009)

- *Combative* vs. *constructive* advertising
- Informative complementary and persuasive advertising
  - Informative: increase awareness, reduce search costs, increase product differentiation
  - Complementary (under comparison): increase utility by signaling social prestige
  - Persuasive: decrease price sensitivity (include combative)
- Consumer response moderates the effect of combative adverting on price competition:
  - It decreases price competition
  - It increases price competition when (1) consumers preferences are biased (firms that advertise have their products favored by the consumers), (2) disfavor firms can't advertise and only respond with price. because advertising war leads to a price war (when firms want to increase their own profitability while collective outcome is worse off).

21.11 Product Differentiation

- Horizontal differentiation: different consumers prefer different products
- Vertical differentiation: where you can say one good is "better" than the other.

Characteristics approach: products are the aggregate of their characteristics.

21.12 Product Quality, Durability, Warranties

Horizontal Differentiation

$$U = V - p - t(\theta - a)^2$$

Vertical Differentiation

$$U_B = \theta s_B - p_B U_A = \theta s_A - p_A$$

Assume that product B has a higher quality

θ is the position of any consumer on the vertical differentiation line.

When $U_A < 0$ then customers would not buy

Point of indifference along the vertical quality line

$$\theta s_B - p_B = \theta s_A - p_A \theta (s_B - s_A) = p_B - p_A \bar{\theta} = \frac{p_B - p_A}{s_B - s_A}$$

If $p_B = p_A$ for every θ, s_B is preferred to s_A

1. Moorthy (1988)

$$\pi_A = (p_A - cs_A^2)(Mktshare_A)\pi_B = (p_B - cs_B^2)(Mktshare_B)U_A = \theta s_A - p_A = 0\bar{\theta}_2 = \frac{p_A}{s_A}$$

2. Wauthy (1996)

$\frac{b}{a}$ = such that market is covered, then

$$2 \leq \frac{b}{a} \leq \frac{2s_2 + s_1}{s_2 - s_1}$$

for the market to be covered

In vertical differentiation model, you **can't** have both $\theta \in [0, 1]$ and full market coverage.

Alternatively, you can also specify $\theta \in [1, 2]; [1, 4]$

$$\theta \in \begin{cases} [1, 4] & \frac{b}{a} = 4 \\ [1, 2] & \frac{b}{a} = 2 \end{cases}$$

Under Asymmetric Information

- Adverse Selection: Before contract: Information is uncertain
- Moral Hazard: After contract, intentions are unknown to at least one of the parties.

Alternative setup of Akerlof's (1970) paper

Used cars quality $\theta \in [0, 1]$

Seller - car of type θ

Buyer = WTP = $\frac{3}{2}\theta$

Both of them can be better if the transaction occurs because buyer's WTP for the car is greater than utility received by seller.

1. Assume quality is observable (both sellers and buyers do know the quality of the cars):

Price as a function of quality $p(\theta)$ where $p(\theta) \in [\theta, 3/2\theta]$ both parties can be better off

2. Assume quality is unobservable (since θ is uniformly distributed) (sellers and buyers do not know the quality of the used cars):

$$E(\theta) = \frac{1}{2}$$

then $E(\theta)$ for sellers is $1/2$

$E(\theta)$ for buyer $= 3/2 \times 1/2 = 3/4$

then market happens when $p \in [1/2, 3/4]$

3. Asymmetric info (if only the sellers know the quality)

Seller knows θ

Buyer knows $\theta \sim [0, 1]$

From seller perspective, he must sell at price $p \geq \theta$ and

From buyer perspective, quality of cars on sale is between $[0, p]$. Then, you will have a smaller distribution than $[0, 1]$

If $E[(\theta)|\theta \leq p] = 0.5p$

Buyers' utility is $3/4p$ but the price he has to pay is p (then market would not happen)

21.12.1 Akerlof (1970)

- This paper is on adverse selection
- The relationship between quality and uncertainty (in automobiles market)
- 2 x 2 (used vs. new, good vs. bad)

q = probability of getting a good car = probability of good cars produced

and $(1 - q)$ is the probability of getting a lemon

- Used car sellers have knowledge about the probability of the car being bad, but buyers don't. And buyers pay the same price for a lemon as for a good car (info asymmetry).
- Gresham's law for good and bad money is not transferable (because the reason why bad money drives out goo d money because of even exchange rate, while buyers of a car cannot tell if it is good or bad).

21.12.1.1 Asymmetrical Info

Demand for used automobiles depends on price quality:

$$Q^d = D(p, \mu)$$

Supply for used cars depends on price

$$S = S(p)$$

and average quality depends on price

$$\mu = \mu(p)$$

In equilibrium

$$S(p) = D(p, \mu(p))$$

At no price will any trade happen

Example:

Assume 2 groups of graders:

1. First group: $U_1 = M = \sum_{i=1}^{n} x_i$ where
    - M is the consumption of goods other than cars
    - x_i is the quality of the i-th car
    - n is the number of cars
2. Second group: $U_2 = M + \sum_{i=1}^{n} \frac{3}{2} x_i$

Group 1's income is Y_1

Group 2's income is Y_2

Demand for first group is

$$\begin{cases} D_1 = \frac{Y_1}{p} & \frac{\mu}{p} > 1 \\ D_1 = 0 & \frac{\mu}{p} < 1 \end{cases}$$

Assume we have uniform distribution of automobile quality.

Supply offered by first group is

$$S_2 = \frac{pN}{2}; p \leq 2$$

with average quality $\mu = p/2$

Demand for second group is

$$\begin{cases} D_2 = \frac{Y_2}{p} & \frac{3\mu}{2} > p \\ D_2 = 0 & \frac{3\mu}{2} < p \end{cases}$$

and supply by second group is $S_2 = 0$

Thus, total demand $D(p, \mu)$ is

$$\begin{cases} D(p,\mu) = (Y_2 + Y_1)/p & \text{if } p < \mu \\ D(p,\mu) = (Y_2)/p & \text{if } \mu < p < 3\mu/2 \\ D(p,\mu) = 0 & \text{if } p > 3\mu/2 \end{cases}$$

With price p, average quality is $p/2$, and thus at no price will any trade happen

21.12.1.2 Symmetric Info

Car quality is uniformly distributed $0 \leq x \leq 2$

Supply

$$\begin{cases} S(p) = N & p > 1 \\ S(p) = 0 \end{cases}$$

Demand

$$\begin{cases} D(p) = (Y_2 + Y_1)/p & p < 1 \\ D(p) = Y_2/p & 1 < p < 3/2 \\ D(p) = 0 & p > 3/2 \end{cases}$$

In equilibrium

$$\begin{cases} p = 1 & \text{if } Y_2 < N \\ p = Y_2/N & \text{if } 2Y_2/3 < N < Y_2 \\ p = 3/2 & \text{if } N < 2Y_2 < 3 \end{cases}$$

This model also applies to (1) insurance case for elders (over 65), (2) the employment of minorities, (3) the costs of dishonesty, (4) credit markets in underdeveloped countries

To counteract the effects of quality uncertainty, we can have

1. Guarantees
2. Brand-name good
3. Chains
4. Licensing practices

21.12.2 Spence (1973)

Built on (Akerlof, 1970) model

Consider 2 employees:

- Employee 1: produces 1 unit of production
- Employee 2: produces 2 units of production

We have α people of type 1, and $1 - \alpha$ people of type 2

Average productivity

$$E(P) = \alpha + 2(1 - \alpha) = 2 - \alpha$$

You can signal via education.

To model cost of education,

Let E to be the cost of education for type 1

E/2 to be the cost education for type 2

If type 1 signals they are high-quality worker, then they have to go through the education and cost is E, and net utility of type 1 worker

$$2 - E < 1 E > 1$$

If type 2 signals they are high-quality worker, then they also have to go through the education and cost is E/2 and net utility of type 2 worker is

$$2 - E/2 > 1 E < 2$$

If we keep $1 < E < 2$, then we have separating equilibrium (to have signal credible enough of education)

21.12.3 Moorthy and Srinivasan (1995)

- Money-back guarantee signals quality
- Transaction cost are those the seller or buyer has to pay when redeeming a money-back guarantee
- Money-back guarantee does not include product return (buyers have to incur expense), but guarantee a full refund of the purchase price.
- If signals are costless, there is no difference between money-back guarantees and price
- But signal are costly,
  - Under homogeneous buyers, low-quality sellers cannot mimic high-quality sellers' strategy (i.e., money-back guarantee)
  - Under heterogeneous buyers,
    * when transaction costs are too high, the seller chooses either not to use money-back guarantee strategy or signal through price.
    * When transaction costs are moderate, there is a critical value of seller transaction costs where
      · below this point, the high-quality sellers' profits increase with transaction costs
      · above this point, the high-quality sellers' profits decrease with transaction costs
- Uninformative advertising ("money-burning") is defined as expenditures that do not affect demand directly. **is never needed**
- Moral hazard:
  - Consumers might exhaust consumption within the money-back guarantee period

Model setup

	High-quality sellers (h)	Low-quality sellers (l)
Cost $c_h > c_l$	c_h	c_l

21.13 Bargaining

Books:

- Abhinay Muthoo - Bargaining Theory with Applications (1999) (check books folder)
- Josh Nash - Nash Bargaining (1950)

Allocation of scare resources

Buyers & Sellers	Type
Many buyers & many sellers	Traditional markets
Many buyers & one seller	Auctions
One buyer & one seller	Bargaining

Allocations of

- Determining the share before game-theoretic bargaining
  - Use a judge/arbitrator
  - Meet-in-the-middle
  - Forced Final: If an agreement is not reached, one party will use take it or leave it
- Art: Negotiation
- Science: Bargaining
- Game theory's contribution: to the rules for the encounter
- Area that is still fertile for research

21.13.1 Non-cooperative

Outline for non-cooperative bargaining

- The rules

Take-it-or-leave-it Offers

- Bargain over a cake
- If you accept, we trade
- If you reject, no one eats
- Under perfect info, there is a simple rollback equilibrium
- In general, bargaining takes on a "take-it-or-counteroffer" procedure
- If time has value, both parties prefer to trade earlier to trade later

- E.g., labor negotiations - later agreements come at a price of strikes, work stoppages

- Delays imply less surplus left to be shared among the parties

Two-stage bargaining

- I offer a proportion, p , of the cake to you
- If rejected, you may counteroffer (and δ of the cake melts)
- Payoffs:
  - In the first period: 1-p, p
  - In second period: $(1-\delta)(1-p),(1-\delta)p$
- Since period 2 is the final period, this is just like a take-it-or-leave-it offer
  - You will offer me the smallest piece that I will accept, leaving you with all of $1-\delta$ and leaving me with almost 0
- Rollback: then in the first period: I am better off by giving player B more than what he would have in period 2 (i.e., give you at least as much surplus)
- You surplus if you accept in the first period is p
- Accept if: your surplus in first period greater than your surplus in second period $p \geq 1-\delta$
- IF there is a second stage, you get $1-\delta$ and I get 0
- You will reject any offer in the first stage that does not offer you at least $1-\delta$
- In the first period, I offer you $1-\delta$
- Note: **the more patient you are (the slower the cake melts) the more you receive now**
- Whether first or second mover has the advantage depends on δ.
  - If δ is high (melting fast), then first mover is better.
  - If δ is low (melting slower), then second mover is better.
- Either way - if both players think, agreement would be reached in the first period
- In any bargaining setting, strike a deal as early as possible.

Why doesn't this happen in reality?

- reputation building
- lack of information

Why bargaining doesn't happen quickly? Information asymmetry

- Likelihood of success (e.g., uncertainty in civil lawsuits)

Lessons:

- Rules of the bargaining game uniquely determine the bargain outcome
- which rules are better for you depends on patience, info
- What is the smallest acceptable piece? Trust your intuition
- delays are always less profitable: Someone must be wrong

Non-monetary Utility

- each side has a reservation price
  - LIke in civil suit: expectation of wining
- The reservation price is unknown
- One must:
  - probabilistically determine best offer
  - but - probability implies a chance that non bargain will take place

Example:

- Company negotiates with a union
- Two types of bargaining:
  - Union makes a take-it-or-leave-it offer
  - Union makes a n offer today. If it's rejected, the Union strikes, then makes another offer
    * A strike costs the company 10% of annual profits.
- Probability that the company is "highly profitable", ie., 200k is p
- If offer wage of \$150k
  - Definitely accepted
  - Expected wage = \$150K
- If offer wage of \$200K
  - Accepted with probability p
  - Expected wage = \$200k(p)
- $p = .9$ (90% chance company is highly profitable
  - best offer: ask for \$200K wage
  - Expected value of offer: $.9 * 200 = 180$

- $p = .1$ (10% chance company is highly profitable
  - best offer: ask for \$200K wage
  - Expected value of offer: $.1 * 200 = 20$
  - If ask for \$10k, get \$150k
  - not worth the risk to ask for more
- If first-period offer is rejected: A strike costs the company 10% of annual profits
- Strike costs a high-value company more than a low value company
- Use this fact to screen
- What if the union asks for \$170k in the first period?
  - Low profit firms (\$150k) rejects - as can't afford to take
  - HIgh profit firm must guess what will happen if it rejects
    * Best case: union strikes and then asks for only \$140k (willing to pay for some cost of strike), but not all)
    * In the mean time: strike cost the company \$20K
- High-profit firm accepts

Separating equilibrium

- only high-profit firms accept the first period
- If offer is rejected, Union knows that it is facing a low-profit firm
- Ask for \$140k

What's happening

- Union lowers price after a rejection
  - Looks like giving in
  - looks like bargaining
- Actually, the union is screening its bargaining partner
  - Different "types" of firms have different values for the future
  - Use these different values to screen
  - Time is used as a screening device

21.13.2 Cooperative

two people diving cash

- If they do not agree, they each get nothing
- They cant divide up more than the whole thing

21.13.3 Nash (1950)

- Bargaining, bilateral monopoly (nonzero-sum two -person game).
- Non action taken by one individual (without the consent of the other) can affect the other's gain.
- Assumptions:
  - Rational individuals (maximize gain)
  - Full knowledge: tastes and preferences are known
  - Transitive Ordering: $A > C$ when $A > B$, $B > C$. Also related to substitutability if two events are of equal probability
  - Continuity assumption
- Properties:
  - $u(A) > u(B)$ means A is more desirable than B where u is a utility function
  - Linearity property: If $0 \leq p \leq 1$, then $u(pA + (1-p)B) = pu(A) + (1-p)u(B)$
    - For two person: $p[A, B] + (1-p)[C, D] = [pA - (1-p)C, pB + (1-p)D]$
- Anticipation $= pA - (1-p)B$ where
  - p is the prob of getting A
  - A and B are two events.

u_1, u_2 are utility function

$c(s)$ is the solution point in a set S (compact, convex, with 0)

Assumptions:

- If $\alpha \in S$ s.t there is $\beta \in S$ where $u_1(\beta) > u_2(\alpha)$ and $u_2(\beta) > u_2(\alpha)$ then $\alpha \neq c(S)$
  - People try to maximize utility
- If $S \in T$, c(T) is in S then $c(T) = c(S)$

- If S is symmetric with respect to the line $u_1 = u_2$, then $c(S)$ is on the line $u_1 = u_2$
  - Equality of bargaining

21.13.4 Iyer and Villas-Boas (2003)

- Presence of a powerful retailer (e.g., Walmart) might be beneficial to all channel members.

21.13.5 Desai and Purohit (2004)

- 2 customers segment: hagglers, nonhagglers.
- When the proportion of nonhagglers is sufficient high, a haggling policy can be more profitable than a fixed-price policy

21.14 Pricing and Search Theory

21.14.1 Varian and Purohit (1980)

- From Stigler's seminar paper (Stiglitz and Salop, 1982; Salop and Stiglitz, 1977), model of equilibrium price dispersion is born
  - Spatial price dispersion: assume uninformed and informed consumers
    * Since consumers can learn from experience, the result does not hold over time
  - Temporal price dispersion: sales
- This paper is based on
  - Stiglitz: assume informed (choose lowest price store) and uninformed consumers (choose stores at random)
  - Shilony (Shilony, 1977): randomized pricing strategies

$I > 0$ is the number of informed consumers

$M > 0$ is the number of uninformed consumers

n is the number of stores

$U = M/n$ is the number of uninformed consumers per store

Each store has a density function $f(p)$ indicating the prob it charges price p

Stores choose a price based on $f(p)$

- Succeeds if it has the lowest price among n prices, then it has $I + U$ customers
- Fails then only has U customers

- Stores charge the same lowest price will share equal size of informed customers

$c(q)$ is the cost curve

$p^* = \frac{c(I+U)}{(I+U}$ is the average cost with the maximum number of customers a store can get

Prop 1: $f(p) = 0$ for $p > r$ or $p < p^*$

Prop 2: No symmetric equilibrium when stores charge the same price

Prop 3: No point masses in the equilibrium pricing strategies

Prop 4: If $f(p) > 0$, then

$$\pi_s(p)(1 - F(p))^{n-1} + \pi_f(p)[1 - (1 - F(p))^{n-1}] = 0$$

Prop 5: $\pi_f(p)(\pi_f(p) - \pi_s(p))$ is strictly decreasing in p

Prop 6: $F(p^*_\epsilon) > 0$ for any $\epsilon > 0$

Prop 7: $F(r - \epsilon) < 1$ for any $\epsilon > 0$

Prop 8: No gap (p_1, p_2) where $f(p) \equiv 0$

Decision to be informed can be endogenous, and depends on the "full price" (search costs + fixed cost)

21.14.2 Lazear (1984)

Retail pricing and clearance sales

- Goods' characteristics affect pricing behaviors
- Market's thinness can affect price volatility
- Relationship between uniqueness of a goods and its price
- Price reduction policies as a function of shelf time

Set up

Single period model

V = the price of the only buyer who is willing to purchase the product

$f(V)$ is the density of V (firm's prior)

$F(V)^2$ is its distribution function

Firms try to

$$\max_R R[1 - F(R)]$$

where R is the price

$1-F(R)$ is the prob that $V>R$

Assume V is uniform $[0,1]$ then

$F(R)=R$ so that the optimum is $R=0.5$ with expected profits of 0.25

Two-period model

Failures in period 1 implies $V<R_1$.

Hence, based on Bayes' theorem, the posterior distribution in period 2 is $[0,R_1]$

$F_2(V)=V/R_1$ (posterior distribution)

R_1 affect (1) sales in period 1, (2) info in period 2

Then, firms want to choose R_1,R_2 .Firms try to

$$\max_{R_1,R_2} R_1[1-F(R_1)]+R_2[1-F_2(R_2)]F(R_1)$$

Then, in period 2, the firms try to

$$\max_{R_2} R_2[1-F_2(R_2)]$$

Based on Bayes' Theorem

$$F_2(R_2)=\begin{cases} F(R_2)/F(R_1) & \text{for } R_2<R_1 \\ 1 & \text{otherwise} \end{cases}$$

Due to FOC, second period price is always lower than first price price

Expected profits are higher than that of one-period due to higher expected probability of a sale in the two-period problem.

But this model assume

- no brand recognition
- no contagion or network effects

In thin markets and heterogeneous consumers

we have N customers examine the good with the prior probability P of being shoppers, and $1-P$ being buyers who are willing to buy at V

There are 3 types of people

1. customers = all those who inspect the good
2. buyers = those whose value equal V

3. shoppers = those who value equal 0

An individual does not know if he or she is a buyer or shopper until he or she is a customer (i.e., inspect the goods)

Then, firms try to

$$\max_{R_1,R_2} R_1(\text{prob sale in 1}) + R_2(\text{Posterior prob sale in 2}) \times (\text{Prob no sale in 1})$$
$$R_1 \times (-F(R_1))(1-P^N) + R_2\{(1-F_2(R_2))(1-P^N)\} \times \{1-[(1-F(R_1))(1-P^N)]\}$$

Based on Bayes' Theorem, the density for period 2 is

$$f_2(V) = \begin{cases} \frac{1}{R_1(1-P^N)+P^N} & \text{for } V \leq R_1 \\ \frac{P^N}{R_1(1-P^N)+P^N} & \text{for } V > R_1 \end{cases}$$

Conclusion:

As $P^N \to 1$ (almost all customers are shoppers), there is not much info to be gained. Hence, 2-period is no different than 2 independent one-period problems. Hence, the solution in this case is identical to that of one-period problem.

When P^N is small, prices start higher and fall more rapid as time unsold increases

When $P^N \to 1$, prices tend to be constant.

P^N can also be thought of as search cost and info.

Observable Time patterns of price and quantity

Pricing is a function of

- The number of customers N
- The proportion of shoppers P
- The firm's beliefs about the market (parameterized through the prior on V)

Markets where prices fall rapidly as time passes, the probability that the good will go unsold is low.

Goods with high initial price are likely to sell because high initial price reflects low P^N - low shoppers

Heterogeneity among goods

The more disperse prior leads to a higher expected price for a given mean. And because of longer time on shelf, expected revenues for such a product can be lower.

Fashion, Obsolescence, and discounting the future

The more obsolete, the more anxious is the seller

Goods that are "classic", have a higher initial price, and its price is less sensitive to inventory (compared to fashion goods)

Discounting is irrelevant to the pricing condition due to constant discount rate (not like increasing obsolescence rate)

For non-unique good, the solution is identical to that of the one-period problem.

Simple model

Customer's Valuation $\in [0, 1]$

Firm's decision is to choose a price p (label - R_1)

One-period model

Buy if $V > R_1$ prob $= 1 - R_1$

Not buy if $V < R_1$ probability $= R_1$

$\max_{R_1} [R_1][1 - R_1]$ hence, FOC $R_1 = 1/2$, then total $\pi = 1/2 - (1/2)^2 = 1/4$

Two-period model

Two prices R_1, R_2

$R_1 \in [0, 1]$

$R_2 \in [0, R_1]$

$$\max_{R_1} [R_1][1 - R_1] + R_2(1 - R_2)(R_1)$$

$$\max_{R_1} [R_2][\frac{R_1 - R_2}{R_1}]$$

FOC $R_2 = R_1/2$

$$\max_{R_1} R_1(1 - R_1) + \frac{R_1}{2}(1 - \frac{R_1}{2})(R_1)$$

FOC: $R_1 = 2/3$ then $R_2 = 1/3$

N customers

Each customer could be a

- shopper with probability p with $V < 0$
- buyer with probability $1 - p$ with $V >$ price

Modify equation 1 to incorporate types of consumers

$$R_1(1-R_1)(1-p^N) + R_2(1-R_2)R_1(1-p^N)[1-(1-R_1)(1-p^N)]$$

Reduce costs by

- Economy of scale c(number of units)
- Economy of scope c(number of types of products) (typically, due to the transfer of knowledge)
- Experience effect c(time) (is a superset of economy of scale)

Lal and Sarvary (1999)

- Conventional idea: lower search cost (e.g., Internet) will increase price competition.
- Assumptions:
  - Demand side: info about product attributes:
    * digital attributes (those can be communicated via the Internet)
    * nondigital attributes (those can't)
  - Supply side: firms have both traditional and Internet stores.
- Monopoly pricing can happen when
  - high proportion of Internet users
  - Not overwhelming nondigital attributes
  - Favor familiar brands
  - destination shopping
- Monopoly pricing can lead to higher prices and discourage consumer from searching
- Stores serve as acquiring tools, while Internet maintain loyal customers.

Kuksov (2004)

- For products that cannot be changed easily (design),lower search costs lead to **higher price competition**
- For those that can be easily changed, lower search costs lead to higher product differentiation, which in turn **decreases price competition**, lower social welfare, higher industry profits.

(Salop and Stiglitz, 1977)

21.15 Pricing and Promotions

- Extensively studied
- Issue of Everyday Low price vs Hi/Lo pricing
- Short-term price discounts
  - offering trade-deals
  - consumer promotions
    - shelf-price discounts (used by everybody)
    - cents-off coupons (some consumers whose value of time is relatively low)

Loyalty is similar to inform under analytic modeling:

- Uninformed = loyal
- Informed = non-loyal

30 years back, few companies use price promotions

Effects of Short-term price discounts

- measured effects (84%) (Gupta, 1988)
  - Brand switching (14%)
  - purchase acceleration (2%)
  - quantity purchased
  - elasticity of ST price changes is an order of magnitude higher
- Other effects:
  - general trial (traditional reason)
  - encourages consumers to carry inventory hence increase consumption
  - higher sales of complementary products
  - small effect on store switching
  - Asymmetric effect (based on brand strength) (bigger firms always benefit more)
    - expect of store brands

Negative Effects

- Expectations of future promotions
- Lowering of Reference Price
- Increase in price sensitivity

- Post-promotional dip

Trade Discounts

Short-term discounts offered to the trade:

- Advantages
  - Incentivize the trade to push our product
  - gets attention of sales force
- Disadvantages
  - might not be passed onto the consumer
  - trade forward buys (hurts production plans)
  - hard to forecast demand
  - trade expects discounts in the future (cost of doing business)

Scanback can help increase retail pass through (i.e., encourage retailers to have consumer discounts)

Determinants of pass through

- Higher when
  - Consumer elasticity is higher
  - promoting brand is stronger
  - shape of the demand function
  - lower frequency of promotions

(Online) Shelf-price discounts (Raju et al., 1990)

- If you are a stronger brand you can discount infrequently because the weaker brands cannot predict when the stronger brands will promote. Hence, it has to promote more frequently

Coupons

- Little over 1% get redeemed each year
- The ability of cents-off coupons to price distribution has reduced considerably because of their very wide availability
- Sales increases required to make free-standing-insert coupons profitable are not attainable

Coupon Design

- Expiration dates

  - Long vs short expiration dates: Stronger brands should have shorter windows (because a lot more of your loyalty customer base will utilize the coupons).

- Method of distribution
  - In-store (is better)
  - Through the package
  - Targeted promotions

Package Coupons acquisition and retention trade-offs

3 types of package coupons:

- Peel-off (lots more use the coupons) lowest profits for the firm
- in-packs (fewer customers will buy the products in the first period)
- on-packs (customers buy the products and they redeem in the next period) **best approach**

Summary

- Trade and consumer promotion are necessary
- Consumer promotion (avoid shelf price discount/news paper coupons, use package coupons
- strong interaction between advertising and promotion (area for more research)

3 degrees price discrimination

1. First-degree: based on willingness to pay
2. Second-degree: based on quantity
3. Third-degree: based on memberships
4. Fourth-degree: based on cost to serve

21.15.1 Narasimhan (1988)

- Marketing tools to promote products:
  - Advertising
  - Trade promotions
  - Consumer promotions
- Pricing promotions:
  - Price deals
  - Cents-off labels
  - coupons

    - Rebates

- Brand loyalty can help explain the variation in prices (in competitive markets)
- Firms try to make optimal trade-off between
    - attracting brand switchers
    - loss of profits from loyal customers.
- Deviation from the maximum price = promotion

Assumptions:

1. Firms with identical products, and cost structures (constant or declining). Non-cooperative game.
2. Same reservation price
3. Three consumer segments:
    1. Loyal to firm 1 with size $\alpha_1 (0 < \alpha_1 < 1)$
    2. Loyal to firm 2 with size $\alpha_2 (0 < \alpha_2 < \alpha_1)$ **(asymmetric firm)**
    3. Switchers with size $\beta (0 < \beta = 1 - \alpha_1 - \alpha_2)$
4. Costless price change, no intertemporal effects (in quantity or loyalty)

To model β either

1. $d \in (-b, a)$ is switch cost (individual parameter)

$$\begin{cases} \text{buy brand 1} & \text{if } P_1 \leq P_2 - d \\ \text{buy brand 2} & \text{if } P_1 > P_2 - d \end{cases}$$

2. Identical switchers (same d)
3. $d = 0$ (extremely price sensitive)

For case 1, there is a pure strategy, while case 2 and 3 have no pure strategies, only mixed strategies

Details for case 3:

Profit function

$$\Pi_i(P_i, P_j) = \alpha_i P_i + \delta_{ij} \beta P_i$$

where

$$\delta_{ij} = \begin{cases} 1 & \text{if } P_i < P_j \\ 1/2 & \text{if } P_i = P_j \\ 0 & \text{if } P_i > P_j \end{cases}$$

and $i = 1, 2, i \neq j$

Prop 1: no pure Nash equilibrium

Mixed Strategy profit function

$$\Pi_i(P_i) = \alpha_i P_i + Prob(P_j > P_i)\beta P_i + Prob(P_j = P_i)\frac{\beta}{2}P_i$$

where $P_i \in S_i^*, i \neq j; i, j = 1, 2$

Then the expected profit functions of the two-player game is

$$\max_{F_i} E(\Pi_i) = \int \Pi_i(P_i) dF_i(P_i)$$

$P_i \in S_i^*$

such that

$$\Pi_i \geq \alpha_i r \int dF_i(P_i) = 1 P_i \in S_i^*$$

21.15.2 Balachander et al. (2010)

- Bundle discounts can be more profitable than price promotions (in a competitive market) due to increased loyalty (which will reduce promotional competition intensity).

21.15.3 Goić et al. (2011)

- Cross-market discounts, purchases in a source market can get you a price discounts redeemable in a target market.
  - Increase prices and sales in the source market.

21.16 Market Entry Decisions and Diffusion

Golder and Tellis (1993)

Golder and Tellis (2004)

Boulding and Christen (2003)

Van den Bulte and Joshi (2007)

21.17 Principal-agent Models and Salesforce Compensation

21.17.1 Gerstner and Hess (1987)

21.17.2 Basu et al. (1985)

21.17.3 Raju and Srinivasan (1996)

Compare to (Basu et al., 1985), basic quota plan is superior in terms of implementation

Different from (Basu et al., 1985), basic quota plan has

1. Shape-induced nonoptimality: not a general curvilinear form
2. Heterogeneity-induced nonoptimality: common rate across salesforce

However, only 1% of cases in simulation shows up with nonoptimality. Hence, minimal loss in optimality

Basic quota plan is a also robust against changes in

- salesperson switching territory
- territorial changes (e.g., business condition)

Heterogeneity stems from

- Salesperson: effectiveness, risk level, disutility for effort, and alternative opportunity
- Territory: Sales potential and volatility

Adjusting quotas can accommodate the heterogeneity

To assess nonoptimality, following Basu and Kalyanaram (1990)

Moral hazard: cannot assess salesperson's true effort.

Assumptions:

- The salesperson reacts to the compensation scheme by deciding on an effort level that maximizes his overall utility, i.e., the expected utility from the (stochastic) compensation minus the effort distuility.
- Firm wants to maximize its profit
- compensation is greater than saleperson's alternative.
- Dollar sales $x_i \sim Gamma$ (because sales are non-negative and standard deviation getting proportionately larger as the mean increases) with density $f_i(x_i|t_i)$

Expected sales per period

$$E[x_i|t_i] = h_i + k_i t_i, (h_i > 0, k_i > 0)$$

where

- h_i = base sales level
- k_i = effectiveness of effort

and $1/\sqrt{c}$ = uncertainty in sales (coefficient of variation) = standard deviation / mean where $c \to \infty$ means perfect certainty

salesperson's overall utility

$$U_i[s_i(x_i)] - V_i(t_i) = \frac{1}{\delta_i}[s_i(x_i)]^{\delta_i} - d_i t_i^{\gamma_i}$$

where

- $0 < \delta_i < 1$ (greater δ means less risk-averse salesperson)
- $\gamma_i > 1$ (greater γ means more effort)
- $V_i(t_i) = d_i t_i^{\gamma_i}$ is the increasing disutility function (convex)

21.17.4 Lal and Staelin (1986)

- A menu of compensation plans (salesperson can select, which depends on their own perspective)
- Proposes conditions when it's optimal to offer a menu
- Under (Basu et al., 1985), they assume
  - Salespeople have identical risk characteristics
  - identical reservation utility
  - identical information about the environment
- When this paper relaxes these assumptions, menu of contract makes sense
- If you cannot distinguish (or have a selection mechanisms) between high performer and low performer, a menu is recommended. but if you can, you only need 1 contract like (Basu et al., 1985)

21.17.5 Simester and Zhang (2010)

21.18 Branding

Wernerfelt (1988)

- Umbrella branding

Chu and Chu (1994)

- retailer reputation
-

21.19 Marketing Resource Allocation Models

This section is based on (Mantrala et al., 1992)

21.19.1 Case study 1

Concave sales response function

- Optimal vs. proportional at different investment levels
- Profit maximization perspective of aggregate function

$$s_i = k_i(1 - e^{-b_i x_i})$$

where

- s_i = current-period sales response (dollars / period)
- x_i = amount of resource allocated to submarket i
- b_i = rate at which sales approach saturation
- k_i = sales potential

Allocation functions

- Fixed proportion
  - R_i = Investment level (dollars/period)
  - w_i = fixed proportion or weights

$$\hat{x}_i = w_i R; \sum_{t=1}^{2} w_t = 1; 0 < w_t < 1$$

- Informed allocator
  - optimal allocations via marginal analysis (maximize profits)

$$maxC = m\sum_{i=1}^{2} k_i(1-e^{-b_i x_i}) x_1+x_2 \leq R; x_i \geq 0 \text{ for } i = 1, 2 x_1 = \frac{1}{(b_1 + b_2)(b_2 R + \ln(\frac{k_1 b_1}{k_2 b_2})} x_2 = \frac{}{(b_1 + b_2)(b}$$

21.19.2 Case study 2

S-shaped sales response function:

- Optimal vs. proportional at different investment levels
- Profit maximization perspective of aggregate function

21.19.3 Case study 3

Quadratic-form stochastic response function

- Optimal allocation only with risk averse and risk neutral investors.

21.20 Mixed Strategies

Games with finite number, and finite strategy for each player, there will always be a Nash equilibrium (might not be pure Nash, but always mixed)

Extended game

- Suppose we allow each player to choose randomizing strategies
- For example, the server might serve left half of the time, and right half of the time
- In general, suppose the server serves left a fraction p of the time
- What is the receiver's best response?

Best Responses

If $p = 1$, the receiver should defend to the left

$p = 0$, the receiver should defend to the right

The expected payoff to the receiver is

$p \times 3/4 + (1 - p) \times 1/4$ if defending left

$p \times 1/4 + (1 - p) \times 3/4$ if defending right

Hence, she should defend left if

$p \times 3/4 - (1 - p) \times 1/4 > p \times 1/4 + (1 - p) \times 3/4$

We said to defend left whenever

$$p \times 3/4 - (1 - p) \times 1/4 > p \times 1/4 + (1 - p) \times 3/4$$

Server's Best response

- Suppose that the receiver goes left with probability q
- if $q = 1$, the server should serve right

- If $q = 0$, the server should server left
- Hence, serve left if $1/4 \times q + 3/4 \times (1 - q) > 3/4 \times q + 1/4 \times (1 - q)$
- Simplifying, he should serve left if $q < 1/2$

Mixed strategy equilibrium:

- A mixed strategy equilibrium is a pair of mixed strategies that are mutual best responses
- In the tennis example, this occurred when each play chose a 50-50 mixture of left and right
- Your best strategy is when you make the option given to your opponent is obsolete.
- A player chooses his strategy to make his rival indifferent
- A player earns the same expected payoff for each pure strategy chosen with positive probability
- Important property: When player's own payoff form a pure strategy goes up (or down), his mixture does not change.

21.21 Bundling

	Equipment	Installation
Customer Type 1	$8,000	$2,000
Customer Type 2	$5,000	$3,000

Say we have equal numbers of type 1 and type 2, then you would like to charge $5,000 for the equipment and $2,000 for installation when considering equipment and installment equally . If you price it separately, then your total profit is 14,000.

But you bundle, you get $16,000.

If we know that bundles work. But we don't see every company does ti?

Because it depends on the **number of type 1 and 2 customers, and negative correlation in willingness to pay**.

For example:

- Information Products
  - margin cost is close to 0.
  - Bundling of info products is very easy
  - hence always bundle

21.22 Market Entry and Diffusion

Product Life Cycle model

Bass (1969)

- Discussion of sales has 2 types
  - Innovators
  - Imitators

$p =$ coefficient of innovation (fraction of innovators of the untapped market who buy in that period)

$q =$ coefficient of imitation (fraction of the interaction which lead to sales in that period)

$M =$ market potential

$N(t) =$ cumulative sales till time t

$M - N(t) =$ the untapped market

Sales in any time is People buying because of the pure benefits of the product, plus people buy the product after interacting with people who owned the product.

$$S(t) = p(M - N(t)) + q\frac{N(t)}{M}[M - N(t)] = pM + (q-p)N(t) - \frac{q}{M}[N(t)]^2$$

one can estimate p, q, M from data

$q > p$ (coefficient of imitation > coefficient of innovation) means that you have life cycle (bell-shaped curve)

Golder and Tellis (1993)

Previous use

- limited databases (PIM and ASSESOR) (Urban et al., 1986)
- exclusion of nonsurvivors
- single-informant self-report

New dataset overcomes these limitations and show 50% of the market pioneers fail, and their mean share is much lower

Early market leaders have greater long-term success and enter on average 13 years after pioneers.

Definitions (p. 159)

1. Inventor: firms that develop patent or important technologies in a new product category
2. Product pioneer: the first firm to develop a working model or sample in a new product category
3. Market pioneer is the first firm to sell in a new product category
4. Product category: a group of close substitutes

Boulding and Christen (2003)

At the business level, being the leader can give you long-term profit disadvantage from the samples of consumer and industrial goods.

First-to-market leads to an initial profit advantage, which last about 12 to 14 years before becoming long-term disadvantage.

Consumer learning (education), market position (strong vs. weak) and patent protection can moderate the effect of first-mover on profit.

Golder and Tellis (2004)

- Research on product life cycle (PLC)
- Consumer durables typically grow 45 per year over 8 years, then slowdown when sales decline by 15%, and stay below those of the previous peak for 5 years.
- Slowdown typically happens when the product penetrates 35-50% of the market
- large sales increases (at first) will have larger sales declines (at slowdown).
- Leisure-enhancing products tend to have higher growth rate and shorter growth stages than non leisure-enhancing products
- Time-saving products have lower growth rates and longer growth stages than non time-saving products
- Lower likelihood of slowdown correlates with steeper price reduction, lower penetration, and higher economic growth
- A hazard model gives reasonable prediction of the slowdown and takeoff.

Van den Bulte and Joshi (2007)

- Innovations market have two segments:
  - Influentials: aware of new developments an affect imitators
  - Imitators: model after influentials.
- This market structure is reasonable because it exhibits consistent evidence with the prior research and market (e.g., dip between the early and later parts fo the diffusion curve).

- " Erroneously specifying a mixed-influence model to a mixture process where influentials act independently from each other can generate systematic changes in the parameter values reported in earlier research."
- Two-segments model performs better than the standard mixed-influence, the Gamma/Shifted Gompertz, the Weibull-Gamma models, and similar to the Karmeshu-Goswami mixed influence model.

21.23 Principal-Agent Models and Salesforce Compensation

Key Question:

1. Ensuring agents exert effort
2. Design compensation plans such that workers exert high effort?

Designing contracts:

- Effort can be monitored
- Monitoring costs are too high

Timing

1. Manger designs the construct
2. manager offers the construct and worker chooses to accept
3. Worker decides the extent of effort
4. Outcome is observed and wage is given to the worker

Scenario 1: Certainty

e = effort put in by worker

2 levels of e

- 2 if he works hard
- 0 if he shirks

Reservation utility = 10 (other alternative: can work somewhere else, or private money allows them not to work)

Agent's Utility

$$U = \begin{cases} w - e & \text{if he exerts effort e} \\ 10 & \text{if he works somewhere else} \end{cases}$$

Revenue is a function of effort

$$R(e) = \begin{cases} H & \text{if } e = 2 \\ L & \text{if } e = 0 \end{cases}$$

Contract

w^H = wage if $R(e) = H$

w^L = wage if $R(e) = L$

Constraints:

- Worker has to participate in this labor market - participation constraint $w^H - 2 \geq 10$
- Incentive compatibility constraint (ensure that the works always put in the effort and the manager always pay for the higher wage): $w^H - 2 \geq w^L - 0$

Hence,

$$w^H = 12w^L = 10$$

Thus, contract is simple because of monitoring

Scenario 2: Under uncertainty

$$R(2) = \begin{cases} H & \text{w/ prob } 0.8 \\ L & \text{w/ prob } 0.2 \end{cases} R(0) = \begin{cases} H & \text{w/ prob } 0.4 \\ L & \text{w/ prob } 0.6 \end{cases}$$

Agent Utility

$$U = \begin{cases} E(w) - e & \text{if effort e is put} \\ 10 & \text{if they choose outside option} \end{cases}$$

Constraints:

- Participation Constraint: $0.8w^H + 0.2w^L - 2 \geq 10$
- Incentive compatibility constraint: $0.8w^H + 0.2w^L - 2 \geq 0.4w^H + 0.6w^L - 0$

Thus,

$$w^H = 13w^L = 8$$

Expected wage bill that the manager has to pay:

$$13 \times 0.8 + 8 \times 0.2 = 12$$

Hence, the expected money the manager has to pay is the same for both cases (certainty vs. uncertainty)

Scenario 3: Asymmetric Information

Degrees of risk aversion

Manger perspective

$$R(2) = \begin{cases} H & \text{w/ prob 0.8} \\ L & \text{w/ prob 0.2} \end{cases}$$

Worker perspective (the number for worker is always lower, because they are more risk averse, managers are more risk neural) (the manager also knows this).

$$R(2) = \begin{cases} H & \text{w/ prob 0.7} \\ L & \text{w/ prob 0.3} \end{cases}$$

Participation Constraint

$$0.7w^H + 0.3w^L - 2 \geq 10$$

Incentive Compatibility Constraint

$$0.6w^H + 0.3w^L - 2 \geq 0.4w^H + 0.6w^L - 0$$

(take R(0) from scenario 2)

$$0.7w^H + 0.3w^L = 120.3w^H - 0.3w^L = 2$$

Hence,

$$w^H = 14w^L = 22/3$$

Expected wage bill for the manager is

$$14 * 0.8 + 22/3 * 0.2 = 12.66$$

Hence, expected wage bill is higher than scenario 2

Risk aversion from the worker forces the manager to pay higher wage

		Salesperson	
		Risk neutral	Risk averse
Effort	Observable	Any mix desired effort	All salary Desired effort

		Salesperson	
	Not observable	All commission Desired effort	Specific mix (S+C) Salesperson shirks

Grossman and Hart (1986)

- landmark paper for principal agent model

21.23.1 Basu et al. (1985)

Types of compensation plan:

- Independent of salesperson's performance (e.g., salary only)
- Partly dependent on output (e.g., salary with commissions)
- In comparison to others (e.g., sales contests)
- Options for salesperson to choose the compensation plan

In the first 2 categories, the 3 major schemes:

1. Straight salary
2. Straight commissions
3. Combination of base salary and commission

Compensation Type	Best when	Limitation
Straight salary	Long-term objective Hard to measure performance	Less effort
Straight commission	Easy-to-measure performance	Effort-reward ratio is emphasized High risk (uncertainty) to the salesperson
Combination		

Dimensions that affect the proportion of salary tot total pay (p. 270, table 1)

Previous research assumes deterministic relationship between sales and effort, but this study says otherwise (stochastic relationship between sales and effort).

Assumptions:

- Firm: Risk neutral: maximize expected profits
- Salesperson: Risk averse . Hence, diminishing marginal utility for income $U(s) \geq 0; U'(s) > 0, U''(s) < 0$
- Expected utility of the salesperson for this job > alternative

- Utility function of the salesperson: additively separable: $U(s) - V(t)$ where s = salary, and t = effort (time)
- Marginal disutility for effort increases with effort $V(t) \geq 0, V'(t) > 0, V''(t) > 0$
- Constant marginal cost of production and distribution c
- Known utility function and sales-effort response function (both principal and agent)
- dollar sales $x \sim Gamma, Binom$

Expected profit for the firm

$$\pi = \int [(1-c)x - s(x)] f(x|t) dx$$

Objective of the firm is to

$$\max_{s(x)} \int [(1-c)x - s(x)] f(x|t) dx$$

subject to (agent's best alternative e.g., other job offer - m)

$$\int [U(s(x))] f(x|t) dx - V(t) \geq m$$

and the agent wants to maximize the the utility

$$\max_{t} \int [U(s(x))] f(x|t) dx - V(t)$$

21.23.2 Lal and Staelin (1986)

21.23.3 Raju and Srinivasan (1996)

Compare quota-based compensation with (Basu et al., 1985) curvilinear compensation, the basic quota plan is simpler, and only in specical cases (about 1% in simulation) that differs from (Basu et al., 1985). And it's easier to adapt to changes in moving salesperson and changing territory, unlike (Basu et al., 1985)'s plan where the whole commission rate structure needs to be changed.

Heterogeneity stems from:

- Salesperson: disutility effort level, risk level, effectiveness, alternative opportunity
- Territory: Sales potential an volatility

Adjusting the quota (per territory) can accommodate the heterogeneity

Quota-based < BLSS (in terms of profits)

Constraints:

1. quota-based from curve (between total compensation and sales) (i.e., shape-induced nonoptimality)
2. common salary and commission rate across salesforce (i.e., heterogeneity-induced nonoptimality)

To assess the shape-induced nonoptimality following

21.23.4 Joseph and Thevaranjan (1998)

21.23.5 Simester and Zhang (2010)

- Tradeoff: Motivating manager effort and info sharing.

21.24 Meta-analyses of Econometric Marketing Models

21.25 Dynamic Advertising Effects and Spending Models

21.26 Marketing Mix Optimization Models

Check this post for implementation in Python

21.27 New Product Diffusion Models

21.28 Two-sided Platform Marketing Models

Example of Marketing Mix Model in practice: link

Chapter 22

Empirical Models

22.1 Attribution Models

22.1.1 Ordered Shapley

Based on (Zhao et al., 2018) (to access paper) Cooperative game theory: look at the marginal contribution of each player in the game, where **Shapley value** (i..e, the credit assigned to each individual) is the expected value value of the marginal contribution over all possible permutations (e.g., all possible sequences) of the players.

Shapely value considered:

- marginal contribution of each player (i.e., channel)
- sequence of joining the coalition (i.e., customer journey).

Typically, we can't apply the Shapley Value method due to computational burden (you need all possible permutations). And a drawback is that all the credit must be divided among your channels, if you have missing channels, then it will distort the estimates of other channels' estimates.

It's hard to use Shapley value model for model comparison since we have no "ground truth"

Marketing application:

- Pardot Einstein features and more

```
library("GameTheory")
```

packages reference

22.1.2 Markov Model

Markov chains maps the movement and gives a probability distribution, for moving from one state to another state. A Markov Chain has three properties:

- **State space** – set of all the states in which process could potentially exist
- **Transition operator** –the probability of moving from one state to other state
- **Current state probability distribution** – probability distribution of being in any one of the states at the start of the process

In mathematically sense

$$w_{ij} = P(X_t = s_j | X_{t-1} = s_i), 0 \leq w_{ij} \leq 1, \sum_{j=1}^{N} w_{ij} = 1 \forall i$$

where

- The Transition Probability (w_{ij}) = The Probability of the Previous State (X_{t-1}) Given the Current State (X_t)
- The Transition Probability (w_{ij}) is No Less Than 0 and No Greater Than 1
- The Sum of the Transition Probabilities Equals 1 (i.e., Everyone Must Go Somewhere)

To examine a particular node in the Markov graph, we use **removal effect** (s_i) to see its contribution to a conversion. In another word, the Removal Effect is the probability of converting when a step is completely removed; all sequences that had to go through that step are now sent directly to the null node

Each node is called **transition states**
The probability of moving from one channel to another channel is called **transition probability**.

first-order or "memory-free" Markov graph is called "memory-free" because the probability of reaching one state depends only on the previous state visited.

- Order 0: Do not care about where the you came from or what step the you are on, only the probability of going to any state.
- Order 1: Looks back zero steps. You are currently at a state. The probability of going anywhere is based on being at that step.
- Order 2: Looks back one step. You came from state A and are currently at state B. The probability of going anywhere is based on where you were and where you are.
- Order 3: Looks back two steps. You came from state A after state B and are currently at state C. The probability of going anywhere is based on where you were and where you are.

- Order 4: Looks back three steps. You came from state A after B after C and are currently at state D. The probability of going anywhere is based on where you were and where you are.

22.1.2.1 Example 1

This section is by Analytics Vidhya

data link

```
# #Install the libraries
# install.packages("ChannelAttribution")
# install.packages("ggplot2")
# install.packages("reshape")
# install.packages("dplyr")
# install.packages("plyr")
# install.packages("reshape2")
# install.packages("markovchain")
# install.packages("plotly")

#Load the libraries
library("ChannelAttribution")
library("ggplot2")
library("reshape")
library("dplyr")
library("plyr")
library("reshape2")
# library("markovchain")
library("plotly")

#Read the data into R
channel = read.csv("images/Channel_attribution.csv", header = T) %>%
    select(-c(Output))
head(channel, n = 2)
```

```
##   R05A.01 R05A.02 R05A.03 R05A.04 R05A.05 R05A.06 R05A.07 R05A.08 R05A.09
## 1      16       4       3       5      10       8       6       8      13
## 2       2       1       9      10       1       4       3      21      NA
##   R05A.10 R05A.11 R05A.12 R05A.13 R05A.14 R05A.15 R05A.16 R05A.17 R05A.18
## 1      20      21      NA      NA      NA      NA      NA      NA      NA
## 2      NA      NA      NA      NA      NA      NA      NA      NA      NA
##   R05A.19 R05A.20
## 1      NA      NA
## 2      NA      NA
```

The number represents:

- 1-19 are various channels

- 20 – customer has decided which device to buy;
- 21 – customer has made the final purchase, and;
- 22 – customer hasn't decided yet.

Pre-processing

```
for (row in 1:nrow(channel)){
    if (21 %in% channel[row,]){
        channel$convert = 1
    }
}

column = colnames(channel)
channel$path = do.call(paste, c(channel, sep = " > "))
head(channel$path)
```

```
## [1] "16 > 4 > 3 > 5 > 10 > 8 > 6 > 8 > 13 > 20 > 21 > NA > NA > NA > NA > NA > NA >
## [2] "2 > 1 > 9 > 10 > 1 > 4 > 3 > 21 > NA > NA > NA > NA > NA > NA > NA > NA > NA >
## [3] "9 > 13 > 20 > 16 > 15 > 21 > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA >
## [4] "8 > 15 > 20 > 21 > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA >
## [5] "16 > 9 > 13 > 20 > 21 > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA >
## [6] "1 > 11 > 8 > 4 > 9 > 21 > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA > NA
```

```
for(row in 1:nrow(channel)){
  channel$path[row] = strsplit(channel$path[row], " > 21")[[1]][1]
}
channel_fin = channel[,c(22,21)]
channel_fin = ddply(channel_fin,~path,summarise, conversion= sum(convert))
head(channel_fin)
```

```
##                             path conversion
## 1               1 > 1 > 1 > 20          1
## 2              1 > 1 > 12 > 12          1
## 3    1 > 1 > 14 > 13 > 12 > 20          1
## 4      1 > 1 > 3 > 13 > 3 > 20          1
## 5          1 > 1 > 3 > 17 > 17          1
## 6 1 > 1 > 6 > 1 > 12 > 20 > 12          1
```

```
Data = channel_fin
head(Data)
```

```
##                             path conversion
## 1               1 > 1 > 1 > 20          1
## 2              1 > 1 > 12 > 12          1
## 3    1 > 1 > 14 > 13 > 12 > 20          1
## 4      1 > 1 > 3 > 13 > 3 > 20          1
## 5          1 > 1 > 3 > 17 > 17          1
## 6 1 > 1 > 6 > 1 > 12 > 20 > 12          1
```

heuristic model

```r
H <- heuristic_models(Data, 'path', 'conversion', var_value='conversion')
```

```
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for free! Visit
```

```r
H
```

```
##    channel_name first_touch_conversions first_touch_value
## 1             1                     130               130
## 2            20                       0                 0
## 3            12                      75                75
## 4            14                      34                34
## 5            13                     320               320
## 6             3                     168               168
## 7            17                      31                31
## 8             6                      50                50
## 9             8                      56                56
## 10           10                     547               547
## 11           11                      66                66
## 12           16                     111               111
## 13            2                     199               199
## 14            4                     231               231
## 15            7                      26                26
## 16            5                      62                62
## 17            9                     250               250
## 18           15                      22                22
## 19           18                       4                 4
## 20           19                      10                10
##    last_touch_conversions last_touch_value linear_touch_conversions
## 1                      18               18                73.773661
## 2                    1701             1701               473.998171
## 3                      23               23                76.127863
## 4                      25               25                56.335744
## 5                      76               76               204.039552
## 6                      21               21               117.609677
## 7                      47               47                76.583847
## 8                      20               20                54.707124
## 9                      17               17                53.677862
## 10                     42               42               211.822393
## 11                     33               33               107.109048
## 12                     95               95               156.049086
## 13                     18               18                94.111668
## 14                     88               88               250.784033
## 15                     15               15                33.435991
## 16                     23               23                74.900402
## 17                     71               71               194.071690
```

```
## 18                     47               47                 65.159225
## 19                      2                2                  5.026587
## 20                     10               10                 12.676375
##    linear_touch_value
## 1           73.773661
## 2          473.998171
## 3           76.127863
## 4           56.335744
## 5          204.039552
## 6          117.609677
## 7           76.583847
## 8           54.707124
## 9           53.677862
## 10         211.822393
## 11         107.109048
## 12         156.049086
## 13          94.111668
## 14         250.784033
## 15          33.435991
## 16          74.900402
## 17         194.071690
## 18          65.159225
## 19           5.026587
## 20          12.676375
```

- First Touch Conversion: credit is given to the first touch point.
- Last Touch Conversion: credit is given to the last touch point.
- Linear Touch Conversion: All channels/touch points are given equal credit in the conversion.

Markov model

```r
M <- markov_model(Data, 'path', 'conversion', var_value='conversion', order = 1)
```

```
##
## Number of simulations: 100000 - Convergence reached: 2.05% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
##
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for f
```

```r
M
```

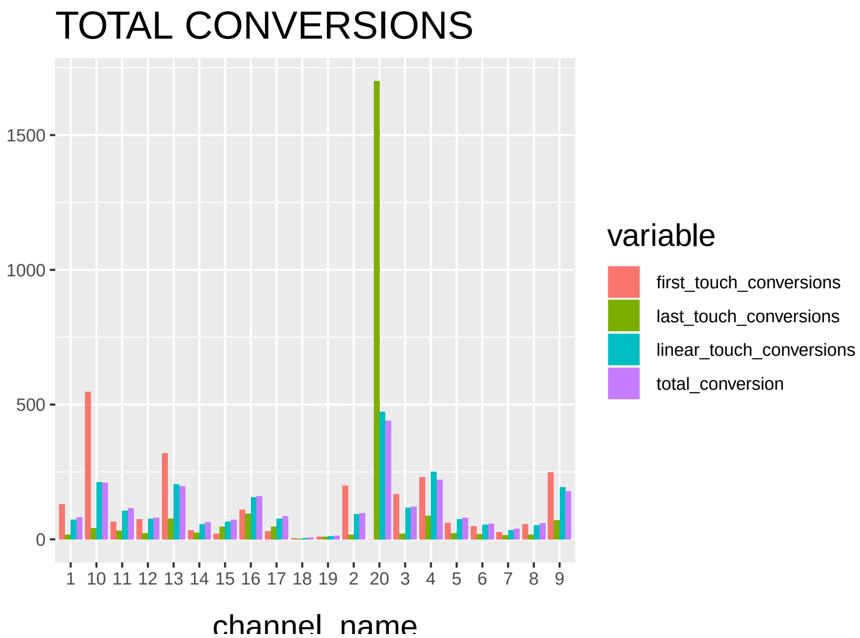
```
##    channel_name total_conversion total_conversion_value
## 1             1        82.805970              82.805970
## 2            20       439.582090             439.582090
## 3            12        81.253731              81.253731
```

```
## 4           14       64.238806               64.238806
## 5           13      197.791045              197.791045
## 6            3      122.328358              122.328358
## 7           17       86.985075               86.985075
## 8            6       58.985075               58.985075
## 9            8       60.656716               60.656716
## 10          10      209.850746              209.850746
## 11          11      115.402985              115.402985
## 12          16      159.820896              159.820896
## 13           2       97.074627               97.074627
## 14           4      222.149254              222.149254
## 15           7       40.597015               40.597015
## 16           5       80.537313               80.537313
## 17           9      178.865672              178.865672
## 18          15       72.358209               72.358209
## 19          18        6.567164                6.567164
## 20          19       14.149254               14.149254
```

combine the two models

```r
# Merges the two data frames on the "channel_name" column.
R <- merge(H, M, by='channel_name')

# Select only relevant columns
R1 <- R[, (colnames(R) %in% c('channel_name', 'first_touch_conversions', 'last_touch_conversions'

# Transforms the dataset into a data frame that ggplot2 can use to plot the outcomes
R1 <- melt(R1, id='channel_name')
```

```r
# Plot the total conversions
ggplot(R1, aes(channel_name, value, fill = variable)) +
  geom_bar(stat='identity', position='dodge') +
  ggtitle('TOTAL CONVERSIONS') +
  theme(axis.title.x = element_text(vjust = -2)) +
  theme(axis.title.y = element_text(vjust = +2)) +
  theme(title = element_text(size = 16)) +
  theme(plot.title=element_text(size = 20)) +
  ylab("")
```



and then check the final results.

22.1.2.2 Example 2

Example code by Sergey Bryl'

```
library(dplyr)
library(reshape2)
library(ggplot2)
library(ggthemes)
library(ggrepel)
library(RColorBrewer)
library(ChannelAttribution)
library(markovchain)

##### simple example #####
# creating a data sample
df1 <- data.frame(path = c('c1 > c2 > c3', 'c1', 'c2 > c3'), conv = c(1, 0, 0), conv_nu

# calculating the model
mod1 <- markov_model(df1,
                     var_path = 'path',
                     var_conv = 'conv',
                     var_null = 'conv_null',
                     out_more = TRUE)
```

```
# extracting the results of attribution
df_res1 <- mod1$result

# extracting a transition matrix
df_trans1 <- mod1$transition_matrix
df_trans1 <- dcast(df_trans1, channel_from ~ channel_to, value.var = 'transition_probability')

### plotting the Markov graph ###
df_trans <- mod1$transition_matrix

# adding dummies in order to plot the graph
df_dummy <- data.frame(channel_from = c('(start)', '(conversion)', '(null)'),
                       channel_to = c('(start)', '(conversion)', '(null)'),
                       transition_probability = c(0, 1, 1))
df_trans <- rbind(df_trans, df_dummy)

# ordering channels
df_trans$channel_from <- factor(df_trans$channel_from,levels = c('(start)','(conversion)', '(null
df_trans$channel_to <- factor(df_trans$channel_to,levels = c('(start)', '(conversion)', '(null)',
df_trans <- dcast(df_trans, channel_from ~ channel_to, value.var ='transition_probability')

# creating the markovchain object
trans_matrix <- matrix(data = as.matrix(df_trans[, -1]),nrow = nrow(df_trans[, -1]), ncol = ncol(
trans_matrix[is.na(trans_matrix)] <- 0
# trans_matrix1 <- new("markovchain", transitionMatrix = trans_matrix)
#
# # plotting the graph
# plot(trans_matrix1, edge.arrow.size = 0.35)
```

```
# simulating the "real" data
set.seed(354)
df2 <- data.frame(client_id = sample(c(1:1000), 5000, replace = TRUE),
                  date = sample(c(1:32), 5000, replace = TRUE),
                  channel = sample(c(0:9), 5000, replace = TRUE,
                                   prob = c(0.1, 0.15, 0.05, 0.07, 0.11, 0.07, 0.13, 0.1, 0.06, 0
df2$date <- as.Date(df2$date, origin = "2015-01-01")
df2$channel <- paste0('channel_', df2$channel)

# aggregating channels to the paths for each customer
df2 <- df2 %>%
        arrange(client_id, date) %>%
        group_by(client_id) %>%
        summarise(path = paste(channel, collapse = ' > '),
                  # assume that all paths were finished with conversion
                  conv = 1,
```

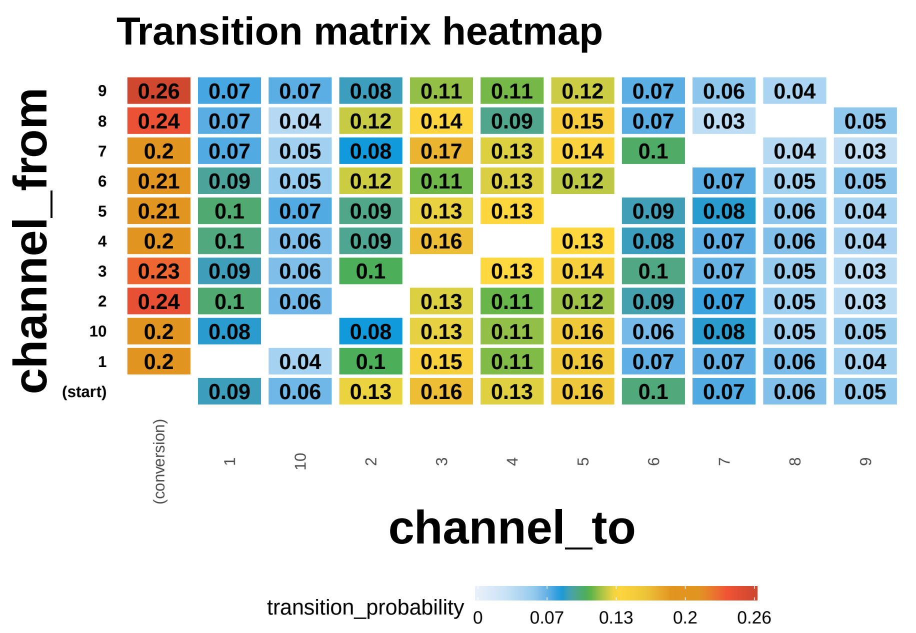
```
                   conv_null = 0) %>%
        ungroup()

# calculating the models (Markov and heuristics)
mod2 <- markov_model(df2,
                      var_path = 'path',
                      var_conv = 'conv',
                      var_null = 'conv_null',
                      out_more = TRUE)
```

```
##
## Number of simulations: 100000 - Convergence reached: 1.40% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
##
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for fr
```

```
# heuristic_models() function doesn't work for me, therefore I used the manual calcula
# instead of:
#h_mod2 <- heuristic_models(df2, var_path = 'path', var_conv = 'conv')

df_hm <- df2 %>%
        mutate(channel_name_ft = sub('>.*', '', path),
               channel_name_ft = sub(' ', '', channel_name_ft),
               channel_name_lt = sub('.*>', '', path),
               channel_name_lt = sub(' ', '', channel_name_lt))
# first-touch conversions
df_ft <- df_hm %>%
        group_by(channel_name_ft) %>%
        summarise(first_touch_conversions = sum(conv)) %>%
        ungroup()
# last-touch conversions
df_lt <- df_hm %>%
        group_by(channel_name_lt) %>%
        summarise(last_touch_conversions = sum(conv)) %>%
        ungroup()

h_mod2 <- merge(df_ft, df_lt, by.x = 'channel_name_ft', by.y = 'channel_name_lt')

# merging all models
all_models <- merge(h_mod2, mod2$result, by.x = 'channel_name_ft', by.y = 'channel_name
colnames(all_models)[c(1, 4)] <- c('channel_name', 'attrib_model_conversions')
```

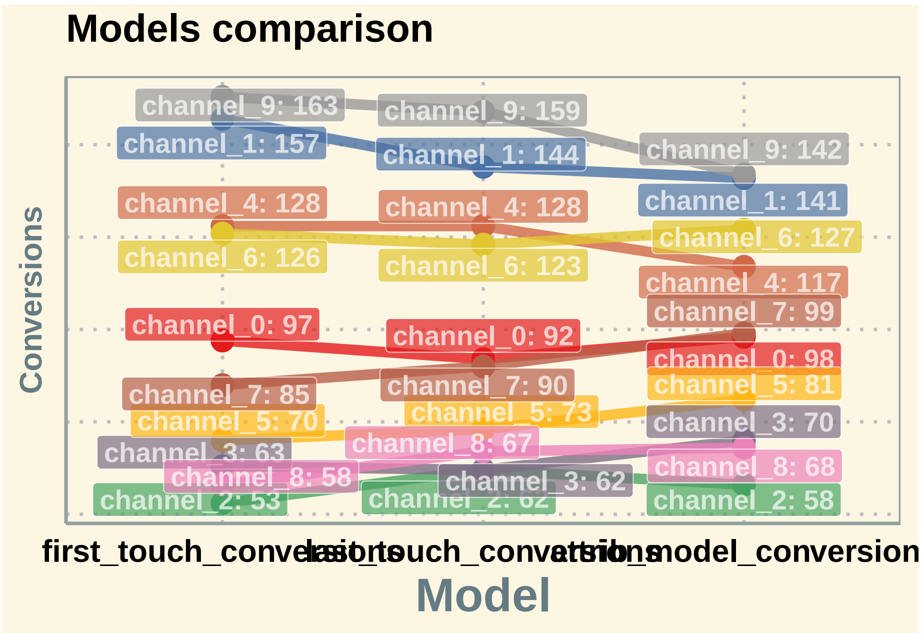
```
library("RColorBrewer")
library("ggthemes")
library("ggrepel")
```

```
############# visualizations #############
# transition matrix heatmap for "real" data
df_plot_trans <- mod2$transition_matrix

cols <- c("#e7f0fa", "#c9e2f6", "#95cbee", "#0099dc", "#4ab04a", "#ffd73e", "#eec73a",
          "#e29421", "#e29421", "#f05336", "#ce472e")
t <- max(df_plot_trans$transition_probability)

ggplot(df_plot_trans, aes(y = channel_from, x = channel_to, fill = transition_probability)) +
        theme_minimal() +
        geom_tile(colour = "white", width = .9, height = .9) +
        scale_fill_gradientn(colours = cols, limits = c(0, t),
                             breaks = seq(0, t, by = t/4),
                             labels = c("0", round(t/4*1, 2), round(t/4*2, 2), round(t/4*3, 2), r
                             guide = guide_colourbar(ticks = T, nbin = 50, barheight = .5, label
        geom_text(aes(label = round(transition_probability, 2)), fontface = "bold", size = 4) +
        theme(legend.position = 'bottom',
              legend.direction = "horizontal",
              panel.grid.major = element_blank(),
              panel.grid.minor = element_blank(),
              plot.title = element_text(size = 20, face = "bold", vjust = 2, color = 'black', lin
              axis.title.x = element_text(size = 24, face = "bold"),
              axis.title.y = element_text(size = 24, face = "bold"),
              axis.text.y = element_text(size = 8, face = "bold", color = 'black'),
              axis.text.x = element_text(size = 8, angle = 90, hjust = 0.5, vjust = 0.5, face = '
        ggtitle("Transition matrix heatmap")
```




```
# models comparison
all_mod_plot <- reshape2::melt(all_models, id.vars = 'channel_name', variable.name = '
all_mod_plot$value <- round(all_mod_plot$value)
# slope chart
pal <- colorRampPalette(brewer.pal(10, "Set1"))
ggplot(all_mod_plot, aes(x = conv_type, y = value, group = channel_name)) +
        theme_solarized(base_size = 18, base_family = "", light = TRUE) +
        scale_color_manual(values = pal(10)) +
        scale_fill_manual(values = pal(10)) +
        geom_line(aes(color = channel_name), size = 2.5, alpha = 0.8) +
        geom_point(aes(color = channel_name), size = 5) +
        geom_label_repel(aes(label = paste0(channel_name, ': ', value), fill = factor(
                         alpha = 0.7,
                         fontface = 'bold', color = 'white', size = 5,
                         box.padding = unit(0.25, 'lines'), point.padding = unit(0.5,
                         max.iter = 100) +
        theme(legend.position = 'none',
              legend.title = element_text(size = 16, color = 'black'),
              legend.text = element_text(size = 16, vjust = 2, color = 'black'),
              plot.title = element_text(size = 20, face = "bold", vjust = 2, color = '
              axis.title.x = element_text(size = 24, face = "bold"),
              axis.title.y = element_text(size = 16, face = "bold"),
              axis.text.x = element_text(size = 16, face = "bold", color = 'black'),
              axis.text.y = element_blank(),
              axis.ticks.x = element_blank(),
```

```
            axis.ticks.y = element_blank(),
            panel.border = element_blank(),
            panel.grid.major = element_line(colour = "grey", linetype = "dotted"),
            panel.grid.minor = element_blank(),
            strip.text = element_text(size = 16, hjust = 0.5, vjust = 0.5, face = "bold", color
            strip.background = element_rect(fill = "#f0b35f")) +
      labs(x = 'Model', y = 'Conversions') +
      ggtitle('Models comparison') +
      guides(colour = guide_legend(override.aes = list(size = 4)))
```



Additional concerns:

```
library(tidyverse)
library(reshape2)
library(ggthemes)
library(ggrepel)
library(RColorBrewer)
library(ChannelAttribution)
# library(markovchain)
library(visNetwork)
library(expm)
library(stringr)
library(purrr)
library(purrrlyr)
```

```
##### simulating the "real" data #####
set.seed(454)
df_raw <- data.frame(customer_id = paste0('id', sample(c(1:20000), replace = TRUE)), da
        group_by(customer_id) %>%
        mutate(conversion = sample(c(0, 1), n(), prob = c(0.975, 0.025), replace = TRU
        ungroup() %>%
        dmap_at(c(1, 3), as.character) %>%
        arrange(customer_id, date)

df_raw <- df_raw %>%
        mutate(channel = ifelse(channel == 'channel_2', NA, channel))
head(df_raw, n = 2)
```

```
## # A tibble: 2 x 4
##   customer_id date       channel   conversion
##   <chr>       <date>     <chr>          <dbl>
## 1 id1         2016-01-02 channel_7          0
## 2 id1         2016-01-09 channel_4          0
```

22.1.2.2.1 1. Customers will be at different stage of purchase journey after each conversion. First-time buyer's journey will look different from n-times buyer's (e.g., he will not start at website)

You can create your own code to split data into customers in different stages.

```
##### splitting paths #####
df_paths <- df_raw %>%
        group_by(customer_id) %>%
        mutate(path_no = ifelse(is.na(lag(cumsum(conversion))), 0, lag(cumsum(conversi
        ungroup()
head(df_paths)
```

```
## # A tibble: 6 x 5
##   customer_id date       channel   conversion path_no
##   <chr>       <date>     <chr>          <dbl>   <dbl>
## 1 id1         2016-01-02 channel_7          0       1
## 2 id1         2016-01-09 channel_4          0       1
## 3 id1         2016-01-18 channel_5          1       1
## 4 id1         2016-01-20 channel_4          1       2
## 5 id100       2016-01-01 channel_0          0       1
## 6 id100       2016-01-01 channel_0          0       1
```

attribution path for first-time buyers:

```
df_paths_1 <- df_paths %>%
        filter(path_no == 1) %>%
        select(-path_no)
```

22.1.2.2.2 2. Handle missing data We might have missing data on the channel or do not want to attribute a path (e.g., Direct Channel). We can either

- Remove NA/Channel
- Use the previous channel in its place.

In the first-order Markov chains, the results are unchanged because duplicated channels don't affect the calculation.

```
##### replace some channels #####
df_path_1_clean <- df_paths_1 %>%
        # removing NAs
        filter(!is.na(channel)) %>%

        # adding order of channels in the path
        group_by(customer_id) %>%
        mutate(ord = c(1:n()),
               is_non_direct = ifelse(channel == 'channel_6', 0, 1),
               is_non_direct_cum = cumsum(is_non_direct)) %>%

        # removing Direct (channel_6) when it is the first in the path
        filter(is_non_direct_cum != 0) %>%

        # replacing Direct (channel_6) with the previous touch point
        mutate(channel = ifelse(channel == 'channel_6', channel[which(channel != 'channel_6')][is

        ungroup() %>%
        select(-ord, -is_non_direct, -is_non_direct_cum)
```

22.1.2.2.3 3. one vs. multi-channel paths We need to calculate the weighted importance for each channel because the sum of the Removal Effects doesn't equal to 1. In case we have a path with a unique channel, the Removal Effect and importance of this channel for that exact path is 1. However, weighting with other multi-channel paths will decrease the importance of one-channel occurrences. That means that, in case we have a channel that occurs in one-channel paths, usually it will be underestimated if attributed with multi-channel paths.

There is also a pretty straight logic behind splitting – for one-channel paths, we definitely know the channel that brought a conversion and we don't need to distribute that value into other channels.

To account for one-channel path:

1. Split data for paths with one or more unique channels
2. Calculate total conversions for one-channel paths and compute the Markov model for multi-channel paths
3. Summarize results for each channel.

```
##### one- and multi-channel paths #####
df_path_1_clean <- df_path_1_clean %>%
        group_by(customer_id) %>%
        mutate(uniq_channel_tag = ifelse(length(unique(channel)) == 1, TRUE, FALSE)) %>
        ungroup()

df_path_1_clean_uniq <- df_path_1_clean %>%
        filter(uniq_channel_tag == TRUE) %>%
        select(-uniq_channel_tag)

df_path_1_clean_multi <- df_path_1_clean %>%
        filter(uniq_channel_tag == FALSE) %>%
        select(-uniq_channel_tag)

### experiment ###
# attribution model for all paths
df_all_paths <- df_path_1_clean %>%
        group_by(customer_id) %>%
        summarise(path = paste(channel, collapse = ' > '),
                  conversion = sum(conversion)) %>%
        ungroup() %>%
        filter(conversion == 1)

mod_attrib <- markov_model(df_all_paths,
                           var_path = 'path',
                           var_conv = 'conversion',
                           out_more = TRUE)
```

```
##
## Number of simulations: 100000 - Convergence reached: 1.28% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
##
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for f
```

```
mod_attrib$removal_effects
```

```
##   channel_name removal_effects
## 1    channel_7       0.2812250
## 2    channel_4       0.4284428
## 3    channel_5       0.6056845
## 4    channel_0       0.5367294
## 5    channel_1       0.3820056
## 6    channel_3       0.2535028
```

```r
mod_attrib$result
```

```
##   channel_name total_conversions
## 1    channel_7          192.8653
## 2    channel_4          293.8279
## 3    channel_5          415.3811
## 4    channel_0          368.0913
## 5    channel_1          261.9811
## 6    channel_3          173.8533
```

```r
d_all <- data.frame(mod_attrib$result)

# attribution model for splitted multi and unique channel paths
df_multi_paths <- df_path_1_clean_multi %>%
        group_by(customer_id) %>%
        summarise(path = paste(channel, collapse = ' > '),
                  conversion = sum(conversion)) %>%
        ungroup() %>%
        filter(conversion == 1)

mod_attrib_alt <- markov_model(df_multi_paths,
                               var_path = 'path',
                               var_conv = 'conversion',
                               out_more = TRUE)
```

```
##
## Number of simulations: 100000 - Convergence reached: 1.21% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps (19) is rea
##
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for free! Visit
```

```r
mod_attrib_alt$removal_effects
```

```
##   channel_name removal_effects
## 1    channel_7       0.3265696
## 2    channel_4       0.4844802
## 3    channel_5       0.6526369
## 4    channel_0       0.5814164
## 5    channel_1       0.4343546
## 6    channel_3       0.2898041
```

```r
mod_attrib_alt$result
```

```
##   channel_name total_conversions
## 1    channel_7          150.9460
## 2    channel_4          223.9350
## 3    channel_5          301.6599
```

```
## 4    channel_0          268.7406
## 5    channel_1          200.7661
## 6    channel_3          133.9524
```

```
# adding unique paths
df_uniq_paths <- df_path_1_clean_uniq %>%
        filter(conversion == 1) %>%
        group_by(channel) %>%
        summarise(conversions = sum(conversion)) %>%
        ungroup()

d_multi <- data.frame(mod_attrib_alt$result)

d_split <- full_join(d_multi, df_uniq_paths, by = c('channel_name' = 'channel')) %>%
        mutate(result = total_conversions + conversions)

sum(d_all$total_conversions)
```

```
## [1] 1706
```

```
sum(d_split$result)
```

```
## [1] 1706
```

22.1.2.2.4 4. Higher Order Markov Chains Since the transition matrix stays the same in the first order Markov, having duplicates will not affect the result. But starting from the second order order Markov, you will have different results when skipping duplicates. In order to check the effect of skipping duplicates in the first-order Markov chain, we will use my script for "manual" calculation because the package skips duplicates automatically.

```
##### Higher order of Markov chains and consequent duplicated channels in the path ###

# computing transition matrix - 'manual' way
df_multi_paths_m <- df_multi_paths %>%
        mutate(path = paste0('(start) > ', path, ' > (conversion)'))
m <- max(str_count(df_multi_paths_m$path, '>')) + 1 # maximum path length

df_multi_paths_cols <- reshape2::colsplit(string = df_multi_paths_m$path, pattern = ' >
colnames(df_multi_paths_cols) <- paste0('ord_', c(1:m))
df_multi_paths_cols[df_multi_paths_cols == ''] <- NA

df_res <- vector('list', ncol(df_multi_paths_cols) - 1)

for (i in c(1:(ncol(df_multi_paths_cols) - 1))) {

        df_cache <- df_multi_paths_cols %>%
```

```
                select(num_range("ord_", c(i, i+1))) %>%
                na.omit() %>%
                group_by_(.dot = c(paste0("ord_", c(i, i+1)))) %>%
                summarise(n = n()) %>%
                ungroup()

        colnames(df_cache)[c(1, 2)] <- c('channel_from', 'channel_to')
        df_res[[i]] <- df_cache
}

df_res <- do.call('rbind', df_res)

df_res_tot <- df_res %>%
        group_by(channel_from, channel_to) %>%
        summarise(n = sum(n)) %>%
        ungroup() %>%
        group_by(channel_from) %>%
        mutate(tot_n = sum(n),
               perc = n / tot_n) %>%
        ungroup()

df_dummy <- data.frame(channel_from = c('(start)', '(conversion)', '(null)'),
                       channel_to = c('(start)', '(conversion)', '(null)'),
                       n = c(0, 0, 0),
                       tot_n = c(0, 0, 0),
                       perc = c(0, 1, 1))

df_res_tot <- rbind(df_res_tot, df_dummy)

# comparing transition matrices
trans_matrix_prob_m <- dcast(df_res_tot, channel_from ~ channel_to, value.var = 'perc', fun.aggre
trans_matrix_prob <- data.frame(mod_attrib_alt$transition_matrix)
trans_matrix_prob <- dcast(trans_matrix_prob, channel_from ~ channel_to, value.var = 'transition_

# computing attribution - 'manual' way
channels_list <- df_path_1_clean_multi %>%
        filter(conversion == 1) %>%
        distinct(channel)
channels_list <- c(channels_list$channel)

df_res_ini <- df_res_tot %>% select(channel_from, channel_to)
df_attrib <- vector('list', length(channels_list))

for (i in c(1:length(channels_list))) {
```

```
        channel <- channels_list[i]

        df_res1 <- df_res %>%
                mutate(channel_from = ifelse(channel_from == channel, NA, channel_from)
                       channel_to = ifelse(channel_to == channel, '(null)', channel_to)
                na.omit()

        df_res_tot1 <- df_res1 %>%
                group_by(channel_from, channel_to) %>%
                summarise(n = sum(n)) %>%
                ungroup() %>%

                group_by(channel_from) %>%
                mutate(tot_n = sum(n),
                       perc = n / tot_n) %>%
                ungroup()

        df_res_tot1 <- rbind(df_res_tot1, df_dummy) # adding (start), (conversion) and

        df_res_tot1 <- left_join(df_res_ini, df_res_tot1, by = c('channel_from', 'chann
        df_res_tot1[is.na(df_res_tot1)] <- 0

        df_trans1 <- dcast(df_res_tot1, channel_from ~ channel_to, value.var = 'perc',

        trans_matrix_1 <- df_trans1
        rownames(trans_matrix_1) <- trans_matrix_1$channel_from
        trans_matrix_1 <- as.matrix(trans_matrix_1[, -1])

        inist_n1 <- dcast(df_res_tot1, channel_from ~ channel_to, value.var = 'n', fun
        rownames(inist_n1) <- inist_n1$channel_from
        inist_n1 <- as.matrix(inist_n1[, -1])
        inist_n1[is.na(inist_n1)] <- 0
        inist_n1 <- inist_n1['(start)', ]

        res_num1 <- inist_n1 %*% (trans_matrix_1 %^% 100000)

        df_cache <- data.frame(channel_name = channel,
                               conversions = as.numeric(res_num1[1, 1]))

        df_attrib[[i]] <- df_cache
}

df_attrib <- do.call('rbind', df_attrib)

# computing removal effect and results
```

```
tot_conv <- sum(df_multi_paths_m$conversion)

df_attrib <- df_attrib %>%
        mutate(tot_conversions = sum(df_multi_paths_m$conversion),
               impact = (tot_conversions - conversions) / tot_conversions,
               tot_impact = sum(impact),
               weighted_impact = impact / tot_impact,
               attrib_model_conversions = round(tot_conversions * weighted_impact)
        ) %>%
        select(channel_name, attrib_model_conversions)
```

Since with different transition matrices, the removal effects and attribution results stay the same, in practice we skip duplicates.

22.1.2.2.5 5. Non-conversion paths We incorporate null paths in this analysis.

```
##### Generic Probabilistic Model #####
df_all_paths_compl <- df_path_1_clean %>%
        group_by(customer_id) %>%
        summarise(path = paste(channel, collapse = ' > '),
                  conversion = sum(conversion)) %>%
        ungroup() %>%
        mutate(null_conversion = ifelse(conversion == 1, 0, 1))

mod_attrib_complete <- markov_model(
        df_all_paths_compl,
        var_path = 'path',
        var_conv = 'conversion',
        var_null = 'null_conversion',
        out_more = TRUE
)
```

```
##
## Number of simulations: 100000 - Convergence reached: 4.05% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps (27) is rea
##
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for free! Visit
```

```
trans_matrix_prob <- mod_attrib_complete$transition_matrix %>%
        dmap_at(c(1, 2), as.character)

##### viz #####
edges <-
        data.frame(
                from = trans_matrix_prob$channel_from,
```

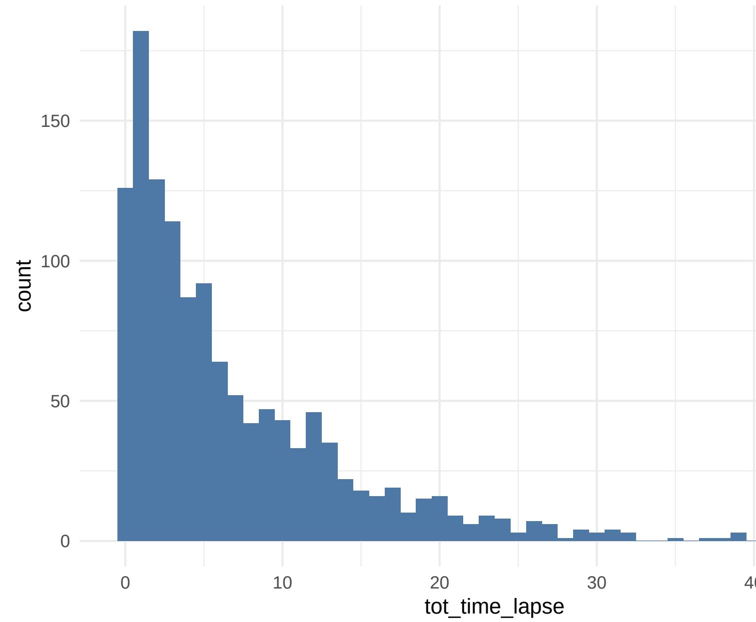
```
                to = trans_matrix_prob$channel_to,
                label = round(trans_matrix_prob$transition_probability, 2),
                font.size = trans_matrix_prob$transition_probability * 100,
                width = trans_matrix_prob$transition_probability * 15,
                shadow = TRUE,
                arrows = "to",
                color = list(color = "#95cbee", highlight = "red")
        )

nodes <- data_frame(id = c( c(trans_matrix_prob$channel_from), c(trans_matrix_prob$char
        distinct(id) %>%
        arrange(id) %>%
        mutate(
                label = id,
                color = ifelse(
                        label %in% c('(start)', '(conversion)'),
                        '#4ab04a',
                        ifelse(label == '(null)', '#ce472e', '#ffd73e')
                ),
                shadow = TRUE,
                shape = "box"
        )


visNetwork(nodes,
           edges,
           height = "2000px",
           width = "100%",
           main = "Generic Probabilistic model's Transition Matrix") %>%
        visIgraphLayout(randomSeed = 123) %>%
        visNodes(size = 5) %>%
        visOptions(highlightNearest = TRUE)
```

Generic Probabilistic model's Transition Matrix

```
##### modeling states and conversions #####
# transition matrix preprocessing
trans_matrix_complete <- mod_attrib_complete$transition_matrix
trans_matrix_complete <- rbind(trans_matrix_complete, df_dummy %>%
                                      mutate(transition_probability = perc) %>%
                                      select(channel_from, channel_to, transition_probability))
trans_matrix_complete$channel_to <- factor(trans_matrix_complete$channel_to, levels = c(levels(tr
trans_matrix_complete <- dcast(trans_matrix_complete, channel_from ~ channel_to, value.var = 'tra
```

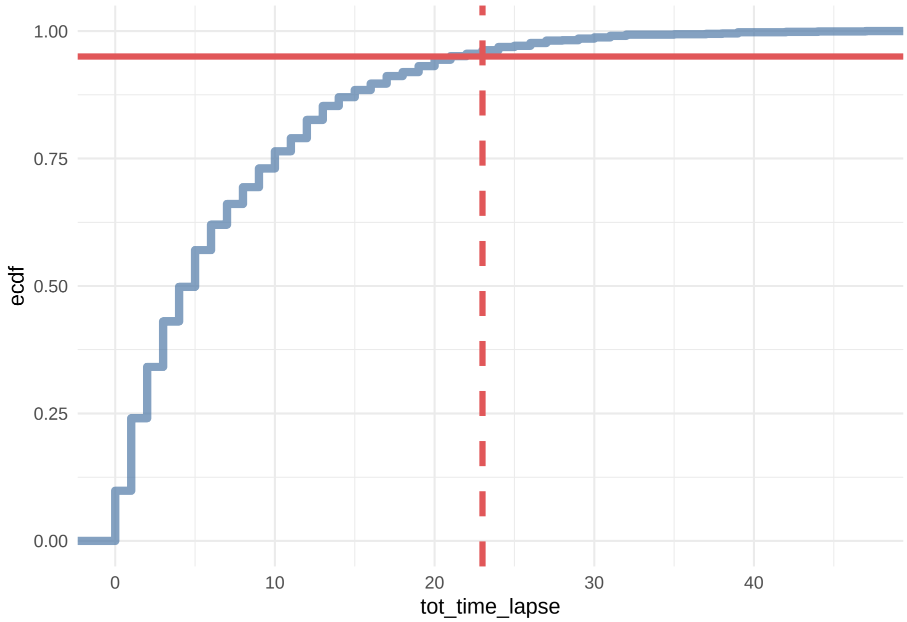
```
trans_matrix_complete[is.na(trans_matrix_complete)] <- 0
rownames(trans_matrix_complete) <- trans_matrix_complete$channel_from
trans_matrix_complete <- as.matrix(trans_matrix_complete[, -1])


# creating empty matrix for modeling
model_mtrx <- matrix(data = 0,
                     nrow = nrow(trans_matrix_complete), ncol = 1,
                     dimnames = list(c(rownames(trans_matrix_complete)), '(start)'))
# adding modeling number of visits
model_mtrx['channel_5', ] <- 1000

c(model_mtrx) %*% (trans_matrix_complete %^% 5) # after 5 steps
c(model_mtrx) %*% (trans_matrix_complete %^% 100000) # after 100000 steps
```

```
##### Customer journey duration #####
# computing time lapses from the first contact to conversion/last contact
df_multi_paths_tl <- df_path_1_clean_multi %>%
        group_by(customer_id) %>%
        summarise(path = paste(channel, collapse = ' > '),
                  first_touch_date = min(date),
                  last_touch_date = max(date),
                  tot_time_lapse = round(as.numeric(last_touch_date - first_touch_date
                  conversion = sum(conversion)) %>%
        ungroup()

# distribution plot
ggplot(df_multi_paths_tl %>% filter(conversion == 1), aes(x = tot_time_lapse)) +
        theme_minimal() +
        geom_histogram(fill = '#4e79a7', binwidth = 1)
```

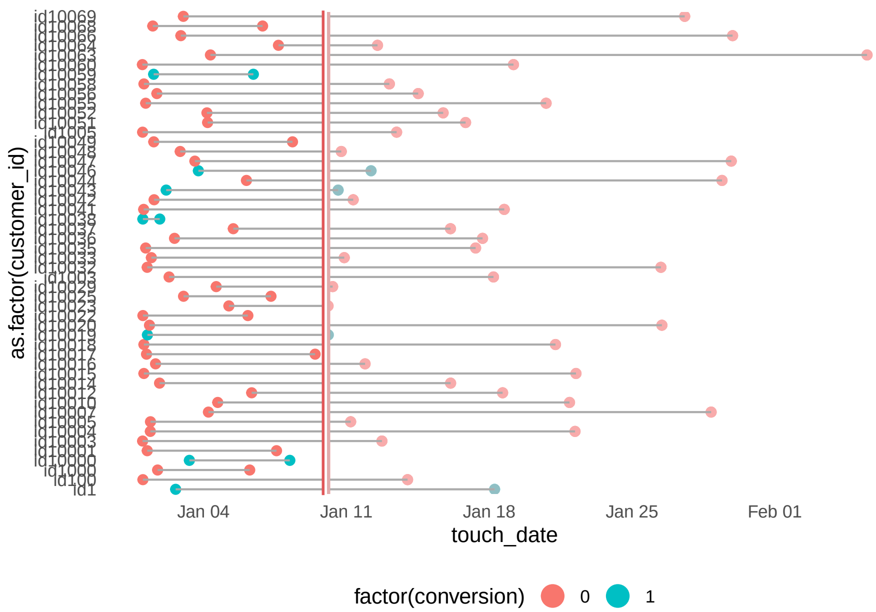


22.1.2.2.6 6. Customer Journey Duration

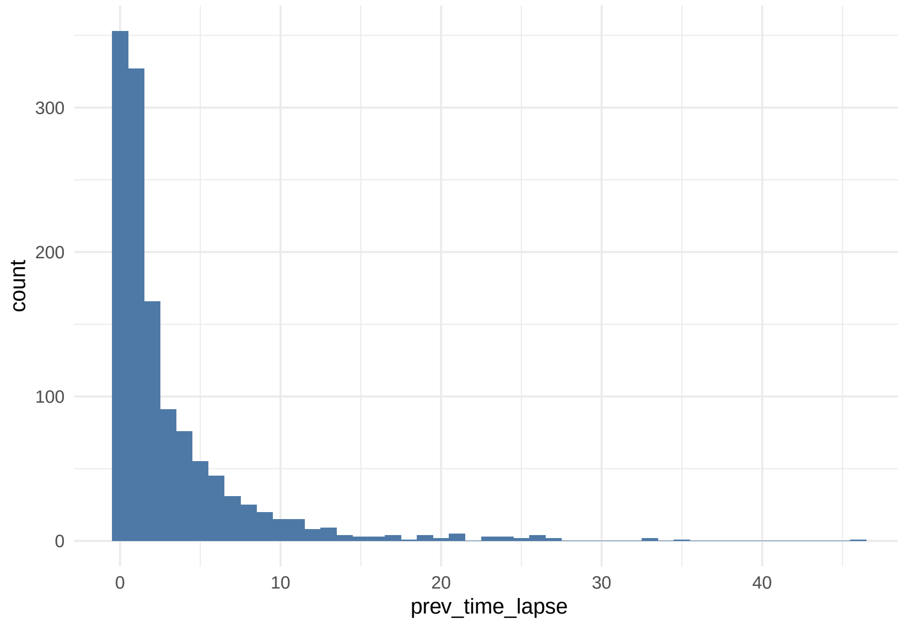
```
# cumulative distribution plot
ggplot(df_multi_paths_tl %>% filter(conversion == 1), aes(x = tot_time_lapse)) +
        theme_minimal() +
        stat_ecdf(geom = 'step', color = '#4e79a7', size = 2, alpha = 0.7) +
        geom_hline(yintercept = 0.95, color = '#e15759', size = 1.5) +
        geom_vline(xintercept = 23, color = '#e15759', size = 1.5, linetype = 2)
```



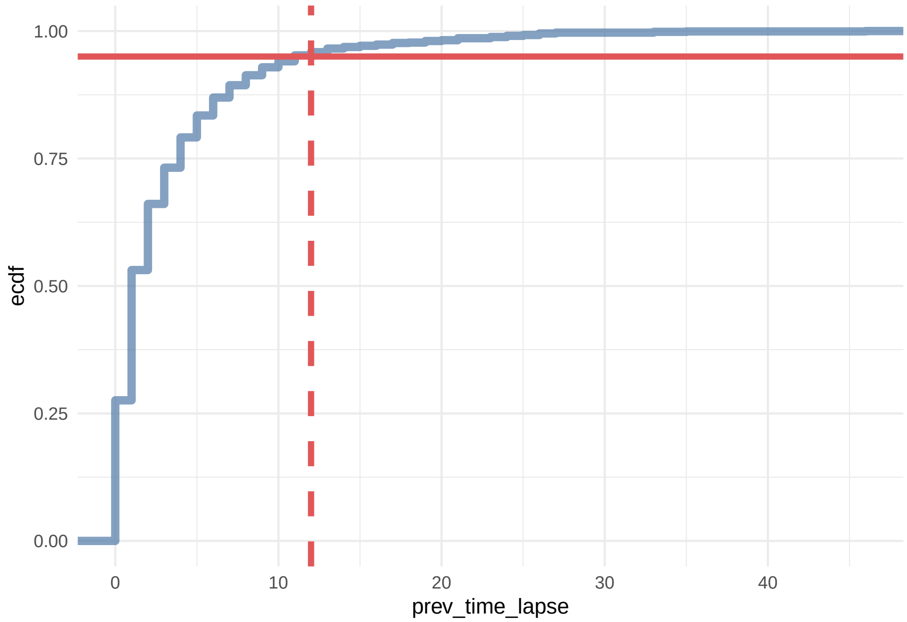

```
### for generic probabilistic model ###
df_multi_paths_tl_1 <- reshape2::melt(df_multi_paths_tl[c(1:50), ] %>% select(customer_
                    id.vars = c('customer_id', 'conversion'),
                    value.name = 'touch_date') %>%
        arrange(customer_id)
rep_date <- as.Date('2016-01-10', format = '%Y-%m-%d')

ggplot(df_multi_paths_tl_1, aes(x = as.factor(customer_id), y = touch_date, color = fac
        theme_minimal() +
        coord_flip() +
        geom_point(size = 2) +
        geom_line(size = 0.5, color = 'darkgrey') +
        geom_hline(yintercept = as.numeric(rep_date), color = '#e15759', size = 2) +
        geom_rect(xmin = -Inf, xmax = Inf, ymin = as.numeric(rep_date), ymax = Inf, alp
        theme(legend.position = 'bottom',
              panel.border = element_blank(),
              panel.grid.major = element_blank(),
              panel.grid.minor = element_blank(),
              axis.ticks.x = element_blank(),
              axis.ticks.y = element_blank()) +
        guides(colour = guide_legend(override.aes = list(size = 5)))
```




```
df_multi_paths_tl_2 <- df_path_1_clean_multi %>%
        group_by(customer_id) %>%
        mutate(prev_touch_date = lag(date)) %>%
        ungroup() %>%
        filter(conversion == 1) %>%
        mutate(prev_time_lapse = round(as.numeric(date - prev_touch_date)))

# distribution
ggplot(df_multi_paths_tl_2, aes(x = prev_time_lapse)) +
        theme_minimal() +
        geom_histogram(fill = '#4e79a7', binwidth = 1)
```




```r
# cumulative distribution
ggplot(df_multi_paths_tl_2, aes(x = prev_time_lapse)) +
        theme_minimal() +
        stat_ecdf(geom = 'step', color = '#4e79a7', size = 2, alpha = 0.7) +
        geom_hline(yintercept = 0.95, color = '#e15759', size = 1.5) +
        geom_vline(xintercept = 12, color = '#e15759', size = 1.5, linetype = 2)
```



In conclusion, we say that if a customer made contact with a marketing channel the first time for more than 23 days and/or hasn't made contact with a marketing channel for the last 12 days, then it is a fruitless path.

```
# extracting data for generic model
df_multi_paths_tl_3 <- df_path_1_clean_multi %>%
        group_by(customer_id) %>%
        mutate(prev_time_lapse = round(as.numeric(date - lag(date)))) %>%
        summarise(path = paste(channel, collapse = ' > '),
                  tot_time_lapse = round(as.numeric(max(date) - min(date))),
                  prev_touch_tl = prev_time_lapse[which(max(date) == date)],
                  conversion = sum(conversion)) %>%
        ungroup() %>%
        mutate(is_fruitless = ifelse(conversion == 0 & tot_time_lapse > 20 & prev_touch_tl > 10,
        filter(conversion == 1 | is_fruitless == TRUE)
```

22.1.2.2.7 7. Channel Comparisons We can use `markov_model` with `var_value` to compare the gross margin among channels.

22.1.2.3 Example 3

This example is from Bounteus

```
# Install these libraries (only do this once)
# install.packages("ChannelAttribution")
# install.packages("reshape")
```

```
# install.packages("ggplot2")

# Load these libraries (every time you start RStudio)
library(ChannelAttribution)
library(reshape)
library(ggplot2)

# This loads the demo data. You can load your own data by importing a dataset or readi
data(PathData)
```

- Path Variable – The steps a user takes across sessions to comprise the sequences.
- Conversion Variable – How many times a user converted.
- Value Variable – The monetary value of each marketing channel.
- Null Variable – How many times a user exited.

Build the simple heuristic models (First Click / first_touch, Last Click / last_touch, and Linear Attribution / linear_touch):

```
H <- heuristic_models(Data, 'path', 'total_conversions', var_value='total_conversion_va
```

```
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for fr
```

Markov model

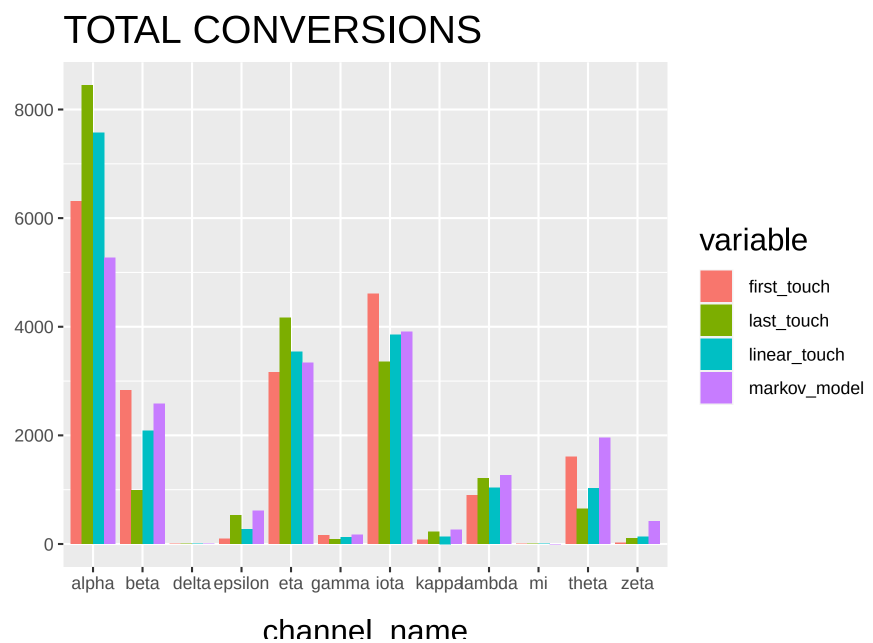
```
M <- markov_model(Data, 'path', 'total_conversions', var_value='total_conversion_value
```

```
##
## Number of simulations: 100000 - Convergence reached: 1.46% < 5.00%
##
## Percentage of simulated paths that successfully end before maximum number of steps
##
## [1] "*** Looking to run more advanced attribution? Try ChannelAttribution Pro for fr
```

```
# Merges the two data frames on the "channel_name" column.
R <- merge(H, M, by='channel_name')

# Selects only relevant columns
R1 <- R[, (colnames(R)%in%c('channel_name', 'first_touch_conversions', 'last_touch_conv

# Renames the columns
colnames(R1) <- c('channel_name', 'first_touch', 'last_touch', 'linear_touch', 'markov_

# Transforms the dataset into a data frame that ggplot2 can use to graph the outcomes
R1 <- melt(R1, id='channel_name')
```

Plot the total conversions

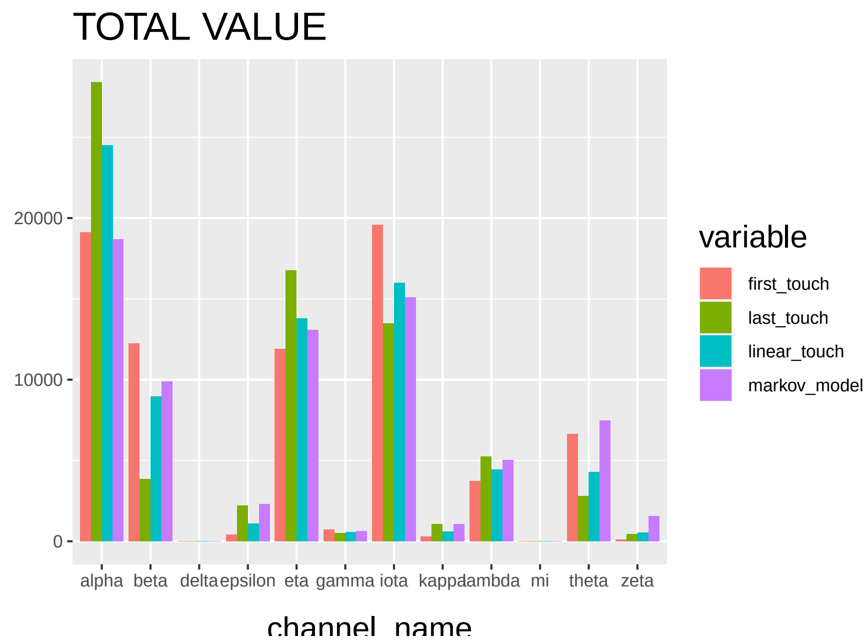
```
ggplot(R1, aes(channel_name, value, fill = variable)) +
  geom_bar(stat='identity', position='dodge') +
  ggtitle('TOTAL CONVERSIONS') +
  theme(axis.title.x = element_text(vjust = -2)) +
  theme(axis.title.y = element_text(vjust = +2)) +
  theme(title = element_text(size = 16)) +
  theme(plot.title=element_text(size = 20)) +
  ylab("")
```



The "Total Conversions" bar chart shows you how many conversions were attributed to each channel (i.e. alpha, beta, etc.) for each method (i.e. first_touch, last_touch, etc.).

```
R2 <- R[, (colnames(R)%in%c('channel_name', 'first_touch_value', 'last_touch_value', 'linear_touc

colnames(R2) <- c('channel_name', 'first_touch', 'last_touch', 'linear_touch', 'markov_model')

R2 <- melt(R2, id='channel_name')

ggplot(R2, aes(channel_name, value, fill = variable)) +
  geom_bar(stat='identity', position='dodge') +
  ggtitle('TOTAL VALUE') +
  theme(axis.title.x = element_text(vjust = -2)) +
  theme(axis.title.y = element_text(vjust = +2)) +
  theme(title = element_text(size = 16)) +
```

```
theme(plot.title=element_text(size = 20)) +
ylab("")
```



The "Total Conversion Value" bar chart shows you monetary value that can be attributed to each channel from a conversion.

22.2 Sales Funnel

22.2.1 Example 1

This example is based on Sergey Bryl

$$Awareness \rightarrow Interest \rightarrow Desire \rightarrow Action$$

Step in the funnel:

- 0 step (necessary condition) – customer visits a site for the first time
- 1st step (awareness) – visits two site's pages
- 2nd step (interest) – reviews a product page
- 3rd step (desire) – adds a product to the shopping cart
- 4th step (action) – completes purchase

Simulate data

```
library(tidyverse)
library(purrrlyr)
library(reshape2)

##### simulating the "real" data #####
set.seed(454)
df_raw <-
  data.frame(
    customer_id = paste0('id', sample(c(1:5000), replace = TRUE)),
    date = as.POSIXct(
      rbeta(10000, 0.7, 10) * 10000000,
      origin = '2017-01-01',
      tz = "UTC"
    ),
    channel = paste0('channel_', sample(
      c(0:7),
      10000,
      replace = TRUE,
      prob = c(0.2, 0.12, 0.03, 0.07, 0.15, 0.25, 0.1, 0.08)
    )),
    site_visit = 1
  ) %>%

  mutate(
    two_pages_visit = sample(c(0, 1), 10000, replace = TRUE, prob = c(0.8, 0.2)),
    product_page_visit = ifelse(
      two_pages_visit == 1,
      sample(
        c(0, 1),
        length(two_pages_visit[which(two_pages_visit == 1)]),
        replace = TRUE,
        prob = c(0.75, 0.25)
      ),
      0
    ),
    add_to_cart = ifelse(
      product_page_visit == 1,
      sample(
        c(0, 1),
        length(product_page_visit[which(product_page_visit == 1)]),
        replace = TRUE,
        prob = c(0.1, 0.9)
      ),
      0
    ),
```

```
    purchase = ifelse(add_to_cart == 1,
                      sample(
                        c(0, 1),
                        length(add_to_cart[which(add_to_cart == 1)]),
                        replace = TRUE,
                        prob = c(0.02, 0.98)
                      ),
                      0)
  ) %>%
  dmap_at(c('customer_id', 'channel'), as.character) %>%
  arrange(date) %>%
  mutate(session_id = row_number()) %>%
  arrange(customer_id, session_id)
df_raw <-
  reshape2::melt(
    df_raw,
    id.vars = c('customer_id', 'date', 'channel', 'session_id'),
    value.name = "trigger",
    variable.name = 'event'
  ) %>%
  filter(trigger == 1) %>%
  select(-trigger) %>%
  arrange(customer_id, date)
```

Preprocessing

```
### removing not first events ###
df_customers <- df_raw %>%
  group_by(customer_id, event) %>%
  filter(date == min(date)) %>%
  ungroup()
```

Assumption: all customers are first-time buyers. Hence, every next purchase as an event will be removed with the above code.

Calculate channel probability

```
### Sales Funnel probabilities ###
sf_probs <- df_customers %>%

    group_by(event) %>%
    summarise(customers_on_step = n()) %>%
    ungroup() %>%

    mutate(
        sf_probs = round(customers_on_step / customers_on_step[event == 'site_visit'],
        sf_probs_step = round(customers_on_step / lag(customers_on_step), 3),
```

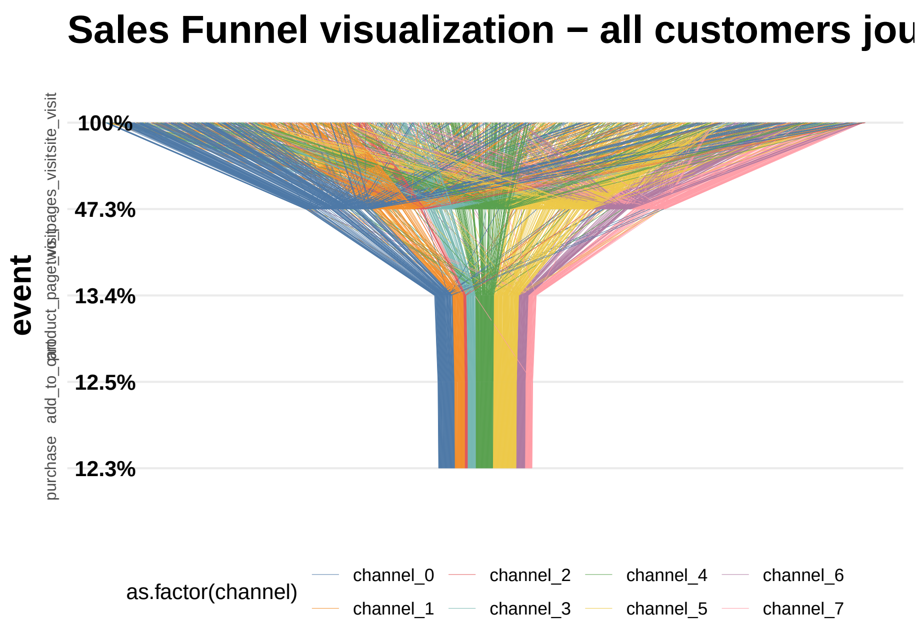
```
        sf_probs_step = ifelse(is.na(sf_probs_step) == TRUE, 1, sf_probs_step),
        sf_importance = 1 - sf_probs_step,
        sf_importance_weighted = sf_importance / sum(sf_importance)
    )
```

Visualization

```
### Sales Funnel visualization ###
df_customers_plot <- df_customers %>%

    group_by(event) %>%
    arrange(channel) %>%
    mutate(pl = row_number()) %>%
    ungroup() %>%

    mutate(
        pl_new = case_when(
            event == 'two_pages_visit' ~ round((max(pl[event == 'site_visit']) - max(pl[event ==
            event == 'product_page_visit' ~ round((max(pl[event == 'site_visit']) - max(pl[event
            event == 'add_to_cart' ~ round((max(pl[event == 'site_visit']) - max(pl[event == 'add
            event == 'purchase' ~ round((max(pl[event == 'site_visit']) - max(pl[event == 'purcha
            TRUE ~ 0
        ),
        pl = pl + pl_new
    )

df_customers_plot$event <-
    factor(
        df_customers_plot$event,
        levels = c(
            'purchase',
            'add_to_cart',
            'product_page_visit',
            'two_pages_visit',
            'site_visit'
        )
    )

# color palette
cols <- c(
    '#4e79a7',
    '#f28e2b',
    '#e15759',
    '#76b7b2',
    '#59a14f',
    '#edc948',
```

```
    '#b07aa1',
    '#ff9da7',
    '#9c755f',
    '#bab0ac'
)

ggplot(df_customers_plot, aes(x = event, y = pl)) +
    theme_minimal() +
    scale_colour_manual(values = cols) +
    coord_flip() +
    geom_line(aes(group = customer_id, color = as.factor(channel)), size = 0.05) +
    geom_text(
        data = sf_probs,
        aes(
            x = event,
            y = 1,
            label = paste0(sf_probs * 100, '%')
        ),
        size = 4,
        fontface = 'bold'
    ) +
    guides(color = guide_legend(override.aes = list(size = 2))) +
    theme(
        legend.position = 'bottom',
        legend.direction = "horizontal",
        panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        plot.title = element_text(
            size = 20,
            face = "bold",
            vjust = 2,
            color = 'black',
            lineheight = 0.8
        ),
        axis.title.y = element_text(size = 16, face = "bold"),
        axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        axis.text.y = element_text(
            size = 8,
            angle = 90,
            hjust = 0.5,
            vjust = 0.5,
            face = "plain"
        )
    ) +
```

```
    ggtitle("Sales Funnel visualization - all customers journeys")
```



Calculate attribution

```
### computing attribution ###
df_attrib <- df_customers %>%
    # removing customers without purchase
    group_by(customer_id) %>%
    filter(any(as.character(event) == 'purchase')) %>%
    ungroup() %>%

    # joining step's importances
    left_join(., sf_probs %>% select(event, sf_importance_weighted), by = 'event') %>%

    group_by(channel) %>%
    summarise(tot_attribution = sum(sf_importance_weighted)) %>%
    ungroup()
```

22.2.2 Example 2

Code from Sergey Bryl

```
library(dplyr)
library(ggplot2)
library(reshape2)
```

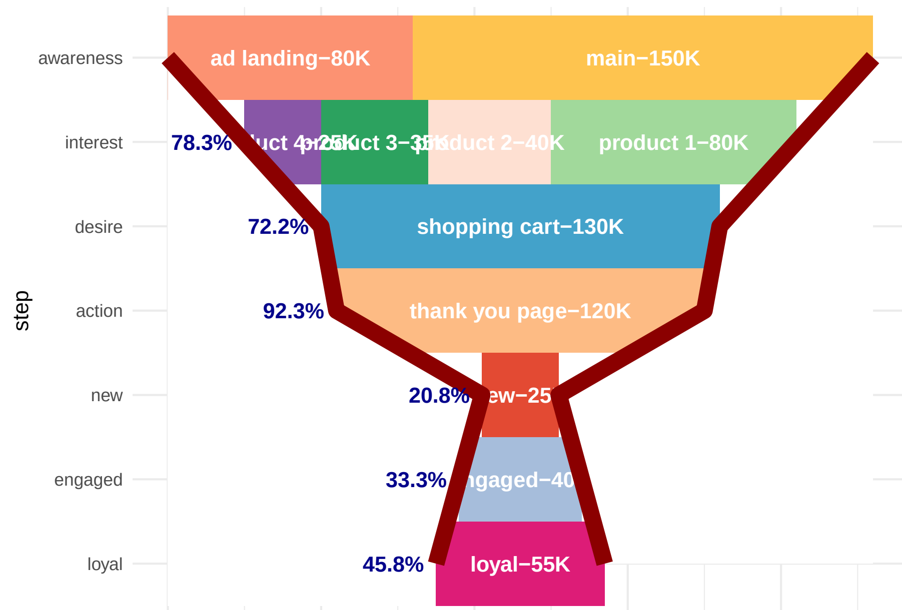
```r
# creating a data samples
# content
df.content <- data.frame(
    content = c(
        'main',
        'ad landing',
        'product 1',
        'product 2',
        'product 3',
        'product 4',
        'shopping cart',
        'thank you page'
    ),
    step = c(
        'awareness',
        'awareness',
        'interest',
        'interest',
        'interest',
        'interest',
        'desire',
        'action'
    ),
    number = c(150000, 80000,
               80000, 40000, 35000, 25000,
               130000,
               120000)
)
# customers
df.customers <- data.frame(
    content = c('new', 'engaged', 'loyal'),
    step = c('new', 'engaged', 'loyal'),
    number = c(25000, 40000, 55000)
)
# combining two data sets
df.all <- rbind(df.content, df.customers)

# calculating dummies, max and min values of X for plotting
df.all <- df.all %>%
    group_by(step) %>%
    mutate(totnum = sum(number)) %>%
    ungroup() %>%
    mutate(dum = (max(totnum) - totnum) / 2,
           maxx = totnum + dum,
           minx = dum)
```

```r
# data frame for plotting funnel lines
df.lines <- df.all %>%
    distinct(step, maxx, minx)

# data frame with dummies
df.dum <- df.all %>%
    distinct(step, dum) %>%
    mutate(content = 'dummy',
           number = dum) %>%
    select(content, step, number)

# data frame with rates
conv <- df.all$totnum[df.all$step == 'action']

df.rates <- df.all %>%
    distinct(step, totnum) %>%
    mutate(
        prevnum = lag(totnum),
        rate = ifelse(
            step == 'new' | step == 'engaged' | step == 'loyal',
            round(totnum / conv, 3),
            round(totnum / prevnum, 3)
        )
    ) %>%
    select(step, rate)
df.rates <- na.omit(df.rates)

# creting final data frame
df.all <- df.all %>%
    select(content, step, number)

df.all <- rbind(df.all, df.dum)

# defining order of steps
df.all$step <-
    factor(
        df.all$step,
        levels = c(
            'loyal',
            'engaged',
            'new',
            'action',
            'desire',
            'interest',
            'awareness'
```

```
        )
    )
df.all <- df.all %>%
    arrange(desc(step))
list1 <- df.all %>% distinct(content) %>%
    filter(content != 'dummy')
df.all$content <-
    factor(df.all$content, levels = c(as.character(list1$content), 'dummy'))

# calculating position of labels
df.all <- df.all %>%
    arrange(step, desc(content)) %>%
    group_by(step) %>%
    mutate(pos = cumsum(number) - 0.5 * number) %>%
    ungroup()

# creating custom palette with 'white' color for dummies
cols <- c(
    "#fec44f",
    "#fc9272",
    "#a1d99b",
    "#fee0d2",
    "#2ca25f",
    "#8856a7",
    "#43a2ca",
    "#fdbb84",
    "#e34a33",
    "#a6bddb",
    "#dd1c77",
    "#ffffff"
)

# plotting chart
ggplot() +
    theme_minimal() +
    coord_flip() +
    scale_fill_manual(values = cols) +
    geom_bar(
        data = df.all,
        aes(x = step, y = number, fill = content),
        stat = "identity",
        width = 1
    ) +
    geom_text(
        data = df.all[df.all$content != 'dummy',],
```

```
        aes(
            x = step,
            y = pos,
            label = paste0(content, '-', number / 1000, 'K')
        ),
        size = 4,
        color = 'white',
        fontface = "bold"
    ) +
    geom_ribbon(data = df.lines,
                aes(
                    x = step,
                    ymax = max(maxx),
                    ymin = maxx,
                    group = 1
                ),
                fill = 'white') +
    geom_line(
        data = df.lines,
        aes(x = step, y = maxx, group = 1),
        color = 'darkred',
        size = 4
    ) +
    geom_ribbon(data = df.lines,
                aes(
                    x = step,
                    ymax = minx,
                    ymin = min(minx),
                    group = 1
                ),
                fill = 'white') +
    geom_line(
        data = df.lines,
        aes(x = step, y = minx, group = 1),
        color = 'darkred',
        size = 4
    ) +
    geom_text(
        data = df.rates,
        aes(
            x = step,
            y = (df.lines$minx[-1]),
            label = paste0(rate * 100, '%')
        ),
        hjust = 1.2,
```

```
        color = 'darkblue',
        fontface = "bold"
    ) +
    theme(
        legend.position = 'none',
        axis.ticks = element_blank(),
        axis.text.x = element_blank(),
        axis.title.x = element_blank()
    )
```



22.3 RFM

RFM is calculated as:

- A recency score is assigned to each customer based on date of most recent purchase.
- A frequency ranking is assigned based on frequency of purchases
- Monetary score is assigned based on the total revenue generated by the customer in the period under consideration for the analysis

```
library("rfm")
rfm_data_customer
```

```
## # A tibble: 39,999 x 5
##    customer_id revenue most_recent_visit number_of_orders recency_days
```

```
##          <dbl>   <dbl> <date>                         <dbl>        <dbl>
##  1       22086     777 2006-05-14                         9          232
##  2        2290    1555 2006-09-08                        16          115
##  3       26377     336 2006-11-19                         5           43
##  4       24650    1189 2006-10-29                        12           64
##  5       12883    1229 2006-12-09                        12           23
##  6        2119     929 2006-10-21                        11           72
##  7       31283    1569 2006-09-11                        17          112
##  8       33815     778 2006-08-12                        11          142
##  9       15972     641 2006-11-19                         9           43
## 10       27650     970 2006-08-23                        10          131
## # i 39,989 more rows
```

```r
# a unique customer id
# number of transaction/order
# total revenue from the customer
# number of days since the last visit


rfm_data_orders # to generate data_orders, use rfm_table_order()
```

```
## # A tibble: 4,906 x 3
##    customer_id         order_date revenue
##    <chr>               <date>       <dbl>
##  1 Mr. Brion Stark Sr. 2004-12-20      32
##  2 Ethyl Botsford      2005-05-02      36
##  3 Hosteen Jacobi      2004-03-06     116
##  4 Mr. Edw Frami       2006-03-15      99
##  5 Josef Lemke         2006-08-14      76
##  6 Julisa Halvorson    2005-05-28      56
##  7 Judyth Lueilwitz    2005-03-09     108
##  8 Mr. Mekhi Goyette   2005-09-23     183
##  9 Hansford Moen PhD   2005-09-07      30
## 10 Fount Flatley       2006-04-12      13
## # i 4,896 more rows
```

```r
# unique customer id
# date of transaction
# and amount
# customer_id: name of the customer id column
# order_date: name of the transaction date column
# revenue: name of the transaction amount column
# analysis_date: date of analysis
# recency_bins: number of rankings for recency score (default is 5)
# frequency_bins: number of rankings for frequency score (default is 5)
# monetary_bins: number of rankings for monetary score (default is 5)
```

```r
analysis_date <- lubridate::as_date('2007-01-01')
rfm_result <-
    rfm_table_customer(
        rfm_data_customer,
        customer_id,
        number_of_orders,
        recency_days,
        revenue,
        analysis_date
    )
rfm_result
```

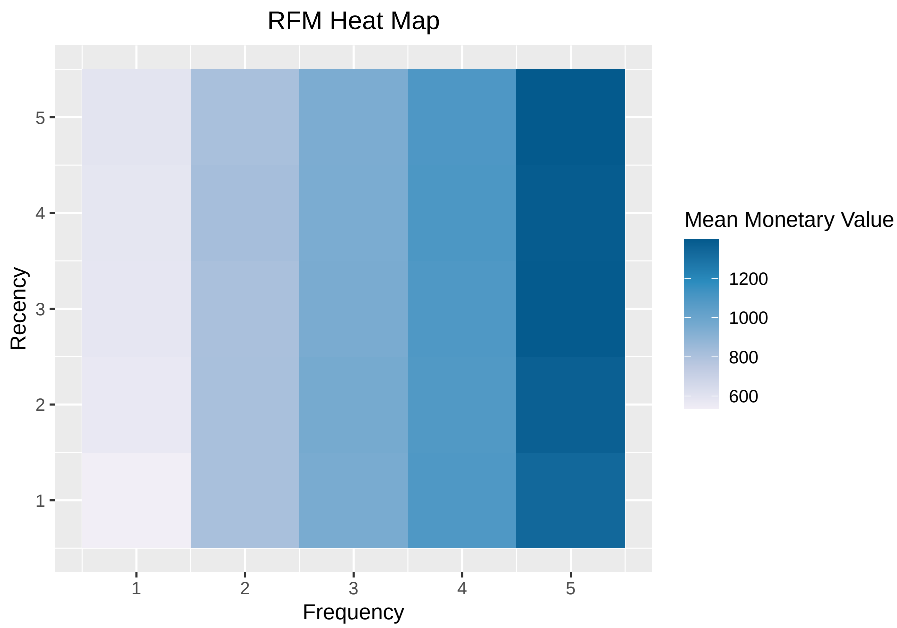
```
## # A tibble: 39,999 x 8
##    customer_id recency_days transaction_count amount recency_score
##          <dbl>        <dbl>             <dbl>  <dbl>         <int>
##  1       22086          232                 9    777             2
##  2        2290          115                16   1555             4
##  3       26377           43                 5    336             5
##  4       24650           64                12   1189             5
##  5       12883           23                12   1229             5
##  6        2119           72                11    929             5
##  7       31283          112                17   1569             4
##  8       33815          142                11    778             3
##  9       15972           43                 9    641             5
## 10       27650          131                10    970             3
## # i 39,989 more rows
## # i 3 more variables: frequency_score <int>, monetary_score <int>,
## #   rfm_score <dbl>
```

```r
# customer_id: unique customer id
# date_most_recent: date of most recent visit
# recency_days: days since the most recent visit
# transaction_count: number of transactions of the customer
# amount: total revenue generated by the customer
# recency_score: recency score of the customer
# frequency_score: frequency score of the customer
# monetary_score: monetary score of the customer
# rfm_score: RFM score of the customer
```

22.3.1 Visualization

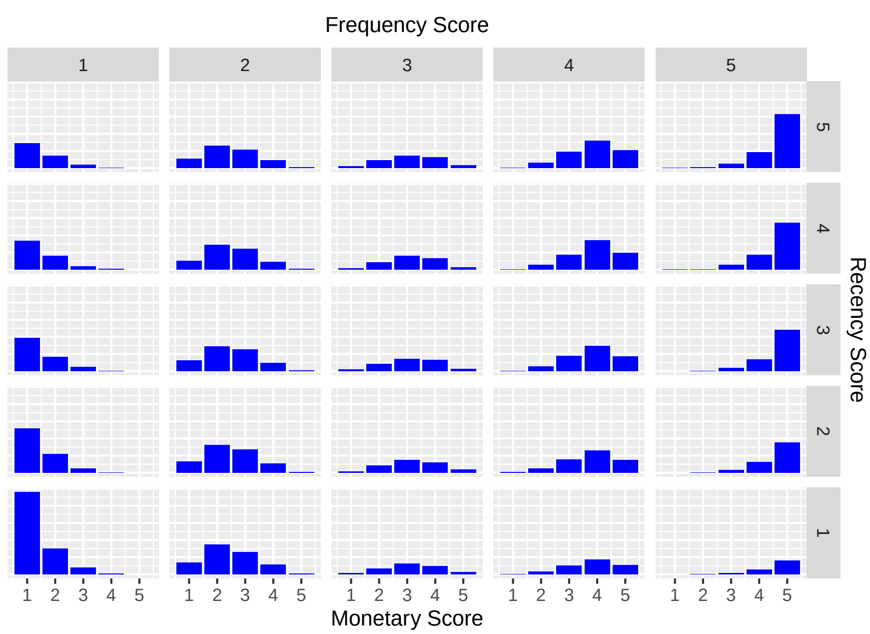
heat map shows the average monetary value for different categories of recency and frequency scores

```r
rfm_heatmap(rfm_result)
```



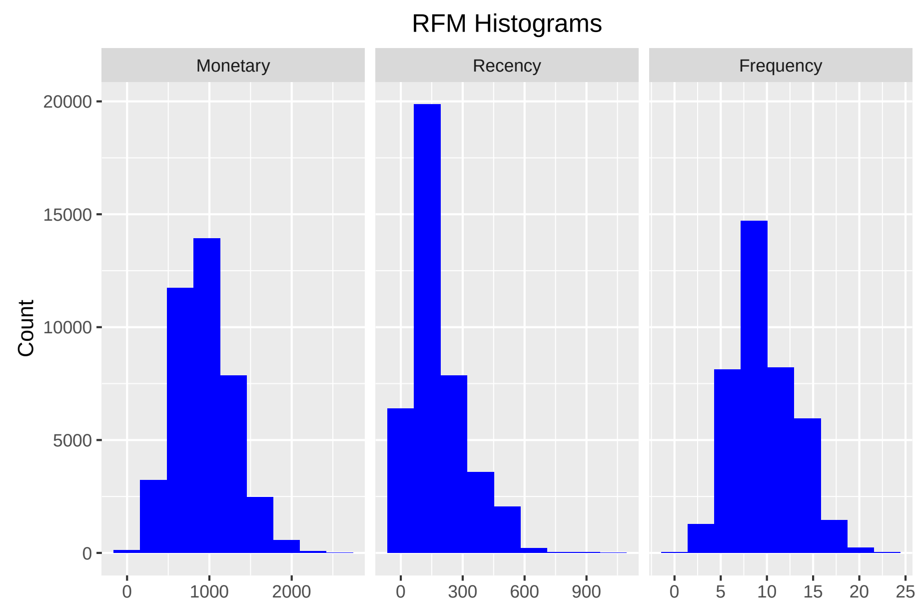
bar chart

```
rfm_bar_chart(rfm_result)
```



histogram

```r
rfm_histograms(rfm_result)
```

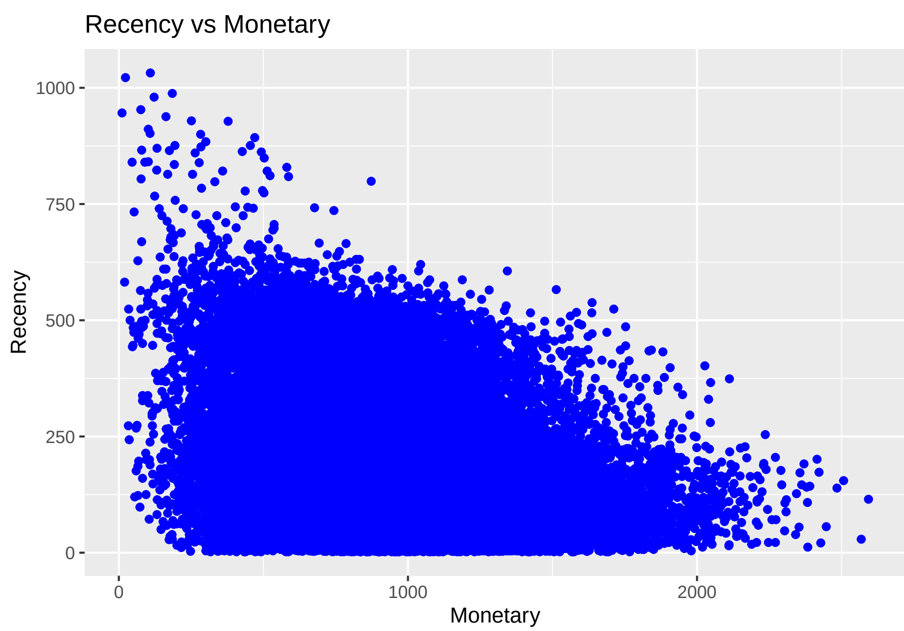


Customers by Orders

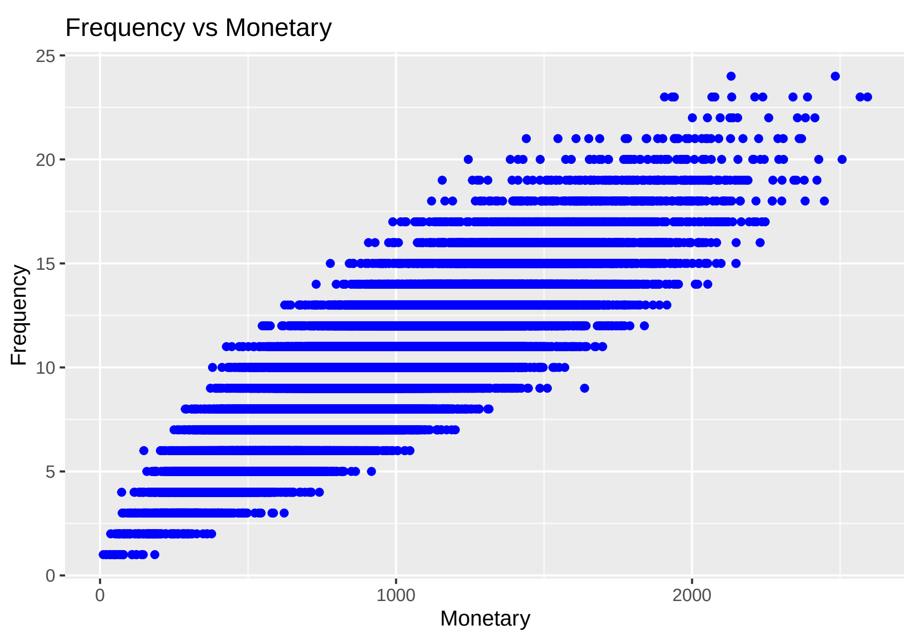
```r
rfm_order_dist(rfm_result)
```

Scatter Plots

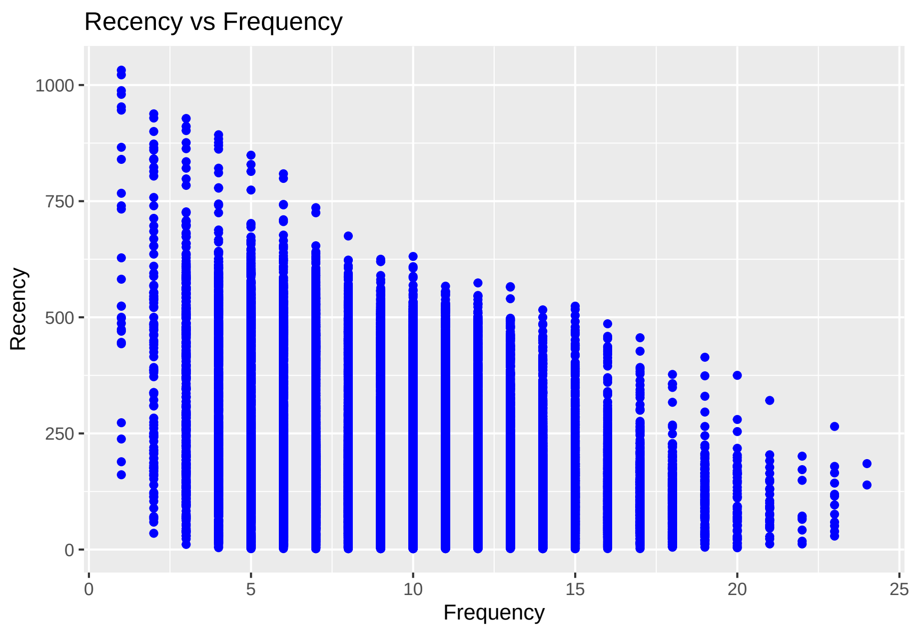
```
rfm_rm_plot(rfm_result)
```




```
rfm_fm_plot(rfm_result)
```



```
rfm_rf_plot(rfm_result)
```



22.3.2 RFMC

(C) clumpiness is defined as the degree of nonconformity to equal spacing (Zhang et al., 2015)

In finance, clumpiness can indicate high growth potential but large risk, Hence, it can be incorporated into firm acquisition decision. Originated from sports phenomenon - hot hand effect - where success leads to more success.

In statistics, clumpiness is the serial dependence or "non-constant propensity, specifically temporary elevations of propensity— i.e. periods during which one event is more likely to occur than the average level." (Zhang et al., 2013)

Properties of clumpiness:

- Min (max) if events are equally spaced (close to one another)
- Continuity
- Convergence

22.4 Customer Segmentation

22.4.1 Example 1

Continue from the RFM

```r
segment_names <-
    c(
        "Premium",
        "Loyal Customers",
        "Potential Loyalist",
        "New Customers",
        "Promising",
        "Need Attention",
        "About To Churn",
        "At Risk",
        "High Value Churners/Resurrection",
        "Low Value Churners"
    )

recency_lower <- c(4, 2, 3, 4, 3, 2, 2, 1, 1, 1)
recency_upper <- c(5, 5, 5, 5, 4, 3, 3, 2, 1, 2)
frequency_lower <- c(4, 3, 1, 1, 1, 2, 1, 2, 4, 1)
frequency_upper <- c(5, 5, 3, 1, 1, 3, 2, 5, 5, 2)
monetary_lower <- c(4, 3, 1, 1, 1, 2, 1, 2, 4, 1)
monetary_upper <- c(5, 5, 3, 1, 1, 3, 2, 5, 5, 2)

rfm_segments <-
    rfm_segment(
        rfm_result,
        segment_names,
        recency_lower,
        recency_upper,
        frequency_lower,
        frequency_upper,
        monetary_lower,
        monetary_upper
    )

head(rfm_segments, n = 5)


rfm_segments %>%
    count(rfm_segments$segment) %>%
    arrange(desc(n)) %>%
    rename(Count = n)

# median recency
rfm_plot_median_recency(rfm_segments)

# median frequency
```

```r
rfm_plot_median_frequency(rfm_segments)

# Median Monetary Value
rfm_plot_median_monetary(rfm_segments)
```

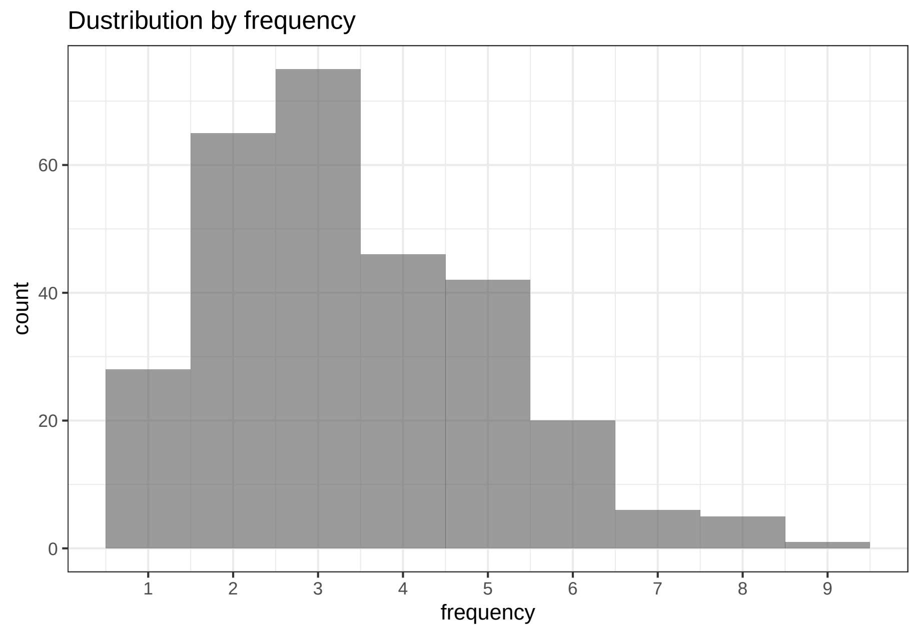
22.4.2 Example 2

Example by Sergey

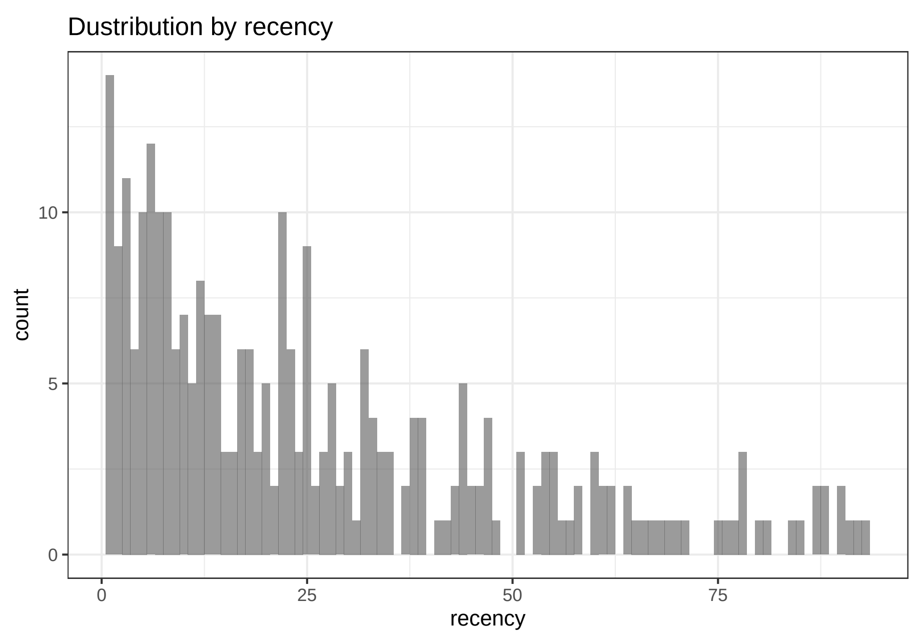
22.4.2.1 LifeCycle Grids

```r
# loading libraries
library(dplyr)
library(reshape2)
library(ggplot2)

# creating data sample
set.seed(10)
data <- data.frame(
    orderId = sample(c(1:1000), 5000, replace = TRUE),
    product = sample(
        c('NULL', 'a', 'b', 'c'),
        5000,
        replace = TRUE,
        prob = c(0.15, 0.65, 0.3, 0.15)
    )
)
order <- data.frame(orderId = c(1:1000),
                    clientId = sample(c(1:300), 1000, replace = TRUE))
gender <- data.frame(clientId = c(1:300),
                     gender = sample(
                         c('male', 'female'),
                         300,
                         replace = TRUE,
                         prob = c(0.40, 0.60)
                     ))
date <- data.frame(orderId = c(1:1000),
                   orderdate = sample((1:100), 1000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, gender, by = 'clientId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL',]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")
rm(data, date, order, gender)
```

```r
# reporting date
today <- as.Date('2012-04-11', format = '%Y-%m-%d')

# processing data
orders <-
    dcast(
        orders,
        orderId + clientId + gender + orderdate ~ product,
        value.var = 'product',
        fun.aggregate = length
    )

orders <- orders %>%
    group_by(clientId) %>%
    mutate(frequency = n(),
           recency = as.numeric(today - orderdate)) %>%
    filter(orderdate == max(orderdate)) %>%
    filter(orderId == max(orderId)) %>%
    ungroup()

# exploratory analysis
ggplot(orders, aes(x = frequency)) +
    theme_bw() +
    scale_x_continuous(breaks = c(1:10)) +
    geom_bar(alpha = 0.6, width = 1) +
    ggtitle("Dustribution by frequency")
```



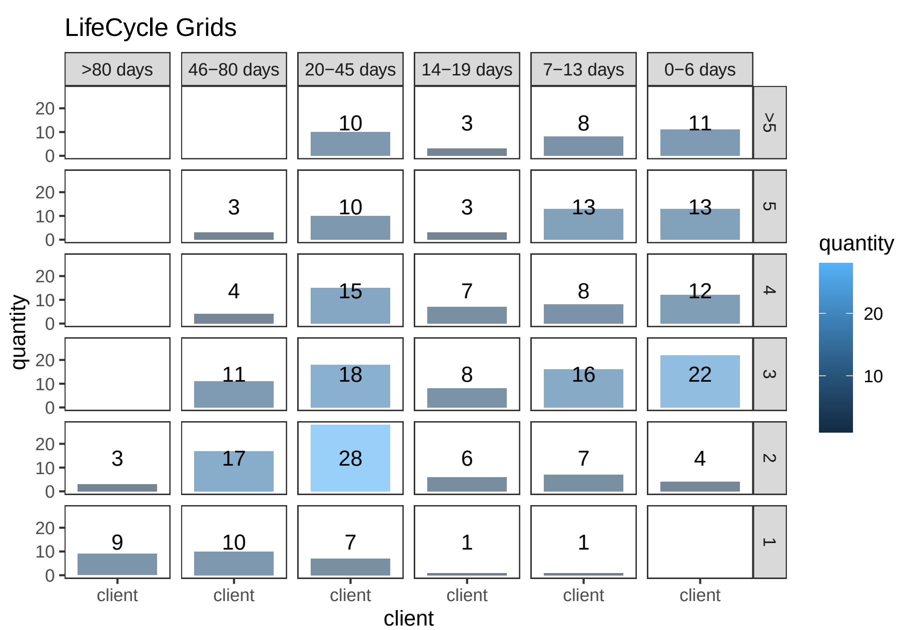

```
ggplot(orders, aes(x = recency)) +
    theme_bw() +
    geom_bar(alpha = 0.6, width = 1) +
    ggtitle("Dustribution by recency")
```



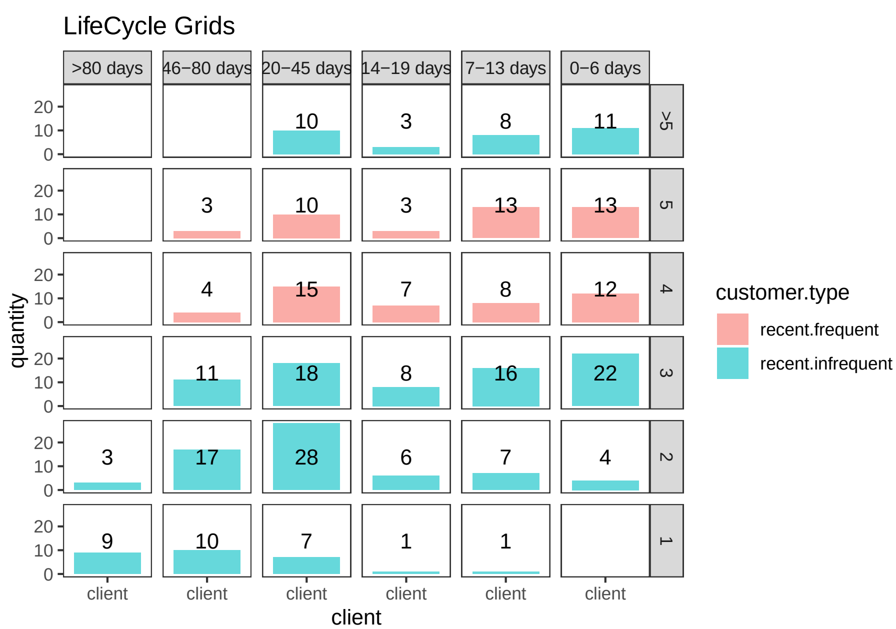
```
orders.segm <- orders %>%
    mutate(segm.freq = ifelse(between(frequency, 1, 1), '1',
                              ifelse(
                                  between(frequency, 2, 2), '2',
                                  ifelse(between(frequency, 3, 3), '3',
                                         ifelse(
                                             between(frequency, 4, 4), '4',
                                             ifelse(between(frequency, 5, 5), '5', '>5')
                                         ))
                              ))) %>%
    mutate(segm.rec = ifelse(
        between(recency, 0, 6),
        '0-6 days',
        ifelse(
            between(recency, 7, 13),
            '7-13 days',
            ifelse(
                between(recency, 14, 19),
                '14-19 days',
                ifelse(
                    between(recency, 20, 45),
                    '20-45 days',
                    ifelse(between(recency, 46, 80), '46-80 days', '>80 days')
                )
            )
        )
    )) %>%
    # creating last cart feature
    mutate(cart = paste(
        ifelse(a != 0, 'a', ''),
        ifelse(b != 0, 'b', ''),
        ifelse(c != 0, 'c', ''),
        sep = ''
    )) %>%
    arrange(clientId)

# defining order of boundaries
orders.segm$segm.freq <-
    factor(orders.segm$segm.freq, levels = c('>5', '5', '4', '3', '2', '1'))
orders.segm$segm.rec <-
    factor(
        orders.segm$segm.rec,
        levels = c(
            '>80 days',
            '46-80 days',
```

```
                '20-45 days',
                '14-19 days',
                '7-13 days',
                '0-6 days'
            )
    )
```

```
lcg <- orders.segm %>%
    group_by(segm.rec, segm.freq) %>%
    summarise(quantity = n()) %>%
    mutate(client = 'client') %>%
    ungroup()
lcg.matrix <-
    dcast(lcg,
          segm.freq ~ segm.rec,
          value.var = 'quantity',
          fun.aggregate = sum)

ggplot(lcg, aes(x = client, y = quantity, fill = quantity)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(stat = 'identity', alpha = 0.6) +
    geom_text(aes(y = max(quantity) / 2, label = quantity), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    ggtitle("LifeCycle Grids")
```

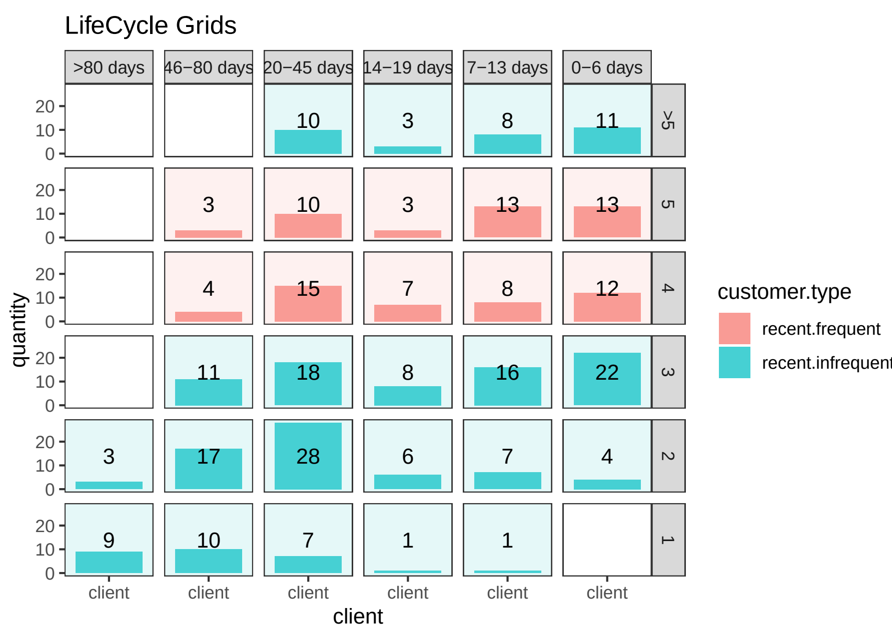


```
lcg.adv <- lcg %>%
    mutate(
        rec.type = ifelse(
            segm.rec %in% c("> 80 days", "46 - 80 days", "20 - 45 days"),
            "not recent",
            "recent"
        ),
        freq.type = ifelse(segm.freq %in% c(" >
          5", "5", "4"), "frequent", "infrequent"),
        customer.type = interaction(rec.type, freq.type)
    )

ggplot(lcg.adv, aes(x = client, y = quantity, fill = customer.type)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    facet_grid(segm.freq ~ segm.rec) +
    geom_bar(stat = 'identity', alpha = 0.6) +
    geom_text(aes(y = max(quantity) / 2, label = quantity), size = 4) +
    ggtitle("LifeCycle Grids")
```



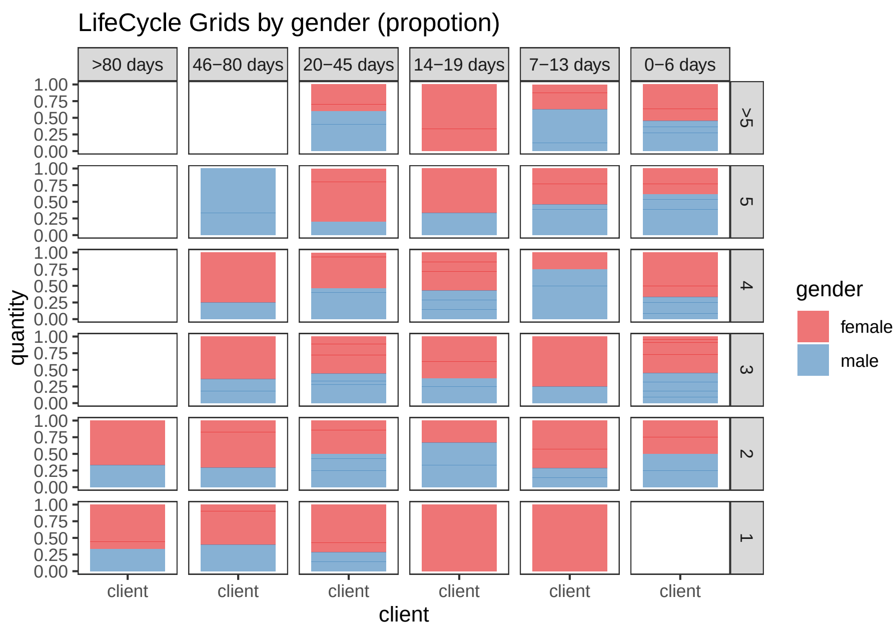

```
# with background
ggplot(lcg.adv, aes(x = client, y = quantity, fill = customer.type)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_rect(
```

```
        aes(fill = customer.type),
        xmin = -Inf,
        xmax = Inf,
        ymin = -Inf,
        ymax = Inf,
        alpha = 0.1
    ) +
    facet_grid(segm.freq ~ segm.rec) +
    geom_bar(stat = 'identity', alpha = 0.7) +
    geom_text(aes(y = max(quantity) / 2, label = quantity), size = 4) +
    ggtitle("LifeCycle Grids")
```

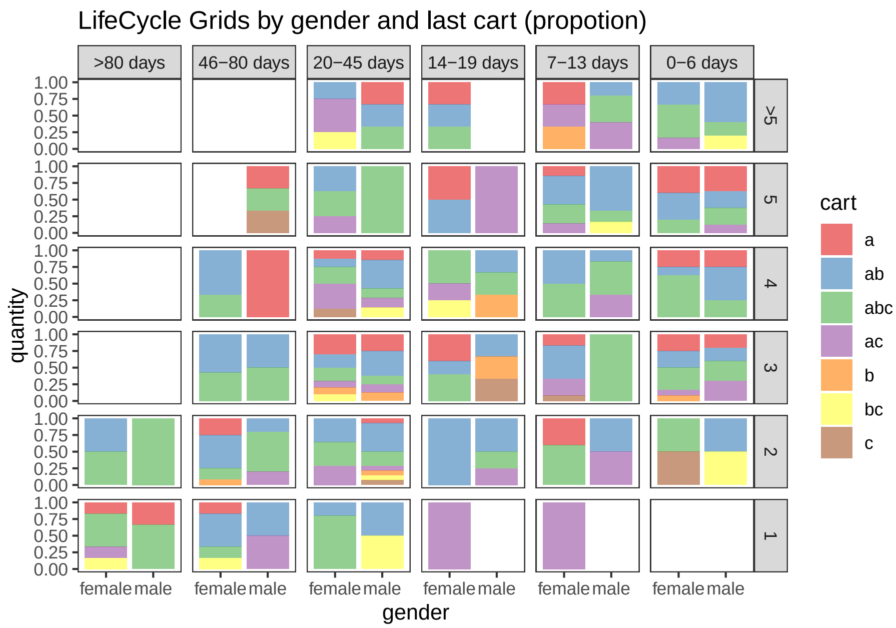



```
lcg.sub <- orders.segm %>%
    group_by(gender, cart, segm.rec, segm.freq) %>%
    summarise(quantity = n()) %>%
    mutate(client = 'client') %>%
    ungroup()

ggplot(lcg.sub, aes(x = client, y = quantity, fill = gender)) +
    theme_bw() +
    scale_fill_brewer(palette = 'Set1') +
    theme(panel.grid = element_blank()) +
    geom_bar(stat = 'identity',
             position = 'fill' ,
             alpha = 0.6) +
```

```
    facet_grid(segm.freq ~ segm.rec) +
    ggtitle("LifeCycle Grids by gender (propotion)")
```




```
ggplot(lcg.sub, aes(x = gender, y = quantity, fill = cart)) +
    theme_bw() +
    scale_fill_brewer(palette = 'Set1') +
    theme(panel.grid = element_blank()) +
    geom_bar(stat = 'identity',
             position = 'fill' ,
             alpha = 0.6) +
    facet_grid(segm.freq ~ segm.rec) +
    ggtitle("LifeCycle Grids by gender and last cart (propotion)")
```



22.4.2.2 CLV & CAC

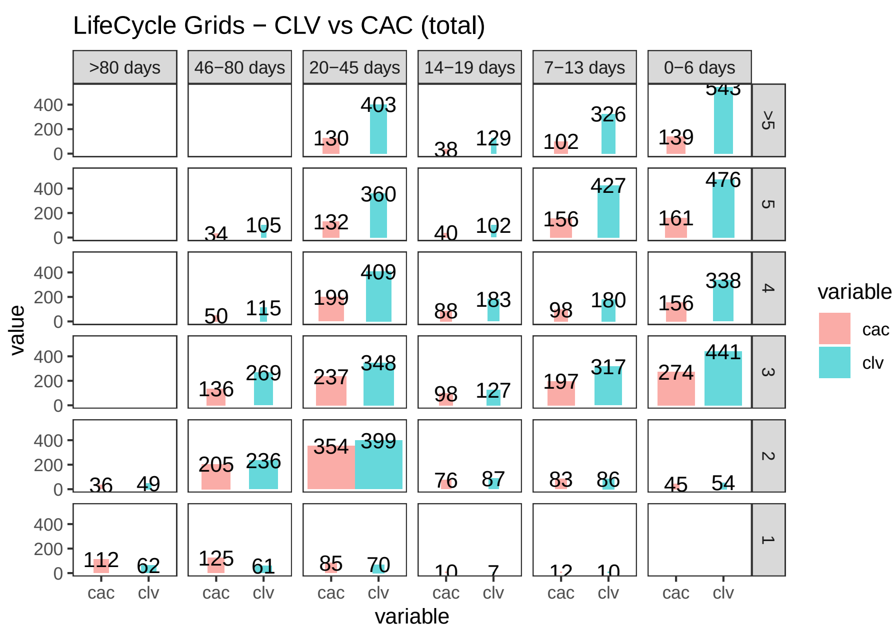
calculate customer acquisition cost (CAC) and customer lifetime value (CLV)

```
# loading libraries
library(dplyr)
library(reshape2)
library(ggplot2)

# creating data sample
set.seed(10)
data <- data.frame(
    orderId = sample(c(1:1000), 5000, replace = TRUE),
    product = sample(
        c('NULL', 'a', 'b', 'c'),
        5000,
        replace = TRUE,
        prob = c(0.15, 0.65, 0.3, 0.15)
    )
)
order <- data.frame(orderId = c(1:1000),
                    clientId = sample(c(1:300), 1000, replace = TRUE))
gender <- data.frame(clientId = c(1:300),
                     gender = sample(
                         c('male', 'female'),
```

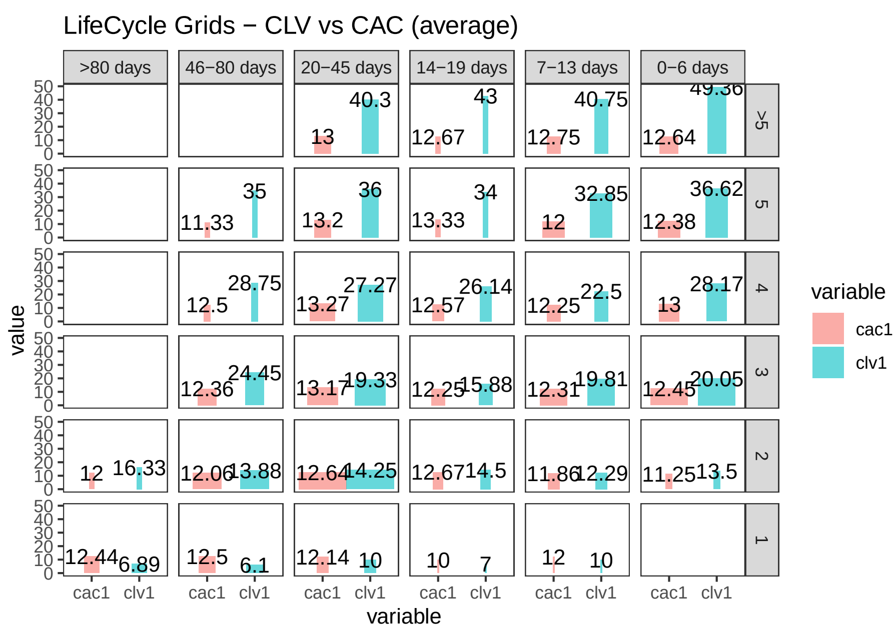
```
                         300,
                         replace = TRUE,
                         prob = c(0.40, 0.60)
                     ))
date <- data.frame(orderId = c(1:1000),
                   orderdate = sample((1:100), 1000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, gender, by = 'clientId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL', ]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")

# creating data frames with CAC and Gross margin
cac <-
    data.frame(clientId = unique(orders$clientId),
               cac = sample(c(10:15), 288, replace = TRUE))
gr.margin <-
    data.frame(product = c('a', 'b', 'c'),
               grossmarg = c(1, 2, 3))

rm(data, date, order, gender)

# reporting date
today <- as.Date('2012-04-11', format = '%Y-%m-%d')

# calculating customer lifetime value
orders <- merge(orders, gr.margin, by = 'product')

clv <- orders %>%
    group_by(clientId) %>%
    summarise(clv = sum(grossmarg)) %>%
    ungroup()

# processing data
orders <-
    dcast(
        orders,
        orderId + clientId + gender + orderdate ~ product,
        value.var = 'product',
        fun.aggregate = length
    )

orders <- orders %>%
    group_by(clientId) %>%
    mutate(frequency = n(),
```

```
          recency = as.numeric(today - orderdate)) %>%
    filter(orderdate == max(orderdate)) %>%
    filter(orderId == max(orderId)) %>%
    ungroup()

orders.segm <- orders %>%
    mutate(segm.freq = ifelse(between(frequency, 1, 1), '1',
                              ifelse(
                                  between(frequency, 2, 2), '2',
                                  ifelse(between(frequency, 3, 3), '3',
                                         ifelse(
                                             between(frequency, 4, 4), '4',
                                             ifelse(between(frequency, 5, 5), '5', '>5
                                         ))
                              ))) %>%
    mutate(segm.rec = ifelse(
        between(recency, 0, 6),
        '0-6 days',
        ifelse(
            between(recency, 7, 13),
            '7-13 days',
            ifelse(
                between(recency, 14, 19),
                '14-19 days',
                ifelse(
                    between(recency, 20, 45),
                    '20-45 days',
                    ifelse(between(recency, 46, 80), '46-80 days', '>80 days')
                )
            )
        )
    )) %>%
    # creating last cart feature
    mutate(cart = paste(
        ifelse(a != 0, 'a', ''),
        ifelse(b != 0, 'b', ''),
        ifelse(c != 0, 'c', ''),
        sep = ''
    )) %>%
    arrange(clientId)

# defining order of boundaries
orders.segm$segm.freq <-
    factor(orders.segm$segm.freq, levels = c('>5', '5', '4', '3', '2', '1'))
orders.segm$segm.rec <-
```

```
    factor(
        orders.segm$segm.rec,
        levels = c(
            '>80 days',
            '46-80 days',
            '20-45 days',
            '14-19 days',
            '7-13 days',
            '0-6 days'
        )
    )

orders.segm <- merge(orders.segm, cac, by = 'clientId')
orders.segm <- merge(orders.segm, clv, by = 'clientId')

lcg.clv <- orders.segm %>%
    group_by(segm.rec, segm.freq) %>%
    summarise(quantity = n(),
              # calculating cumulative CAC and CLV
              cac = sum(cac),
              clv = sum(clv)) %>%
    ungroup() %>%
    # calculating CAC and CLV per client
    mutate(cac1 = round(cac / quantity, 2),
           clv1 = round(clv / quantity, 2))

lcg.clv <-
    reshape2::melt(lcg.clv, id.vars = c('segm.rec', 'segm.freq', 'quantity'))

ggplot(lcg.clv[lcg.clv$variable %in% c('clv', 'cac'), ], aes(x = variable, y =
                                                                 value, fill = variable)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(stat = 'identity', alpha = 0.6, aes(width = quantity / max(quantity))) +
    geom_text(aes(y = value, label = value), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    ggtitle("LifeCycle Grids - CLV vs CAC (total)")
```




```
ggplot(lcg.clv[lcg.clv$variable %in% c('clv1', 'cac1'), ], aes(x = variable, y =
                                                                 value, fill = varial

    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(stat = 'identity', alpha = 0.6, aes(width = quantity / max(quantity))) +
    geom_text(aes(y = value, label = value), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    ggtitle("LifeCycle Grids - CLV vs CAC (average)")
```



22.4.2.3 Cohort Analysis

link

combine customers through common characteristics to split customers into homogeneous groups

```r
# loading libraries
library(dplyr)
library(reshape2)
library(ggplot2)
library(googleVis)

set.seed(10)
# creating orders data sample
data <- data.frame(
    orderId = sample(c(1:5000), 25000, replace = TRUE),
    product = sample(
        c('NULL', 'a', 'b', 'c'),
        25000,
        replace = TRUE,
        prob = c(0.15, 0.65, 0.3, 0.15)
    )
)
order <- data.frame(orderId = c(1:5000),
                    clientId = sample(c(1:1500), 5000, replace = TRUE))
```

```
date <- data.frame(orderId = c(1:5000),
                   orderdate = sample((1:500), 5000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL',]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")
rm(data, date, order)
# creating data frames with CAC, Gross margin, Campaigns and Potential CLV
gr.margin <-
    data.frame(product = c('a', 'b', 'c'),
               grossmarg = c(1, 2, 3))
campaign <- data.frame(clientId = c(1:1500),
                       campaign = paste('campaign', sample(c(1:7), 1500, replace = TRU
                                        ' '))
cac <-
    data.frame(campaign = unique(campaign$campaign),
               cac = sample(c(10:15), 7, replace = TRUE))
campaign <- merge(campaign, cac, by = 'campaign')
potential <- data.frame(clientId = c(1:1500),
                        clv.p = sample(c(0:50), 1500, replace = TRUE))
rm(cac)

# reporting date
today <- as.Date('2013-05-16', format = '%Y-%m-%d')
```

where

- campaign, which includes campaign name and customer acquisition cost for each customer,
- margin, which includes gross margin for each product,
- potential, which includes potential values / predicted CLV for each client,
- orders, which includes orders from our customers with products and order dates.

```
# calculating CLV, frequency, recency, average time lapses between purchases and defin

orders <- merge(orders, gr.margin, by = 'product')

customers <- orders %>%
    # combining products and summarising gross margin
    group_by(orderId, clientId, orderdate) %>%
    summarise(grossmarg = sum(grossmarg)) %>%
    ungroup() %>%
    # calculating frequency, recency, average time lapses between purchases and defini
    group_by(clientId) %>%
    mutate(
```

```
        frequency = n(),
        recency = as.numeric(today - max(orderdate)),
        av.gap = round(as.numeric(max(orderdate) - min(orderdate)) / frequency, 0),
        cohort = format(min(orderdate), format = '%Y-%m')
    ) %>%
    ungroup() %>%
    # calculating CLV to date
    group_by(clientId, cohort, frequency, recency, av.gap) %>%
    summarise(clv = sum(grossmarg)) %>%
    arrange(clientId) %>%
    ungroup()
# calculating potential CLV and CAC
customers <- merge(customers, campaign, by = 'clientId')
customers <- merge(customers, potential, by = 'clientId')
# leading the potential value to more or less real value
customers$clv.p <-
    round(customers$clv.p / sqrt(customers$recency) * customers$frequency,
          2)

rm(potential, gr.margin, today)
# adding segments
customers <- customers %>%
    mutate(segm.freq = ifelse(between(frequency, 1, 1), '1',
                              ifelse(
                                  between(frequency, 2, 2), '2',
                                  ifelse(between(frequency, 3, 3), '3',
                                         ifelse(
                                             between(frequency, 4, 4), '4',
                                             ifelse(between(frequency, 5, 5), '5', '>5')
                                         ))
                              ))) %>%
    mutate(segm.rec = ifelse(
        between(recency, 0, 30),
        '0-30 days',
        ifelse(
            between(recency, 31, 60),
            '31-60 days',
            ifelse(
                between(recency, 61, 90),
                '61-90 days',
                ifelse(
                    between(recency, 91, 120),
                    '91-120 days',
                    ifelse(between(recency, 121, 180), '121-180 days', '>180 days')
                )
```

```
            )
        )
    ))

# defining order of boundaries
customers$segm.freq <-
    factor(customers$segm.freq, levels = c('>5', '5', '4', '3', '2', '1'))
customers$segm.rec <-
    factor(
        customers$segm.rec,
        levels = c(
            '>180 days',
            '121-180 days',
            '91-120 days',
            '61-90 days',
            '31-60 days',
            '0-30 days'
        )
    )
```

```
lcg.coh <- customers %>%
    group_by(cohort, segm.rec, segm.freq) %>%
    # calculating cumulative values
    summarise(
        quantity = n(),
        cac = sum(cac),
        clv = sum(clv),
        clv.p = sum(clv.p),
        av.gap = sum(av.gap)
    ) %>%
    ungroup() %>%
    # calculating average values
    mutate(
        av.cac = round(cac / quantity, 2),
        av.clv = round(clv / quantity, 2),
        av.clv.p = round(clv.p / quantity, 2),
        av.clv.tot = av.clv + av.clv.p,
        av.gap = round(av.gap / quantity, 2),
        diff = av.clv - av.cac
    )

# 1. Structure of averages and comparison cohorts

ggplot(lcg.coh, aes(x = cohort, fill = cohort)) +
```

```
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(aes(y = diff), stat = 'identity', alpha = 0.5) +
    geom_text(aes(y = diff, label = round(diff, 0)), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = .5,
        vjust = .5,
        face = "plain"
    )) +
    ggtitle("Cohorts in LifeCycle Grids - difference between av.CLV to date and av.CAC")

ggplot(lcg.coh, aes(x = cohort, fill = cohort)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(aes(y = av.clv.tot), stat = 'identity', alpha = 0.2) +
    geom_text(aes(
        y = av.clv.tot + 10,
        label = round(av.clv.tot, 0),
        color = cohort
    ), size = 4) +
    geom_bar(aes(y = av.clv), stat = 'identity', alpha = 0.7) +
    geom_errorbar(aes(y = av.cac, ymax = av.cac, ymin = av.cac),
                  color = 'red',
                  size = 1.2) +
    geom_text(
        aes(y = av.cac, label = round(av.cac, 0)),
        size = 4,
        color = 'darkred',
        vjust = -.5
    ) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = .5,
        vjust = .5,
        face = "plain"
    )) +
    ggtitle("Cohorts in LifeCycle Grids - total av.CLV and av.CAC")

# 2. Analyzing customer flows
# customers flows analysis (FPD cohorts)

# defining cohort and reporting dates
```

```
coh <- '2012-09'
report.dates <- c('2012-10-01', '2013-01-01', '2013-04-01')
report.dates <- as.Date(report.dates, format = '%Y-%m-%d')

# defining segments for each cohort's customer for reporting dates
df.sankey <- data.frame()

for (i in 1:length(report.dates)) {
    orders.cache <- orders %>%
        filter(orderdate < report.dates[i])

    customers.cache <- orders.cache %>%
        select(-product,-grossmarg) %>%
        unique() %>%
        group_by(clientId) %>%
        mutate(
            frequency = n(),
            recency = as.numeric(report.dates[i] - max(orderdate)),
            cohort = format(min(orderdate), format = '%Y-%m')
        ) %>%
        ungroup() %>%
        select(clientId, frequency, recency, cohort) %>%
        unique() %>%
        filter(cohort == coh) %>%
        mutate(segm.freq = ifelse(
            between(frequency, 1, 1),
            '1 purch',
            ifelse(
                between(frequency, 2, 2),
                '2 purch',
                ifelse(
                    between(frequency, 3, 3),
                    '3 purch',
                    ifelse(
                        between(frequency, 4, 4),
                        '4 purch',
                        ifelse(between(frequency, 5, 5), '5 purch', '>5 purch')
                    )
                )
            )
        )) %>%
        mutate(segm.rec = ifelse(
            between(recency, 0, 30),
            '0-30 days',
            ifelse(
```

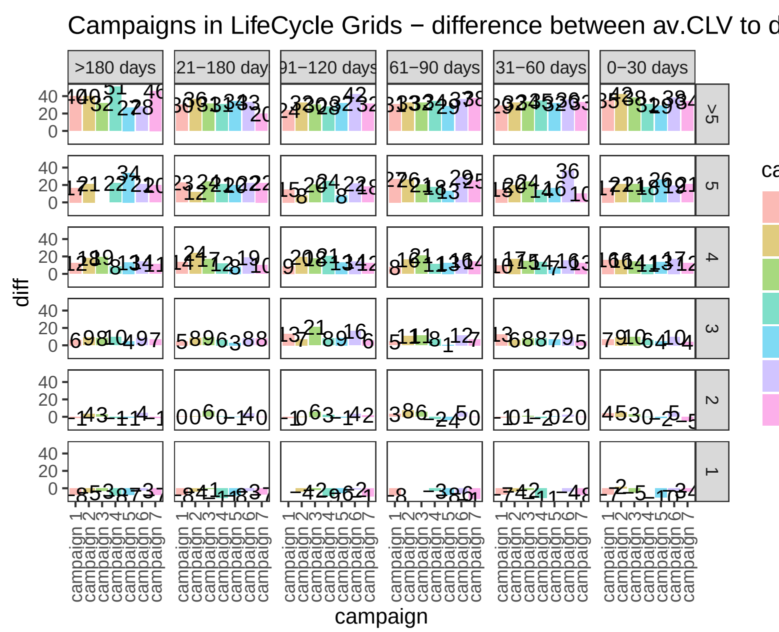
```
                between(recency, 31, 60),
                '31-60 days',
                ifelse(
                    between(recency, 61, 90),
                    '61-90 days',
                    ifelse(
                        between(recency, 91, 120),
                        '91-120 days',
                        ifelse(between(recency, 121, 180), '121-180 days', '>180 days')
                    )
                )
            )
        )) %>%
        mutate(
            cohort.segm = paste(cohort, segm.rec, segm.freq, sep = ' : '),
            report.date = report.dates[i]
        ) %>%
        select(clientId, cohort.segm, report.date)

    df.sankey <- rbind(df.sankey, customers.cache)
}

# processing data for Sankey diagram format
df.sankey <-
    dcast(df.sankey,
          clientId ~ report.date,
          value.var = 'cohort.segm',
          fun.aggregate = NULL)
write.csv(df.sankey, 'customers_path.csv', row.names = FALSE)
df.sankey <- df.sankey %>% select(-clientId)

df.sankey.plot <- data.frame()
for (i in 2:ncol(df.sankey)) {
    df.sankey.cache <- df.sankey %>%
        group_by(df.sankey[, i - 1], df.sankey[, i]) %>%
        summarise(n = n()) %>%
        ungroup()

    colnames(df.sankey.cache)[1:2] <- c('from', 'to')

    df.sankey.cache$from <-
        paste(df.sankey.cache$from, ' (', report.dates[i - 1], ')', sep = '')
    df.sankey.cache$to <-
        paste(df.sankey.cache$to, ' (', report.dates[i], ')', sep = '')
```

```
    df.sankey.plot <- rbind(df.sankey.plot, df.sankey.cache)
}

# plotting
plot(gvisSankey(
    df.sankey.plot,
    from = 'from',
    to = 'to',
    weight = 'n',
    options = list(
        height = 900,
        width = 1800,
        sankey = "{link:{color:{fill:'lightblue'}}}"
    )
))

# purchasing pace

ggplot(lcg.coh, aes(x = cohort, fill = cohort)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(aes(y = av.gap), stat = 'identity', alpha = 0.6) +
    geom_text(aes(y = av.gap, label = round(av.gap, 0)), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = .5,
        vjust = .5,
        face = "plain"
    )) +
    ggtitle("Cohorts in LifeCycle Grids - average time lapses between purchases")
```

22.4.2.3.1 First-purchase date cohort

```
# campaign cohorts
lcg.camp <- customers %>%
    group_by(campaign, segm.rec, segm.freq) %>%
    # calculating cumulative values
    summarise(
        quantity = n(),
        cac = sum(cac),
        clv = sum(clv),
        clv.p = sum(clv.p),
        av.gap = sum(av.gap)
```

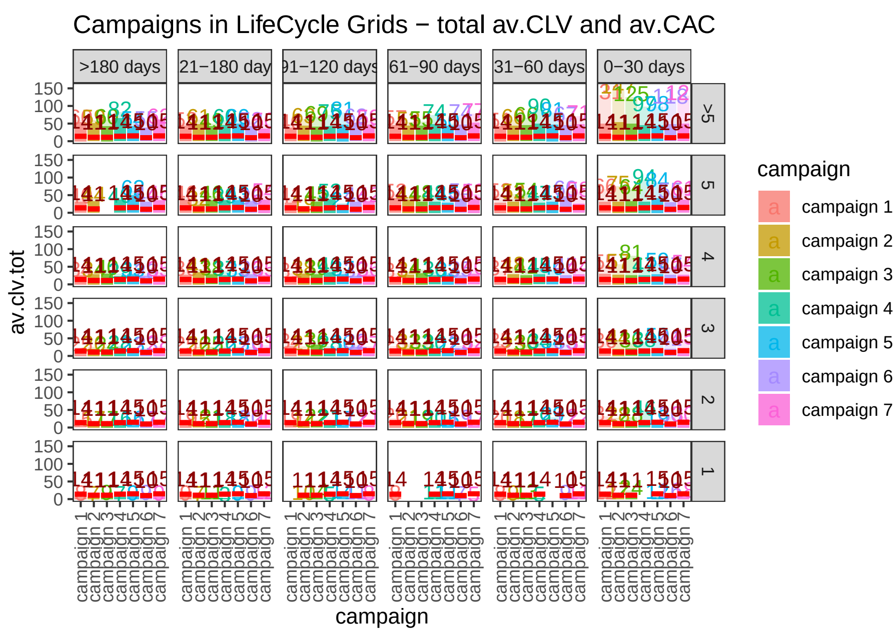
```
    ) %>%
    ungroup() %>%
    # calculating average values
    mutate(
        av.cac = round(cac / quantity, 2),
        av.clv = round(clv / quantity, 2),
        av.clv.p = round(clv.p / quantity, 2),
        av.clv.tot = av.clv + av.clv.p,
        av.gap = round(av.gap / quantity, 2),
        diff = av.clv - av.cac
    )

ggplot(lcg.camp, aes(x = campaign, fill = campaign)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(aes(y = diff), stat = 'identity', alpha = 0.5) +
    geom_text(aes(y = diff, label = round(diff, 0)), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = .5,
        vjust = .5,
        face = "plain"
    )) +
    ggtitle("Campaigns in LifeCycle Grids - difference between av.CLV to date and av.CAC")
```



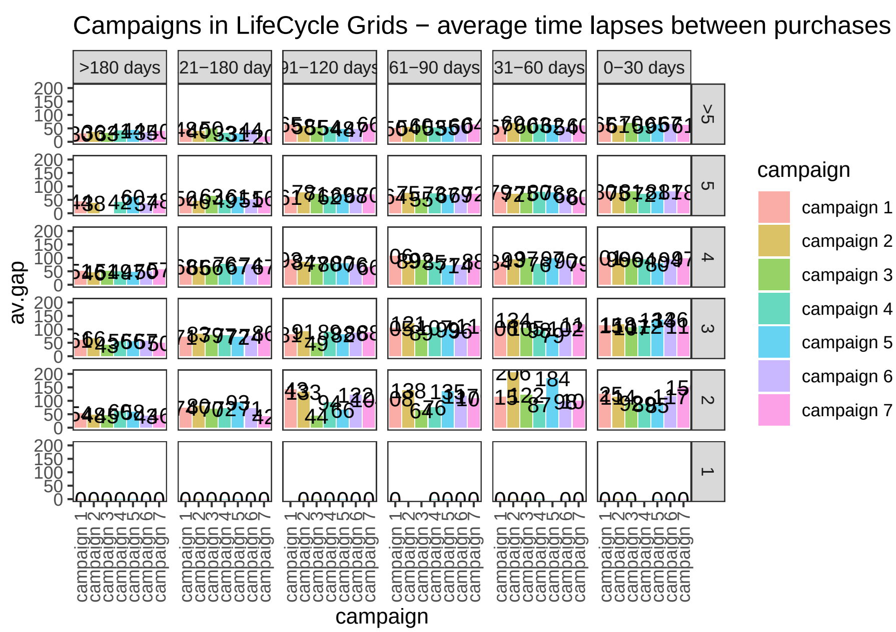
22.4.2.3.2 Campaign Cohorts

```
ggplot(lcg.camp, aes(x = campaign, fill = campaign)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(aes(y = av.clv.tot), stat = 'identity', alpha = 0.2) +
    geom_text(aes(
        y = av.clv.tot + 10,
        label = round(av.clv.tot, 0),
        color = campaign
    ), size = 4) +
    geom_bar(aes(y = av.clv), stat = 'identity', alpha = 0.7) +
    geom_errorbar(aes(y = av.cac, ymax = av.cac, ymin = av.cac),
                  color = 'red',
                  size = 1.2) +
    geom_text(
        aes(y = av.cac, label = round(av.cac, 0)),
        size = 4,
        color = 'darkred',
        vjust = -.5
    ) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = .5,
        vjust = .5,
        face = "plain"
```

```
    )) +
    ggtitle("Campaigns in LifeCycle Grids - total av.CLV and av.CAC")
```




```
ggplot(lcg.camp, aes(x = campaign, fill = campaign)) +
    theme_bw() +
    theme(panel.grid = element_blank()) +
    geom_bar(aes(y = av.gap), stat = 'identity', alpha = 0.6) +
    geom_text(aes(y = av.gap, label = round(av.gap, 0)), size = 4) +
    facet_grid(segm.freq ~ segm.rec) +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = .5,
        vjust = .5,
        face = "plain"
    )) +
    ggtitle("Campaigns in LifeCycle Grids - average time lapses between purchases")
```



22.4.2.3.3 Retention Rate

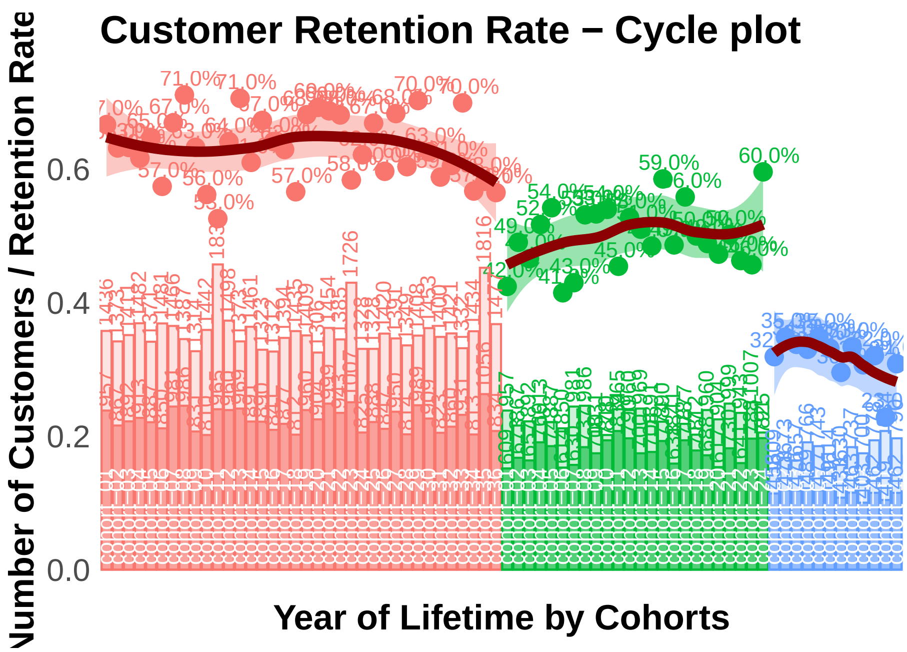
Customer Retention Rate

```
# loading libraries
library(dplyr)
library(reshape2)
library(ggplot2)
library(scales)
library(gridExtra)
# creating data sample
set.seed(10)
cohorts <-
    data.frame(
        cohort = paste('cohort', formatC(
            c(1:36),
            width = 2,
            format = 'd',
            flag = '0'
        ), sep = '_'),
        Y_00 = sample(c(1300:1500), 36, replace = TRUE),
        Y_01 = c(sample(c(800:1000), 36, replace = TRUE)),
        Y_02 = c(sample(c(600:800), 24, replace = TRUE), rep(NA, 12)),
        Y_03 = c(sample(c(400:500), 12, replace = TRUE), rep(NA, 24))
    )
# simulating seasonality (Black Friday)
cohorts[c(11, 23, 35), 2] <-
```

```
    as.integer(cohorts[c(11, 23, 35), 2] * 1.25)
cohorts[c(11, 23, 35), 3] <-
    as.integer(cohorts[c(11, 23, 35), 3] * 1.10)
cohorts[c(11, 23, 35), 4] <-
    as.integer(cohorts[c(11, 23, 35), 4] * 1.07)

# calculating retention rate and preparing data for plotting
df_plot <-
    reshape2::melt(
        cohorts,
        id.vars = 'cohort',
        value.name = 'number',
        variable.name = "year_of_LT"
    )

df_plot <- df_plot %>%
    group_by(cohort) %>%
    arrange(year_of_LT) %>%
    mutate(number_prev_year = lag(number),
           number_Y_00 = number[which(year_of_LT == 'Y_00')]) %>%
    ungroup() %>%
    mutate(
        ret_rate_prev_year = number / number_prev_year,
        ret_rate = number / number_Y_00,
        year_cohort = paste(year_of_LT, cohort, sep = '-')
    )

##### The first way for plotting cycle plot via scaling
# calculating the coefficient for scaling 2nd axis
k <-
    max(df_plot$number_prev_year[df_plot$year_of_LT == 'Y_01'] * 1.15) / min(df_plot$ret_rate[df_

# retention rate cycle plot
ggplot(
    na.omit(df_plot),
    aes(
        x = year_cohort,
        y = ret_rate,
        group = year_of_LT,
        color = year_of_LT
    )
) +
    theme_bw() +
    geom_point(size = 4) +
    geom_text(
```

```
        aes(label = percent(round(ret_rate, 2))),
        size = 4,
        hjust = 0.4,
        vjust = -0.6,
        fontface = "plain"
    ) +
    # smooth method can be changed (e.g. for "lm")
    geom_smooth(
        size = 2.5,
        method = 'loess',
        color = 'darkred',
        aes(fill = year_of_LT)
    ) +
    geom_bar(aes(y = number_prev_year / k, fill = year_of_LT),
             alpha = 0.2,
             stat = 'identity') +
    geom_bar(aes(y = number / k, fill = year_of_LT),
             alpha = 0.6,
             stat = 'identity') +
    geom_text(
        aes(y = 0, label = cohort),
        color = 'white',
        angle = 90,
        size = 4,
        hjust = -0.05,
        vjust = 0.4
    ) +
    geom_text(
        aes(y = number_prev_year / k, label = number_prev_year),
        angle = 90,
        size = 4,
        hjust = -0.1,
        vjust = 0.4
    ) +
    geom_text(
        aes(y = number / k, label = number),
        angle = 90,
        size = 4,
        hjust = -0.1,
        vjust = 0.4
    ) +
    theme(
        legend.position = 'none',
        plot.title = element_text(size = 20, face = "bold", vjust = 2),
        axis.title.x = element_text(size = 18, face = "bold"),
```

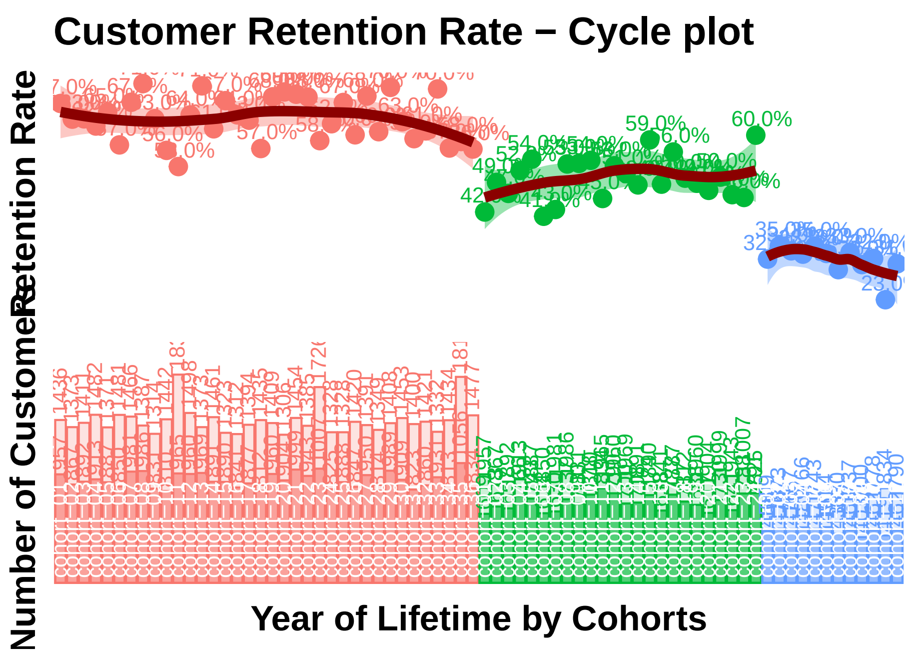
```
        axis.title.y = element_text(size = 18, face = "bold"),
        axis.text = element_text(size = 16),
        axis.text.x = element_blank(),
        axis.ticks.x = element_blank(),
        axis.ticks.y = element_blank(),
        panel.border = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank()
    ) +
    labs(x = 'Year of Lifetime by Cohorts', y = 'Number of Customers / Retention Rate') +
    ggtitle("Customer Retention Rate - Cycle plot")
```




```
##### The second way for plotting cycle plot via multi-plotting
# plot #1 - Retention rate
p1 <-
    ggplot(
        na.omit(df_plot),
        aes(
            x = year_cohort,
            y = ret_rate,
            group = year_of_LT,
            color = year_of_LT
        )
    ) +
    theme_bw() +
```

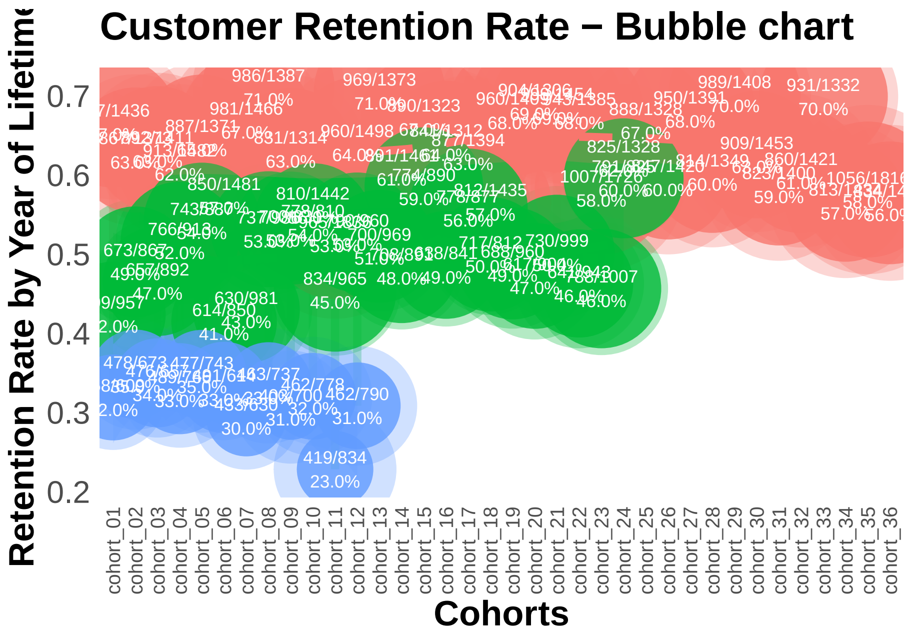
```
    geom_point(size = 4) +
    geom_text(
        aes(label = percent(round(ret_rate, 2))),
        size = 4,
        hjust = 0.4,
        vjust = -0.6,
        fontface = "plain"
    ) +
    geom_smooth(
        size = 2.5,
        method = 'loess',
        color = 'darkred',
        aes(fill = year_of_LT)
    ) +
    theme(
        legend.position = 'none',
        plot.title = element_text(size = 20, face = "bold", vjust = 2),
        axis.title.x = element_blank(),
        axis.title.y = element_text(size = 18, face = "bold"),
        axis.text = element_blank(),
        axis.ticks.x = element_blank(),
        axis.ticks.y = element_blank(),
        panel.border = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank()
    ) +
    labs(y = 'Retention Rate') +
    ggtitle("Customer Retention Rate - Cycle plot")

# plot #2 - number of customers
p2 <-
    ggplot(na.omit(df_plot),
           aes(x = year_cohort, group = year_of_LT, color = year_of_LT)) +
    theme_bw() +
    geom_bar(aes(y = number_prev_year, fill = year_of_LT),
             alpha = 0.2,
             stat = 'identity') +
    geom_bar(aes(y = number, fill = year_of_LT),
             alpha = 0.6,
             stat = 'identity') +
    geom_text(
        aes(y = number_prev_year, label = number_prev_year),
        angle = 90,
        size = 4,
        hjust = -0.1,
```

```
        vjust = 0.4
    ) +
    geom_text(
        aes(y = number, label = number),
        angle = 90,
        size = 4,
        hjust = -0.1,
        vjust = 0.4
    ) +
    geom_text(
        aes(y = 0, label = cohort),
        color = 'white',
        angle = 90,
        size = 4,
        hjust = -0.05,
        vjust = 0.4
    ) +
    theme(
        legend.position = 'none',
        plot.title = element_text(size = 20, face = "bold", vjust = 2),
        axis.title.x = element_text(size = 18, face = "bold"),
        axis.title.y = element_text(size = 18, face = "bold"),
        axis.text = element_blank(),
        axis.ticks.x = element_blank(),
        axis.ticks.y = element_blank(),
        panel.border = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank()
    ) +
    scale_y_continuous(limits = c(0, max(df_plot$number_Y_00 * 1.1))) +
    labs(x = 'Year of Lifetime by Cohorts', y = 'Number of Customers')

# multiplot
grid.arrange(p1, p2, ncol = 1)
```



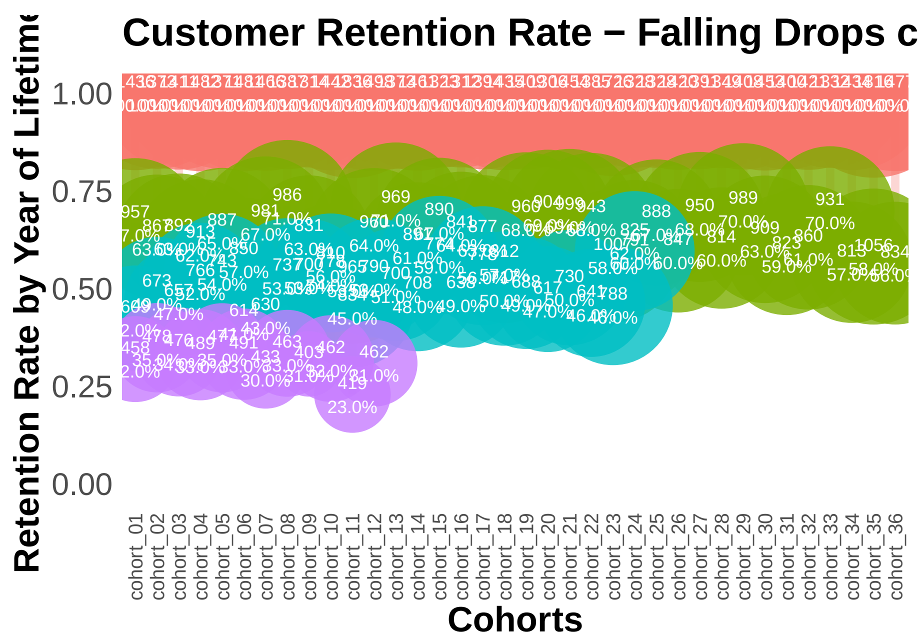

```
# retention rate bubble chart
ggplot(na.omit(df_plot),
       aes(
           x = cohort,
           y = ret_rate,
           group = cohort,
           color = year_of_LT
       )) +
    theme_bw() +
    scale_size(range = c(15, 40)) +
    geom_line(size = 2, alpha = 0.3) +
    geom_point(aes(size = number_prev_year), alpha = 0.3) +
    geom_point(aes(size = number), alpha = 0.8) +
    geom_smooth(
        linetype = 2,
        size = 2,
        method = 'loess',
        aes(group = year_of_LT, fill = year_of_LT),
        alpha = 0.2
    ) +
    geom_text(
        aes(label = paste0(
            number, '/', number_prev_year, '\n', percent(round(ret_rate, 2))
        )),
        color = 'white',
```

```
        size = 3,
        hjust = 0.5,
        vjust = 0.5,
        fontface = "plain"
    ) +
    theme(
        legend.position = 'none',
        plot.title = element_text(size = 20, face = "bold", vjust = 2),
        axis.title.x = element_text(size = 18, face = "bold"),
        axis.title.y = element_text(size = 18, face = "bold"),
        axis.text = element_text(size = 16),
        axis.text.x = element_text(
            size = 10,
            angle = 90,
            hjust = .5,
            vjust = .5,
            face = "plain"
        ),
        axis.ticks.x = element_blank(),
        axis.ticks.y = element_blank(),
        panel.border = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank()
    ) +
    labs(x = 'Cohorts', y = 'Retention Rate by Year of Lifetime') +
    ggtitle("Customer Retention Rate - Bubble chart")
```




```r
# retention rate falling drops chart
ggplot(df_plot,
       aes(
           x = cohort,
           y = ret_rate,
           group = cohort,
           color = year_of_LT
       )) +
    theme_bw() +
    scale_size(range = c(15, 40)) +
    scale_y_continuous(limits = c(0, 1)) +
    geom_line(size = 2, alpha = 0.3) +
    geom_point(aes(size = number), alpha = 0.8) +
    geom_text(
        aes(label = paste0(number, '\n', percent(round(
            ret_rate, 2
        )))),
        color = 'white',
        size = 3,
        hjust = 0.5,
        vjust = 0.5,
        fontface = "plain"
    ) +
    theme(
        legend.position = 'none',
```

```
        plot.title = element_text(size = 20, face = "bold", vjust = 2),
        axis.title.x = element_text(size = 18, face = "bold"),
        axis.title.y = element_text(size = 18, face = "bold"),
        axis.text = element_text(size = 16),
        axis.text.x = element_text(
            size = 10,
            angle = 90,
            hjust = .5,
            vjust = .5,
            face = "plain"
        ),
        axis.ticks.x = element_blank(),
        axis.ticks.y = element_blank(),
        panel.border = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank()
    ) +
    labs(x = 'Cohorts', y = 'Retention Rate by Year of Lifetime') +
    ggtitle("Customer Retention Rate - Falling Drops chart")
```



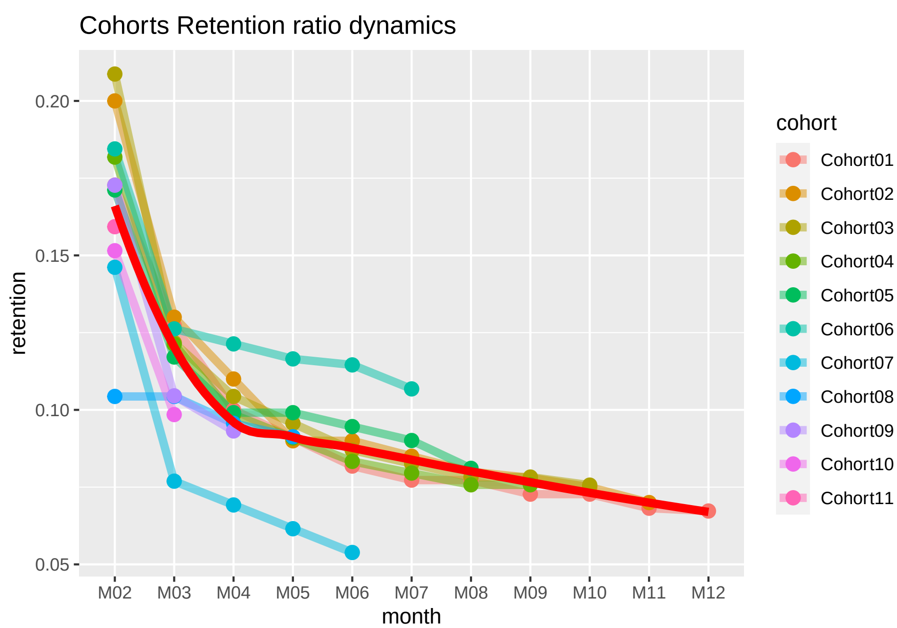
22.4.2.3.4 Retention Charts

Retention charts

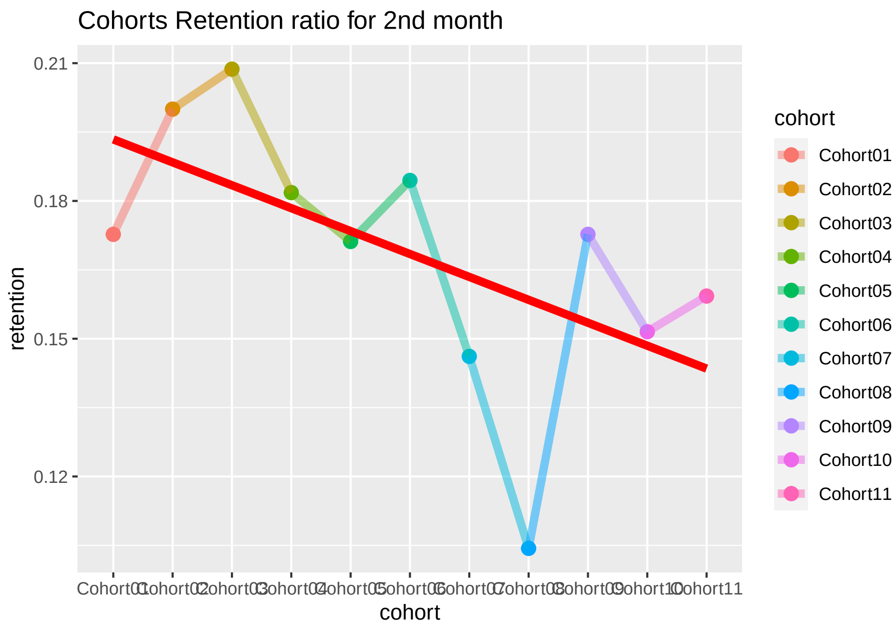
```
# libraries
library(dplyr)
library(ggplot2)
```

```
library(reshape2)

cohort.clients <- data.frame(
    cohort = c(
        'Cohort01',
        'Cohort02',
        'Cohort03',
        'Cohort04',
        'Cohort05',
        'Cohort06',
        'Cohort07',
        'Cohort08',
        'Cohort09',
        'Cohort10',
        'Cohort11',
        'Cohort12'
    ),
    M01 = c(11000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    M02 = c(1900, 10000, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    M03 = c(1400, 2000, 11500, 0, 0, 0, 0, 0, 0, 0, 0, 0),
    M04 = c(1100, 1300, 2400, 13200, 0, 0, 0, 0, 0, 0, 0, 0),
    M05 = c(1000, 1100, 1400, 2400, 11100, 0, 0, 0, 0, 0, 0, 0),
    M06 = c(900, 900, 1200, 1600, 1900, 10300, 0, 0, 0, 0, 0, 0),
    M07 = c(850, 900, 1100, 1300, 1300, 1900, 13000, 0, 0, 0, 0, 0),
    M08 = c(850, 850, 1000, 1200, 1100, 1300, 1900, 11500, 0, 0, 0, 0),
    M09 = c(800, 800, 950, 1100, 1100, 1250, 1000, 1200, 11000, 0, 0, 0),
    M10 = c(800, 780, 900, 1050, 1050, 1200, 900, 1200, 1900, 13200, 0, 0),
    M11 = c(750, 750, 900, 1000, 1000, 1180, 800, 1100, 1150, 2000, 11300, 0),
    M12 = c(740, 700, 870, 1000, 900, 1100, 700, 1050, 1025, 1300, 1800, 20000)
)

cohort.clients.r <- cohort.clients #create new data frame
totcols <-
    ncol(cohort.clients.r) #count number of columns in data set
for (i in 1:nrow(cohort.clients.r)) {
    #for loop for shifting each row
    df <- cohort.clients.r[i,] #select row from data frame
    df <- df[, !df[] == 0] #remove columns with zeros
    partcols <-
        ncol(df) #count number of columns in row (w/o zeros)
    #fill columns after values by zeros
    if (partcols < totcols)
        df[, c((partcols + 1):totcols)] <- 0
    cohort.clients.r[i,] <- df #replace initial row by new one
}
```

```
# Retention ratio = # clients in particular month / # clients in 1st month of life-time

#calculate retention (1)
x <- cohort.clients.r[, c(2:13)]
y <- cohort.clients.r[, 2]
reten.r <- apply(x, 2, function(x)
    x / y)
reten.r <- data.frame(cohort = (cohort.clients.r$cohort), reten.r)

#calculate retention (2)
c <- ncol(cohort.clients.r)
reten.r <- cohort.clients.r
for (i in 2:c) {
    reten.r[, (c + i - 1)] <- reten.r[, i] / reten.r[, 2]
}
reten.r <- reten.r[,-c(2:c)]
colnames(reten.r) <- colnames(cohort.clients.r)




#charts
reten.r <- reten.r[,-2] #remove M01 data because it is always 100%
#dynamics analysis chart
cohort.chart1 <- melt(reten.r, id.vars = 'cohort')
colnames(cohort.chart1) <- c('cohort', 'month', 'retention')
cohort.chart1 <- filter(cohort.chart1, retention != 0)
p <-
    ggplot(cohort.chart1,
           aes(
               x = month,
               y = retention,
               group = cohort,
               colour = cohort
           ))
p + geom_line(size = 2, alpha = 1 / 2) +
    geom_point(size = 3, alpha = 1) +
    geom_smooth(
        aes(group = 1),
        method = 'loess',
        size = 2,
        colour = 'red',
        se = FALSE
    ) +
    labs(title = "Cohorts Retention ratio dynamics")
```

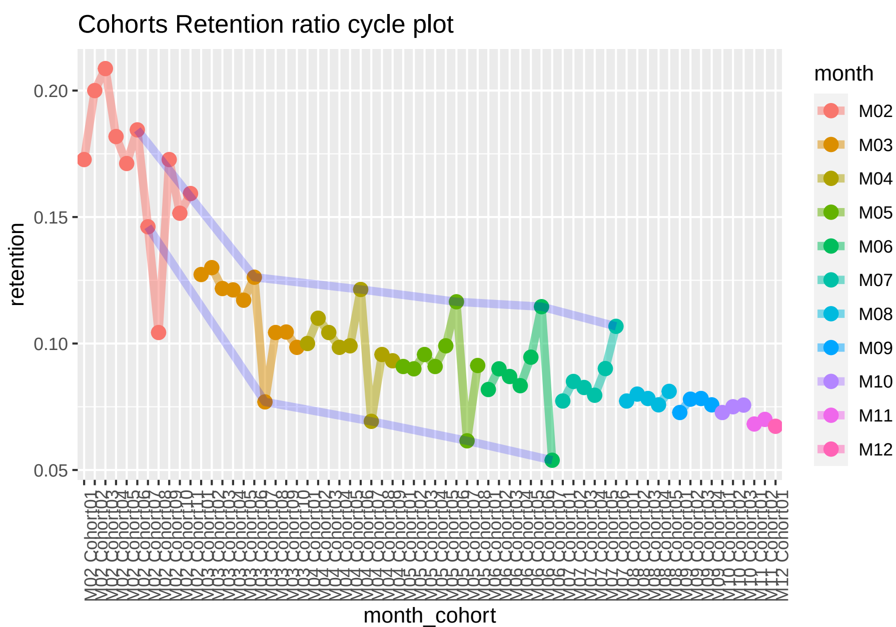



```
#second month analysis chart
cohort.chart2 <-
    filter(cohort.chart1, month == 'M02') #choose any month instead of M02
p <-
    ggplot(cohort.chart2, aes(x = cohort, y = retention, colour = cohort))
p + geom_point(size = 3) +
    geom_line(aes(group = 1), size = 2, alpha = 1 / 2) +
    geom_smooth(
        aes(group = 1),
        size = 2,
        colour = 'red',
        method = 'lm',
        se = FALSE
    ) +
    labs(title = "Cohorts Retention ratio for 2nd month")
```




```
#cycle plot
cohort.chart3 <- cohort.chart1
cohort.chart3 <-
    mutate(cohort.chart3, month_cohort = paste(month, cohort))
p <-
    ggplot(cohort.chart3,
           aes(
               x = month_cohort,
               y = retention,
               group = month,
               colour = month
           ))
#choose any cohorts instead of Cohort07 and Cohort06
m1 <- filter(cohort.chart3, cohort == 'Cohort07')
m2 <- filter(cohort.chart3, cohort == 'Cohort06')
p + geom_point(size = 3) +
    geom_line(aes(group = month), size = 2, alpha = 1 / 2) +
    labs(title = "Cohorts Retention ratio cycle plot") +
    geom_line(
        data = m1,
        aes(group = 1),
        colour = 'blue',
        size = 2,
        alpha = 1 / 5
    ) +
```

```
    geom_line(
        data = m2,
        aes(group = 1),
        colour = 'blue',
        size = 2,
        alpha = 1 / 5
    ) +
    theme(axis.text.x = element_text(angle = 90, hjust = 1))
```



22.4.2.4 Lifecycle phase sequential analysis

- analyze the path patterns of each cohort
- identify cohorts that attracted customers with the path we prefer to make offers.

```
library(TraMineR)

min.date <- min(orders$orderdate)
max.date <- max(orders$orderdate)

l <-
    c(seq(0, as.numeric(max.date - min.date), 10), as.numeric(max.date - min.date))

df <- data.frame()
```

```
for (i in l) {
    cur.date <- min.date + i
    print(cur.date)

    orders.cache <- orders %>%
        filter(orderdate <= cur.date)

    customers.cache <- orders.cache %>%
        select(-product,-grossmarg) %>%
        unique() %>%
        group_by(clientId) %>%
        mutate(frequency = n(),
               recency = as.numeric(cur.date - max(orderdate))) %>%
        ungroup() %>%
        select(clientId, frequency, recency) %>%
        unique() %>%

        mutate(segm =
                   ifelse(
                       between(frequency, 1, 2) & between(recency, 0, 60),
                       'new customer',
                       ifelse(
                           between(frequency, 1, 2) &
                               between(recency, 61, 180),
                           'under risk new customer',
                           ifelse(
                               between(frequency, 1, 2) & recency > 180,
                               '1x buyer',

                               ifelse(
                                   between(frequency, 3, 4) &
                                       between(recency, 0, 60),
                                   'engaged customer',
                                   ifelse(
                                       between(frequency, 3, 4) &
                                           between(recency, 61, 180),
                                       'under risk engaged customer',
                                       ifelse(
                                           between(frequency, 3, 4) & recency > 180,
                                           'former engaged customer',

                                           ifelse(
                                               frequency > 4 & between(recency, 0, 60),
                                               'best customer',
                                               ifelse(
```

```
                                                   frequency > 4 &
                                                       between(recency, 61, 180),
                                                   'under risk best customer',
                                                   ifelse(frequency > 4 &
                                                              recency > 180, 'former be
                                               )
                                           )
                                       )
                                   )
                               )
                           )
                       )
                   )) %>%

        mutate(report.date = i) %>%
        select(clientId, segm, report.date)

    df <- rbind(df, customers.cache)
}

# converting data to the sequence format
df <-
    dcast(df,
          clientId ~ report.date,
          value.var = 'segm',
          fun.aggregate = NULL)
df.seq <- seqdef(df,
                 2:ncol(df),
                 left = 'DEL',
                 right = 'DEL',
                 xtstep = 10)

# creating df with first purch.date and campaign cohort features
feat <- df %>% select(clientId)
feat <- merge(feat, campaign[, 1:2], by = 'clientId')
feat <- merge(feat, customers[, 1:2], by = 'clientId')

par(mar = c(1, 1, 1, 1))

# plotting the 10 most frequent sequences based on campaign
seqfplot(df.seq, border = NA, group = feat$campaign)

# plotting the 10 most frequent sequences based on campaign
seqfplot(
    df.seq,
```

```
    border = NA,
    group = feat$campaign,
    cex.legend = 0.9
)

# plotting the 10 most frequent sequences based on first purch.date cohort
coh.list <- sort(unique(feat$cohort))
# defining cohorts for plotting
feat.coh.list <- feat[feat$cohort %in% coh.list[1:6] ,]
df.coh <- df %>% filter(clientId %in% c(feat.coh.list$clientId))
df.seq.coh <-
    seqdef(
        df.coh,
        2:ncol(df.coh),
        left = 'DEL',
        right = 'DEL',
        xtstep = 10
    )
seqfplot(
    df.seq.coh,
    border = NA,
    group = feat.coh.list$cohort,
    cex.legend = 0.9
)
```

22.5 Shopping carts analysis

22.5.1 Multi-layer pie chart

Example by Sergey Bryl

```
# loading libraries
library(dplyr)
library(tidyverse)
library(reshape2)
library(plotrix)

# Simulate of orders
set.seed(15)
df <- data.frame(
    orderId = sample(c(1:1000), 5000, replace = TRUE),
    product = sample(
        c('NULL', 'a', 'b', 'c', 'd'),
        5000,
        replace = TRUE,
```

```
        prob = c(0.15, 0.65, 0.3, 0.15, 0.1)
    )
)
df <- df[df$product != 'NULL',]
head(df)
```

```
##   orderId product
## 1     549       b
## 3     874       a
## 4     674       d
## 5     505       a
## 6     294       a
## 7     177       a
```

```
# processing initial data
# we need to be sure that product's names are unique
df$product <- paste0("#", df$product, "#")

prod.matrix <- df %>%
    # removing duplicated products from each order (exclude the effect of quantity)
    group_by(orderId, product) %>%
    arrange(product) %>%
    unique() %>%
    # combining products to cart and calculating number of products
    group_by(orderId) %>%
    summarise(cart = paste(product, collapse = ";"),
              prod.num = n()) %>%
    # calculating number of carts
    group_by(cart, prod.num) %>%
    summarise(num = n()) %>%
    ungroup()

head(prod.matrix)
```

```
## # A tibble: 6 x 3
##   cart            prod.num   num
##   <chr>              <int> <int>
## 1 #a#                    1   115
## 2 #a#;#b#                2   248
## 3 #a#;#b#;#c#            3   172
## 4 #a#;#b#;#c#;#d#        4    78
## 5 #a#;#b#;#d#            3   119
## 6 #a#;#c#                2    98
```

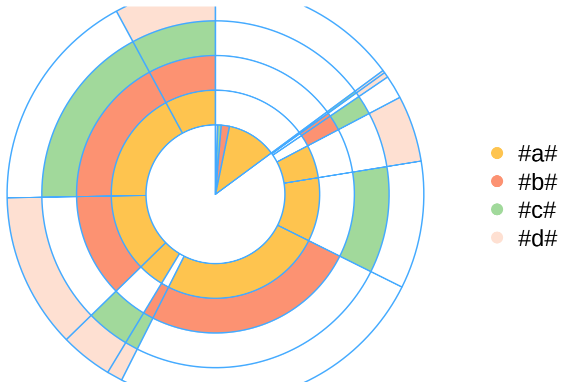
```
# calculating total number of orders/carts
tot <- sum(prod.matrix$num)
```

```r
# spliting orders for sets with 1 product and more than 1 product
one.prod <- prod.matrix %>% filter(prod.num == 1)

sev.prod <- prod.matrix %>%
    filter(prod.num > 1) %>%
    arrange(desc(prod.num))
```

```r
# defining parameters for pie chart
iniR <- 0.2 # initial radius
cols <- c(
    "#ffffff",
    "#fec44f",
    "#fc9272",
    "#a1d99b",
    "#fee0d2",
    "#2ca25f",
    "#8856a7",
    "#43a2ca",
    "#fdbb84",
    "#e34a33",
    "#a6bddb",
    "#dd1c77",
    "#ffeda0",
    "#756bb1"
)
prod <- df %>%
    select(product) %>%
    arrange(product) %>%
    unique()
prod <- c('NO', c(prod$product))
colors <- as.list(setNames(cols[c(1:(length(prod)))], prod))
```

```r
# 0 circle: blank
pie(
    1,
    radius = iniR,
    init.angle = 90,
    col = c('white'),
    border = NA,
    labels = ''
)

# drawing circles from last to 2nd
for (i in length(prod):2) {
    p <- grep(prod[i], sev.prod$cart)
    col <- rep('NO', times = nrow(sev.prod))
```

```
    col[p] <- prod[i]
    floating.pie(
        0,
        0,
        c(sev.prod$num, tot - sum(sev.prod$num)),
        radius = (1 + i) * iniR,
        startpos = pi / 2,
        col = as.character(colors [c(col, 'NO')]),
        border = "#44aaff"
    )
}

# 1 circle: orders with 1 product
floating.pie(
    0,
    0,
    c(tot - sum(one.prod$num), one.prod$num),
    radius = 2 * iniR,
    startpos = pi / 2,
    col = as.character(colors [c('NO', one.prod$cart)]),
    border = "#44aaff"
)

# legend
legend(
    1.5,
    2 * iniR,
    gsub("_", " ", names(colors)[-1]),
    col = as.character(colors [-1]),
    pch = 19,
    bty = 'n',
    ncol = 1
)
```




```
# creating a table with the stats
stat.tab <- prod.matrix %>%
    select(-prod.num) %>%
    mutate(share = num / tot) %>%
    arrange(desc(num))

library(scales)
stat.tab$share <-
    percent(stat.tab$share) # converting values to percents

# adding a table with the stats
# addtable2plot(
#     -2.5,
#     -1.5,
#     stat.tab,
#     bty = "n",
#     display.rownames = FALSE,
#     hlines = FALSE,
#     vlines = FALSE,
#     title = "The stats"
# )
```

22.5.2 Sankey Diagram

```
# loading libraries
library(googleVis)
library(dplyr)
library(reshape2)

# creating an example of orders
set.seed(15)
df <- data.frame(
    orderId = c(1:1000),
    clientId = sample(c(1:300), 1000, replace = TRUE),
    prod1 = sample(
        c('NULL', 'a'),
        1000,
        replace = TRUE,
        prob = c(0.15, 0.5)
    ),
    prod2 = sample(
        c('NULL', 'b'),
        1000,
        replace = TRUE,
        prob = c(0.15, 0.3)
    ),
    prod3 = sample(
        c('NULL', 'c'),
        1000,
        replace = TRUE,
        prob = c(0.15, 0.2)
    )
)

# combining products
df$cart <- paste(df$prod1, df$prod2, df$prod3, sep = ';')
df$cart <- gsub('NULL;|;NULL', '', df$cart)
df <- df[df$cart != 'NULL',]

df <- df %>%
    select(orderId, clientId, cart) %>%
    arrange(clientId, orderId, cart)

head(df)
```

```
##   orderId clientId  cart
## 1     181        1     b
## 2     282        1     a
```

```
## 3     748        1 a;b;c
## 4      27        2   a;b
## 5     209        2   b;c
## 6     244        2 a;b;c
```

```
orders <- df %>%
    group_by(clientId) %>%
    mutate(n.ord = paste('ord', c(1:n()), sep = '')) %>%
    ungroup()

head(orders)
```

```
## # A tibble: 6 x 4
##   orderId clientId cart  n.ord
##     <int>    <int> <chr> <chr>
## 1     181        1 b     ord1
## 2     282        1 a     ord2
## 3     748        1 a;b;c ord3
## 4      27        2 a;b   ord1
## 5     209        2 b;c   ord2
## 6     244        2 a;b;c ord3
```

```
orders <-
    dcast(orders,
          clientId ~ n.ord,
          value.var = 'cart',
          fun.aggregate = NULL)


# choose a number of carts/orders in the sequence we want to analyze
orders <- orders %>%
    select(ord1, ord2, ord3, ord4, ord5)

orders.plot <- data.frame()

for (i in 2:ncol(orders)) {
    ord.cache <- orders %>%
        group_by(orders[, i - 1], orders[, i]) %>%
        summarise(n = n()) %>%
        ungroup()

    colnames(ord.cache)[1:2] <- c('from', 'to')

    # adding tags to carts
    ord.cache$from <- paste(ord.cache$from, '(', i - 1, ')', sep = '')
    ord.cache$to <- paste(ord.cache$to, '(', i, ')', sep = '')
```

```
    orders.plot <- rbind(orders.plot, ord.cache)

}

plot(gvisSankey(
    orders.plot,
    from = 'from',
    to = 'to',
    weight = 'n',
    options = list(
        height = 900,
        width = 1800,
        sankey = "{link:{color:{fill:'lightblue'}}}"
    )
))

# The bandwidths correspond to the weight of sequence
```

22.5.3 Sequence in-depth analysis

Example by Sergey Bryl

- To understand customer's behavior and churn based on purchase sequence. For example,
    - If the customer has left or did not make the next purchase
    - Understand duration between purchases

```
library(dplyr)
library(TraMineR)
library(reshape2)
library(googleVis)

# creating an example of shopping carts
set.seed(10)
data <- data.frame(
    orderId = sample(c(1:1000), 5000, replace = TRUE),
    product = sample(
        c('NULL', 'a', 'b', 'c'), # assume we have only 3 products
        5000,
        replace = TRUE,
        prob = c(0.15, 0.65, 0.3, 0.15)
    )
)

# we also know customers' purchase order
```

```
order <- data.frame(orderId = c(1:1000),
                    clientId = sample(c(1:300), 1000, replace = TRUE))

# suppose we know customers' gender
sex <- data.frame(clientId = c(1:300),
                  sex = sample(
                      c('male', 'female'),
                      300,
                      replace = TRUE,
                      prob = c(0.40, 0.60)
                  ))


date <- data.frame(orderId = c(1:1000),
                   orderdate = sample((1:90), 1000, replace = TRUE))
orders <- merge(data, order, by = 'orderId')
orders <- merge(orders, sex, by = 'clientId')
orders <- merge(orders, date, by = 'orderId')
orders <- orders[orders$product != 'NULL',]
orders$orderdate <- as.Date(orders$orderdate, origin = "2012-01-01")
rm(data, date, order, sex)

head(orders)
```

```
##   orderId clientId product    sex  orderdate
## 1       1      204       a female 2012-02-11
## 2       1      204       a female 2012-02-11
## 3       1      204       a female 2012-02-11
## 4       1      204       a female 2012-02-11
## 5       2       71       a   male 2012-02-27
## 6       2       71       a   male 2012-02-27
```

```
# combining products to the cart (include cases where customers make 2 visits in a day)
df <- orders %>%
    arrange(product) %>%
    select(-orderId) %>%
    unique() %>%
    group_by(clientId, sex, orderdate) %>%
    summarise(cart = paste(product, collapse = ";")) %>%
    ungroup()

head(df)
```

```
## # A tibble: 6 x 4
##   clientId sex    orderdate  cart
##      <int> <chr>  <date>     <chr>
```

```
## 1        1 female 2012-02-18 a;c
## 2        1 female 2012-03-19 a;b;c
## 3        1 female 2012-03-21 a;b;c
## 4        2 female 2012-02-14 a;b
## 5        2 female 2012-02-27 a;b
## 6        2 female 2012-03-06 a
```

```
max.date <- max(df$orderdate) + 1
ids <- unique(df$clientId)
df.new <- data.frame()

for (i in 1:length(ids)) {
    df.cache <- df %>%
        filter(clientId == ids[i])

    ifelse(nrow(df.cache) == 1,
           av.dur <- 30,
           av.dur <-
               round(((
                   max(df.cache$orderdate) - min(df.cache$orderdate)
               ) / (
                   nrow(df.cache) - 1
               )) * 1.5, 0))

    df.cache <-
        rbind(
            df.cache,
            data.frame(
                clientId = df.cache$clientId[nrow(df.cache)],
                sex = df.cache$sex[nrow(df.cache)],
                orderdate = max(df.cache$orderdate) + av.dur,
                cart = 'nopurch'
            )
        )
    ifelse(max(df.cache$orderdate) > max.date,
           df.cache$orderdate[which.max(df.cache$orderdate)] <- max.date,
           NA)

    df.cache$to <- c(df.cache$orderdate[2:nrow(df.cache)] - 1, max.date)

    # order# for Sankey diagram
    df.cache <- df.cache %>%
        mutate(ord = paste('ord', c(1:nrow(df.cache)), sep = ''))

    df.new <- rbind(df.new, df.cache)
}
```

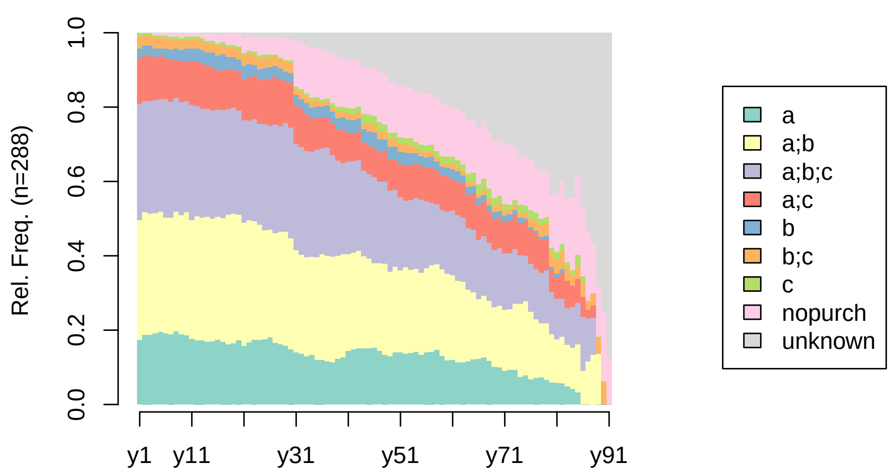
```
# filtering dummies
df.new <- df.new %>%
    filter(cart != 'nopurch' | to != orderdate)
rm(orders, df, df.cache, i, ids, max.date, av.dur)

##### Sankey diagram #######

df.sankey <- df.new %>%
    select(clientId, cart, ord)

df.sankey <-
    dcast(df.sankey,
          clientId ~ ord,
          value.var = 'cart',
          fun.aggregate = NULL)

df.sankey[is.na(df.sankey)] <- 'unknown'

# chosing a length of sequence
df.sankey <- df.sankey %>%
    select(ord1, ord2, ord3, ord4)

# replacing NAs after 'nopurch' for 'nopurch'
df.sankey[df.sankey[, 2] == 'nopurch', 3] <- 'nopurch'
df.sankey[df.sankey[, 3] == 'nopurch', 4] <- 'nopurch'

df.sankey.plot <- data.frame()
for (i in 2:ncol(df.sankey)) {
    df.sankey.cache <- df.sankey %>%
        group_by(df.sankey[, i - 1], df.sankey[, i]) %>%
        summarise(n = n()) %>%
        ungroup()

    colnames(df.sankey.cache)[1:2] <- c('from', 'to')

    # adding tags to carts
    df.sankey.cache$from <-
        paste(df.sankey.cache$from, '(', i - 1, ')', sep = '')
    df.sankey.cache$to <- paste(df.sankey.cache$to, '(', i, ')', sep = '')

    df.sankey.plot <- rbind(df.sankey.plot, df.sankey.cache)
}

plot(gvisSankey(
    df.sankey.plot,
```

```
    from = 'from',
    to = 'to',
    weight = 'n',
    options = list(
        height = 900,
        width = 1800,
        sankey = "{link:{color:{fill:'lightblue'}}}"
    )
))

rm(df.sankey, df.sankey.cache, df.sankey.plot, i)


df.new <- df.new %>%
    # chosing a length of sequence
    filter(ord %in% c('ord1', 'ord2', 'ord3', 'ord4')) %>%
    select(-ord)

# converting dates to numbers
min.date <- as.Date(min(df.new$orderdate), format = "%Y-%m-%d")
df.new$orderdate <- as.numeric(df.new$orderdate - min.date + 1)
df.new$to <- as.numeric(df.new$to - min.date + 1)

df.form <-
    seqformat(
        as.data.frame(df.new),
        id = 'clientId',
        begin = 'orderdate',
        end = 'to',
        status = 'cart',
        from = 'SPELL',
        to = 'STS',
        process = FALSE
    )

df.seq <-
    seqdef(df.form,
           left = 'DEL',
           right = 'unknown',
           xtstep = 10) # xtstep - step between ticks (days)
summary(df.seq)
```

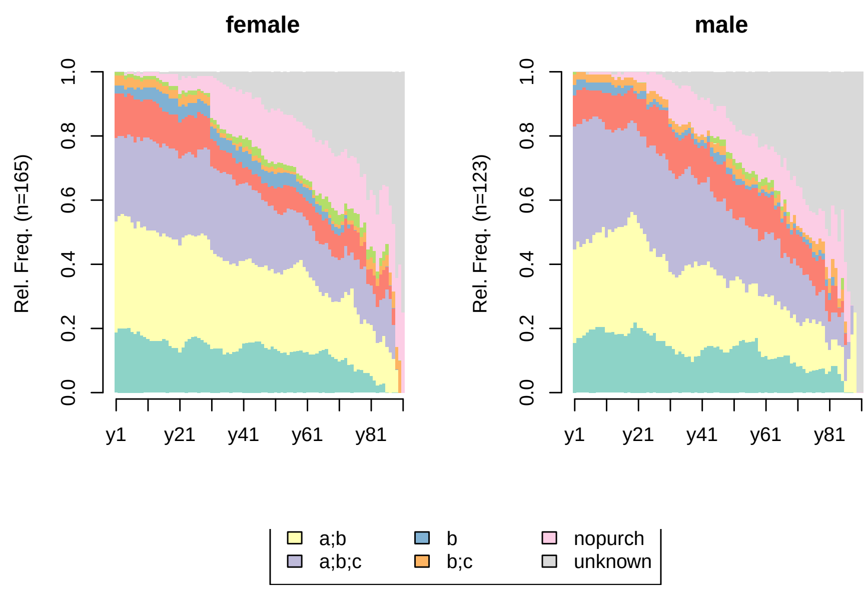
```
##  [>] sequence object created with TraMineR version 2.2-7
##  [>] 288 sequences in the data set, 286 unique
##  [>] min/max sequence length: 4/91
```

```
##   [>] alphabet (state labels):
##        1=a (a)
##        2=a;b (a;b)
##        3=a;b;c (a;b;c)
##        4=a;c (a;c)
##        5=b (b)
##        6=b;c (b;c)
##        7=c (c)
##        8=nopurch (nopurch)
##        9=unknown (unknown)
##   [>] dimensionality of the sequence space: 728
##   [>] colors: 1=#8DD3C7 2=#FFFFB3 3=#BEBADA 4=#FB8072 5=#80B1D3 6=#FDB462 7=#B3DE69 8=#FCCDE5 9
##   [>] symbol for void element: %
```

```
df.feat <- unique(df.new[, c('clientId', 'sex')])

# distribution analysis
seqdplot(df.seq, border = NA, withlegend = 'right')
```




```
seqdplot(df.seq, border = NA, group = df.feat$sex) # distribution based on gender
```




```r
seqstatd(df.seq)
```

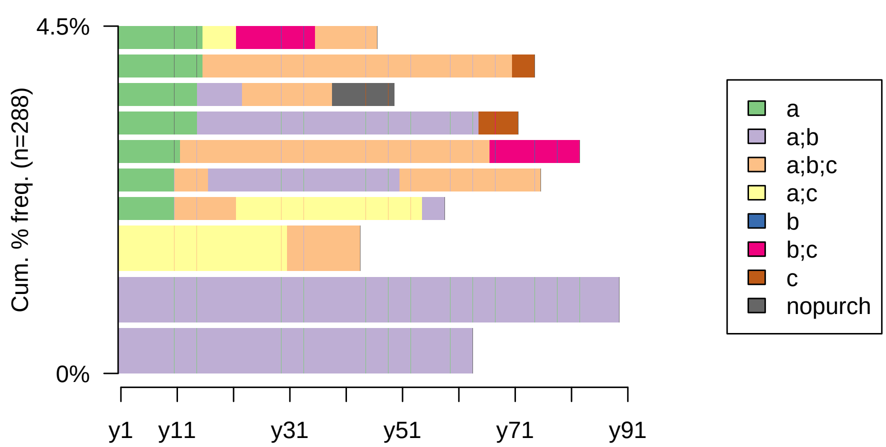
```
##                  [State frequencies]
##             y1     y2     y3     y4     y5    y6     y7    y8     y9   y10   y11
## a        0.174 0.1875 0.1875 0.1910 0.1951 0.192 0.1888 0.196 0.1895 0.187 0.176
## a;b      0.323 0.3299 0.3264 0.3229 0.3240 0.311 0.3147 0.323 0.3193 0.331 0.320
## a;b;c    0.312 0.2986 0.3021 0.3056 0.3031 0.318 0.3112 0.305 0.3053 0.299 0.310
## a;c      0.122 0.1215 0.1215 0.1146 0.1150 0.112 0.1119 0.102 0.1088 0.106 0.116
## b        0.028 0.0278 0.0278 0.0243 0.0209 0.024 0.0280 0.032 0.0316 0.035 0.035
## b;c      0.031 0.0278 0.0278 0.0278 0.0279 0.028 0.0280 0.025 0.0246 0.025 0.025
## c        0.010 0.0069 0.0069 0.0069 0.0070 0.007 0.0070 0.007 0.0070 0.007 0.007
## nopurch 0.000 0.0000 0.0000 0.0035 0.0035 0.007 0.0070 0.011 0.0105 0.011 0.011
## unknown 0.000 0.0000 0.0000 0.0035 0.0035 0.000 0.0035 0.000 0.0035 0.000 0.000
##            y12    y13    y14    y15    y16    y17    y18    y19    y20    y21
## a        0.173 0.1725 0.1696 0.1702 0.1744 0.1685 0.1619 0.1655 0.1727 0.1583
## a;b      0.335 0.3310 0.3357 0.3298 0.3310 0.3333 0.3489 0.3489 0.3381 0.3309
## a;b;c    0.296 0.2923 0.2898 0.2908 0.2883 0.2903 0.2842 0.2842 0.2806 0.2770
## a;c      0.120 0.1197 0.1166 0.1170 0.1032 0.1075 0.1043 0.1043 0.1043 0.1079
## b        0.035 0.0387 0.0353 0.0390 0.0427 0.0430 0.0360 0.0324 0.0324 0.0360
## b;c      0.025 0.0246 0.0247 0.0248 0.0249 0.0251 0.0252 0.0252 0.0288 0.0288
## c        0.007 0.0070 0.0071 0.0071 0.0071 0.0072 0.0072 0.0072 0.0072 0.0072
## nopurch 0.011 0.0106 0.0177 0.0177 0.0214 0.0215 0.0288 0.0288 0.0324 0.0396
## unknown 0.000 0.0035 0.0035 0.0035 0.0071 0.0036 0.0036 0.0036 0.0036 0.0144
##             y22    y23    y24    y25    y26    y27    y28    y29    y30    y31
## a        0.1667 0.1739 0.1745 0.1758 0.1801 0.1661 0.1624 0.1587 0.1481 0.1407
```

```
## a;b     0.3297 0.3188 0.3091 0.2930 0.2904 0.2952 0.3026 0.3063 0.3000 0.2741
## a;b;c   0.2681 0.2754 0.2727 0.2857 0.2794 0.2915 0.2841 0.2915 0.2963 0.2852
## a;c     0.1159 0.1159 0.1164 0.1209 0.1250 0.1292 0.1255 0.1144 0.1222 0.1074
## b       0.0362 0.0290 0.0291 0.0293 0.0331 0.0295 0.0295 0.0258 0.0259 0.0259
## b;c     0.0290 0.0290 0.0291 0.0293 0.0257 0.0258 0.0258 0.0258 0.0259 0.0148
## c       0.0072 0.0072 0.0073 0.0073 0.0074 0.0037 0.0037 0.0037 0.0074 0.0074
## nopurch 0.0399 0.0399 0.0509 0.0476 0.0478 0.0480 0.0554 0.0590 0.0556 0.1259
## unknown 0.0072 0.0109 0.0109 0.0110 0.0110 0.0111 0.0111 0.0148 0.0185 0.0185
##            y32    y33    y34    y35    y36    y37   y38   y39   y40   y41   y42
## a       0.1370 0.1296 0.1338 0.1199 0.1199 0.1170 0.114 0.122 0.127 0.144 0.149
## a;b     0.2667 0.2667 0.2639 0.2772 0.2846 0.2830 0.284 0.279 0.277 0.261 0.259
## a;b;c   0.2889 0.2852 0.2825 0.2884 0.2846 0.2906 0.273 0.256 0.246 0.249 0.247
## a;c     0.1037 0.1000 0.0892 0.0861 0.0824 0.0830 0.087 0.084 0.088 0.078 0.075
## b       0.0259 0.0296 0.0297 0.0300 0.0300 0.0302 0.030 0.031 0.035 0.035 0.035
## b;c     0.0185 0.0185 0.0186 0.0187 0.0112 0.0113 0.011 0.019 0.015 0.016 0.016
## c       0.0074 0.0074 0.0074 0.0075 0.0075 0.0075 0.011 0.011 0.012 0.016 0.016
## nopurch 0.1259 0.1296 0.1338 0.1311 0.1348 0.1283 0.136 0.134 0.131 0.128 0.129
## unknown 0.0259 0.0333 0.0409 0.0412 0.0449 0.0491 0.053 0.065 0.069 0.074 0.075
##           y43   y44   y45   y46   y47   y48   y49   y50   y51   y52   y53   y54
## a       0.151 0.151 0.151 0.153 0.145 0.133 0.130 0.140 0.140 0.137 0.140 0.142
## a;b     0.263 0.247 0.241 0.227 0.236 0.246 0.227 0.230 0.221 0.232 0.223 0.222
## a;b;c   0.243 0.231 0.229 0.231 0.219 0.221 0.218 0.209 0.196 0.180 0.188 0.191
## a;c     0.076 0.076 0.078 0.079 0.079 0.083 0.084 0.081 0.089 0.094 0.092 0.093
## b       0.036 0.036 0.037 0.041 0.037 0.029 0.034 0.034 0.034 0.034 0.035 0.027
## b;c     0.016 0.016 0.020 0.021 0.021 0.021 0.021 0.021 0.021 0.021 0.022 0.013
## c       0.016 0.024 0.024 0.025 0.025 0.021 0.017 0.017 0.017 0.017 0.017 0.018
## nopurch 0.127 0.127 0.122 0.132 0.136 0.133 0.134 0.136 0.140 0.137 0.135 0.133
## unknown 0.072 0.092 0.098 0.091 0.103 0.112 0.134 0.132 0.140 0.146 0.148 0.160
##           y55   y56   y57    y58    y59    y60   y61    y62    y63   y64   y65
## a       0.135 0.140 0.142 0.1475 0.1308 0.1190 0.121 0.1127 0.1133 0.116 0.122
## a;b     0.220 0.226 0.233 0.2304 0.2336 0.2333 0.227 0.2206 0.2167 0.192 0.180
## a;b;c   0.197 0.181 0.169 0.1613 0.1589 0.1667 0.174 0.1765 0.1724 0.167 0.169
## a;c     0.090 0.090 0.096 0.0922 0.0935 0.0952 0.082 0.0882 0.0936 0.091 0.095
## b       0.027 0.027 0.027 0.0230 0.0234 0.0238 0.029 0.0294 0.0197 0.020 0.021
## b;c     0.013 0.014 0.014 0.0092 0.0093 0.0095 0.014 0.0098 0.0099 0.015 0.011
## c       0.018 0.018 0.018 0.0184 0.0187 0.0190 0.019 0.0196 0.0197 0.020 0.026
## nopurch 0.139 0.140 0.137 0.1429 0.1402 0.1381 0.135 0.1373 0.1379 0.146 0.143
## unknown 0.161 0.163 0.164 0.1751 0.1916 0.1952 0.198 0.2059 0.2167 0.232 0.233
##           y66   y67   y68   y69   y70   y71   y72   y73   y74    y75    y76
## a       0.123 0.127 0.117 0.102 0.101 0.091 0.094 0.093 0.075 0.0786 0.0682
## a;b     0.160 0.166 0.162 0.159 0.166 0.164 0.163 0.179 0.197 0.2000 0.1818
## a;b;c   0.160 0.160 0.156 0.153 0.154 0.152 0.150 0.146 0.129 0.1214 0.1288
## a;c     0.091 0.094 0.084 0.085 0.083 0.085 0.087 0.093 0.088 0.0929 0.0909
## b       0.021 0.017 0.017 0.017 0.018 0.018 0.019 0.013 0.014 0.0071 0.0076
## b;c     0.011 0.017 0.017 0.017 0.018 0.012 0.012 0.013 0.014 0.0143 0.0227
## c       0.027 0.028 0.028 0.023 0.024 0.018 0.012 0.013 0.020 0.0214 0.0227
```

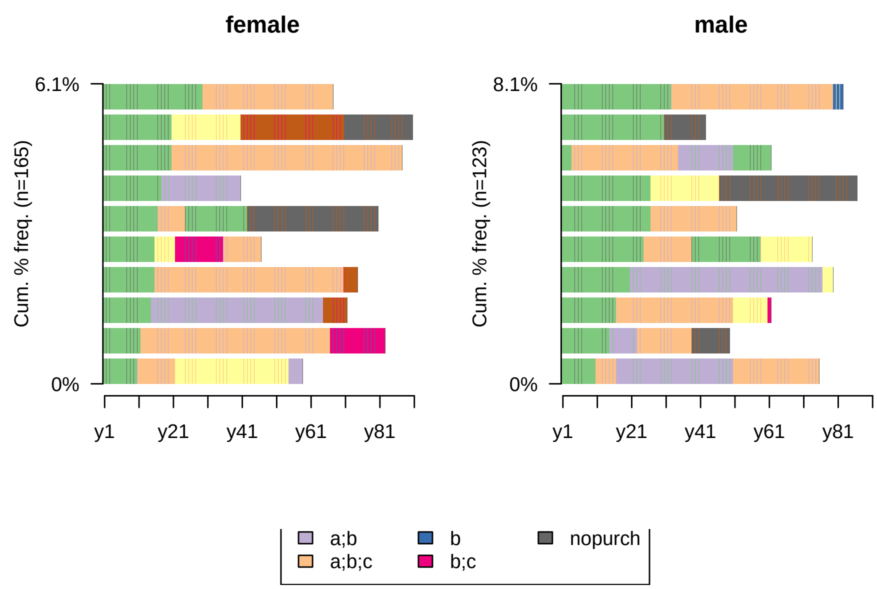
```
## nopurch 0.150 0.155 0.151 0.153 0.148 0.158 0.163 0.139 0.129 0.1286 0.1364
## unknown 0.257 0.238 0.268 0.290 0.290 0.303 0.300 0.311 0.333 0.3357 0.3409
##            y77    y78    y79    y80    y81   y82   y83   y84   y85   y86   y87
## a       0.0714 0.0726 0.0672 0.0603 0.0588 0.057 0.049 0.042 0.032 0.000 0.000
## a;b     0.1587 0.1452 0.1513 0.1293 0.1176 0.125 0.111 0.111 0.129 0.091 0.116
## a;b;c   0.1349 0.1371 0.1429 0.1121 0.1078 0.102 0.099 0.111 0.113 0.145 0.116
## a;c     0.0952 0.0887 0.0840 0.0603 0.0588 0.068 0.074 0.056 0.065 0.055 0.023
## b       0.0079 0.0081 0.0084 0.0086 0.0098 0.011 0.000 0.000 0.000 0.000 0.000
## b;c     0.0238 0.0323 0.0336 0.0345 0.0392 0.045 0.037 0.028 0.032 0.036 0.023
## c       0.0238 0.0161 0.0168 0.0172 0.0196 0.023 0.012 0.014 0.032 0.018 0.000
## nopurch 0.1270 0.1290 0.1261 0.1379 0.1569 0.170 0.173 0.194 0.210 0.182 0.186
## unknown 0.3571 0.3710 0.3697 0.4397 0.4314 0.398 0.444 0.444 0.387 0.473 0.535
##           y88   y89   y90  y91
## a       0.000 0.000 0.000 0.00
## a;b     0.133 0.136 0.000 0.00
## a;b;c   0.100 0.000 0.000 0.00
## a;c     0.033 0.000 0.000 0.00
## b       0.000 0.000 0.000 0.00
## b;c     0.033 0.045 0.062 0.00
## c       0.000 0.000 0.000 0.00
## nopurch 0.133 0.136 0.188 0.12
## unknown 0.567 0.682 0.750 0.88
##
##               [Valid states]
##          y1  y2  y3  y4  y5  y6  y7  y8  y9 y10 y11 y12 y13 y14 y15 y16 y17 y18
## N       288 288 288 288 287 286 286 285 285 284 284 284 284 283 282 281 279 278
##         y19 y20 y21 y22 y23 y24 y25 y26 y27 y28 y29 y30 y31 y32 y33 y34 y35 y36
## N       278 278 278 276 276 275 273 272 271 271 271 270 270 270 270 269 267 267
##         y37 y38 y39 y40 y41 y42 y43 y44 y45 y46 y47 y48 y49 y50 y51 y52 y53 y54
## N       265 264 262 260 257 255 251 251 245 242 242 240 238 235 235 233 229 225
##         y55 y56 y57 y58 y59 y60 y61 y62 y63 y64 y65 y66 y67 y68 y69 y70 y71 y72
## N       223 221 219 217 214 210 207 204 203 198 189 187 181 179 176 169 165 160
##         y73 y74 y75 y76 y77 y78 y79 y80 y81 y82 y83 y84 y85 y86 y87 y88 y89 y90
## N       151 147 140 132 126 124 119 116 102  88  81  72  62  55  43  30  22  16
##         y91
## N         8
##
##               [Entropy index]
##          y1  y2  y3   y4   y5   y6   y7   y8   y9  y10  y11  y12  y13  y14  y15
## H       0.7 0.7 0.7 0.71 0.71 0.71 0.72 0.71 0.72 0.71 0.72 0.72 0.73 0.73 0.74
##          y16  y17  y18  y19  y20  y21  y22  y23  y24  y25  y26  y27  y28  y29
## H       0.75 0.74 0.74 0.74 0.75 0.77 0.77 0.77 0.78 0.78 0.78 0.77 0.78 0.77
##          y30 y31  y32  y33  y34  y35  y36  y37  y38  y39  y40  y41  y42  y43
## H       0.78 0.8 0.81 0.82 0.82 0.82 0.81 0.81 0.82 0.84 0.85 0.85 0.86 0.86
##          y44  y45  y46  y47  y48  y49  y50  y51  y52 y53  y54  y55  y56  y57
## H       0.87 0.88 0.89 0.89 0.88 0.89 0.89 0.89 0.89 0.9 0.88 0.88 0.88 0.88
```

```
##          y58  y59  y60  y61  y62  y63  y64  y65  y66  y67  y68  y69  y70  y71
## H       0.87 0.87 0.87 0.88 0.88 0.87 0.88 0.88 0.88 0.88 0.87 0.86 0.86 0.85
##          y72  y73  y74  y75  y76  y77  y78  y79  y80  y81  y82  y83  y84  y85
## H       0.84 0.84 0.83 0.82 0.83 0.83 0.82 0.82 0.78 0.79 0.81 0.75 0.74 0.78
##          y86 y87 y88  y89  y90  y91
## H       0.69 0.6 0.6 0.43 0.32 0.17
```

```r
df.seq <- seqdef(df.form,
                 left = 'DEL',
                 right = 'DEL',
                 xtstep = 10)


# the 10 most frequent sequences
seqfplot(df.seq, border = NA, withlegend = 'right')
```




```r
# the 10 most frequent sequences based on gender
seqfplot(df.seq, group = df.feat$sex, border = NA)
```




```r
# returning the frequency stats
seqtab(df.seq) # frequency table
```

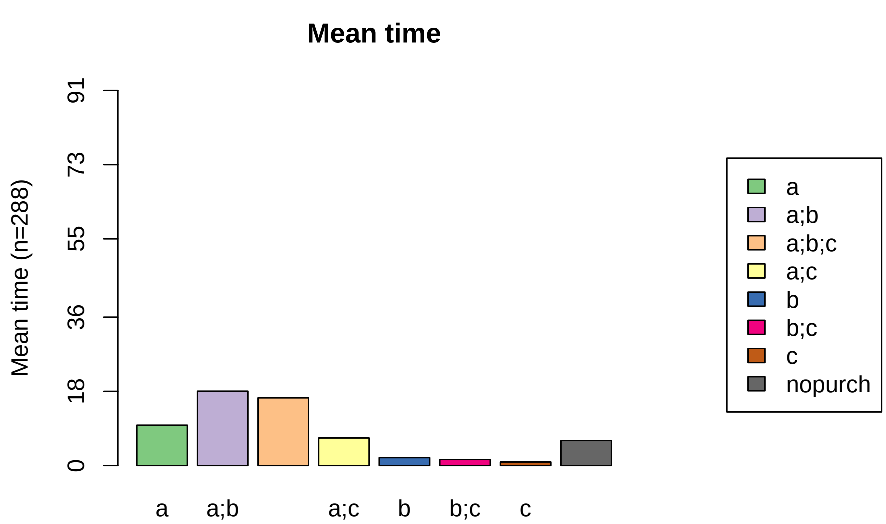
```
##                                        Freq Percent
## a;b/63                                    2    0.69
## a;b/89                                    2    0.69
## a;c/30-a;b;c/13                           2    0.69
## a/10-a;b;c/11-a;c/33-a;b/4                1    0.35
## a/10-a;b;c/6-a;b/34-a;b;c/25              1    0.35
## a/11-a;b;c/55-b;c/16                      1    0.35
## a/14-a;b/50-c/7                           1    0.35
## a/14-a;b/8-a;b;c/16-nopurch/11            1    0.35
## a/15-a;b;c/55-c/4                         1    0.35
## a/15-a;c/6-b;c/14-a;b;c/11                1    0.35
```

```r
seqtab(df.seq[, 1:30]) # frequency table for 1st month
```

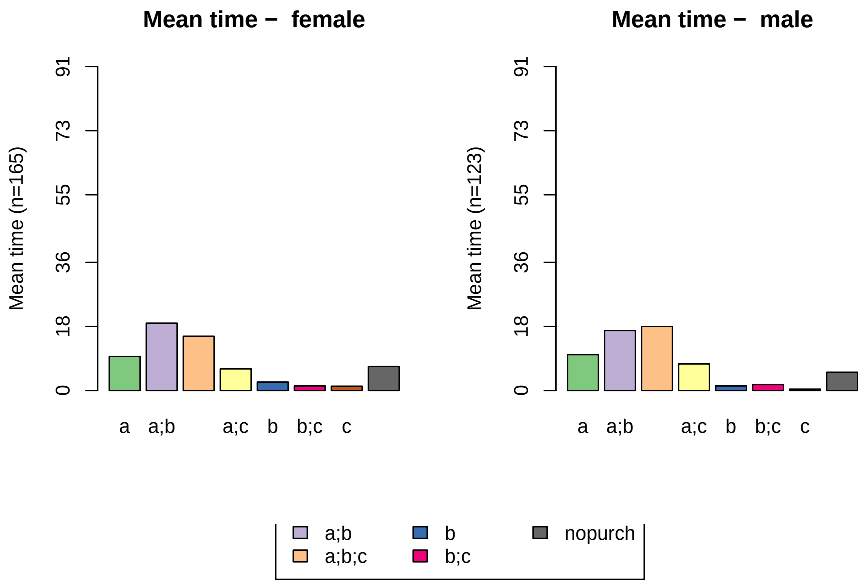
```
##                  Freq Percent
## a;b;c/30           41   14.24
## a;b/30             40   13.89
## a/30               24    8.33
## a;c/30             15    5.21
## b;c/30              4    1.39
## a;b/1-a;b;c/29      2    0.69
## a;b;c/28-a;b/2      2    0.69
## a;c/15-a;b/15       2    0.69
```

```
## a;c/7-a;b/23      2    0.69
## b/30              2    0.69
```

```
# mean time spent in each state
seqmtplot(df.seq, title = 'Mean time', withlegend = 'right')
```



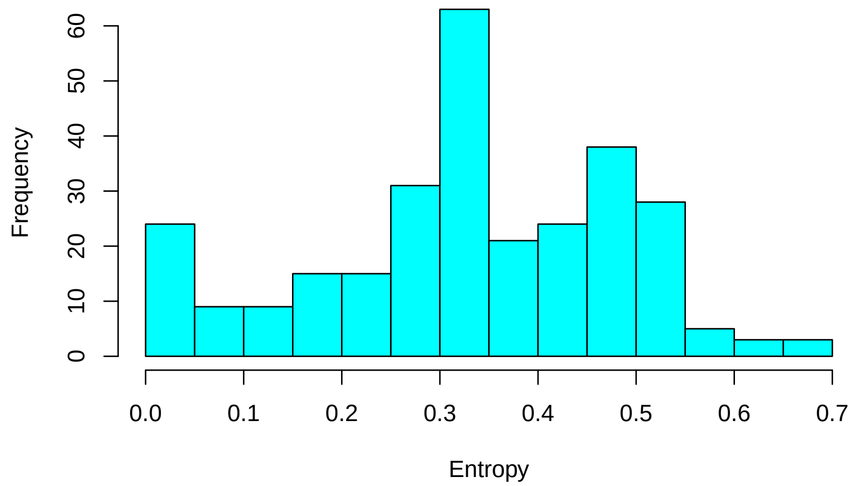

```
seqmtplot(df.seq, group = df.feat$sex, title = 'Mean time')
```



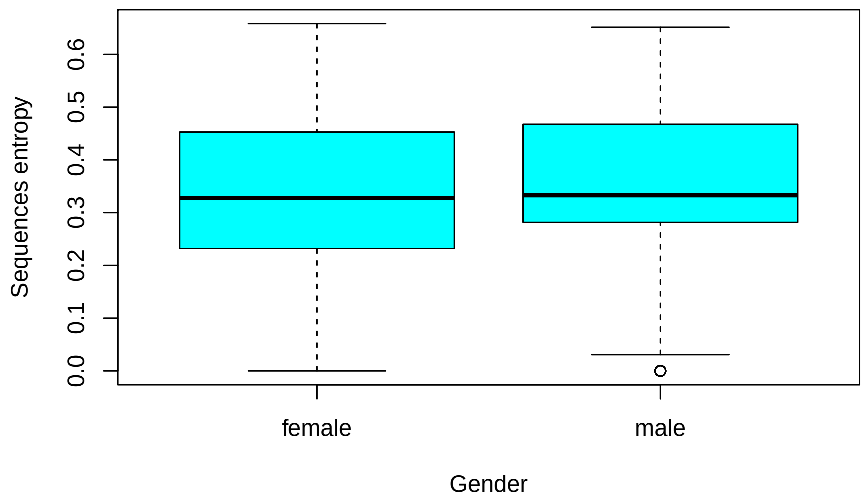

```
statd <-
    seqistatd(df.seq) #function returns for each sequence the time spent in the differe
apply(statd, 2, mean) #We may be interested in the mean time spent in each state
```

```
##          a        a;b      a;b;c        a;c          b        b;c          c
##  9.7916667 18.0416667 16.4201389  6.6840278  1.9236111  1.4583333  0.8611111
##    nopurch
##  6.0694444
```

```
# calculating entropy
df.ient <- seqient(df.seq)
hist(df.ient,
     col = 'cyan',
     main = NULL,
     xlab = 'Entropy') # plot an histogram of the within entropy of the sequences
```




```
# entrophy distribution based on gender
df.ent <- cbind(df.seq, df.ient)
boxplot(
    Entropy ~ df.feat$sex,
    data = df.ent,
    xlab = 'Gender',
    ylab = 'Sequences entropy',
    col = 'cyan'
)
```



22.6 Geodemographic Classification

Example by Nick Bearman

```
#Load libraries
library(scales)
library(ggplot2)


#set up data and data frame
oac_names <-
    c(
        "Blue Collar Communities",
        "City Living",
        "Countryside",
        "Prospering Suburbs",
        "Constrained by Circumstances",
        "Typical Traits",
        "Multicultural"
    )

broadsheets <- c(73.2, 144, 103.9, 109.1, 78.2, 97.1, 120.2)
oac_broadsheets <- data.frame(oac_names, broadsheets)
#convert the percentage values (e.g. 144%) to decimal increase or decrease (e.g. 0.44)
```

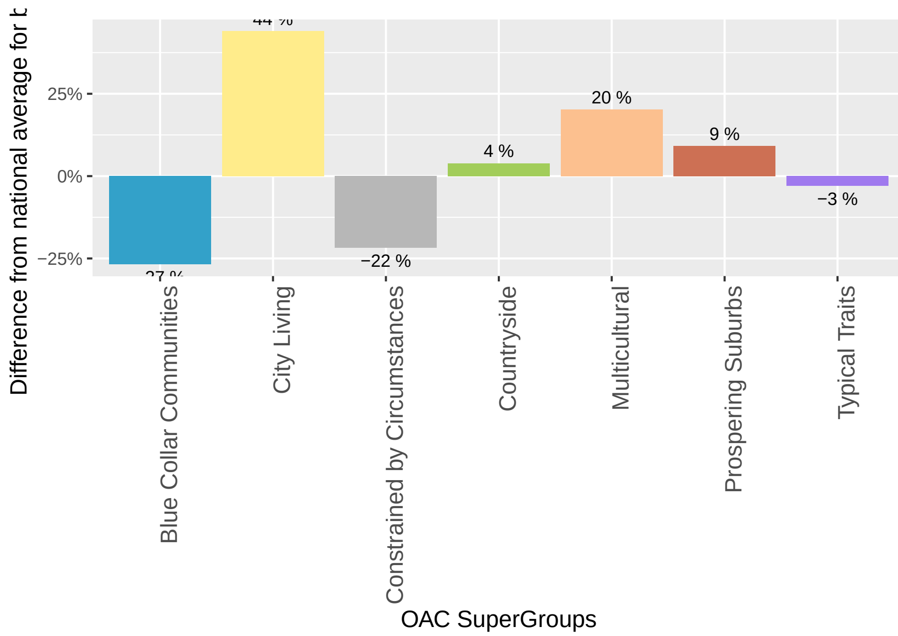
```
oac_broadsheets$broadsheets <-  broadsheets / 100 - 1

oac_broadsheets
```

```
##                      oac_names broadsheets
## 1      Blue Collar Communities      -0.268
## 2                 City Living       0.440
## 3                 Countryside       0.039
## 4           Prospering Suburbs       0.091
## 5 Constrained by Circumstances      -0.218
## 6               Typical Traits      -0.029
## 7                 Multicultural       0.202
```

plot each group's percentage difference from the mean

```
#select the colours we are going to use
my_colour <-
    c("#33A1C9",
      "#FFEC8B",
      "#A2CD5A",
      "#CD7054",
      "#B7B7B7",
      "#9F79EE",
      "#FCC08F")

#plot the graph - this has several bits to it
#the first three lines setup the data and type of graph
ggplot(oac_broadsheets, aes(oac_names, broadsheets)) +
    geom_bar(stat = "identity",
             fill = my_colour,
             position = "identity") +
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = 1,
        vjust = 1,
        size = 12
    )) +
    #this line add the lables to each bar
    geom_text(aes(
        label = paste(round(broadsheets * 100, digits = 0), "%"),
        vjust = ifelse(broadsheets >= 0,-0.5, 1.5)
    ), size = 3) +
    #these lines as the axis labels and these fonts
    theme(axis.title.x = element_text(size = 12)) +
    theme(axis.title.y = element_text(size = 12)) +
    scale_y_continuous("Difference from national average for broadsheet", labels = percent_format
```

```
    scale_x_discrete("OAC SuperGroups")
```



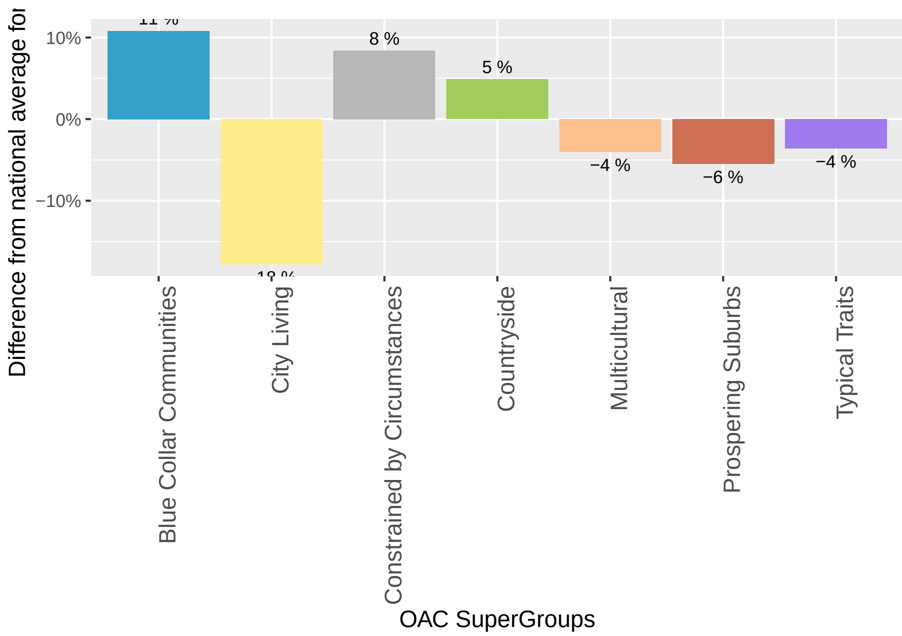
Create plot for another variable

```
tabloids <- c(110.8, 82.2, 104.9, 94.5, 108.4, 96.4, 96.0)
oac_tabloids <- data.frame(oac_names, tabloids)

#convert the percentage values (e.g. 144%) to decimal increase or decrease (e.g. 0.44)
oac_tabloids$tabloids <-  tabloids / 100 - 1

oac_tabloids
```

```
##                      oac_names tabloids
## 1      Blue Collar Communities    0.108
## 2                  City Living   -0.178
## 3                  Countryside    0.049
## 4           Prospering Suburbs   -0.055
## 5 Constrained by Circumstances    0.084
## 6               Typical Traits   -0.036
## 7                 Multicultural   -0.040
```

```
# plot
ggplot(oac_tabloids, aes(oac_names, tabloids)) +
    geom_bar(stat = "identity",
             fill = my_colour,
             position = "identity") +
```

```
    theme(axis.text.x = element_text(
        angle = 90,
        hjust = 1,
        vjust = 1,
        size = 12
    )) +
    #this line add the lables to each bar
    geom_text(aes(
        label = paste(round(tabloids * 100, digits = 0), "%"),
        vjust = ifelse(tabloids >= 0, -0.5, 1.5)
    ), size = 3) +
    #these lines as the axis labels and these fonts
    theme(axis.title.x = element_text(size = 12)) +
    theme(axis.title.y = element_text(size = 12)) +
    scale_y_continuous("Difference from national average for tabloids", labels = percent_format()
    scale_x_discrete("OAC SuperGroups")
```

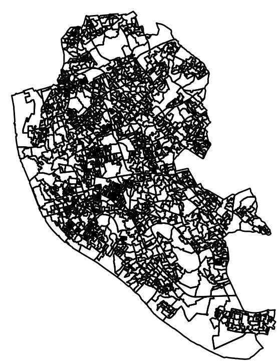


To visualize how these two variables are distributed in a location such as Liverpool in this case.

We load the shapefile

```
#load library
library(maptools)
```

```r
#download file
# download.file("https://raw.githubusercontent.com/nickbearman/r-geodemographic-analys

#unzip file
unzip("images/liverpool_OA.zip")


#read in shapefile
liverpool <- readShapeSpatial('liverpool_OA/liverpool', proj4string = CRS("+init=epsg:2

plot(liverpool)
```



The dataset can be attained here select 2011 OAC Clusters and Names csv (1.1 Mb ZIP)

```r
library(tidyverse)

#read in OAC by OA csv file
OAC <- rio::import("images/2011oacclustersandnamescsvv2.zip") %>%
    select(
        "Output Area Code",
        "Supergroup Name",
        "Supergroup Code",
        "Group Name",
        "Group Code",
```

```
        "Subgroup Name",
        "Subgroup Code"
    ) %>%
    rename("OA_CODE" = "Output Area Code")
```

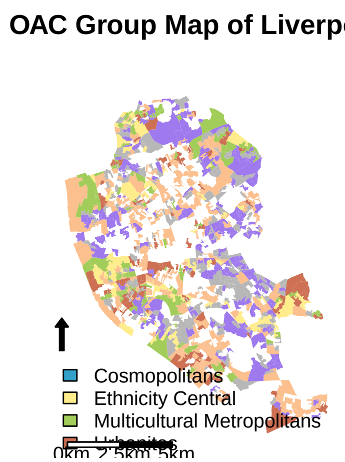
Merge data (OAC) to its location (liverpool)

```
# check dataset
# head(liverpool@data)
# head(OAC)

#Join OAC classification on to LSOA shapefile
liverpool@data = data.frame(liverpool@data, OAC[match(liverpool@data[, "OA01CD"], OAC[, "OA_CODE'
#Show head of liverpool
head(liverpool@data)
```

```
##      OA01CD  OA01CDOLD   OA_CODE   Supergroup.Name Supergroup.Code
## 2 E00032987 00BYFA0003 E00032987 Ethnicity Central               3
## 3 E00032988 00BYFA0004 E00032988     Cosmopolitans               2
## 4 E00032989 00BYFA0005 E00032989 Ethnicity Central               3
## 5 E00032990 00BYFA0006 E00032990 Ethnicity Central               3
## 6 E00032991 00BYFA0007 E00032991 Ethnicity Central               3
## 7 E00032992 00BYFA0008 E00032992 Ethnicity Central               3
##                Group.Name Group.Code                           Subgroup.Name
## 2      Ethnic Family Life         3a            Established Renting Families
## 3   Students Around Campus        2a              Students and Professionals
## 4 Endeavouring Ethnic Mix         3b Multi-Ethnic Professional Service Workers
## 5    Aspirational Techies         3d                     Old EU Tech Workers
## 6 Endeavouring Ethnic Mix         3b Multi-Ethnic Professional Service Workers
## 7 Endeavouring Ethnic Mix         3b                 Striving Service Workers
##   Subgroup.Code
## 2           3a1
## 3           2a3
## 4           3b3
## 5           3d3
## 6           3b3
## 7           3b1
```

```
#Define a set of colours, one for each of the OAC supergroups
my_colour <-
    c("#33A1C9",
      "#FFEC8B",
      "#A2CD5A",
      "#CD7054",
      "#B7B7B7",
      "#9F79EE",
      "#FCC08F")
```

```
#Create a basic OAC choropleth map
plot(liverpool,
     col = my_colour[liverpool@data$Supergroup.Code],
     axes = FALSE,
     border = NA)


#Name the groups we've used
oac_names <-
    liverpool@data %>% select(Supergroup.Name, Supergroup.Code) %>% unique() %>% arrang


#Add the legend (the oac_names object was created earlier)
legend(
    x = 332210,
    y = 385752,
    legend = oac_names,
    fill = my_colour,
    bty = "n",
    cex = .8,
    ncol = 1
)
#Add North Arrow
SpatialPolygonsRescale(
    layout.north.arrow(2),
    offset = c(332610, 385852),
    scale = 1600,
    plot.grid = F
)
#Add Scale Bar
SpatialPolygonsRescale(
    layout.scale.bar(),
    offset = c(333210, 381252),
    scale = 5000,
    fill = c("white", "black"),
    plot.grid = F
)
#Add text to scale bar
text(333410, 380952, "0km", cex = .8)
text(333410 + 2500, 380952, "2.5km", cex = .8)
text(333410 + 5000, 380952, "5km", cex = .8)
#Add a title
title("OAC Group Map of Liverpool")
```


OAC Group Map of Liverpool
Cosmopolitans
Ethnicity Central
Multicultural Metropolitans

Chapter 23

Model Building

$$U_A = V - P_A - t|x - a| U_B = V - P_B - t|b - x|$$

where

- $\bar{x}$ = marginal consumer = indifferent consumer
- $-t|x - a|$ = Disutility of consuming a product away from ideal
  - transportation cost parameters, weights on the disuility incur when they purchase a product different from the ideal point.
- V = reservation price = max max that the customer is willing to take
- x = location of marginal customer - customer who is indifference from buying from firm A and firm B.

To find the point that a consumer is indifference from buying from firm A and buying from firm B, we equate $U_A = U_B$

$$V - P_A - t(\bar{x} - a) = V - P_B - t(b - \bar{x}) P_B - P_A = 2t\bar{x} + at - tb\bar{x} = \frac{b - a}{2} + \frac{P_B - P_A}{2t}$$

Market Share for firm A = $\bar{x}$

Market Share for firm B = $1 - \bar{x}$

$$\pi_A = P_A(\frac{b - a}{2} + \frac{P_B - P_A}{2t}) \pi_B = P_B(1 - \frac{b - a}{2} - \frac{P_B - P_A}{2t})$$

$$\frac{d\pi_A}{dP_A} = \frac{p\pi_B}{dP_B} = 0$$

$$\frac{P_B}{2t} - \frac{2P_A}{2t} + \frac{b-a}{2} = 0$$

and

$$\frac{d^2\pi_A}{dP_A^2} = \frac{1}{t} < 0$$

$$1 - \frac{b-a}{2} - \frac{P_B}{t} + \frac{P_A}{2t} = 0 1/2 + \frac{b-a}{4} - \frac{3P_A}{4t} = 0 \frac{t(2+b-a)}{3} = P_A \frac{1}{2} - \frac{b-a}{4} + \frac{2+b-a}{12} = \frac{P_B}{2t} \frac{4/3}{-} \frac{b-a}{3} = \frac{P}{-}$$

Chapter 24

Structural Models

Structural modeling in marketing refers to the use of statistical models to depict the relationships between different variables and factors in a marketing context. The goal is to understand how certain factors influence others, allowing for more accurate prediction of outcomes and more effective decision making.

Structural models can be quite complex, incorporating many variables and accounting for different potential interactions. These might include customer behaviors, market trends, pricing strategies, promotional effects, and many other factors.

Examples of structural modeling papers in marketing:

- (Misra and Nair, 2011): This paper uses structural modeling to understand the complex interactions between sales force compensation, motivation, and performance. The authors create a model that takes into account various factors including incentive levels, job difficulty, and salesperson effort.
- (Song and Chintagunta, 2003a)
- (Goh et al., 2013)
- (Chintagunta et al., 2006) This paper reviews the field of structural modeling in marketing, highlighting its strengths and weaknesses and discussing how it has evolved over time. The authors discuss a range of models and their applications, providing a comprehensive overview of the field.

24.1 Top Seminal Papers

- (Allenby et al., 1999) This paper uses a structural model to understand and predict consumer purchase timing in direct marketing scenarios.

- (Misra and Nair, 2011): This work models the dynamics of sales force compensation, motivation, and performance, providing valuable insights into sales team management.
- (Montgomery and Rossi, 1999): This paper focuses on using structural models to estimate price elasticities, offering insights into pricing strategies.
- (Rutz et al., 2011): This paper uses structural modeling to understand the indirect effects of paid search advertising, with implications for online marketing strategies.
- (Netzer et al., 2008): This paper presents a structural model for understanding the dynamic nature of customer relationships over time.
- (Goh et al., 2013): This paper models the co-evolution of user behavior in social media, enabling a better understanding of social media trends.
- (Gowrisankaran and Rysman, 2012): This work models how consumers' preferences evolve over time, with applications to durable goods markets.
- (Kadiyali et al., 2001): This work presents methods for using structural modeling to understand competitive behavior in markets.
- (Heiss, 2002): This paper shows how to use nested logit models, a type of structural model, to analyze consumer choice behavior.
- (Elberg et al., 2019): This paper investigates the dynamic effects of price promotions using a structural model.
- (Erdem et al., 2008): This paper uses a structural model to understand how advertising influences consumer price sensitivity in experience goods markets.
- (Kim et al., 2022): This work models how a multitasking salesforce operates, providing insights into salesforce compensation and customer relationship management.
- (Hitsch, 2006): This paper models how firms should optimally launch and exit products under demand uncertainty.
- (Dubé et al., 2002): This paper provides insights into how discrete choice models, a type of structural model, can be applied in a variety of marketing contexts.
- (Kadiyali et al., 2001) This work presents methods for using structural modeling to understand competitive behavior in markets.
- (Kamakura et al., 2005): This work presents a choice model for understanding customer relationship management strategies.

- (Gupta et al., 2006): This paper uses a structural model to understand and predict customer lifetime value, with implications for customer relationship management.

24.2 To get started in this area

24.2.1 Books

- (Greene, 2003): This book covers a range of econometric methods, including many that are relevant for structural modeling. It's a classic text in the field of econometrics.
- (Cameron and Trivedi, 2005): This book goes in-depth into econometric methods used in micro-level data analysis – these are very important for structural modeling in marketing.
- (Hensher et al., 2005) A lot of structural modeling in marketing has to do with modeling consumer choice, and this book gives a thorough overview of different choice modeling techniques.
- (Diamantopoulos et al., 2013): It provides insights into modern applications of quantitative models in marketing
- (Heckman and Vytlacil, 2007a, Heckman and Vytlacil (2007b), Abbring and Heckman (2007))

24.3 Structural modeling and Causal Inference

Structural modeling and causal inference are both valuable tools in economics and social sciences. In a nutshell, both methods try to understand relationships between variables; however, the objectives and methodologies can differ.

- Causal inference focuses on the identification and estimation of causal relationships from observational data. It employs strategies like randomized controlled trials, natural experiments, matching, instrumental variables, difference-in-differences, regression discontinuity, etc., to estimate the causal effect of a treatment on an outcome, while trying to control for confounding.
- On the other hand, structural modeling refers to the practice of using economic theory to guide the specification of statistical models. Structural models explicitly model the decision-making process of agents (consumers, firms, etc.), often taking into account optimization behavior and equilibrium conditions.

While causal inference mainly focuses on "reduced-form" relationships (i.e., direct associations between variables, without necessarily modeling the underlying process), structural modeling aims to uncover the "deep parameters" of the underlying process that generates the data, which represent preferences, technologies, or strategic interactions.

As such, structural models have more demanding data requirements and often require stronger assumptions. However, they can be more flexible in extrapolating beyond the observed data (i.e., for policy analysis or prediction), because they're designed to model the underlying process that generates the data. In other words, while causal inference asks "what is the effect of X on Y?", structural modeling often asks "how does the system work?"

To transition from causal inference to structural modeling, it might be helpful to focus on these aspects:

- Learning more about optimization theory and game theory: These are the foundations of a lot of structural models.
- Understanding how to estimate structural models: This usually involves techniques like maximum likelihood estimation or generalized method of moments, which are more complex than the regression-based methods often used in causal inference.
- Studying some of the seminal papers in structural modeling (like the ones listed above), to see how they specify and estimate their models.
- Practicing with simple structural models, such as the linear demand and supply model, before moving on to more complex models.
- Understanding the strengths and weaknesses of structural modeling as compared to causal inference. For example, structural models often require stronger assumptions, but they allow for counterfactual analysis and policy simulations.

It's also worth noting that the two methods can be complementary. For instance, results from causal inference can be used to test or validate a structural model, and a structural model can be used to guide the search for causal relationships. So, having a background in causal inference can be a big advantage as you're learning about structural modeling.

1. **Simple Models (Linear Demand and Supply):**

   A basic model of linear demand and supply involves modeling how quantity demanded and supplied depend on price. For instance, the demand function might be Qd = a - bP, where Qd is the quantity demanded, P is the price, and a and b are parameters to be estimated. Similarly, the supply function might be Qs = c + dP, where Qs is the quantity supplied, and c and d are parameters to be estimated. By solving these two equations, we can find the equilibrium price and quantity. This model is simple, but it forms the basis for more complex structural models.

2. **Strong Assumptions in Structural Models:**

   Structural models often involve assumptions about:

- The functional form of relationships between variables. For example, is demand linear or non-linear in price?
- The decision-making process of agents. For instance, do consumers always buy the product that gives them the highest utility?
- The information available to agents. Do consumers know everything about all products when making a choice, or do they face uncertainty?
- Equilibrium conditions. For example, in a market model, we might assume that the market always clears (demand equals supply).

These assumptions are often necessary to make the model tractable and to allow for estimation, but they can also be a source of bias if they're incorrect.

3. **Counterfactual Analysis and Policy Simulations:**

   Counterfactual analysis involves asking "what if" questions about scenarios that did not actually occur. For example, "what would have happened to sales if we had set a different price?" Policy simulations involve asking similar questions about potential future policies. For example, "how would a change in our pricing strategy affect future sales?"

   An example: (Gowrisankaran and Rysman, 2020)

4. **Complementarity of Causal Inference and Structural Modeling:**

   While causal inference focuses on estimating the effect of a particular treatment, structural modeling aims to understand the underlying process that generates the data. Therefore, the results from a causal analysis can provide useful information for specifying or validating a structural model. For example, a causal analysis might reveal that price has a negative effect on demand, which could be used to specify the demand function in a structural model.

   Conversely, a structural model can help guide causal analysis. For example, it can help identify potential sources of endogeneity or omitted variable bias, and suggest instrumental variables or other strategies for causal identification.

   An example is (Conlon and Mortimer, 2013). The authors use structural modeling to address endogeneity in product availability, and use these structural estimates to perform a counterfactual analysis.

Chapter 25

Qualitative Research

25.1 Inter-rate reliability methods

Calculation to judge the degree of agreement between the choices made by two or more independent judges

Other packages are

- `vcd` for visualization
- `DescTools`

25.1.1 Percent Agreement

$$\frac{\text{number of agreement}}{\text{number of total}} \times 100$$

```r
library("irr")
```

```
## Loading required package: lpSolve
```

```r
data("diagnoses", package = "irr")
data("anxiety", package = "irr")
head(diagnoses,10)
```

```
##                     rater1                  rater2                  rater3
## 1              4. Neurosis             4. Neurosis             4. Neurosis
## 2  2. Personality Disorder 2. Personality Disorder 2. Personality Disorder
## 3  2. Personality Disorder        3. Schizophrenia        3. Schizophrenia
## 4                 5. Other                5. Other                5. Other
## 5  2. Personality Disorder 2. Personality Disorder 2. Personality Disorder
## 6            1. Depression           1. Depression        3. Schizophrenia
## 7         3. Schizophrenia        3. Schizophrenia        3. Schizophrenia
```

```
## 8            1. Depression           1. Depression        3. Schizophrenia
## 9            1. Depression           1. Depression             4. Neurosis
## 10                5. Other                5. Other                5. Other
##               rater4           rater5           rater6
## 1       4. Neurosis      4. Neurosis      4. Neurosis
## 2          5. Other         5. Other         5. Other
## 3  3. Schizophrenia 3. Schizophrenia         5. Other
## 4          5. Other         5. Other         5. Other
## 5       4. Neurosis      4. Neurosis      4. Neurosis
## 6  3. Schizophrenia 3. Schizophrenia 3. Schizophrenia
## 7  3. Schizophrenia         5. Other         5. Other
## 8  3. Schizophrenia 3. Schizophrenia      4. Neurosis
## 9       4. Neurosis      4. Neurosis      4. Neurosis
## 10         5. Other         5. Other         5. Other
```

```
agree(diagnoses)
```

```
##  Percentage agreement (Tolerance=0)
##
##  Subjects = 30
##    Raters = 6
##   %-agree = 16.7
```

```
# library(vcd)
# # Create the plot
# p <- agreementplot(anxiety)
```

25.1.2 Cohen's Kappa

(Cohen, 1960)

$$k = \frac{p_o - p_e}{1 - p_e} = 1 - \frac{1 - p_o}{1 - p_e}$$

where

- p_o = relative observed agreement among raters
- p_e = hypothetical probability of chance agreement

strict agreements between raters

appropriate in case of 2 ordinal or nominal varibles

Based on (Landis and Koch, 1977)'s guide, we have

Degree	Decision
0.01 – 0.20	slight agreement

Degree	Decision
0.21 – 0.40	fair agreement
0.41 – 0.60	moderate agreement
0.61 – 0.80	substantial agreement
0.81 – 1.00	almost perfect or perfect agreement

```r
# Unweighted kappa for 2 nominal or 2 ordinal categorical
kappa2(diagnoses[, c("rater1", "rater2")], weight = "unweighted") # two ordinal variables only, c
```

```
##  Cohen's Kappa for 2 Raters (Weights: unweighted)
##
##  Subjects = 30
##    Raters = 2
##     Kappa = 0.651
##
##         z = 7
##   p-value = 2.63e-12
```

```r
# Weighted kappa ordinal scales
kappa2(diagnoses[, c("rater1", "rater2")], weight = "equal") # linear weightes of the differences
```

```
##  Cohen's Kappa for 2 Raters (Weights: equal)
##
##  Subjects = 30
##    Raters = 2
##     Kappa = 0.633
##
##         z = 5.43
##   p-value = 5.52e-08
```

```r
kappa2(diagnoses[, c("rater1", "rater2")], weight = "squared") # squared weightes of the differen
```

```
##  Cohen's Kappa for 2 Raters (Weights: squared)
##
##  Subjects = 30
##    Raters = 2
##     Kappa = 0.655
##
##         z = 3.91
##   p-value = 9.37e-05
```

p-value less than 0.05, mean that raters agree more than what you would expect by chance.

25.1.3 Fleiss'kappa

for two or more categorical variables (nominal or ordinal) for three or more raters

- 0 = no agreement
- 1 = perfect agreement

```
# no assumption of same raters for all subjects
kappam.fleiss(diagnoses[, 1:3])
```

```
##  Fleiss' Kappa for m Raters
##
##  Subjects = 30
##    Raters = 3
##     Kappa = 0.534
##
##         z = 9.89
##   p-value = 0
```

25.2 Krippendorff's Alpha

calculates disagreement among raters.

$$\frac{\text{observed disagreement}}{\text{expected disagreement}}$$

Rules of thumb by (Shelley and Krippendorff, 1984)

- $\alpha \geq 0.08$ = good
- $\alpha \geq 0.667$ = acceptable lower limit

```
nmm <-
    matrix(
        c(
            1,
            1,
            NA,
            1,
            2,
            2,
            3,
            2,
            3,
            3,
            3,
            3,
            3,
            3,
```

```r
            3,
            3,
            2,
            2,
            2,
            2,
            1,
            2,
            3,
            4,
            4,
            4,
            4,
            4,
            1,
            1,
            2,
            1,
            2,
            2,
            2,
            2,
            NA,
            5,
            5,
            5,
            NA,
            NA,
            1,
            1,
            NA,
            NA,
            3,
            NA
        ),
nrow = 4
)
 # first assume the default nominal classification
 kripp.alpha(nmm)
```

```
##  Krippendorff's alpha
##
##  Subjects = 12
##    Raters = 4
##     alpha = 0.743
```

```
# now use the same data with the other three methods
kripp.alpha(nmm, "ordinal")
```

```
##  Krippendorff's alpha
##
##  Subjects = 12
##    Raters = 4
##     alpha = 0.815
```

```
kripp.alpha(nmm, "interval")
```

```
##  Krippendorff's alpha
##
##  Subjects = 12
##    Raters = 4
##     alpha = 0.849
```

```
kripp.alpha(nmm, "ratio")
```

```
##  Krippendorff's alpha
##
##  Subjects = 12
##    Raters = 4
##     alpha = 0.797
```

25.2.1 Kendall's W

Continuous ordinal ratings

```
rtr1 <- c(1, 6, 3, 2, 5, 4)
rtr2 <- c(1, 5, 6, 2, 4, 3)
rtr3 <- c(2, 3, 6, 5, 4, 1)
ratings <- cbind(rtr1, rtr2, rtr3)
library(DescTools)
KendallW(ratings, test=TRUE)
```

```
##
##  Kendall's coefficient of concordance W
##
## data:  ratings
## Kendall chi-squared = 8.5238, df = 5, subjects = 6, raters = 3, p-value
## = 0.1296
## alternative hypothesis: W is greater 0
## sample estimates:
##        W
## 0.568254
```

25.2.2 Intraclass correlation coefficients

(Shrout and Fleiss, 1979)

25.2.2.1 Continuous scales

Decision:

- model:
  - `"oneway"` when subjects are random effects
  - `"twoway"` when subjects and raters are random effects
- type:
  - `"agreement"` differences in mean ratings among raters are of interest
  - `"consistency"` is default
- unit
  - `"single"` single value
  - `"average"` mean of several ratings

```
icc(anxiety,
    model = "twoway", # can be "oneway"
    type = "agreement", # can be "consistency"
    unit = "single" # can be "average"
    )
```

```
##  Single Score Intraclass Correlation
##
##    Model: twoway
##    Type : agreement
##
##    Subjects = 20
##      Raters = 3
##    ICC(A,1) = 0.198
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(19,39.7) = 1.83 , p = 0.0543
##
##  95%-Confidence Interval for ICC Population Values:
##   -0.039 < ICC < 0.494
```

`DescTools` package

```
rtr1 <- c(9, 6, 8, 7, 10, 6)
rtr2 <- c(2, 1, 4, 1,  5, 2)
rtr3 <- c(5, 3, 6, 2,  6, 4)
rtr4 <- c(8, 2, 8, 6,  9, 7)
```

```r
ratings <- cbind(rtr1, rtr2, rtr3, rtr4)
DescTools::ICC(ratings)
```

```
##
## Intraclass correlation coefficients
##                          type   est F-val df1 df2    p-val lwr.ci upr.ci
## Single_raters_absolute   ICC1 0.166  1.79   5  18 0.164769     NA     NA
## Single_random_raters     ICC2 0.290 11.03   5  15 0.000135     NA     NA
## Single_fixed_raters      ICC3 0.715 11.03   5  15 0.000135     NA     NA
## Average_raters_absolute ICC1k 0.443  1.79   5  18 0.164769     NA     NA
## Average_random_raters   ICC2k 0.620 11.03   5  15 0.000135     NA     NA
## Average_fixed_raters    ICC3k 0.909 11.03   5  15 0.000135     NA     NA
##
##  Number of subjects = 6     Number of raters = 4
```

25.2.3 Light's kappa

- average of Cohen's Kappa if using more than two categorical variables

```r
# Light's kappa: multiple raters
kappam.light(diagnoses[, 1:3])
```

```
## Warning in sqrt(varkappa): NaNs produced
```

```
##  Light's Kappa for m Raters
##
##  Subjects = 30
##    Raters = 3
##     Kappa = 0.555
##
##         z = NaN
##   p-value = NaN
```

Chapter 26

Measurement Scales

- **Handbook of Marketing Scales: Multi-Item Measures for Marketing and Consumer Behavior Research** (Bearden et al., 2011)
- **Marketing Scales Handbook, Volume IV: Consumer Behavior** (Stewart et al., 1993)
- Check "books" folder

Chapter 27

Preference Measurement

Reservation price = price a consumer willing to pay for a product

Estimation for reservation price:

1. Survey question direct asks for consumer reservation price (but this estimation method tends to bias downward because participants don't want people to know that they are price sensitive).
2. From the coefficients of conjoint analysis.

Assumptions to get consumer utility from regression:

1. Consumption utility equals sum of utilities for the product product attributes

Let N be the number of non-price attributes and each attributes has more than 1 level.

We use fractional factorial design to derive different profiles for consumers to choose with the goal of least number of profiles.

Then the utility function can be represented as

$$U = \beta_0 + \beta_p \times P + \sum_{k=1}^{N} \beta_k \times A_k + \epsilon$$

where

- U = consumer rating or utility
- β's = utility weights
- β_0 = heterogeneity in the range of the ratings scale per consumer.
- P = price

- A = non-price attributes (categorical variables)

Then the reservation price (at the individual level) for product P is

$$r(P) = \frac{\Delta p}{\beta_p}(\sum_{k=1}^{N} \frac{\beta_k}{\Delta A_k} A_k)$$

where

- Δp = difference in price between two levels (of the same attribute)
- ΔA_k = difference in attribute levels

and the dollar value for each unit of utility is

$$\frac{\Delta p}{\beta_p}$$

and for each attribute k, the utility a consumer can derive is

$$(\frac{\beta_k}{\Delta A_k})A_k$$

Hence, we can see that the reservation price for a product (i.e., total dollar value a consumer assigns on product P) is just the sum of utility across all attributes times dollar value for each unit of utility

Netzer et al. (2008)

- Problem
  - Companies: Product development, pricing segmentation, positioning, advertising, project selection, and investment decisions
  - Consumers: recommendation agents
  - Policy makers and health care professionals:
  - Academic researchers;
- Design and Data collection
  - Original goals: maximizing A-efficiency and D-efficiency (based on covariance matrix of the estimates of the partworths)
  - New: M-efficiency measure (accounts for managers' considerations).
  - New Adaptive method for data collection:
    * Commercially available: Adaption conjoint analysis (ACA), Fast Polyhedral approach, Adaptive Self-explicated approach
- Model Specification, Estimation, and Action

27.1 Conjoint Analysis

also known as conjoint measurements, which stems from mathematical psychology and psychometrics

Aims:

- Willingness to pay: measure what individual are willing to give up to obtain certain product benefits
- Distribution of willing to pay : measure how consumer differ

Advantages:

- Improves ability to assess customers' true wants and needs (especially price)
- Dissect "everything is important"

Perception vs. preference

- Perception (beliefs about the market) is largely homogeneous across customers
- Preference (what product they buy) is heterogeneous across customers.

Conjoint analysis is a tool for preference assessment.

Attributes in conjoint:

- Not too many attributes
- Attributes should be actionable
- Need to choose reasonable range of attributes
  - within range under consideration
  - All levels should be feasible
- Number of attribute levels
  - Too many can be confusing
  - Halo effect of number of attributes (or the order of levels)

Number of profiles equal number of dummy variables with the assumption of independence

If you suspect that there might be interaction effect (violation of independence), you can either

- use higher-level attribute
- increase the number of profiles

Ranking and Rating conjoint analysis are not different Kalish and Nelson (1991)

(Green and Wind, 1975)

Basics:

- Use orthogonal array (in experimental design) to test combinations where independent contributions of all factors (attributes) are balanced.
- Then rank all the reduced combinations in order

Uses:

- New product development
- Package design
- Pricing and brand alternatives
- Descriptions of new products or services
- Alternative service design
- Also organizations as consumers

Outcome variable can be:

- Preference
- Likelihood to purchase
- best value for the money
- Convenience of use
- Suitability for a specific job
- Psychological images (e.g., ruggedness, distinctiveness, conservativeness)

Can be used in conjunction with

- factor analysis: firm-level decision variables (attribute/factor), dimension reduction technique
- perceptual mapping: multidimensional scaling
- cluster analysis: customer-level variables, segmentation technique

Suggested packages (in order of preference):

- `AlgDesign`
- `conjoint`
- `faisalconjoint`
- `mlogit`
- `conf.design`

- prefmod

27.1.1 Full-Profile

```
library(conjoint)
data(tea)
caModel(y = tprefm[1,],x = tprof)
```

```
##
## Call:
## lm(formula = frml)
##
## Residuals:
##       1       2       3       4       5       6       7       8       9      10
##  1.1345 -1.4897  0.3103 -0.2655  0.3103  0.1931  1.5931 -1.4310 -1.4310  1.1207
##      11      12      13
##  0.3690  1.1931 -1.6069
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)           3.3937     0.5439   6.240  0.00155 **
## factor(x$price)1    -1.5172     0.7944  -1.910  0.11440
## factor(x$price)2    -1.1414     0.6889  -1.657  0.15844
## factor(x$variety)1  -0.4747     0.6889  -0.689  0.52141
## factor(x$variety)2  -0.6747     0.6889  -0.979  0.37234
## factor(x$kind)1      0.6586     0.6889   0.956  0.38293
## factor(x$kind)2     -1.5172     0.7944  -1.910  0.11440
## factor(x$aroma)1     0.6293     0.5093   1.236  0.27150
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.78 on 5 degrees of freedom
## Multiple R-squared:  0.8184, Adjusted R-squared:  0.5642
## F-statistic:  3.22 on 7 and 5 DF,  p-value: 0.1082
```

27.1.2 Choice-based

Also known as Discrete-choice conjoint analysis

27.1.3 Adaptive

- Varies the choice sets based on respondent's preference

- Reduces the survey length

27.1.4 Hybrid

- Combination of Full-Profile and Adaptive

27.1.5 Max-Diff

- Assortment under most preferred and least preferred scenario.

27.1.6 Self-explicated

27.1.7 Hierarchical Bayes analysis

27.1.8 Application

```r
library(conjoint)
Experiment = expand.grid(weight = c("a","b"),
                         Price = c("Low","High"),
                         Warranty = c("2","3"),
                         Battery = c("10","20"))

partial_design = caFactorialDesign(Experiment,type = "orthogonal",seed = 123)
partial_profile = caEncodedDesign(partial_design)

levels <- c("a","b","Low","High","2","3","10","20")
lev.df <- data.frame(levels)

data = c("")
```

Example by Martin Muller

```r
# Declaration of features and feature values
feature_names <- c("LENGTH", "ILLUSTRATION", "CLAPS")
feature_values <- list()
feature_values[[1]] <- c("2min", "7min", "20min")
feature_values[[2]] <- c("several images", "one image", "no image")
feature_values[[3]] <- c("+500 claps", "less than 500 claps")

# All concept generation
articles <- expand.grid(LENGTH = feature_values[[1]],
                        ILLUSTRATION = feature_values[[2]],
                        CLAPS = feature_values[[3]])

library (conjoint)
maxNumberOfArticles = 8

# Selection of relevant concepts
selectedArticles <- caFactorialDesign(data = articles,
```

```
                                    type = 'fractional',
                                    cards = maxNumberOfArticles)

# Checking if selected concepts are relevant for study
corrSelectedArticles <- caEncodedDesign(selectedArticles)
#print(cor(corr_design_mails))
print(cor(corrSelectedArticles))
```

```
##                 LENGTH ILLUSTRATION     CLAPS
## LENGTH       1.0000000            0 0.1240347
## ILLUSTRATION 0.0000000            1 0.0000000
## CLAPS        0.1240347            0 1.0000000
```

```
# List of rates of each article allowing to compare them
ranking = c(7 / 7, 3 / 7, 3 / 7, 5 / 7, 2 / 7, 6 / 7, 3 / 7, 0 / 7)

# Performing conjoint analysis
Conjoint(ranking, selectedArticles, unlist(feature_values))
```

```
##
## Call:
## lm(formula = frml)
##
## Residuals:
##        1        2        3        4        5        6        7        8
## -0,03401 -0,02041  0,07483  0,10884 -0,12925  0,05442 -0,07483  0,02041
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)              0,44898    0,05651   7,945   0,0155 *
## factor(x$LENGTH)1        0,25850    0,07534   3,431   0,0755 .
## factor(x$LENGTH)2        0,06803    0,07534   0,903   0,4619
## factor(x$ILLUSTRATION)1  0,30612    0,07534   4,063   0,0556 .
## factor(x$ILLUSTRATION)2 -0,18367    0,08835  -2,079   0,1732
## factor(x$CLAPS)1         0,02041    0,05651   0,361   0,7526
## ---
## Signif. codes:  0 '***' 0,001 '**' 0,01 '*' 0,05 '.' 0,1 ' ' 1
##
## Residual standard error: 0,1495 on 2 degrees of freedom
## Multiple R-squared:  0,9389, Adjusted R-squared:  0,7863
## F-statistic: 6,151 on 5 and 2 DF,  p-value: 0,1457
```

```
## [1] "Part worths (utilities) of levels (model parameters for whole sample):"
##               levnms    utls
## 1          intercept   0,449
## 2               2min  0,2585
## 3               7min   0,068
```

```
## 4               20min -0,3265
## 5      several images  0,3061
## 6           one image -0,1837
## 7            no image -0,1224
## 8          +500 claps  0,0204
## 9 less than 500 claps -0,0204
## [1] "Average importance of factors (attributes):"
## [1] 52,44 43,90  3,66
## [1] Sum of average importance:  100
## [1] "Chart of average factors importance"
```

Chapter 28

Image Processing

Transdermal Optical Imaging:

- https://www.nuralogix.ai/technology
- extract facial blood flow information
- https://github.com/nuralogix
- https://www.deepaffex.ai/
- https://www.deepaffex.ai/media

Chapter 29

Surveys

Chandon et al. (2005) posits that "self-generated validity" can obscure the link between purchase intentions and purchase behavior. On average, the link between latent intentions and purchase behavior is 58% stronger among survey consumers that that of nonsurveyed consumers.

Sheppard et al. (1988) found that the link between intentions and behavior is about 0.53

A little more improvement can be achieved in predictive power if we use segmentation before forecasting sales based on historical purchases and purchase intention Morwitz and Schmittlein (1992)

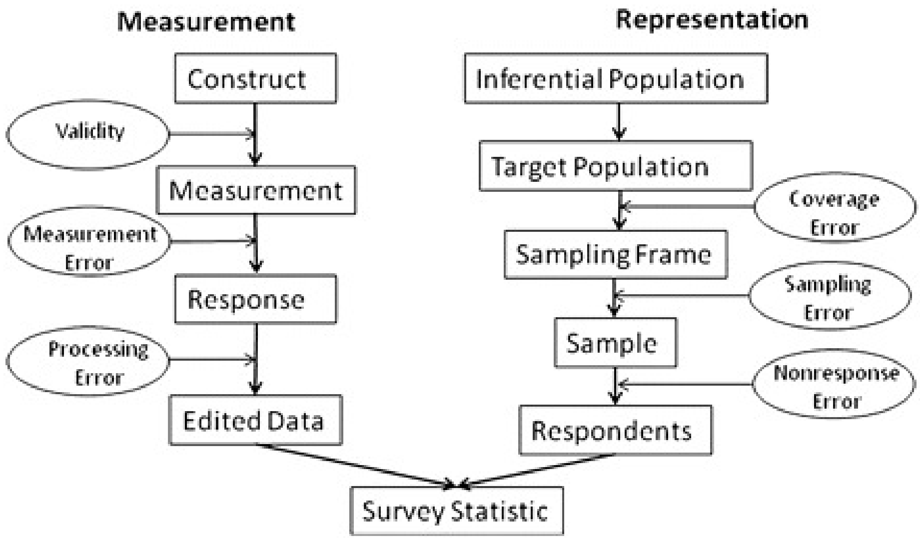
Table 29.1: SICSS Summer 2020 -Survey Research in the Digital Age by professor **Matthew Salganik**

	Sampling	Interviews	Data environment
1st era	area probability	face-to-face	stand-alone
2nd ear	random digital dial probability	telephone	stand-alone
3rd era	non-probability	computer-administered	linked

Total survey error framework (Groves and Lyberg, 2010)

Insight:

1. Errors can come from bias or variance
2. Total survey error = Measurement error + representation error



$@Groves_2010, Fig.3$

Probability and Non-probability Sampling

- Probability sample: every unit from a frame population has a known and non-zero probability of inclusion
- With weighting, we can recover bias in your sampling.
- Non-response problem

Horvitz-Thompson estimator (or bias estimator):

$$\hat{\bar{y}} = \frac{\sum_{i \in s} y_i / \pi_i}{N}$$

where π_i = person i's probability of inclusion (we have to estimate)

Wiki Survey

- Create a survey that leverages the power of people

Mass Collaboration

- Human Computation: Train People -> Train Lots of People -> Train Machine
  - Cleaning
  - De-biasing
  - Combining
- Open Call:

  - solutions are easier to check than generate
  - required specialized skills

- Distributed Data Collection:
  - People go out and collect data
  - quality check

Fragile family challenges

29.1 Anchoring Vignettes

- Problem of interpersonal incomparability
- Resources:
  - (KING et al., 2004)
  - (King and Wand, 2007)
  - (Hopkins and King, 2010)
  - Examples
- Help with 2 questions:
  - Different respondents understand the same question differently: Incomparaability in Survey Responses ("DIF"). Agreement on theoretical concept is almost nearly impossible.
  - How can we measure concepts that can only be defined by examples
- Measure like usually, then subtract the incomparable portion. (i.e., using the assessment from the same respondents for a particular example/case to correct/adjust for the self-assessment).
- Varying vignette assessments give us DIF (i.e., **differential item functioning**)
- Since we created the anchors (i.e., examples), we know the true vignette assessments are fixed over respondents

29.1.1 Nonparametric method

Code the relative ranking of self-assessment in accordance to vignettes.

Inconsistencies would be considered ties.

Measurement Assumptions:

- **Response consistency**: Each responder approaches the self-assessment and vignette categories in a same manner across questions.

- **Vignette Equivalence**: For every vignette, the real level is the same for all respondents.

Used Ordered Probit to estimate.

29.1.2 Parametric method

The more vignettes that we have better identification. But it will introduce measurement errors.

- Also use an ordinal probit model
- with varying thresholds and a random effect.

```r
# install.packages("anchors")
library(anchors)

# Example from the package's authors
data("freedom")
head(freedom)
a1 <-
    anchors(self ~ vign2 + vign3 + vign4 + vign5 + vign6, freedom, method = "C")
summary(a1)
```

Chapter 30

Experiment

Things to consider:

- Validity
    - statistical conclusion validity
    - internal validity
    - construct validity: whether you operationalize your construct correctly
    - external validity: generalization
- Heterogeneity of treatment effects
    - average (aggregate) effect can be artificially constructed.
- Mechanisms

More things to consider

- cost
- control
- realism
- ethics

The Principles of Humane Experimental Technique by Russell and Burch (1959): 3Rs

- Replace: with natural (less invasive)
- Refine: less harmful
- Reduce: min number of participants

Before running experiments, to make sure that you did not tamper with the data, or hypotheses (i.e., change them after the fact), you should always preregister your research. A website that allows you to do this and automatically provides you with time stamp is aspredicted

This way then you can show to reviewers that you did hypothesize before running analysis.

To combat replication crisis in psychology, we can advocate for open research in which researchers public their data and analyses. Popular websites include:

- ResearchBox
- https://osf.io/

To see whether authors temper with their data, we can use p-curve to inspect. 1 2 3 4 5

Interaction Effects Need Interaction Controls (basically, you can't just control for main effects).

Part IV

OTHERS

Chapter 31

Report

1. Introduction
    1. Note on managerial relevance/ context
    2. Note on the gap in current managerial understanding and academic literature
    3. State your research questions
    4. Summarize your intended contributions to both theory and practice (at least 3 contributions)

Chapter 32

Review Process

Communicate your review by O. C. Ferrell:

1. Paragraph
    1. contextualize the research
    2. convey the key message of the research
    3. be positive about what was done well
2. Paragraph:
    1. Is the paper well written, interesting and important?
    2. Are the methods, data, etc. appropriate?
    3. Does the data support the findings?
3. Paragraphs: Numbers, topics-examples of an issue:
    1. Factual errors
    2. Invalid arguments
    3. length/journal guidance
    4. Paths to improve the research

Why review?

- Contribution/ Give back to the community
- Get to the fore front of current research
- Understand what a strong paper looks like

Documenting your reviews

- Use publons.com to document manuscripts you reviewed for promotion and CV
- to verify your review, send the "thank you for reviewing" email to reviews@publons.com

The review process:

- Journals always look for reviewers
  - Get involved and email your CV to editor, or area editor.
- Do not wait to answer whether you review or not.

Responding to reviews

- Turn around in 3 months
- R & R: reject and resubmit (new). The turn around will be sent to a new set of reviewers
- R & R: review and resubmit.

32.1 Review at JM

For data replication submission, read here

For detail, read here

JM articles should have:

1. New knowledge
2. Real-world marketing topics and problems (i.e., relevance to stakeholders)
3. Validity (i.e., good approximation of the truth)

Expected quality review process:

1. Fairness
2. Reviewers should have reasonable knowledge
3. No conflicts of interests
4. Consistency
    1. Interesting and important
    2. Empirical rigor
    3. Conceptual rigor
    4. Relevance
    5. Constructive
    6. Risk-taking
5. Not a vote-counting exercise (i.e., number of rejections from reviewers)

6. Responsible
    1. Respect author's original objective
    2. Explain fatal flaws
    3. Privacy
7. Timely: reviewers (three weeks after the review invitation) and AEs (within two weeks)
8. Roadmap

Format

- 2 single-spaced pages
- reviewers number their comments.
- Suggested structure:
    - A short synopsis of the paper and its findings
    - Evaluate the contribution: Suggestions to improve contribution(if needed)
    - Identify major strengths, weaknesses, recommendations regarding
        * conceptual
        * empirical
    - Readability
    - Minor comments and suggestions

32.2 Review Process Back end

When you submit an article

- Plagiarism check
- Repeat submission check
- An editor reviews the paper for suitability (might be desk-reject)
- An area/associate editor is assigned
- Reviewers (2-3) are assigned

Reviewers are chosen based on

- your reference list
- your suggested list of reviewers
- the editor's knowledge of the domain

- Candidate reviewers that declined to review
- web search
- a publisher's database

Chapter 33

Scientific Writing

Section	Length (words)	Purpose	Verb Tense	Elements
Title	8 - 15			
Abstract	200-250	Paper's summary	Simple past (work done) Present (general fact, paper itself or analysis)	• Objectives • Methods • Results • Conclusions
Introduction	500-1000	Rationale for the paper	Present (current state-of-art, general background) Present perfect (previous research)	• The problem
Literature Review	1000-2000	Understanding of the current knowledge landscape	Present (current state-of-art)	• Review of the literature • Hypothesis

Section	Length (words)	Purpose	Verb Tense	Elements
Methods	500-1000	What was done	Simple past (work done) Present (explain diagrams/figures)	• Approach/methods used • Data description • Procedure description
Results	500-1500	presents data, facts	Simple past (what was found) present (explain diagrams/figures)	• Results of analysis
Discussion	1000-1500	discuss relationships among constructs, and insights	Present (established knowledge) past (summarize findings)	• Relationships • Outliers • Comparison to previous studies
Conclusions	500	summarize findings	Present (implications and future search) Past (completed research)	

Rules of thumb:

- Past: events that have been completed
  - We found
  - Method proposed by X was followed
- Present Perfect: events that started in the past but are still ongoing
  - Many studied have researched on
- Present: events that are general fact, and current

Journal paper types:

1. Research Paper: hypotheses and findings
2. Methods Paper: new method
3. Review Paper: consolidate a domain
4. Discussion Paper: opinions regarding other papers

Chapter 34

CB Seminar

"Cognitive perspective-taking refers to the ability to make inferences about others' thoughts and beliefs."

"Affective perspective-taking is the ability to make inferences about others' emotions and feelings" [@healey2018]

34.1 Overview

(Simonson et al., 2001):

- Behavioral Decision Theory (i.e., study judgment and decisions) (Kahneman & Tversky) vs Social Cognition
    - Both use mediation and path analysis
- Positivist (focuses on causation) Versus Postmordern (focuses on interpretation)

	Behavioral Decision Theory	Social Cognition
Underlying model	decision-making model, (e.g., determinants of choice)	response hierarchy model (i.e., how judgments and attitudes are formed)
Phenomena/Task	Stimulus-based	Memory-based
Measures	Information acquisition, verbal protocols, and response time	Cognitive response

Types fo Consumer Research:

- Theory Development
- Theory Application
    - limited impact on managerial practice

Theory-Testing vs. Substantive Phenomena-Driven Consumer Research

(Janiszewski et al., 2016)

- Deductive-Conceptual Knowledge
    - A theory is "a statement of concepts and their interrelationships that shows how and/or why a phenomenon occurs." (Corley and Gioia, 2011)
    - Knowledge Creation Strategies:
        * Bridging disciplines
        * Challenging assumptions
        * Introducing mediators and moderators
        * Contrastive explanation
        * Borrowing and blending
    - Knowledge Appreciation
        * Quality of the contribution
            · depends on rigor, and execution of research method
        * Benefit of the contribution
            · Originality / Novelty
            · Interestingness
            · Changes in core understanding/beliefs
- Integration (i.e., tree of knowledge)

34.2 Social Influence

(McFerran et al., 2010)

- body type of others influences one's food consumption
    - moderated by body type (undesirable reference group) of the other consumers
    - low appearance self-esteem or cognitive load also moderate this relationship
- Anchoring and adjustment processes

- Social Influence and Food Choice
  - people conform to group average (anchor)
- Study 1: Social influence effects is present regardless of food perception (healthy vs. unhealthy). People eat more when there is another person, but the magnitude is moderated by the other's body type (consume less when seeing obese others - undesirable group). The effect persists after social influence.
- Study 2: Presence of other increases food quantity.
  - High anchor (obese person takes high quantity), participants take less.
  - Low anchor (obese person takes small quantity), participants take more
- Study 3: People with low appearance self-esteem and high processing resources will attenuate more

(Moreau and Herd, 2010)

- Social comparison: people want to gain self-knowledge.
  - Self-designed products are based on:
    * comparison with the characteristics of other comparable products
    * comparison with skills, talents, and expertise of the other designers.
- Since professional designers are expert, consumer designers face upward comparison
- Study 1:
  - Self-deigned product evaluation is lower when compared to professional default product, than consumer designed default design
  - Firm guidance moderates this effect
  - There is a premium for self-designed product.
- Study 2: Defensive vs. Nondefensive processing
  - need to protect self-esteem/ self-image.
  - At low defensive processing, evaluations of self-designed products are lower when the default product is professionally designed than consumer designed
  - defensive processing reduces negative comparison
- Study 3: The behavioral consequences of social comparison in self-design.

    - Public prize (visibility) enhance self-evaluations.
    - Repair opportunity affects the evaluation of self-designed products

(Hamilton, 2003)

- Influence based on context effects (suggesting other unlikely alternative)
    - moderated by the belief whether others are trying to manipulate the choice context.
- People understand context effects (e.g., choice set construction).
- To influence others, people construct a set of unattractive alternatives (attraction effect), or a compromise between two other alternatives (compromise effect).
- Author hypothesized: People correct the contextual effects when they know constructors' intentions (to influence them).
- Results: "Subjects' beliefs that menus had been created **(by their friends)** to influence their choices seemed to enhance rather than limit the effectiveness of the menus." (p. 498) likely due to homophily bias.
    - Evaluated the target more favorably.
- Source credibility, expertise, trustworthiness, attractiveness, and similarity can influence the effectiveness of persuasion attempt (Wilson and Sherrell, 1993)
- Hypothesis: ulterior motive of persuader moderates increase the resistant effect of chooser than group-oriented motive.
    - Subjects are likely to agree (choose) when they know the menu is created by their friends
    - knowing that the menu has been created by a stranger, under compromise strategy, subjects still choose the middle alternative.
- Attraction strategy is more likely to be used than compromise strategy
- Even if subjects understand contextual effects (manipulation), they do not resist because
    - it's hard to recognize characteristics of a local set from the characteristics of a global set. (Simonson et al., 1993)
    - If subjects assume that the most attractive option has been eliminated, they have to choose the best among alternatives.

(Argo et al., 2005)

- Non interactive social presence in the consumption context.
- social forces has greatest influence (Latan? and Wolf, 1981):

  - number (social size) (large > small):
    - emotions and behaviors: (e.g., stage fright, crowding)
  - immediacy (proximity) (closer > far)
  - social source strength (importance) (high > low)

- Social Impact Theory
- Study 1:
  - 0 -> 1 person increased positive emotions
  - 1 -> 3 person decreased positive emotions
- Study 2:
  - proximity moderates the impact of social size on emotions and brand selection.

34.3 Interpersonal perception and consumer lay beliefs

(Folkes and Patrick, 2003)

- Info valence: **Negativity bias for products** (negative info about a product has greater influence on brand perceptions than positive info). (Herr et al., 1991)
- **Positivity bias for services** (positive experience from a service provider can lead customers to infer positivity about the brand as a whole than negative experience).
  - Could be because service is more heterogeneous.
  - this effect is greater for novice consumer.
- Study 1: First time car insurance buyers
  - Alternative explanation: Subtyping (Weber and Crocker, 1983), or occupational stereotype consistency, question order are all ruled out.
- Study 2: replicate study 1: compare the effect of info about individual service provider to that of others who is not from the firm.
  - Subjects perceive focal firm employee more positively and consider negative experience from the other firms' agents as outliers.
- Because consumers are predisposed to think that a service encounter in general should be more positive than negative (Fornell, 2005), this assumption could explain why consumers consider positive experience as typical and generalize it to the brand, and negative experience as outliers.

- 3 sources that affect consumers' beliefs:
  - general perceptions of services
  - beliefs specific to a firm
  - beliefs specific to an occupation
- Study 3 used a critical incident method to examine the following hypothesis: the positive effect is moderated by consumer experience (e.g., stronger for less experience consumers).
  - Valence affects perceived typicality , but experience with a firm did not influence typicality (cant find support for H3)
- Study 4: Replicate study 1 and 2 in a natural setting

(Brough et al., 2016)

- Men are less likely to buy environmentally friendly products than women because the close association between green behavior and femininity (perceived by both men and women, and both users and observers).
  - Prior research attribute this gap to personality traits (women are more prosocial altruistic (Lee and Holden, 1999), and stronger ethic of care (Zelezny et al., 2000))
  - The link between greenness and femininity could be explained by
    - target of green marketing usually involved women (e.g., cleaning, laundry).
    - caring and nurturing from greenness are feminine traits
- Since men are more concerned with gender -identity maintenance (which is kinda surprising from my view), they avoid purchasing green products which might jeopardize their macho image.
  - From social-identity theory: self-concept is also derived from perceived group membership (Turner and Oakes, 1986)

(Wang et al., 2016)

- Smile affects 2 social judgments: warmth and competence
  - Broader smile leads to more warmth, but less competent
    - Consumers focus moderates this effect
      - Promotion-focused consumers (and low-risk consumption) like bigger smile on warmth
      - Prevention-focused consumers (and high-risk consumption) like slight smile (signal competent)

- Based on stereotype content model (SCM) of social judgments (Fiske et al., 2002)
- Facial configuration has evolutionary proposes.

(Haws et al., 2017)

- Consumers believe healthier foods are more expensive than less healthy foods.
- Due to the dual process model, intuitions based on biased information (processed heuristically) will lead consumers evaluating healthy claims with higher standard (e.g., higher standard for intuition-inconsistent than consistent claim).
- Lay theories or lay beliefs about food = expensive intuition affects consumer decision making process, because this process is low involvement one.

34.4 Emotions, mood and affect

(Bagozzi et al., 1999)

- Emotions are "mental states of readiness that arise from appraisals of events or one's own thoughts" (p. 184)
- Affect is defined as "a set of more specific mental processes including emotions, moods, and (possibly) attitudes." (p. 184)
- Moods (also known as affect transfer) are longer lasting and lower in intensity and non-intentional than emotions (intentional)
- Attitudes are instances of affect defined as an evaluative judgments (also known as appraisals and under appraisal theories in psychology), and have two components:
  - affective
  - cognitive (evaluative)
- Under appraisals theory, there are two appraisal at the emotion formation stage (Lazarus 1991
  - emotions and adaption):
  - goal relevance
  - goal congruence

(Pham et al., 2001)

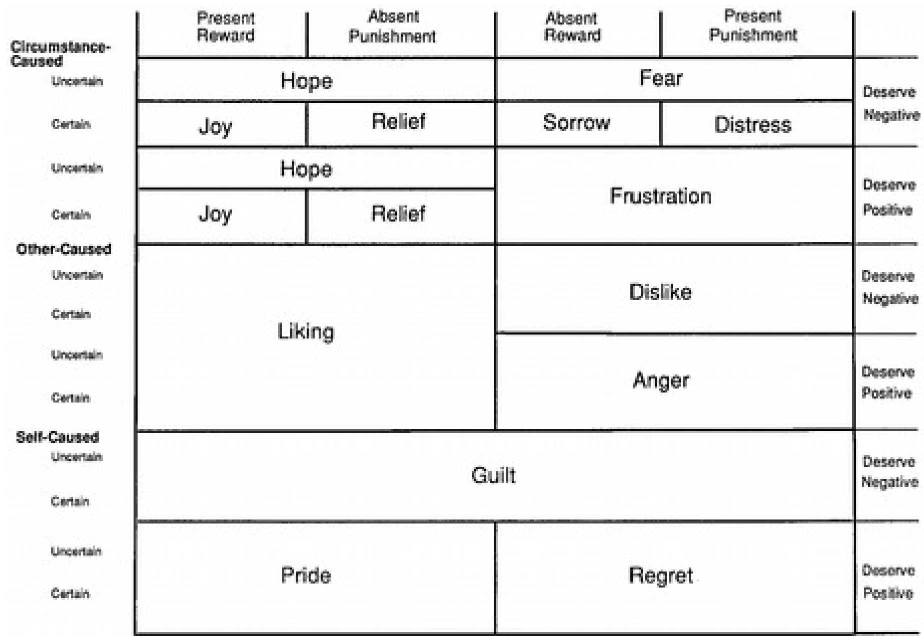


Figure 34.1: From p. 186

- Under moderately complex and consciously accessible stimuli, conscious monitoring of feelings facilitates (1) faster, (2) more stable and homogeneous (among individuals), and (3) more predictive of the quantity and valance of people's thoughts than reason-based assessment.
- Affect-as-information framework posits that feelings contain valuable judgmental information (Schwarz, 1990). People knowingly create overall evaluation based on momentary feelings toward the target. These feelings are consequences of either
  - mental representation fo the target (produced integrally)
  - preexisting o contextually-induced mood produced by experience (produced incidentally)
- Two components of attitudes:
  - cognitive component (also known as utilitarian or instrumental)
  - affective component (also known as hedonic or consummatory).
- Two types of affect:
  - Type I affect: innate programs
  - Type II affect: conditioned stimulus that triggers emotional response

- Type III affect: controlled appraisal of the stimulus

- Integral Feelings vs. Reason-based assessment (also known as descriptively-based evaluation
    - Relative speed: feeling faster than reason-based assessment (Type I affect, and Type II), but not under type III
    - Interpersonal Agreement: people more likely to agree on their feeling-based responses as compared to reason-based responses (under type I and type II, not type III)
    - Relation to thought generation: number and valence are better predicted by feeling responses.

(Isen, 2001)

- Positive affect increases problem solving and facilitates decision making in the sense of flexibility, innovation, and creativity, efficiency.
- Positive affect also increases helping, generosity, and interpersonal understanding.

(Raghunathan and Irwin, 2001)

- Pleasantness of the product context is "the degree to which a product context makes the respondent happy." (p. 355)
- Domain match is "the extent to which the target product is from the same domain (or category) as that of the product context. (p. 355-356)
- Under domain match, exposure to pleasant and improving -sequence (compared to less pleasant and worsening) will lower happiness with the target product.
- Under domain mismatch, exposure to pleasant and improving will increase happiness with the target product

34.5 Persuasion and attitude change

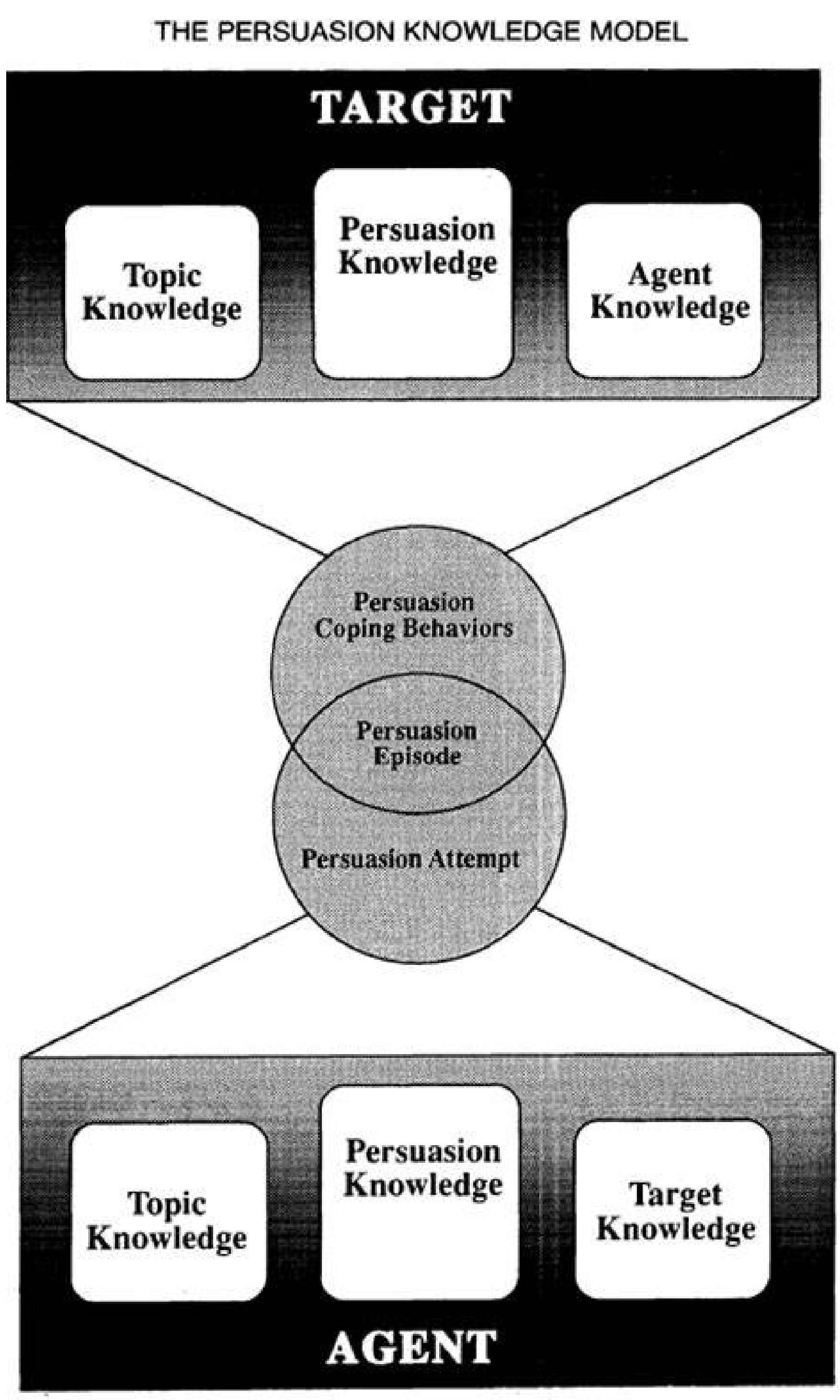
Friestad and Wright (1994)

- The persuasion knowledge model
- Target (Audience), Agent (persuasion message creator)
- People can switch roles fluently between target and agent but the knowledge remain intact.
- Beliefs about:
    - Psychological mediators
    - Marketers' Tactics

- The Effectiveness and Appropriateness of Marketers' Tactics

- Marketers' persuasion goals and one's own coping goals

- two dimension of overall persuasion competence evaluation:
  - Perceived effectiveness
  - perceived appropriateness of the persuasion tactic


THE PERSUASION KNOWLEDGE MODEL
TARGET
Topic Knowledge
Persuasion Knowledge
Agent Knowledge
Persuasion Coping Behaviors
Persuasion Episode
Persuasion Attempt
Topic Knowledge
Persuasion Knowledge
Target Knowledge
AGENT

Campbell and Kirmani (2000)

- Persuasion knowledge refers to "consumers' theories about persuasion and includes beliefs about marketers' motives, strategies, an tactics" (p. 69)
- Both accessibility and cognitive capacity affect inferences of persuasion motives, which then affect the perception of agent from consumers' perspective.
  - Cognitive capacity:
    * Characterization stage = perceptual and automatic
    * Correction stage = higher-order attributional processing
  - Usually, target has more mental load and less cognitive capacity than observers
  - Accessibility of ulterior motives, affected by:
    * expectations
    * strength of association
    * frequency of activation
    * recency of activation
- Under high ulterior persuasion motive, both cognitively busy and unbusy target use persuasion knowledge to evaluate
- Under low ulterior motive, cognitively busy targets are less likely to use persuasion knowledge when evaluating the salesperson.
- If the persuasion tactic is recognized as having ulterior motive, then it's persuasion knowledge.

Ahluwalia (2000)

Baca-Motes et al. (2013)

- Commitment in this research was only symbolic (given a lapel pin to symbolize guests' commitment). But guests were more likely to engage in environmentally friendly behavior.
- "Attitude change does not always equate to behavior change" (Ajzen and Fishbein 1977)
- Proposed solution to change guests' behaviors:
  - Reciprocity concern
  - referencing social norms
- This study posits that individuals' personal values have better predictive power than beliefs in external norms (Viscusi, Huber, and Bell 2011)

(p. 1071) based on the principle internal consistency to avoid cognitive dissonance

- Using signaling theory, the authors prime consumers to stay consistent with their signaled commitment.
- They have competing needs
  - They want to be lazy
  - But they also want to be perceived as environmentally friendly because I said to others already.

Thompson and Malaviya (2013)

- Propose **skepticism-identification model** of ad creator influence
- There are two opposing effects:
  - Skepticism about the creator's competence
  - Identification with the creator
- Factors that affect the two effects:
  - Cognitive resources (to scrutinize the message)
  - Increased source similarity
  - loyalty toward the brand

Isaac and Grayson (2016)

Campbell et al. (2013)

34.6 Judgment and decision making and behavioral pricing

Tversky and Kahneman (1974)

3 heuristics that are used to assess probabilities and predict values:

- **Representativeness** heuristic: "the degree to which A resembles B." (p. 1124) (also known as similarity)
  - **Insensitivity to prior probability (base rate) of outcomes**: (neglected and only take into consideration similarity). With no specific evidence, prior prob is considered, with worthless (misleading) evidence, prior prob is ignored
  - **Insensitivity to sample size**: (1) smaller hospitals are more likely to stray from 50% prop of boys (hence, more likely to obtain more than 60% boys), (2) Even with smaller sample, if the proportion is

greater, one can still infer posterior prob to be higher (unaccounted for equal prior prob), which is known as "conservatism."

- **Misconceptions of chance**: local representativeness (local characteristics represent global ones). *Gambler's fallacy*: "chance is commonly viewed as a self-correcting process in which a deviation in one direction induces a deviation in the opposite direction to restore the equilibrium." (p. 1125) *Law of small numbers*: "even small samples are highly representative of the populations from which they are drawn." (p. 1125)

- **Insensitivity to predictability**: predictions based on favorability of the description (or attributes), but it violates the *normative statistical theory* (predictability here is strictly related to information, the range of predictions and extremeness should depend on the range of information, not on uninformative info).

- **The illusion of validity**: people are confidence in their prediction when they are presented with strong similarity between input and the selected outcomes), which persists even in cases that the judge understands factors that limit his predictions' accuracy. This illusion can be caused by *internal consistency* (highly consistent patterns). Redundancy (or correlation) among inputs decreases accuracy while it increases confidence.

- **Misconceptions of regression**: regression toward the mean (e.g., flight instructors praise or scold on previous landing can "predict" the next landing).

• **Availability** heuristic: The ease at which instances or concurrences come to mind.

  - **Biases due to the retrievability of instances**: Factors affect retrievability are familiarity, and salience (p. 1127) (recent occurrence > remote, and more vivid instance > less vivid).

  - **Biases due to the effectiveness of a search set**: how easy it is for you to recall your search set matters. (e.g., abstract vs. concrete words (Galbraith and Underwood, 1973))

  - **Biases of imaginability**: the ease with which an instance can be constructed. (evidence in constructing committee based on imagination).

  - **Illusory correlation** (frequency with which two events co-occur): "the associative connection between events are strengthened when the events frequently co-occur." (p. 1128).

• **Adjustment** and **anchoring**:

- **Insufficient adjustment**: under anchoring effect, (e.g., African countries in UN, and long calculations based on insufficient calculation).
- **Biases in the evaluation of conjunctive and disjunctive events**: people prefer conjunctive events over simple events (because overestimate of conjunctive events), and simple events over disjunctive events (because underestimate of disjunctive events), which might lead to underestimate the length in planning (because it usually involves successive events - conjunctive events). Similarly, evaluation of risk are conjunctive events, which we will underestimate. "The chain-like structure of conjunctions leads to overestimation, the funnel-like structure of disjunctions leads to underestimation." (p. 1129).
- **Anchoring in the assessment of subjective probability distributions**: regardless of expertise (e.g., sophisticated or native subjects), people usually overly narrow confidence intervals (which reflect more certainty), than is justified by knowledge about the assessed quantities (p. 1129). People have 2 ways to derive subjective probability distributions for a quantity:
  1. Select values that correspond to specified percentiles of his probability distribution
  2. Asses the probability that the true value of the quantity will exceed some specified values

Interestingly, the second procedure yields less extreme odds than the first procedure.

Thaler (1985)

Internal accounting

Let $z = \{z_1, \dots, z_i, \dots, z_n\}$ be vector of goods at prices $p = \{p_1, \dots, p_i, \dots, p_n\}$

Consider consumer's utility function

$$\max_z U(z) \text{ s.t.} \sum p_i z_i \leq I$$

where I is the individual's wealth.

Using Lagrange multipliers

$$\max_z U(z) - \lambda(\sum p_i z_i - I)$$

which is the original model of economic which takes into account of prices and products characteristics (normative prescriptions of economic theory), while ignoring the "framing" effects proposed by (Tversky and Kahneman, 1981)

This paper proposes:

1. **Value function** $v(.)$ from prospect theory
2. Price is changed into **reference price**
3. Relax the principle of **fungibility**

Assumptions:

1. "People respond more to perceived changes than to absolute levels." (p. 201). Hence, we have reference point to infer losses and gains.
2. Concave for gains ($v''(x) < 0, x > 0$) and convex for losses ($v''(x) > 0, x < 0$)
3. Losses loom larger than gains (also known as endowment effect): loss function is steeper than the gain function $v(x) < -v(-x), x > 0$

Joint outcome (x, y) can be valued

1. Jointly as $v(x + y)$ known as integrated
2. Separately as $v(x) + v(y)$ known as segregated.

And the joint outcome (x, y) can have 4 combinations:

Outcomes	Math	Preference
Multiple gains	$x > 0, y > 0$ $v(x) + v(y) > v(x + y)$ (v is concave	Segregation
Multiple losses	$v(-x) + v(-y) < v(-(x + y))$	Integration
Mixed gain (cancellation)	Consider $(x, -y)$ where $x > y$ (net gain). $v(x) + v(-y) < v(x - y)$	Integration
Mixed loss ("silver lining" principle)	Consider $(x, -y)$ where $x < y$ (net loss) $v(x) + v(-y) \geq\leq v(x - y)$ (undetermined)	Segregation if $v(x) > v(x - y) - v(-y)$

To incorporate the notice of reference outcomes (or reference price), instead of y, we will substitute it with Δx (which is changes in the expected outcome).

We label reference outcome as $(x + \Delta x : x)$

When individuals encountered a decision (transaction) as a pleasure maximizing machine, they will

- Stage 1: evaluate potential transactions (i.e., judgment process)

- Stage 2: whether to approve each potential transaction (i.e., decision process)

Transactions

3 price concepts regarding good z:

1. p = actual price
2. $\bar{p}$ = value equivalent of z (i.e.., money that the individual would be indifferent between receiving $\bar{p}$ or z as a gift)
3. p^* = reference price (i.e., expected or "just" price for z), determined by:
    - Fairness

2 kinds of utility:

1. **Acquisition utility**: "value of the good received compared to the outlay"
    - Acquisition utility equals the net utility from the trade of p to obtain z (valued at $\bar{p}$)
    - Compound outcome $(z, -p) = (\bar{p}, -p)$, with value function $v(\bar{p}, -p)$ (usually coded as **integrated**, hence cost of good is not treated as a loss)
2. **Transaction utility**: perceived merits
    - Transaction utility equals price paid compared to reference price
    - Reference outcome $v(-p : -p^*)$
3. **Total utility** equals the sum of acquisition utility and transaction utility
    - $w(z, p, p^*) = v(p, -p) + \beta v(-p : -p^*)$
    - where β is the weight of transaction utility (can be thought of as consumer surplus in the standard theory). Hence, pathological bargain hunters have $\beta > 1$

For multiple accounts (i.e., multiple goods categories), people will purchase iff

$$\frac{w(z_i, p_i, p_i^*)}{p_i} \geq k$$

where k is a constant (similar to λ in the standard model)

- High k's are observed for seductive or additive goods (apply to luxury goods as gifts as well)
- Low k's are observed for goods that are desirable in the long run (e.g., exercise or education).

and individuals seek to maximize $\sum w(.)$ subject to $\sum p_i z_i \leq I$

Due to **local optimization** (i.e., constraint of time and category specific budget constraints, decision process is that people will buy good z at price p iff

$$\frac{(w, p, p^*)}{p} > k_{it}$$

where k_{it} is the budget constraint for category i in time period t

Alternatively, **global optimization** would require k_{it} to be constant (but ppl don't act this way).

3 implications:

1. Compounding rule
2. Transaction utility:
    1. Sellouts and scalping
    2. Methods of raising price:
        1. Increase the perceived reference price
        2. increase the min purchase required or tie the sale to something else
        3. obscure reference price (hence, transaction disutility less salient)
        4. Suggested retail price
3. Budgeting: Theory of gift giving (give something that recipients already consume in positive quantities)

Lichtenstein and Bearden (1989)

- Consistency and distinctiveness influence internal price standard and purchase evaluations.

Simonson and Tversky (1992)

- Effect of context on choice:
    - Tradeoff contrast: preference for an alternative is enhanced based on the tradeoffs withing the set under consideration are favorable.
    - Extremeness aversion: the attractiveness of an option is increased if is an intermediate option, while decreased if it is an extreme option.

Weber and Johnson (2009)

- Summary of mindful judgment and decision making

Sokolova et al. (2020)

- Left-digit bias (e.g., Difference between $4 and $2.99 is larger than between $4.01 and $3) is a cognitive bias we all have (independent of culture).
- Left-digit bias is stronger in stimulus based price evaluation (people see focal and reference price at the same time) because people rely on perceptual representation of prices without rounding them
- Left-digit bias is weaker in memory-based price evaluations (people can retrieve at least one price from memory) because people rely on conceptual representations, which makes them more likely to round the prices.
- Perceptually, 2.99 is represented as a sequence of digits, so it's represented as 2.99 itself.
- Conceptually, 2.99 is coded as 3 (nearest accessible round number)

34.7 Goals and Motivation

Difference between goal and wants: you make progress (strive) for goal, but you do not necessarily make progress on your want (in the literature, there isn't want0

Difference between goal and motivation: goal is regarding progress, while motivation is related to wish, and desire. (similar to wants). In the literature, sometimes interchangeable. But used to be that goal (end state), motivation (the drive to the end state)

Goal can be salient or not.

Kunda (1990)

- Motivated reasoning
- Motivation is mediated cognitive processes to affect reasoning.
- Motivation is defined as "any wish, desire, or preference that concerns the outcome for a given reasoning task." (p. 480)
- Reasoning tasks include: forming impressions, determining one's beliefs and attitudes, evaluating evidence, and making decisions.
- Motivated reasoning phenomena include:
  - those in which the motive is to arrive at an accurate conclusion
  - those in which the motive is to arrive at a particular, directional conclusions.
- Both kinds of goal "affect reasoning by including the choice of beliefs and strategies applied to a given problem." (p. 481)
  - Accuracy goal uses most appropriate beliefs and strategies

  - Directional goal uses those that most likely yield the desired conclusion

- Reasoning Driven by **Accuracy** goals:
  - People are more driven in terms of cognitive effort when you want accuracy-driven reasoning
  - People are motivated to be accurate when (p.481)
    * They are evaluated
    * expected to justify their judgments
    * expected their judgments to be made public
    * expected their evaluation to affect th evaluated person's life
  - But it does not remove bias or improve reasoning

- Reasoning Driven by **Directional** goals:
  - people have "illusion of objectivity" to search for or create accessed knowledge to construct beliefs that support the desired conclusion

- Biased Accessing of Beliefs
  - Dissonance research: when holding two contradictory cognition, people encounter unpleasant state of cognitive dissonance that they strive to reduce by changing or or more of the relevant conditions (Festinger, 1957). But studies on dissonance can also be reinterpreted from self-perception: subject infer their attitudes from their behaviors due to limited access to attitudes (but people still accept attitude change in dissonance experiments are from motivation). **Arousal is important for motivated reasoning**

- Cognitive processes can't fully account for self-serving biases, but it helps motivation affect reasoning.

Bagozzi and Dholakia (1999)

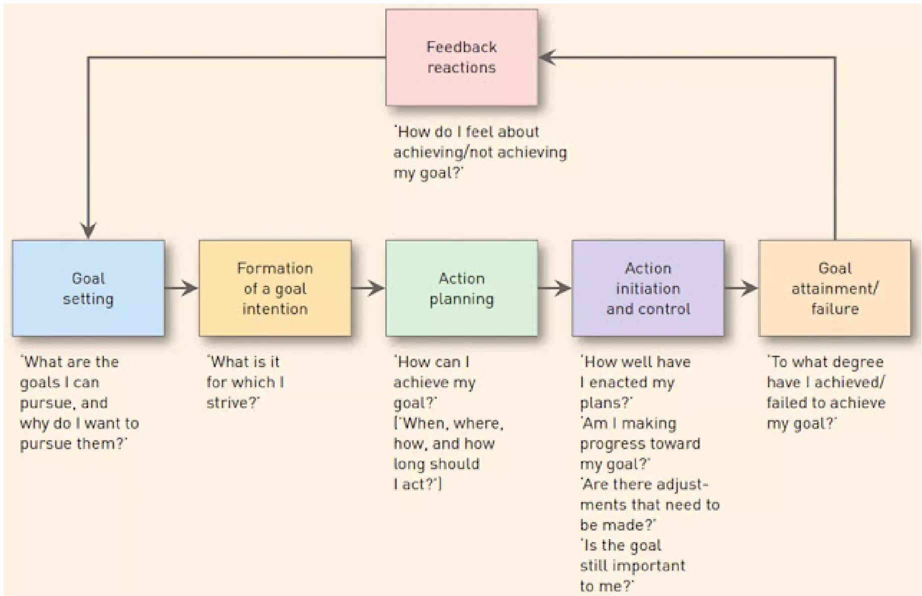
- **Goal Setting**: answers (1) What are the goals I can pursue?" (2) Why do I want or not want to pursue them?", which can be activated either
  - Externally: opportunities present
  - Internally (or imperative): consumers construct goal schemas, or choose form self-generated alternatives.

- Types of goal pursuit activities:
  - **Habitual** goal-directed behavior (though deliberative processing or learning via classical or operant conditioning)
  - **Impulsive** goal-directed behavior

  - **Volitional** goal-directed behavior: Goal intentions as
    * End performances: (use theory of reasoned action, but the author of the theory stated that it should not be used for outcome for end-state goals (Ajzen and Fishbein 1980, p 29-30)), also termed "behavioral intention"
    * Instrumental acts or implementation intention (defined as "intention to perform a goal-directed behavior given that future contingencies occur" (p. 21))

- Goal Setting can occur
  - **Consciously**: can arise from
    * Coercion or reward power, or virtue of others' position
    * Automatically due to biological, emotional or ethical forces
    * reasoned reactions to external stimuli, or internal stijli.
  - **Unconsciously**: see Bargh (1990) auto-motive model, which include habituated actions
- Knowledge in the cognitive system is derived from Barsalou (1991)
  - Exemplar learning (bottom-up, automatic process)
  - Conceptual combination (manipulation of existing knowledge) : where goal-derived categories come from.
- Goal Striving
  - Intention is the the bridge between goal setting and goal striving
- When decisions are fulfilled long after intention was formed, we called delayed intentions: which requires
  - prospective memory: remember to perform an action at a future point in time
  - retrospective memory remember the content of the action to perform and the conditions for its execution.
- General representation of goal hierarchy: subordinate goals → Focal goal → Superordinate goals

(p. 20)



Fishbach and Dhar (2005)

- Goals are seen as "cognitive structures that can be represented in terms of movement and progress toward some abstract and desirable end state or in terms of commitment to a fixed and desirable end state" (p. 370).
- Consumer have multiple goals (instead of a single goal from the original self-regulation process literature)
- "**actions that are used to infer goal progress** act to liberate the individual and thereby **increase the likelihood of pursuing incongruent actions**, whereas the same actions interpreted in terms of goal commitment elicit a tendency to subsequently maintain the pursuit of the focal goal." (p. 370)
- Goals as excuses or guides: goal progress vs. goal commitment.
  - Goal commitment is defined as " an inference concerning the strength of a goal" (p. 370)
  - Goal progress refers to "the pursuit of a previously defined goal." (p. 370)
  - Goals include long-term objectives and salient short-term temptations (Trope and Fishbach, 2000)
- The order of actions to pursue different goals does not matter, but future progress are subject to cognitive and motivational biases, then we overestimate their future goal progress, then more likely to switch to pursuing another goal.

Chernev (2004)

- Choice between either
  - Preserve the status quo
  - Departure from the status quo
- Preference for the status is a function of goal orientation, which is more pronounced fro prevention-focused than for promotion-focused consumers.
- Goal orientation on status quo preference is independent of loss aversion

Dalton and Spiller (2012)

- Implemental planning increase perceived difficulty of executing multiple goals, which undermines commitment to goals, and their success.
- Framing the execution of multiple goals can reduce this negative effect

Etkin et al. (2015)

- Perceived greater conflict between goals increases the feeling of time constrain (mediated by stress and anxiety)
- Solution (questionable):
  - Slow breathing
  - Anxiety reappraisal
- People naturally encounter goal conflict
- Goal conflict increases with stress and anxiety (Riediger and Freund, 2004)
- Stress and anxiety increases feel pressed for time (p. 395)
- Feeling time constrained increase consumers' valuation of their time more highly (more willing to pay to save time)
- Authors used Mturk for study 1 and 2

34.8 Culture and consumer behavior

Markus and Kitayama (1991)

- In Asian cultures, people prefer relatedness and interdependence, while American culture values independence (i.e., focus on the self and expressive of inner attributes)

Aaker and Maheswaran (1997)

- Prediction of dual process models is evaluated in the context of cross-culture: it's robust across culture
  - Elaboration LIkelihood Model (Petty and Cacioppo, 1979)

  - Heuristic-Systematic Model (Chaiken, 1980)
  - The concurrent occurrence of both are called "additivity". It happens when both both heuristic cue-related and attribute-related thoughts are generated (which likely under cases where one does not have contradictory info).
  - When systematic processing dominates (i.e., overriding heuristics), we call "attenuation"
  - Heuristics cues can be consensus info (which is more prevalent in collectivist culture)

- "Perceptual differences in cue diagnosticity account for systematic differences in persuasive effects."
- In layman's term, the model is transferable across cultures, but should incorporate cue diagnosticity in the model.
  - Cue diagnosticity refers to "the extent to which consumers perceive that inferences based on the information alone would be adequate to achieve their objective." (p. 322)
- Individualism-Collectivism (Cousins, 1989; Singelis, 1994; Triandis, 1989)
  - Individualism: separateness, internal attributes, uniqueness of individuals
  - connectedness, social context, relationship

Rodas et al. (2021)

- Paradox brands (i.e., brands that have contradictory brand meanings) are more appealing to bicultural consumers (due to greater cognitive flexibility, especially those who use an acculturation strategy).
- "a paradox brand as a brand identity that includes brand associations that appear to be contradictory in nature." (p. 2)
  - where associations can be either specific (e.g., product benefits) or abstract/symbolic (e.g., brand personalities, value, image).
- "brand identity" refers to "a set of brand associations that are selected by marketers to represent what the brand stands for and/or aspires to be." (p. 2)
- brand identity (only positive association) is from firms' perspectives, whereas the brand image (both positive and negative association) is from consumers' perspectives.
- Contradiction can emerge from the unlikelihood of certain brand associations co-occurrence.
- Brand values are defined under Schwartz (1992) (Value structure)

- Greater cognitively flexible can lead to greater engagement in paradox brands even for monocultural consumers.

Chen et al. (2005)

- Cross-cultural effect on future evaluation: Westerners are more impatient thus discount the future more than Easterners. In other words, Westerners value immediate consumption more than Easterners.
- "Easterners are faced with the threat of a delay in receiving a product (i.e., a prevention loss), they are more impatient, whereas when Westerners are faced with the threat of not being able to enjoy a product early (i.e., a promotion loss), their impatience increases." (p. 291)

Triandis et al. (1988)

- U.S. individualism is reflected in
  - self-reliance with competition
  - Low concern for ingroups
  - Distance from ingroups
- While allocentric person feel that they receive more and better quality from their social support, idiocentric person feel more lonely .

Nisbett et al. (2001)

- East Asians are more holistic with focus on assigning causality to the whole field and disregard categories and formal logic because they rely on "dialectical" reasoning
- Westerners are more analytic because they focus on object and its category, and use formal logic.
- "Social organization and social practices can directly affect the plausibility of metaphysical assumptions" (p. 292)
- "Social organization and social practices can influence directly the development and use of cognitive processes" (p. 292)

34.9 Prosocial behavior and morality

Sachdeva et al. (2009)

- An internal balancing of moral self-wroth can result in moral or immoral behavior.
  - If people perceive their moral identity is affirmed (i.e., perceive that they are good), then they feel that they are justified to act immorally

- If people perceive their moral identity is threatened (i.e., perceive that they are immoral), then they want to regain moral identity by performing moral behavior

Yin et al. (2020)

- Based on field and lab experiments, the effectiveness of pregiving incentives (PGIs) (e.g., either low-value monetary - coins, and non-monetary - greeting cards) depends on the organization's goals.
- For direct mail campaigns, monetary PGIs increase response rates (in donor acquisition campaigns) and likelihood to open a read acquisition letter (vs. a non-monetary or no PGIs)
- ROI and average donations for monetary PGIs (for direct mail campaigns) is lower than (that of non-monetary or no PGI) due to the fact that monetary PGIs increase exchange norms, and decrease communal norms (even accounting for manipulative intent, anchoring and adjustment heuristics).

Chernev and Blair (2015)

- CSR increases not only public relations and customer goodwill, but also customer product evaluation.
- Consumers perceive that products from companies performed CSR to be better performing (even though they can experience and observe the products or when CSRs are unrelated to the company's core business).
- This effect is lowered when consumers believe that company's behavior is driven by self-interest as compared to benevolence.
- Hence, acting good (and people don't know your true intent) can lead to better company's performance.

Graham et al. (2009)

- 5 sets of moral intuitions:
  - Harm/care
  - Fairness/reciprocity
  - Ingroup/loyalty
  - Authority/respect
  - Purity/sanctity
- Liberals put more weights on the first two foundations (i.e., Harm/care, and Fairness/reciprocity)
- Conservatives weight all 5 foundations equally.
- This difference is due to abstract assessments of violence or loyalty, moral judgments of statements and scenarios , etc.

Haidt (2001)

- From the perspective of rationalists, moral judgment is the result of moral reasoning.
- This article argues that moral reasoning is a post-hoc construction of judgments.
- From the intuitionists' view, reasoning is not only based on individual assessment, but also social and cultural influences.

Tangney et al. (2007)

- Moral emotions can influence the link between moral standards and moral behavior.
- Negatively valenced "self-conscious" emotions include shame, guilt, and embarrassment.
- A review on shame and guilt.

Kristofferson et al. (2013)

- Subsequent helping behaviors after initial support token depends on:
  - Impression management
  - Desire to be consistent with one's own values.
- Under private initial support, consumers are more likely to exhibit greater helping later as compared to public display.

34.10 Consumer well-being & Food Consumption Decisions

(Norton et al., 2012)

- The IKEA effect: increase in volaution of self-made products
- People evaluate their own creation similar in value to that of experts, and expect other to have similar opinions
- Labor leads to love only when one has successful task completion
  - When the task is destroyed or incomplete, there is no IKEA effects

Raghunathan et al. (2006)

- The portrayal of unhealthy product increases food's inferred, and actual taste, and more preferable when the is more hedonically salient.
- This result is robust among those who believe the negative correlation between healthiness and tastiness as well as those who do believe in such correlation.

- To change this negative correlation beliefs, authors suggest that we can educate consumers with better information about the definition of "healthy"
- The unhealthy = tasty intuition can come from
  - Internal source (e.g., personal; experience and self-observation). There is an inverse relationship (compensatory) between wholeness and hedonic potential
  - External source (e.g., environmental cues)

Shah et al. (2014)

- (Price) Surcharge or Unhealthy label along cannot change the demand for unhealthy food
- The combination of both can reduce demand for unhealthy food.
- Among women, unhealthy label can be as effective as unhealthy label + surcharge
- Among men, unhealthy label increases the demand for unhealthy foods as compared to unhealthy label + surcharge.

Berry et al. (2017)

- Since there is no formal definition of "natural", companies exploited this loopholes.
- based on activation theory and inferential processing, the mediation path from natural claims to product evaluation is via consumers' attribute inferences.

Liu et al. (2019)

- Consumers use type as the primary dimension and quantity as the secondary dimension in judging food's healthiness

Woolley and Liu (2020)

- Magnitude estimates refers to when "consumers judge whether something has"very few" to "many" calories" (p. 147).
  - Under which, people think that a smaller portion of healthy food has more calories than a larger portion of healthier food
  - Sensitive to type (healthy vs. unhealthy)
- Numeric estimates refers when "consumers estimate a number of calories" (p. 147)
  - Under which , people think that a larger portion of healthier food has more calories than a small proportion of healthy food.
  - Sensitive to type (healthy vs. unhealthy) and quantity (large vs. small)

- Food healthiness is processed before quantity.
- The two modes will converge if quantity information is made first (primary) or in an intuitive way.

Moorman (1990)

- Consumers characteristics (familiarity and motivation) and stimulus characteristics (info format and content) influence information processing and decision quality of how they use nutrition information

Haws et al. (2019)

- People choose any-size-same-price beverage because they think they can get more value (in financial terms). This effect is so strong that with calorie postings, this demand is still intact.
    - Finding is robust under diet vs. non-diet beverage (rule out that customers don't see value in getting more calories, but only from saving money).
- Graphic health intervention can still have an effect on the appeal of larger sizes

Other resources:

- Robitaille et al. (2021)
- Longoni et al. (2019)
- VanEpps et al. (2021)
- Woolley and Liu (2020)

34.11 Digital marketing and WOM

Berger and Milkman (2012)

- Positive emotional valence content is more likely to be shared
- Activation = physiological arousal induces action.
    - Low arousal = deactivation = relaxation (Feldman Barrett and Russell, 1998)
    - HIgh arousal = activation = activity (Heilman, 1997)

1. Examine 7000 articles from thew New York Times
    1. Examine emotionality, prominent features, interest evoked can affect likelihood to make the most email list. (controlling for practically content usefulness, interestingness, surprise, release timing and author fame (using hits for first author's full name from the number of

   Google hits), writing complexity, author gender, article length and day dummies).

   2. Robustness: control for article's general topic.

2. Lab experiments

   1. Amusement case (fictitious): high arousal

   2. Anger case (real): high arousal

   3. Sadness (real vs. fictitious): low arousal

All hypotheses are confirmed

Potential confounders: structural virality

How likely they would share a story? (no social risk involved - risk to other weak ties, hence the effect might be inflated, the same thing with the New York Times study)

All experiments have low participant numbers

Tellis et al. (2019)

Two field studies

- Information-focused content is less likely to be shared (exception risky contexts)
- Positive emotions (e.g., amusement, excitement, inspiration, warmth) are more likely to be shared
- Drama elements (e.g., surprise, plot, characters, babies, animals, celebrities) increase arousal, which in turn increases sharing.
- Prominent placement of brand name (brand prominence)
- Emotional ads are shared more on general platforms (Facebook, Twitter) as compared to professional one (e.g., LinkedIn), while informational ads are more likely to be shared on professional ones.
- Optimal length is 1.2 to 1.7 min ads.

Third study: identifies predictors of sharing

Lovett et al. (2013)

Melumad and Pham (2020)

Vosoughi et al. (2018)

Moore (2012)

Naylor et al. (2012)

Nguyen et al. (2020)

34.12 Experiential consumption and time

Weingarten and Goodman (2020)

- When compared to material items, purchasing experiences bring consumers more enjoyment.
- The meta analysis found that it's possible that the experiential advantage has less to do with happiness and willingness to pay and more to do with relatedness.
- Negative experiences, isolated experiences, lower socioeconomic status consumers, and when experiences deliver a similar amount of utilitarian advantages as material items lessen the experiential advantage.

Oh and Pham (2021)

- Experience of fun depends on
  - Hedonic engagement
  - A sense of liberation
- 4 situational facilitators:
  - Novelty
  - Social Connectedness
  - Spontaneity
  - Spatial/temporal boundedness
- Fun is different from happiness (can view this article for arguments)

Huang et al. (2009)

- The Internet blurs the lines between search and experience goods by lowering the costs of acquiring and sharing information and providing new methods to learn about products before purchasing them. Similarly, disparities in the types of information sought for search and experience products can lead to differences in the way customers gather information and make decisions online. In traditional retail locations, there are significant variations in customers' perceived capacity to evaluate product quality before purchase between search and experience items, but these distinctions are obscured in online environments
- Consumers spend equal amounts of time online obtaining information for both search and experiential products, but their browsing and purchasing behavior for these two types of goods differs significantly.
- Experience goods need more depth (time per page) and less breadth (total number of pages) of searching than search goods. Furthermore, for experience products, free riding (buying from a shop other than the primary

source of product information) is less common than for search goods. Finally, the existence of other consumers' product reviews and multimedia that allow consumers to interact with products prior to purchase has a higher impact on consumer decision-making.

- The inclusion of other customers' product reviews and multimedia that allows consumers to interact with products prior to purchase has a bigger impact on consumer search and purchase behavior for experiences than for search goods.

Gilovich et al. (2015)

- Experiential purchases provide more satisfaction and happiness than material purchases:
  - Purchases of experiences improve social relations more quickly and effectively than purchases of tangible stuff.
  - Experiential purchases are important in shaping a person's identity.
  - Experiential purchases are judged on their own merits and do not elicit as many social comparisons as material purchases.

Chan and Mogilner (2016)

- Regardless of whether the gift giver and recipient consume the item together, experiential gifts increase relationship strength more than financial gifts.
- The boost in relationships that receivers get from experience gifts is due to the depth of emotion created when they consume the presents, rather than when they get them.
- As a result, giving experiential gifts has been highlighted as a particularly successful kind of prosocial spending.

Tully and Sharma (2017)

- Even though experiential purchases have a shorter physical lifespan, consumers are more likely to borrow for them than for material expenditures.
- When a transaction is framed as more experience than material, people are more inclined to borrow.
  - This effect emerges because customers' sensitivity to missing out on scheduled consumption makes purchase time more essential for experience purchases.

Dai et al. (2019)

- Based on Amazon reviews and experiments, Consumer reviews are less likely to be trusted for experiential purchases than they are for material ones.

- This impact stems from the idea that ratings of experiences are less representative of the purchase's objective quality than reviews of actual objects. These findings reveal not only how **word of mouth** influences different types of purchases, but also the psychological mechanisms that underpin customers' reliance on consumer reviews. Furthermore, these findings imply that people are less responsive to being instructed what to do than what to have, as one of the first examinations into how people choose among numerous experience and material buying possibilities.

Chapter 35

Marketing Mix Models

Taxonomy of models:

1. Descriptive: describe relationships between observables
2. Structural: estimate features of a data generating process (i.e., a model) that are unaffected by the changes in the variables of interest

Any type of causal inference is a form of structural estimation. (All causal identification/estimation/inference is always special case structural identification/estimation/inference).

All quantity of interest (TT, ATE, LATE, etc.) are always under some assumptions, there is no such thing as "model free"

Reduced Form: is a functional/stochastic mapping for which the inputs are

1. Exogenous variables
2. Unobservables ("structural errors")

and outputs are endogenous variables (satisfy independence condition wrt unobservables) (e..g, $Y = f(X, Z, U)$)

Reduced form is obtain by solving a structural model for each endogenous variable as a function of the exogenous observables and structural errors

For example,

The perfectly competitive supply and demand

$$
\begin{cases}
Q &= D(P, X, U_d) \\
P &= MC(Q, Z, U_s)
\end{cases}
$$

after solving for equilibrium, we derive the reduced form relations

$$\begin{cases} P &= p(Z, X, U_s, U_d) \\ Q &= q(Z, X, U_s, U_d) \end{cases}$$

Reduced Form is also under structural models. Some also say "reduced form structural model"

Reduced form is not recommended in applied econ because it's either wrong or logically incoherent

You always need to name your unobservable(s) under the error terms.

Identification (p. 29):

- a **structure** S is a data generating process (i.e., a set of probabilistic or functional relationships between the observable and latent variables that implies ("generates") a joint distribution of the observables
- Let $\mathcal{S}$ be the set of all structures; $S_0 \in \mathcal{S}$ the true structure
- a **hypothesis** is any nonempty subset of $\mathcal{S}$
- hypothesis $\mathcal{H}$ is true if $S_0 \in \mathcal{H}$
- a structural feature $\theta(S_0)$ is a functional of the true structure
- A structural feature $\theta(S_0)$ is **identified** under the hypothesis $\mathcal{H}$ if $\theta(S_0)$ is uniquely determined within the set $\{\theta(S) : S \in \mathcal{H}\}$ by the joint distribution of observables

Analyze

1. Transaction history:
    - Examples:
        - Purchase incident, brand choice, quantity. The product of these three things can give demand.
        - Prescription history -> Demand
        - Casino visits, spending in casino -> Demand

Modeling Issues. Why are we doing this?:

1. We model demand because we want to relate demand to marketing activity.
2. We also want to optimize and target

Heterogeneity is at the heart of marketing (especially in the brand choice models)

- Unobserved heterogeneity:
    - Differences in taste and **preferences**

    - Differences in **responsiveness** to marketing lever
    - Structural heterogeneity: The decision making process are not the same (newer one).

- A priori segmentation: estimation problem is easy (if you don't the problem of $p >> n$ and you actually observed heterogeneity)
    - Firms segment based on characteristics, and they are different.

Two ways to handle heterogeneity

1. Latent Class Models
2. Individual Level Heterogeneity

Endogeneity

- structural component allows for jointly modeling demand and supply

35.1 Discrete Choice Models and Continuous Heterogeneity

- Discrete and Continuous Heterogeneity
- Price Customization
- Targeting

Models:

- Random Coefficients Logit
- Purchase Incidence
- Brand Choice Models

(Chintagunta and Nair, 2011)

- Goals of demand analysis (which affect model-form)
    - Forecast: (not much causal inference)
        - Examples:
            - Aggregate: (Bass, 1969); (Dekimpe and Hanssens, 1995)
            - Individual: (Guadagni and Little, 1983)
        - Descriptive model in stable environments
        - Structural for **radically different counterfactuals**
    - Measurement
        - usually used under experiments and causal inference

    * Structural models
    * Reduced-form, causal effects models
  - Testing
    * Reduced-form, causal effects models

- Demand, supply, and marketing mix are endogenously determined.
  - Best case: find exogenous shocks to the system to estimate
  - Impose supply model into the demand estimation step (p. 980)
- Counterintuitive to assume utility maximization for estimating consumer-level models, instead of firms. But we observe evidence of well-fitting model for the demand-side, but not yet in the supply side. But lack thereof evidence still does not mean that it's wrong, it's just mean we need more development.
- Building blocks of individual-level demand models
  - Direct utility specification of demand system
  - Indirect utility specification of demand systems.

(Lehmann et al., 2011)

- Tradeoff between rigor (sophistication) and relevance
- Basic discipline migrated and viewed as more sophisticated, which lead to arms race. (cascade more and more sophisticated)
- Execution rigor > idea quality. We should view analytical rigor and substantive content equally.
- Impact:
  - Citation
    * Breadth and reach (to other disciplines)
    * Game the system: cite reviewers.
- A good research paper should be (p.162)
  - reasonably realistic/general
  - relatively simple and robust
  - insightful
  - reasonably communicable
- More complex methods are only appropriate when (p. 163)

(Chintagunta et al., 2005)

- Try to understand the role of potential weekly brand-specific characteristics that influence consumer choices, but they are unobserved
- Endogeneity
- Inclusion of the UBC

UBC: they are the first guys to do it in the dis aggregate model.

(Zhang and Wedel, 2009)

(Dong et al., 2009)

- What is the benefit of individual-level targeting in the presence of strategic behavior by other firms?
- Setting
  - Pharmaceutical industry
  - Individual-level targeting to physicians
  - Targeted ad (i.e., detailing)
- Model
  - Physician response: capture the responsiveness each physician to targeting
  - Firm detailing choices: firms strategically target and how much ad

(Nair et al., 2017)

35.2 Structural Models, Endogeneity

Good Empirical Research requires

1. Good Data
   1. Original
   2. Cool results
   3. Exogenous
2. Good Theory
   1. Interesting Hypotheses
3. Cool new approach

Analysis

1. Descriptive
2. Predictive
3. Causal

4. Prescriptive or policy-oriented.

Both 3 and 4, you need structural or experimental research.

Structural Equation Modeling is different Structural Equation

Causal as in experiments, there

In the structural model, we still want to make causal inference.

- Endogeneity

Models:

- Instrumental Variables
- Joint Estimation of Supply and Demand Models
- Empirical Bargaining Models

Ask:

- Bounds analyses:
  - (Ellickson and Misra, 2011)
  - (Manski and Tamer, 2002)

35.2.1 Background

(Reiss, 2011)

Types of Empirical models in marketing

- Descriptive (no need to concern for endogeneity): covert data into info
  1. Statements about facts
  2. High-quality and relevant data
  3. Accurate Interpretation
- Structural (also known as latent/ path models)
- Experimental (including quasi-experimental)

The data and research questions should always determine methodological approach.

Under structural models, we rely on

1. Formal formal specification linking Y and X
2. Stochastic specification connects theoretical model to data. Ex: heterogeneity helps explain the imperfect fit by including
   1. Consumer preference
   2. Consumer decision-making errors

3. Measurement errors

Structural models help recover counterfactuals.

Structural models differ from descriptive models because it can recover the structural parameters using reduced from.

Reduced form regression means that you know the structure of the data generating process.

A reduced from only exist with an underlying structural model. When researchers say they use "reduced-form analysis" when they only do regression: They erroneously assign a causal interpretation to the regression coefficients.

(Rossi, 2014)

- IV methods even with valid instruments can still have poor sampling properties (finite sample bias, large sampling errors).
- Problems with Instrumental variables in marketing
- It's hard to find instrument for **advertising** and **promotional** variables
- Lagged marketing variables are invalid instruments when advertising and promotional variables are unobserved.
- Control functions can still work under nonlinear demand model (e.g., choice model).
- Endogenous variables in marketing:
  - Price, advertising, promotion, entry order, distributions, market structure, market share, revenue, networks
  - Instruments:
    - lagged variables,
    - costs (input and wholesale prices).
      - Cost input: Theoretically good instrument for endogenous price, but hard to measure (especially marginal cost measured by BLS that has high measurement error) (p. 666).
      - Wholesales price to deal with price endogeneity is plausible (but people can still argue that wholesalers set price in anticipation of adveritsing and promotion). But they have less variation (frequency of changes is lower than retail price) hence using wholesale price as an instrument, you account for the difference between long-run and short-run effects of price, instead of endogeneity.
    - other products. Good instrument for endogenous price when unobserved demand shocks (that vary by market and time, for those shocks that only vary by market, but not only time, FE

can only fix) are uncorrelated across market (exogeneity), but costs are correlated across market (relevance).

    * fixed effects (brand, time dummies). Good but only for linear models.

        · Price endogeneity: (Villas-Boas and Winer, 1999) (another flaw - no heterogeniety and state-dependence for packaged goods panel) uses lag price as instruments, but it is bad (unmatched time) and is not supported.

    * demographics (bad instruments),
    * product characteristics (Berry et al., 1995),
    * price indices,
    * display and features.

  - People tend to use lagged variables to fix endogenous price (price correlates with unobserved quality, which induces downward endogenous in price sensitivity).

- The Hausman test can only be used to determine the validity of one set of instruments based on the validity of another set of instruments.

35.2.2 Examples

(Berry et al., 2004)

- Second-choice data as an instrument: if consumers hadn't purchased their cars, what would have been their second choice. But you still need high variation in this variable to estimate the model
- General Motors data set: second choice = substitution pattern. (This might only help with non-parametric estimate)
- Prior models: To estimate substitution coefficient (pattern): match consumer attributes to consumer choices (observables).
  - Identification: estimation based on **changes across markets (or across time)**.
  - Assume the distribution of consumers' underlying tastes, conditional on an observed distribution of consumer incomes and demographics (i.e., observables) is **constant** across markets and time.
  - Hence, substitution coefficient is estimated from the data on changes in (1) characteristics and number of product, and (2) changes in observed consumer attributes across markets.
  - In other words, estimation is based on

    * 1 switchers of consumers (i.e., people buy different product when there are changes in product prices, choice set, or other characteristics).
    * 2 different people (distribution of consumer attributes) will choose different product for a set of product.

- But the prior models are without unobserved heterogeneity and only with observed consumer attributes are actually bad at replicating the substitution pattern observed in the second-choice data.
- This paper identification strategy is based on the second-choice data
  - Advantages:
    * (1): direct data-driven substitution pattern.
    * 2 more identification power without the exogenous changes in choice sets.
  - Disadvantages:
    * Since second-choice data is available for single market (i.e., not across market), we can't estimate across-market pattern of substitution.
- Future research:
  - Combine across market second-choice data (i.e., SUVs switch to minivan).

Baseline model (Berry et al., 1995)

$$u_{ij} = \sum_k x_{jk}\tilde{\beta}_{ik} + \xi_j + \epsilon_{ij}$$

where

- u_{ij} = linear utility of consumer i consuming product j ($j \in [0, J]$ where $j = 0$ means the consumer did not buy from any of the competing market
- k = observed product characteristics
- r = observed household attributes.
- x_{jk} = observed product characteristics
- ξ_j = unobserved product characteristics (pick up all the impact that weren't observed, but it might also correlate with the observe, in which case results in small price elasticities).
- ϵ_{ij} = individual preferences (independent of the product attributes and each other).

- $\tilde{\beta}_{ik} = \bar{\beta}_k + + \sum_r \mathbf{z}_{ir} \beta^o_{kr} + \beta^u_k \mathbf{v}_{ik}$ (consumer taste)
  - $\mathbf{z}_i$ = vectors of observed consumer attributes
  - $\mathbf{v}_{ik}$ = vector of unobserved consumer attributes
  - This model also assumes that there is only one unobserved characteristics (i.e., without subscript r) per household.

Substitute the above two equation

$$u_{ij} = \delta_j + \sum_{kr} x_{jk} \mathbf{z}_{ir} \beta^o_{kr} + \sum_k x_{jk} \mathbf{v}_{ik} \beta^u_k + \ _{ij}$$

where

- $\delta_j = \sum_k x_{jk} \bar{\beta}_k + \xi_j$ (choice-specific constant). (equation 4)

Without any additional assumption on ξ (i.e., product characteristics), we can have consistent estimators of $\ = (\ , \ ^{\mathbf{o}}, \ ^{\mathbf{u}})$

But we need to know the identifying assumption of ξ_j to be able to estimate $\bar{\beta}$:

- ϵ_j are mean independent of the *nonprice* characteristics of all the products.

Estimation

- 2 choices to estimate ξ_j:
  1. Estimate $\ = (\ ^{\mathbf{o}}, \ ^{\mathbf{u}}, \ )$ (always consistent)
  2. Restrict the joint distribution of $(\xi, \mathbf{x})$ and estimate only $(\ ^{\mathbf{o}}, \ ^{\mathbf{u}}, \bar{\ })$ (efficient if there the restrictions are true, but inconsistent if the restrictions are wrong). Hence, better off with first choice.
- Choice of estimation methods:
  - ML: computationally costly
  - Method of moments: matched on 3 sets of moments
    1. Covariances of the observed first-choice product characteristics ($\mathbf{x}$)with the observed consumer attributes ($\mathbf{z}$) for estimating $\ ^o$: help identify $\ ^o, \mathbf{x}, \mathbf{z}$
    2. Covariances of first choice product characteristics and second-choice product characteristics: help identify unobserved consumer characteristics.
    3. Market share of J products: help identify (choice-specific constant).

(BLP) (Berry et al., 1995)

Question:

- Hand-waving: "For computational simplicity, ..., ϵ_{ij} have an independently and identically distributed extreme value"double exponential" distribution". Basically it was modeled this way to have a tractable form of the model's choice probabilities conditional on $(\mathbf{z}, \mathbf{v})$: $P(y_i^1 = j|\mathbf{z}_i, \mathbf{v}_i, \ , \mathbf{x})$
  - Closed-form solution: pretty close to the normal distribution (see MacFadden).
- To construct the choice set: the car characteristics: the authors only used modal vehicle (combinations of options that was most commonly purchased). and price was average price of the model vehicle.
  - Defensible thing to do
- Python implementation of this paper: (Conlon and Gortmaker, 2020)

(Draganska et al., 2010)

- How do we measure power in the distribution channel?
- Between manufacturers and retailers
  - Manufacturers
    * Bargain over profit margins with retailer
    * Bounded by agreement with retailer
    * Bargaining power comes form size of manufacturer and supplying product for retailers
  - Retailers:
    * Intense composition in mature coffee market
    * bounded by consumer price sensitivity
- A shift of bargaining power from manufacturers to retailers
- Standard models are good to measure distribution channel power.
- Bargaining position: stand to lose more (endogenously determined by the substitution patterns on the demand side)
- Bargaining power: negation skills, patience, risk tolerance (exogenous - depends on negotiation partners).
- Channel margin and split = f(bargaining position, bargaining power)
- Contributions:
  - Bargaining power is still with manufacturer (manufacturer gets over half of the pie).
  - Overall profit of the distribution channel is not a zero-sum game
  - Quantify the effects of bargaining power on channel profits

      * Bargaining power predominantly affects manufacturers
      * Bargaining power weakly affects retailers. retailer margins tied down by pricing power over consumers

(Ozturk et al., 2019)

- Impact of Chapter 11 on consumer demand for the bankrupt firms' competitors
- Possibilities:
  - Consumers go to the competitions (competitive effect)
  - reduced demand also fro the competitors (negative info about the industry: contagion effect)
- Research question: temporally local effect of chapter 11 on demand for rival firms
- Data: dealer-model-day level
- Challenge:
  - General decline in economic condition: Great Recession
  - "Cash for Clunkers" program: anticipation for the program may decrease demand
- Remedies: regression discontinuity in time (RDiT)
  - Control variables (price, ads, recalls, Macroeconomic conditions)
  - Competitors' sales patterns in Canada (where Chrysler didn't file)
- Results: Negative effect on competitors.
- The mechanism:
  - Increased consumer uncertainty about car purchases
  - Decreased cross-traffic form the bankrupt firm's dealers to competitors' dealers

Jayarajan et al. (2021) Changing the Power Equation: A Structural Analysis of the Impact of Used Car Markets on the Automobile Retail Channel

Main idea: study the automobile retail channel where retailers sell new and used cars

Structural model:

- Demand: used and new cars, heterogeneity, price endogeneity (IV)
- Supply: Oligopolistic structure with multiple retailers and dealers

Outcomes: profits, margins and power in the distribution channel

Counterfactual analysis: What if we change used cars' quality and availability?

Main result: selling used cars are important for retailers profits and bargain power.

35.3 Cross-Category and Store Choice Models

- **Models: Restricted Boltzman Machine Learning Models**

How would you name the topic for this week?

Store Choice Model -> Category Choice Model -> Brand choice -> Quantity

35.3.1 Background

(Seetharaman et al., 2005)

- Typically outcome variables of interest:
  - store choice (Which store visited?)
  - Incidence (whether the product category was purchased)
  - brand choice (which brand)
  - quantity (how many?)

Incidence Outcomes in Multiple Categories

1. Multi-category "whether to Buy" models

- Base Model:
  - (Manchanda et al., 1999): assumed **joint** distribution (not independent normal dist from the binary probit model) of two products (underestimate cross-category correlation and overestimates the effectiveness of the marketing mix as compared to (Chib et al., 2002))
  - (Chib et al., 2002): 12 products category, and find that accounting the effects of unobserved heterogeneity across households can recover the overestimated cross-category correlation and underestimated effectiveness of marketing mix.
  - (Ma et al., 2012) (publish 5 years later) address the spurious correlation due to 0 outcome (i.e., no purchase) by the multivariate logit model.

2. Multi-category "When to to Buy" models

- Multivariate Hazard model

    - (Chintagunta and Haldar, 1998): bivariate hazard model with only positive correlation between two timing outcomes
    - Ma and Seetharaman (2004) used Multivariate Proportional Hazard Model to account for both positive and negative pair-wise correlations in the outcomes.

3. Bundle Choice Models

- whether or not to buy a bundle
    - (Chung and Rao, 2003) uses nested logit with error terms follow a joint Gumbel distribution, assumes:
        - Degree of comparability among product categories
            - Fully comparable attributes (e..g, brand reliability)
            - Partially comparable attributes
            - Non-comparable attributes
        - Two types of attributes:
            - Non-balancing attributes
            - balancing attributes
    - (Jedidi et al., 2003): consumer's (random) utility = sum of reservation price + random component
        - Multinomial probit

Brand choice outcome models in multiple categories

1. Correlated marketing mix sensitivities across categories

- (Ainslie and Rossi, 1998): Multinomial Probit model of brand choice. Found correlation between responsiveness to price and feature advertising across product categories
- (Seetharaman et al., 1999) found household inertia is correlated among product categories
- (Iyengar et al., 2003): high coefficients across categories, leveraging info across categories (one observed, focal wasn't)

2. Correlated Brand Preferences across categories

(Russell and Kamakura, 1997): Poisson model for brand's purchase volume, they found Inter-category correlation in purchase volume

(Erdem, 1998) (Erdem and Winer, 1998):using multinational logit brand choice model: signaling theory of umbrella branding explains correlated quality perceptions among product categories

Other papers: (Singh et al., 2005) (Hansen et al., 2006)

Models of Multiple Outcomes in Multiple Categories

1. Incidence and Brand Choice
    1. Incidence as an alternative in a multiple choice model:
        1. Deepak et al. (2002): used Multivariate Probit (MVP) of incidence and brand choice outcomes.
        2. (Manchanda et al., 1999) found cross-category correlations in marketing mix sensitivities of household
        3. Ma, Seetharaman and Narasimhan (2005): used Multivariate Logit Model to model incidence and brand choice outcome.
    2. Incidence and Brand choice as 2 decision stages:
        1. (Mehta, 2007): Simultaneous model of incidence and brand choice
        2. Chib et al. (2005): Brand choice within each product category
2. Incidence and Quantity
    1. (Niraj et al., 2008) Two-stage bivariate logit model
3. Incidence, brand choice and quantity
    1. (Song and Chintagunta, 2007): simultaneous model: cross-category effects come from the incidence and brand choice outcomes, not from the quantity outcomes

Estimation: Bayesian framework is a better fit for this type of models. (see (Albert and Chib, 1993))

Store Choice Outcomes:

- (Bell and Lattin, 1998)
- (Bell et al., 1998)

35.3.2 Examples

35.3.2.1 (Bucklin et al., 2008)

- Changes in the intensity of mature distribution networks (by car make) influence consumer choice.
- Three measures for intensity level (for each make)
    - Dealer **accessibility** (buyer's distance to the nearest outlet): prefer closer
    - Dealer **concentration** (i.e.,the distance required to encircle a given number of same make dealers around a given buyer) (number of dealers near a buyer): prefer more dealers

  - Dealer **spread** (dispersion of the multiple dealers relative to the buyer's locations): prefer skewed toward the buyer (think of the circle). Using Gini coefficient from the Lorenz curve).

- Used logit choice model to model the correlation of the three measure with new car choices.
- found significant correlation between measures and car choice.
- Motivations:
  - Want to infer causation between distribution coverage/ intensity and sales
    * It's hard. It might depend on product categories (e..g, convenience, shopping or specialty goods).
- Focus: relationship between distribution intensity and buyer choice in consumer durables market
- Leveraging slow changes in the distribution channel, the authors probe the effect of distribution intensity on choice.
- But because it was cross sectional, need to include constant heterogeneity in preferences and other marketing mix effects to avoid confounds.
- Data: individual-level purchase record by Power Information Network (PIN), under J. Power and Associates from 1997 to 2004 in Cali.
- Different from previous literature: instead of store choice, brand choice was modeled as a function of outlet locations.

Utility:

$$U_{it}^h = \alpha_i^h + \Sigma_j \beta_j^h X_{ijt}^h$$

where

- U_{it}^h = buyer h's utility for i at time t
- X_{ijt}^h = attribute j's value at time t by buyer h
- α_i^h = product-specific constant (vary by household) (i.e., brand preference)

Heterogeneity is modeled at the zip-code level (buyers in the same zip code share α, β

Endogeneity:

1. Measurement Level: individual data, less measurement error.
2. Simultaneity: Not much changes in distribution network (with empirical evidence). Hence, unlikely

3. Sample selection: large and representative sample of Cali market.
4. Omitted variable bias:
    1. Include heterogeneity at the dis aggregate level (capture unobserved geographical effects)
    2. Since model at the make level, we have less correlation with the unobserved model-level factors
    3. Individual makes have less correlation with manufacturer unobserved variables.

Logit choice probability

$$P_{it}^h = \frac{\exp(U_{it}^h)}{\sum_k \exp(U_{kt}^h)}$$

Using Hierarchical Bayes

Choice probability buyer h in zip code z pick make i at time t

$$\text{Prob}_t^h(i|\ ^z, X_{it}^h) = \frac{\exp(\ ^{\mathbf{z}} X_{it}^h)}{\sum_j \exp(\ ^{\mathbf{z}} \mathbf{X}_{jt}^h)}$$

where

- $^{\mathbf{z}}$ = a zip-code-specific parameter vector ($^{\mathbf{z}} \sim MVN(\ , \ )$
    - $\sim MVN(\ , \mathbf{C})$
    - $^{-1} \sim \text{Wishart}[(\rho R)^{-1}, \rho]$

(Ngwe, 2017)

- Structural model:
    - Demand: sensitivity to travel distance and taste for new product
    - Supply: responses to changes in store locations.
- Outlets focus on lower-value consumers with lower desire for newness (correlation between travel sensitivity and taste for new products).
- Outlets help regular store introduce more new products (possibly improve quality).

35.3.2.2 (Donnelly et al., 2021)

- Model for estimating single product choice from alternatives:
    - Heterogeneity in Individual preferences for product attributes and price sensitivity (across products).

- Account for time-varying product attributes, and out-of-stock.

- Improvement from traditional model due to:
  - estimate heterogeneity in individual preferences.
  - estimate preferences of infrequent (purchase) custeomers

35.3.2.3 (Gabel and Timoshenko, 2021)

- Deep network model accounts for
  - cross-product relationships,
  - time-series filters to capture purchase dynamics for product with varying inter-purchase times

35.4 Policy Applications of Discrete Choice Models

35.4.1 (Khan et al., 2016)

- Variation: prices vary wiht fat content level (
- Price is determined at a regional level, and independent of local demand conditions (i.e., exogenous shocks)
- Examine price sensitivity and substitution patterns (heterogeneous for different socioeconomic groups).
- Higher price leads to more likely consumption of lower calorie milk.
  - Especially for low-income households.
- Recommendation: tax scheme based on relative prices of healthier options.
- Interesting choice of presenting data in the introduction section
- Data: IRI

35.4.2 (Rao and Wang, 2017)

- Demand reduced after the termination of the claims, (12 - 67 % monthly loss in revenue)
  - The decline effects come mainly from newcomers.

35.4.3 (Tuchman, 2019)

- Descriptive evidence for e-cig ads reducing traditional cig (i.e., e-cig is a sub of traditional cig)

- From structural models, propose counterfactual evidence for banning e-cig ad (but might increase traditional cig demand again)

35.4.4 (Seiler et al., 2021)

- Examine the impact of sugar-sweetened beverages (SBB) tax on Philadelphia, where they found that cross-shopping to stores outside the area accounted for half the reduction in sales and decreases the net reduction in sales 22%
- Key findings:
  - Tax pass through at an average rate of 97% (i.e., 34% price increase)
  - Price increase reduce quantity purchased by 46% (but half went to other stores outside of the city). Hence, the net sales of SSB decreased by 22%
  - Bottled water is not a substitute for SSB, but natural juices might.
  - Low income neighborhood just decreased demand (no increase in cross-shopping) due to limitation of transportation.
- Counterfactuals:
  - 15 cents per ounce is close to the revenue-maximizing tax rate (but 2 cents higher could be optimal because it lowers sales while costs marginally to tax revenue).
  - Initial plan of 3 cents per ounce could be detrimental (tax revenue decreases by 75%)
- The authors have to argue for the paper's contribution above the one studied in Berkeley and other places (i.e., representative demographics, and results)
- Tax on distributors and only on artificial sweetener because of financial purposes
- Data: IRI retail point-of-sale data 2015-2018, tax date = Jan 2017
- Product aggregate at the brand/diet status/pack size level. (i.e., total units sold and quantity-weighted prices at the product/store/week level.
  - 861 products (489 taxed, 372 untaxed)
  - data cover 28% of sales of taxed beverages
- Demo data from Census Bureau and obesity rates from the CDC
- Dif-n-dif research design:
  - Treatment; tax area
  - Control: 3-digit surrounding zipcode - 6-mile away (non-taxed)

- Parallel trend pretax data.

$$y_{st} = \alpha(\text{Philly}_s \times \text{AfterTax}_t) + \gamma_s + \delta_t + \epsilon_{st}$$

where

- y_{st} = quantity sold and price
- γ_s = store fixed effect
- δ_t = week fixed effect
- ϵ_{st} = error
- α = dif-in-dif coefficient

To assess heterogeneity

$$y_{st} = \tilde{\alpha}_0(\text{Philly}_s \times \text{AfterTax}_t) + (\text{Philly}_s \times \text{AfterTax}_t \times \mathbf{X}_s)'\tilde{\alpha}_1 + (\text{afterTax}_t \times \mathbf{X}_s)'\tilde{} + \tilde{\epsilon}_{st}$$

where

- $\tilde{\gamma}_s$ = store fixed effects
- $\tilde{\delta}_t$ = week fixed effects
- $\mathbf{X}_s$ = a set of store characteristics
- $\tilde{}$ = vector of coefficients capturing the change in the outcome in stores outside of Philly after the tax took effect as a function of $\mathbf{X}_s$
- $\tilde{}_1$ = the differential change in the outcome in Philly stores relative to control group as a function of $\mathbf{X}_s$
- $\tilde{}_0$ = baseline (i.e., uninteracted dif-in-dif estimate)
- two-way clustered SE at the store and the week level

No single-term $\mathbf{X}_s$ because fixed store effects already absorb all store characteristics.

Quantity:

The reason why drugstores and convenience stores experience modest to no decrease in quantity sold is because

1. They already have higher pretax price level
2. Consumers who buy at those places are less price sensitive

"Quantity decreases more in high-income areas" (contrary to intuition, high-income should respond less to changes in price, may be because of lower transportation costs).

"Obesity rates do not predict a differential quantity response."

Provided evidence for revenue maximum relating quantity sold and price elasticity.

35.5 Frontier Papers

35.5.1 (Neumann et al., 2019)

- 19 data brokers , 6 buying platforms, 90 third-party segments
- Descriptive Analysis
- Study 1:
  - Examine performance of an ad campaign with the support of data (to target customers)
  - Automated system can only delivery 59% to the target market.
  - Audience accuracy varies between platforms.
- Study 2:
  - Examine the optimization of DSPs (Demand-side platforms) for selecting data sources and ad placements.
  - Delivering performance = f(audience selection, quality of the profiles by data brokers, and other factors).
  - This study only focuses on the quality of profiles by data brokers.
  - Optimization is worse than random selection (because average accuracy of identifying the true subject is 24.4% which is less than 26.5% according to the natural distribution of the two attributes - age and gender).
  - Households with children significantly reduce the performance accuracy (due to potential usage by multiple members)
- Study 3:
  - Audience interest-based data are the new type of target (besides age and gender)
    - Sports interested
    - fitness interested
    - travel interested
  - High accuracy for this interest-based (but still variation by data brokers)
- Cost-benefit analysis

  - Cost = fixed (third-party audience info) + variable costs (cost-per-mille of online ads)
  - Ad optimization is more costly than banner (about 151% more), but compared to the gain, third party solution is still economical.

35.6 Advertising Response Measurement

- Structural, Experimental and Quasi Experimental Approaches

35.6.1 (Terui et al., 2011)

- Previous studies assume that advertising has a direct and lagged effect on consumer utility
- This study found evidence that there is no direct effect of advertising on consumer utility for mature brands.
- Data: scanner panel (laundry detergent and instant coffee)
- Advertising affect consideration sets, not the marginal utility of offering (i.e., previous studies did not account for consideration set formation, and just take the advertising effect on customer utility, later underestimate the effect of advertising on sales)
  - Hence, we should use **brand consideration as the dependent variable when studying the advertising effect**.
- Periodic advertising is still beneficial because it raises the advertising stock to be above the threshold level for brand inclusion in the consideration set.
- Contribution:
  - Account for heterogeneous consumer response to advertising and consideration set formation
  - Include a hard constraint on brand inclusion in the consideration set which helps distinguish considerations from choice in the model likelihood.
- Base model: (Gilbride and Allenby, 2004)
- Future research: can use this paper for structural models of consideration for price.
- Question: is it applicable to high-involvement products?

Model Development:

Let N be the number of choice alternatives

Consumer h has advertising stock AS_{jht} for each alternative $(j = 1, \dots, N)$

Alternative j can be in the consideration set, $C_{ht}^{AS_{jht} \ge r_h}$, of consumer h at time t when $AS_{jht} \ge r_h$ (where r_h is the threshold value of consumer h across choice alternative and time invariant. also known as **effective advertising stock**)

Elements in $C_{ht}^{AS_{jht} \ge r_h}$ can change over time with changes in AS

Consumer h utility for the alternatives in the consideration set is

$$u_{jht} = x'_{jht}\beta_h + \epsilon_{jht}$$

where

- $\epsilon_{jht} \sim N(0, \sigma_j^2 = 1)$ for $j \in C_{ht}^{AS_{jht} \ge r_h}$

The choice probability of an alternative in the consideration set is

$$P(j)_{ht} = P\{u_{jht} = \max\{u_{kht} : k \in C_{ht}^{AS_{jht} \ge r_h}\}\}$$

To make the model solvable, if a person did not watch any ad, but still purchase a brand, then his or her $r_h \approx 0$

Advertising stock is modeled based on (Bass and Clarke, 1972), (Clarke, 1976):

$$AS_{jht} = \sum_{g=0}^{\infty} \alpha_{jht-g}\rho_h^g$$

where

- α_{jht-g} is when consumer h is exposed to adverting for brand j at time $t-g$
- ρ_h is advertising diminishing effect ($0 \le \rho_h < 1$)

Advertising effect occurs instantly and diminished exponentially (to the g order), which was evidenced in experiential research design (Lodish et al., 1995) (Little, 1979)

Two other stock variables:

Brand Loyalty (Guadagni and Little, 2008) (Erdem, 1996):

$$BL_{jht} = \sum_{g=1}^{\infty} y_{jht-g}\tau_h^g$$

where

- y_{jht-g} is the purchase variable for brand j
- $0 \le \tau < 1$

- Threshold λ_h

Display Stock

$$DS_{jht} = \sum_{g=9}^{\infty} d_{jht-g} \phi_h^g$$

where

- $0 \leq \phi_h < 1$
- Threshold κ_h

35.6.2 (Narayanan and Kalyanam, 2015)

- Causal effect of position in search engine advertising listing on click-through rates and sales
- Because of selection bias, causal inference is difficult (experiments can't model bidding behavior).
- Without addressing for these selection biases, position effects on click-through rates and sales are huge, but with RD design, the estimates are smaller.
- Position effects are
  - stronger for small advertiser, or consumer with little experience with the keyword for the advertiser.
  - weaker brand or production info is included in the keyword, on weekends compared to weekdays.
- Position could affect click-through rate and purchase behavior via
  - signalling: advertising expenses signal product quality
  - consumer expectation
  - sequential search: learned experience by costumers that better results are higher in the search engine.
  - attention: consumers only pay attention certain parts of the screen.
- Endogeneity problems:
  - Brands target keywords with high conversions. (inflate the causal effect of viewing ad on conversions)
  - Position is determined by online auction. (randomization of bid would not lead to randomization of position)
    * Cannot use parametric selection equations because positioning is determined by complex processes

- Solution: RD
  - Running variable: adrank = f(advertisers' bids, quality score) -> (sharp cutoff)
  - nonobservability of competitors' adrank prevents selection into treatment by the focal firms. Hence, unless you have both focal and competitors bids and Adrank, you can't do RD here (but the authors they have both a focal advertiser and its main competitors - before M&A).
- Moderators (Decision by the advertisers):
  - Match: Exact vs. Broad (for keywords)
  - Advertisers: e.g., higher vs. lower quality firms
  - Experience and advertising are substitute (Narayanan and Manchanda, 2009): recent consumers are not going to change their probability of buying when exposed to ads, as compared to those who have not recently experienced the product.
  - Category vs. brand terms: prior literature shows category terms precede brand terms (people use broad search terms = novices = rely more on ad position).
  - Weekday vs. weekend: search cost lower on the weekends. Thus position effects are stronger on weekdays.
- Selection Issues:
  - Selection on observables:
    * Differences in keywords, match types, advertisers
  - Selection on unobserveable:
    * Bidding behaviors by advertisers (both ways: positive - higher CRT invest more and negative - higher CTR invest less).
    * Competition:
- Possible Solutions:
  - Experiments: but cannot control/randomize competitors.
  - Model selection parametrically: hard to believe
  - Latent Instrument: but rely on a single latent instrument, outcomes are normal (hard to believe)
- RD:
  - Assumptions:

        * Brands can't manipulate its position: unobservability of competitors Adrank even ex-post
        * Forcing variable is continuous: Adrank
    - Procedure:
        * Selection of observation (those close to the cutoff)
        * Selection of the bandwidth (how wide the window, bias and variance trade-off)
        * Use local linear regression within the bandwidth
        * Test different bandwidth using "leave-one-out cross validation"

- Data: 28.5 mil daily obs -> 13.1 mils (with 2 firms involved) -> 414,310 obs with adjacent observations.
- Results:
    - Both OLS and Fixed Effect inflate the effect of position.

Table 2 **Position Effects on CTRs**

	Naive estimates (CTR %)				RD estimates (CTR %)		
Position	Baseline	Num. of obs.	OLS estimate (*p*-value)	Fixed eff. estimate (*p*-value)	Baseline	Num. of obs.	Estimate (*p*-value)
2 to 1	1.8325	141,492	2.4413 (0.0000)	0.4286 (1.39e–23)	2.3260	4,736	0.4796 (0.0049)
3 to 2	1.1597	192,346	0.9217 (0.0000)	0.0643 (0.0138)	1.2864	14,134	0.0597 (0.2884)
4 to 3	0.8732	124,488	0.5588 (0.0000)	0.0470 (0.0690)	1.0348	14,350	0.1109 (0.0147)
5 to 4	0.7280	63,754	0.2642 (7.82e–09)	−0.0406 (0.1679)	0.8621	9,318	0.0570 (0.2069)
6 to 5	0.6405	26,564	0.1394 (0.0198)	−0.0318 (0.4126)	0.7746	4,532	0.1294 (0.0147)
7 to 6	0.4998	11,258	0.1458 (0.1729)	0.0479 (0.4286)	0.6345	1,864	0.1239 (0.0474)
8 to 7	0.2740	4,960	0.1708 (0.0703)	0.0583 (0.3185)	0.3855	426	−0.0058 (0.9601)

Figure 35.1: Table 2: Positive Effect on CTRs (p. 400)

- OLS estimates are positively biased (selection on observables and unobservables),
- Fixed effects correct for selection of observables (a little lower than OLS) (selection on unobservables causes negative bias)
- With varying selection bias by position, it's unlikely that parametric approaches or instrumental variables can accommodate for this.
- The effect of position on CTR and later on sales is not straight forward. And only moving from 6 to 5 has a significant difference to sales. (which is right above the page fold, or it might be due to consumer perceive top 5 as higher quality).

Table 3 Position Effects on Number of Sales Orders

	Naive estimates (CTR %)				RD estimates (CTR %)		
Position	Baseline	Num. of obs.	OLS estimate (*p*-value)	Fixed eff. estimate (*p*-value)	Baseline	Num. of obs.	Estimate (*p*-value)
2 to 1	0.0045	141,492	0.0049 (4.75e–13)	0.0009 (0.0550)	0.0042	4,338	0.0006 (0.7299)
3 to 2	0.0023	192,346	0.0036 (8.93e–15)	0.0005 (0.1757)	0.0023	12,116	0.0007 (0.3656)
4 to 3	0.0021	124,488	0.0018 (0.0001)	−0.0002 (0.5034)	0.0018	13,332	−0.0009 (0.1549)
5 to 4	0.0019	63,754	0.0006 (0.2628)	−0.0002 (0.6777)	0.0017	9,206	0.0011 (0.1695)
6 to 5	0.0011	26,564	0.0001 (0.8470)	−0.0005 (0.3203)	0.0008	4,630	0.0017 (0.0409)

Figure 35.2: Table 3: Position Effects on Number of Sales Order (p. 401)

35.6.3 (Lewis and Rao, 2015)

- Individual sales data are volatile which leads to high experiments cost to require precise estimate.
- Data on 25 field experiments (cost \$2.8 mil in digital marketing)
- Evidence that observational methods (i.e., control for observables) are untrustworthy to measure returns to advertising.
- Economic universe
- Weak evidence of advertising effectiveness.

35.6.4 (Gordon et al., 2019b)

- Compare experiments results with observational models, where observational methods do not show the same effect as the randomized experiments.
- Demand (click-through rate) universe

Chapter 36

Strategic Dynamic Models

(Erdem and Keane, 1996) is a good paper to think of structural modeling in marketing

What is interesting and impactful?

- Correctness is not king
- Challenge audience assumptions
    - Too strong = absurd
    - Took weak = not interesting
    - Sweet spot

Pitfall in Empirical Approach

- Selective (biased) sample
- Omit competition
- Ignore
    - Dynamics
    - Heterogeneity
    - Endogeneity

Marketing Complexity

- Sales response to a single marketing instrument
- Marketing Mix Interaction
- Competitive Effects
- Delayed Response

- Multiple Territories
- Multiple products
- Functional Interactions
- Multiple Goals

Methodology

- Verbal Model
- Mathematical Model

Purpose

- Measurement models
    - Conjoint model
- Decision support models
- Theoretical models

36.1 Market Entry

Pioneering paradox

- Market entry massively important
    - Big decision
    - Start of business strategy
    - Perennial conflicts:
        * Pioneer vs. 2nd move vs. late entry
        * Incumbent vs. Entrant
    - Huge payoff if played well
- One explanation: Fixation
    - Fixation: focus on micro hurdle /breakthrough
    - Entrenchment: hang on to /perfect early success
    - Marketing Myopia
    - Baggage: routines. bureaucracy hinders vision
- Another explanation: high failure rate of ideas
- Third Explanation: Trend Projection Hot hand bias

Anything can be wrong. As a reviewer you have to say why you have a better explanation for a result

36.1.1 (Golder and Tellis, 1993)

- Downfall of previous research using PIMS and ASSESSOR or business press:
  - survivorship bias
  - single-informant self-reports: measurement errors
- Half of market pioneers fail and mean market share is lower (compared to previous studies)
- Early market leaders have greater long-term success and enter about 13 years after the first pioneers
- Theories of pioneer advantages
  - Consumer-based:
    - Uncertainty in trying later entrants
    - Consumer stable preferences
    - Learning theory: pioneer = standard
    - Positioning advantage
    - Consumer with high switching costs will stay
  - Product-based:
    - Barrier to entry: economies of scale + learning + technological leadership + limited suppliers
- Theories of pioneer disadvantages
  - Free-riders: late entrants can come in at lower cost
  - Shifts in technology, customer needs
  - Incumbent inertia
  - Improper positioning (late entrants can pick optimal position later because pioneers' high cost of switching)
  - changing resource requirement
  - insufficient investments
- Data: historical analysis based on all publicly available sources of info.
  - Prospective contrast to retrospective (from database)
  - Might be less biased because of multiple sources (instead of single informants).
  - Examples: business week, advertising age

- Criteria for selection:
    * Competence
    * Objectivity
    * Reliability
    * Corroboration: Confirmation Bias?

- Sampling (have to justify you chose what you choose): before sampling was drawn.
  - Sample 1: consumer goods + new product categories and its extensions.
  - Sample 2: categories from Advertising Age
  - Sample 3: acknowledged pioneers
- Limitation:
  - Did not consider marketing mix
  - Customer-oriented definition of product category = arbitrary
  - Sample selection
  - Uncertainty regarding survivorship bias

36.1.2 (Johnson and Tellis, 2008)

- Market entry into China and India
- Smaller firms are more successful than larger firms
- Markets that are more open have less success rate.
- Success is greater for companies (1) enter earlier, (2) have greater control of entry mode, (3) similar to the host country.
- India is a tougher market than China (i.e., less successes)
- Drivers of Entry success:
  1. Firm differentiation
     - Firm strategy
       * Entry mode: export, license and franchise, alliance, joint venture, wholly owned subsidiary (related to degrees of control over its marketing resources from lowest to highest). Opposite prediction
         · Resource-based: degree of control increases with success likelihood, and help control resource leakage, and complementary resources.

        - Transactions cost: cost increases with degree of control (high investment -> high levels of investment to break even).
      - Entry timing:
        - Early entry: lock up key resources (e.g., distribution channels + suppliers), create standard, consumer preferences, exploit governmental incentives.
        - Late entry: pioneers usually don't have long-term success (Golder and Tellis, 1993), learn lesson from early entrants, lower learning curve
    - Firm resources: Firm size
      - Larger > Smaller: more resources, more product- and marketing-specific knowledge, can absorb more negative periods
      - Smaller > larger: less bureaucracy, which lower innovative ability (Chandy and Tellis, 2000)
  2. Country differentiation
    - Host-country characteristics:
      - Openness: lack of regulatory and obstacles to entry
        - Good: increase demand, competition on quality, higher efficiency and lower prices
        - Bad: increase competition from foreign entrants (thin margins, high cost of purchases, hiring of talent).
      - Country risk: negatively affect entry success
        - Political: tariffs, regulations
        - Financial + Economic: recession, currency crises, inflation.
  3. Host-home location
    - Cultural distance: closer better
    - Economic distance:
      - Closer better: similar market segments (transformable market demand knowledge), similar physical infrastructure (greater efficiency in operations, lowering costs), more market knowledge
- Data: historical analysis where data meet the following criteria:
  - Competence

  - Neutrality / Objectivity
  - Reliability
  - Corroboration
  - Contemporaneity
- Small sample size
  - 192 from China
  - 64 from India

Variable	Measure	Source
Success	Degree of success numerical rating	Historical Analysis from LexisNexis and ABI/INFORM
Entry mode	6 points scale based on (Anderson and Gatignon, 1986)	Archival data
Entry timing	Arbitrary: China: 1978, India 1991.	Archival data
Firm size	year-end sales for the focal firm	Compustat, Mergent Online
Economic distance	(Mitra and Golder, 2002)	International Financial Statistics yearbook
Cultural distance	Follow (Kogut and Singh, 1988)	Hofstede (1991, 2001)
Openness	Fraction of foreign direct investment over the host country's GDP	International Monetary Fund
Country Risk	Based on International Country Risk Guide (Erb et al., 1996)	International Country Risk Guide

36.1.3 (Zervas et al., 2017)

- Use DiD identification strategy
- sharing economy decreases demand for hotel via less aggressive hotel room pricing.
  - Those with low price and don't cater to business travelers suffer most.
- Data: from Airbnb (using review history) and 300 hotels in Texas (Texas Comptroller of Public Accounts),
- Dependent variables:
  - Cumulative measure
  - Instantaneous measure

- 10% increases in the market share of Airbnb lead to .39% decrease in hotel room revenue

36.2 Product Adoption and Diffusion

36.2.1 Background

- Every new thing either diffuses through population or fails
- Researchers are interested in the shape and processes of diffusion
- Bass is the first to model in marketing

Diffusion in different fields:

- Demography
- Archaeology
- Geography
- Epidemiology
- Sociology
- Linguistics
- Physics
- Cosmology

Models of Diffusion

- Negative Exponential
- Bass
- FDA
- Network

Levels of analysis:

- Class:
- Category
- Technology
- Brand

Classic model

- does not account fro marketing mix
- requires peak sales for stable estimates (if you have the peaks, you don't need the model)

- no repurscrhsases
- no multiple generation
- does not fit viral patterns

36.2.1.1 (Chandrasekaran and Tellis, 2007) A review of new products diffusion

- Products = idea, person, good, or service
- New product ≠ innovation

	In econ	In marketing
Diffusion	"the spread of an innovation across social groups over time (p. 39)	"the communication of an innovation through the population"
	Phenomenon (spread of a product) ≠ drivers (communication)	Phenomenon (spread of a product) = driver (communication)

This paper focuses on the econ definition

Product's life cycle stages:

1. Commercialization: when the product was first sold
2. **Takeoff**: dramatic and sustained increase in sales
3. Introduction: between commercialization and takeoff
4. **Slowdown**: decreasing in sales
5. Growth: between takeoff and slowdown
6. Maturity: Slowdown until decline.

Generalizations:

- **Shape of the Diffusion Curve**: cumulative sales over time is S-shaped curve.
- **Parameters of the Bass model**:
  - Coefficient of innovation or external influence (p)
    * mean between 0.0007 and 0.03
    * mean for developed countries is 0.001 and developing countries is 0.0003
  - Coefficient of imitation or internal influence (q)
    * mean between 0.38 and 0.53
    * industrial/medical innovation > consumer durables

    * 0.51 for developed countries and 0.56 for developing countries
  - the market potential (α or m)
    * 0.52 for developed countries and 0.17 for developing countries.
- **Cautions regarding the parameters**:
  - Time to peak sales: 19 years for developing and 16 for developed countries.
  - Biases in parameter estimation: static models (e.g., Bass) lead to downward biases in market potential and innovation while upward bias in imitation.
- **Drivers**: WOM, communication, economics, marketing mix variables (e.g., prices, consumer heterogeneity, consumer learning), purchasing power parity adjusted per capita income, international trade.
- Turning points of the diffusion curve
  - Takeoff
    * Time to takeoff: 6-10 years (varies by countries,products, time).
    * Drivers: price decrease
  - Slowdown
    * Sales decline by 15-32%
    * Drivers: price decline, market penetration, wealth (GNP), and info cascades (fast takeoff = fast decline)
- Findings across stages
  - Duration:
    * introduction: 6-10 years
    * growth: 8-10 years
    * early maturity: 5 years
    * duration of growth:
      · time saving products > non-time saving products
      · leisure enhancing products < non-leisure enhancing products
    * introduction and early maturity duration get shorter over time (but not growth)
  - Price: price reduction is getting larger as time progresses (for both introduction nd growth).
  - Growth rates:

    * Introduction: 31%
    * Takeoff: 428%
    * Growth: 45%
    * Slowdown: -15%
    * Early maturity: -25%
    * Late maturity: 3.7%

Future Research:

- Measurement: When to start or stop, or takeoff, differentiation between first purchases and repurchases, demand is better than supply measure,
- Theories: no reconciliation yet
- Models: comprehensive (from commercialization to takeoff, growth, and slowdown)
- Findings: More fine-tune subgroups, include failed diffusion, and consider other countries.

Specification

The probability that an individual will purchase at time T is a function of the number of previous buyers.

$$P(t) = \frac{f(t)}{1 - F(t)} = p + \frac{q}{m} Y(t)$$

where

- $P(t)$ = hazard rate
- $Y(t)$ = cumulative number of adopters at t
- p = probability of an initial purchase at time 0 (when $Y(0) = 0$) (also known as innovators importance).
- $\frac{q}{m} Y(t)$ = pressure of prior adopters on imitators
- m = number of initial purchases before any replacement purchases (i.e., market size)
- $F(t)$ = cumulative fraction of adopters at time t
- $f(t)$ = likelihood of purchase at time t

Rearrange the formula to get the likelihood of purchase at time t

$$f(t) = (p + qF(t))[1 - F(t)]$$

The number of adoptions at time t is

$$S(t) = mf(t) = pm + (q - p)Y(t) - \frac{q}{m}Y^2(t)$$

then Bass solves the differential equation:

$$dt = \frac{dF}{p + (q - p)F - qF^2}$$

to obtain cumulative adoption at time t

$$F(t) = \frac{1 - e^{-(p+q)t}}{q + (q/p)e^{-(p+q)t}}$$

Hence, the cumulative number of adopters is

$$Y(t) = m\frac{1 - e^{-(p+q)t}}{q + (q/p)e^{-(p+q)t}}$$

Rewriting the number of adoptions at time t

$$S_t = a + bY_{t-1} + cY_{t-1}^2, t = 2, 3, ...$$

where

- S_t = sales at time t
- Y_{t-1} = cumulative sales through period $t - 1$
- $a = p \times m$
- $b = q - p$
- $c = -q/m$

Equivalently,

$$p = a/mq = -cmm = (-b \pm (b^2 - 4ac)^{1/2})/2c$$

Strengths

- Good fit to the S-shaped curve (thank to the quadratic term)
- Appealing interpretations:
  - p = coefficient of innovation (i.e., spontaneous rate of adoption in the population) or external influence (e.g., mass -media communications)

  - q = coefficient of imitation (i.e., effect of prior cumulative adopters on adoption) or internal influence (e.g., interpersonal communication influence from prior adopters).

- Good application: time (t) or magnitude ($S(t)$) of peak sales.

$$t^* = \frac{1}{p+q} \times \ln(\frac{q}{p}) S(t)^* = m \times \frac{(p+q)^2}{4q}$$

- Incorporated prior literature
  - If $p = 0$, the Bass model is a logistic diffusion function (driven only be imitation adoption)
  - If $q = 0$, the Bass model is an exponential function (driven only innovation adoption)

Limitations

- Bass requires 2 most important events that we want to predict in the first place: takeoff and slowdown to have stable estimates.
- Unstable estimates after incorporating new observations.
- Do not directly account for marketing mix variables (price, promotion), but indirectly capture by m, p
- Assumes product definition is static (no growth or changes in product as time progresses)
- Using OLS which can cause
  - Multicollinearity between Y_{t-1}, Y_{t-1}^2 (making the estimates unstable)
  - Do not estimate the SE for p, q, m
  - Time interval bias (model uses discrete time series data to estimate a continuous model)
- Hard to determine starting and ending points of the the sales time.
  - Supposedly, we need to use first adoptions of new product as sales (S_t), but data could not capture this, only both first purchases and repurchases
  - Sales should start from the first year of commercialization, but usually we only have reports when products are selling well already
  - No clear stopping rule for the time interval.

Improvements

- **Incorporating marketing mix**

- Price: affect market potential (m) and probability of adoption ($P(t)$) and heterogeneous across products
- Advertising
- Distribution: 2 adoption processes: retailer and consumer, where number of retailers who affect determine the market potential m for consumers

(Bass et al., 1994) incorporate both price and promotion to the Generalized Bass model

$$\frac{f(t)}{1-F(t)} = (p + qF(t))x(t)$$

where $x(t)$ is the current marketing effort (sum of advertising and price) on the conditional probability of product adoption at time t such that

$$x(t) = 1 + \beta_1 \frac{\Delta P(t)}{P(t-1)} + \beta_2 \frac{\Delta A(t)}{A(t-1)}$$

where

- $\Delta P(t) = P(t) - P(t-1)$ rate of changes in price
- $\Delta A(t) = A(t) - A(t-1)$ rate of changes in advertising

When prices and advertising remain constant, GB model reduces to Bass model. But it seems like they only stop at 2 variables (not all marketing mix variables or macro and micro econ variables - income changes).

- **Incorporate supply restrictions**
  - Include another stage between potential adopter to adopters which is waiting applicants.

$$\frac{dA(t)}{dt} = [p + \frac{q_1}{m}A(t) + \frac{q_2}{m}N(t)][m - A(t) - N(t)] - c(t)A(t) = \text{[Waiting population + Adopters] - conversion rate of appl}$$

and

$\frac{dN(t)}{dt} = c(t)A(t)$

where

- $d(A)/dt$ is the rate of changes of waiting applicants
- $c(t)$ is the supply coefficient
- the second equation is the impact of supply restrictions on adoption rate

The growth of new applicants is

$$\frac{dZ(t)}{dt} = \frac{dA(t)}{dt} + \frac{dN(t)}{dt} = (p + \frac{q_1}{m}A(t) + \frac{q_2}{m}N(t))(m - A(t) - N(t))$$

To incorporate waiting applicants abandoning their adoption decision after some time see (Ho et al., 2002)

- **Incorporate competitive effects**
  - Instead of using product category as the unit of analysis, we can model at the brand level (different brand might have different rate of diffusion).
  - A new brand can
    * increase the entire market potential (m) (by increased promotion and product variety)
    * compete in the existing market potential (interfere the diffusion process of other brands)
  - Diffusion depends on the order of entry and competition.
- **Incorporate complementary effects**
  - In market that has indirect network externalities, co-diffusion exists and asymmetric
- **Incorporate technological generations** for successive generations of the same product (i.e., substitution effects).

$$S_1(t) = m_1 F_1(t) - m_1 F_1(t) F_2(t - r_2)$$

where r_2 is the introduction time of the next-generation product.

$$S_2(t) = F_2(t - r_2)[m_2 + F_1(t) m_1]$$

where

- $S_i(t)$ = sales of generation i
- $F_i(t)$ = fraction of adoption for each generation
- m_i = market potential for each generation

Leapfrogging behavior is possible (i.e., skip a generation to buy the next one) (Mahajan and Muller, 1996)

- **Incorporate time-varying parameters**

  - Model market potential (m) as a function of time-varying exogenous and endogenous variables (Mahajan and Peterson, 1978)
  - Model coefficient of imitation to be time-varying (Easingwood et al., 1983)

$$\frac{dF(t)}{dt} = [p + qF(t)^\delta][1 - F(t)]$$

where δ is the nonuniform influence

- when $\delta = 1$, the model becomes the Bass model
- When $\delta \in [0, 1]$, means high initial coefficient of imitation,
- When $\delta > 1$, means delay in influence -> lower and later peak.

Different adopters could influence later adopters differently (people who adopted more recently are more vocal) (Sharma and Bhargava, 1994)

- **Incorporate replacement and mufti-unit purchases**

(Balasubramanian and Kamakura, 1989)

$$y(t) = [a + bX(t)][\alpha \text{Population}(t) P^\beta(t) - X(t)] + r(t) + e(t)$$

where

- $y(t)$ = sales
- $P(t)$ = price index
- $X(t)$ = total units in use at the beginning of year t with dead units are replaced already
- $r(t)$ = number o units that have died or need replacement at year t
- a = coefficient of innovation
- b = coefficient imitation
- β = price change effect on ultimate penetration
- α = ultimate penetration (price is at its original level)

(Steffens, 2003) models multiple units purchase by a single household.

- Incorporate trail-repeat purchases
- Incorporate variations across countries

Evaluation:

- All of the improvements still rest on the assumption of one driving mechanism: knowledge dispersion through WOM.

Improvements in estimation

- MLE: avoid time-interval bias, but underestimates the SE (Schmittlein and Mahajan, 1982)
- Non linear least squares: (Srinivasan and Mason, 1986) need lots of obs
  - Estimates are more flexible
  - No time-interval bias
  - valid SE
- Hierarchical Bayesian method
  - Incorporate parameter updating
  - Problem with definition of similar products (fixed by (Bayus, 1993) with product segmentation scheme)
- Adaptive techniques: stochastic techniques (parameter vary over time) ((Xie et al., 1997)augmented Kalman filter)
- Genetic algorithms:
  - can find global optimum
  - better estimate (less bias).

Alternative models of diffusion

- Alternative drivers:
  - Affordability: (Golder and Tellis, 1998) model as Cobb-Douglas model:
    * $S = P^{\beta_1} \times I^{\beta-2} \times CS^{\beta_3} \times MP^{\beta_4} \times e^{\epsilon}$
    * Sales = product (price, income, consumer sentiment, market presence)
    * (Horsky, 1990) incoproates both price and income and WOM on sales growth.
  - Heterogeneity: aggregate level diffusion models: (Roberts and Urban, 1988), (Oren and Schwartz, 1988), (Chatterjee and Eliashberg, 1990), (Bemmaor, 1984) (Song and Chintagunta, 2003b), (Sinha and Chandrashekaran, 1992) (Karshenas and Stoneman, 1993)
  - Strategy: model supply side: (market entry, marketing mix, location) (Dekimpe et al., 2000), (den Bulte and Lilien, 2001),(Bronnenberg and Mela, 2004)
- Alternative phenomena:
  - Spatial diffusion (Mahajan and Peterson, 1979), (Redmond, 2003), (Garber et al., 2004)

    * Contagious diffusion (infectious diseases)
    * Expansion diffusion (one source like wildfire)
    * Hierarchical diffusion (ordered series of classes)
    * Relocation diffusion:

  - Diffusion of entertainment products: follow exponential decay (Eliashberg and Sawhney, 1994), (Eliashberg et al., 2000), (Elberse and Eliashberg, 2003), (Moe and Fader, 2002), (Lee et al., 2003)

Modeling the turning points in diffusion

- Takeoff: follow (Golder and Tellis, 1997) definition: "point of transition from the introduction stage to the growth stage"
  - Measurement
    * (Golder and Tellis, 1997): threshold takeoff (compare to other in the categories)
    * Logistic curve rule: first turning point of the logistic curve (max of the 2nd derivative) (hindsight only)
    * Maximum growth rule: largest sales increases within 3 years (not size invariant)
    * (Agarwal and Bayus, 2002) measure based on annual percentage change in sales
    * (Stremersch and Tellis, 2004) adapted the threshold method for international markets
    * (Garber et al., 2004) rule of thumb: 10-20 market penetration
  - Drivers
    * (Golder and Tellis, 1997) price declines lead to takeoff
    * (Agarwal and Bayus, 2002) increase in firm entry lead to better product quality, marketing infrastructures
    * (Tellis et al., 2003) venturesome culture lead to takeoff
  - Model: either proportional hazards (Golder and Tellis, 1997) or log-logistic hazard (Tellis et al., 2003)
  - Evaluation: Only model successful innovation so far.
- Slowdown: point of transition from the growth stage to the maturity stage (Golder and Tellis, 1997)
  - Measurement: (Golder and Tellis, 2004) "operationalize as the first year of two consecutive years after takeoff in which sales are lower than the highest previous sales." (p.72)

- Explanation:
  - Dual-market phenomenon: early adopters vs. early majority (Goldenberg et al., 2001)
  - Informational cascades: negative cascades (Golder and Tellis, 2004)
  - Affordability(Golder and Tellis, 2004)
- Modeling:
  - Cellular automata models: (Goldenberg et al., 2001)
  - Hazard models: (Golder and Tellis, 2004)
- Evaluation: still new can have more research

36.2.1.2 (Bass, 1969)

Assumption:

- The timing of a consumer's initial purchase is correlated with the number of previous
- This paper looks at new class of products (not new brands or new models of older products)
- Focus on infrequently purchased products

Theory of Adoption and Diffusion

- **Innovators**: adopt independently (regardless of others' opinions): pressure to adopt does not increase with the growth of the adoption.
- **Imitators** (include early adopters, early majority, late majority): adoption depends on the timing of adoption (i.e., influenced by the decisions of others to adopt.
- Laggards

"The probability that an initial purchase will be made at T given that no purchase has yet been made is a linear function of the number of previous buyers" (p. 216)

$$P(T) = p + \frac{q}{m} Y(T)$$

where p and q/m are constants

$Y(T)$ is the number of previous buyers.

When $Y(T) = 0$, p represents the probability of an initial purchase at $T = 0$

$(q/m)Y(T)$ is the pressures on imitators to adopt.

Model Assumptions:

36.2.2 Discussion

36.2.2.1 (Sood et al., 2009)

- Functional regression
- Contributions:
  - Theoretically sound (integrate info across categorizes)
  - Augmented Functional regression outperforms existing models
  - Product-specific effects are more helpful in predicting penetration than country-specific effects.
- They use yearly cumulative penetration of each category as the unit of analysis (i.e., curve/ function).
- 3 functional data analysis techniques:
  - Functional principal components
  - functional regression
  - functional cluster analysis
- To treat discrete intervals: use smoothing spline to generate continuous smooth curves
- Even though the spline approach requires a lot of data to smooth, other appearances to create smoothness are still available. Hence, you can still use function regression and or cluster with 2 or 3 time points.
- Advantage s of functional regression:
  - incorporate info from other products
  - nonparametric fitting procedure
  - uses the functional nature of the penetration curves.
- Predictions on: number of years to take off, peak marginal penetration and the level of peak marginal penetration
- **Good**: tell a story from simple to more sophisticated model to justify their improvements in the paper.
- 2 dimensions that are not captured by simple extrapolation models:
  - info from prior history of the new product
  - intrinsic info across products and countries.
- Classic Bass model ignores:

  - other categories (fixed by meta-bass and augmented meta-bass)
  - uses parametric methods.
- **Questions**:
  - Technically could redo the analysis with new dataset (including 2009 till now) to see the out of sample performance.
  - No hypothesis, just model and probable explanation
  - Use only curves under the same category to predict the new product (not all categories).

36.2.2.2 (Appel et al., 2019)

- Growth, Popularity and the Long Tail: Evidence from Digital Markets
- part of MSI's working paper series and MSI insights
- Context: digitized markets (long-tail markets)
- Most popular products do have S-shaped curve, but lower-popularity products exponential-like decline ("slide") or a combination of slide and bell (S&B) are more common.
- Shortcomings of previous research:
  - Pro-innovation bias: success correlates with importance in the new product development research
- Data: SourceForge (exclude inactive and less than 200 downloads): 5 years with high Gini coefficient - 0.96 (i.e., high concentration).
- Dominant patterns:
  - A bell-shaped pattern: bell (popular products)
    - Caveat in the movie market: popular products decline over time.
  - An exponential-like decline beginning at launch: slide
  - Combination of the first 2: S&B
- Proposed model: inception model (inception effect = heightened external growth).
- Long-tail market:
  - Supply side: low cost of inventory, stocking, efficient delivery, and low cost of new products development.
  - Demand-side: easy to search, recommendation system, social networks and online communities.
- Popularity = extent of demand = number of downloads.

- The shape of new product growth: previous literature says S-shaped
- Non-S-shaped markets:
  - r-shaped cumulative curve: because of
    * Large budget for promotion: movies
    * Pre-launch buzz: on social media
- The role of popularity on the shape of growth: was ignored in the literature
- Free and Open-Source Software (FOSS)
- Data Analysis:
  - Stage 1: To facilitate comparison, scale pattern to a (0,1) by dividing each observation by the total sum of downloads, and smooth the graph using Hodrick-Prescott filter
  - Stage 2: Use peaks-and-troughs algorithm for the classification
- Descriptive: the S-shaped curve is representative for more popular products, while for those that are not as popular, we have a blend of S&B and slide as well.
- Try to observe the same pattern with smartphone app download (data provided by Mobility - an anonymous app providers for businesses)
- Drivers of Multi-pattern Growth
  - Analogy to movies (characterized by an exponential decline): not similar because
    * different product types (utilitarian vs. entertainment)
    * Different pattern exhibited by popular and unpopular: while in movies the exponential decline is from blockbuster, and sleepers has a bell shape, under this dataset, less popular product has the exponential decline, while the popular products are bell-shaped.
  - Analogy to supermarkets: not good because FDP is affected by social influence, supermarkets are usually under large investments and not much social influence.
  - The inception alternative: 2 influences of new product growth
    * Internal: from previous adopters
    * External (not from previous adopters): marketing mix, social media posts, recommendation, expert opinions, influencers. Expected to stronger early on and decay. (i.e., inception effect - external influence as a function of time with an initial external influence parameter $p(t) = pe^{\delta t}$)

- The relationship between inception and popularity: The higher the product's popularity, the lower the share of adoptions due to the inception effects (i.e., products with high initial investment that failed to reach critical is less popular).
- Inception is typically a necessary but not sufficient condition to reach popularity.

36.2.2.3 (Tellis et al., 2020)

- No awards (nominated only)
- Emotion is more effective than information
- brand hurts, but branding is used a lot
- surprise and humor are good, but videos don't use
- Limitation: Because these emotions are rare, maybe that why they are effective. But if everyone starts using these tactics, maybe that they wont' work anymore.

36.2.2.4 (Chandrasekaran et al., 2020)

- Was rejected 5 times.
- Leapfrogging, Cannibalization, and survival during disruptive technological change
- 2 types of dilemma when it comes to new technology:
  - Incumbent: invest in new technology or old or both
  - Entrant: target niche or mass.
  - Solution: relation between new technology and old one (i.e., high rate of disengagement - cannibalization or low rate of disengagement-coexistence)
- Data:
  - Successive technology penetration across multiple countries and years
  - Sales of contemporaneous pair across multiples countries
  - Case analyses
- "Disruption occurs if the incumbent focuses on the old technology to the exclusion of the new one" (p. 4)
- Definitions:
  - Successive/New technology: not new version/generations of the same product

- Cannibalization: "the extent to which the successive technology"eats" into real or potential sales (or penetration) of the old technology due to substitution."(p. 5)
- Rate of disengagement F_{12}: (account for partial substitution)
- Adopter segments for a new successive technology:
    - Leapfroggers: adopt new, but would never have adopted the old
    - Switchers: Adopted old, but switch to new once it's introduced
    - Opportunists: wait for the old, but end up with the new one.
    - Dual users: both technologies
- Models: based on (Norton and Bass, 1987)

$$S_1(t) = m_1 F_1(t)(1 - F_{12}(t - \tau_2 + 1)) S_2(t) = F_2(T - \tau_2 + 1)(m_2 + m_1 F_1(t))$$

where

- $S_i(t)$ = penetration of technology i in period t
- m_1 = long-run penetration for technology 1
- $m_1 + m_2$ = long-run penetration for technology 2

The fraction of all potential technology_g consumers for each technology (g = technology 1 or 2)

$$F_g(t) = \frac{p_g(1 - e^{-(p_g+q_g)^t})}{p_g + q_g e^{-(p_g+q_g)t}}$$

where

- $t \geq 0$
- $g = 1, 2$
- p = innovation coefficient
- q = imitation coefficient
- p_{12}, q_{12} = disengagement coefficients
- F_1, F_2, F_{12} = adoption rate of technology 1, technology 2, and disengagement rate at which technology 1 customers abandon to get technology 2

Model contributions:

- Model the adoption rate of technology 2 different from disengagement rate of technology 1 ($F_2 \neq F_{12}$)

- Varying p, q (for different technologies)
- F_1 has the same function form as F_1, F_2 (because it fits the data well, and reduces to previous model which matches previous literature)
- Model can be applied to both generational and technology diffusion

Model Estimation

Using nonlinear least squares to estimate the parameters that that minimize

$$\sum_{i=1}^{n}(s_{i1}-m_1F_1(t_i))(1-F_{12}(t_i-\tau_2+1))^2+\sum_{i=1}^{n}(s_{i2}-F_2(t_i-\tau_2+1)(m_2+m_1F_1(t_i)))^2$$

Segments of adopters

$$S_2(t) = L_2(t) + DU_2(t) + SW_2(t) + O_2(t)$$

while

$$S_1(t) = L_1(t) - CAN_2(t) = L_1(t) - (SW_2(t) + O_2(t))$$

where

- SW = switchers
- O = Opportunists
- CAN = Canalization
- L = Leapfroggers
- DU = dual -users

Market growth segment = sum(leapfroggers, dual users)

Cannibalization = sum(switchers, opportunists).

36.2.2.5 (Prins and Verhoef, 2007) Marketing effects on adoption timing

- Studies the effects of direct marketing and mass marketing on adoption timing (in the context of a new e-service among existing customers)
- Data: 6k customers of a Dutch telecom operator over 25 months
- Findings:
  - advertising shortens the time to adoption (including those by competitors)

  - Mass marketing has a greater effect on loyal customers (compared direct marketing)

- Related literature:

  - Adoption

  - customer management

- Adoption timing is defined as "the time between the introduction and the adoption of the new service" (p. 170) following (Steenkamp and Gielens, 2003)

- Switchers to competitive services are considered as non-adopters (even if they adopt comeptitor's new service). It's valid when the focus is on the adoption of the folca company's new service among existing customers.

- (Donkers et al., 2003) demonstrates that if oversampling is not accompanied by stratfied sampling on the independent variables, it should not affect the parameter estimates or SE for are event in binnary choice models.

- Meausres of Time to adoption: For each tiem period t, a customer can either adopt the new serive or not. The time to adoption for each customer is the time elasped in t since the intro of the service. Dependent vairable = indivudal time to adoption.

36.3 Take-off Disruption

Marginal Prob vs. Hazard of Death (what is the conditional probability of dying conditional on you are alive)

Sometimes we study takeoff instead of sales of new products because new products either takeoff or die, wee dont' see flat salles. (managerial implication: invest if takeoff)

We have to wait at least till the peak of the hazard function (5 years)

Pervasiveness of disruption: US

36.3.1 Disruptive Technologies

- Companies stay too close to their current customers, without accounting for future ones.

- For each industry, there is performance trajectory that help track new technology performance in comparison with old ones'.

  - Sustaining technology: maintain the rate of improvement

  - Disruptive technology:

- Solution to cultivate disruptive technologies:
  - Is the technology disruptive or sustaining?
  - What is the strategic significance of the disruptive technology?
  - Where is the initial market for the disruptive technology?
  - There should be a separate organization or business that handle disruptive technology

36.3.2 (Golder and Tellis, 1997) takeooff

- Key issues:
  - How long does it typically take a product to take off?
  - Is there a takeoff pattern?
  - Can we predict takeoff?
- If the baseline sales is small, it takes a large increase in sales to takeoff, but if the baseline sales is big, it takes only a small increase in sales to takeoff. Hence, there is a **threshold** for takeoff
- **Definition of takeoff**: "the first year in which an individual category's growth rate relative to base sales crosses this threshold." (p. 256) or "the point of transition from the introductory stage to the growth stage of the product file cycle." (p. 257)
  - Metric: the first large increase in sales in the new category (still don't quite understand)
- **Operational definition of takeoff**: "threshold for takeoff as a plot of the percentage increase in sales relative to its base sales that demarcates the takeoff." (p. 259)
- **Independent variables**: price, year of introduction, market penetration (percentage of households that have purchased a new product), and controls (product specific, and economic variables)
- Found:
  - price at takeoff is lower than price at the introduction stage
  - Average time to takeoff is 6 years
  - penetration at takeoff is 1.7%
  - Products usually takeoff around 3 price points: \$1000, \$500, \$100
- Model: Cox's proportional hazard mode

$$h_i(t) = h(t; z_{it}) = h_0(t) \times e^{z_{it}\beta}$$

where

- $h_0(t)$ is the baseline hazard function
- z_{it} are the independent variables
- β is the same for all categories (questionable choice)
- Do not include unbosomed heterogeneity because each event is unique (non repeated)

Samples:

1. 11 consumer durables (usually studied in diffusion research)
2. 10 recently introduced consumer durables
3. 10 categories during the review process.

Model performance

- U^2 measure reduction in uncertainty
- Forecasts: (1) at introduction (2) one year ahead

36.3.3 (Chandy and Tellis, 2000) Incumbent's curse

- Present this paper
- Definition: "A radical product innovation is a new product that incorporates a substantially different core technology and provides substantially higher customer benefits relative to previous products in the industry" (Chandy and Tellis, 1998).
- Theory of S-curves: figure 1
- Reasons incumbents don't like radical innovations:
  - Perceived incentives: prospect theory (incumbents stand to lose, innovators stand to gain)
  - Organizational filter: resources are invested in important tasks that yield money.
  - Organizational routines: repetitive tasks are very efficient.
  - Opportunities of incumbents: market capabilities (customer knowledge, customer franchise, market power)
- Size and incumbency are positively correlated
  - Theory of (bureaucratic) inertia: it's hard to get new idea through a large firm because of filtering and screening + no incentives to do so.
  - Opportunities of large firms: financial and technical capabilities
- There are more nonincumbents (i.e., small firms) as innovators in the US than other countries (e.g., Japan, or Western Europe) because of (1) institution (2) culture
- Historical analysis: 1 author + 9 assistants over 4 years
- Sample frame:
  - Product classes: consumer durables + office products
  - High unit sales (> 1 mil) (from Predicasts)

  - Radically new technology: (1) identify the most significant product innvoaitosn in each product category (2) 3 experts rate the radicalness
- Measures
  - Radical innovation means (1) differences in core technology: utilizing a distinct core technology (2) superiority in user benefits:gives a lot more value to the customer than the first product in the same category.
  - Firm size: employees, sales volumn, value of asset from Moody's Industrial Manual and S&P manual, for private firms: company directories - Industrial laboratories Directory, Edison Electric Light Co.
  - Innovator (firm that first commercialized the radical innovation) and incumbent (firms that sell previous generation product on the introduction date)
- Results: 64 out of 93 innovations have data.
- Categorical Analysis:
  - Large firms are more likely to be incumbents
  - Small firms were more radical in their innovation before the World War 2, large firms are radical in their innovation recently.
  - US innovators are from non-incumbent. Before the World War II, the US innovation were likely to come from smaller firms, but recent US innovation tend to come from large firms.
- Multivariate
  - While larger organizations have historically introduced fewer innovative inventions, the tendency in recent years has been the polar opposite.
  - In recent years, US corporations have developed more radical ideas than non-US firms.
- Further Analyses
  - Relevant Population: Large firms account for a significantly higher proportion of radical innovations when compared to its total number of firms in the economy. In any product class (incumbent vs. non), the number of incumbent is much smaller than non incumbents, but incumbents still account for half of the nubmer of radical innovations.
  - Alternative measure of firm size
  - Radical Innovator: but what if incumbents can be early entrants?

36.3.4 (Tellis et al., 2003) International Takeoff

- 137 products across 10 categories inn 16 countries
- Parametric hazard model
- Takeoff in Europe (e.g., 6 years after introductionn) is different from those in US

- Time-to-takeoff varies by countries and categories
- Not much evidence for the effect of culture and economic factors on inter-country differences in time-to-takeoff
- Use waterfall strategy when going international.
- Countries with less uncertainty avoidance will have greater adoption
- Countries with higher education will have greater adoption

36.3.5 (Hauser et al., 2006)Review on Innovation

5 fields

- Consumer response to innovation
- Organzattion and innovation
- Market entry strategies
- prescriptive technique for product development processes
- Defense against market entry

36.3.6 (Chandrasekaran and Tellis, 2008) Global Takeoff

- 16 products in 31 countries
- Parametric hazard model
- Economic variable (developed vs. developing) (isn't this kinda contradict (Tellis et al., 2003), product types (work vs. fun), cultural clusters, calendar time can affect takeoff time
- Takeoff is getting shorter over time

36.3.7 (Sood and Tellis, 2011) Predict takeoff

36.3.8 (Zhang and Luo, 2016) Restaurant survival from Yelp

36.4 Advertising Response (Effectiveness)

Consumer response to advertising

Key issues

- Does advertising work?
- When, where, why and for how long?

5 effects of ad exposure

- Short
- Sleeper
- Hysteresis
- Long
- Instant

Simple model of ad response

$$S_t = \alpha + \beta A_t + \mu_t$$

- Does not capture the carryover effect

Using (Koyck, 1954) model captures carryover

$$S_t + \alpha + \beta A_t + \beta \lambda A_{t-1} + \dots + \epsilon_t$$

This is a moving average model with an infinite lag that precisely captures carryover effect of advertising

Then, we need the Koyck transformation, lag on period and multiply by λ (carryover effect) $(0 < \lambda < 1)$

Then

$$\lambda S_{t-1} = \alpha \lambda + \beta \lambda A_{t-1} + \dots + \epsilon_t \lambda$$

With subtraction,

$$\begin{aligned}
S_t - \lambda S_{t-1} &= \alpha - \alpha\lambda + \beta A_t + \epsilon_t - \epsilon_t \lambda \\
S_t &= \alpha - \alpha\lambda + \lambda S_{t-1} + \beta A_t + \epsilon_t - \epsilon_t \lambda \\
S_t &= \alpha + \lambda S_{t-1} + \beta A_t + u_t
\end{aligned}$$

Pros:

- An infinite lag series turns to 1 period auto-regressive model
- easy to estimate
- λ is the carryover or decay in effect of advertising

$\beta =$ current effect of ad

$\beta\lambda/(1-\lambda)$ carryover effect of ad

$\beta/(1-\lambda) =$ total effect advertising

p% duration interval $= \log(1-p)/\log\lambda$

If include a lagged ad term

$$S_t = \alpha + \lambda S_{t-1} + \beta A_t + \beta_1 A_{t-1} + \mu_t$$

- Separate inertia from ad carryover
- separate out decay from multiple independent variables
- identify shape of decay

(Clarke, 1976) found major limitation of Koyck model

- Aggregation bias: the larger the data interval: the larger the estimated λ, the larger the estimated carryover effect, the longer the estimated duration of ad
- People used to think the best data interval time is the inter-purchase time. But (Tellis and Franses, 2006) showed that unit exposure time is the optimal data interval (the smallest interval within which advertising occurs only once and at the same time every period)

General Autoregressive distributed Lag Model (ADL, ARMA)

$$S_t = \alpha + \lambda S_{t-1} + \lambda S_{t-2} + \dots + \beta A_t + \beta A_{t-1} + \dots + \mu_t$$

pros:

- rich variety of decay shapes
  - β affect number and position of bumps
  - λ affect speed of decay
- precursor to Vector Autoregressive model (VAR)

cons:

- aggregate data at population level and time cannot identify ad exposure
- aggregate time cannot identify treated period
- reverse causality: ad set on expected sales
- multicollinearity

Major advances in ad response modeling:

- Dis-aggregate data
  - modeling at individual household, consumer
  - modeling by day, hour
  - modeling moment-to-moment
  - modeling exposure (not $)

- quasi-experiments
  - DID
  - Synthetic control

36.4.1 (Tellis et al., 2000) Direct TV ad

Study Context

- A referral is "a call by a customer for the firm's service" (p. 33)
- Theory of message repetition:
  - A current effect on behavior
  - A carryover effect on behavior
  - A non behavior effect on attitude and memory
- Research questions:
  - Given current brand equity, what is the effect of advertising on referrals?
    * Ad placement
    * Creatives
    * Time period
    * Age and repetition
  - Is marginal benefit greater than marginal cost for advertising?

Model

$$R_t = \alpha + \gamma_1 R_{t-1} + \gamma_2 R_{t-2} + \gamma_3 R_{t-3} + ... + \beta_0 A_t + \beta_1 A_{t-1} + \beta A_{t-2} + \dots + \epsilon$$

where

- A = advertising
- R = referral

Controls: Opening hour + time of the day.

Expect:

- Morning ads have longer decay than other time
- Differences in creatives

Transfer function analysis

- temporal patterns: auto correlations + partial auto-correlation show patterns at the hourly and weekly level
- Lag structure: 3 lags on the dependent, and 4 lags on the independent (advertising)
  - Why there are lags of the dependent variable:
    * Algebraic: if didn't have of the dependent, the independent lag would be infinite
    * Intuitive: separate the effect of carry over effect of advertising and inertia.
- Error patterns:

$$R_t = \alpha + v(\mathbf{B})A_t + N_t$$

where

- R_t, A_t stationary
- $v(\mathbf{B})$ transfer function of advertising on referrals where $v(\mathbf{B}) = Cw(B)B^b/\delta(B)$
- $N_t = [\theta(B)/\phi(B)](1-B)^d a_t$ where $a_t \sim N(0)$

Advertising Effects (decay)

Total effects of advertising = sum of ad coefficients divided by (1 - sum of lag-referral coefficients)

$$\text{Total Effect} = \frac{\sum_{l=0}^{n} \beta_l}{(1 - \sum_{j=1}^{p} \lambda_l)}$$

where l is the index for the time lag

and the partial advertising effect at each time period is

$$TA_{t-l} = \beta_l A_{t-l} + \sum_{j=0}^{l} \lambda_j TA_{t-l+j}$$

Results

- Advertising effect dissipate after 8 hours
- Ad Effectiveness varies by station
- Creatives also varies

36.4.2 (Tellis and Franses, 2006) Optimal Data Interval for estimating ad response (on sales)

- Such a seminal paper
- This could also be applied to firm optimal interval for estimating announcement effect on stock performance.

Too disaggregate does not lead to disaggregate bias

Optimal interval is unit exposure time (not inter-purchase time)

To get the true estimates, it depends on the unit exposure time (instead of assumption of the advertising process)

Definition:

Term	Definition
Data Interval	temporal level of the records
Inter purchase time	Smallest calendar time between any two consumer purchases
Duration Interval	Length of time that advertising effect lasts
Calendar time	Discrete time period
Exposure time	Moment a pulse of ad first hits a consumer
p% duration interval	length of time that accounts for p% of the advertising effect
Current effect of ad	portion of the total advertising effect that occurs in the same time period as the exposure
Duration interval bias	carryover effect estimated at the true interval - estimated on aggregate data

Optimal interval balances between storage cost and estimate unbiasedness

Koyck model

- s_t, a_t are sales and ad at the true microdata interval

$$s_t = \mu + \beta a_t + \beta \lambda a_{t-1} + \beta \lambda^2 a_{t-2} + \dots + \epsilon_t$$

where

- $\epsilon \sim N(0, \sigma^2_\epsilon)$
- β = current effect of advertising
- $\beta/(1-\lambda)$ = carryover effect
- λ determines the duration interval (what do we call this term)

Using (Koyck, 1954) transformation (i.e., multiply both sides by $1-\lambda L$ where L is the familiar lag operator $L^k y_t = y_{t-k}$) then

$$s_t = \lambda s_{t-1} + \beta a_t + \epsilon_t - \lambda \epsilon_{t-1}$$

For aggregate data, denote S_T as the aggregate sales series from aggregating sales in the K periods from the current to the $K-1$ prior period that are sampled at the current period

$$\begin{aligned} S_T &= s_t + s_{t-1} + s_{t-2} + \dots + s_{t-(K-1)} \\ &= (1 + L + L^2 + \dots + L^{K-1}) s_t \end{aligned}$$

Hence,

$$A_T = (1+L+L^2+\dots+L^{K-1}) a_t \epsilon_T = (1+L+L^2+\dots+L^{K-1}) \epsilon_t S_{T-1} = (1+L+L^2+\dots+L^{K-1}) s_{t-K}$$

The true aggregate form of the micromodel

$$S_T = \lambda^K S_{T-1} + \beta A_T + \beta\lambda(1+\lambda L+\lambda^2 L^2+\dots+\lambda^{K-1} L^{K-1}) \times (1+L+\dots+L_{K-1}) a_{t-1} + \epsilon_T - \lambda^K \epsilon_{T-1}$$

The bias stem from the fact that

$$A_{T-1} \neq (1 + \lambda L + \lambda^2 L^2 + \dots + \lambda^{K-1} L^{K-1}) \times (1 + L + \dots + L_{K-1}) a_{t-1}$$

because it was lost in aggregation

With optimal data interval (1 exposure pulse per interval), we can recover the **carryover effect**

$$\frac{\beta_1 + \beta_2}{1 - \lambda^K}$$

and the **true duration interval** is

$$\sqrt[K]{\hat{\lambda}^K}$$

the the **current effect** is β

When we have even more dis aggregate data than the optimal interval, we just have to adjust the formula to recover the true effects.

36.4.3 (Teixeira et al., 2010) Ad Pulsing to prevent consumer ad avoidance

- Model: probit with MCMC
- Data: eye-tracking on 31 commercials for 2000 participants.
- New metric to predict attention dispersion based on eye-tracking data.
- Optimization of ads:
  - problem: minimize avoidance subject to a given level of brand activity level
  - Solution: Pulsing

36.4.4 (Sethuraman et al., 2011) Advertising effectiveness meta-analysis

Data: 1960 - 2008, 56 studies.

Average short-term ad elasticity is .12

a decline in the advertising elasticity over time.

advertising elasticity is higher

- for durable goods (vs. nondurables)
- in the early stage than the mature stage of the life cycle
- yearly data than quarterly data
- ad is measured in gross rating points than monetary terms

Long-term ad elasticity is .24

36.4.5 (Liaukonyte et al., 2015) TV advertising on online shopping

- Impression merging process: human coders
- Data: \$3.4 bil spending by 20 brands, consists of traffic and transactions and content measures for 1,2224 commercials.
- Dif-n-dif: 2 mins pre/post windows of time. (similar to regression discontinuity)
- Action-focus content increases direct website traffic and sales conditional on visitation
- Info and emotion-focus content reduce web traffic while increases purchases, and positive net effect on sales for most brands.
- Imagery-focus ad content decreases direct traffic to the website

- After the tv ad
    1. consumer choose whether to visit the website
    2. consumer then determine whether to buy a product
- Data:
    - Online traffic: **comScore Media Metrix**
        * Direct traffic
        * Search engine referrals
        * Transaction Count
    - TV Ad Data: Kantar Media
- Argument for no endogeneity problem is that brands can't manipulate the exact time the ad will air. (since hte ad will be placed in a 15-min window while the research design looks at the 4 minutes windows). For the case that the authors look at the 2-hour window, they use the dif-n-dif design where they pick the largest brands within each product category that did not advertise

36.4.6 (Tirunillai and Tellis, 2017) TV ad on Online chatter: synthetic control

Raw metrics

1. Reviews: from Amazon, Epinions, cnet, twitter, YouTube, Facebook
    1. Volume of reviews
    2. valence of the review (positive vs. negative)
    3. Polarity (entropy)
2. Blogs: from **Spinn3r**
    1. Volume
    2. In-degree (links) of the brand website
    3. In-degree (links) of blog posts
    4. Volume of blogs that gain/lose rank

Using Dynamic factor analysis

$$Y_t = \xi f_t + \epsilon_t f_t = \Psi f_{t-1} + \eta_t$$

where

- Y_t raw measure of reviews and blogs
- f_t is the underlying factors
- ξ is the factors loadings
- ϵ idiosyncratic error

- η = white noise where $E(\epsilon_t \eta'_{t-k}) = 0$

Dimension of chatter (using dynamic factor analysis)

- Content-based dimensions:
  - Popularity: loads on volume of reviews and blogs
  - Negativity: loads positively on positive valence and polarity and negatively on positive valence
- Information spread dimensions:
  - Visibility: loads on the volume of blogs and the in-degree links of the brand website
  - **Virality**: loads on volume of blogs that gained rank and in-degree of the blogs

TV ad causally increases a short positive effect on online chatter (info-spread > content-based)

Ad can reduce the negativity in online chatter in the short-term.

Ad can

- simulate conversation online
- trigger brand recall
- Interpreting experience: give more favorable assessment toward the brand
- Refute negatives: greater credibility and persuasiveness

Empirical Setting: A campaign: Let's Do Amazing (ad duration). 20 days after the campaign date)>

Method:

- Synthetic control (synthetic brand): **the difference might already account for the spillover effect of the focal brands on other brands in the same industry** (authors argue that there was no spillover effect).
- **No justification for 70 days before and 20 days after**
- To make sure YouTube did not affect much, the authors use data from Visible Measures to assess viewership, and TV viewership from https://tvlistings.zap2it.com/?aid=gapzap and Nielsen TV Ratings and **Stradegy** (need to ask about this company).
- Authors also use Vector Auto-regressive model to examine the short-term and long-term dynamics between the dependent (chatter metrics) and independent variables (advertising).

36.5 Marketing Return

Event Analysis

Nature of series

1. Continuous
   1. Univariate: Class Bass, Classic FDA
   2. Multivariate Unidirectional: functional regression, classic Koyck, ADL, ARIMA
   3. Multivariate Multidirectional: VAR, VARX, PVAR, Simultaneous Equation
2. Punctuated
   1. Event is dependent: Hazard models, split hazard, bivariate hazard
   2. Evident is independent: Event analysis, synthetic control, DID

Decreasing rigor of causal inference

1. Lab Experiment
2. Field Experiment
3. Nature Experiment
4. Instrumental Variables
5. Granger causality (improves with shocks)
6. Times series regression (improves with shocks)
7. Cross-sectional regression

Levels of testing causality in field

- Correlation
- Multiple regression: control for other plausible causes
- Times series model (use of current and past values: Koyck, ADL, ARIMA)
- First differences (effect of changes)
- Lag of first differences (Arellano & Bond)
- Granger causality (use of only past values of independent variables + control of past values of dependent variables (VAR), preferably in differences).
- Intervention or event analysis
- Natural experiments
- RCT

Concept of Abnormal Return:

- Stock price (P_t) = random walk
- Return = $P_t - P_{t-1}$ = white noise

Panel Regression

- Sample similar firms, j
- Identify each of their similar events: First stage regression (WRDS)
- Estimate abnormal returns of each of these firms associated with each of those events e_{jt}

2nd stage: equation

- Pool abnormal returns
- Estimate factors that may affect the distribution of e_{jt}

Strength of event analysis

- Increases with clearly defined event, narrow window of treatment, removal of confounding events
- Long time series for baseline
- large number of firms
- diverse contexts of treatments
- Extraction effects of known predictors
- temporal dependent series (returns)
- punctuated independent series: event
- Focus on effects of event on series of returns
- simulates a natural experiment
- Define: a natural or artificial shock

Types of natural experiments:

- Compare treated vs. untreated
- compared before and after
- DiD
- Synthetic control

Types of pre-temporal controls

- One prior period
- baseline of prior period
- synthetic control
- function of known factors (Fama-French 4)
- Cross-over (treated becomes control and rev)

Time capsule in Marketing

Event	Source
market Entry	Factiva, Lexis-Nexis
new product	Factiva, Thomson Reuters
Consumers satisfaction	CSI
Innovation activities	Factiva, Cap IQ
Acquisitions	Factiva, SDC platinum
Quality	Web chat, product reviews
Advertising	TNS Stradegy, YouTube
Recalls	Govt web, others
Sales	Yahoo fin, 10k GFK, euromonitor, Nielsen
Earnings	SEC Filings
Stock Prices	CRSP, WRDS

36.5.1 (Fornell et al., 2006) Customer satisfaction and stock return

- Historically, people understand that customer satisfaction affects firm economic performance. But we haven't studied the relationship between customer satisfaction and stock performance.
- People don't incorporate the info about customer satisfaction into the stock price right away (market is not so efficient)
- From the literature, we understand that there are 4 determinants of a company's market value
  - Acceleration of cash flow: speed of buyer response marketing efforts
  - increase in cash cash flows: repeat business and low marginal costs of sales
  - reduction in cash flow risk: lower by satisfaction
  - increase in the residual value of the business
- Data: Compsutat + American Customer Satisfaction Index

Regression (correlation) analysis

$$\ln Marketvalue = \alpha + \beta_1 \ln Bookvalue + \beta_2 \ln Bookvalueliability + \beta_3 \ln ACSI$$

There is evidence for a correlation market value and customer satisfaction.

However, investors don't always respond positively to increased satisfaction news

1. The firms is giving away consumer surplus

2. firms that already have leads over competition
3. Why trade-off between satisfaction and productivity
4. reverse causality
5. timing expectation (i.e., measurement of satisfaction)

36.5.1.1 Event study

- Suing market model to estimate abnormal return

$$AR_{jt} = R_{jt} - (\alpha_j + \beta_j R_{mt})$$

where j = firm, and t = day

- estimation period = 255 days ending 46 days before the event date (McWilliams and Siegel, 1997)
- one-day event period = day when Wall Street Journal publish ACSI announcement.
- 5 days before and after event to rule out other news (PR Newswire, Dow Jones, Business Wires)
  - M&A, Spin-offs, stock splits
  - CEO or CFO changes,
  - Layoffs, restructurings, earnings announcements, lawsuits
- **No evidence for the effect of ACSI on CAR**

36.5.1.2 Portfolio study

- 2 portfolios: hypothetical portfolio, and real-world portfolio
- **Customer satisfaction helps portfolio earn higher return (for both up and down market)**

36.5.2 (Srinivasan and Hanssens, 2009) Marketing and Firm Value

- Marketing investments don't always translate to firm value readily.
- Marketing investments are typically intangible:
  - brand equity
  - customer equity
  - customer satisfaction
  - R&D

  - product quality
  - specific marketing-mix actions
- Market is not so efficient: e.g.
  - Intangible-intensive firms are usually undervalued (Lev, 1989)

Market Valuation Modeling:

- Fame-French factor explains excess returns come from
  - market risk factor: excess return on a broad market portfolio
  - size risk factor: difference in return between a large and small cap portfolio
  - value risk factor: difference in return between high and low book-to-market stocks
  - Momentum: Carhart (1997)
- Metrics:
  - Top-line (revenue)
  - bottom-line (earnings) surprises
- Methods: 4-factor model can still have omitted variables

Metrics on Marketing and Firm value

- Market cap: need to
  - isolate the book value (using Tobin's q)
  - Incorporate random-walk behavior in stock prices (first difference of log(stock price))
- stock returns

Table 36.5: Table 1 Adapted from the Overview of research approaches (p. 295)

Method	Characteristics	Litimations	Examples	Dependent/Independent
Four Factor Model	Assume efficient market theory	sensitive to benchmark portfolio correlation analysis can contain omitted variable bias examine cross-sectional variation only	(Rao et al., 2004) (Barth et al., 1998) (Madden et al., 2002)	Tobin's q/ Branding strategy Firm val/ brand value estimates Stock returns/ brand valuation
Event Study	Assume efficient market Causal Analysis	can't measure long-term effect	(Horsky and Swyngedouw, 1987): name change (Chaney et al., 1991): new product intro (Lane and Jacobson, 1995): brand extension (Geyskens et al., 2002)	Stock returns/ name events Stock returns/ new product intro Stock returns/ brand extensions Stock returns/ Internet channel

Method	Characteristics	Litimations	Examples	Dependent/Independent
Calendar protfolio	Include firms with certain to measure long-term impact more accurate than event studies	Can't measure per event effect might be sensitive to benchmark prtofolio	(Sorescu et al., 2007)	Stock returns/ new product
Stock return response model	based on Carhart (1997) and EMH account dynamic properties of stock returns incorporate continuous events	detailed data at the brand so business unit level marketing info must be public single equation model without temporal chain	(Aaker and Jacobson, 1994) (Aaker and Jacobson, 2001) (Mizik and Jacobson, 2003) (Srinivasan et al., 2009)	Stock returns/ perceived quality Stock return / brand attitude stock return/ strategic shifts Stock returns/ marketing actions
Persistency modeling	system of equations: consumer (demand equation), manager (decision rule equation), competition, (competitive reaction equation), investor (stock price equation) VAR: examines both short-term and long-term robust to deviations from stationarity incorporate dynamic feedback loops	detailed data at the business unit level time-series over a long horizon reduced-form models	(Pauwels et al., 2018) (Joshi and Hanssens, 2010)	Firm value/ new product intro, sales promotions stock returns/ advertising

4 factor model:

$$R_{it} - R_{rf,t} = \alpha_i + \beta_i(R_{mt} - R_{rf,t}) + s_i SMB_t h_i HML_t + u_i UMD_t + \epsilon - it$$

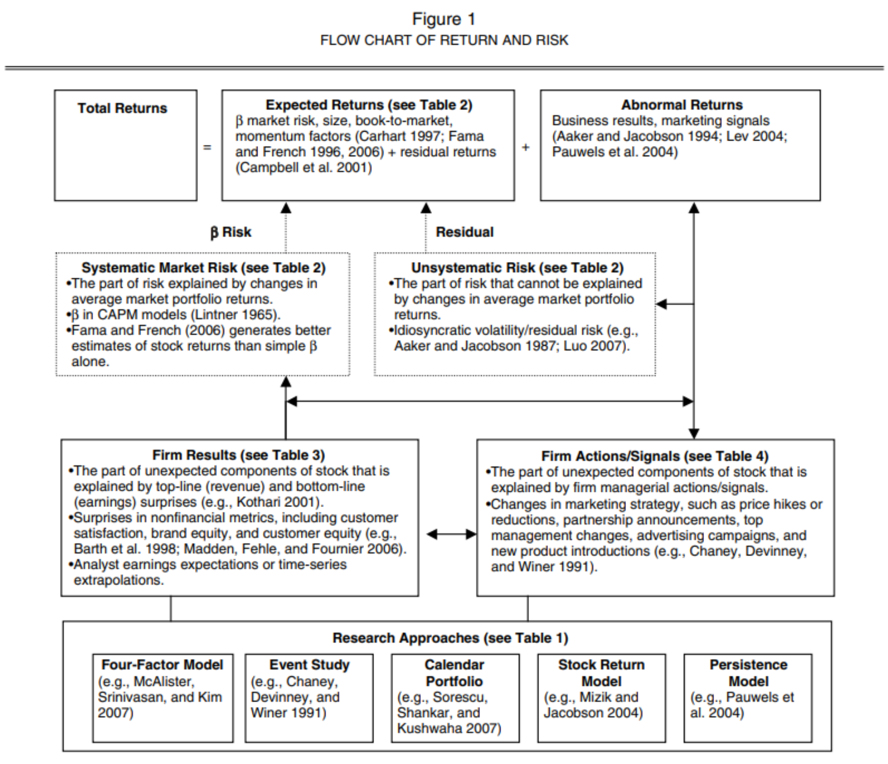


Figure 36.1: Figure 1: Flow chart of return and risk p. 297)

where

- R_{it} = stock return for firm i at time t
- $R_{rf,t}$ = risk-free rate in period t
- market factor = R_{mt} = market return in period t
- Size factor = SMB_t = return on a value-weighted portfolio of small stocks - the return of big stocks
- Value factor = HML_t = return on a vlaue-weighted portfolio of high book-to-market stocks - return on a value-wegihted portfolio of low book-to-market stocks
- Momentum factor UMD_t = average return on 2 high prior-return portfolio - the average return on two low prior return portfolio

36.5.3 (Sood and Tellis, 2009) Innovation and Stock Return

- Innovation is important for firms
- But firms are cautious when investing in R&D (long-term effect hard to justify)
- Finding: innovations effect on stock prices is underestimated when events are distinct vs. aggregate

3 types of innovation activities

1. Initiation: alliance, funding, expansions
2. Development: Prototypes, patents
3. Commercialization: Porudct Launch, awards

Takeaways

- Total market returns to an innovation project: 643 mil (compared to 49 mil the return to an average event in the innovation project)
- Positive events increase returns for all three types of events
- Negative events decrease return for development and commercialization stages only
- The absolute value of the market returns is higher for negative announcements than for positive announcements

36.5.4 (Jacobson and Mizik, 2009a)

- Disagreeing with previous research conclusion that there was a systemic mispricing of customer satisfaction into the stock price (Fornell et al., 2006) (Aksoy et al., 2008), the anomaly stem from only a small group of satisfaction leaders in the computer and internet sector. (i.e., sampling bias).

- This study is consistent with (O'Sullivan et al., 2009)

36.5.5 (Jacobson and Mizik, 2009b)

36.5.6 (Borah and Tellis, 2014) Choice of Payoff from announcements (Innovations)

- Whether a firm should make, buy or ally regarding new technologies

Innovation phases:

1. Initiation
    1. Make
    2. Buy
    3. Ally
2. Development
3. Commercialization
    1. New product launch
    2. initial shipments
    3. new app and markets for the new products
    4. awards

Models

1. Model of returns
2. Model of investment choice: multinomial logit model
3. Model of payoffs:

36.5.7 (Tirunillai and Tellis, 2012) Chatter effect on stock performance

Research questions:

- Cor(UGC, stock performance)
- What is the direction of causality
- Among the UGC metrics, which best relates to stock performance
- What are the dynamics of the relationship in terms of wear-in, war-out, and duration?

Data: 4 years, 6 markets , 15 firms

Findings:

- Volume of chatter increases abnormal returns by a few day (using Granger causality tests) and trading volume

- Positive UGC has no effect on abnormal returns
- Negative UGC has negative effect on abnormal returns with a short "wear-in" and long "wear-out"
- Interaction between chatter volume and negative chatter have a positive effect on trading volume
- negative UGC positively correlates with idiosyncratic risk
- Positive UGC has no effect on the idiosyncratic risk
- Offline ad also increases the volume of chatter and decreases negative chatter

UGC:

- Product reviews + product ratings

Stock performance:

- A measure of shareholder value
- Available at the daily level

Assumption:

- Market is not efficient: it takes time for the market to reflect info about UGC.
- Asymmetric response across UGC metrics:
  - Losses loom larger than gain
  - investors discount positive info because it's unreliable
  - Positive messages are usually influenced by the firms, but not negative

Sampling:

- Product categories that have rich data on UGC (digital, high tech and popular consumer durable)
- Product categories that reviews are related to sales
- Public firm only
- No M&A during the period
- The sample markets should be representative of the whole market.

Time: June 2005 - Jan 2010

Media:

- Product reviews instead of text or videos, etc because intuitively people use this form to express their opinion

- Consumer reviews instead of evaluations, blogs, forums, because it's more focused and greater signal-to-noise ratio
- Consumer reviews instead of expert review because of wisdom of the crowds
- 3 popular websites: Amazon.com, Epinions.com, Yahoo! Shopping.
- ratings + text reviews

Measures

- UGC: ratings, volume chatter, positive valence, negative valence
- Stock market performance
  - Abnormal returns: Fame-French (1993) three-factor + Carhart 1997 momentum factor.
  - Idiosyncratic risk: same model as abnormal returns
  - Trading volume: = daily turnover = volume of trade / shares outstanding at the end of the day

Using EGARCH specification:

$$R_{i,t}-R_{f,t} = \alpha_i+\beta_{i,MKT}(R_{MKT,t}-R_{f,t})+\beta_{i,SMB}SMB_t+\beta_{i,HML}HML_t+\beta_{i,MOM}MOM_t+\epsilon_{i,t}$$

where

- $\epsilon_{i,t} \sim N(0, \sigma_{i,t})$

$$\ln(\sigma^2_{i,t}) = a_i + \sum_{j=1}^{p} b_{i,j} \ln(\sigma^2_{i,t-j}) + \sum_{k=1}^{q} c_{i,k}\{\Theta(\frac{\epsilon_{i,t-k}}{\sigma_{i,t-k}}) + \Gamma(|\frac{\epsilon_{i,t-k}}{\sigma_{i,t-k}}| - (\frac{2}{\pi})^{1/2})\}$$

Control Variables

- Analysts' Forecasts: IBES Database
- Advertising: TV ad from TNS media Intelligence
- Media Citations: Number of articles in print media from LexisNexis (with relevancy score above 60%) and Factiva (using company tag)
- New product Announcement: also LexisNexis and Factiva (following (Sood et al., 2009))

Models

Vector Auto-regression (VAR)

- can handle continuous events (instead of discrete events used in event studies)
- account for immediate and lagged-term of the independent variables
- capture the carryover effects over time with the generalized impulse response function
- Controls for trends, seasonality, non-stationary, serial correlation, and reserve causality (Luo, 2009)

Procedure

1. Estimate the stationary (unit roots + co-integration) properties of stock performance and UGC
    1. Stationarity test: Augmented Dickey-Fuller test + Kwiatkowski-Philips-Schmidt-Shin test
    2. Co-integration: Johansen's procedure (Johansen et al., 1992)
2. Granger causality test
3. Estimate dynamics of carryover effect using impulse response function
    - Not sensitive to the causal ordering to the causal ordering of the variable in the system of equations
4. Estimate the effect of UGC using variance decomposition: relative importance of metrics of UGC

36.6 Creativity

Implications of social media

1. Wisdom of the Crowds
2. Advertising almost free

36.6.1 (Bayus, 2013) Crowdsourcing New Product Ideas over Time

- from dell's IdeaStorm community, serial ideators are more likely to have 1 idea that the organization will implement, but they don't repeat this success.
- Negative effect of past success can be mitigated for idators with more diverse commenting activity
    - Fixation effect = unconscious plagiarism (or cryptomnesia) (Marsh and Landau, 1995) (Marsh et al., 1999)
- Good
    - First paper to study crowdfunding of ideas
    - Good theory: fixation effect

  - Good descriptive analysis

- Cons
  - Model: not taken into account rare events.

36.6.2 (Toubia and Netzer, 2017) Idea generation, creativity, prototypicality

Creativity = balance(novelty , familiarity)

Beauty in avergeness effect

Automate read ideas to identify promising ones

Research questions

1. How novelty and familiarity defined in the idea generation context? From literature using Geneplore
   1. "novelty is the association of word stems that do not appear frequently together in text related to the topic under consideration" (p. 3)
   2. "familiarity is the association of word stems that appear frequently together" (p. 3)
2. How should novelty and familiarity be measured? semantic network co-word analysis (by the combinations of word stem instead of the word itself)
3. What is the optimal balance between novelty and familiarity? beauty in averageness effect

idea = "a document made of words that attempts to add value given a particular idea generation topic" p. 2

Automatically recommend words to improve idea

Baseline for semantic network:

- Pre-test idea: consumers generate initial set of ideas on a topic
- Google results: top search (might be biased to high-quality contents)

Used Jaccard index for edge weights

Control variables: (Barrat et al., 2007)

- Frequencies of nodes in the network: average edge weight, coefficient of variation of edge weights, minimum edge weight, maximum edge weight, average node frequency, coefficient of variation of node frequencies, minimum node frequency, maximum node frequency, and the number of nodes in the subnetwork, length of the idea using number of characters
- Clustering coefficients of the nodes in the network; average node clustering coefficient, coefficient of variation of node clustering coefficients, minimum node clustering coefficient, and maximum node clustering coefficient.

Prototypical distribution of edge weights using mean of the prototypical distribution

Measure distance between two distributions - The Kolmogorov-Smirnov statistics (2 cdfs). Alternatively could use Kullback-Leibler divergence

Idea evaluation: manual with 4 dimension: creativity, purchase interest, predicted popularity, writing quality

Alternative measure to edge weight distributions: Info retrieval literature: vector space representation: each document as a vector with dimensionality equal to the number of word stems in our dictionary (i.e., number of nodes in our semantic network

Specification of the baseline semantic network is dangerous to the sub-network distribution.

Robust to synonyms

Strengths:

- Good way to measure a complex and highly qualitative construct
- Good connection between the theory and method
- Robust
  - Different measures, ideas, evaluators, baseline networks.

Cons

- With other representations, the results do not hold

36.6.3 (Wei et al., 2021) Machine leaning creativity

- Crowdfunding: for both finance and marketing (market reaction, advertise ideas)
- Combinatorial theory:
- measure novelty, overshooting and undershooting, measure styles of imitation
- Research questions
  - How to measure the similarity between all the projects on crowdfunding sites in an objective and automated way?
  - The relationship between the similarity pattern and funding performance
    * Can previous successful projects that are similar product a new project's success?
    * Do people value novelty?

        * whether to overshoot or undershoot the funds raised?
        * Do people value atypicality?
    - Recommendation from the similarity measure

- Data: 98,058 Kickstarter projects from 2009 - 2017 (from 3 categories: Film & Video, Music and publishing. only English.
- Techniques: Semantic Similarity
    - Word2vec: word-level similarity
    - Word Mover's Distance (WMD): Document-level similarity $w_{ij} = \delta^{|t_i - t_j|} \times L(\gamma_0 - \gamma_1 d_{ij})$ where $0 < \delta \leq 1$ is the decay factor, d_{ij} is the WMD between 2 projects and L is the logistic function and γ are chosen based
- Similarity network where each node is a project,and the strength of a link
    - Increases with degree of similarity
    - decreases with the time lapse between 2 projects
- Funding performance
    - Whether the funding is successful
    - How much money is raised
- Findings
    - The average level of success by prior projects is a good predictor of the current project's funding performance
    - High novelty means less similar to all previous projects, good projects are balanced of being novel and appearing familiar to investors
    - Goals should be set close to the number by prior similar projects
    - An inverted U-shaped relation between atypicality (borrow from another stream) and funding performance
- Recommendations:
    - goals should be benched marked by other previous projects ($\pm 10\%$ goal adjustment)
    - project should be similar to prior projects

Combinotorial

- Geneplore framework
    - Generation process: retrieve prior info and recombine in a creative way

- Exploration process: these recombinations will be elaborated

Results are robust against unweighted network whether link is present when it passes certain threshold.

Network-based metrics

- Amount of prior similarity: degree of similarity
- Prior success rate: weighted average of previous similar projects.
- Prior success residual: reweigh the success rate with other control variables
- Goal overshoot: difference between the focal project' funding goal and the average of previous project funding goal (in log)
- Atypicality: use unweighted network (using the cutoff of .5), atypicality = proportion of isolated in i subnetwork.

Control variables

1. Project-related features
    1. Log funding goal
    2. log number of images
    3. Dummy for video
    4. Log length of the project depreciation text
    5. Dummy for project category
    6. Time trend and quarter dummies
2. Creator-related features
    1. Dummy for prior project
    2. Average success rate of the creator's prior projects

Models

- Success: logistic
- Fund raised: regression

Information weighting: $I_i \equiv \log(1 + \sum_{j:T_j<t_i} w_{ij})$ choosing this specification because

1. when there is no similarity between the focal project and prior projects, the information weight should be 0
2. Under the Bayesian framework, there is a diminishing return of more signals.

Info weight is used for all metrics except similarity and atypicality

36.6.4 Can AI do ideation? 2022

Basic research question: How to screen ideas

Based on 3 models:

- Word Colocation
- Content Atypicality
- Inspiration Redundancy

Prediciton mode

- LASSO
- Random Forest
- RuleFit

36.6.5 (Berger and Packard, 2018) Content Atypicality

- Ideas are better if they are different from other in the same contest.

36.6.6 (Stephen et al., 2016) The Effects of Network Structure on Redundancy of Ideas

- Ideators with more diverse background tend to have better idea.

36.7 Quality

- Fundamental concept in many disciplines: policy, economics, consumer behavior, marketing strategy
- Quality: attribute on which all (most) consumers prefer more to less (e.g., speed, reliability, durability, power). (Tellis and Wernerfelt, 1987)
- Market for quality (Klein and Leffler, 1981): why quality commands a premium

Measurement of objective quality

- Consumer reports (historically, until 2010)
  - Since 1935
  - Blind experiments with products
  - evaluated by experts
  - Problem: quality is multi-dimensional, composite quality depends on choice of dimensions and weights to combine them.
- Solutions:
  - (Kopalle and Hoffman, 1992): ranking products on quality not too noisy even if the weights are uncorrelated. but you still need attributes of quality to be positively correlated.

  - (Tellis and Johnson, 2007): expert reviews: published quality ratings are good indicator of quality
  - (Tirunillai and Tellis, 2014): wisdom of the crowds

36.7.1 (Tellis et al., 2009) Network effects and quality in high tech

- Evidence for market efficiency (defined as the best quality brand should have the largest market share)
- Both quality and network effect affect market share flows (network effect > quality)
- Network effect: "the increase in a consumer's utility from a product when the number of other users of that product increases." (p. 135)
- Quality is defined as "a composite of a brand's attributes, on each of which all consumers prefer more to less." (p. 136) (e.g., reliability, performance, convenience).
- Quality seems to be the driving force of the market (market share, return on investment, premium prices charged, advertising, perception of quality, stock market return, p. 136)

Theoretical cases: table 1

Sampling: Personal computer

Data: from International Data Corporation and Dataquest

36.7.2 (Golder et al., 2012) An Integrative Framework for Quality

Quality processes:

- Quality production process: focus on firms.. depedns on attribute design, process design, resoruce inptus and methods of controlling the production process.
- Quality experience process: focus on customers
  - What the firm deliver and what the customer perceive can be different (relative to expectation) depends on
    * customer measurement knowledge
    * motivation
    * emotions
  - Experienced Attribute Quality vs. Delivered Attribute

- Quality evaluation process: based on transactional and global judgments
  - "is the conversion of perceived attributes into an aggregated evaluation of quality, which is a summary jdugment of the customer's experience of the firm's offering." (p. 9)
  - Evaluated aggregated quality is based on customer expertise and attribute characteristics
  - Customer Expectations: (1) "Will" expectation (2) "Ideal" expectation (3) "Should" expectation (perceived quality and fairness)

Quality is defined as " a set of three distinct states of an offering's attributes' relative performance generated while producing, experiencing, and evaluating the offering." (p. 2)

Figure 1 shows the framework

Typology of attribute types:

1. Customer preference: homogeneous vs. heterogneous
2. Measures ambiguity: unambiguous vs. ambiguous

		Customer preference	
		Homogenous	Heterogeneous
Measure ambiguity	Unambiguous	Universal attributes (flight delay)	Preference attributes (meal cuisine type, cabin temperature)
	Ambiguous		Idiosyncratic attributes (art, beauty)

36.7.3 (Tirunillai and Tellis, 2014) Mining Quality from Consumer Reviews

- use unsupervised LDA to measure quality dimensions in UGC
- Data: 350,000 consumer reviews from (Tirunillai and Tellis, 2012)
- Results
  - Dynamic analysis allows marketers to track the value of variables over time and dynamically map competitive brand positions on those dimensions.

Market	Dimension	Across markets	Heterogeneity	Stability
Vertically differentiated (computer)	Objective dimensions dominate	Similiar	Low across dimensions	high over time
Horizontally differentiated (Shoes, toys)	Subjective dimensions dominate	Vary	High across dimensions	Low over time

36.7.4 (Borah and Tellis, 2016) Spillover Effects in Social Media

- Perverse halo (negative spillover): negative chatter about one nameplate increases negative chatter for another nameplate. And affect both sales and stock performance.
  - Depends on the similarity between the focal and rival brand's market shares (dominant brand's spillover is stronger) and countries of origin (similar COO suffers more).
- Apology ad is harmful on both recalled brand and its rival
- Online chatter amplifies the negative effect of recalls on downstream sales by 4.5 times.
- Definitions:
  - Brand = makes of the automobiles (e.g., Toyota)
  - Subbrand = automobiles with their own name (Toyota, Lexus)
  - nameplate = name of the automobile model under the subbrand (Corolla or Camry)
  - brand dominance = higher market share means higher dominance
- Based on the accessibility-diagnosticity theory by (Feldman and Lynch, 1988), one brand's perceptions can be used to make inference aobut another brand's perception if they are simialrin the consumer's mind.
- Data:
  - Industry context: automobile
  - Time: Jan 2009 - April 2010 (can only obtain chatter through 2010)
  - Include both voluntary and involuntary recalls. Using Granger-causality, do not find temporal causality from negative chatter to recalls (evidence, but not strong)
- Measures of endogenous variables

  - Online chatter: only negative online chatter
  - Media citations: in print media per day on LexisNexis with 60% relevancy score (similar to (Tirunillai and Tellis, 2012)
  - ABC news coverage; because the network broke the news from LexisNexis
  - Negative events in Toyota's acceleration crisis: 1 for negative event day.
  - Advertising: from Kantar using 4 types: general, promotional, leasing, and advertisements with only apology ad.
  - Key developments: earnings announcements, acquisition, strategic alliances, awards using data from brand's websites and S&P capital IQ data
- Measures of exogenous variables
  - Recalls: units of recalls. with evidence from Granger causality that recall is unlikely to be endogenous
  - New product intro: Use the brand website and Capital IQ and can't find evidence that new product negative online chatter Granger-caused new product introductions.
- Modeling:
  - VARX:
    * Estimates Granger Causality
    * Robust to nonstationarity, spurious causal, endogeneity, serial correlation, and reverse causality
    * Estimate the long-term or cumulative effects of causal variables using the impulse response functions
- Results
  - Perverse halo exists in online chatter
  - Perverse halo is stronger for brands from the same country
  - Perverse halo is stronger from dominant brands to less dominant brands
  - perverse halo has a one-day wear-in period and wear-out six days
  - Within-brand perverse halo exists because consumers are aware of the family brand
  - Apology ads increase concerns (negative chatter)

- Concerns about the focal nameplate significant decrease the nameplate's sales and rival's sales
- Using the forecast error variance decomposition, concerns about the focal nameplate explain more of the variance of the focal nameplate's sales than that of the nearest rival.
- Increase in concerns will decrease Toyota's stock performance and reach its lowest point on the fourth day. But mixed results on the significant effect on rival brands due to the country of origin effect.

Chapter 37

WashU Analytical Model

Interview:

- Don't even show your findings first (people will attack them)
- Never deny that you have a problem (always accept or have good answers for your problem).
- Only present summary, main results, and leave extension in the end if there is no question asked.

37.1 Platforms

37.1.1 (Jiang et al., 2011) Amazon's mid-tail

- Firm strategies in the "mid tail" of platform-based retailing
- For two sellers on Amazon (high vs. low demand sellers), the threat to entry from the platform benefit them.
- Upon entry, the platform decreases consumer surplus in the early period, while increases in later period
- Intuitively, Amazon should sell high-volume products and independent sellers should sell low-volume long-tail products.
- For mid-tail sellers, if they sell too good, they will be taken over by Amazon. Hence, reducing sales can avoid Amazon's notice.
- Findings:
  - After observing demand, the platform can separate the high demand and low-demand seller by setting high fee. Thus only high-demand seller can remain, and later the platform will take the market share

of these ones. With low probability of high demand, the platform can set low fee (which allows the high-demand sellers to mask).

  - The platform can forgo the option to independently sell altogether (so that both types of seller will provide high level of service and subsequently high sales). But both types of sellers prefer the threat of entry because of the low per-unit fee for ex ante probability of high demand.
  - If the platform enters, it will reduce consumer surplus at first and increase it later.
  - With the consumer reviews (to proxy seller service quality), the platform can increase its optimal sales fee if the ex ante probability of a high-demand type is low and vice versa.

- Related literature:
  - Internet retailing
  - Two-sided market
  - Distribution channel
  - Asymmetric information games
- Model

Amazon's fixed cost: $F > 0$

Seller fixed cost $= 0$ (sunk)

Demand in each period

$$q^{(i)}(p, e) = \gamma + e(i) - bp^{(i)}$$

where

- p is the price of the product
- $e =$ service level
- γ, b are constant

Uncertainty about product (= uncertainty about the seller's type:

1. High with probability θ
2. Low with probability $1 - \theta$

Per-unit marginal cost is $s(e) = ke^2$ where k is constant

There are two conditions for Amazon to make decisions:

1. If the fixed cost is too high, Amazon will let the sellers to sell

2. If the fixed cost is low enough, and Amazon know the true demand is high ($\gamma = \gamma_H$) then it will sell directly

Two-stage game:

1. Nature determines the seller's type, Amazon charge a per-unit fee f (constant in 2-period). Sellers choose to sell or not (both service level $e^{(1)}$ and price $p_t^{(1)}$ where $t \in \{L, H\}$)
2. Amazon updates beliefs about the seller's type, Amazon decides to sell directly or not.

3 case studies:

1. Amazon with full info
2. Amazon promises to not enter: without threat, both types of seller will have the same service levels for both periods
3. Amazon with incomplete info and decide whether to enter
    1. Separating equilibrium, based on the fee f
        1. H-type could mimic L-type to avoid Amazon notice
        2. Only H-type can sell on Amazon. Hence, it makes profit in the first period and 0 in the second.
    2. Pooling equilibrium

37.1.2 (Zou and Zhou, 2021) search neutrality

- Counter-intuitive idea: Search neutrality hurts consumers, but good for platforms (due to reduced price competition)
    - Search neutrality decreases seller's price competition (only apply to online platform with personalized search, not to offline store with uniform display)
    - it decreases consumer surplus but increases consumer's search relevancy
- Search neutrality is different from net neutrality
- Related literature:
    - Intermediary biases (e.g., recommendation results, copycat products, etc.)
    - Consumer search

Model

2 groups of consumers:

- Consumer prefer first-party products than third-party products
- Consumer prefer third-party over first-party

Consumer dynamics (sequentially):

1. Pre-search:
    1. no costs
    2. Consumer learn expected valuations from pre-search
2. Search:
    1. Search costs to determine match and exact value

Consumers' action set

1. buy the prominent product
2. end search (no purchase)
3. incur search cost to inspect non-prominent product

3 stages:

1. Set prices
2. Personalized search rankings
3. Consumer purchase

- It's still hard to implement the search neutrality policy (similar to the loan policy is not based on race).
- Tradeoff between commission (profits from true ranking) and product sales from fake order. (but in the paper, they already assumes that the platform capture higher profits from selling the first-party products)
-

37.1.3 Diao et al. 2021 P2P rideshare vs. taxi

- Competition between P2P ridesharing platforms and traditional taxis

37.1.4 (Gal-Or and Shi, 2022) Subscription platform

- Designing entry strategies for subscription platforms
- Platforms like ClassPass (fitness) or MoviePass, MealPass, Inspirato Pass (hotel and resorts) offer physical service subscriptions.
- Reason for limited offer is the transportation cost
- 2 conditions:
    - Consumers: variety-seeking
    - High transportation cost

Assumptions:

- One platform offers a bundle of two vendors (on the Hotelling's line)

Possible variables in contract:

- Quality of the service

- Transfer payments between the platform and the vendor: variable fee per customer + fixed fee
- The number of platform customers that vendors agree to serve.

No deal if there are only 2 variables: a fixed transfer payment + quality of service (due to intensified competition in prices). Only when we add either (1) transfer fee per customer or (2) the number of customers the that vendors agree to serve.

Additional requirement: balanced bargaining positions between vendors and the platform.

Recommendations:

- Enter markets where vendors are differentiated (variety-seeking consumer will be enticed -> higher industry profits)
- Weak competitions between vendors (e.g., local vendors are local monopoly because of high transportation cots)
- Appropriate tools to avoid price competitions:
  - transfer fee per customer
  - impose restrictions on the quantity
  - Do not: offer lower quality service for platform-user as compared to vendor-users. (restrictions on quantity is better than restriction on quality).

Contributions:

- Bundling: Different from retailer's bundles, because platform (bundle) competes against the independent sellers as well. Different from information goods bundling, this study provides suggestion on the bundling of physical services
- Platforms: different types of platform (e.g., as a marketplace, as a retailer), this study focuses on competition between a platform (not an independent seller) and vendors.
- Channel management: possible only when there is little price competition.

Model
2 vendors, 1 platform

- p_i = price charged by vendor i
- s = subscription fee charged by the platform.
- v = gross utility
- $(1+\alpha)v$ is the gross utility when the consumer wants variety where $\alpha \in (0,1]$

  - α denotes the preference for variety

- If we assume that the vendors offer $0 \leq k_i \leq 1$ (lower quality service) to the platforms subscribers, the gross utility will be $[(k_1 + k_2)/2](1 + \alpha)v$ (i.e., the gross utility reduced by the average quality offered by vendors). (not too sure why the authors introduce this assumption so early)
- Say that a subscriber after purchasing the pass, only utilize one, her utility is $k_i v$

The net utility when a consumer pick only i is

$$U(x_i, \text{only vendor } i) = \max\{[v - tx_i - p_i], [k_i v - tx_i - s]\}$$

Either the customer has to purchase directly, or via the platform

The net utility when a consumer want to choose both

$$U(x_i, \text{both vendors}) = \max\{[(\frac{k_1 + k_2}{2}(1+\alpha)v - t - s], [(1-\alpha)v - t - (p_1 + p_2)]\}$$

again, either purchase from the platform, or directly from the service providers.

Notice difference in transportation costs between one or two services utilization.

Restriction for model tractability: $(1 + \alpha)v - t > 0$ (see for both vendors) means that consumers always derive a net positive (not including total price paid). (questionable assumptions)

In a market that has both vendors and platform (i.e., positive market share), their prices have to satisfy:

$$p_1 + p_2 - v(1 + \alpha)[1 - \frac{k_1 + k_2}{2}] \geq s \geq p_i - (1 - k_i)v$$

- Lower bound: if a consumer only buys from one vendor, she will directly purchase it from the vendor
- Upper bound: if a consumer wants to buy both, she will subscribe.

Defensible assumptions:

1. Vendors continue to offer services independent of the platform
2. If there is no market share for vendors (i.e., vertical integration), it will attract antitrust law suit.

Market share:

$$MS_i = x_i^* = \frac{t + s - p_i - v[(\frac{k_1+k_2}{2})(1 + \alpha) - 1]}{t}$$

where $i, j = 1, 2; i \neq j$

and

$$MS_p = \frac{v[(k_1 + k_2)(1+\alpha) - 2] - t - 2s + p_1 + p_2}{t}$$

Stages:

1. The platform decides whether to negotiate with the vendors
2. The platform negotiates with the vendors separately (i.e., different terms, different contracts)
    1. Pairwise negation between the platform and the individual vendors, use Generalized Nash bargaining solution
3. The parties (vendors + platform) set prices noncooperatively and independently

2 regimes (common variables: The quality of the service k_i and the fixed payment F_i):

1. Variable transfer fee: Fee per customer m_i -> higher platform fee -> lower market share
2. Quantity restriction: Number of customers agree to serve y_i -> directly lower market share

Pairwise negotiations:

1. Platform vs. vendor 1: k_1, F_1, m_1 (or y_1)
2. Platform vs. vendor 2: k_2, F_2, m_2 (or y_2)

To establish the Generalized Nash Bargaining solution, we have to see the outside options for each party

1. Platform = 0: cannot be formed if there is no agree with either vendor
2. Vendor (pick only the equilibrium that yields the highest aggregate industry profits because of allowance for side payments): $w_i = f(\alpha, v, t)$

Lemma 1: The best possible outside option for both vendors ($w_1 + w_2$) to be highest satisfy 4 conditions

When	w_i	Implication
$\frac{1}{1+\alpha} < \frac{v}{t} < 1$ Since the transportation cost is so high, the market is not covered (i.e., those far away drop out, and those in also go to one vendor)	$w_i = \frac{v^2}{4t}$ Each vendor becomes local monopoly ($\pi_i = w_i$)	Without the platform, the lower the transportation cost, the more intense the competition on price is.

When	w_i	Implication
$[\alpha < \frac{2\sqrt{2}}{e} \& 1 \leq \frac{v}{t} \leq \frac{3}{2}]$ or $[\alpha \geq \frac{2\sqrt{2}}{3} \& 1 \leq \frac{v}{t} \leq \frac{1-\sqrt{1-\alpha^2}}{\alpha^2}]$ Market is covered Consumer only buys from 1 vendor	$w_i = \frac{1}{2}(v - \frac{t}{2})$ Multiple equilibrium satisfying sum of vendors prices equal $2v - t$	
$\alpha < \frac{2\sqrt{2}}{3}$ and $\frac{3}{2} \leq \frac{v}{t} < \frac{\sqrt{2}}{\alpha}$ Low transportation cost and low variety seeking parameter $\alpha \leq (2\sqrt{2})/3$ Consumer only buys from 1 vendor	$w_i = \frac{t}{2}$	Price competition is intense
$[\alpha < \frac{2\sqrt{2}}{e} \& \frac{\sqrt{2}}{3} \leq \frac{v}{t} \leq \frac{2}{\alpha}]$ or $[\alpha \geq \frac{2\sqrt{2}}{\alpha} \& \frac{1+\sqrt{1-\alpha^2}}{\alpha^2} \leq \frac{v}{t} \leq \frac{2}{\alpha}]$	$w_i = \frac{(\alpha v)^2}{4t}$	Most intense competition

Then the Generalized Nash Bargaining solution can be formulated

under the "variable transfer fee"

$$\max_{k_i, m_i, F_i} NB_{ip} = [MS_i p_i + m_i MS_p + F_i - w_i]^{\beta}[MS_p(s - m_i - m_j) - F_i - F_j]^{1-\beta}$$
$$= [\text{net vendor i account for outside option}][\text{net for the platform}]$$

where $i, j = 1, 2; i \neq j$

under the "quantity restriction"

$$\max_{k_i, y_i, F_i} NB_{ip} = [MS_i p_i + F_i - w_i]^{\beta}[MS_p s - F_i - F_j]^{1-\beta}$$
$$= [\text{net vendor i account for outside option}][\text{net for the platform}]$$

where $i, j = 1, 2; i \neq j$

and β = negotiating power of vendor i and

$1 - \beta$ = negotiating power of the platform

Prerequisites for the existence of the platform

1. The total surplus upon agreement with both vendors is higher than the sum of outsides options

$$TS = sMS_p + p_1MS_1 + p_2MS_2 > w_1 + w_2$$

(shouldn't it only be $p_1MS_1 + p_2MS_2 > w_1 + w_2$, why do we need to include sMS_p)

2. Under the "variable transfer fee" regime, parties can make decisions on quality of service k_i and fee per customer m_i of vendor i and the platform (given the deal with j is done already)

$$\max_{k_i, m_i} TS_{ip} = (s - m_j)MS_p + p_iMS_i - F_j$$

(This should be the total surplus $TS = (s - m_1 - m_2)MS_p + p_1MS_1 + p_2MS_2 - F_1 - F_2$

3. Under the quantity restriction regime, parties can negotiate on quality of the service k_i and number of customer agree to serve y_i for vendor i and platform (given the deal is done with j already)

$$\max_{k_i, y_i} TS_{ip} = sMS_p + p_iMS_i - F_j$$

Introduction of the platform can increase the industry total profit because it widens to pie to include people who value variety and those who are farther away from either vendor.

Proposition 1: With only two variables: (1) k_i (quality service) (2) F_i (fixed payment), there will be no agreement.

But with either regime, agreement can be reached

1. **Variable transfer fee**

Maximization problems (i.e., setting prices to maximize profits):

$$\begin{aligned}
\max_{p_i} \pi_i &= p_iMS_i + m_iMS_p + F_i \\
&= \frac{p_i[t + s - p_i - v[(\frac{k_1+k_2}{2})(1+\alpha) - 1]}{t} \\
&+ \frac{m_i[v[(k_1 + k_2)(1+\alpha) - 2] - t - 2s + p_1 + p_2]}{t} \\
&+ F_i
\end{aligned}$$

where $i = 1, 2$

and

$$\begin{aligned}\max_s \pi_p &= (s - m_i - m_j)(MS_p) - F_i - F_j \\ &= \frac{(s - m_i - m_j)[v[(k_1 + k_2)(1 + \alpha) - 2] - t - 2s + p_1 + p_2]}{t} - F_i - F_j\end{aligned}$$

After solving

$$p_i = \frac{6t + 11m_i + 5m_j - 2v[(1 + \alpha)(k_1 + k_2) - 2]}{12}$$

$$MS_i = \frac{6t - m_i + 5m_j - 2v[(1 + \alpha)(k_1 + k_2) - 2]}{12}$$

and

$$s = \frac{v[(1 + \alpha)(k_1 + k_2) - 2] + 5(m_1 + m_2)}{6}$$

$$MS_p = \frac{v(1 + \alpha)(k_1 + k_2) - 2] - (m_1 + m_2)}{3t}$$

Both p_i, s increase with m_i (variable transfer fee) while if we were to negotiate on ly on k_i (quality), only p_i will increase while s will decrease. Allowing for negotiation based on variable transfer fee can facilitate agreement.

Since in the second stage, we have to make that the indifferent customers can buy from both the vendor and the platform. And the net payoff of platform customers are equal (because of same transportation cost)

$$\max_{k_i, m_i} TS_{ip} = p_i MS_i + (s - m_j) MS_p - F_j$$

where $i, j = 1, 2$ subject to (non negative utility for the indifferent customer)

$$U(x_i^*) = (1 + \alpha)v - t - S \geq 0$$

Proposition 2:

With the addition of the "variable transfer", the platform can enter the market. And it has the following properties:

1. $k_1 = k_2 = 1$ (same quality both platform and vendor users) (both parties have no interest to offer different quality because of cannibalization) and the whole market is covered.

2. For the entry of platform, the variety parameter α has to be big. If the negotiating position of the platform increase (greater value of α), the negotiating powers of each vendor should improve as well (i.e., weak competition between the vendors).

Isn't big value of the variety parameter is the same thing as the two vendors are very much differentiated? No, because two vendors can be local monopoly that has no incentive to agree with the platform (unless there is a segment of consumers that you could serve additionally).

3. When before entry competition was weak, the surplus of platform customers will be realized upon entry
4. When before entry competition was strong, the platform customer only derive a positive surplus only when the competition continues to be strong (big value of $\frac{v}{t}$)
5. The payoff for the parties are

$$pi_i = \frac{\beta}{1+\beta}[TS + \frac{w_i(1-\beta)}{\beta}]$$

and

$$\pi_p = \frac{1-\beta}{1+\beta}[TS - 2w_i]$$

2. **Quantity Restriction Regime**

To specify y_i as the number of customers vendor i agrees to serve, we can either use (1) $(y_1 + y_2)/2$ or $\min(y_1, y_2)$ (the first specification is easier to understand and solve while both yield the same results)

When 2 vendors pick the price and platform inherit its price:

Since the market share is $MS_1 + MS_2 + MS_p = 1$ and $MS_p = (y_1 + y_2)/2$, the subscription fee is

$$s = \frac{p_1 + p_2 + v[(k_1 + k_2)(1+\alpha) - 2]}{2} - \frac{t(2 + y_1 + y_2)}{4}$$

(the result is similar when one vendor and the platform picks the price)

Proposition 3: with quantity restriction, an agreement can be reached where

1. $k_1 = k_2 1$ and the market is covered
2. When before entry, vendors were local monopoly the variety parameter needs to be sufficiently high
3. With moderatle before-entry competition, consumers can still yield benefit from the platform
4. Under very strong competition that consumers always yield positive payoff, and there is no agreement.

Proposition 4: (only under the regions where both regimes exist) industry total profit is higher for the variable transfer fee if $\frac{9}{4+6\alpha} < v/t$ (i.e., when the transportation cost is small and variety parameter is big), otherwise the restriction on quantity regime is better.

Future research:

- Assume: sellers with different market power
- Sellers could also bundle products (subscription)
- Learning model: customers only use platform to figure out their preference, then pick 1 vendor.
- Multiple round negotiation.

37.1.5 (Long et al., 2022) Design Amazon marketplace

- 2 sellers on a marketplace with search query that consists of sponsored and organic listings.
- Consumers preferences are jointly determined by vertical and horizontal aspects.
  - Vertical component: asymmetric info that are known only by the sellers (product quality $q_i \sim U(0,1), i \in (1,2)$).
  - Horizontal component: asymmetric info that are known only by the platforms about the consumer preference (personal fit $\lambda_{ki} \in U(0,1), i \in (1,2)$)
- Products and sellers will either be match ($\sim Ber(m_{ki})$) or no match

The match probability is

$$m_{ki} = \theta_k q_i + (1 - \theta_k)\lambda_{ki}$$

where

- θ_k (consumer's private info) the relative weight that consumer k put on the vertical vs. horizontal components.
- q_i = seller's private info
- λ_{ki} = the platform's private info
- All three parameters follow $\sim U(0,1)$

Platform chooses seller to put in the organic slot with the highest value of

$T\tilde{q}_i + (1 - T)\lambda_{ki}$

where

- T = info weight (i.e., info sharing parameter)

- $\tilde{q}_i$ = partial info on product quality from the auction
- λ_{ki} = platform's personal fit info
- Assuming that the bidding price from seller to the ad place reveals some info about their product quality

Platform's revenue can come from either

1. Sales commissions: $\phi \in [0,1]$ (percentage charge per transaction)
2. Ad Revenue (either pay-per-click payment rule or pay-per-impression)

Timeline

1. Platform decides on T (info weight) and ϕ (commission rate)
2. Sellers decide to join
3. Consumers decide θ_k (personal preference), sellers decide on q_i (quality), platform decides on λ_{ki} (platform's private info)
4. Sellers bids b_i
5. Platform infers $\tilde{q}_i$ from b_i and decides who gets the ad placement
6. Consumers decide on m_i (matching probability), search

Analysis

Additional assumption: seller's bid $b(q_i)$ strictly increases in q_i (verifiable in the seller's optimal bidding behavior)

Then, the platform can infer the true value of q_i from sellers' bids and places the seller with the higher $Tq_i + (1-T)\lambda_{ki}$ in the organic result

Hence, consumers also infer that ad position is seller with higher product quality (q_i) and organic place is the seller with higher $Tq_i + (1-T)\lambda_{ki}$

Consumer choice:

- When the same seller appears in the ad and organic result (she is the higher-quality seller) (results don't change with multiple slot
- When the ad slot seller (i) (higher quality product) is different from the organic slot (j) (higher personal fit)
  - i is chosen, if consumer value quality more
  - j is chosen, if consumer value personal fit more.

Seller's Ad Bids: optimal bid verifies the bidding behavior of sellers (seller with higher product quality is going to bid higher)

Platform's Sales Commission and Ad revenue

Sales Commission: $\phi D(T)$

Ad revenue: $2\int_0^1 \int_0^{q_i} b(q_i) dq_i dq_j$

Proposition 2: Sales Commission revenue is concave while Ad revenue is increasing.

Sales is concave (information effect of strategic listing) because if

1. T is too small (personal fit is overvalued), consumers will not benefit from the product quality info revealed from sellers' bids
2. T is too large (product quality is overvalued), consumers will not benefit from the personal fit info, in turn worsen the match probability.

Ad revenue is increasing (competition effect of strategic listing):

- As T increases, platform ad revenue increases (because sellers' probability of being in the ad slot as well as the organic slot increases)

Platform's design decision

1. Platform's choice of commission rate (fixed platform's info weight T): the total revenue increases with the commission rate because commission revenue (from both organic and ad slots) dominates the losses in ad revenue (from only the ad slot)
2. Platform's choice of commission rate (fixed platform's info weight T) and the sellers' have an outside option u_0 is $\phi^* = 1 - \frac{u_0}{U(T,0)}$, meaning as the optimal commission rate decreases as T increases because a larger T increase the bids and increases advertising fee.
3. With varying info weight and commission rate
    1. If seller's outside option is small (first region), the optimal commission rate decreases as seller's outside option increases
    2. If the sellers' outside potion is intermediate, the optimal information weight decreases as seller's outside option increases, and the optimal commission rate is 0
    3. If the outside option is large enough, sellers will not participate in the first place.

Extensions:

1. Comparison with Independent listing (no strategic listing - using info from the bidding side to list organic search): When comparing with the benchmark that there is no strategic listing behavior, the commission ate under strategic listing leads to lower commission revenue, but higher advertising revenue. Hence, overall the platform benefit from strategic listings
2. Comparison with Independent Listing (Under Exogenous Commission rate)
    1. If the commission rate is small, strategic listing leads to higher to both higher commission and advertising revenue
    2. If the commission rate is intermediate, strategic listing leads to lower commission and ad revenues
    3. If the commission rate is large, both listing strategies lead to 0 revenue.

3. When seller's information q_i and platform's info λ_{ki} overlap (correlated), results do not change

37.1.6 (Hajihashemi et al., 2021) Personalized Pricing with Network Effects

- Read (Katz and Shapiro, 1985) for fulfilled expectations equilibrium

Stages

Without personalized pricing

1. Firms know the type of consumers, set uniform price p
2. Consumer i knows p_i, s_i

With personalized pricing

1. Firms know the type of consumers, set price p_L, p_H
2. Consumer i only knows

Proposition 1: Without network effect, price personalization leads to a lower equilibrium price for consumer L, but with network effect, price personalization leas to higher equilibrium for consumer L

- If the network effect is too small, the marginal effect of price decreases on the reservation price is minimal.

Proposition 4: Consumer Surplus: price personalization can decrease the unconditional expected surplus of consumer L

37.2 Dynamic Pricing and Bundling

- A monopolist sells products in a bundle to homogenize consumer heterogeneity in willingness to pay for individual products. and to sell individual products to price discriminate

37.2.1 (Prasad et al., 2017) Bundling

- 3 static bundling strategies:
  - Products are sold separately (pure components PC)
  - Products are sold only in a bundle (pure bundling PB)
  - Products are sold under both (mixed bundling MB) second-degree price discrimination
- Selling products under the
  - pure-bundling in two stages is optimal when the marginal costs are low

  - pure components in two stages is optimal when marginal costs are high (with lots of **myopic** consumers)
  - pure bundling then pure components is optimal when marginal costs are moderate (with lots of **strategic** consumers)
  - mixed bundling in both stages when the market is mostly **strategic** consumers.

- Scenario:
  - PB-PC: bundle first then component
  - PC-PB: component then bundle
- Assumptions:
  - Monopoly with either high or low marginal cost
  - Heterogeneous consumers (regarding product preference)
  - 2 types in the population: myopic (α) (maximize at each stage independently) and strategic ($1-\alpha$) (maximize both stages)
  - Marginal cost of the bundle = sum(component costs)
- Interesting
  - optimal bundle price can be higher than the sum of individual products
- Literature:
  - Bundling
  - Inter-temporal pricing

PC-PC

Demand for each product from myopic consumers in the first stage is $(1-P)\alpha$

$P-\delta$ is the price for the second stage where δ is the discount.

Demand for each product from myopic consumer in the second stage is $\alpha\delta$

Demand for each product from strategic consumer is $(1-\alpha)(1-P+\delta)$

For the seller, the first stage profit is $2(P-c)(1-P)\alpha$

and the second stage profit is $2(P-\delta-c)[\alpha\delta+(1-\alpha)(1-P+\delta)]$

Hence, optimal prices are

$$P_1 = \frac{1+c}{2} + \frac{(2-\alpha)(1-c)}{2(4-\alpha)}$$

and

$$P_2 = \frac{1+c}{2} - \frac{\alpha(1-c)}{2(4-\alpha)}$$

the total profit is

$$\frac{2(1-c)^2}{4-\alpha}$$

37.2.2 (Diao et al., 2019) Intertemporal price, consumer fairness concern

- Consumer fairness concerns (unfair due to increased price) lead to lower retail prices in both periods while the wholesale prices might be higher in the first period.
- Consumer fairness concerns can decrease the first and second period retail prices.
  - The intuition that retailers increase in the first period and decrease in the second period only hold under a centralized system (wholesale to consumer), not under a decentralized one.
- If the demand intercept does not raise much in the second period, the cost reduction incentive usually prevails over fairness-mitigation incentive.
- A larger second-period demand can cause both the producer and the store to lower their pricing.
- When consumers have fairness costs, the second-period equilibrium unit sales is actually still higher than when they don't, because fairness concerns alleviate the double-marginalization problem.
- To model fairness concern heterogeneity, they model 2 types of consumers (those with fairness concern and those without), where α is the fraction of customer with fairness concerns. And it should affect the retailer's second-period demand function.

37.2.3 (Dana and Williams, 2022) Intertemporal price

- Oligopoly model
- Pricing on capacity:
  - fixed capacity, firms dynamically set price.
- Stage:
  - Choose capacity
  - Compete in prices

- Inventory control (i.e., sales limit restriction) can help mitigate price competition.
- Contributions:
  - Equilibrium prices are flat or uniform over time
  - Intetemporal price discrimination is possible when firms adopt inventory controls (if the demand is more inelastic over time).
- Firm choices: capacity and 2 prices (2 periods)

37.2.4 (Kolay and Tyagi, 2022) Event bundle

- Two events sold as a bundle in 2 periods.
- Contribution to current literature:
  - Consumer expectations of future prices
    - Durable goods pricing
    - Product obsolescence
  - Loyalty programs
  - Consumer uncertainty regarding valuations
- Features:
  - Events with different popularity
  - Popular event in stage 1 or 2
  - Consumers can be uncertain about their valuation of the second event
  - Uncertainty in seller's commitment to the price of a future event
  - Consumers valuations are independently distributed or positively correlated
- Examine when it's optimal to sell
  - event independently
  - as a bundle
  - or mixed
- Events are different from products in that are
  - perishable
  - temporal
- If the seller can choose the order of introduction, popular events should be saved for later. ("save the best for last"). Similar to that we introduce

the lower quality products before the higher quality ones. (Moorthy and Png, 1992)

Model

Consumer valuation of the less popular event L is $v_L \sim U(0,1)$ and of the more popular event H is $v_H \sim U(0, 1+x)$ where x represents how more popular the second one is (i.e., the proportion of consumers preferring H over L).

Consumers understand the order and importance of events before the game begins.

Demand function depends on seller's strategies

- No bundling at all: consumers only buy products L, H can only be purchased separately
- Pure bundling: Either buy bundle or not
- Full and partial mixed bundling: can buy either L, H separately or in a bundle.

37.3 Consumer Fairness

Also see (Diao et al., 2019)

37.3.1 (Fu et al., 2021) Unfair machine learning algorithms

- Compares the consequences of Equal treatment (ET) and equal opportunity (EO)
  - Regular and protected groups are better off if we change from ET to EO?
  - Optimal learning efforts under ET and EO?
  - Is the firm better off if ET to EO?
- Strategic behavior of a firm: decide the amount of leaning effort. The firm invest for better ML algorithm to predict the outcomes.

Settings

Consumers

- Good consumer (non-defaulters) and bad consumers (defaulters)
- Consumers are in either regular or protected group

Firm

- Risk-neutral, and want to select good consumers

- Invest in ML by paying the learning costs to separate the good and bad consumers
- Set approval thresholds for each group
- A fairness constraint: either ET or EO

Main Results:

- Firm less motivated to invest in learning under EO than under ET
- Profit is lower under EO because it's hard to separate the good and bad consumers in the protected group
- EO makes everyone (regular + protected) worse off
  - Regular group has higher threshold under EO (than under ET)
  - Firm invests less on the algorithm under EO

Model

Firm

- accepts or rejects an application
- A consumer can either be good or bad
- the profit/loss (α, β) where it comes from good/bad consumer
- firm exerts leaning efforts (s) for better separability between the two groups

Consumer

- Either regular or protected (size = 1)
- % of bad consumer in each group is d
- Expected gains:
  - $\alpha_p = \alpha(1 - d_p), \beta_p = \beta d_p$
  - $\alpha_r = \alpha(1 - d_r), \beta_r = \beta d_r$

The bank uses ML and assigns scores to candidates to represent their goodness where γ_r, γ_p are learning efficiency

Summary

- The optimal threshold and the profit depend on β_p

37.4 Decentralized Channels

37.4.1 (Jiang and Tian, 2022) Reactive capacity on product quality and profitability in uncertain markets

- Under "pull" contracts, the retailer designs the product quality and makes the wholesales and retail pricing decisions, but the risk of excess production is assumed by the upstream suppliers or contract manufacturers, rather than brand-owning retailers or OEMs.
- How much improvement in reactive capacity will benefit the supplier and how much it will benefit the retailer?

Model

Production Setup

- A supply chain where before the selling season, a brand-owning retailer designs a production quality q and offers a wholesale price w to outsource its production to ta supplier.
- At the beginning of the selling seasons, the retailer will decide how many units Q of the product to order from the supplier and choose its retail price p
- Supplier's marginal cost $= kq^2$

Market uncertainty setup

- Then consumer utility from product with quality q at price p is $u = q\theta - p$
- where θ is the consumer's WTP
- Assume that the market uncertainty is about the distribution of the consumer's rather than about the total number of consumers in the market. Then a fraction of $\alpha \in (0, 1)$ of consumers have $\theta = \theta_H$

Time setup

- Supplier's production lead time $T =$ total production time + shipping and handling
- $T = T_S + \frac{N}{c}$ quantity independent time + quantity dependent time
- Denote A as the acceptable delivery time. The supplier will have to manufacture the product before the selling season if $T_s > A$ and could produce some units during the selling season other wise
- The market outcome $T_S > A$ is equal to $T_S = A$

Model

- The supplier's reactive capacity is measured by $\bar{Q}$, which is the maximum quantity (number of units) of the product that the supplier can produce and deliver to the retailer within the selling season
- $T = T_S + \frac{Q}{c} \leq A$ Hence, $\bar{Q} = (A - T_s)c$
- The supplier's reactive capacity, $\bar{Q}$ is driven by both the amount of production capacity c and how reactive and flexible that capacity is (T_s)
- Whether is it the retailer or the supplier that bears the inventory risk?

Timeline

- Before selling seasons
  - Retailer decides product quality and wholesale price
  - Supplier: pre-season inventory I
- Within the selling seasons
  - α is known to the retailers
  - Retailer: product quantity Q
  - Supplier:
    - $Q \leq I$ revenue wQ
    - $Q > I$ decide $I' \leq \bar{Q}$ to produce within A
  - Retailer: selling price
  - Consumer: purchase decisions

Analysis

- Without market uncertainty ($\alpha = \alpha_g = \alpha_b$)
- With market uncertainty
  - Extreme cases
    - When $\alpha_b \geq \frac{\theta_L}{\theta_H}$ even the bad market has many high-valuation consumers, the retailer will find it optimal to charge a high price to target only high-valuation consumer regardless of the market state
  - In-between cases
    - Low reactive capacity
    - Medium reactive capacity
      - Opportunistic targeting with stock-out: the retailer sets the wholesale price low at the supplier's marginal cost and the supplier will produce the risk-free inventory of $I = ag$

  · Opportunistic targeting with overstock: the retailer chooses a high wholesale price to induce the supplier to optimally produce a higher pre-season inventory of $I = 1 - \bar{Q}$

 * High reactive capacity: the supplier has enough time to produce during the selling season within the acceptable delivery time. Anticipating the supplier's high ability to fulfill its order, the retailers will target only high valuation costumers at a high retail price

37.4.2 (Hu et al., 2022) Wholesale vs. Agency

Literature

- Distribution channels
  - Previously examine wholesale contract and downstream consequences (double marginalization problem)
  - This paper examines channel efficiency of agency contract.
- Strategic contract choices: wholesale vs. agency
- Retail pass-through

Model setup

$n \geq 2$ competing suppliers selling substitutable products to consumers through a dominant online retailer r

Each supplier has a unique product

Assume same marginal cost across supplier, e-tailer has marginal cost for each unit of products demanded by costumers

Demand structure

Consider the demand of product is given by continuous $q_i(p)$

where $p =$ market prices

Common assumptions:

1. Demand is downward sloping
2. Demand function is symmetric

Let $\gamma > 0$ be the product substitutability for specific demand function

$\lambda(p) = -\frac{q(p)}{q'(p)}$ be a measure of sensitivity of demand

Additional assumptions on $\lambda(p)$

1. profit functions are quasi-concave
2. elasticity of demand decreases with p increases

Game Structure

Wholesale

1. Supplier announces wholesale price w_i
2. E-tailer sets selling price p_i for each product

Agency

1. E-tailer sets revenue share rate $1 - \alpha$
2. Suppliers can choose to accept and sell directly to costumers on e-tailer platform, setting retail prices p_i

37.5 Consumer Search

37.5.1 (Jiang and Zou, 2020) Consumer Search and Filtering on Online Retail Platforms

Findings

- Lower search cost reduces seller's profit margin (due to increased competition), but also attract more consumers to the platform
- If referral fee is exogenously determined, a decrease in search cost can either increase and decrease the platform's profit
  - Hence, platform should compensate by raising its referral fee, but it actually depends on the effect of search cost on demand elasticity:
    * If the search cost increases the demand elasticity, platform should lower its referral fee
    * If the search cost decreases the demand elasticity, the platform should increase its referral fee.
- If referral fee is set optimally, the platform's profit will increase (kinda intuitive)
- With the filtering feature, consumers will be more likely to buy after searching due to reduction in uncertainty about the product match value for consumers

Motivation:

- The search costs tend to decrease overtime
- However, there is not study that study the platform's optimal referral-fee in respect to search costs, filtering.

Research questions:

- How search cost affects optimal referral fee and profit?
- How filtering affects the platform, sellers, and consumers?

- How filtering impact differs from that of a reduction of the search cost?

Results are robust to

- heterogeneous search costs
- heterogeneous product quality
- platform charges a fixed per-unit referral fee (instead of percentage)

Related Literature

- Consumer search: mixed literature on whether to increase or decrease search cost
- Information structure effect on consumer search and firm profits
- This paper:
  - Add filtering (different from target search)
  - search cost affects platform's and sellers' optimal pricing decision
  - search cost affects platform's optimal referral fee

Setup

- Horizontally differentiated products
- After search (cost), consumers learn the price and match value (with the product)
- Match value = f(filterable match value + unfilterable match value)

Consumer utility

$$u_{ij} = M_{ij} + p_i$$

where

- $M_{ij} = \mu_{ij} + m_{ij}$ = match value of product i to consumer j
  - $\mu_{ij} = \delta \times \hat{\mu}$ = filterable match value
    * $\delta > 0$ is a cosntnat scale factor
    * $\hat{\mu}$ = mean-zero discrete random variable with K realizations
  - m_{ij} = unfilterable match value
  - It's ok if μ_{ij} correlates with m_{ij}
- p_i = product price

Consumer search

1. Consumer know her outside option u_{0j} and μ_{ij} (filterable match value)

2. Consumer do not know m_{ij} (match value) and price before hand, only after search cost τ

Consumers decisions

1. Whether to search on the platform
2. Search sellers sequentially and learn m_{ij} and p_i
3. Consumers can buy, search more (to the next product with highest μ_{ij}), and leave

Game Timing

1. Retail platform sets a percentage referral fee $r \in (0, 1)$
2. Sellers set price where profit is $(1 - r)p_i - c$
3. Consumers make search and purchase decisions.

Notes:

- $p^*(r)$ = equilibrium retail price
- r^* = optimal percentage referral fee

In equilibrium, consumers will continue search until the next product expected utility beyond u_{ij} is less than τ (search cost) (Wolinsky, 1986)

Result 1: consumer purchase if $m_{ij} \geq \bar{m}(\tau)$ where $\bar{m}(\tau)$ satisfies $\int_{\bar{m}(\tau)}^{m_{max}} (m - \bar{m}(\tau))dF(m) = \tau$

Result 2: $\bar{m}(\tau)$ decreases with τ

In equilibrium, $\bar{m}(\tau)$ is the equilibrium acceptance threshold for the unfilterable match value

For exogenous referral fee

Lemma 1: retail price, sellers' profit, total demand

Result 3: As the cost of searching reduces, a consumer's expected number of searches before purchasing increases, as does the expected aggregate match value M_{ij} of the product she will buy.

Lemma 2: Market demand and sellers' profits change with the search cost (τ). Lower search cost intensify sellers' competition, which reduces their prices and increase profits due to a demand-expanding effect (increase the whole pie). The platform's equilibrium profit can also increase or decrease as the consumer's search cost decreases

Endogenous referral fee

Proposition 1: A decrease in search cost increases the platform's expected profit

Proposition 2: As the consumer's search cost τ decreases, the platform will

1. Increase its referral fee if the search does not increase the demand elasticity (but communication is important here)

2. Decreases its referral fee if the search cost increase the demand elasticity

Effects of Filtering

- Consumer's optimal search strategy
  - Lemma 3: Filtering increases the consumers' acceptance threshold for the aggregate match value by less than μ_K
    - Filtering allows consumers to have smaller consideration set, which raises acceptance threshold for the aggregate match value
    - Filtering reduces the direct benefit of searching. Hence, consumers are more likely to stop searching at a lower acceptance threshold.
    - But the first effect > the second effect.
  - Proposition 3: filtering increases a consumer's probability of buying a product after searching it, and reduces the consumer's expected number of search.
- Seller's pricing decisions: filtering reduces retail price
  - Proposition 4: Filtering reduces consumer's uncertainty about the product's match value
    - Consumers only search products with the highest filterable attributes, reducing the match-value differentiation among product (increase price competition, and lower equilibrium price)
    - filtering decreases consumers' equilibrium probability of continuing search after searching a product, reducing competition and higher equilibrium prices.
  - Proposition 5: Filtering increases the platform's demand, seller's profit, and platform's profit, and consumer surplus under small scale factor for the match value and referral fee is exogenous

Conclusion

- Reduction in search cost makes consumers search more, while filtering reduces the benefit of search making consumers search power products
- Decrease in search cost intensifies seller's competition, leading to lower retail prices. While filtering softens seller competition, and lead to higher retail prices
- When the referral fee is exogenous, a lower search cost can either be good or bad for the platform, and consumers. Whereas filtering will always benefit the sellers, the platform, and consumers if filtering only reveals a small amount of the variations in the match value.

Heterogeneous search costs:

- Proposition 6: Optimal retail price is the weighted average of the homogeneous-search cost equilibrium prices

Heterogeneous Product quality:

- Premium sellers charge a higher price (than non-premium sellers), but the difference is smaller than their quality difference

Fixed Per-unit referral fee:

- When the platform's referral fee is exogenous, a reduction in search costs will always increase the platform's profit under a set per-unit referral price, but it will lower the platform's profit under a percentage referral fee.

37.5.2 (Chen et al., 2021) Consumer Search with blind buying

Objective and Motivation

- Consumers don't have relevant info and must invest effort to purchase a suitable product
- Previous search literature, consumers pay a fixed search cost to visit a store and fully informed about the price and match value at the store
- These models do not capture
  - Online search technology that lower the cost of visiting stores and collecting info about prices
  - Consumers cannot physically inspect and try products.
- This paper designs a sequential search model to capture these two aspects

Sequence:

- Before searching, consumers observe the prices and prior values of both firms, and they pay a search cost to learn the match value
- Consumers have to decide product to search firs, and whethe to continue searching after inspecting the first product. If consumers decide to stop searching, they can either buy the insepcted product or bldinly buy an unispected product

Model construction

- Duopoly market, 2 firms sell horizontally differentiated products in price competition.
- The random utility (net of nay search cost) fo consumer l, who purchases the product from firm i is $\tilde{u}_{il} = \tilde{\eta}_{il} + \tilde{\epsilon}_{il} - p_i$
  - $\tilde{\eta}_{il}$ = observable match value

- $\tilde{\epsilon}_{il}$ = unobservable match value

The key feature is that consumers can blindly buy a product

- At any stage of search process, consumers have the option of purchasing a product without inspecting its match value $\tilde{\epsilon}_{il}$ where no search cost is incurred
- The expected utility for consumer blind search is $E(\tilde{u}_{il}) = \eta_{il} + E(\tilde{\epsilon}_{il}) - p_i$

Search is assumed to be costly

Game sequence

1. Firms determine their prices
2. Each consumer costlessly observes both the prices and the prior values of the products and decides whether to blindly buy a product. the game ends if a blind buying takes place
3. The game continue and after learning product match value, the costumer decides among three 3 options
    1. buy product without uncertainty
    2. blindly buy product without inspecting its match value
    3. discover both realized value

Results

- The effects of blind buying on equilibrium prices
    - In price-directed search models without the blind buying option, search friction has 2 effect on prices competition.
- When the match value has a symmetric distribution, both consumer and firms are indifferent to the search order, conditioning on searching
- An increase int eh first sample search cost has no effect on market prices in existing price-directed search models, reducing equilibirum prices in the model

37.5.3 (Li and Xu, 2022) Superior knowledge, price discrimination, and customer inspection

- Customer inspection : learn about their intrinsic preferences.
- Superior knowledge: firms know customers preferences

Setup

A monopolistic firm selling a product to a continuum of customers of unit measure

Customer have heterogeneous match values with the firm's product (either high or low)

the prior probability of a high match value is α

Information structure:

- Customers know their expected match value, they are ex ante uninformed about their idiosyncratic shock and do not know their match value beyond its prior distribution
- Customer inspection: the cost that consumer make for an informed purchase. The inspection cost is independent of rational inferences
- firm knows both its product characteristics and preference of individual customers.

Timeline

1. Each customer has their match
2. Firm offers each customer a personalized price according to her preference. The firm chooses the probability distribution of its price
3. Observing the personalized price, customer decides whether to inspect to find out her true preference by incurring an inspection cost with probability
4. Based on the inspection outcome, if the customer chooses to inspect, the customer decides whether to make a purchase

37.6 Review paper

- To reject, find major flaws
- If you propose alternative specification or robustness, provide rationale why it's necessary.

How to deal with rejections

- Let it sit for a few days before/after you read it
- Vent your anger with your coauthors
- Examine main points of rejection and focus on improvement
- Submit to another journal and forget about it

How to deal with revisions

- Let it sit for a few days
- Vent with coauthors
- Examine main concerns (for those papers that you can't address reviewer's concern, consider submitting to another journal). If you don't send back within certain window, editors assume you are not going to submit it.
- For major revisions, you don't send it back right away in a month

Results

- Rejection: don't ever send it back
- Reject and resubmit: If you do these certain thing, we treat it as a new paper
- Major revision: major problems, but if you revise it, the outcome is still uncertain, but you pass the first round (about 50% will be rejected in the second round)
- Minor revision: problems that are easily solvable (less uncertainty)
- Conditional acceptance: some small things, wordings (no risk of being rejected)
- Accept as is

Chapter 38

Marketing Strategy

38.1 Intro

(Morgan et al., 2018)

Marketing strategy = central construct of strategic marketing, where strategic marketing = the general field of study

Marketing strategy is "an organization's integrated pattern of decisions that specify its crucial choices concerning products, markets, marketing activities and marketing resources in the creation, communication and/or delivery of products that offer value to customers in exchanges with the organization and thereby enables the organization to achieve specific objectives." (Varadarajan 2010, p. 119)

marketing strategy "encompasses the"what" strategy decisions and actions and "how" strategy-making and realization processes concerning a firm's desired goals." (p. 7)

Methods:

Contributions

1. In the past 19 years, marketing strategy as a field focuses on marketing tactics, resources, capabilities
2. 4 sub-domains under marketing strategy
    1. formulation-content: figuring out what to do + strategy decisions: studies goals that a marketing strategy is designed to deliver and the major broad strategic decisions related to how these are to be achieved.
    2. formulation-process: figuring out what to do + strategy making and strategy realization: studies the mechanism used to develop marketing strategy goals and pick the broad strategic means (i.e., market target, required value proposition, desired positioning, timing).

3. implementation-content: doing it + strategy decisions: studies the detailed integrated marketing program tactics decisions, actions, and resource deployments.
4. implementation-process: doing it + strategy making and strategy realization: studies "the mechanism used to identify, select, and realize integrated marketing program tactics designed to deliver marketing strategy content decisions."

Insights:

- Most studies focused on individual marketing mix (i.e., individual tactics), not marketing strategies and integrated marketing programs.
- Half of the studies are logic or data-driven. Hence, lack of theory.
- Papers typically use multiple theories, instead of a single theory.
- Qualitative method is rare and trending to 0.

(Zeithaml et al., 2019)

What is Theories-in-use?

- A TIU is "a person's mental model of how things work in a particular context." (Argyris and Schon, 1974)
- individual's TIU is a set of "if-then" relationships among actions and outcomes. (Zaltman et al., 1982)
- Espoused theories are "the mental models individuals claim or purport to have." It's different from TIU in the sense that individuals might have TIU, but they can't articulate it.
- TIU is grounded in both the researcher and the interviewees' mindsets.

When TIU approach is appropriate?

- To construct organic marketing theories
- To extend extant perspectives and address ambiguities
- To guide empirical efforts

Implementing a TIU approach

Intro

- Most cited papers in marketing typically use the TIU approach. However, marketing as a field usually borrow theory from other disciplines such as econ or psych.

Questions:

- What is the different between grounded theory and TIU? Is it that TIU just a special case of grounded theory, while there are other approaches such as case study and ethnography?

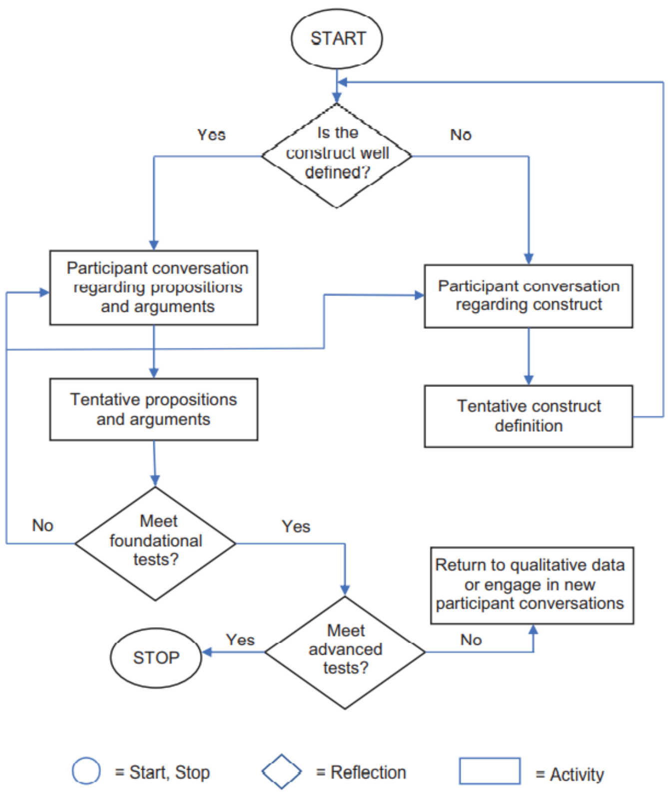


Figure 38.1: Figure 1 The TIU research process (p. 37)

Advanced Guidelines:

- Construct development - Abstraction:
  - Identify elements that are part of a a broader, more abstract element
  - Identify elements that likely co-occur (or covary)
  - Identify elements that are neither subset of one of the elements nor do they necessarily co-occur.
- Proposition development:
  - Axial coding: linking two or more concepts
- Argument development
- Pull it all together: choose a parsimonious set of proposition - selective coding.

TIU Tradecraft

- Extensive Iteration
- Active Listening
- Brief Suspension
- Depth over Breadth
- Openness to New Issues
- Conflict Appreciation
- Mosaic Filling
- Bias Recognition
- Demarcation of TIU Study Limits
- Crafting Research papers

Evaluate rigor in TIU research:

TIU limitation:

- not suited for theory testing
- Participants' lack of knowledge (but can still have espouse theories)

Future research

- Meta domains:
  - An underlying assumption that may be reexamined
  - Conflicting firm and customer viewpoints
  - Conflict among key stakeholders

Table 6. Rigor in TIU Research.

Type of TIU Rigor	Analog to Theory-Testing Research	Meaning of Rigor Type in TIU Research	Demonstrating Rigor
Credibility	Internal validity	The extent to which a new theory's if-then propositions are plausible	• Provide arguments to support the new if-then propositions • Document the inclusion of range-spanning questions in participant conversations • Document data from multiple participants which suggest the same theory
Transferability	External validity	The extent to which a new theory's constructs and if-then propositions are valid in contexts not sampled for the research	• Document similarities in theory emerging from participants sampled from multiple contexts
Dependability	Reliability	The extent to which multiple researchers find the same constructs and if-then propositions from the same data	• Document similarity of constructs and if-then propositions surfaced by multiple researchers processing the same data
Confirmability	Objectivity	The extent to which a new theory's constructs and if-then propositions can be independently certified as emerging from the data (rather than from researcher dispositions)	• Document consistency of theory with participant views through participant checks • Document interjudge reliability: agreement between independent (external) judges about the fit between data and constructs and propositions • Provide thick descriptions of data to allow readers to directly assess consistency of if-then propositions with data
Distinctiveness	Discriminant validity	The extent to which a new theory's constructs and if-then propositions are different from existing constructs and if-then propositions in the literature	• Describe differences in definitions of new constructs relative to those of most similar constructs in literature • Describe differences in the new if-then propositions relative to most similar/close propositions in the literature

Figure 38.2: Table 6 (p. 44)

- Possible content domains:
  - Role of marketing in the firm
  - Digital transformation of the firm
  - consumer privacy
  - Health care policy
  - Government involvement and regulation
- TIU method:
  - Optimal sample size
  - Eliciting causal connection
  - Alternative probing techniques

(Moorman and Day, 2016)

- Marketing excellence is "a superior ability to perform essential customer-facing activities that improve customer, financial, stock market, and societal outcomes." (p. 6)
  - superior ability to perform the 7As
- 4 elements of marketing organization (MARORG)

  - Capabilities: "complex bundles of firm-level skills and knowledge that carry out marketing tasks and firm adaptation to marketplace changes." (p. 6)
  - Configuration: the organizational structures, metrics, and incentives/control systems that shape marketing activities." (p. 6)
  - Human capital: "creates, implement,s and evaluates a firm's strategy." (p. 6)
  - Culture: "guides thinking and actions throughout the firm." (p. 6). Forms of organizational culture:
    * Culture as the pattern of shared values and beliefs
    * Culture as behaviors
    * Culture presented by cultural artifacts (e.g., stories, rituals).
- 4 elements are mobilized through 7 marketing activities:
  - **anticipating** marketplace changes
  - **adapting** the firm to such changes
  - **aligning** processes, structures, and people
  - **activating** efficient and effective individual and organizational behaviors
  - creating **accountability** for marketing performance
  - **attracting** important financial, human and other resources
  - engaging in **asset management** that develops and deploys marketing assets

(Kohli and Jaworski, 1990)

- Market orientation "is the implementation of the marketing concept" (p. 1)
- Market orientation is "the organizationwide **generation** of market intelligence pertaining to current and future customer needs, **dissemination** of the intelligence across departments, and **responsiveness** to it." (p. 6)
- Core themes / pillars of the marketing concept
  - Custoemr focus: Market intelligence is a broad concept that includes
    * exogenous market factors that affect customer needs and preferences
    * current and future needs of customers
  - Coordinated marketing: one or more departments engaging in activities geared towards developing an understanding across departments

  - Profitability: is a consequence of a market orientation rather than a part of it

- Marketing orientation < market orientation (less politically charged and focus on market).

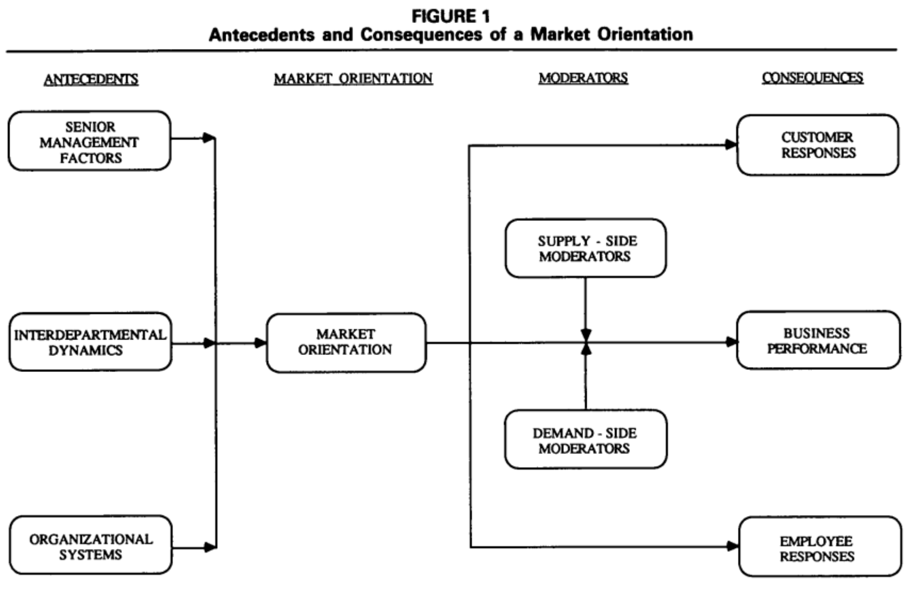


Figure 38.3: Figure 1 (p. 7)

38.2 Branding

(Swaminathan et al., 2022)

- Branding-finance literature review:
  - (Srivastava et al., 1998a)
  - (Keller and Lehmann, 2006)
  - (Edeling and Fischer, 2016)
  - (Edeling et al., 2021)
- From a comprehensive search process with 4 criteria, 114 journal articles that look at the brand-related actions and stock market-based outcomes and 38 more with accounting-based metrics

Measuring the financial impact of brands

- Stock returns (event studies)

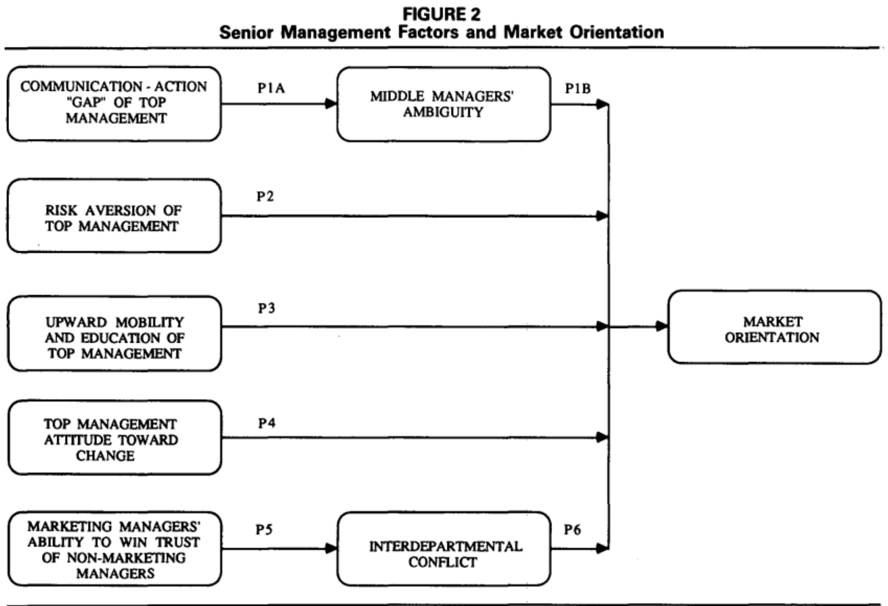


Figure 38.4: Figure 2 (p. 8)

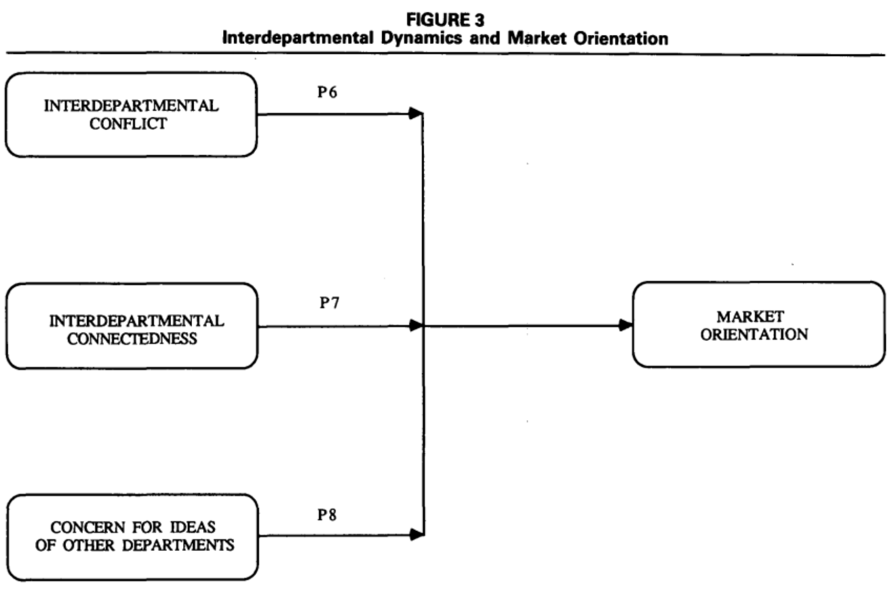


Figure 38.5: Figure 3 (p. 10)

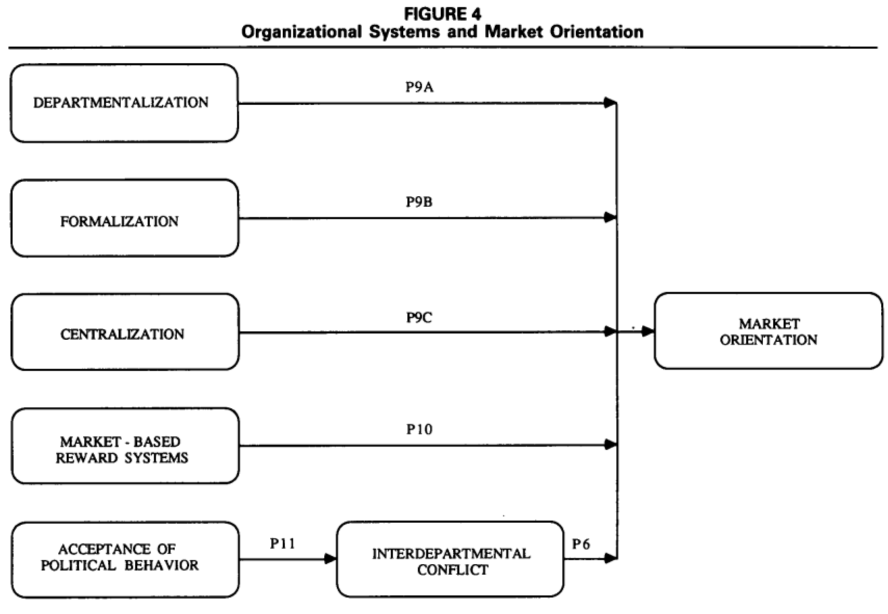


Figure 38.6: Figure 4 (p. 11)

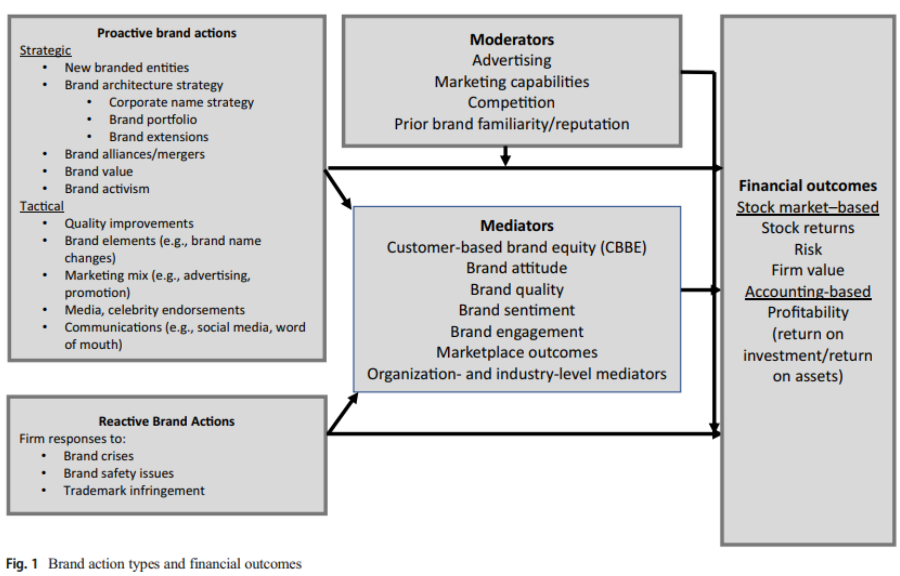


Figure 38.7: p. 640

- Stock returns (response models) to examine continuous events.
- Idiosyncratic and systematic risk
- Firm value (Tobin's q) = the ratio of the market value of a firm to the replacement cost of its assets. Even though criticism regarding this metric has been raised (Bendle and Butt, 2018), there are answers (Peters and Taylor, 2017). Factors impact brand Tobin's q:
  - corporate branding strategy (Rao et al., 2004)
  - characteristics of a brand portfolio (Morgan and Rego, 2009a)
  - advertising (Wang et al., 2008)
- Accounting-based performance and related outcomes are important for understanding the indirect effect f brand actions on financial value.

Categorizing brand actions by scope and cause

- Cause (proactive vs. reactive)
  - Proactive actions can be more positive than reactive actions.
  - Even though managers are paid to be proactive, but is it always good to be proactive? Researchers have over-researched actions that can be taken by managers, but what if we play by the status quo (idea by Taleb)?
    * bias toward intervention (naive interventionism: intervention that produces small visible gains, while creating the possibility off large - later harm): restraint from inaction is not likely to be rewarded.
- Scope (strategic vs. tactical)
  - Strategic actions focus on broader and longer-term goals (targeting, segmentation, diversification, new product introduction, rebranding, brand architecture, and brand acquisitions)
  - Tactical actions focus on short-term expected implications (e.g., marketing mix)

Brand-finance interface:

- Theoretical perspectives:
  - Signaling theory/ info economics theory: brands are signals of quality to both investors and consumers (Connelly et al., 2010). (call for research on signal environment and signal feedback dimension)
    * Signaler: Types of actions and firms
    * Signal: signal quality (credibility)
    * receiver of signal: type of signal

        * signaling environment (e.g., social media platforms)
        * feedback from receiver to signaler
    - Resource-based theory: A business's ability to generate value depends on its resources, whether they are valuable, rare, and costly to replicate, and if the organization is organized to get value from them. And if a resource has these components, it can create sustained competitive advantage.

- Mediators
    - Brand quality perceptions
    - attitudes
    - engagement: receives less attention
    - marketplace outcomes (e.g., customer lifetime value, market share)
    - organization-focused processes:
    - industry-based processes (e.g., barriers to entry, threat of competition, analyst converge, relative market share): receives less attention
- Moderators (then what is the difference between the organization factors as mediators and moderators):
    - advertising (and other marketing mix variables)
    - Prior brand strength
    - organization factors (e..g., marketing capabilities)
    - industry factors (e.g., competition)

Financial consequences of brand actions based on the scope and cause of actions

- Proactive-strategic brand actions
    - brand introduction and brand innovation: are value-creating activities (on revenue, profitability, firm value)
    - brand architecture (e.g., new products in new and existing markets, brand naming strategy, target market)
        * 2 corporate naming strategies
            · House of brands
            · Branded house: greater efficiency and lower customization and cannibalization, more potential risk(Rao et al., 2004)
    - leverage brand (e..g, leverage brand reputation to appeal to newer segments of consumers)
    - Brand image and reputational capital:

    * reputational capital: "an organization's stock of perceptual and social assets." (p. 649)
    * Corporate social performance: "doing well while doing good." But the impact of its on firm performance is still under debate, which means we might be able to find moderators for this relationship.
    * Sociopolitical activism (CSA): can be negative.
  - Further research: new brand type (e.g., digital brand, sharing platform), or global branding, cocreation.
- Proactive-tactical brand actions
  - Brand tactical actions (Keller and Swaminathan, 2019):
    * choice of brand elements (e.g., brand names, URLs, logos, symbols, characters, spokespeople, slogans, jingles, packages,signage).
    * marketing-mix activities (4Ps)
    * secondary associations (linking to other people, places, or things).
  - Brand tactical actions by this paper:
    * Name changes: good signal for changes in strategic direction.
    * Brand quality improvement: positive effect on firm perofrmnace
    * Advertising: direct (signalling and spillover on investors) and indirect effect (brand meaning and brand strength) on firm performance
    * Social Media Communications: earned and owned social media influence brand awareness, purchase intent, and customer satisfaction then stock performance.
    * WOM communication: using social tagging to assess brand image.
    * pricing and channels: pricing and brand image have inverse relationship, while distribution intensity has a positive impacts on brand equity.
    * celebrity endorsement: can have positive and negative impact on firm performance
  - Further research: mostly stem from info rich environment.
- Reactive brand actions

  - Product recalls: negatively affect stock return, but prior reputation and advertising strength can mitigate this negative impact. And it can spillover to competitors as well. Prior reputation can serve as both a benefit (dampen unintentional impact), but a liability (exacerbate intentional impact - e.g., corporate social irresponsibility).
  - Trademark disputes and protection: always good for plaintiff (regardless of thee verdict), bad for defendant in casae of unfavorable verdict.
  - Brand crises: e.g., #DeleteUber
  - Further Research: resolve double-edge sword of prior reputation; when and how advertising can mitigate crises; whether and when firms should be proactive vs. reactive) in regard to brand crises.
- Additional research on contingent factors
  - Number and strength of competitors can be a moderator of the impact of brand actions on CBBE (in my research, I look at both the impact of competition on brand actions e.g., social media adoption, and as a moderator between brand action and brand reputation).
  - The quality of brand actions
  - The speed of brand actions: (e.g., first -mover advantage)

(Swaminathan et al., 2020)

38.3 Channels and Customer Management

(Hadida et al., 2018) The temporary marketing organization

- A temporary organization is defined as "temporally bounded group of interdependent actors formed to perform a complex task." (Burke and Morley, 2016, p. 1237)
  - "temporally bounded" = "institutionalized termination"
- "Stand-Alone" Temporary Organization: Discrete time horizon (also known as stand-alone)
  - Different from Joint venture (no past), Start-up (no past), Permanent organization.
  - Absent of (1) history of interaction between members (might induce info asymmetry regarding task ability -> self-selection to send signals, and require more enforcement mechanisms, but also because of the discreteness nature, this can be hard) (2) future.
- Hybrids and fully embedded temporary organizations: the role of context

- Temporary can acquire a post or a future using
    * its members' prior relationships
    * Being embedded within a permanent organization with organizational context (but have to be careful, because "strong" prior ties can have limited information access(Granovetter, 1973), but can also serve as a good quality control - enforcement mechanisms, reducing principal-agent discrepancy)

- Temporary organization drivers
  - Task: reason for being: "The greater the novelty of the task, the greater the need for organization-specific selection and enforcement mechanism, and the higher the likelihood of using a stand-alone temporary organization relative to hybrid and fully embedded forms." (p. 7)
  - Time "The shorter the duration of the task, the lower the feasibility of crafting organization-specific selection and enforcement mechanisms and the higher the likelihood of using a hybrid temporary organization relative to standalone and fully embedded forms." (p. 8)
  - Team (composition - heterogeneity of its members): "The greater the heterogeneity of the team, the greater the need for enforcement through low-powered incentives and the higher the likelihood of using a fully embedded temporary organization relative to hybrid and stand-alone forms

- Portability and Exogenous Selection and Enforcement Effects
  - Portability: "selection and enforcement needs of a temporary organization may be portable or supplied exogenous, either from preexisting agent relationships (hybrid) or form the features of a permanent organization (fully embedded)" (p. 8).
  - "The greater the novelty of the task, the lower the portability of the selection benefits from a hybrid temporary organization and the greater the need for organization-specific selection mechanisms." (p. 9)
  - "For a hybrid temporary organization, enforcement benefits are portable regardless of the nature of the task." (p. 9)
  - "The portability of a fully embedded temporary organization's selection and enforcement benefits is weaker than for those of a hybrid." (p. 10)

- Performance Outcomes of the Temporary Marketing Organization
  - Output Creativity (inherently linked to selection): "The closer the match between the novelty of a temporary marketing organization's

task and its selection efforts, the greater the creativity of its output." (p. 10)

- Decision-making speed (inherently linked to enforcement): ""The closer the match between the novelty of a temporary marketing organization's task and its enforcement efforts, the greater its decision-making speed." (p. 11)
  * The over-organization in term of enforcement can have adverse consequences on decision-making speed (due to reactance - lower instrinsic motivation).

- Managerial Implications: Task novelty, time pressures, team heterogeneity, as temporary organization form's inputs.
- Future Research Directions
  - Different forms of temporary organizations: current study's classification based on dyadic vs. organization-levels norms might not be sufficient for future fine-grained analysis.
  - Structural embeddedness: different from relational embeddedness (between a principal and an agent) where structural embeddedness is about connections among mutual contacts.
  - Multilevel relationships (e.g., upstream suppliers, producers, downstream customer).
  - Micro level proprieties and processes:

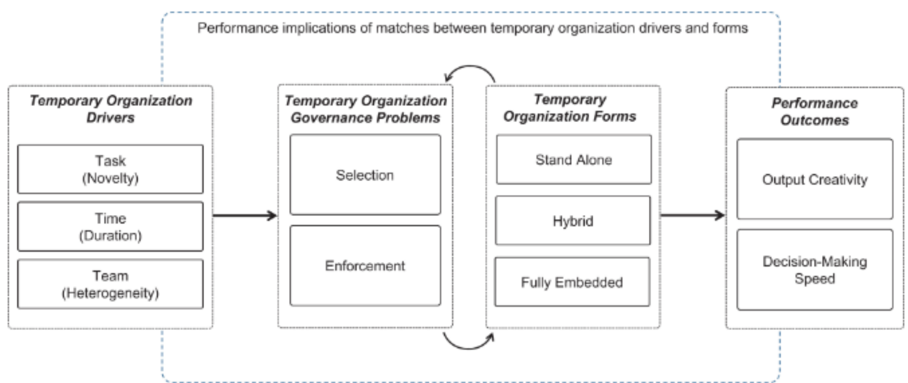


Figure 1. Conceptual framework.

Figure 38.8: Figure 1 (p. 6)

(Rindfleisch and Heide, 1997)

- TCA has many applications in marketing because

  - it focuses on exchange
  - survey-data
- Problems to underutilized
  - Refinements to the theory are no well known
  - Empirical works are not well integrated
- TCA is under the "New Insititutional Economics" paradigm (supplanted neoclassical economics)
- TCA views firm as a governance structure
- Assumptions

Behavioral Assumptions	Transactional dimension
Bounded rationality	Environmental uncertainty
Opportunism	Asset specificity
Risk Neutraliity	Transaction Frequency

- Logic
  - If adaptability, performance evaluation, and safeguarding costs are absent or minimal, economic actors will choose market governance.
  - If internal organization expenses exceed market production cost benefits, enterprises will choose this.
- Contexts to apply TCA
  - Vertical integration
  - Vertical inter-organizational relationships
  - Horizontal inter-organizational relationships

(Palmatier et al., 2013)

- Commitment velocity: the rate and direction of change in commitment
- Data
- Study 1
  - Data: longitudinal (6 years)
  - Method: latent growth curve analysis
  - Commitment velocity is a predictor of performance
- Study :
  - Data: Matched multiple-source data to see the antecedents of commitment velocity

  - Found customer trust and dynamic capabilities as antecedents, but less impactful as a relationship ages where as investment capabilities become more important.

- Tenets

  - Exchange performance is influenced by both static and dynamic elements of relationship constructs, although dynamic elements are more important than static components for forecasting future behaviors and performance.

  - Depending on the relative contributions of a construct-specific collection of underlying time-varying processes, the growth trajectories of relational constructs (such as trust, commitment, and relational norms) are distinctive and path-dependent.

  - As relationships change, so do (a) the connections between relational constructs and (b) the relative weight of relational constructs in determining how trade outcomes are influenced.

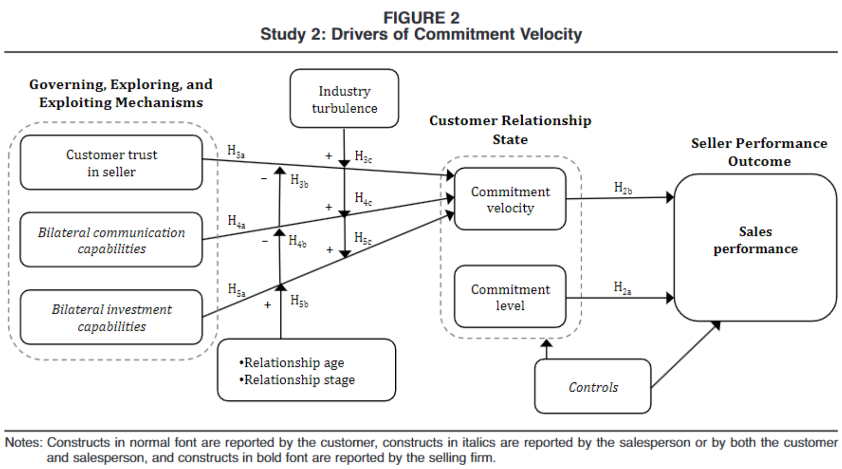


Figure 38.9: Figure 2 (p. 21)

(Harmeling et al., 2016)

- Customer engagement marketing: "a firm's deliberate effort to motivate, empower, and measure customer contributions to marketing functions" (p. 312)

38.4 Human Capital

(Nath and Mahajan, 2008) CMO presence

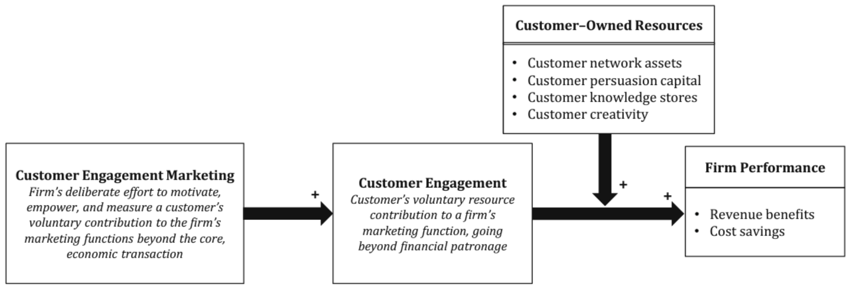


Figure 38.10: Figure 1 (p. 314)

Table 2 Typology of customer-owned resources

Types of customer-owned resources	Descriptions	Value to firm
Customer network assets	The number, diversity, and structure of a customer's interpersonal ties within his or her social network	• Increases reach of engagement marketing initiative • Provides access to particularly influential individuals or unique subgroups
Customer persuasion capital	The degree of trust, goodwill, and influence a customer has with other existing or potential customers	• Increases the influence of the content shared over other customers' purchase decisions
Customer knowledge stores	A customer's accumulation of knowledge about the product, brand, firm, and other customers	• Improves the quality and relevance of the content shared throughcustomer engagement behaviors (e.g., blogging, writhing reviews) • Aids in the development, management and dissemination of the brand narrative • Improves customer-to-customer support and customer contributions to new product development
Customer creativity	"Production, conceptualization, or development of novel, useful ideas, processes, or solutions to problems" (Kozinets et al. 2008, p. 341)	• Provides unique insights into marketing functions (e.g., new product development, product usages)

Figure 38.11: Table 2 (p. 317)

Table 3 Five key differences between engagement marketing and other marketing strategies

Engagement marketing	Promotion marketing	Relationship marketing
A firm's deliberate effort to motivate, empower, and measure a customer's voluntary contribution to the firm's marketing functions beyond the core, economic transaction (i.e., customer engagement)	The use of a special offer to raise a customer's interest and influence the purchase of the focal product versus competitors' products (Wierenga and Soethoudt 2010)	"All marketing activities directed towards establishing, developing, and maintaining successful relational exchanges" (Morgan and Hunt 1994, p. 22)
1. Objective of the marketing initiative		
Encourage a customer's active participation in and contribution to the firm's marketing functions	Induce a single transactiom with the focal firm versus a competitive firm	Retain the focal customer and motivate future, repeat transaction with the customer
2. Assesment of customer value		
Customer-owned resources and potential future contributions to the firm's marketing functions	Purchasing power and customer share of wallet	Customer lifetime value from past customer transactions
3. Flow of information		
Networked communication among the customer, other customers, and the firm	One-way communication from the firm to the customer	Bilateral communication between the customer and the firm
4. Firm-directed customer learning		
Training a customer how to contribute to the firm's marketing functions	Teaching the customer how to buy and use the focal product	Understanding the idiosyncratic norms of the exchange relationship
5. Customer control over value creation		
Customer exercises high control, which can affect outcomes relevant to the broader customer population	Customer has no control over value creation and is a receiver of marketing	Customer control is negotiated with the firm, which affects outcomes relevant to the focal customer-firm relationship

Figure 38.12: Table 3 (p. 318)

Fig. 2 Conceptual model of the effects of customer engagement marketing on customer engagement

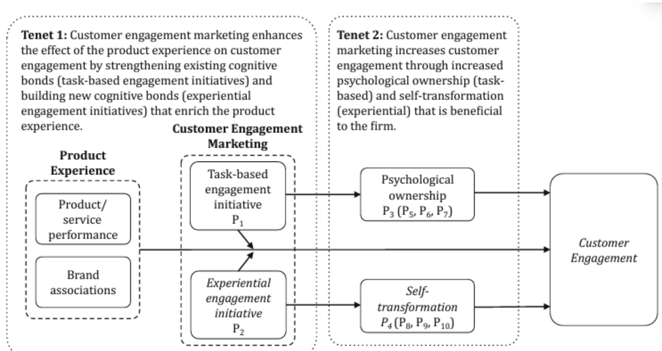


Figure 38.13: Fig 2 (p. 323)

(Germann et al., 2015)

- Performance (Tobin's q) of firms that have a CMO is 15% greater than that of firms without a CMO
- An extension of (Nath and Mahajan, 2008)
  - Time horizon: more years
  - Industry: more industries
  - Models: panel data and instrumental variables

Overview

	Study 1a	Study 1b	Study 2	Study 3	Study 4
Industries sampled	Cross section of industries	Same as Study 1a	Studies 1a and 1b + additional industries	Same as Studies 1a and 1b	Same as Study 2
Firms sampled	Firms that report both advertising and R&D with sales of at least $250 million in 2002	Firms that report both advertising and R&D with sales of at least $250 million in 2002	Firms that report both advertising and R&D with sales of at least $250 million in 2002	Firms that report both advertising and R&D with sales of at least $250 million in 2002	Firms that report both advertising and R&D with sales of at least $250 million in 2002
Firm observations	123 firms	123 firms	155 firms	123 firms	155 firms
Time frame	2000–2004	2000–2004	2000–2004	2000–2011	2000–2011
Firm–year observations	615 (Sample 1)	615 (Sample 1)	775 (Sample 2)	1258 (Sample 3)	1604 (Sample 4)
Models used	Between-effects regression	Between-effects OLS Random-effects Fixed-effects OLS with lagged DV 2SLSRE Random effects with control functions HT[a] BB[b]	Between-effects OLS Random-effects Fixed-effects OLS with lagged DV 2SLSRE Random effects with control functions HT[a] BB[b]	Between-effects OLS Random-effects Fixed-effects OLS with lagged DV 2SLSRE Random effects with control functions HT[a] BB[b]	Between-effects OLS Random-effects Fixed-effects OLS with lagged DV 2SLSRE Random effects with control functions HT[a] BB[b]
CMO effect on Tobin's q	Not significant	Significant and positive in some models	Significant and positive in all but the between-effects model	Significant and positive in all but the between-effects model	Significant and positive in all but the between-effects model

[a]Hausman and Taylor (1981).
[b]Blundell and Bond (1998, 1999).
Notes: In Study 1a, we attempted to replicate N&M as closely as possible. R&D = research and development, OLS = ordinary least squares, DV = dependent variable, and 2SLSRE = two-stage least-squares random effects.

Figure 38.14: Table 1 (p. 2)

Chapter 39

Others

(Li et al., 2015) Article citability and its dynamic effects

Objective:

- Investigate the factors influencing the impact and influence of published research papers, considering it as a marketing challenge for academic works.

New Proposition:

- Beyond acknowledged overt factors that influence research paper influence (like the journal's reputation, the content of the article, the author's prominence, and the so-called "Matthew effects" where those with initial advantages gain more), there's a latent factor: "Citability."

What is Citability?

- A latent, or hidden, variable representing how relatable or relevant an article is to its academic field over time. It fluctuates, changing the degree to which a paper is likely to be cited by others.

Methodology:

- **Model Used:** A discretized Tobit model combined with hidden Markov processes.
- This approach indicates that citability has two distinct states, which in turn affect a paper's influence.
- **Data Source:** Articles published in seven premier marketing journals from 1996 to 2003.

Key Findings:

1. **Dynamic Nature of Citability:** The impact of a research paper isn't static; it evolves over time due to the changing states of its citability.

2. **Importance of Citability:** A paper's influence is not only determined by overt factors like where it's published or who wrote it. The underlying relevance or citability of a paper plays a crucial role.

3. **Effect of "Uncitedness":** The model captures the effect of a paper remaining uncited, which many other models do not consider. Not being cited can have implications for a paper's influence and citability.

Differences from Prior Research:

- Traditional models assume static effects, like the influence of a journal's reputation or an author's prominence. In contrast, this study's model acknowledges the shifting nature of a paper's impact.

- Existing models overlook the evolving dynamics of a paper's influence and how various influence drivers may have different effects based on the latent state of the paper.

Implications:

- Recognizing the role of citability can help academics and institutions strategize research dissemination, ensuring that relevant papers don't lose their deserved spotlight.

- The model offers a more holistic view of what determines a paper's influence, which can influence publishing decisions and how researchers approach their work.

(Jedidi et al., 2021) created an index of relevance (called **R2M** - Relevance to marketing), which measures the relevance of marketing publications to marketing practice (dimensions: topics, time).

(Lin and Misra, 2022) illustrates that consumer identity fragmentation (i.e., a person using multiple devices with different tracking systems) can induce bias the estimate of quantities of interest (e.g., advertising effects). The bias comes from 3 sources:

1. Outcome fragmentation: measurement errors that create attenuation bias. Due to an artificial increased in the number of observations (i.e., one person with two devices are counted as 2 identities), but the total outcome variation remains constant, while the measured outcome is attenuated. This bias only goes away if the true effect (e.g., advertising effect) is 0.
2. Exposure fragmentation: similar to omitted variable bias because the cross-device exposure is not taken into account. Hence, only when exposure to one device will not induce a consumer to purchase a product on another device, then this bias will go away (very hard to convince).
3. Spurious Covariance: from device-level activity bias (i.e., patterns to use one device for browsing but another for purchasing - differential device usage preferences and differential exposure levels), and cross-device sustitution.

1. Using a separate model for each device type can eliminate the device-level activity bias.

Solutions to identity fragmentation include

1. Identity linking: but partially linked data can create more problems than solution
2. Experiment-based estimator adjustment by (Coey and Bailey, 2016) to recover the true effect using experiments with an additional assumption of symmetric and independent exposures (SIE). But it can still be wrong due to its inability to capture cross-device variations (e.g., activity bias).
3. Stratified Aggregation: first group by geographic location, then put each group into a bin of consumer characteristics' combinations (e.g., sex, gender, education).

(Feng et al., 2015) Marketing department power and firm performance

- **Objective:** Examine the evolution of marketing department power within U.S. firms from 1993 to 2008 and determine its implications for firm performance.

- **Methodology:**

  - Deployed a novel objective metric to assess marketing department power.

  - Analyzed a cross-industry cohort consisting of 612 public U.S. firms.

- **Key Findings:**

  - There was a discernible augmentation in the power of marketing departments over the specified timeframe.

  - Elevated marketing department power yielded:

    * Enhanced long-term total shareholder returns, surpassing its influence on short-term return on assets (ROA).

    * Mediating effects on long-run market-based-asset-building and short-run market-based-asset-leveraging capabilities were observed. These capabilities partially mediated the influence on long-term shareholder value and fully mediated the effect on short-term ROA performance.

(Feng et al., 2017) Firm capabilities and growth: the moderating role of market conditions

- **Objective:** Utilizing the contingency theory framework, the study delves into the role of diverse firm-level capabilities and their interplay on firm growth, contextualized within varying market scenarios.

- **Data & Methodology:**

  - Analyzed panel data from 612 public U.S. firms spanning 16 years across 60 industries.
  - Focuses on the empirical exploration of the interaction between three pivotal firm capabilities: marketing, R&D, and operations, and their cumulative effect on revenue and profit growth trajectories.
  - Additionally, the research probes into external boundary conditions, namely market munificence and competitive dynamism, and their modulating effect on the growth implications of the aforestated capabilities.
- **Key Findings:**
  - R&D capabilities amplify the growth effects of marketing capabilities, whereas operational capabilities attenuate them.
  - The strength and direction of these interactions are not static but fluctuate depending on prevailing market conditions.

(Bhattacharya et al., 2022) why and when market share drives firm profit

- **Objective:** Delve into the underpinnings of the market share-profit relationship, addressing gaps in empirical understanding and seeking to discern the key mechanisms influencing this relationship.
- **Background:** Firms conventionally employ market share as a metric for setting marketing objectives and gauging performance. Despite recent meta-analytic revelations on market share's economic implications and influential factors, a holistic understanding remains elusive.
- **Methodology:**
  - Conducted a simultaneous analysis of the trio of primary theoretical mechanisms believed to interlink firm market share and profit.
- **Key Findings:**
  - Predominantly, market power and quality signaling emerged as robust explicators of the variance in market share's positive impact on profit. Conversely, operating efficiency presented minimal explanatory potency.
  - This predominant role of the three mechanisms sustained even in scenarios where market share bore a negative correlation with profit, as observed with niche firms and those tactically "purchasing" market share.

(Feng et al., 2020) Unprofitable customer management strategies on shareholder value

- **Objective:** Examine the ramifications of unprofitable customer management (UCM) strategies on shareholder value and identify influencing factors.
- **Background:** While a significant segment of a firm's clientele may be unprofitable, and the management of such customers has been researched from customer and managerial perspectives, the impact on shareholder value remains uncharted.
- **Methodology:**
  - Employed an event study approach to investigate stock market reactions following announcements related to firms' UCM strategic decisions.
- **Key Findings:**
  - UCM strategy disclosures yielded an average short-term abnormal stock return of -0.53%.
  - Employing signaling theory, multiple facets of the signal (UCM strategy), signaler (the firm in question), and the signaling milieu were scrutinized to elucidate shareholder value effects:
    * Indirect UCM strategies were more favorably received by investors compared to direct customer divestment strategies.
    * Specific iterations of indirect UCM strategies and the underlying strategic intent during adoption, bolstered by firm's robust marketing capabilities and positive media exposure, can offset the generally observed negative abnormal stock returns.

(Luo et al., 2006) Cross-functional "coopetition": The simultaneous role of cooperation and competition within firms

- **Objective:** Explore the phenomenon of "coopetition" across functional areas within firms, breaking away from the conventional dichotomy in marketing literature that views cross-functional relationships as either purely cooperative or competitive.
- **Conceptual Focus:** The term "coopetition" encapsulates a simultaneous blend of cooperation and competition in inter-departmental dynamics.
- **Methodology:**
  - Collected input from mid-level managerial personnel and top-tier executives.
- **Key Findings:**
  - Cross-functional coopetition, contrary to prevailing conceptions, bolsters both customer and financial outcomes for a firm.

  - This performance enhancement is not direct but rather facilitated by market learning. Thus, the beneficial repercussions of cross-functional coopetition are channeled through an intrinsic learning process.

(Olsen et al., 2014) Green Claims and Message Frames: How Green New Products Change Brand Attitude

- **Objective:** Analyze the brand-level ramifications of introducing environmentally sustainable ("green") new products, with the aim of discerning how these introductions influence consumer attitudes towards the brand.
- **Background:** With environmental sustainability ranking prominently in global consumer trends, firms are gravitating towards the introduction of green products, committing substantial resources to this endeavor.
- **Theoretical Base:** The study integrates insights from both social identity and framing theories. This helps in deconstructing the motives behind green product introductions and understanding the conditional effects of message framing, source credibility, and product typology.
- **Methodology:**
  - Deployed a three-stage least squares modeling technique.
  - Evaluated green product introductions from 75 brands over a four-year span (2009–2012).
- **Key Findings:**
  - Introducing green products can indeed enhance consumer attitude towards the brand.
  - The strategic positioning of the brand and product category substantially dictates the launch of green products.
  - Multiple factors, such as the volume of green-centric messaging, the nature of the product, and the credibility of the information source, jointly determine the magnitude of change in brand attitude resulting from green product introductions.

(Slotegraaf and Dickson, 2004) Paradox in Marketing Planning

- **Context:**
  - Ongoing debate among strategy scholars about the benefits of formal planning.
  - Research provides inconsistent evidence about planning's impact on firm performance.
- **Framework:**
  - Utilizes the resource-based view of the firm to analyze the paradox.

- **Findings:**
  - Strong marketing planning capability can:
    1. Decrease postplan improvisation.
    2. Introduce inherent process rigidity.
  - Both outcomes have potential to enhance firm performance.

(Slotegraaf et al., 2003)

- **Current Research Landscape:**
  - Two predominant approaches in marketing research:
    1. Focus on a firm's resources.
    2. Focus on a firm's strategic/tactical actions.
- **Authors' Perspective:**
  - Neither approach alone encapsulates the complete drivers of competitive advantage.
- **Concept Introduced:**
  - "Market Deployment": Refers to specific marketing actions taken by a firm.
- **Research Objective:**
  - Understand the relationship between a firm's resources and the efficacy of its marketing actions (market deployment).
- **Methodology:**
  - Secondary data approach.
  - Employed a series of sequentially estimated hierarchical regression models.
- **Key Findings:**
  1. The possession of resources directly influences the returns on market deployment.
  2. High levels of intangible marketing resources and technological resources enhance the success of market deployment activities, especially distribution and coupon activities.
  3. Contrarily, higher levels of financial resources decrease the efficiency of distribution and coupon-based market deployment activities.
- **Implication:**

- The nature and magnitude of resources a firm possesses can either amplify or diminish the returns on specific marketing actions. Managers should strategically consider the resource base when planning and executing marketing strategies.

Part V

CASE STUDIES

Chapter 40

Case Studies in Branding

1. [Apple: Innovation and Design as Brand Identity]
2. [Nike: Building a Global Brand Through Storytelling and Innovation]
3. [Tesla: Revolutionizing the Automotive Industry Through Innovation and Sustainability]
4. [Amazon: Transforming Retail and Beyond]
5. [Zoom: Connecting the World Through Video Communications]
6. [Beyond Meat: A Plant-Based Revolution]
7. [TikTok: A Dance with Global Success]
8. [Coca-Cola: Quenching the World's Thirst for Over a Century]
9. [Netflix: Redefining the Future of Entertainment]
10. [Airbnb: Disrupting the Hospitality Industry]
11. [Starbucks: Brewing Success Through Innovation and Responsibility]
12. [The Walt Disney Company: A Kingdom of Creativity and Innovation]
13. [McDonald's: Serving Success with a Side of Innovation]
14. [Dove (Unilever): Crafting Beauty and Confidence]
15. [IKEA: A Symphony of Design, Affordability, and Sustainability]
16. [LEGO: Building Blocks of Innovation and Success]
17. [Slack: Revolutionizing Workplace Communication]
18. [Patagonia: A Case Study in Sustainable Business Practices]
19. [Spotify: Transitioning from music sales to subscription streaming]

20. [Warby Parker: Disrupting the traditional eyewear market with an online-first approach]

21. [Allbirds: A Case Study in Sustainable Footwear Innovation]

40.1 Apple: Innovation and Design as Brand Identity

1. Introduction:

Apple Inc., known for its revolutionary technology and design, has built its brand on innovation and a unique user experience. What began as a garage startup in 1976 has become one of the world's most valuable companies. Let's explore how Apple achieved this success.

2. Background:

- **Founding and Early Years:** Founded by Steve Jobs, Steve Wozniak, and Ronald Wayne, Apple started as a computer manufacturer. The launch of the Apple I computer in 1976 marked the company's debut, and the subsequent Apple II became a significant success.

- **Rise to Prominence:** With the introduction of the Macintosh in 1984, Apple emphasized graphical user interface, leading the way in user-friendly computing. The iPod, iPhone, iPad, and MacBook line have since become iconic products.

3. Marketing and Branding Strategies:
   - Innovation
     - **Product Development:** Regularly updating products to include the latest technology.
     - **Software Ecosystem:** Creating a seamless software environment that ties different Apple products together.
   - **Design Focus**
     - **Aesthetic Appeal:** Sleek and modern design across all products.
     - **User Experience:** Emphasizing intuitive interfaces.
   - **Cultivating Loyalty**
     - **Apple Ecosystem:** The interoperability of products encourages customers to stay within the Apple brand.
     - **Customer Service:** Apple's customer support, including the Genius Bar in Apple Stores, provides personalized service.
   - **Retail Experience**
     - **Store Design:** Apple Stores are known for their minimalist design and layout.
     - **In-Store Experience:** Offering hands-on experience with products and one-on-one customer service.

4. Challenges and Criticisms:
   - **Economic**
     - **High Pricing Strategy:** Apple's premium pricing limits accessibility for many consumers.
     - **Dependence on Key Products:** A significant reliance on the iPhone, which generates a large portion of revenue.
   - **Ethical**
     - **Manufacturing Practices:** Criticisms regarding working conditions in factories.
     - **Environmental Concerns:** Issues related to recycling and waste management.
5. Cultural Impact and Legacy:

Apple's marketing has not only sold products but also shaped culture.

- **Think Different Campaign:** This campaign emphasized Apple's image as a company for creative and unconventional thinkers.
- **Influence on Music Industry:** With the iPod and iTunes, Apple changed how people buy and listen to music.
- **Smartphone Revolution:** The iPhone transformed mobile communication.

6. Conclusion:

Apple's brand is more than just a logo; it's a symbol of innovation, quality, and a unique customer experience. By consistently focusing on design and innovation, Apple has maintained a strong brand identity that resonates with consumers globally. Its success offers essential insights into how a focus on innovation, design, and customer experience can build a powerful and enduring brand. The company's challenges and criticisms also provide a nuanced understanding of the complexities of operating at the forefront of technology.

7. Further Exploration:

- **Apple's Advertising:** Analyzing various Apple advertising campaigns over the years.
- **Competitor Analysis:** Understanding how Apple's branding strategies compare with competitors like Samsung, Google, and Microsoft.
- **Future Outlook:** Speculating on Apple's future in an ever-changing technology landscape.

This extended case study provides a comprehensive view of Apple's branding, suitable for students who want to delve deeply into branding's multifaceted nature. It includes various aspects of branding, marketing, challenges, and impact, allowing for a rich understanding of how a brand can shape not only a company's success but also influence broader culture and industry trends.

40.2 Nike: Building a Global Brand Through Storytelling and Innovation

1. Introduction:

Nike, Inc. is a household name synonymous with athleticism, performance, and innovation. Through its creative marketing strategies and commitment to design, Nike has become a leader in the sports apparel industry. This case study will explore Nike's rise to prominence and the branding strategies that have kept it at the forefront of the sports industry.

2. Background:
   - **Founding and Early Years:** Founded as Blue Ribbon Sports in 1964 by Bill Bowerman and Phil Knight, the company changed its name to Nike, Inc. in 1971. The famous swoosh logo and the "Just Do It" slogan became integral parts of the brand's identity.
   - **Growth and Expansion:** With an initial focus on running shoes, Nike expanded into various sports, including basketball, soccer, and golf, becoming a multi-sport brand.
3. Marketing and Branding Strategies:
   - **Athlete Endorsements**
     - **Historical Partnerships:** Nike's collaboration with athletes like Michael Jordan led to the creation of the Air Jordan line.
     - **Global Ambassadors:** Associating with top athletes like Serena Williams, Cristiano Ronaldo, and LeBron James.
   - **Storytelling and Advertising**
     - **Emotional Connection:** Creating ads that resonate emotionally with consumers, such as the "Find Your Greatness" campaign.
     - **Social Commentary:** Engaging in cultural conversations, like the Colin Kaepernick campaign.
   - **Product Innovation**
     - **Technological Advancements:** Such as Nike Air cushioning technology and Flyknit fabric.
     - **Customization:** Allowing consumers to personalize products through the NIKEiD platform.
   - **Community Engagement**
     - **Nike Run Clubs:** Building a community around the brand through running clubs and apps.
     - **Sustainability Initiatives:** Such as the "Move to Zero" campaign focusing on reducing environmental impact.
4. Challenges and Criticisms:
   - **Economic**
     - **Market Competition:** Competition from brands like Adidas and Under Armour.
     - **Pricing Strategies:** Balancing premium pricing with accessi-

bility for a broader audience.

- **Ethical**
  - **Labor Practices:** Historical criticisms regarding factory working conditions.
  - **Sustainability Challenges:** Managing environmental impacts across the supply chain.

5. Cultural Impact and Legacy:

Nike's influence goes beyond sports apparel.

- **Influence on Streetwear:** Collaborations with designers like Virgil Abloh have made Nike relevant in fashion circles.

- **Promotion of Women's Sports:** Marketing campaigns focusing on female athletes.

- **Global Reach:** Establishing a presence in various global markets and sports.

6. Conclusion:

Nike's brand success lies in its ability to intertwine sports, culture, and personal aspiration. Its collaborations with athletes, investment in storytelling, and commitment to innovation have made it a leader in the sports apparel industry. The challenges and criticisms it has faced provide insight into the complexities of maintaining a global brand. Understanding Nike's branding strategies offers an exciting exploration into how a brand can connect with consumers on multiple levels and across diverse markets.

7. Further Exploration:

- **Analyzing Advertising Campaigns:** Students may explore various campaigns to understand how Nike connects with different demographics.

- **Competitor Analysis:** Comparing Nike's strategies with competitors to understand market dynamics.

- **Future of Sports Branding:** Speculating on the future of branding in the sports industry and how Nike may continue to innovate.

This comprehensive case study provides a deep understanding of Nike's branding strategies and allows students to appreciate the multifaceted nature of branding in the modern market. The connections between sports, culture, innovation, and marketing weave together to create a compelling story that offers valuable insights for anyone interested in branding, marketing, or the sports industry.

40.3 Tesla: Revolutionizing the Automotive Industry Through Innovation and Sustainability

1. Introduction:

Tesla, Inc. is not just a car manufacturer; it's a technology company with a mission to accelerate the world's transition to sustainable energy. Founded by a group of engineers, including Elon Musk, who became the public face of the company, Tesla has become a symbol of innovation and environmental responsibility. This case study explores how Tesla achieved this status.

2. Background:
    - **Founding and Early Years:** Founded in 2003 by Martin Eberhard and Marc Tarpenning, and later joined by Elon Musk, JB Straubel, and Ian Wright, Tesla started with a vision to create electric cars that didn't compromise on performance.
    - **Road to Success:** The launch of the Tesla Roadster in 2008 proved that electric cars could be both stylish and powerful. Subsequent models, including the Model S, Model X, Model 3, and Model Y, diversified the product line.
3. Marketing and Branding Strategies:
    - **Innovation and Technology**
        - **Autopilot:** Developing self-driving technology.
        - **Battery Technology:** Pioneering advancements in battery efficiency and lifespan.
    - **Sustainability Focus**
        - **Clean Energy Products:** Including solar panels and the Powerwall for energy storage.
        - **Sustainable Manufacturing:** Efforts to minimize environmental impact in production.
    - **Direct Sales Model**
        - **Online Sales:** Bypassing traditional dealerships, selling directly to consumers online.
        - **Customer Experience:** Creating unique showrooms and offering test drives.
    - **Public Relations and Social Media**
        - **Elon Musk's Twitter Presence:** Utilizing social media to promote and defend the brand.
        - **Product Launches:** Hosting grand events to unveil new products.
4. Challenges and Criticisms:
    - **Economic**
        - **Production Challenges:** Meeting demand and managing quality control.
        - **Market Competition:** Growing competition from traditional

automakers entering the EV market.

- **Ethical**
  - **Labor Practices:** Controversies related to factory conditions.
  - **Autopilot Safety Concerns:** Debates over the safety of Tesla's self-driving technology.

5. Cultural Impact and Legacy:

- **Changing Automotive Industry:** Pushing the entire automotive industry towards electric vehicles.
- **Energy Conversation:** Shaping dialogues about renewable energy and climate change.
- **Stock Market Phenomenon:** Tesla's unique position in the stock market as a technology/automotive company.

6. Conclusion:

Tesla's brand represents a fusion of technology, sustainability, and luxury. Through innovative products, a focus on environmental responsibility, and disruptive sales models, Tesla has not only built a successful brand but has also changed the landscape of the automotive industry. Analyzing Tesla's strategies, challenges, and impacts provides valuable insights into how a brand can be a catalyst for industry-wide change.

7. Further Exploration:

- **Comparative Analysis:** Understanding how Tesla's branding strategies differ from traditional automotive brands.
- **Future of Mobility:** Speculating on the future of electric vehicles, autonomous driving, and Tesla's role in shaping that future.
- **Global Expansion:** Exploring Tesla's efforts to expand into various global markets, such as China and Europe.

40.4 Amazon: Transforming Retail and Beyond

1. Introduction:

Amazon, founded by Jeff Bezos in 1994, started as an online bookstore and quickly expanded into a vast e-commerce platform that sells virtually everything. Beyond retail, Amazon has also entered cloud computing, entertainment, and even healthcare. This case study will explore Amazon's diverse business activities and how they've contributed to its colossal success.

2. **Background:**
   - **Early Years:** Started in a garage, focusing on books, before expanding into other categories.

- **Global Expansion:** Rapid growth into international markets and diversified product offerings.

3. Strategies and Innovations:
    - **E-commerce Dominance**
        - **Customer Experience:** One-click ordering, personalized recommendations, and fast shipping.
        - **Amazon Prime:** Subscription model offering free shipping, video streaming, and more.
        - **Amazon Marketplace:** Allowing third-party sellers to reach Amazon's vast customer base.
    - **Technology and Cloud Computing**
        - **Amazon Web Services (AWS):** A leading provider of cloud computing services.
        - **Voice Technology:** Introduction of Alexa and Echo smart speakers.
    - **Entertainment and Media**
        - **Amazon Studios:** Producing and distributing original content.
        - **Twitch Acquisition:** Engaging the gaming community.
    - **Entry into New Markets**
        - **Whole Foods Acquisition:** Entering the brick-and-mortar retail space.
        - **Amazon Pharmacy:** Expanding into the healthcare sector.
4. Challenges and Criticisms:
    - **Economic**
        - **Market Power:** Criticisms related to monopolistic practices.
        - **Tax Practices:** Scrutiny over tax strategies and contributions.
    - **Ethical**
        - **Working Conditions:** Concerns over conditions in warehouses and treatment of employees.
        - **Environmental Impact:** Criticisms related to packaging and carbon footprint.
5. Cultural Impact and Legacy:
    - **Changing Retail Landscape:** Influencing consumer expectations and competitors' strategies.
    - **Innovation Leader:** Setting standards in technology, logistics, and customer service.
6. Conclusion:

Amazon's success story is a testament to innovation, diversification, and relentless focus on customer experience. By continuously expanding into new areas, Amazon has not only transformed retail but also various other industries. Examining Amazon's strategies, challenges, and cultural impact provides a deep understanding of modern business dynamics and the role of branding in shaping industry landscapes.

7. Further Exploration:

- **Competitive Analysis:** Understanding Amazon's position among global tech giants.
- **Future Projections:** Exploring potential new markets and technologies for Amazon.
- **Regulatory Landscape:** Analyzing potential legal and regulatory challenges.

This extensive case study offers students a multifaceted exploration of one of the world's most impactful brands. From e-commerce to entertainment, Amazon's influence is felt across multiple sectors. Understanding its success and challenges provides insights into innovation, strategy, ethics, and the complex dynamics of modern business environments.

40.5 Zoom: Connecting the World Through Video Communications

1. Introduction:

Zoom Video Communications, known simply as Zoom, played a pivotal role in connecting people during a time of global upheaval. Founded by Eric Yuan in 2011, Zoom quickly rose to prominence as a leading platform for video conferencing, webinars, and collaboration. This case study explores Zoom's exponential growth, the strategies that propelled it, and the challenges it faced along the way.

2. Background:
    - **Founding Vision:** Eric Yuan, a former Cisco executive, founded Zoom with a mission to make video communication frictionless and reliable.
    - **Early Growth:** Despite entering a competitive market, Zoom differentiated itself through ease of use and robust performance.
3. Strategies and Success Factors:
    - **Focus on User Experience**
        - **Ease of Use:** Simple interface, quick setup, and no user account required for joining meetings.
        - **Quality and Reliability:** Consistent video and audio quality across various devices and internet connections.
    - **Targeting Various Market Segments**
        - **Business and Enterprise Solutions:** Offering scalable solutions for organizations of all sizes.
        - **Education Sector:** Customized features for virtual classrooms and administrative meetings.
        - **Healthcare Integration:** Compliance with healthcare regulations for telemedicine use.
    - **Global Expansion**

    - **Localization:** Tailoring offerings to different regions and languages.
    - **Strategic Partnerships:** Collaborating with hardware vendors and integrators for seamless user experience.
  - **Adaptation During COVID-19**
    - **Free Access for Schools:** Providing free access to educational institutions during lockdowns.
    - **Scaling Infrastructure:** Rapidly expanding server capacity to handle surging demand.
    - **Security Enhancements:** Addressing early security concerns with significant updates and transparency.
4. Challenges and Criticisms:
  - **Security and Privacy**
    - **"Zoombombing" Incidents:** Unwanted intrusions into meetings raised questions about security.
    - **Data Privacy Concerns:** Scrutiny over encryption and data handling practices.
  - **Competition and Market Pressure**
    - **Competing Platforms:** Navigating competition from established players like Microsoft and new entrants like Google.
    - **Sustaining Growth:** Challenges in maintaining growth rates as restrictions lift and in-person meetings resume.
5. Cultural Impact and Legacy:
  - **Changing Work Culture:** Enabling remote work, hybrid models, and global collaboration.
  - **Social Connections:** Facilitating social interactions, virtual family gatherings, and online events.
  - **Redefining Communication:** Setting new standards for video communication and online engagement.
6. Conclusion:

Zoom's journey is a compelling study in understanding customer needs, agile adaptation, and effective scaling. From a startup competing against tech giants to becoming a household name, Zoom's story offers valuable lessons in innovation, strategic planning, crisis management, and ethical considerations. Analyzing Zoom's branding, growth strategies, challenges, and cultural impact provides rich insights into the dynamics of technology-driven market disruption and the responsibilities that come with rapid success.

7. Further Exploration:

- **Competitive Landscape Analysis:** Understanding Zoom's position in a fast-evolving market.

- **Ethical and Regulatory Considerations:** Analyzing Zoom's response to security and privacy concerns.

- **Long-term Strategy and Sustainability:** Evaluating Zoom's plans to sustain growth and diversify offerings.

40.6 Beyond Meat: A Plant-Based Revolution

1. Introduction:

Beyond Meat has become a synonym for the plant-based food movement, leading the way in creating meat alternatives that cater to a growing global demand for sustainable and ethical eating. This case study explores the company's journey, its innovative products, market strategies, and the broader impact on the food industry.

2. Background:
    - **Founding Vision:** Established by Ethan Brown in 2009, Beyond Meat aimed to address environmental, health, and ethical concerns related to animal agriculture.
    - **Product Innovation:** The development of plant-based meat substitutes that mimic the taste, texture, and appearance of traditional meat.
3. Market Strategies:
    - **Targeting a Broader Audience**
        - **Not Just for Vegetarians:** Positioning products to appeal to meat-eaters looking to reduce meat consumption.
        - **Retail and Food Service Partnerships:** Collaborations with supermarkets, fast-food chains, and restaurants.
    - **Branding and Marketing**
        - **Celebrity Endorsements:** Engaging well-known advocates of plant-based diets, such as Bill Gates and Leonardo DiCaprio.
        - **Sustainability Messaging:** Emphasizing the environmental and health benefits of plant-based foods.
    - **Global Expansion**
        - **Adaptation to Local Tastes:** Developing products tailored to various global markets and cuisines.
        - **Regulatory Compliance:** Navigating complex food regulations in different countries.
4. Challenges and Competitors:
    - **Market Competition**
        - **Rising Competitors:** Facing competition from both traditional food companies and new entrants in the plant-based sector.
        - **Product Differentiation:** Striving to stand out in an increasingly crowded market.
    - **Consumer Perceptions**
        - **Taste and Texture Expectations:** Meeting consumer expectations for flavors and textures similar to traditional meat.
        - **Price Barriers:** Addressing price competitiveness with animal-

based products.

- **Ethical and Environmental Considerations**
  - **Transparency in Ingredients:** Providing clear information about ingredients and processing methods.
  - **Life Cycle Analysis:** Assessing the full environmental impact of products, from production to consumption.

5. Cultural and Industry Impact:
   - **Changing Consumer Habits:** Influencing a shift in dietary preferences towards plant-based options.
   - **Industry Collaboration:** Collaborations with traditional meat producers and food service providers.
   - **Impact on Animal Agriculture:** Contributing to debates about the sustainability and ethics of conventional meat production.
6. Conclusion:

Beyond Meat's story represents a transformative moment in the food industry, reflecting a broader cultural shift towards sustainability and conscious consumption. By analyzing Beyond Meat's product innovation, market strategies, challenges, and cultural impact, students can gain insights into how a company can both lead and adapt to changing consumer values and industry dynamics. This case encourages critical thinking about innovation, branding, competition, ethics, and the interplay between business and societal needs.

7. Further Exploration:

- **Comparative Analysis with Competitors:** Examining strategies and approaches of other players in the plant-based food market.

- **Consumer Behavior Study:** Investigating consumer attitudes towards plant-based alternatives.

- **Sustainability Assessment:** Conducting a comprehensive analysis of the sustainability aspects of plant-based foods.

40.7 TikTok: A Dance with Global Success

1. Introduction:

TikTok, a social media app developed by Chinese tech company ByteDance, has quickly become a sensation, particularly among younger users. This case study examines TikTok's rapid growth, innovative content delivery, competition, and the complex regulatory landscape it navigates.

2. Background:
   - **Launch and Growth:** TikTok was launched in 2016 and merged with Musical.ly in 2018 to expand its reach in the U.S. market.
   - **Algorithm Magic:** TikTok's unique algorithm offers personalized content, leading to higher engagement and user retention.

3. Unique Features:
    - **Content Creation and Discovery**
        - **Short Video Format:** Users create engaging 15-second videos with a wide array of editing tools.
        - **Personalized Feed:** The "For You Page" algorithm provides a customized content feed, enhancing user experience.
    - **Community Engagement**
        - **Hashtag Challenges:** Promoting user-generated content through viral challenges.
        - **Collaborations and Duets:** Enabling collaboration between users to foster community.
4. Market Strategies:
    - **Targeting the Youth Demographic**
        - **Music and Dance Focus:** Strong emphasis on music and dance-related content.
        - **Influencer Partnerships:** Collaborating with youth influencers to drive adoption.
    - **Global Expansion**
        - **Local Content Adaptation:** Encouraging content that resonates with local cultures and trends.
        - **Strategic Advertising:** Utilizing in-app advertising and partnerships with brands.
5. Challenges and Competitors:
    - **Regulatory and Privacy Concerns**
        - **Data Security Issues:** Ongoing debates over data privacy and national security.
        - **Regulatory Scrutiny:** Challenges related to compliance with international regulations.
    - **Competition with Other Platforms**
        - **Competing for Attention:** A battle with platforms like Instagram, Snapchat, and YouTube.
        - **Intellectual Property Concerns:** Issues related to copyright and content ownership.
6. Social and Cultural Impact:
    - **Democratizing Content Creation:** Empowering individuals to become content creators.
    - **Cultural Influence:** Fostering global cultural exchange and trends.
7. Conclusion:

TikTok's story is a fascinating example of how a social media platform can become a global phenomenon through innovative technology, strategic targeting, community engagement, and adaptability to local cultures. This case allows students to explore various aspects of social media business, including algorithms, user engagement, competition, regulation, and cultural impact.

8. Further Exploration:

- **Algorithm Analysis:** Delve into how TikTok's algorithm works and compare it with other platforms.
- **Regulatory Compliance Study:** Investigate TikTok's compliance with different countries' regulatory frameworks.
- **Cultural Impact Research:** Explore how TikTok influences and reflects cultural trends across the globe.

40.8 Coca-Cola: Quenching the World's Thirst for Over a Century

1. Introduction:

Coca-Cola, founded in 1886, has grown to become one of the world's leading beverage companies. This case study explores Coca-Cola's brand legacy, marketing innovations, product diversity, sustainability initiatives, and the challenges and opportunities in an ever-changing global beverage market.

2. Background:
   - **Founding and Early Years:** From a pharmacy concoction to a global brand.
   - **Iconic Advertising Campaigns:** A look at some of Coca-Cola's most memorable marketing efforts.
3. Market Strategies:
   - **Global Brand Presence**
     - **Logo and Packaging:** The evolution of Coca-Cola's iconic logo and bottle design.
     - **Sponsorships and Partnerships:** Coca-Cola's association with sports events, entertainment, and charities.
   - **Targeted Marketing Strategies**
     - **Local Market Adaptation:** Customizing products and campaigns to fit regional tastes and cultures.
     - **Digital Engagement:** Leveraging social media and technology for customer engagement.
   - Product Diversification:
     - **Beverage Portfolio:** Introduction to Coca-Cola's diverse product line, including soft drinks, water, and juices.
     - **Health-Conscious Offerings:** Response to changing consumer preferences towards healthier options.
   - Sustainability Efforts:
     - **Water Stewardship:** Initiatives to reduce water usage and support community water projects.
     - **Recycling and Packaging:** Commitment to reducing plastic waste through recycling and innovative packaging.
4. Challenges and Opportunities:

- **Competition and Health Concerns**
  - **Market Competition:** An overview of competitors like PepsiCo and changing consumer tastes.
  - **Health and Regulatory Scrutiny:** Challenges related to sugar content and obesity concerns.
- **Global Expansion and Economic Factors**
  - **Emerging Markets:** Strategies and challenges in entering and thriving in new markets.
  - **Economic Sensitivities:** How global economic fluctuations affect sales and operations.

5. Conclusion:

Coca-Cola's story offers an inspiring journey into the world of branding, marketing, innovation, and corporate responsibility. The brand's ability to adapt, innovate, and remain socially responsible provides valuable insights for anyone interested in business, marketing, and sustainability.

6. Further Exploration:

- **Marketing Analysis:** Investigate how Coca-Cola has maintained its brand appeal over time.
- **Sustainability Evaluation:** Examine Coca-Cola's efforts in promoting environmental stewardship.
- **Global Business Study:** Analyze Coca-Cola's strategies in adapting to different cultures and markets.

This student version of the Coca-Cola case study serves as an engaging educational resource for courses related to business, marketing, branding, sustainability, and global commerce. Through exploration, discussion, and critical analysis, students can uncover the multifaceted dynamics that have shaped Coca-Cola's success and its continued relevance in today's competitive and evolving marketplace. It invites learners to reflect on the power of branding, the importance of innovation, the challenges of global expansion, and the growing significance of corporate social responsibility in modern business.

40.9 Netflix: Redefining the Future of Entertainment

1. Introduction:

Netflix, founded in 1997, has transformed from a DVD rental service to a global streaming giant. With over 200 million subscribers worldwide, Netflix has redefined the way people consume entertainment. This case study explores Netflix's growth, innovation, content strategy, and the challenges it faces in a competitive market.

2. Background:
   - **Founding and Early Growth:** From a mail-order DVD service to streaming pioneer.
   - **Subscription Model:** Introduction of the subscription model that revolutionized content consumption.
3. Marketing and Branding
   - Innovation and Technology:
     - **Streaming Technology:** Development of cutting-edge streaming technology to deliver content seamlessly.
     - **Personalized Recommendations:** Utilization of algorithms to tailor content suggestions to individual viewers.
   - Content Strategy:
     - **Original Content Creation:** Investment in exclusive shows and movies to differentiate from competitors.
     - **Content Licensing:** Acquiring rights to popular shows and movies to broaden the content library.
   - Global Expansion:
     - **Localization Strategy:** Adapting content to suit diverse cultural tastes and regulatory requirements.
     - **Emerging Markets Growth:** Expanding into developing regions with unique pricing and content strategies.
4. Competition and Challenges:
   - **Streaming Wars:** Competition with other streaming platforms like Amazon Prime, Disney+, and HBO Max.
   - **Regulatory and Legal Hurdles:** Navigating complex international laws and content regulations.
   - **Content Piracy Concerns:** Efforts to combat unauthorized sharing and illegal streaming of content.
5. Conclusion:

Netflix's story is a testament to innovation, adaptability, and the power of a customer-centric approach. The lessons drawn from Netflix's success and ongoing challenges provide valuable insights for those interested in technology, media, marketing, and global business strategy.

6. Further Exploration:

- **Technology Analysis:** Investigate how Netflix's technological advancements have shaped its success.

- **Content Strategy Evaluation:** Examine how Netflix's original content creation has redefined the entertainment industry.

- **Global Business Study:** Analyze Netflix's strategies for entering and thriving in diverse global markets.

40.10 Airbnb: Disrupting the Hospitality Industry

1. Introduction:

Airbnb, established in 2008, has emerged as a disruptive force in the global hospitality industry. This platform connects hosts and travelers, providing unique accommodations and experiences. This case study examines Airbnb's innovation, growth, and the challenges it faces, providing comprehensive insights for students interested in entrepreneurship, technology, law, and global business.

2. Background:
    - **Founding Story:** How an idea to rent air mattresses turned into a revolutionary business concept.
    - **Peer-to-Peer Model:** Airbnb's model of connecting hosts with travelers and its impact on traditional lodging.
3. Marketing and Branding:
    - Business Model and Innovation:
        - **Platform Design:** Exploration of the user-friendly design, including search functionality, booking process, and communication between hosts and guests.
        - **Trust and Community Building:** Methods of establishing trust through reviews, verification processes, host education, community guidelines, and conflict resolution.
        - **Revenue Model:** Understanding Airbnb's commission-based revenue model, pricing strategies, and value proposition for hosts and guests.
    - Growth and Expansion:
        - **Global Growth Strategy:** Airbnb's rapid expansion into various cities and countries, including marketing strategies, partnerships, and local engagement.
        - **Experiences and Diversification:** Introduction of Airbnb Experiences, business travel accommodations, and other extensions of the platform.
        - **Challenges in Scaling:** Examination of the obstacles faced during rapid growth, including maintaining quality, customer support, and local adaptation.
4. Regulatory Challenges and Social Impact:
    - **Local Regulations and Compliance:** Encounters with legal issues, zoning laws, city ordinances, and ongoing battles with regulators and the traditional hotel industry.
    - **Impact on Housing Markets:** Exploration of criticisms and studies on Airbnb's effect on local housing prices, availability, gentrification, and neighborhood dynamics.
    - **Safety and Liability Concerns:** Analysis of safety measures, insurance policies, host responsibilities, and incidents that have raised

concerns.

5. Sustainability and Social Responsibility:
    - **Sustainable Travel Initiatives:** Airbnb's efforts to promote eco-friendly travel practices, partnerships with local communities, and support for responsible hosting.
    - **Community Outreach and Disaster Response:** Airbnb's involvement in community development and providing emergency accommodations during natural disasters or crises.
6. Marketing and Branding:
    - **Brand Identity and Positioning:** Examination of Airbnb's brand evolution, advertising campaigns, social media presence, and efforts to differentiate itself from competitors.
    - **Customer Segmentation and Personalization:** Strategies for targeting different customer segments and personalizing the user experience through algorithms and data analysis.
7. Conclusion:

Airbnb's transformation of the hospitality industry offers an in-depth look into technology-driven disruption, entrepreneurial innovation, community engagement, legal complexities, and social impact. The multifaceted nature of Airbnb's journey provides a rich context for exploring diverse business concepts.

8. Further Exploration and Assignments:

- **Platform Analysis Project:** Students analyze Airbnb's platform functionality, user experience, and technological innovations.
- **Regulatory Environment Study:** Research and debates on the legal and ethical aspects of Airbnb's operations in different regions.
- **Global Strategy Simulation:** Group exercise to plan Airbnb's entry into a new market, considering cultural, legal, and market dynamics.
- **Social Impact Assessment:** Critical evaluation of Airbnb's social responsibility efforts, community impact, and sustainability initiatives.

40.11 Starbucks: Brewing Success Through Innovation and Responsibility

1. Introduction:

Starbucks, founded in 1971 in Seattle, Washington, has become a global coffee icon, known for its premium quality coffee, unique store ambiance, and commitment to social responsibility. This case study examines Starbucks' journey from a single store to an international chain, focusing on its strategic decisions, marketing practices, innovations, and challenges.

2. Background:

- **Founding and Early Years:** How Starbucks transformed from a single store selling quality coffee beans into a global coffeehouse chain.
- **Mission and Vision:** An examination of Starbucks' commitment to inspiring and nurturing the human spirit, one cup at a time.

3. Business Model and Strategies:
   - **Retail Innovation:** An exploration of Starbucks' unique store designs, customer experience, and the introduction of the "third place" concept.
   - **Product Diversification:** Starbucks' expansion into various products, including specialty beverages, food, packaged products, and even non-coffee items.
   - **Global Expansion:** Strategies and challenges in entering new markets across different continents.
4. Marketing and Branding:
   - **Brand Building and Positioning:** How Starbucks built a strong brand that emphasizes quality, community, and ethical sourcing.
   - **Loyalty Programs:** The impact and success of Starbucks' rewards program in enhancing customer loyalty and retention.
   - **Digital Engagement:** Utilizing mobile apps, social media, and digital marketing to engage customers.
5. Sustainability and Social Responsibility:
   - **Ethical Sourcing:** Commitment to sourcing ethically produced coffee through fair trade practices and farmer support.
   - **Environmental Initiatives:** Efforts in reducing waste, conserving energy, and promoting reusable products.
   - **Community Engagement:** Investing in local communities through education, volunteerism, and support for local causes.
6. Challenges and Criticisms:
   - **Market Saturation:** The challenge of maintaining growth amid increasing competition and market saturation.
   - **Cultural Sensitivity:** Navigating cultural differences in global markets and occasional backlashes.
   - **Economic Factors:** Responding to economic downturns and changes in consumer spending habits.
7. Innovation and Technology:
   - **Mobile Ordering:** Implementing mobile ordering and payment systems to enhance convenience.
   - **Data Analytics:** Leveraging data to personalize marketing and enhance customer experiences.
   - **Partnerships with Technology Companies:** Collaborations to expand reach and offer new products.
8. Conclusion:

Starbucks' story offers valuable insights into brand building, global expansion, innovation, social responsibility, and resilience in the face of challenges. Its journey from a single store to a global chain showcases the importance of strategic decision-making, adaptability, and commitment to core values.

9. Further Exploration:

- **Supply Chain Analysis:** Investigate Starbucks' complex supply chain and its approach to ensuring quality and ethical practices.
- **Competitive Landscape Study:** Analyze Starbucks' competitive positioning and the dynamics of the coffeehouse industry.
- **Crisis Management Review:** Examine Starbucks' response to various challenges and crises over the years.

40.12 The Walt Disney Company: A Kingdom of Creativity and Innovation

1. Introduction:

The Walt Disney Company, founded in 1923 by Walt and Roy O. Disney, has grown from a small animation studio to a global entertainment conglomerate. This case study delves into Disney's storied history, business diversification, technological leadership, and strategies that have made it a symbol of creativity and imagination.

2. Background:
    - **Founding and Early Success:** The birth of Mickey Mouse, the creation of the first synchronized sound and full-color cartoons, and the groundbreaking "Snow White and the Seven Dwarfs."
    - **Expanding the Magic Kingdom:** Disney's foray into theme parks, beginning with Disneyland in 1955 and followed by a global expansion.
3. Business Model and Strategies:
    - **Diversification:** Exploration of Disney's diversification into various entertainment sectors, including movies, television, theme parks, merchandise, and media networks.
    - **Content Creation and Distribution:** Examination of Disney's strategies in producing and distributing content through various channels, including streaming services like Disney+.
    - **Global Expansion:** Analysis of Disney's strategies to enter and thrive in international markets, including China and Europe.
4. Marketing and Branding:
    - **Brand Building:** How Disney built a universally loved brand based on storytelling, characters, and immersive experiences.
    - **Synergy:** Understanding how Disney leverages its characters and stories across multiple business segments.
    - **Digital Engagement:** Exploration of Disney's digital marketing efforts, social media presence, and engagement with younger audiences.

5. Technological Innovation:
    - **Revolutionizing Animation:** Disney's pioneering role in animation technology, including the introduction of CGI.
    - **Immersive Experiences:** The integration of technology in theme parks for personalized and interactive experiences.
6. Acquisitions and Collaborations:
    - **Strategic Acquisitions:** Insight into Disney's acquisitions, including Pixar, Marvel, Lucasfilm, and 21st Century Fox.
    - **Collaborations and Partnerships:** Exploration of Disney's collaborations with other companies to enhance its product offerings and reach.
7. Social Responsibility and Ethics:
    - **Corporate Social Responsibility (CSR):** Disney's efforts in environmental conservation, community support, and ethical sourcing.
    - **Content and Cultural Sensitivity:** Balancing storytelling with cultural respect and inclusiveness.
8. Challenges and Future Prospects:
    - **Market Saturation and Competition:** Navigating an increasingly competitive media and entertainment landscape.
    - **Regulatory and Legal Challenges:** Adhering to varying regulations across global markets.
    - **Pandemic Response:** Adaptation and response to the COVID-19 pandemic's impact on various business segments.
9. Conclusion:

The Walt Disney Company's journey offers a captivating exploration of creativity, innovation, strategic thinking, and adaptability. From pioneering animation to building global theme parks, launching streaming services, and acquiring leading entertainment brands, Disney's story is a rich lesson in entrepreneurship, marketing, technology, and global business strategies.

10. Further Exploration:

- **Leadership Analysis:** Investigate Disney's leadership strategies and the role of key leaders in shaping the company.

- **Competitive Landscape Study:** Analyze Disney's competitive positioning and the dynamics of the entertainment industry.

- **Crisis Management Review:** Examine Disney's response to various challenges, including economic downturns and unexpected crises.

40.13 McDonald's: Serving Success with a Side of Innovation

1. Introduction:

McDonald's is more than just a fast-food chain; it's a global phenomenon that has shaped the way people eat around the world. Founded in 1940 by Richard and Maurice McDonald, the company has since evolved into a multi-billion-dollar giant with thousands of locations worldwide. This case study examines the key ingredients behind McDonald's success.

2. Background:
    - **Founding and Early Growth:** A look at McDonald's beginnings, from a single drive-in to the creation of the Speedee Service System, a precursor to the modern fast-food restaurant.
    - **Global Expansion:** How McDonald's turned the Golden Arches into an international symbol, adapting to various cultures and tastes.
3. Business Model and Strategies:
    - **Franchising:** Exploration of McDonald's franchising model and how it fueled the company's rapid growth.
    - **Menu Innovation:** How McDonald's constantly innovates its menu to meet consumer demands and local preferences.
    - **Supply Chain Management:** Examination of McDonald's logistical prowess in sourcing and distributing ingredients across the globe.
    - **Sustainability Efforts:** An insight into McDonald's initiatives to reduce environmental impact and promote sustainable practices.
4. Marketing and Branding:
    - **Iconic Branding:** Understanding how the Golden Arches and characters like Ronald McDonald became global icons.
    - **Advertising and Promotions:** A review of memorable ad campaigns and marketing strategies that resonate with various demographics.
    - **Customer Experience:** How McDonald's focuses on customer satisfaction through services like McDelivery and the recent digital transformation.
5. Technological Innovation:
    - **Digital Ordering and Mobile Apps:** Exploration of McDonald's embrace of technology to enhance customer convenience.
    - **Smart Restaurants:** How technology is changing the in-store experience, from kiosks to AI-powered drive-thrus.
6. Challenges and Controversies:
    - **Health Concerns:** Analysis of criticisms regarding the nutritional content of McDonald's food and the company's response.
    - **Labor Practices:** Discussion of challenges related to employee wages, benefits, and working conditions.
    - **Competitive Landscape:** Examination of the fast-food market competition and how McDonald's maintains its edge.
7. Future Prospects:
    - **Adaptation to Changing Consumer Preferences:** The shift towards healthier options and how McDonald's is responding.
    - **Investments in Technology:** Future technological innovations

that may shape the McDonald's experience.
   - **Sustainability Goals:** Long-term objectives in minimizing environmental impact and promoting social responsibility.
8. Conclusion:

McDonald's journey offers a multifaceted case study in entrepreneurship, innovation, marketing, global expansion, and adaptability. From flipping burgers in a single location to flipping the script on fast food worldwide, the company continues to evolve, facing new challenges and seizing opportunities.

40.14 Dove (Unilever): Crafting Beauty and Confidence

1. Introduction:

Dove, a personal care brand owned by Unilever, has become synonymous with beauty and self-esteem through its innovative products and socially conscious campaigns. This case study invites you to explore Dove's journey and its commitment to promoting a more inclusive and positive depiction of beauty.

2. Background:
   - **Dove's Inception:** A look at the brand's origins in 1957 with the launch of the Dove Beauty Bar.
   - **Product Portfolio:** Overview of Dove's wide range of personal care products, including body wash, hair care, and skincare.
3. Marketing and Branding:
   - **The "Real Beauty" Campaign:** Examination of Dove's groundbreaking campaign that challenged conventional beauty standards.
   - **Customer Engagement:** Insights into Dove's interaction with customers through social media, events, and community outreach.
   - **Global Expansion:** Strategies behind Dove's growth into various international markets and adaptation to different cultures.
4. Product Innovation:
   - **Research and Development:** A look at how Dove constantly innovates its product line through scientific research and consumer insights.
   - **Sustainability Initiatives:** Understanding Dove's efforts in reducing environmental impact and promoting ethical sourcing.
5. Social Campaigns and Responsibility:
   - **Promoting Self-Esteem:** Analysis of Dove's initiatives to enhance self-esteem, particularly among young women, through education and advertising.
   - **Partnerships and Collaborations:** How Dove collaborates with NGOs, influencers, and other stakeholders to amplify social messages.
6. Challenges and Controversies:

- **Market Competition:** Assessment of the competitive landscape and how Dove differentiates itself.
- **Advertising Backlash:** Discussion of certain advertising missteps and how the brand managed the fallout.

7. Future Prospects:
    - **Trend Adaptation:** Exploration of how Dove aligns with emerging beauty and wellness trends.
    - **Technology Integration:** How Dove leverages technology, including AI and data analytics, for product development and personalized experiences.
    - **Sustainability Goals:** Examination of Dove's long-term commitment to environmental sustainability and ethical practices.
8. Conclusion:

Dove's journey presents an engaging case study that goes beyond products and marketing to encompass social values, consumer connection, innovation, and global reach. The brand's commitment to challenging beauty norms and promoting self-esteem has set it apart in a crowded market.

40.15 IKEA: A Symphony of Design, Affordability, and Sustainability

1. Introduction:
    - **Founding and Mission:** Founded in Sweden in 1943 by Ingvar Kamprad, IKEA's mission is to "create a better everyday life for many people." It emphasizes affordability, design, and functionality.
    - **Overview of Offerings:** IKEA offers a wide range of home furnishings, including furniture, kitchen appliances, decor, and accessories.
    - **Global Presence:** With over 400 stores in 50 countries, IKEA has become a global leader in the home furnishing industry.
2. IKEA's Business Model:
    1. **Product Design and Development:** IKEA's products are known for minimalist design, functionality, and ease of assembly. Collaboration with designers worldwide keeps its offerings fresh and innovative.
    2. **Supply Chain and Manufacturing:** A well-integrated supply chain with close relationships to over 1,000 suppliers allows IKEA to maintain low costs while ensuring quality and sustainability.
    3. **Retail Experience:** The IKEA in-store experience is distinctive with showrooms, self-service warehouses, and in-store restaurants offering Swedish cuisine.
    4. **Pricing Strategy:** IKEA's cost-conscious approach means designing products from the price tag up, ensuring affordability without compromising on quality.
    5. **Digitalization and E-commerce:** With a strong online presence, IKEA provides customers with online shopping options, planning

tools, and virtual product previews.

3. Marketing and Branding:
    1. **Advertising Campaigns:** IKEA uses creative and often humorous advertising to appeal to a broad customer base, focusing on life improvement and solutions.
    2. **Online Engagement:** Digital catalogs, apps, and social media keep IKEA's audience engaged and provide valuable customer insights.
    3. **In-store Promotions:** Seasonal displays and in-store events promote new products and encourage customer interaction.
    4. **Brand Identity and Values:** IKEA's brand emphasizes sustainability, inclusiveness, and accessibility.
4. Sustainability Initiatives:
    1. **Environmental Practices:** Commitment to sustainable sourcing, waste reduction, and energy efficiency are core to IKEA's operations.
    2. **Renewable Energy Projects:** IKEA invests in wind and solar energy, aiming to produce as much renewable energy as it consumes in its operations by 2030.
    3. **Social Responsibility:** The IKEA Foundation supports initiatives related to children's education, refugee support, and climate change.
    4. **Sustainable Product Lines:** IKEA offers products that promote sustainable living, from energy-efficient appliances to recycled materials.
5. Global Expansion:
    1. **Cultural Adaptation:** IKEA adapts its product lines and marketing to reflect local tastes, customs, and living conditions.
    2. **Market Entry Strategies:** IKEA studies each market carefully, adapting its store format and product selection to local needs.
    3. **Challenges in Different Markets:** Navigating regulations, cultural differences, and local competition has posed challenges in some markets.
6. Challenges and Controversies:
    1. **Competition and Market Pressures:** IKEA faces competition from both traditional furniture stores and online platforms.
    2. **Cultural Missteps:** Some global marketing campaigns have been criticized for insensitivity to local cultures.
    3. **Quality Concerns:** IKEA's emphasis on low cost has sometimes led to perceived quality issues.
7. Future Prospects and Innovation:
    1. **Emerging Markets:** Expansion into new markets like India and South America presents opportunities and challenges.
    2. **Technological Innovations:** IKEA is exploring augmented reality, artificial intelligence, and smart home technologies.
    3. **Sustainability Goals:** Commitment to further sustainability through its entire value chain.
    4. **Collaborations and Partnerships:** IKEA's collaboration with designers, tech companies, and even other retailers fuels innovation.

8. Conclusion:

IKEA's unique blend of design, affordability, sustainability, and global reach has made it a standout brand in the home furnishing industry. The company's multifaceted approach offers a rich study of modern retail, branding, international business, and corporate responsibility. The complexities and successes of IKEA's model provide invaluable insights and inspiration for students across various disciplines.

40.16 LEGO: Building Blocks of Innovation and Success

1. Introduction:
    1. **Founding and History:** LEGO was founded in 1932 by Ole Kirk Christiansen in Billund, Denmark. The LEGO brick, as we know it today, was launched in 1958.
    2. **Product Portfolio:** Beyond the iconic bricks, LEGO's products include themed sets, video games, movies, and educational tools.
    3. **Mission and Values:** LEGO's mission is to "Inspire and develop the builders of tomorrow" through creative play and learning.
2. Product Development:
    1. **Innovation in Design:** LEGO constantly innovates its product line, incorporating new themes and licensed partnerships (e.g., Star Wars, Marvel).
    2. **Quality and Precision:** The manufacturing process emphasizes precision and quality, ensuring compatibility across generations of LEGO bricks.
    3. **Digital Expansion:** LEGO has embraced digital gaming and augmented reality experiences, extending the brand into the digital realm.
3. Marketing and Community Engagement:
    1. **Brand Building:** LEGO's brand revolves around creativity, imagination, learning, and fun.
    2. **Advertising and Promotion:** Utilizing various channels, LEGO engages customers through inventive advertising campaigns and social media.
    3. **Community Engagement:** LEGO Ideas invites fans to submit and vote on new product ideas. The LEGO community is actively engaged in product development, events, and online forums.
4. Retail and Distribution:
    1. **Retail Experience:** LEGO stores offer hands-on experiences with play areas, workshops, and exclusive products.
    2. **Online Shopping:** The online store provides an extensive product selection, customization options, and exclusive membership benefits.
    3. **Global Distribution:** LEGO products are available in more than

140 countries through various retail channels.

5. Education and Social Responsibility:
    1. **LEGO Education:** Through LEGO Education, the company offers learning solutions that encourage hands-on, playful learning in schools.
    2. **Charitable Activities:** The LEGO Foundation supports children's development and learning through various global initiatives.
    3. **Environmental Sustainability:** LEGO is committed to reducing its environmental impact, including the goal to produce all products and packaging with sustainable materials by 2030.
6. Challenges and Controversies:
    1. **Market Pressures:** Facing competition from both traditional toys and digital games, LEGO has had to continuously innovate and adapt.
    2. **Intellectual Property Issues:** LEGO has faced legal challenges around patents and copyrights, particularly concerning the design of its bricks.
    3. **Economic Fluctuations:** Economic downturns and shifts in consumer behavior have influenced LEGO's sales and growth strategies.
7. Global Expansion:
    1. **Adaptation to Local Markets:** LEGO tailors its marketing and product strategies to different cultures and consumer preferences.
    2. **Challenges in Emerging Markets:** Entering new markets such as China has presented both opportunities and challenges, including issues related to counterfeiting.
8. Future Prospects and Strategic Directions:
    1. **Technological Innovation:** LEGO continues to explore new technologies, such as 3D printing and artificial intelligence.
    2. **Collaborations and Licensing:** Partnerships with entertainment franchises and designers fuel creativity and market reach.
    3. **Focus on Adult Fans:** LEGO has been expanding its appeal to adult fans through complex sets and themes that cater to various interests.
9. Conclusion:

LEGO's journey from a small carpentry shop to a global brand is a study in innovation, adaptability, community engagement, and brand stewardship. Its commitment to quality, creativity, and social responsibility offers a multifaceted case study with insights into product development, marketing, sustainability, global business strategy, and more. The story of LEGO inspires aspiring entrepreneurs, marketers, designers, and leaders to think creatively and act with purpose and integrity.

40.17 Slack: Revolutionizing Workplace Communication

1. Introduction:
    - **Founding and Background:** Launched in 2013 by Stewart Butterfield, Eric Costello, Cal Henderson, and Serguei Mourachov, Slack has quickly become one of the leading tools for team communication.
    - **Business Model:** Slack offers a freemium model where basic features are free, with paid plans for more functionality.
2. Product Development:
    1. **Key Features:** Slack provides channels, direct messaging, file sharing, integrations with other tools, and more to enhance team communication.
    2. **Innovation and Updates:** Continual updates and feature enhancements have kept Slack at the forefront of workplace communication tools.
    3. **User-Centric Design:** Slack's interface is designed for ease of use and collaboration, reducing email overload.
3. Marketing and Growth:
    1. **Target Audience:** Primarily targeting businesses, both small and large, Slack has also found usage in communities and other groups.
    2. **Growth Strategies:** Referral programs, partnerships, and effective content marketing have contributed to Slack's rapid adoption.
    3. **Customer Engagement:** Slack has utilized community engagement, feedback, and customer support to foster loyalty and improve its product.
4. Competition and Challenges:
    1. **Competitors:** Major competitors include Microsoft Teams, Zoom, and others offering communication and collaboration tools.
    2. **Differentiation:** Slack's integrations, customization, and user experience have been key differentiators.
    3. **Security Concerns:** As with many digital platforms, security and privacy have been challenges, and Slack has implemented measures to ensure data protection.
5. Business Model and Monetization:
    1. **Freemium to Premium:** The free version attracts users, while additional features and support drive customers to paid plans.
    2. **Enterprise Solutions:** Slack's Enterprise Grid offers solutions tailored to large organizations, including advanced security and administrative features.
6. Global Expansion:
    1. **Localization and Cultural Adaptation:** Slack has localized its product for various markets and cultures to drive global adoption.
    2. **Challenges in Emerging Markets:** Issues such as local compliance, competition, and connectivity can present challenges in various

regions.

7. Impact of Remote Work:
    1. **Pandemic Response:** The shift to remote work during the COVID-19 pandemic led to a surge in Slack usage, adapting to new work patterns.
    2. **Long-term Trends:** Remote and hybrid work trends may shape Slack's future development and market positioning.
8. Acquisitions and Partnerships:
    1. **Strategic Acquisitions:** Acquiring companies like Rimeto added capabilities to Slack's portfolio.
    2. **Partnerships:** Collaborations with companies like Google, Salesforce, and others have extended Slack's functionality.
9. Future Prospects:
    1. **Salesforce Acquisition:** The pending acquisition by Salesforce as of the cut-off knowledge date may significantly shape Slack's future direction.
    2. **Continued Innovation:** Slack continues to explore new features, integrations, and market opportunities.
10. Conclusion:

Slack's story offers insights into the fast-paced world of technology startups, product development, global expansion, and market competition. Its response to changing work patterns and its strategic acquisitions and partnerships make it a rich subject for study. The lessons from Slack's journey are relevant to aspiring entrepreneurs, product managers, marketers, and others interested in technology, innovation, and the future of work.

40.18 Patagonia: A Case Study in Sustainable Business Practices

1. Introduction:
    1. **Background:** Patagonia, founded in 1973 by Yvon Chouinard, is an outdoor clothing and gear retailer known for its commitment to environmental sustainability.
    2. **Mission:** "Build the best product, cause no unnecessary harm, use business to inspire and implement solutions to the environmental crisis."
2. Product Development:
    1. **Innovation:** Patagonia has been a leader in developing sustainable fabrics and materials.
    2. **Quality & Durability:** Emphasizing long-lasting products to reduce consumerism.
    3. **Recycling & Repairing:** Offering repair services and encouraging recycling of products through programs like "Worn Wear."
3. Marketing and Branding:

   1. **Transparency:** Publicly sharing supply chain information and environmental impacts.
   2. **Activism Marketing:** Taking strong stances on environmental and social issues.
   3. **Community Engagement:** Collaborating with NGOs and community organizations.
4. Sustainability Initiatives:
   1. **Supply Chain:** Focusing on ethical production, fair labor practices, and organic materials.
   2. **Environmental Activism:** Regularly donating to environmental causes and supporting conservation efforts.
   3. **B Corp Certification:** Patagonia is a certified B Corporation, aligning profit with purpose.
5. Financial Performance:
   1. **Profit vs. Purpose:** Balancing strong financial growth with a commitment to environmental and social responsibility.
   2. **Investing in Sustainability:** Reinvesting profits in sustainable initiatives and environmental causes.
6. Challenges:
   1. **Market Competition:** Navigating a competitive market while maintaining ethical standards.
   2. **Scale and Growth:** Balancing growth and scalability with sustainability commitments.
   3. **Greenwashing Accusations:** Managing perceptions and criticisms related to authenticity and impact.
7. Global Presence:
   1. **International Expansion:** Adapting sustainable practices across diverse markets and cultures.
   2. **Global Partnerships:** Collaborating with global organizations to expand environmental initiatives.
8. Corporate Culture:
   1. **Employee Engagement:** Fostering a workplace culture that aligns with company values.
   2. **Leadership and Governance:** Maintaining leadership that embodies the brand's ethos.
9. Impact on Industry:
   1. **Influencing Other Brands:** Patagonia's practices have influenced other companies to consider sustainability.
   2. **Industry Collaboration:** Working with competitors on common goals such as responsible sourcing.
10. Future Prospects:
    1. **Adaptation to Climate Change:** Developing strategies to mitigate and adapt to the impacts of climate change.
    2. **New Market Opportunities:** Exploring new product lines and markets while adhering to core values.
11. Conclusion:

Patagonia serves as a compelling example of a company that has successfully integrated sustainability, ethical considerations, and environmental activism into every aspect of its business. From innovative product development to bold marketing strategies and influential industry leadership, Patagonia's case study offers valuable insights for those interested in business ethics, environmental stewardship, social entrepreneurship, and innovative brand management. The brand's ongoing challenges and successes provide rich material for analysis and reflection on the future of sustainable business practices.

40.19 Spotify: Transitioning from music sales to subscription streaming

1. Introduction:
    1. **Background:** Spotify, founded in 2006 by Daniel Ek and Martin Lorentzon, transformed the way people access and enjoy music.
    2. **Mission:** "To unlock the potential of human creativity—by giving a million creative artists the opportunity to live off their art and billions of fans the opportunity to enjoy and be inspired by it."
2. Product and Technology:
    1. **Streaming Model:** Spotify's on-demand streaming model allows users to access millions of songs and podcasts.
    2. **Algorithm & Personalization:** The use of algorithms to create personalized playlists and recommendations.
3. Business Model:
    1. **Freemium Model:** Free, ad-supported tier alongside premium subscriptions.
    2. **Revenue Streams:** Subscriptions, advertising, and partnerships.
4. Marketing and Branding:
    1. **User Engagement:** Innovative playlists like "Discover Weekly" engage users.
    2. **Collaborations:** Partnerships with artists, labels, and other brands.
5. Global Expansion:
    1. **International Reach:** Spotify has expanded to numerous countries, adapting to various markets and regulations.
    2. **Localized Content:** Offering content that resonates with local cultures and tastes.
6. Competition and Challenges:
    1. **Market Competitors:** Facing competitors like Apple Music, Amazon Music, and YouTube Music.
    2. **Royalty Disputes:** Navigating complex relationships with labels, artists, and rights holders.
7. Sustainability and Social Impact:
    1. **Environmental Footprint:** Efforts to reduce carbon footprint and promote sustainable practices.

   2. **Supporting Artists:** Initiatives to support emerging artists and creatives.
8. Innovation and Future Prospects:
   1. **New Features:** Continual innovation in features and user experience.
   2. **Podcasts and Original Content:** Investing in podcasts and original content to diversify offerings.
   3. **Technology Investments:** Exploring technologies like AI to enhance user experience.
9. Impact on the Industry:
   1. **Changing Consumer Behavior:** Transforming the way people consume and interact with music.
   2. **Influence on the Music Industry:** Affecting record labels, artists, and music distribution.
10. Conclusion:

Spotify's rise as a leading music streaming platform offers a multifaceted case study encompassing technology innovation, marketing strategies, global expansion, and industry impact. From navigating complex licensing agreements to crafting personalized user experiences, Spotify's journey provides valuable insights into digital transformation, competitive strategy, customer engagement, and the future of entertainment. It serves as a valuable example for understanding modern business dynamics in the digital age, including the ongoing challenges and opportunities of operating in a rapidly evolving industry.

40.20 Warby Parker: Disrupting the traditional eyewear market with an online-first approach

1. Introduction:
   1. **Background:** Founded in 2010, Warby Parker aimed to offer designer eyewear at a fraction of the price through a direct-to-consumer model.
   2. **Mission:** "To offer designer eyewear at a revolutionary price, while leading the way for socially conscious businesses."
2. Product and Customer Experience:
   1. **Design:** In-house design leading to unique and affordable eyewear.
   2. **Home Try-On:** A free program allowing customers to try on glasses at home before purchasing.
3. Business Model:
   1. **Direct-to-Consumer:** Selling directly to customers through e-commerce and physical stores, cutting out intermediaries.
   2. **Social Responsibility:** "Buy a Pair, Give a Pair" program donates glasses to those in need.

4. Marketing and Branding:
   1. **Digital Marketing:** Effective use of social media and content marketing.
   2. **Community Engagement:** Building brand loyalty through community events and collaborations.
5. Global Expansion and Retail Strategy:
   1. **Physical Stores:** Combining e-commerce with brick-and-mortar stores for an omnichannel experience.
   2. **International Growth:** Expanding to Canada and other markets, adapting to local regulations and preferences.
6. Competition and Challenges:
   1. **Traditional Competitors:** Competition with traditional eyewear brands and retailers.
   2. **Copycat Brands:** Managing competition from similar direct-to-consumer eyewear startups.
7. Sustainability Initiatives:
   1. **Environmentally Conscious Manufacturing:** Commitment to using sustainable materials.
   2. **Carbon Neutrality:** Efforts to reduce and offset carbon emissions.
8. Innovation and Technology:
   1. **Virtual Try-On:** Use of augmented reality for virtual try-ons via mobile app.
   2. **Telehealth Services:** Offering eye exams and prescriptions through telehealth technology.
9. Impact on the Industry:
   1. **Disrupting Traditional Retail:** Changing the way people shop for glasses.
   2. **Promoting Social Responsibility:** Encouraging other brands to adopt socially responsible practices.
10. Conclusion:

Warby Parker's innovative approach to eyewear retail has not only disrupted traditional industry practices but also set new standards in customer experience, social responsibility, and sustainability. Through its unique business model, commitment to social causes, and use of technology, Warby Parker has carved out a unique position in the market. The case study of Warby Parker offers valuable insights into how innovative thinking, customer-centric approaches, and ethical business practices can create a strong brand identity and successful business in today's competitive retail landscape. It's an exemplary story for understanding modern entrepreneurship, retail strategies, marketing, and social entrepreneurship.

40.21 Allbirds: A Case Study in Sustainable Footwear Innovation

1. Introduction:
   1. **Background:** Allbirds, founded in 2016 by Tim Brown and Joey Zwillinger, aimed to create comfortable and sustainable footwear.
   2. **Mission:** "To tread lighter on the planet while making better things people love to wear."
2. Product Innovation:
   1. **Sustainable Materials:** Allbirds uses renewable materials like merino wool and eucalyptus fiber.
   2. **Comfort and Design:** Combining sustainable materials with comfortable and aesthetically appealing design.
3. Business Model:
   1. **Direct-to-Consumer:** Selling directly to customers to reduce costs and improve accessibility.
   2. **Ethical Sourcing:** Ensuring the ethical treatment of animals and workers in the supply chain.
4. Marketing and Branding:
   1. **Storytelling:** Emphasizing the brand's commitment to sustainability and innovative materials.
   2. **Word-of-Mouth:** Leveraging satisfied customers as brand advocates.
5. Global Expansion:
   1. **International Presence:** Expanding into international markets while staying true to the brand's values.
   2. **Localized Initiatives:** Tailoring products and marketing to suit local preferences.
6. Competition and Challenges:
   1. **Market Competitors:** Competing with established footwear brands and other sustainable startups.
   2. **Scale and Sustainability:** Balancing growth with maintaining eco-friendly practices.
7. Sustainability Initiatives:
   1. **Carbon Footprint:** Measuring and reducing the brand's carbon footprint.
   2. **Circular Economy:** Exploring ways to make footwear more recyclable and sustainable.
8. Social Impact:
   1. **Transparency:** Sharing information about the supply chain and material sources.
   2. **Community Engagement:** Partnering with organizations for social and environmental causes.
9. Innovation and Future Prospects:
   1. **Research and Development:** Continuing to innovate with new

   materials and product lines.
   2. **Market Expansion:** Exploring new markets and consumer segments.
10. Impact on the Industry:
    1. **Changing Consumer Behavior:** Influencing the way consumers think about sustainable products.
    2. **Inspiring Competitors:** Encouraging other brands to prioritize sustainability.
11. Conclusion:

Allbirds' unique approach to footwear production, blending innovation, comfort, and sustainability, has positioned it as a leader in the sustainable fashion movement. The Allbirds case study provides a valuable window into the world of sustainable business, marketing, and product innovation. By exploring Allbirds' strategies and challenges, students can gain insights into how a commitment to ethical practices, environmental consciousness, and customer satisfaction can drive success in today's competitive market. The case offers lessons for those interested in entrepreneurship, sustainable business practices, and ethical consumerism.

Chapter 41

Case Studies in Advertising

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